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José M. Mazón

Elements of Functional Analysis

From Banach Spaces to Elliptic Problems

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José M. Mazón

Elements of Functional Analysis

From Banach Spaces to Elliptic Problems

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José M. Mazón
Departamento de Análisis Matemático
Universitat de Valencia
Burjassot, Valencia, Spain

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Preface

Functional Analysis can be understood as the change that occurs in *Mathematical Analysis* when moving from the study of individual functions to the study of *function spaces*, sets of such spaces, the various structures present in these spaces and the mappings between them. This theory, developed primarily during the twentieth century, has been and continues to be essential for the study of *Partial Differential Equations*.

In this textbook, we first present the general theory: *Banach spaces*, *Hilbert spaces*, and the *spectral theory of operators*. We then proceed to a thorough study of *Schwartz's Theory of Distributions* and some of its applications, such as the *fundamental solutions of classical operators in Physics*, the *Malgrange-Ehrenpreis Theorem*, *hypoelliptic operators*, and the *Schrödinger Equation*.

We dedicate an extensive chapter to the study of *Sobolev spaces*, including a section on *functions of bounded variation*. These spaces provide the appropriate framework for the study of *elliptic boundary value problems*, which is the focus of the final chapter.

The first chapter is devoted to some historical notes on the development of Functional Analysis. In addition, each chapter begins with a historical overview of the topic addressed therein. I consider it very important that the student knows the historical background and motivations of the topics to be studied.

This textbook compiles, in a more elaborate form, various courses given at the University of Valencia for more than 30 years. A subject as classical as the one treated here lends itself little to originality; nevertheless, we hope to have contributed our personal perspective and experience in this field.

The only prerequisites necessary to follow this text are the basic elements of General Topology, Measure Theory, and Differential Calculus. The results used from these theories are given in the two Appendices that conclude this textbook.

In the bibliography, we list the source used in the preparation of this textbook, including classical references on this subject. However, it may be far from complete and I apologize for any omission.

Burjassot, València, Spain José M. Mazón
Agosto 2025

Declarations

Competing Interests The author has no competing interests to declare that are relevant to the content of this manuscript.

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Chapter 1
Brief Historical Notes



We understand *Functional Analysis* as the transformation that occurs in *Mathematical Analysis* when we move from considering individual functions to function spaces, sets of such spaces, the different structures that exist in these spaces, and the mappings between them. The idea of a *"function space"* dates back to the famous doctoral dissertation[1] of Bernhard Riemann in 1851, where he states:

> *The totality of such functions forms a connected domain closed in itself.*

Later, in a paper presented to the Academy of Sciences of Paris in 1861, Riemann formulated the idea of a function space of infinite dimension in the following way:

> *But there also exist manifolds in which the determination of position requires not a finite number, but an infinite sequence or a continuous manifold of determinations of magnitudes.*

Some authors believe that Functional Analysis originated in Italy, with the *risorgimento*, at the end of the nineteenth century, when Italian mathematics flourished once again. In addition to the great geometers: Betti, Beltrami, and Cremona, a major school of analysts also emerged, associated with the names of: Ascoli, Arzelà, Dini, Peano, and Volterra. Probably it was the classic Ascoli–Arzelà Theorem, published in 1884, which states that

> *If a sequence* $\{f_n\}$ *of real functions on* $[0, 1]$ *is uniformly bounded and equicontinuous on* $[0, 1]$, *then it has a subsequence that converges uniformly on* $[0, 1]$,

a result which generalizes the classical Bolzano–Weierstrass Theorem to the infinite-dimensional space $C([0, 1])$, the first result properly belonging to Functional Analysis. Among these Italian mathematicians, the one who had the greatest influence on the birth of Functional Analysis as a discipline was Vito Volterra, who, in 1887, in a series of papers[2] motivated by problems in the Calculus of Variations,

[1] Grundlagen für eine allgemeine Theorie einer veränderlichen komplexen Größe.

[2] Rendiconti della Accademia dei Lincei **3**(1887), 97–105; 141–146; 153–158.

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investigated a special class of functions which he called line functions (*fonctions de lignes*).

A line function was, for Volterra, a real function F, defined on certain continuous functions $y(x)$ defined on an interval $[a, b]$ with values in $\mathbb{R}^N$. These works by Volterra from 1887 are considered the origin of Functional Analysis. Although some of the ideas in Volterra's work were criticized by J. Hadamard in an article[3] from 1902, Hadamard was one of the great promoters of Functional Analysis. His first paper on Functional Analysis was a short note presented at the First International Congress of Mathematicians held in Zürich in 1897, where he considered curves as points of a set. From 1897 to 1906, Hadamard and his student Maurice Fréchet developed the ideas of Cantor's Set Theory as a new tool in Functional Analysis.

Just like Volterra, motivated by problems in the Calculus of Variations, Hadamard, in a note[4] to the Academy of Sciences of Paris, introduced the concept of a *functional* (fonctionelle) and proved that every functional T (a bounded linear map) on the space $C([a, b])$ of real continuous functions on the interval $[a, b]$, can be represented in the form

$$T(f) = \lim_{n\to\infty} \int_a^b f(x)\alpha_n(x)dx,$$

where α_n are continuous functions on $[a, b]$.

Fréchet quickly developed Hadamard's ideas in three papers[5] which culminated in his famous 1906 thesis[6] titled *"Sur quelques points du calcul fonctionnel"*, where he introduced the concept of a *metric space*, using the term *"écart"* to denote the metric (the term "metric space" was later coined by Hausdorff), as well as the key notions of *compactness, completeness*, and *separability*. Fréchet also considered functionals, which he referred to as *"opérations fonctionnelles"*, for which he defined continuity and semicontinuity, extending to functionals the concept of semicontinuity introduced by R. Baire for functions in 1899. In addition to the general theory, Fréchet also studied some specific examples of function spaces, such as the space $C([a, b])$, which he endowed with the "écart"

$$d(f, g) = \max\{|f(x) - g(x)| \ : \ x \in [a, b]\}, \quad f, g \in C([a, b]).$$

After Fréchet's thesis, both Hadamard and Fréchet continued to promote *"le calcul fonctionnel"*. The article that had a major influence was one of a more

[3] Bull. Soc. Math. de France **30**(1902), 40–43.

[4] Comptes Rendus Paris **136**(1903), 351–345.

[5] Trans. Amer. Math. Soc. **5** (1904), 493–499; **6** (1905), 134–140 and 435–449.

[6] Rendiconti del Circolo Matematico di Palermo **22** (1906), 1–71.

philosophical than technical nature, published by Hadamard in *L'Enseignement Mathématique* in 1912, where he wrote:

> *The calculus of variations is to functional operations what differential calculus is to functions.*

Following Hadamard's ideas, Fréchet introduced what is now known as the *Fréchet differential* in a series of articles, the first of which was published in 1909 and the last in 1925.[7] Another follower of Hadamard's ideas was R. Gateaux, who wrote several works on functionals and abstract differentiation in 1913–1914, where the concept now known as *Gâteaux differentiability* appears. Gateaux died in the early months of the First World War, and his manuscripts were published in 1919[8] by Paul Lévy, another student of Hadamard. It was precisely P. Lévy who was the first to publish, in 1922, a textbook titled *Leçon d'Analyse Fonctionnelle* (Gauthier–Villars, Paris, 1922), where for the first time the name Functional Analysis appeared.

One of the followers of Hadamard's work in *L'Enseignement Mathématique* was the Italian mathematician Leonida Tonelli. After writing a large number of papers on the Calculus of Variations from the viewpoint of functionals, he published a major work on the subject, *Fondamenti di calcolo delle variazioni* (2 volumes, 1922, 1924), where he developed the so-called *direct method* in the calculus of variations, which shifts the classical approach of using the Euler–Lagrange equation to that of directly minimizing the energy functional, using the concept of lower semicontinuity, since such functionals are not usually continuous.

Another branch of Analysis that had a decisive influence on the birth of Functional Analysis was the theory of integral equations, which began with the works of Liouville in 1837 on what are now called *Sturm–Liouville problems*, that is, problems of the form:

$$L[u] + \lambda u = 0,$$

with

$$L[u] = (pu')' + qu, \quad p > 0,$$

and with homogeneous boundary conditions

$$\alpha_1 u(a) + \alpha_2 u'(a) = 0, \quad \beta_1 u(b) + \beta_2 u'(b) = 0.$$

The connection with integral equations lies in the fact that if G is the Green's function of the operator L, solving the eigenvalue problem

$$\lambda\phi(x) = -\int_a^b G(x, y)\phi(y)\, dy,$$

[7] Ann. Sci. École Norm. Sup. (3) **42** (1925), 293–323.

[8] Bull. Soc. Math. de France **47** (1919), 70–96; **50** (1922), 1–37.

the solution to $L[u] + \lambda u = f$ satisfies the integral equation

$$u(x) + \frac{1}{\lambda} \int_a^b G(x, y)\phi(y)\, dy = \frac{1}{\lambda} f(x).$$

Another important class of integral equations appears in the works of Carl Neumann from 1870 on the existence of solutions for the *Dirichlet problem*. In this work, applying the method of successive approximations to solve the integral equation, Neumann expresses the solution as a series, today known as the *Neumann series*, which is of great importance in Spectral Theory. Although many works on particular integral equations appeared during this period—such as those by Paul du Bois-Reymond, H. Poincaré, and H. A. Schwarz—the first to establish existence and uniqueness for a general class of integral equations were J. M. Le Roux in 1895 and V. Volterra in 1896. The latter created a general theory for integral equations of the form

$$g(x) - \int_a^x k(x, y)g(y)\, dy = f(x),$$

which are now known as *Volterra integral equations*.

Now, the decisive works on integral equations are those by the Swedish mathematician Ivar Fredholm[9] in 1903, where he developed his famous theory on the equations that Hilbert later called *Fredholm equations of the second kind*, that is, integral equations of the form

$$g(x) - \lambda \int_a^b k(x, y)g(y)\, dy = f(x).$$

Fredholm's fundamental idea for solving this type of integral equation was to replace the integral with the corresponding Riemann sums, solve the resulting linear system using determinants, and then pass to the limit. In this way, he arrived at what is now known as the *Fredholm Alternative Theorem*.

Precisely, these ideas of Fredholm—moving from the finite to the infinite—were the starting point for a series of works by D. Hilbert, published in the *Göttinger Nachrichten* from 1904 to 1910, which would lead to the birth of one of the most powerful branches of Functional Analysis: Spectral Theory. Hilbert developed a general theory of continuous bilinear and quadratic forms, independent of integral equations but with applications to them. In these works, the space ℓ^2 of square-summable sequences of real numbers appears for the first time, along with the

[9] Acta Mathematica **27** (1903), 365–390.

concept of a *completely continuous* form on such spaces. He called an infinite symmetric quadratic form

$$Q(x) = \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} k_{ij} x_i x_j \qquad (k_{ij} = k_{ji})$$

completely continuous when

$$\lim_{n\to\infty} Q_n(x) = Q(x), \quad \text{where } Q_n(x) = \sum_{i=1}^{n} \sum_{j=1}^{n} k_{ij} x_i x_j$$

uniformly for all $x = (x_i) \in \ell^2$, with $\sum_{i=1}^{\infty} x_i^2 \leq 1$. He called it *bounded* when

$$|Q_n(x)| \leq M < \infty, \quad \forall\, n \in \mathbb{N}, \text{ and } \forall\, x = (x_i) \in \ell^2, \text{ with } \sum_{i=1}^{\infty} x_i^2 \leq 1.$$

As a consequence of his spectral theory, Hilbert proved that for a bounded form Q, the corresponding infinite system

$$x_i - \lambda \sum_{j=1}^{n} k_{ij} x_i x_j = 0, \qquad i = 1, 2, \ldots$$

has a solution in ℓ^2 if and only if λ is an eigenvalue.

Hilbert's ideas were further developed by his many outstanding students. In the period between 1900 and 1910 alone, he supervised approximately 40 doctoral theses, including those of G. Hamel, O. D. Kellogg, E. Schmidt, E. Hellinger, H. Weyl, A. Haar, R. Courant, and H. Steinhaus. Among all these students, it was Erhard Schmidt who made the most significant contributions to the emerging theory of Functional Analysis. Although Hilbert had considered the space ℓ^2, he did not go so far as to regard these sequences as coordinates of a point in a space, nor did he use geometric language. This task was carried out by E. Schmidt, who, in addition to considering sequences of complex numbers for the elements of ℓ^2, introduced the notation $\|z\|$ for

$$\sqrt{\sum_{i=1}^{\infty} z_i\, \overline{z_i}}, \qquad z = (z_i) \in \ell^2,$$

which was later called the norm of z; he used the notation (z, w) for the inner product $\sum_{i=1}^{\infty} z_i\, \overline{w_i}$, defined the concept of orthogonality, and established the generalized form of the Pythagorean Theorem, as well as Bessel's inequality and the Projection Theorem.

Another founder of Functional Analysis was the Hungarian mathematician Frédéric Riesz. One of his first results is what is now known as the Riesz–Fischer Theorem, since it was also established by E. Fischer at almost the same time,[10] a result which states the following:

Given a sequence $\{a_n\}$ of real numbers and $\{\phi_n\}$ an orthonormal system in $L^2([a,b])$, there exists a function $f \in L^2([a,b])$ which has the a_n as its Fourier coefficients, that is,

$$\int_a^b f(x)\phi_n(x)\,dx = a_n \quad n = 1, 2, \ldots$$

if and only if $\{a_n\} \in \ell^2$.

From this result it follows that if we consider the space $L^2([a,b])$ of all square-integrable functions on $[a,b]$ in the Lebesgue sense, endowed with the metric

$$d(f,g) = \sqrt{\int_a^b |f(x) - g(x)|^2\,dx},$$

then this space is complete and isomorphic to the Hilbert space ℓ^2. Riesz was one of the first to use in Functional Analysis the powerful tool of the Lebesgue integral, which Lebesgue had developed in his 1902 dissertation. In 1907, Riesz[11] and Fréchet,[12] another major promoter of the Lebesgue integral, independently proved that for every continuous linear functional T on $L^2([a,b])$, there exists a unique function $u \in L^2([a,b])$ such that

$$T(f) = \int_a^b f(x)u(x)\,dx, \qquad \text{for all } \ f \in L^2([a,b]),$$

a result that generalizes the one obtained by Hadamard in 1903. The general form of the representation theorem, i.e., in the context of abstract Hilbert spaces, was established by Riesz himself in 1934.[13]

In 1909, Riesz[14] made a major breakthrough in duality theory, solving the problem of representing continuous linear functionals on the space $C([a,b])$, which Hadamard and Fréchet had left unresolved (an excellent article on the history of the Riesz representation theorem is that of J. D. Gray [24]). Riesz showed that such

[10] Hilbert submitted Riesz's result to the *Göttinger Nachrichten* on March 9, 1907, while Fischer presented it in a seminar at the University of Brünn on March 5 of the same year.

[11] Comptes Rendus (Paris) **144** (1907), 1409–1411.

[12] Trans. Amer. Math. Soc. **8** (1907), 433–446.

[13] Acta Szeged **7**, 34–38.

[14] Comptes Rendus (Paris) **149** (1909), 974–977.

functionals can be represented by a Stieltjes integral of the form

$$T(f) = \int_a^b f(x)\,\alpha(x),$$

where α is a function of bounded variation on $[a, b]$.

In 1910, Riesz, once again using the great power of the Lebesgue integral, introduced the spaces L^p, $1 < p < \infty$, and extended Schmidt's results for $p = 2$. Without using the word dual, Riesz proved that the spaces $L^p([a, b])$ and $L^q([a, b])$ are dual to each other when $\frac{1}{p} + \frac{1}{q} = 1$, with $p > 1$. To do so, he established and used Hölder's and Minkowski's inequalities. In this same work,[15] Riesz defined the strong convergence of a sequence $\{f_n\}$ to f in $L^p([a, b])$ as

$$\lim_{n\to\infty} \int_a^b |f_n(x) - f(x)|^p\,dx = 0,$$

and weak convergence as

$$\lim_{n\to\infty} \int_a^b f_n(x)g(x)\,dx = \int_a^b f(x)g(x)\,dx, \quad \forall\, g \in L^q([a, b]).$$

He defined completely continuous transformations between L^p spaces as those that transform every weakly convergent sequence into a strongly convergent one, and noted their equivalence with the previous definition of complete continuity given by Hilbert.

The advent of the Lebesgue integral and the general topology developed by Fréchet and Moore were the two fundamental ingredients for the development of Functional Analysis. Regarding the development of general topology, in addition to Riesz's contributions—introducing the notions of accumulation point, neighborhood, interior point, open set, etc.—the establishment of general topology as a discipline is due to Felix Hausdorff, with his monograph *Grundzüge der Mengenlehre* from 1914, and to the doctoral thesis of C. Kuratowski from 1922,[16] where the concept of the closure of a set is defined. The axiomatic definition of open set was given by H. Tietze in 1923[17] and by P. Alexandroff in 1925.[18]

At the International Congress of Mathematicians in 1928, held in Bologna, almost all the pioneers of Functional Analysis participated: Volterra, Hadamard, Fréchet, Hilbert, and Riesz. Except for Hilbert, who spoke about the foundations of mathematics, the others gave talks related to Functional Analysis. Hadamard remarked that *"Functional Analysis has a higher level of abstraction than the theory*

[15] Math. Ann. **69** (1910), 449–497.

[16] Fundamenta Mathematica **3** (1922), 182–199.

[17] Math. Ann. **88** (1923), 290–312.

[18] Math. Ann. **94** (1923), 296–308.

of functions, since it considers the functions themselves as variables." Fréchet spoke about *"General analysis and abstract spaces"*, and Riesz on *"the decomposition of a linear functional into its positive and negative parts"*, a talk that would give rise to the modern theory of vector lattices, also known as *Riesz spaces*. Volterra's lecture focused more on the applications of Functional Analysis than on its purely theoretical character.

Functional Analysis was definitively established as a branch of Analysis in 1932 with the publication in that year of the following textbooks:

Théorie des Opérations Linéaires by S. Banach,
Mathematische Grundlagen der Quantenmechanik by J. von Neumann,
Linear Transformations of Hilbert Spaces by M. H. Stone,

which laid the basic foundations of the theory.

Let us pause here with these brief historical notes on the birth of Functional Analysis, which we will revisit in the introduction of each chapter, where we will provide a short historical overview of each topic. We believe it is very important to understand the origins and problems that motivated the development of each concept. For readers who wish to delve deeper into the origin of Functional Analysis, we recommend the excellent work by G. Birkhoff and E. Kreyszig [5],Siegmund-Schultze [55] as well as the textbook by A. F. Monna [43] and Pier [44], from which part of the above notes have been extracted.

Although the origin of Functional Analysis lies in problems arising from Partial Differential Equations, the application of the new theory to such equations is not the main focus of the three textbooks that established it.

In Banach's book, one finds only a single application of the Closed Graph Theorem to harmonic functions, and in Stone's, a spectral study of Sturm–Liouville operators. However, in the subsequent development of Functional Analysis, one of its most important motivations has been the problems posed by partial differential equations. Clear evidence of this can be seen in the theories developed by Friedrichs (extension of operators), Schauder and Leray (fixed point theorems), Sobolev (Sobolev spaces), Hille and Phillips (semigroups of operators), and Schwartz (theory of distributions), among others. It is not our intention here to give a historical account of the relationship between Functional Analysis and Partial Differential Equations. A good article on the subject is that of L. Gårding, *Functional Analysis in the Theory of Partial Differential Equations*.[19]

One of the guiding threads we have followed in preparing this textbook has been to study the topics of Functional Analysis with an emphasis not only on their intrinsic interest (the Bourbakist viewpoint), but also on their origins and applications—especially to Partial Differential Equations. We believe that this is the most suitable method for understanding the reasons behind this abstract theory.

[19] Memoirs. Amer. Math. Soc. **48**, no. 298 (1984), 13–23.

We share the following words of Felix Browder,[20] which summarize the ideas just mentioned:

> *There is no magic recipe emanating from the axioms or the basic theory of infinite dimensional topological vector spaces that can automatically solve every problem of significance. Nor do I think that the mathematicians who are involved in any such concrete investigation believe this to be the case. However, it is clear from any careful consideration that in the areas of greatest interest and concern, the spaces of functions and the operations upon these spaces which are the principal concern of the functional analysis viewpoint are already implicit in a very natural and intrinsic way in the central problems of analysis, and bringing them into explicit consideration clarifies and sharpens one's understanding of the phenomena being considered and makes possible a serious technical sharpening of the attack upon these problems. This is fairly obvious in such major fields of analysis as the theory of ordinary and partial differential equations, the calculus of variations, Fourier analysis, and many other areas.*

[20] Historia Mathematica **2** (1975), p. 579.

Chapter 2
Banach Spaces

2.1 Introduction

This concept of a norm had been in the air since 1903, when Hadamard began the study of topological duality by characterizing continuous linear forms in $C([a,b])$ via a Stieltjes integral, a work that was completed in 1909 by F. Riesz. In 1910, Riesz himself began the study of spaces $L^p([a,b])$ motivated by results on infinite systems of linear equations studied by Schmidt in the space ℓ^2.

Although Riesz did not use the word *dual*, he proved that the spaces $L^p([a,b])$ and $L^q([a,b])$ are duals when $\frac{1}{p}+\frac{1}{q}=1$ and $1<p,q<\infty$. Specifically, he established the following result:

> *If to each function $f(x)$ in the class $[L^{\frac{p}{p-1}}]$ a number A_f is assigned, and this assignment satisfies the conditions*
>
> $$A_{\mu_1 f_1+\mu_2 f_2}=\mu_1 A_{f_1}+\mu_2 A_{f_2};$$
>
> $$|A_f|\le M\left(\int_a^b |f(x)|^{\frac{p}{p-1}}\,dx\right)^{\frac{p-1}{p}},$$
>
> *where the bound M depends only on the assignment, then there exists a function $a(x)$ uniquely determined up to a set of measure zero in the class $[L^p]$ such that for all $f(x)$ we have*
>
> $$A_f=\int_a^b a(x)f(x)\,dx.$$

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In this work,[1] Riesz already demonstrates the need for the general structure of a normed space. He writes:

> *In this work, the assumption of square integrability is replaced by the integrability of* $|f(x)|^p$. *The role of class* $[L^2]$ *is here taken up by the classes* $[L^p]$ *and* $[L^{\frac{p}{p-1}}]$. *The investigation of these classes of functions will shed light on the real and apparent advantages of the exponent* $p = 2$; *and one may also state that they will be useful material for an axiomatic study of these spaces of functions*.

In 1912, the Austrian mathematician E. Helly provided an interpretation of the problem addressed by Riesz concerning infinite systems, in which what is now known as the Hahn–Banach Extension Theorem is implicit. In 1921, Helly himself wrote an article[2] in which, for the first time, instead of working with concrete spaces such as L^p, ℓ^p, or $C([a, b])$, he studied general normed sequence spaces and gave the definition of a norm. As a consequence of the work of Riesz and Helly, it was natural to generalize the definition of a norm to an arbitrary vector space and not only to spaces of functions or sequences.

This was done independently by Hans Hahn in 1922,[3] by Norbert Wiener in 1922,[4] and by Stefan Banach in his doctoral dissertation of 1920, *"Sur les opérations dans les ensembles abstraits et leur application aux équations intégrales"*, published in 1922,[5] which he completed under the supervision of Hilbert's student Hugo Steinhaus at the University of Lwów, where Steinhaus founded one of the first schools of Functional Analysis and the journal *Studia Mathematica*, which became the vehicle for the dissemination of new results in the emerging field.

Although the work of these mathematicians overlaps considerably, making it difficult to establish priority, Banach's contributions have had the greatest influence. Wiener soon lost interest in this theory; regarding this topic, he writes in his autobiography:[6]

> *It is so in an aesthetic rather than in any strictly logical sense that, in those years after Strasbourg, Banach spaces did not seem to have the physical and mathematical texture I wanted for a theory on which I was to stake a large part of my future reputation. Nowadays, it seems to me that some aspects of the theory of Banach spaces are taking on a sufficiently rich texture and have been endowed with a sufficiently unobvious body of theorems to come closer to satisfying me in this respect. At that time, however, the theory seemed to me to contain for the immediate future nothing more than some decades of rather formal and thin work. By this I do not mean to reproach the work of Banach himself but that of many inferior writers, hungry for easy doctor's theses, who were drawn to it. As I foresaw, it was this class of writers that was first attracted to the theory of Banach spaces. The chief factor which led me to abandon the theory of Banach spaces, after a few desultory papers.*

[1] Math. Ann. **69** (1910), 449–497.

[2] Monatshefte für Mathematik und Physik **31** (1921), 60–91.

[3] Monatshefte für Mathematik und Physik **32** (1922), 3–88.

[4] Bull. Soc. math. France **150** (1922), 124–134.

[5] Fundamenta Mathematicae **3** (1922), 131–181.

[6] N. Wiener. *I Am a Mathematician*, Doubleday, Garden City—New York, 1956.

It was also Banach who discovered the relationship between the weak compactness of the unit ball and reflexivity, outlining the major lines of duality theory in the context of normed spaces.

In 1927, revisiting the ideas of Helly, Hahn,[7] in the context of complete normed spaces, solved the problem of extending continuous linear functionals and introduced the dual space E' of a complete normed space E. Two years later, Banach,[8] who was unaware of Hahn's work, proved a more general form of Hahn's extension result with a similar proof, establishing what is now known as the *Hahn–Banach Extension Theorem*:

Every continuous linear functional on a subspace Y of a real normed space X can be extended to a continuous linear functional on the whole space X, preserving the same norm.

The complex version of the Hahn–Banach Theorem was first discovered by F. Murray in 1936 for the spaces $L^p([a,b])$, $(p > 1)$, and later generalized by Bohnenblust and Sobczyk in 1938[9] for any complex normed space. The first to establish a geometric version of the Hahn–Banach Theorem was Mazur[10] in 1933, in the form:

If a vector subspace Y of a real normed space X does not intersect a nonempty convex open set C, then there exists a hyperplane H in X containing Y and not intersecting C.

In his 1922 work, Hahn proved the following result.

Let E be a complete normed space and $\{u_n\}$ a sequence of continuous linear forms on E. If for each $x \in E$, the sequence $\{|u_n(x)|\}$ is bounded, then the sequence $\{\|u_n\|\}$ is bounded.

To prove this result, Hahn used a method that has been very fruitful in Functional Analysis: the bump function method, which had already been used by H. Lebesgue in 1905 to construct a continuous function whose Fourier series diverges at one point. In his thesis, Banach proved a more general theorem than Hahn's by considering the u_n as continuous linear maps between complete normed spaces.

Finally, in 1927 Banach and Steinhaus,[11] following a suggestion from Saks, discovered that this theorem could be proved without the bump function method. For their proof, they used a theorem by R. L. Baire from 1899,[12] which states that "*in $\mathbb{R}^N$, the intersection of a sequence of dense open sets is dense.*" In 1932, Banach published his famous monograph *Théorie des Opérations Linéaires*, in which he compiled all known results on normed spaces at that time. As we mentioned in the Preface, two of the basic tools of Functional Analysis are the Lebesgue integral and Fréchet metric spaces. For this reason, the introductory chapter of Banach's monograph is devoted to recalling the most important results from these

[7] Journal für die reine und angewandte Mathematik **157** (1927), 214–219.

[8] Studia Mathematica **1** (1929), 223–239.

[9] Bull. Amer. Math. Soc. **44** (1938), 91–93.

[10] Studia Mathematica **3** (1933), 70–84.

[11] Fundamenta Mathematicae **9** (1927), 50–61.

[12] Ann. di Mat. **3** (1899), 1–123.

two theories. One of the most remarkable results of Banach's monograph is another consequence of Baire's Theorem:

Homomorphism Theorem.– *If T is a continuous linear map from E into F, where E and F are complete normed spaces, then either $T(E)$ is meager in F, or $T(E) = F$.*

As a consequence of this theorem, we obtain a result that is one of the most powerful tools in applications of Functional Analysis:

Closed Graph Theorem.- *If T is a linear map with a closed graph from E to F, where E and F are complete normed spaces, then T is continuous.*

The weak* topology on the dual of a normed space was introduced by S. Banach in 1926.[13] The weak* compactness of the unit ball of the dual of a normed space was announced by L. Alaoglu in 1938[14] and proved in 1940;[15] an alternative proof, using the notions of uniform structure and filter, was given in 1938 by N. Bourbaki.[16]

2.2 General Properties and Examples of Normed Spaces

Let X be a vector space over the field $\mathbb{K}$ ($\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$). A *seminorm* on X is a function $p : X \to [0, +\infty[$ that satisfies:

$$p(x + y) \leq p(x) + p(y), \qquad \forall x, y \in X \tag{2.1}$$

$$p(\alpha x) = |\alpha| p(x), \qquad \forall x \in X, \text{ and } \forall \alpha \in \mathbb{K}. \tag{2.2}$$

It follows from (2.2) that $p(0) = 0$. A *norm* is a seminorm that also satisfies:

$$p(x) = 0 \Longleftrightarrow x = 0. \tag{2.3}$$

A norm is usually denoted by $\| \ \|$, and the pair $(X, \| \ \|)$ is called a *normed space*. It is easy to see that every normed space is a metric space with respect to the metric $d(x, y) := \|x - y\|$. Moreover, this metric is translation invariant and compatible with the linear structure of the space, in the sense that the algebraic operations are continuous. Furthermore, the norm is also continuous, since

$$|\|x\| - \|y\|| \leq \|x - y\|, \qquad \forall x, y \in X. \tag{2.4}$$

[13] Studia Math. **50** (1926), 27–32, 36–43.

[14] Bull. Amer. Math. Soc. **44** (1938), 196.

[15] Ann. Math. **41** (1940), 252–267.

[16] C. R. Acad. Sci Paris **206** (1938), 1701–1704.

We will use the following notation: If $x_0 \in X$ and $r > 0$, the open ball centered at x_0 of radius r will be denoted by $B(x_0, r)$, that is,

$$B(x_0, r) := \{x \in X \ : \ \|x - x_0\| < r\}.$$

For the closed unit ball and the unit sphere, we will use the remarktion $U(X)$ and $S(X)$, respectively, that is,

$$U(X) := \{x \in X \ : \ \|x\| \leq 1\}$$

and

$$S(X) := \{x \in X \ : \ \|x\| = 1\}.$$

The interior of a set $A \subset X$ will be denoted by $\text{int}(A)$; and its closure by $\overline{A}$.

Normed spaces that are complete with respect to the metric induced by their norm are called *Banach spaces*. The following theorem provides a characterization, in terms of series convergence, of when a normed space is a Banach space.

Theorem 2.1 *A necessary and sufficient condition for a normed space* $(X, \| \ \|)$ *to be a Banach space is that for every sequence* $\{x_n\} \subset X$*, the condition* $\sum \|x_n\| < \infty$ *implies that the series* $\sum x_n$ *converges in* X *with respect to the norm.*

Proof Suppose that $(X, \| \ \|)$ is a Banach space and that $\sum \|x_n\| < \infty$. Let $S_n := \sum_{k=1}^{n} x_k$. Then

$$\|S_{n+k} - S_n\| = \|x_{n+1} + \cdots + x_{n+k}\| \leq \|x_{n+1}\| + \cdots + \|x_{n+k}\|.$$

Therefore, $\{S_n\}$ is a Cauchy sequence in X, and since X is complete, it converges. Hence, there exists

$$\sum x_n = \lim_{n \to \infty} S_n \in X.$$

Conversely, suppose that X satisfies the condition. Let $\{x_n\}$ be a Cauchy sequence. It suffices to prove that $\{x_n\}$ has a convergent subsequence. Indeed, since $\{x_n\}$ is a Cauchy sequence, there exists a subsequence $\{x_{n_k}\}$ such that

$$\|x_n - x_{n_k}\| \leq \frac{1}{2^{k+1}} \qquad \forall\, n > n_k.$$

Define

$$y_1 := x_{n_1}, \quad y_k := x_{n_k} - x_{n_{k-1}}, \qquad \forall\, k > 1.$$

Then,

$$\|y_k\| \leq \frac{1}{2^k}, \qquad \forall\, k > 1,$$

which implies that

$$\sum_{k=1}^{\infty} \|y_k\| < \infty.$$

Hence, by hypothesis, there exists

$$\sum_{k=1}^{\infty} y_k = \lim_{k\to\infty} x_{n_k}.$$

□

Let us now give some examples of Banach spaces.

Example 2.1

(1) *Sequence spaces*. For $1 \leq p < \infty$, the vector space is defined as

$$\ell^p := \left\{ (a_n),\ a_n \in \mathbb{K} \ :\ \sum_{n=1}^{\infty} |a_n|^p < \infty \right\}.$$

As we will see in Sect. 2.5, this space, with the norm

$$\|(a_n)\|_p := \left(\sum_{n=1}^{\infty} |a_n|^p \right)^{\frac{1}{p}}$$

is a Banach space. Another classical Banach space is the space of bounded sequences

$$\ell^\infty := \{(a_n),\ a_n \in \mathbb{K} \ :\ (a_n) \text{ is bounded}\},$$

with the norm

$$\|(a_n)\|_\infty := \sup_{n\in\mathbb{N}} |a_n|.$$

Two closed subspaces of ℓ^∞, and hence Banach spaces with the induced norm, are:

$$c := \{(a_n),\ a_n \in \mathbb{K} \ :\ (a_n) \text{ is convergent}\}$$

and

$$c_0 := \left\{(a_n),\ a_n \in \mathbb{K} \ :\ \lim_{n\to\infty} a_n = 0\right\}.$$

(2) *Spaces of continuous functions*. Let Ω be a completely regular topological space, and define

$$C_b(\Omega) := \{f : \Omega \to \mathbb{K} \ :\ f \text{ is continuous and bounded}\}.$$

It is easy to see that

$$\|f\|_\infty := \sup\{|f(x)| \ :\ x \in \Omega\}$$

is a norm on $C_b(\Omega)$. Let us show that this space is complete and hence a Banach space. In fact, let $\{f_n\}$ be a Cauchy sequence in $C_b(\Omega)$. Then, given $\epsilon > 0$, there exists an integer n_0 such that

$$\|f_n - f_m\|_\infty < \epsilon, \qquad \text{for } n, m \geq n_0.$$

In particular, for each $x \in \Omega$, we have

$$|f_n(x) - f_m(x)| \leq \|f_n - f_m\|_\infty < \epsilon, \qquad \text{for } n, m \geq n_0. \tag{2.5}$$

Therefore, $\{f_n(x)\}$ is a Cauchy sequence in $\mathbb{K}$, so for each $x \in \Omega$ there exists $f(x) := \lim_{n\to\infty} f_n(x)$. Then, taking the limit as $m \to \infty$ in (2.5), we obtain $|f(x) - f_n(x)| \leq \epsilon$ for $n \geq n_0$. Since this bound is independent of x, we obtain $\|f - f_n\|_\infty \leq \epsilon$ for each $n \geq n_0$. Thus, $f_n \to f$ uniformly on Ω, and hence f is continuous on Ω and also bounded, since $\|f\|_\infty \leq \|f - f_n\|_\infty + \|f_n\|_\infty < \infty$. Therefore, $f \in C_b(\Omega)$.

In the case when Ω is compact, we have $C_b(\Omega) = C(\Omega)$, so $C(\Omega)$ is a Banach space with respect to the norm $\|\ \|_\infty$.

Definition 2.1 Two norms $\|\ \|_1$ and $\|\ \|_2$ on the same vector space X are said to be *equivalent* if they define the same topology.

We have the following characterization of equivalent norms.

Proposition 2.1 *Two norms $\|\ \|_1$ and $\|\ \|_2$ on the same vector space X are equivalent if and only if there exist positive constants m, M such that*

$$m\|x\|_1 \leq \|x\|_2 \leq M\|x\|_1, \qquad \forall\, x \in X. \tag{2.6}$$

Proof If $\|\ \|_1$ and $\|\ \|_2$ are equivalent, then since they define the same topology, given the neighborhood of zero $\{x \in X \ :\ \|x\|_1 < 1\}$, there exists $m > 0$ such that

$$\{x \in X \ :\ \|x\|_2 < m\} \subset \{x \in X \ :\ \|x\|_1 < 1\},$$

from which it follows that $m\|x\|_1 \leq \|x\|_2$. The other inequality is obtained analogously.

Now suppose that condition (2.6) holds. Given any $x_0 \in X$ and $\epsilon > 0$, we have

$$\left\{x \in X \,:\, \|x - x_0\|_1 < \frac{\epsilon}{M}\right\} \subset \{x \in X \,:\, \|x - x_0\|_2 < \epsilon\}$$

and

$$\{x \in X \,:\, \|x - x_0\|_2 < m\epsilon\} \subset \{x \in X \,:\, \|x - x_0\|_1 < \epsilon\}.$$

Consequently, the two norms define the same topology, and therefore are equivalent. □

In normed spaces, there are two types of properties: topological and metric. Metric properties depend on the specific norm, while topological properties depend on any of the equivalent norms. This gives rise to the following two types of isomorphisms.

Definition 2.2 Two normed spaces $(X_1, \|\ \|_1)$ and $(X_2, \|\ \|_2)$ are said to be *topologically isomorphic* if there exists a linear bijection $T : X_1 \to X_2$ that is a homeomorphism. If, in addition, T is an *isometry*, that is,

$$\|Tx\|_2 = \|x\|_1, \qquad \forall\, x \in X_1,$$

then the spaces are said to be *isometrically isomorphic*.

If $(X, \|\ \|)$ is a normed space and Y is a closed vector subspace of X, then in the quotient space $\frac{X}{Y}$, the function

$$\|\hat{x}\| := \inf\{\|x\| \,:\, x \in \hat{x}\}$$

is a norm. Moreover, it is easy to see that if $(X, \|\ \|)$ is a Banach space, then the quotient is also a Banach space.

Regarding products, we have that if $(X, \|\ \|)$ and $(Y, \|\ \|)$ are two normed spaces, the product space $X \times Y$ is also a normed space with respect to any of the following equivalent norms:

$$\|(x, y)\|_p := \left(\|x\|^p + \|y\|^p\right)^{\frac{1}{p}}, \qquad 1 \leq p < \infty,$$

$$\|(x, y)\|_\infty := \max\{\|x\|, \|y\|\}.$$

We denote by $\ell^p(n)$ the space $\mathbb{K}^n$ equipped with the norm

$$\|(x_1, \ldots, x_n)\|_p := \left(|x_1|^p + \cdots + |x_n|^p\right)^{\frac{1}{p}}.$$

2.3 Finite-Dimensional Normed Spaces

In this section, we will characterize finite-dimensional normed spaces.

Theorem 2.2 *In a finite-dimensional vector space, all norms are equivalent. Moreover, they all endow the space with a Banach space structure.*

Proof It is enough to prove that if $(X, \| \|)$ is a normed space and X has dimension n, then $(X, \| \|)$ is topologically isomorphic to the space $\ell^2(n)$. Indeed: let $x_1, x_2, \ldots, x_n$ be a basis of the space X. We define the linear map $T : \ell^2(n) \to X$ as $T(a_1, \ldots, a_n) := a_1x_1 + \cdots + a_nx_n$. It is easy to see that T is a vector space isomorphism. Moreover, since scalar multiplication and addition are continuous operations, we have that T is continuous. Let us finally show that T^{-1} is continuous. Let $B := \{\xi \in \mathbb{K}^n \ : \ \|\xi\|_2 < 1\}$ and $S := \{\xi \in \mathbb{K}^n \ : \ \|\xi\|_2 = 1\}$. By the Heine–Borel Theorem, we know that S is compact, and since T is continuous, $T(S)$ is compact. Then, since $0 \notin T(S)$, there exists $\delta > 0$ such that, if $V = \{x \in X \ : \ \|x\| < \delta\}$, then $V \cap T(S) = \emptyset$. Let us show that $V \subset T(B)$, from which the continuity of T^{-1} will follow, as it is linear. If $x \notin T(B)$, then $x = T(\xi)$ with $\|\xi\|_2 \geq 1$. Then, if we had $x \in V$, we would reach a contradiction, since $\frac{x}{\|\xi\|} \in V$, and $\frac{x}{\|\xi\|} = T\left(\frac{\xi}{\|\xi\|}\right)$. Consequently, $V \subset T(B)$. □

Lemma 2.1 (Riesz's Lemma) *Let $(X, \| \|)$ be a normed space and let X_0 be a closed proper subspace of X. Then, for every $0 < \theta < 1$ there exists $x_\theta \in X$ with $\|x_\theta\| = 1$ and $\|x - x_\theta\| \geq \theta$ for all $x \in X_0$.*

Proof Let $x_1 \in X \setminus X_0$. Since X_0 is closed, we have

$$d := d(x_1, X_0) = \inf_{x \in X_0} \|x - x_1\| > 0.$$

Now, since $d < \frac{d}{\theta}$, there exists $x_0 \in X_0$ such that $\|x_0 - x_1\| \leq \frac{d}{\theta}$. Let

$$x_\theta := \frac{x_1 - x_0}{\|x_1 - x_0\|}.$$

Then, if $x \in X_0$, since $x_0 + \|x_1 - x_0\|x \in X_0$, we have

$$\|x - x_\theta\| = \left\| x - \frac{x_1}{\|x_1 - x_0\|} + \frac{x_0}{\|x_1 - x_0\|} \right\|$$

$$= \frac{1}{\|x_1 - x_0\|} \|x_0 + \|x_1 - x_0\|x - x_1\| \geq \frac{d}{\|x_1 - x_0\|} \geq \theta.$$

□

Bounded sets in a normed space are those that are bounded with respect to the metric induced by the norm, that is, a subset B of a normed space X is said to be

bounded if there exists $M > 0$ such that $\|x\| \leq M$ for all $x \in B$. The following is a topological characterization of finite-dimensional normed spaces.

Theorem 2.3 *A normed space X is finite-dimensional if and only if every bounded subset of X is precompact.*

Proof If X has dimension n, then by Theorem 2.2, we have that X is topologically isomorphic to the space $\ell^2(n)$. Therefore, as a consequence of the Heine–Borel Theorem, every bounded subset of X is precompact.

Conversely, suppose that every bounded subset of X is precompact. Then $S = \{x \in X \ : \ \|x\| = 1\}$ is precompact. Suppose that X is not finite-dimensional, and we will reach a contradiction. Given $x_1 \in X$, we have that $X_1 := \text{span}\{x_1\}$ is a proper closed subspace of X. Therefore, by applying Riesz's Lemma, there exists $x_2 \in S$ such that $\|x_1 - x_2\| \geq \frac{1}{2}$. Let $X_2 := \text{span}\{x_1, x_2\}$. Reasoning as before, there exists $x_3 \in S$ such that $\|x - x_3\| \geq \frac{1}{2}$ for all $x \in X_2$. Proceeding in this way, we find a sequence $\{x_n\} \subset S$ such that

$$\|x_n - x_m\| \geq \frac{1}{2} \quad \text{if } n \neq m,$$

which contradicts the precompactness of the set S. □

2.4 Continuous Linear Operators

We denote by $L(X, Y)$ the vector space of all linear operators between the vector spaces X and Y. In the case where X and Y are normed spaces, we denote by $\mathcal{L}(X, Y)$ the subspace of $L(X, Y)$ consisting of the operators that are continuous. The following characterization of $\mathcal{L}(X, Y)$ is quite useful.

Theorem 2.4 *Let X and Y be two normed spaces, and let $T \in L(X, Y)$. The following are equivalent:*

(i) $T \in \mathcal{L}(X, Y)$.
(ii) *T is continuous at zero.*
(iii) *T is continuous at some point.*
(iv) *There exists a constant $C > 0$ such that $\|Tx\| \leq C\|x\|$ for every $x \in X$.*

Proof Due to the linearity of T, the only nontrivial implication is (ii) $\Rightarrow$ (iv). Let us prove this implication. Given $\epsilon = 1$, by the continuity of T at 0, there exists $\delta > 0$ such that $\|Tx\| \leq 1$ whenever $\|x\| \leq \delta$. Therefore, if $x \neq 0$, we have

$$\frac{\delta}{2} \frac{1}{\|x\|} \|Tx\| = \left\| T\left(\frac{\delta x}{2\|x\|} \right) \right\| \leq 1.$$

Thus, taking $C := \frac{2}{\delta}$, we obtain $\|Tx\| \leq C\|x\|$ for every $x \in X$. □

Due to the previous result, continuous linear operators between normed spaces are often called *bounded operators*.

Theorem 2.5 *Let X and Y be normed spaces. Then, the space $\mathcal{L}(X, Y)$ is a normed space with respect to the norm*

$$\|T\| := \sup\{\|Tx\| \ : \ \|x\| \leq 1\} = \inf\{M \ : \ \|Tx\| \leq M\|x\| \ \ \forall\, x \in X\}.$$

Moreover, this space is a Banach space whenever Y is a Banach space.

Proof Let us first show that

$$\|T\| = \alpha := \inf\{M \ : \ \|Tx\| \leq M\|x\| \ \ \forall\, x \in X\}.$$

Indeed: by definition, $\|Tx\| \leq \|T\|$ for each $x \in X$ with $\|x\| \leq 1$. Therefore, for every $x \neq 0$, we have $\|T(\frac{x}{\|x\|})\| \leq \|T\|$, so $\|Tx\| \leq \|T\|\|x\|$ for every $x \in X$, and consequently, $\alpha \leq \|T\|$.

On the other hand, if $\|Tx\| \leq M\|x\|$ for every $x \in X$, then $\|Tx\| \leq M$ for every $x \in X$ with $\|x\| \leq 1$. Taking the supremum gives $\|T\| \leq M$, and therefore taking the infimum yields $\|T\| \leq \alpha$.

It is straightforward to show that $\mathcal{L}(X, Y)$ is a normed space. Let us now show that it is complete whenever Y is. Let $\{T_n\}$ be a Cauchy sequence in $\mathcal{L}(X, Y)$. Then, for each $x \in X$, since

$$\|T_n x - T_m x\| \leq \|T_n - T_m\|\|x\|,$$

we conclude that $\{T_n x\}$ is a Cauchy sequence in Y. Hence, we can define the linear operator T on X by $Tx := \lim_{n\to\infty} T_n x$. Furthermore, since $\{T_n\}$ is Cauchy, there exists $M > 0$ such that $\|T_n\| \leq M$ for every $n \in \mathbb{N}$. Therefore,

$$\|Tx\| = \lim_{n\to\infty} \|T_n x\| \leq M\|x\|,$$

and by the previous theorem, we conclude that $T \in \mathcal{L}(X, Y)$.

Finally, let us show that $T_n \to T$ in $\mathcal{L}(X, Y)$. Given $\epsilon > 0$, there exists $n_0 \in \mathbb{N}$ such that $\|T_n - T_m\| \leq \epsilon$ for every $n, m \geq n_0$. Then, for every $x \in X$ with $\|x\| \leq 1$, and for every $n, m \geq n_0$, we have $\|T_n x - T_m x\| \leq \epsilon$. Taking the limit as $m \to \infty$, we obtain $\|Tx - T_n x\| \leq \epsilon$ whenever $\|x\| \leq 1$ and $n \geq n_0$, so $\|T - T_n\| \leq \epsilon$ for every $n \geq n_0$. □

The algebraic dual of a vector space X is denoted by X^*,i.e., $X^* = L(X, \mathbb{K})$, and its elements are called *linear functionals*. Given $x \in X$ and $u \in X^*$, it is common to denote $u(x)$ as $\langle u, x\rangle$. If X is a normed space, then the space $\mathcal{L}(X, \mathbb{K})$ is denoted by X' and is called the *topological dual* of X. As a consequence of the previous theorem, we have that X' is a Banach space with respect to the norm:

$$\|u\| = \sup\{|\langle u, x\rangle| \ : \ \|x\| \leq 1\}.$$

Recall that a hyperplane H of the vector space X is a subspace of X of codimension one, that is, $\dim(X/H) = 1$. Given a linear functional $u \in X^*$, we denote its *kernel* as $\operatorname{Ker} u$, that is,

$$\operatorname{Ker} u := \{x \in X \,:\, \langle u, x\rangle = 0\}.$$

We have the following characterization of hyperplanes.

Lemma 2.2 *A vector subspace H of X is a hyperplane if and only if there exists $u \in X^*$, $u \neq 0$, such that $H = \operatorname{Ker} u$.*

Proof Let H be a hyperplane in X, then there exists $a \in X \setminus H$ such that $X = \operatorname{LIN}\{H, a\}$. Then, for any $x \in X$, we can write $x = y + \lambda a$ with $y \in H$ and $\lambda \in \mathbb{K}$. Define $\langle u, y + \lambda a\rangle := \lambda$, then $u \in X^*$ and $H = \operatorname{Ker} u$.

Conversely, suppose that $H = \operatorname{Ker} u$, with $u \in X^*$, $u \neq 0$. Let $a \in X$ be such that $\langle u, a\rangle \neq 0$. Then, for every $x \in X$, we have

$$x - \frac{\langle u, x\rangle}{\langle u, a\rangle}\, a \in H,$$

which implies $X = \operatorname{LIN}\{H, a\}$, so H is a hyperplane in X. □

Since the closure of a vector subspace of a normed space is again a vector subspace, the hyperplanes of a normed space must be either closed or dense. Therefore, as a consequence of the previous lemma, we obtain the following result.

Proposition 2.2 *Let X be a normed space and $u \in X^*$. Then, u is continuous (i.e., $u \in X'$) if and only if* $\operatorname{Ker} u$ *is closed.*

Example 2.2 Let $u : c_0 \to \mathbb{K}$ be defined by $\langle u, (a_n)\rangle := a_1$. Then, since u is continuous, we have that $\operatorname{Ker} u = \{(a_n) \in c_0 \,:\, a_1 = 0\}$ is closed. Now let us consider an example of a dense hyperplane in c_0. For each $i \in \mathbb{N}$, let $e_i := (0, \ldots, 0, 1, 0, \ldots)$. Let $a^0 = (\frac{1}{n}) \in c_0$. Then, $\{a^0, e_1, e_2, \ldots\}$ is a linearly independent set. Therefore, there exists a Hamel basis $\mathcal{B}$ of c_0 containing $\{a^0, e_1, e_2, \ldots\}$, so that $\mathcal{B} = \{a^0, e_1, e_2, \ldots\} \cup \{b_j \,:\, j \in J\}$. Define $u : c_0 \to \mathbb{K}$ as

$$\left\langle u, \alpha_0 a^0 + \sum_{n=1}^{\infty} \alpha_n e_n + \sum_{j \in J} \beta_j b_j \right\rangle := \alpha_0.$$

Then $u \in (c_0)^*$, and since $e_n \in \operatorname{Ker} u$ for each $n \in \mathbb{N}$, we have that $\operatorname{Ker} u$ is dense in c_0, so $u \notin (c_0)'$.

2.5 The Spaces L^p

In this section, we will study one of the most important examples of Banach spaces that appear in applications: the Lebesgue spaces L^p.

Let $(\Omega, \mathcal{B}, \mu)$ be a measure space, and let $1 \leq p < +\infty$. A measurable function $f : \Omega \to \mathbb{K}$ is said to belong to $L^p(\Omega, \mathcal{B}, \mu)$ if the function $|f|^p$ is μ-integrable on Ω. In $L^p(\Omega, \mathcal{B}, \mu)$, we identify functions that are equal μ-almost everywhere, and when there is no risk of confusion, we will denote the space $L^p(\Omega, \mathcal{B}, \mu)$ simply by $L^p(\Omega, \mu)$ or $L^p(\Omega)$. Since for $\alpha, \beta \in \mathbb{K}$, we have that

$$|\alpha f + \beta g|^p \leq 2^p \big(|\alpha|^p |f|^p + |\beta|^p |g|^p\big),$$

it follows that $L^p(\Omega, \mu)$ is a vector space over $\mathbb{K}$. For $f \in L^p(\Omega, \mu)$, we define

$$\|f\|_p := \left(\int_\Omega |f|^p \, d\mu \right)^{\frac{1}{p}}.$$

Now consider the case $p = \infty$. We define

$$L^\infty(\Omega, \mu) := \{ f : \Omega \to \mathbb{K} \ : \ f \text{ measurable and } \exists C \text{ s.t. } |f(x)| \leq C \ \mu\text{-a.e.} \}$$

where, as before, we identify functions that are equal almost everywhere, and we define

$$\|f\|_\infty := \inf\{ C \ : \ |f(x)| \leq C \ \mu\text{-a.e. on } \Omega \}. \tag{2.7}$$

It is easy to see that if $f \in L^\infty(\Omega, \mu)$, then

$$|f(x)| \leq \|f\|_\infty \quad \mu\text{-a.e. on } \Omega.$$

Indeed: let $C_n \to \|f\|_\infty$. Then, for each $n \in \mathbb{N}$, there exists a μ-null set N_n such that $|f(x)| \leq C_n$ for every $x \in \Omega \setminus N_n$. Therefore, if $N = \bigcup_{n \in \mathbb{N}} N_n$, we have that N is μ-null and $|f(x)| \leq C_n$ for every $n \in \mathbb{N}$ and all $x \in \Omega \setminus N$, from which we deduce (2.7).

We are going to show that, for each $1 \leq p \leq \infty$, the space $\big(L^p(\Omega, \mu), \| \cdot \|_p\big)$ is a Banach space. Since in $L^p(\Omega, \mu)$ we identify functions that are equal μ-a.e., we have

$$\|f\|_p = 0 \iff f = 0.$$

Also,

$$\|\alpha f\|_p = |\alpha| \|f\|_p.$$

Given $1 \leq p \leq \infty$, we denote by p' the *conjugate exponent* of p, i.e., the number satisfying $\frac{1}{p} + \frac{1}{p'} = 1$.

Lemma 2.3 (Young's Inequality) *If $1 < p < \infty$, then for every pair of positive real numbers a, b we have*

$$ab \leq \frac{1}{p}a^p + \frac{1}{p'}b^{p'}.$$

Proof Since the function $t \mapsto \log(t)$ is concave on $]0, \infty[$, we have

$$\log\left(\frac{1}{p}a^p + \frac{1}{p'}b^{p'}\right) \geq \frac{1}{p}\log(a^p) + \frac{1}{p'}\log(b^{p'}) = \log(ab),$$

from which the inequality follows. □

Proposition 2.3 (Hölder's Inequality) *Let $f \in L^p(\Omega)$ and $g \in L^{p'}(\Omega)$ with $1 \leq p \leq \infty$. Then $fg \in L^1(\Omega)$ and moreover*

$$\int_\Omega |fg| \leq \|f\|_p \, \|g\|_{p'}. \tag{2.8}$$

Proof The cases $p = 1$ and $p = \infty$ are evident. So assume $1 < p < \infty$. We may suppose that $\|f\|_p > 0$ and $\|g\|_{p'} > 0$. Define $f_1 := \frac{f}{\|f\|_p}$ and $g_1 := \frac{g}{\|g\|_{p'}}$. As a consequence of Young's inequality, we have

$$|f_1(x)||g_1(x)| \leq \frac{1}{p}|f_1(x)|^p + \frac{1}{p'}|g_1(x)|^{p'}.$$

Integrating, we obtain

$$\int_\Omega |f_1||g_1| \, d\mu \leq \frac{1}{p}\|f_1\|_p^p + \frac{1}{p'}\|g_1\|_{p'}^{p'},$$

from which inequality (2.8) follows, and consequently $fg \in L^1(\Omega)$. □

Theorem 2.6 (Riesz–Fischer Theorem) *For every $1 \leq p \leq \infty$,*

$$(L^p(\Omega, \mu), \| \ \|_p)$$

is a Banach space.

Proof That $\| \ \|_p$ is a norm is evident for $p = 1$ and $p = \infty$. To prove it for $1 < p < \infty$ we just need to verify the *Minkowski inequality*

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p \qquad \forall \, f, g \in L^p(\Omega, \mu). \tag{2.9}$$

Let $f, g \in L^p(\Omega, \mu)$. We know that $f + g \in L^p(\Omega, \mu)$, and we have

$$\|f + g\|_p^p = \int_\Omega |f + g|^{p-1}|f + g| \le \int_\Omega |f + g|^{p-1}|f| + \int_\Omega |f + g|^{p-1}|g|.$$

Now, since $|f + g|^{p-1} \in L^{p'}$, applying Hölder's inequality, we get:

$$\|f + g\|_p^p \le \|f + g\|_p^{p-1}\|f\|_p + \|f + g\|_p^{p-1}\|g\|_p,$$

which proves (2.9).

To prove that the spaces are complete, we first consider the case $p = \infty$. Let $\{f_n\}$ be a Cauchy sequence in $L^\infty(\Omega, \mu)$. For each $j \in \mathbb{N}$, there exists n_j such that

$$\|f_n - f_m\|_\infty \le \frac{1}{j} \quad \text{for } n, m \ge n_j.$$

Therefore, there exist μ-null sets N_j such that

$$|f_n(x) - f_m(x)| \le \frac{1}{j} \quad \forall\, x \in \Omega \setminus N_j, \ \ \forall\, n, m \ge n_j. \tag{2.10}$$

Let $N := \cup_{j\in\mathbb{N}} N_j$. Then N is a μ-null set, and for all $x \in \Omega \setminus N$, $\{f_n(x)\}$ is a Cauchy sequence in $\mathbb{K}$. Hence, $f_n(x) \to f(x)$ for each $x \in \Omega \setminus N$. Taking the limit in (2.10) as $m \to \infty$, we obtain

$$|f(x) - f_n(x)| \le \frac{1}{j}, \quad \forall\, x \in \Omega \setminus N_j, \ \text{ and } \ \forall\, n \ge n_j,$$

from which it follows that $f \in L^\infty(\Omega, \mu)$ and that $f_n \to f$ in the norm $\| \ \|_\infty$. Thus, in this case, we have shown that the space is a Banach space.

Now consider the case $1 \le p < \infty$. As a consequence of Theorem 2.1, we need to prove that if $\{f_n\}$ is a sequence in $L^p(\Omega, \mu)$ such that $\sum_{n=1}^\infty \|f_n\|_p = a < \infty$, then the series $\sum_{n=1}^\infty f_n$ converges in $L^p(\Omega, \mu)$. Indeed, let $g_n(x) := \sum_{j=1}^n |f_j(x)|$. By Minkowski's inequality,

$$\|g_n\|_p \le \sum_{j=1}^n \|f_j\|_p \le a,$$

which implies

$$\int_\Omega g_n^p(x)\, d\mu \le a^p.$$

Let $g(x) := \lim_{n\to\infty} g_n(x)$. By Fatou's Lemma (Lemma B.3),

$$\int_\Omega g^p(x)\, d\mu \le a^p.$$

Hence $g \in L^p(\Omega, \mu)$. Now, since $\sum_{n=1}^{\infty} |f_n(x)| < \infty$ μ-a.e., there exists a finite function h μ-a.e. such that $h(x) = \sum_{n=1}^{\infty} f_n(x)$. Finally, since

$$\left| \sum_{j=1}^{n} f_j(x) - h(x) \right|^p \le 2^p \big(g(x)\big)^p,$$

by the Lebesgue Dominated Convergence Theorem (Theorem B.1) we have

$$\left\| \sum_{j=1}^{n} f_j - h \right\|_p \to 0 \quad \text{as } n \to \infty.$$

This completes the proof. □

Observe that when the measure space Ω is the set $\mathbb{N}$ and the measure μ is such that $\mu(\{n\}) = 1$ for every $n \in \mathbb{N}$, the Banach space $L^p(\Omega, \mu)$ is the sequence space ℓ^p introduced in example (1) of 2.1. When Ω is a set of indices I and μ is the measure $\mu(\{i\}) = 1$ for each $i \in I$, the Banach space $L^p(\Omega, \mu)$ will be denoted by $\ell^p(I)$.

To conclude this section, we will study the topological dual of the space $L^p(\Omega, \mu)$. The following result is one of the remarkable contributions of F. Riesz to the theory of Banach spaces. It was proved by him in 1907 in the particular case of the measure space $[0, 1]$ with the Lebesgue measure, when $1 < p < \infty$. The case $p = 1$ was treated by H. Steinhaus in 1919.

Theorem 2.7 (Riesz Representation Theorem) *Let $(\Omega, \mathcal{B}, \mu)$ be a σ-finite measure space and let $1 \le p < \infty$. Given a bounded linear functional T on $L^p(\Omega, \mu)$ (i.e., $T \in \big(L^p(\Omega, \mu)\big)'$), there exists a unique function $g \in L^{p'}(\Omega, \mu)$ such that*

$$\langle T, f \rangle = \int_\Omega f\overline{g}\, d\mu \qquad \forall\, f \in L^p(\Omega, \mu)$$

and

$$\|T\| = \|g\|_{p'}.$$

Proof For simplicity, we will assume that the space is real. First, suppose that μ is finite. Define $\nu(B) := \langle T, \chi_B \rangle$ for each $B \in \mathcal{B}$. Let $B_n \in \mathcal{B}$ be such that $B_n \cap B_m = \emptyset$ for $n \neq m$. Whenever $\alpha_k \in \mathbb{R}$ with $|\alpha_k| = 1$, we have

$$\sum_{k=1}^{n} \alpha_k \chi_{B_k} \to \sum_{k=1}^{\infty} \alpha_k \chi_{B_k} \quad \text{in } L^p(\Omega, \mu).$$

Therefore, since T is continuous, it follows that

$$\sum_{k=1}^{\infty} |\nu(B_k)| = \lim_{n\to\infty} \sum_{k=1}^{n} \left\langle T, \operatorname{sign}(\langle T, \chi_{B_k} \rangle) \chi_{B_k} \right\rangle$$

$$= \left\langle T, \sum_{k=1}^{\infty} \operatorname{sign}(\langle T, \chi_{B_k} \rangle) \chi_{B_k} \right\rangle < \infty.$$

And similarly,

$$\sum_{k=1}^{\infty} \nu(B_k) = \left\langle T, \sum_{k=1}^{\infty} \chi_{B_k} \right\rangle = \left\langle T, \chi_{\bigcup_{k\in\mathbb{N}} B_k} \right\rangle = \nu\left(\bigcup_{k\in\mathbb{N}} B_k \right).$$

Thus, ν is a signed measure on $(\Omega, \mathcal{B})$. Moreover, ν is absolutely continuous with respect to the measure μ, i.e., $\nu(B) = 0$ whenever $\mu(B) = 0$. Therefore, as a consequence of the Radon–Nikodym Theorem (Theorem B.7) there exists a function $g \in L^1(\Omega, \mu)$ such that

$$\nu(B) = \int_B g \, d\mu \qquad \forall\, B \in \mathcal{B}.$$

If f is a simple function, then we have

$$\left| \int_\Omega fg \, d\mu \right| = |\langle T, f \rangle| \leq \|T\| \, \|f\|_p. \tag{2.11}$$

We now show that $g \in L^{p'}(\Omega, \mu)$. For $p = 1$, from (2.11) we have

$$\left| \int_B g \, d\mu \right| \leq M \qquad \forall\, B \in \mathcal{B}.$$

From this, it easily follows that $g \in L^\infty(\Omega, \mu)$. Now consider the case $1 < p < \infty$. Let $\{g_n\}$ be an increasing sequence of non-negative simple functions converging to $|g|^{p'}$ μ-a.e. Define $h_n := (g_n)^{\frac{1}{p}} \frac{g}{|g|}$ when $g \neq 0$, and $h_n = 0$ when $g = 0$. Then

$g_n \leq h_n g$. Hence, from (2.11) we obtain

$$\int_\Omega g_n \leq \|T\| \, \|h_n\|_p = \|T\| \, \|g_n\|_1^{\frac{1}{p}} \leq \|T\| \, \|g\|_1^{\frac{1}{p}}.$$

Therefore, $|g|^{p'}$ is integrable, and consequently $g \in L^{p'}(\Omega, \mu)$.

Let $S : L^p(\Omega, \mu) \to \mathbb{R}$ be the linear functional defined by

$$\langle S, f \rangle := \int_\Omega fg \, d\mu, \qquad f \in L^p(\Omega, \mu).$$

By Hölder's inequality, we have that S is bounded. We thus have that $S - T$ is a bounded linear functional on $L^p(\Omega, \mu)$ that vanishes on simple functions. Then, since simple functions are dense in $L^p(\Omega, \mu)$, we have that $S - T = 0$, and therefore,

$$\langle T, f \rangle := \int_\Omega fg \, d\mu, \quad \forall \, f \in L^p(\Omega, \mu).$$

It easily follows that $\|T\| = \|g\|_{p'}$, and the uniqueness of g.

We now prove the result when μ is σ-finite. In this case, we have that $\Omega = \bigcup_{n \in \mathbb{N}} \Omega_n$, with $\Omega_n \subset \Omega_{n+1}$, and $\mu(\Omega_n) < \infty$ for each $n \in \mathbb{N}$. From the above, there exists a sequence of functions $\{g_n\} \subset L^{p'}(\Omega, \mu)$, with $g_n(x) = 0$ for $x \in \Omega \setminus \Omega_n$, and satisfying

$$\langle T, f \rangle = \int_\Omega f g_n \, d\mu,$$

for each $f \in L^p(\Omega, \mu)$ such that $f(x) = 0$ if $x \in \Omega \setminus \Omega_n$. Moreover, $\|g_n\|_{p'} \leq \|T\|$ for each $n \in \mathbb{N}$. By the uniqueness of the g_n, we can assume that

$$g_{n+1}(x) = g_n(x) \qquad \text{if } \; x \in \Omega_n.$$

We define the function g as

$$g(x) := g_n(x) \qquad \text{if } \; x \in \Omega_n.$$

Then, by the Monotone Convergence Theorem (Theorem B.2),

$$\int_\Omega |g|^{p'} \, d\mu = \lim_{n \to \infty} \int_\Omega |g_n|^{p'} \, d\mu \leq \|T\|^{p'},$$

hence $g \in L^{p'}(\Omega, \mu)$. Now, by the Lebesgue Dominated Convergence Theorem (Theorem B.1), if $f \in L^p(\Omega, \mu)$, then

$$\int_\Omega fg\, d\mu = \lim_{n\to\infty} \int_{\Omega_n} fg\, d\mu = \lim_{n\to\infty} \int_\Omega (f\chi_{\Omega_n}) g_n\, d\mu$$

$$= \lim_{n\to\infty} \langle T, f\chi_{\Omega_n}\rangle = \langle T, f\rangle,$$

from which, as in the previous case, the result follows. □

Remark 2.1 For a different proof of the Riesz Representation Theorem, which does not use the Radon–Nikodym Theorem, see for example [7]. That proof uses the *Clarkson's Inequalities*, such that, the inequalities:

$$\left\| \frac{f+g}{2} \right\|_p^p + \left\| \frac{f-g}{2} \right\|_p^p \leq \frac{1}{2}\left(\|f\|_p^p + \|g\|_p^p\right) \quad \forall\, f, g \in L^p(\Omega, \mu),\ 2 \leq p < \infty.$$

As an immediate consequence of the Riesz Representation Theorem we obtain that we can identify the topological dual of the space $L^p(\Omega, \mu)$ with the space $L^{p'}(\Omega, \mu)$.

Corollary 2.1 *For each $1 \leq p < \infty$, the topological dual of $L^p(\Omega, \mu)$ is isometrically isomorphic to the space $L^{p'}(\Omega, \mu)$.*

To conclude this section, we will state the Riesz–Thorin Interpolation Theorem, also known as the Convexity Theorem, which was one of the first results on interpolation of bounded linear operators. The first version of this result was given in 1926 by M. Riesz,[17] under the restriction $p \leq q$, and for real-valued function spaces. The main tool used by Riesz was Hölder's inequality. In 1939, G. O. Thorin[18] was able to remove the restriction $p \leq q$ and proved the theorem for complex-valued functions. Instead of using Hölder's inequality, Thorin used as his main tool the following consequence of the Maximum Modulus Principle, due to Hadamard.

Lemma 2.4 (Hadamard's Three Lines Theorem) *Let $F(z)$ be an analytic function in the open strip $D = \{z \in \mathbb{C} : 0 < \Re z < 1\}$, bounded and continuous on the closure $\overline{D} = \{z \in \mathbb{C} : 0 \leq \Re z \leq 1\}$. If*

$$|F(it)| \leq M_0, \quad |F(1+it)| \leq M_1, \quad \forall\, t \in \mathbb{R},$$

[17] Acta Math. **49** (1926), 465–497.

[18] "*An extension of a convexity theorem due to M. Riesz*", Kungl. Fys. Saell. i Lund For **8** (1939).

then for every $0 \leq \theta \leq 1$*, one has*

$$\sup_{t\in\mathbb{R}} |F(\theta + it)| \leq M_0^{1-\theta}\, M_1^{\theta}.$$

Proof We may assume that $M_0 \neq 0$ and $M_1 \neq 0$, since, if for instance $M_0 = 0$, we can replace M_0 by $M_0 + \epsilon$, and let $\epsilon \to 0$ at the end of the proof. Then, considering the function

$$\frac{F(z)}{M_0^{z-1} M_1^{z}},$$

it suffices to prove the result for $M_0 = M_1 = 1$. Under this assumption, F is bounded by 1 on the boundary ∂D of D, and by some constant K in D. Thus, we aim to prove that $|F(z)| \leq 1$ for every $z \in D$. For each $\epsilon > 0$, let

$$F_\epsilon(z) := \frac{F(z)}{1 + \epsilon z}.$$

Then we have

$$|F_\epsilon(x + iy)| \leq 1, \quad \text{if } x + iy \in \partial D. \tag{2.12}$$

$$|F_\epsilon(x + iy)| \leq \frac{K}{\epsilon |y|}, \quad \text{if } x + iy \in D. \tag{2.13}$$

Let R be the rectangle with vertices $\pm\frac{iK}{\epsilon}$, $1 \pm \frac{iK}{\epsilon}$. From (2.12) and (2.13) we have $|F_\epsilon(z)| \leq 1$ for $z \in \partial R$. Therefore, applying the Maximum Modulus Principle, we get $|F_\epsilon(z)| \leq 1$ for $z \in R$. Now, from (2.13), we also have $|F_\epsilon(z)| \leq 1$ for $z \in \overline{D} \setminus R$. Hence, $|F_\epsilon(z)| \leq 1$ for all $z \in \overline{D}$. Therefore, $|F(z)| \leq 1 + \epsilon$ for each $z \in \overline{D}$, and letting $\epsilon \to 0$, we obtain $|F(z)| \leq 1$ for all $z \in \overline{D}$. □

Theorem 2.8 (Riesz–Thorin Interpolation Theorem) *Let* (X, μ) *and* (Y, ν) *be* σ*-finite measure spaces. Let* $1 \leq p_0, p_1, q_0, q_1 \leq \infty$*. Suppose* T *is a linear operator such that*

$$T : L^{p_0}(X, \mu) \to L^{q_0}(Y, \nu)$$

is bounded with norm M_0*, and*

$$T : L^{p_1}(X, \mu) \to L^{q_1}(Y, \nu)$$

is bounded with norm M_1*. Then*

$$T : L^{p}(X, \mu) \to L^{q}(Y, \nu)$$

is bounded with norm $M \leq M_0^{1-\theta} M_1^{\theta}$, *whenever* $0 < \theta < 1$, *and*

$$\frac{1}{p} = \frac{1-\theta}{p_0} + \frac{\theta}{p_1}, \quad \frac{1}{q} = \frac{1-\theta}{q_0} + \frac{\theta}{q_1}.$$

Proof If for $z \in \mathbb{C}$, we consider the functions:

$$\alpha(z) := \frac{1-z}{p_0} + \frac{z}{p_1}, \qquad \beta(z) := \frac{1-z}{q_0} + \frac{z}{q_1},$$

we have that

$$\alpha(\theta) = \frac{1}{p}, \qquad \beta(\theta) = \frac{1}{q}.$$

By Hölder's inequality it is easy to see (Problem 2.15) that

$$\|f\|_q = \sup\left\{\left|\int_Y fg \, d\nu\right| : \ \|g\|_{q'} = 1\right\}.$$

Hence,

$$M = \sup\left\{\left|\int_Y Tf \, g \, d\nu\right| : \ \|f\|_p = \|g\|_{q'} = 1\right\}.$$

Now, since $p < \infty$ and $q' < \infty$, the simple functions are dense in $L^p(X, \mu)$ and in $L^{q'}(Y, \nu)$, so it suffices to prove that

$$\left|\int_Y Tf \, g \, d\nu\right| \leq M_0^{1-\theta} M_1^{\theta} \tag{2.14}$$

for f and g simple functions with $\|f\|_p = \|g\|_{q'} = 1$. Suppose that f and g are such functions. Then,

$$f = \sum_{j=1}^{n} a_j \chi_{A_j}, \qquad g = \sum_{k=1}^{m} b_k \chi_{B_k},$$

where A_j are disjoint subsets of X of finite measure μ, and B_k are disjoint subsets of Y of finite measure ν. Moreover, we have that:

$$\sum_{j=1}^{n} |a_j|^p \, \mu(A_j) = \sum_{k=1}^{m} |b_k|^{q'} \, \nu(B_k) = 1. \tag{2.15}$$

For each $z \in \mathbb{C}$, we consider the functions:

$$f_z := |f|^{\frac{\alpha(z)}{\alpha(\theta)}} e^{i \arg(f)},$$

$$g_z := |g|^{\frac{1-\beta(z)}{1-\beta(\theta)}} e^{i \arg(g)},$$

and define

$$F(z) := \int_Y (Tf_z) g_z \, dv, \qquad z \in \mathbb{C}.$$

Then, since $F(\theta) = \displaystyle\int_Y (Tf) g \, dv$, by Hadamard's three lines theorem, to prove (2.14) it suffices to show that F is analytic in $D = \{z \in \mathbb{C} : 0 < \Re z < 1\}$, bounded and continuous on $\overline{D}$, and that:

$$|F(it)| \le M_0, \quad |F(1+it)| \le M_1, \qquad \forall\, t \in \mathbb{R}. \tag{2.16}$$

The analyticity of F is clear, since by the linearity of T we have

$$F(z) = \sum_{j=1}^{n} \sum_{k=1}^{m} |a_j|^{\frac{\alpha(z)}{\alpha(\theta)}} |b_k|^{\frac{1-\beta(z)}{1-\beta(\theta)}} \int_Y (T\chi_{A_j}) \chi_{B_k} \, e^{i(\arg(a_j)+\arg(b_k))} \, dv.$$

Hence, as F is a linear combination of exponential functions, it is an entire function. Moreover, this expression also shows that F is bounded on $\overline{D}$, since the real parts of $\alpha(z)$ and $\beta(z)$ are bounded. Therefore, it remains to prove (2.16).

Applying Hölder's inequality, for each $t \in \mathbb{R}$ we have

$$|F(it)| \le \|T(f_{it})\|_{q_0} \, \|g_{it}\|_{q_{0'}} \le M_0 \|f_{it}\|_{p_0} \, \|g_{it}\|_{q_{0'}}.$$

Now, since $\Re\alpha(it) = \frac{1}{p_0}$ and $\alpha(\theta) = \frac{1}{p}$, we have

$$\|f_{it}\|_{p_0}^{p_0} = \sum_{j=1}^{n} \left| |a_j|^{\frac{\alpha(it)}{\alpha(\theta)}} \right|^{p_0} \mu(A_j) = 1.$$

Similarly,

$$\|g_{it}\|_{q'_0}^{q'_0} = \sum_{k=1}^{m} \left| |b_k|^{\frac{1-\beta(it)}{1-\beta(\theta)}} \right|^{q'_0} \nu(B_k) = 1,$$

so $|F(it)| \leq M_0$ for each $t \in \mathbb{R}$. Using a similar argument, we show that $|F(1 + it)| \leq M_1$ for each $t \in \mathbb{R}$. Thus, we have established (2.16), and conclude the proof of the theorem. □

Remark 2.2 Note that from the proof of the previous theorem, the following version of the Riesz–Thorin Interpolation Theorem follows:

Let (X, μ) and (Y, ν) be σ-finite measure spaces. Let $1 \leq p_0, p_1 < \infty$, $1 \leq q_0, q_1 \leq \infty$. Let D be a dense subspace of $L^{p_0}(X, \mu)$ and of $L^{p_1}(X, \mu)$. Suppose that T_0 is a linear operator such that:

$$T_0 : L^{p_0}(X, \mu) \to L^{q_0}(Y, \nu)$$

is bounded with norm M_0, and

$$T_1 : L^{p_1}(X, \mu) \to L^{q_1}(Y, \nu)$$

is bounded with norm M_1. Assume that $T_0 f = T_1 f$ for every $f \in D$. Then, there exists a bounded linear operator

$$T : L^p(X, \mu) \to L^q(Y, \nu)$$

with norm $M \leq M_0^{1-\theta} M_1^{\theta}$, provided that $0 < \theta < 1$, and

$$\frac{1}{p} = \frac{1-\theta}{p_0} + \frac{\theta}{p_1}, \quad \frac{1}{q} = \frac{1-\theta}{q_0} + \frac{\theta}{q_1},$$

such that $Tf = T_0 f = T_1 f$ for every $f \in D$.

We denote by $L^1_{loc}(\mathbb{R}^n)$ the space of *locally integrable* functions on $\mathbb{R}^n$ with respect to Lebesgue measure, i.e.,

$$L^1_{loc}(\mathbb{R}^n) = \{f : \mathbb{R}^n \to \mathbb{C} \text{ measurable } : \ f \in L^1(K) \ \forall\, K \subset \mathbb{R}^n \text{ compact}\}.$$

Given two functions $f, g \in L^1_{loc}(\mathbb{R}^n)$, their *convolution product* $f \star g$ is defined as the function

$$f \star g(x) = \int_{\mathbb{R}^n} f(x - y) g(y)\, dy$$

when the integral exists.

When defined, it is easy to verify that the convolution product has the following properties:

$$f \star g = g \star f. \tag{2.17}$$

$$(f \star g) \star h = f \star (g \star h). \tag{2.18}$$

As a consequence of the Riesz–Thorin Interpolation Theorem, we will obtain one of the most general results regarding the existence of the convolution product.

Theorem 2.9 (Young's Inequality) *If $f \in L^p(\mathbb{R}^n)$ and $g \in L^q(\mathbb{R}^n)$, with $1 \leq p, q < \infty$, then $f \star g \in L^r(\mathbb{R}^n)$ with $\frac{1}{r} = \frac{1}{p} + \frac{1}{q} - 1$, and moreover,*

$$\|f \star g\|_r \leq \|f\|_p \, \|g\|_q.$$

Proof Fix q $(1 \leq q < \infty)$. Given $g \in L^q(\mathbb{R}^n)$, we define

$$Tf(x) := (f \star g)(x) = \int_{\mathbb{R}^n} f(x-y)g(y)\, dy.$$

As a consequence of Hölder's inequality, we have

$$\|Tf\|_\infty \leq \|g\|_q \, \|f\|_{q'} \quad \forall\, f \in L^{q'}(\mathbb{R}^n). \tag{2.19}$$

Now we show that

$$\|Tf\|_q \leq \|g\|_q \, \|f\|_1 \quad \forall\, f \in L^1(\mathbb{R}^n). \tag{2.20}$$

First, suppose that $q = 1$. Given $f \in L^1(\mathbb{R}^n)$, consider the measurable function $F(x, y) := f(x-y)f(y)$. Then,

$$\int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} |F(x,y)| dx \right) dy = \int_{\mathbb{R}^n} |g(y)| \left(\int_{\mathbb{R}^n} |f(x-y)|\, dx \right) dy$$

$$= \|f\|_1 \, \|g\|_1 < \infty.$$

Applying Tonelli–Hobson's Theorem, we get $F \in L^1(\mathbb{R}^n \times \mathbb{R}^n)$ and $\|F\|_1 = \|f\|_1 \, \|g\|_1$, so $f \star g \in L^1(\mathbb{R}^n)$, and furthermore,

$$\|f \star g\|_1 = \int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} f(x-y)\, g(y)\, dy \right| dx$$

$$\leq \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} |f(x-y)|\, |g(y)|\, dy \right) dx \leq \|f\|_1 \, \|g\|_1.$$

Now suppose $1 < q < \infty$. Since $|g|^q \in L^1(\mathbb{R}^n)$, from the above we know that for almost every $x \in \mathbb{R}^n$, the function

$$y \mapsto |f(x-y)|\, |g(y)|^q$$

is integrable on $\mathbb{R}^n$, and hence the function

$$y \mapsto |f(x-y)|^{\frac{1}{q}}\, |g(y)|$$

belongs to $L^q(\mathbb{R}^n)$. Then, since the function

$$y \mapsto |f(x-y)|^{\frac{1}{q'}}$$

belongs to $L^{q'}(\mathbb{R}^n)$, applying Hölder's inequality, we get that the function

$$y \mapsto |f(x-y)|\,|g(y)|$$

belongs to $L^1(\mathbb{R}^n)$, and furthermore,

$$\int_{\mathbb{R}^n} |f(x-y)|\,|g(y)|\,dy \le \left(\int_{\mathbb{R}^n} |f(x-y)|\,|g(y)|^q\,dy\right)^{\frac{1}{q}} \|f\|_1^{\frac{1}{q'}}.$$

Therefore,

$$|(f\star g)(x)|^q \le \left(\int_{\mathbb{R}^n} |f(x-y)|\,|g(y)|^q\,dy\right)\|f\|_1^{\frac{q}{q'}} = \big(|f|\star|g|^q\big)(x)\,\|f\|_1^{\frac{q}{q'}}.$$

Then, since the result is known for $q = 1$, we have that $f\star g \in L^q(\mathbb{R}^n)$, and moreover,

$$\|f\star g\|_q^q = \int_{\mathbb{R}^n} |f\star g(x)|^q\,dx \le \|f\|_1^{\frac{q}{q'}} \int_{\mathbb{R}^n} \big(|f|\star|g|^q\big)(x)dx$$

$$\le \|f\|_1^{\frac{q}{q'}}\,\|f\|_1\,\||g|^q\|_1 = \|f\|_1^{\frac{q}{q'}+1}\,\|g\|_q^q.$$

Hence,

$$\|f\star g\|_q \le \|f\|_1\,\|g\|_q.$$

Which completes the proof of (2.20).

As a consequence of (2.19), we have that $T : L^{q'}(\mathbb{R}^n) \to L^\infty(\mathbb{R}^n)$ is a bounded operator with norm $\|T\|_{q',\infty} \le \|g\|_q$. And from (2.20), we also have that $T : L^1(\mathbb{R}^n) \to L^q(\mathbb{R}^n)$ is a bounded operator with norm $\|T\|_{1,q} \le \|g\|_q$. Consequently, applying the Riesz–Thorin Interpolation Theorem, we get that $T : L^p(\mathbb{R}^n) \to L^r(\mathbb{R}^n)$ is a bounded operator whenever

$$\frac{1}{p} = \frac{1-\theta}{q'} + \frac{\theta}{1}, \quad \frac{1}{r} = \frac{1-\theta}{\infty} + \frac{\theta}{q}, \tag{2.21}$$

with $0 < \theta < 1$. Moreover, we have

$$\|T\|_{p,r} \le \|g\|_q^{1-\theta}\,\|g\|_q^{\theta} = \|g\|_q. \tag{2.22}$$

Now, eliminating θ from (2.21), we get $\frac{1}{r} = \frac{1}{p} + \frac{1}{q} - 1$. Consequently, we have proved that if $f \in L^p$ and $g \in L^q$, then $f \star g \in L^r$ with $\frac{1}{r} = \frac{1}{p} + \frac{1}{q} - 1$. Moreover, as a consequence of (2.22), we have

$$\|f \star g\|_r \leq \|f\|_p \, \|g\|_q.$$

□

Important particular cases of Young's inequality are the following:

$$\begin{cases} (i)\ p = q = 1, & r = 1 \\ (ii)\ 1 < p < \infty, & fq = 1,\ r = p. \end{cases}$$

2.6 The Hahn–Banach Theorem

One of the fundamental pillars of Functional Analysis is the Hahn–Banach theorem, to which we dedicate this section. This theorem has two formulations: one analytic and the other geometric. We begin with the analytic form, which refers to the extension of linear forms defined on subspaces. The proof uses Zorn's Lemma, whose statement we recall below.

Zorn's Lemma *Every non-empty inductive ordered set $\mathcal{P}$ (i.e., in which every totally ordered subset has an upper bound) has a maximal element, i.e., an element $m \in \mathcal{P}$ such that if $x \in \mathcal{P}$ and $m \leq x$, then $x = m$.*

The Hahn–Banach Theorem requires a separate treatment depending on whether the space is real or complex. We begin with the real case.

Definition 2.3 Let X be a vector space. A *sublinear functional* on X is a map $q : X \to \mathbb{R}$ that satisfies:

(i) $q(x + y) \leq q(x) + q(y)$ for all $x, y \in X$.
(ii) $q(\alpha x) = \alpha q(x)$ for all $x \in X$ and $\alpha \in \mathbb{R}^+$.

Note that every seminorm is a sublinear functional, but the converse is not true in general.

Theorem 2.10 (Hahn–Banach Theorem (Real Case)) *Let X be a real vector space and let q be a sublinear functional on X. Suppose Y is a vector subspace of X and that $f : Y \to \mathbb{R}$ is a linear map such that*

$$f(y) \leq q(y) \qquad \forall\, y \in Y.$$

Then there exists a linear functional $h \in X^*$ that agrees with f on Y and satisfies

$$h(x) \leq q(x) \qquad \forall\, x \in X.$$

Proof First, suppose that Y is a hyperplane. Then there exists $x_0 \in X \setminus Y$ such that

$$X = \{tx_0 + y \ : \ t \in \mathbb{R},\ y \in Y\}.$$

Suppose there exists an extension h as guaranteed by the theorem. Let us examine how h must behave in this case. Let $\alpha_0 = h(x_0)$. If $t > 0$ and $y_1 \in Y$, then

$$h(tx_0 + y_1) = t\alpha_0 + f(y_1) \leq q(tx_0 + y_1).$$

Hence,

$$\alpha_0 \leq -\frac{1}{t} f(y_1) + \frac{1}{t} q(tx_0 + y_1) = -f\left(\frac{y_1}{t}\right) + q\left(x_0 + \frac{y_1}{t}\right).$$

Therefore,

$$\alpha_0 \leq -f(y_1) + q(x_0 + y_1) \qquad \forall\, y_1 \in Y. \tag{2.23}$$

Moreover, if α_0 satisfies (2.23), then by the same reasoning we obtain

$$t\alpha_0 + f(y_1) \leq q(tx_0 + y_1) \qquad \forall\, t \geq 0.$$

On the other hand, if $t \geq 0$ and $y_2 \in Y$,

$$h(-tx_0 + y_2) = -t\alpha_0 + f(y_2) \leq q(-tx_0 + y_2).$$

Then,

$$\alpha_0 \geq f(y_2) - q(-x_0 + y_2) \qquad \forall\, y_2 \in Y. \tag{2.24}$$

Furthermore, if α_0 satisfies (2.24), then we have

$$-t\alpha_0 + f(y_2) \leq q(-tx_0 + y_2) \qquad \forall\, t \geq 0.$$

Therefore, in order for there to exist an α_0 satisfying (2.23) and (2.24), we must show that

$$f(y_2) - q(-x_0 + y_2) \leq -f(y_1) + q(x_0 + y_1) \qquad \forall\, y_1, y_2 \in Y. \tag{2.25}$$

Or equivalently,

$$f(y_1+y_2) \le q(y_1+y_2) = q\big((x_0+y_1)+(-x_0+y_2)\big) \le q(x_0+y_1)+q(-x_0+y_2).$$

Thus, (2.25) holds, and we can choose α_0 such that

$$\sup\{f(y_2) - q(-x_0+y_2) \ : \ y_2 \in Y\} \le \alpha_0$$

$$\le \inf\{-f(y_1) + q(x_0+y_1) \ : \ y_1 \in Y\}.$$

Therefore, with this choice of α_0, if we define $h(tx_0+y) := t\alpha_0 + f(y)$, then h is an extension of f that satisfies $h(x) \le q(x)$ for all $x \in X$. Hence, the result holds for hyperplanes. Let us now consider the general case. To that end, consider the family $\mathcal{F}$ of all pairs (Y_i, f_i), where Y_i is a vector subspace of X containing Y, and $f_i : Y_i \to \mathbb{R}$ is a linear extension of f such that $f_i(x) \le q(x)$ for all $x \in Y_i$. For $(Y_i, f_i), (Y_j, f_j) \in \mathcal{F}$, define the order

$$(Y_i, f_i) \le (Y_j, f_j) \iff Y_i \subset Y_j, \ \text{ and } \ f_j|_{Y_i} = f_i.$$

Let us verify that the ordered set $(\mathcal{F}, \le)$ is inductive. Let

$$C = \{(Y_j, f_j), \ j \in J\}$$

be a totally ordered subset of $\mathcal{F}$. Then it is easy to see that if $Z := \cup_{j\in J} Y_j$ and we define $g : Z \to \mathbb{R}$ by $g(x) := f_j(x)$ if $x \in Y_j$, then $(Z, g) \in \mathcal{F}$ and is an upper bound of C. Therefore, by Zorn's Lemma, $\mathcal{F}$ has a maximal element (M, h). From here, as we have shown in the first part that the result holds for hyperplanes, we conclude that $M = X$, completing the proof of the theorem. □

Evidently, if X is a vector space over $\mathbb{C}$, then X is also a vector space over $\mathbb{R}$. Also, if $f : X \to \mathbb{C}$ is $\mathbb{C}$-linear, then $\Re f$ is $\mathbb{R}$-linear. The following result provides the converse.

Lemma 2.5 *Let X be a vector space over $\mathbb{C}$. The following hold:*

(i) If $f : X \to \mathbb{R}$ is $\mathbb{R}$-linear, then $\tilde{f}(x) := f(x) - if(ix)$ is $\mathbb{C}$-linear and $f = \Re \tilde{f}$.
(ii) If $g : X \to \mathbb{C}$ is $\mathbb{C}$-linear and $f = \Re g$, then $\tilde{f} = g$.
(iii) If p is a seminorm on X and f and $\tilde{f}$ are as in (i), then

$$|f(x)| \le p(x) \ \ \forall\, x \in X \iff |\tilde{f}(x)| \le p(x) \ \ \forall\, x \in X.$$

(iv) If X is a normed space and f and $\tilde{f}$ are as in (i), then $\|f\| = \|\tilde{f}\|$ if they are continuous.

Proof

(i) We have

$$\tilde{f}\big((\alpha + i\beta)x\big) = f(\alpha x + i\beta x) - if(i\alpha x - \beta x)$$

$$= \alpha f(x) + \beta f(ix) - i\alpha f(ix) + i\beta f(x)$$

$$= (\alpha + i\beta) f(x) - i(\alpha + i\beta) f(x) = (\alpha + i\beta)\tilde{f}(x).$$

Hence, $\tilde{f}$ is $\mathbb{C}$-linear and $\Re \tilde{f} = f$.

(ii) Follows analogously to (i).

(iii) Suppose that $|\tilde{f}(x)| \leq p(x)$ for all $x \in X$. Then,

$$f(x) = \Re \tilde{f}(x) \leq |\tilde{f}(x)| \leq p(x)$$

and

$$-f(x) = \Re \tilde{f}(-x) \leq |\tilde{f}(-x)| = |\tilde{f}(x)| \leq p(x).$$

Thus, $|f(x)| \leq p(x)$ for all $x \in X$.

Now suppose that $|f(x)| \leq p(x)$ for all $x \in X$. Given $x \in X$, let θ be such that $\tilde{f}(x) = e^{i\theta}|\tilde{f}(x)|$. Then

$$|\tilde{f}(x)| = \tilde{f}\big(e^{-i\theta}x\big) = \Re \tilde{f}\big(e^{-i\theta}x\big) = f\big(e^{-i\theta}x\big) \leq p\big(e^{-i\theta}x\big) = p(x).$$

(iv) Since $|f(x)| \leq \|f\|\,\|x\|$, as a consequence of (iii) we have $|\tilde{f}(x)| \leq \|f\|\,\|x\|$, and hence $\|\tilde{f}\| \leq \|f\|$. The reverse inequality $\|f\| \leq \|\tilde{f}\|$ follows analogously.

□

Theorem 2.11 (Hahn-Banach Theorem) *Let X be a vector space over $\mathbb{K}$, Y a vector subspace of X, and p a seminorm on X. If $f : Y \to \mathbb{K}$ is a linear functional such that $|f(y)| \leq p(y)$ for every $y \in Y$, then there exists $h \in X^*$, an extension of f, such that $|h(x)| \leq p(x)$ for every $x \in X$.*

Proof If $\mathbb{K} = \mathbb{R}$, since

$$f(y) \leq |f(y)| \leq p(y) \quad \forall\, y \in Y,$$

by Theorem 2.10, there exists a linear extension h of f to X satisfying $h(x) \leq p(x)$ for every $x \in X$. Therefore, $-h(x) = h(-x) \leq p(-x) = p(x)$, which implies $|h(x)| \leq p(x)$ for every $x \in X$.

Now suppose that $\mathbb{K} = \mathbb{C}$. Let $f_1 = \Re f$. Then, by (iii) of the previous lemma, we have $|f_1(y)| \leq p(y)$ for every $y \in Y$. Then, by the above, there exists an $\mathbb{R}$-linear map $h_1 : X \to \mathbb{R}$, an extension of f_1, such that $|h_1(x)| \leq p(x)$ for every $x \in X$.

Now, define $h(x) := h_1(x) - ih_1(ix)$. As a consequence of (iii) of the previous lemma, we obtain $|h(x)| \leq p(x)$ for every $x \in X$, and moreover, $h(y) = f(y)$ for every $y \in Y$. □

Now we present some important consequences of the Hahn-Banach theorem.

Corollary 2.2 *Let X be a normed space. Then:*

(i) If $\langle u, x_0\rangle = 0$ for every $u \in X'$, then $x_0 = 0$.
(ii) If Y is a subspace of X and $u \in Y'$, then there exists $v \in X'$, an extension of u, such that $\|u\| = \|v\|$.
(iii) If $x_1, x_2, \ldots, x_n$ are linearly independent vectors in X and $\alpha_1, \alpha_2, \ldots, \alpha_n$ are scalars, then there exists $u \in X'$ such that

$$\langle u, x_j\rangle = \alpha_j \quad \text{for } j = 1, \ldots, n.$$

Proof Suppose that $x_0 \neq 0$. Let $Y := \text{LIN}\{x_0\}$, and define $u : Y \to \mathbb{K}$ by $\langle u, \alpha x_0\rangle := \alpha\|x_0\|$.

$$|\langle u, \alpha x_0\rangle| = |\alpha\|x_0\| \,| = \|\alpha x_0\|.$$

Therefore, as a consequence of the Hahn-Banach Theorem, there exists an extension $v \in X^*$ of u such that $|\langle v, x\rangle| \leq \|x\|$ for every $x \in X$, which means that $v \in X'$, a contradiction.

(ii) This follows directly from the Hahn–Banach Theorem with $p(x) = \|x\|$.
(iii) Let $Y := \text{LIN}\{x_1, x_2, \ldots, x_n\}$, and define $v : Y \to \mathbb{K}$ by

$$\left\langle v, \sum_{j=1}^{n} \beta_j x_j \right\rangle := \sum_{j=1}^{n} \beta_j \alpha_j.$$

Since Y is finite-dimensional and v is linear, we have $v \in Y'$. Then by (ii), there exists an extension $u \in X'$ of v.

□

Corollary 2.3 *Let X be a normed space. For each $x \in X$ we have*

$$\|x\| = \sup\{|\langle u, x\rangle| \,:\, u \in X', \ \ \|u\| \leq 1\}.$$

Moreover, this supremum is attained, i.e., there exists $u \in X'$ with $\|u\| \leq 1$ such that $\|x\| = \langle u, x\rangle$.

Proof Let

$$\alpha := \sup\{|\langle u, x\rangle| \,:\, u \in X', \ \ \|u\| \leq 1\}.$$

Since $|\langle u, x\rangle| \leq \|u\|\,\|x\|$, we have $\alpha \leq \|x\|$. Let $Y := \mathrm{LIN}\{x\}$ and define $v : Y \to \mathbb{K}$ by $\langle v, \beta x\rangle := \beta\|x\|$. Then, since $v \in Y'$ and $\|v\| = 1$, applying part (ii) of the previous corollary, there exists $u \in X'$ such that $\|u\| = \|v\| = 1$ and $\langle u, x\rangle = \langle v, x\rangle = \|x\|$. □

Recall that by definition, the norm in the dual space is given by

$$\|u\| = \sup\{|\langle u, x\rangle| \,:\, x \in X,\ \ \|x\| \leq 1\}.$$

In the following example we will see that, in this case, the supremum is not attained in general.

Example 2.3 Let $X = c_0$, so we can identify X' with ℓ^1 (see Problem 2.11). Let $u = \left(\frac{1}{2^n}\right) \in \ell^1$. Then,

$$\|u\|_1 = \sum_{n=1}^{\infty} \frac{1}{2^n},$$

so that

$$1 = \sup\left\{ \left| \sum_{n=1}^{\infty} \alpha_n \frac{1}{2^n} \right| \,:\, (\alpha_n) \in c_0,\ \ \|(\alpha_n)\|_\infty \leq 1 \right\}.$$

Let us show that this supremum is not attained. If $(\alpha_n) \in c_0$ with $\|(\alpha_n)\|_\infty \leq 1$, we can choose $n_0 \in \mathbb{N}$, such that $|\alpha_n| < \frac{1}{2}$ for every $n \geq n_0$. Then,

$$\left| \sum_{n=1}^{\infty} \alpha_n \frac{1}{2^n} \right| \leq \sum_{n=1}^{n_0} |\alpha_n \frac{1}{2^n}| + \sum_{n=n_0+1}^{\infty} |\alpha_n \frac{1}{2^n}|$$

$$< \sum_{n=1}^{n_0} \frac{1}{2^n} + \frac{1}{2} \sum_{n=n_0+1}^{\infty} \frac{1}{2^n} < 1.$$

Corollary 2.4 *Let X be a normed space and Y a closed subspace of X. Then, given $x_0 \in X \setminus Y$, there exists $u \in X'$, such that $\langle u, x_0\rangle = 1$, $\langle u, y\rangle = 0$ for every $y \in Y$, and moreover,*

$$\|u\| = \frac{1}{d(x_0, Y)}.$$

Proof Let $\pi : X \to \frac{X}{Y}$ be the quotient map, i.e., $\pi(x) := x + Y$. Since $\|x_0 + Y\| = d(x_0, Y)$, by applying the previous corollary, there exists $f \in \left(\frac{X}{Y}\right)'$ such that $\|f\| = 1$ and $\langle f, x_0 + Y\rangle = d(x_0, Y)$. Let $u := \frac{1}{d(x_0,Y)} f \circ \pi : X \to \mathbb{K}$. Then,

$u \in X'$, $\langle u, x_0\rangle = \frac{\langle f, x_0+Y\rangle}{d(x_0,Y)} = 1$, and $\langle u, y\rangle = 0$ for every $y \in Y$. Moreover,

$$|\langle u, x\rangle| = \frac{1}{d(x_0, Y)}|\langle f, \pi(x)\rangle| \leq \frac{1}{d(x_0, Y)}\|\pi(x)\| \leq \frac{1}{d(x_0, Y)}\|x\|.$$

Therefore, $\|u\| \leq \frac{1}{d(x_0,Y)}$. On the other hand, since $\|f\| = 1$, there exists a sequence $x_n \in X$ with $\|x_n + Y\| < 1$, and $|\langle f, x_n + Y\rangle| \to 1$. Then, if we take $y_n \in Y$ such that $\|x_n + y_n\| \leq 1$, we get

$$|\langle u, x_n + y_n\rangle| = \frac{1}{d(x_0, Y)}|\langle f, x_n + Y\rangle| \to \frac{1}{d(x_0, Y)}.$$

Therefore,

$$\|u\| = \frac{1}{d(x_0, Y)}.$$

□

Corollary 2.5 *Let X be a normed space and Y a subspace of X. Then,*

$$\overline{Y} = \bigcap\{\text{Ker } u \ : \ u \in X', \ \ Y \subset \text{Ker } u\}.$$

In particular, Y is dense in X if and only if the only continuous linear functional on X that vanishes on Y is the zero functional.

Proof Let

$$Z := \bigcap\{\text{Ker } u \ : \ u \in X', \ \ Y \subset \text{Ker } u\}.$$

Since Z is closed and $Y \subset Z$, we have $\overline{Y} \subset Z$. On the other hand, if $x_0 \in X \setminus \overline{Y}$, then by the previous corollary there exists $u \in X'$ such that $\langle u, x_0\rangle = 1$, and $\langle u, y\rangle = 0$ for every $y \in \overline{Y}$. Consequently, $x_0 \notin Z$. □

We will now proceed to examine the geometric form of the Hahn-Banach Theorem. Recall that an *affine hyperplane* H in a vector space X is the translation of a hyperplane passing through the origin. As a consequence of Lemma 2.2, we have that H is an affine hyperplane of X if and only if there exists $u \in X^*$, $u \neq 0$, and $\alpha \in \mathbb{K}$, such that

$$H = \{x \in X \ : \ \langle u, x\rangle = \alpha\}.$$

In this case, we say that H is the hyperplane with equation $\{u = \alpha\}$. As a consequence of Proposition 2.2, we have:

The hyperplane with equation $\{u = \alpha\}$ is closed if and only if $u \in X'$.

Definition 2.4 Let X be a vector space, and let $A, B \subset X$. We say that the hyperplane with equation $\{u = \alpha\}$ *separates* A and B if

$$\langle u, x\rangle \leq \alpha \ \ \forall\, x \in A \quad \text{and} \quad \langle u, x\rangle \geq \alpha \ \ \forall\, x \in B.$$

We say that the hyperplane with equation $\{u = \alpha\}$ *strictly separates* A and B if there exists $\epsilon > 0$ such that

$$\langle u, x\rangle \leq \alpha - \epsilon \ \ \forall\, x \in A \quad \text{and} \quad \langle u, x\rangle \geq \alpha + \epsilon \ \ \forall\, x \in B.$$

Definition 2.5 Let X be a vector space and $A \subset X$. We say that A is a *convex set* if for every $a, b \in A$, the straight line segment joining a and b is contained in A, *i.e.*,

$$[a, b] := \{ta + (1 - t)b \ : \ t \in [0, 1]\} \subset A.$$

It is easy to see that A is convex if and only if for every $a_1, \ldots, a_n \in A$, $t_1, \ldots, t_n \in [0, 1]$ with $\sum_{j=1}^n t_j = 1$, we have $\sum_{j=1}^n t_j a_j \in A$.

Definition 2.6 Let X be a real normed vector space and let C be a convex and open subset of X such that $0 \in C$. The *Minkowski functional* of C is defined as

$$P_C(x) := \inf \left\{ \alpha > 0 \ : \ \frac{x}{\alpha} \in C \right\}.$$

Lemma 2.6 *The Minkowski functional P_C of a convex open set C that contains the origin of a normed space X is a sublinear functional, and there exists a constant M such that*

$$0 \leq P_C(x) \leq M\|x\| \qquad \forall\, x \in X.$$

Moreover,

$$C = \{x \in X \ : \ P_C(x) < 1\}.$$

Proof Since $0 \in C$ and C is open, there exists $r > 0$ such that $\overline{B(0, r)} \subset C$, from which it follows that

$$P_C(x) \leq \frac{1}{r}\|x\| \qquad \forall\, x \in X.$$

Let us show that $C = \{x \in X \ : \ P_C(x) < 1\}$. Let $x \in C$. Since C is open, there exists $\epsilon > 0$ such that $(1 + \epsilon)x \in C$, hence

$$P_C(x) \leq \frac{1}{1 + \epsilon} < 1.$$

Conversely, if $P_C(x) < 1$, then there exists $0 < \alpha < 1$ such that $\frac{1}{\alpha}x \in C$. Then, since $0 \in C$ and C is convex, we have

$$x = \alpha\left(\frac{1}{\alpha}x\right) + (1-\alpha)0 \in C.$$

Clearly, $P_C(\lambda x) = \lambda P_C(x)$ for every $x \in X$ and $\lambda > 0$. Finally, let us show that $P_C(x+y) \leq P_C(x) + P_C(y)$. Let $x, y \in X$, and let $\epsilon > 0$. Then,

$$P_C\left(\frac{x}{P_C(x)+\epsilon}\right) = \frac{1}{P_C(x)+\epsilon}P_C(x) < 1,$$

so $\dfrac{x}{P_C(x)+\epsilon} \in C$. Similarly, $\dfrac{y}{P_C(y)+\epsilon} \in C$. Therefore,

$$\frac{tx}{P_C(x)+\epsilon} + \frac{(1-t)y}{P_C(y)+\epsilon} \in C \quad \forall\, t \in [0,1].$$

In particular, taking

$$t := \frac{P_C(x)+\epsilon}{P_C(x)+P_C(y)+2\epsilon},$$

we get

$$\frac{x+y}{P_C(x)+P_C(y)+2\epsilon} \in C.$$

Then,

$$P_C\left(\frac{x+y}{P_C(x)+P_C(y)+2\epsilon}\right) = \frac{P_C(x+y)}{P_C(x)+P_C(y)+2\epsilon} < 1.$$

Therefore,

$$P_C(x+y) \leq P_C(x) + P_C(y) + 2\epsilon,$$

and since this is true for every $\epsilon > 0$, we conclude that $P_C(x+y) \leq P_C(x)+P_C(y)$. □

Lemma 2.7 *Let X be a real normed space, $C \subset X$ an open convex set, and $x_0 \in X \setminus C$. Then there exists $u \in X'$ such that $\langle u, x\rangle < \langle u, x_0\rangle$ for all $x \in C$. In particular, the hyperplane given by the equation $\{u = \langle u, x_0\rangle\}$ separates $\{x_0\}$ from C.*

Proof By translation, we may assume that $0 \in C$. Let $Y := \mathrm{LIN}\{x_0\}$, and consider the linear functional v on Y defined by $\langle v, tx_0\rangle := t$ for every $t \in \mathbb{R}$. Since $x_0 \notin$

C, we have $\langle v, y\rangle \leq P_C(y)$ for every $y \in Y$. Then, applying the Hahn–Banach theorem, there exists $u \in X^*$ such that $\langle v, y\rangle = \langle u, y\rangle$ for all $y \in Y$, and $\langle u, x\rangle \leq P_C(x)$ for all $x \in X$. By the previous lemma,

$$|\langle u, x\rangle| \leq P_C(x) \leq M\|x\|,$$

so $u \in X'$. Moreover, $\langle u, x_0\rangle = 1$ and $\langle u, x\rangle \leq P_C(x) < 1$ if $x \in C$. □

With all these ingredients, we can now state one of the geometric versions of the Hahn–Banach Theorem.

Theorem 2.12 (First Separation Theorem for Convex Sets) *Let X be a real normed space. Let A and B be two nonempty, disjoint convex subsets of X. Suppose that A is open. Then there exists a closed hyperplane in X that separates A and B.*

Proof It is easy to see that the set

$$C = A - B := \{a - b \,:\, a \in A,\ b \in B\}$$

is convex. Moreover, since

$$C = \bigcup_{b\in B}(A - b),$$

C is open, and $0 \notin C$ because $A \cap B = \emptyset$. Therefore, by applying the previous lemma, there exists $u \in X'$ such that $\langle u, x\rangle < 0$ for every $x \in C$. Then,

$$\langle u, a\rangle < \langle u, b\rangle \quad \forall\, a \in A,\ b \in B.$$

Fix $\alpha \in \mathbb{R}$ such that

$$\sup_{a\in A}\langle u, a\rangle \leq \alpha \leq \inf_{b\in B}\langle u, b\rangle.$$

Then, the hyperplane given by the equation $\{u = \alpha\}$ separates the sets A and B. □

Theorem 2.13 (Second Separation Theorem for Convex Sets) *Let X be a real normed space. Let A and B be two non-empty disjoint convex subsets of X. Suppose that A is closed and that B is compact. Then, there exists a closed hyperplane in X that strictly separates A and B.*

Proof For $\epsilon > 0$ define $A_\epsilon := A + B(0, \epsilon)$ and $B_\epsilon := B + B(0, \epsilon)$. The sets A_ϵ and B_ϵ are open, convex, and nonempty. Moreover, for ϵ sufficiently small, we have $A_\epsilon \cap B_\epsilon = \emptyset$. In fact, if this was not the case, we could find $x_n \in A$, $y_n \in B$ such that $\|x_n - y_n\| \leq \frac{1}{n}$ for each $n \in \mathbb{N}$. Now, since B is compact, there exists a subsequence $y_{n_k} \to y \in B$. Then, $x_{n_k} \to y \in A$, since A is closed, leading to a contradiction. So let $\epsilon > 0$ be such that $A_\epsilon \cap B_\epsilon = \emptyset$. Applying the previous theorem, there exist $u \in X'$ and $\alpha \in \mathbb{R}$ such that the hyperplane $\{u = \alpha\}$ separates the sets A_ϵ and B_ϵ.

Therefore,

$$\langle u, x + \epsilon z\rangle \leq \alpha \leq \langle u, y + \epsilon z\rangle \quad \forall\, x \in A,\ y \in B,\ z \in B(0,1).$$

Consequently,

$$\langle u, x\rangle + \epsilon\|u\| \leq \alpha \leq \langle u, y\rangle + \epsilon\|u\| \quad \forall\, x \in A,\ y \in B,$$

and since $\|u\| \neq 0$, it follows that $\{u = \alpha\}$ strictly separates A and B. □

Corollary 2.6 *Let X be a complex normed space. Let A and B be two non-empty disjoint convex subsets of X. Suppose that A is closed and that B is compact. Then there exist $u \in X'$, $\alpha \in \mathbb{R}$, and $\epsilon > 0$ such that*

$$\Re\langle u, a\rangle \leq \alpha < \alpha + \epsilon \leq \Re\langle u, b\rangle \quad \forall\, a \in A,\ b \in B.$$

In the previous results, the compactness hypothesis on B is necessary. Indeed, consider in $X = \mathbb{R}^2$ the sets

$$A := \{(x, y) \in \mathbb{R}^2 \ :\ x \geq 0,\ y \leq 0\}$$

and

$$B := \left\{(x, y) \in \mathbb{R}^2 \ :\ x \geq 0,\ y \geq \frac{1}{x}\right\}.$$

The sets A and B are disjoint, closed, and convex, but they are not strictly separated.

2.7 The Banach–Steinhaus and Closed Graph Theorems

In this section, we will establish the Banach–Steinhaus and Closed Graph Theorems, which, together with the Hahn–Banach Theorem, form the foundational pillars of Functional Analysis. The proof of these two results is based on the following result due to R.L. Baire.

Theorem 2.14 (Baire's Theorem) *Let (X, d) be a complete metric space, and let $\{F_n\}$ be a sequence of closed subsets of X. Then, if*

$$\operatorname{int}(F_n) = \emptyset \quad \forall\, n \in \mathbb{N},$$

it follows that

$$\operatorname{int}\left(\bigcup_{n\in\mathbb{N}} F_n\right) = \emptyset.$$

Proof Let $F := \bigcup_{n=1}^{\infty} F_n$. We must prove that for any $x_0 \in X$ and $\epsilon > 0$, it holds that $B(x_0, \epsilon) \cap (X \setminus F) \neq \emptyset$. Indeed, since $\text{int}(F_1) = \emptyset$, there exists $0 < \epsilon_1 < 1$ and $x_1 \in B(x_0, \epsilon)$ such that

$$\overline{B(x_1, \epsilon_1)} \subset B(x_0, \epsilon) \quad \text{and} \quad \overline{B(x_1, \epsilon_1)} \cap F_1 = \emptyset.$$

Similarly, since $\text{int}(F_2) = \emptyset$, there exists $0 < \epsilon_2 < \frac{1}{2}$ and $x_2 \in B(x_1, \epsilon_1)$ such that

$$\overline{B(x_2, \epsilon_2)} \subset B(x_2, \epsilon_1) \quad \text{and} \quad \overline{B(x_2, \epsilon_2)} \cap F_2 = \emptyset.$$

Repeating this process, we find a sequence x_n and scalars $0 < \epsilon_n < \frac{1}{2^n}$ such that

$$\overline{B(x_n, \epsilon_n)} \subset B(x_{n-1}, \epsilon_{n-1}) \quad \forall\, n \in \mathbb{N} \tag{2.26}$$

and

$$\overline{B(x_n, \epsilon_n)} \cap F_n = \emptyset \quad \forall\, n \in \mathbb{N}. \tag{2.27}$$

As a consequence of (2.26), if $n < m$, then

$$d(x_n, x_m) \leq d(x_n, x_{n+1}) + d(x_{n+1}, x_{n+2}) + \cdots + d(x_{m-1}, x_m)$$

$$< \frac{1}{2^n} + \frac{1}{2^{n+1}} + \cdots + \frac{1}{2^m} < \frac{1}{2^{n-1}}.$$

Therefore, the sequence $\{x_n\}$ is Cauchy in (X, d). Since the space is complete, it follows that $x_n \to y \in X$. Now, if $n < m$, then

$$d(x_n, y) \leq d(x_n, x_m) + d(x_m, y) \leq \epsilon_n + d(x_m, y) \to \epsilon_n, \quad \text{as } m \to \infty.$$

From this and (2.27), we obtain

$$y \in \bigcap_{n=1}^{\infty} \overline{B(x_n, \epsilon_n)} \subset \bigcap_{n=1}^{\infty} (X \setminus F_n) = X \setminus F.$$

Consequently, $B(x_0, \epsilon) \cap (X \setminus F) \neq \emptyset$. □

The following result, due to S. Banach and H. Steinhaus, is called the *Uniform Boundedness Principle* in many textbooks.

Theorem 2.15 (Banach–Steinhaus Theorem) *Let X be a Banach space and Y a normed space. Let $T_i \in \mathcal{L}(X, Y)$, $i \in I$. If the family $\{T_i : i \in I\}$ is pointwise bounded, i.e., if*

$$\sup_{i \in I} \|T_i(x)\| < \infty \quad \forall\, x \in X, \tag{2.28}$$

then the family $\{T_i \ : \ i \in I\}$ *is uniformly bounded, i.e.,*

$$\sup_{i \in I} \|T_i\| < \infty. \tag{2.29}$$

In other words, there exists a constant $M > 0$ *such that*

$$\|T_i(x)\| \leq M\|x\| \qquad \forall\, x \in X,\ \forall\, i \in I.$$

Proof For each $n \in \mathbb{N}$, let

$$F_n := \{x \in X \ : \ \|T_i(x)\| \leq n, \ \ \forall\, i \in I\}.$$

Each F_n is a closed set in X, and as a consequence of (2.28), we have

$$X = \bigcup_{n \in \mathbb{N}} F_n.$$

Therefore, by applying Baire's theorem, there exists $n_0 \in \mathbb{N}$ such that $\text{int}(F_{n_0}) \neq \emptyset$. Thus, there exist $x_0 \in X$ and $r > 0$ such that $B(x_0, r) \subset F_{n_0}$. Consequently,

$$\|T_i(x_0 + rx)\| \leq n_0, \quad \forall\, x \in B(0,1), \ \ \forall\, i \in I.$$

Then, for each $i \in I$, we have

$$\|T_i\| = \sup\{\|T_i(x)\| \ : \ x \in B(0,1)\}$$

$$= \sup\left\{\left\|\frac{1}{r}\left(T_i(x_0 + rx) - T_i(x_0)\right)\right\| \ : \ x \in B(0,1)\right\}$$

$$\leq \frac{1}{r}n_0 + \frac{1}{r}\|T_i(x_0)\| < \infty.$$

□

Let us now look at some consequences of the Banach-Steinhaus theorem that are very useful in practice.

Corollary 2.7 *Let* X *be a Banach space and* Y *a normed space. Let* $T_n \in \mathcal{L}(X, Y)$*, and suppose that for each* $x \in X$ *the following limit exists:*

$$T(x) := \lim_{n \to \infty} T_n(x).$$

Then,

(i) $\sup_{n \in \mathbb{N}} \|T_n\| < \infty.$

(ii) $T \in \mathcal{L}(X, Y)$.
(iii) $\|T\| \leq \liminf_{n\to\infty} \|T_n\|$.

Proof (i) is an immediate consequence of the previous theorem. As a consequence of (i), there exists $M > 0$ such that

$$\|T_n(x)\| \leq M\|x\|, \qquad \forall\, x \in X,\ \forall\, n \in \mathbb{N}.$$

Therefore,

$$\|T(x)\| \leq M\|x\| \qquad \forall\, x \in X,$$

which proves (ii). Finally, since

$$\|T_n(x)\| \leq \|T_n\|\|x\|, \qquad \forall\, x \in X,\ \forall\, n \in \mathbb{N},$$

we obtain (iii). □

Corollary 2.8 *Let X be a normed space and $B \subset X$. Suppose that for each $u \in X'$, the set $\{\langle u, x\rangle \ :\ x \in B\}$ is bounded in $\mathbb{K}$. Then, B is bounded in X.*

Proof For each $x \in B$, define $T_x : X' \to \mathbb{K}$ by $T_x(u) := \langle u, x\rangle$. Clearly, $T_x \in \mathcal{L}(X', \mathbb{K})$. Moreover, by hypothesis, the set $\{T_x \ :\ x \in B\}$ is pointwise bounded. Then, applying the Banach-Steinhaus Theorem, there exists $M > 0$ such that

$$|\langle u, x\rangle| = |T_x(u)| \leq M\|u\| \quad \forall\, x \in B,\ u \in X'.$$

Hence, for each $x \in B$,

$$\|x\| = \sup\{|\langle u, x\rangle| \ :\ u \in X',\ \|u\| \leq 1\} \leq M,$$

so B is a bounded subset of X. □

A similar argument proves the following dual version of the previous result.

Corollary 2.9 *Let X be a Banach space and $B \subset X'$ such that for each $x \in X$, the set $\{\langle u, x\rangle \ :\ u \in B\}$ is bounded in $\mathbb{K}$. Then, B is bounded in X'.*

The following result, due to S. Banach, also uses the Baire Theorem in its proof.

Theorem 2.16 (Open Mapping Theorem) *Let X and Y be Banach spaces, and let $T \in \mathcal{L}(X, Y)$ be a surjective linear operator. Then, there exists $c > 0$ such that*

$$B_Y(0, c) \subset T\big(B_X(0, 1)\big),$$

i.e., the map T is open.

Proof In a first step, we show that:

$$\begin{cases} \text{If } T \in L(X,Y) \text{ is surjective, then} \\ \\ \text{there exists } c > 0 \text{ such that } B_Y(0,2c) \subset \overline{T\big(B_X(0,1)\big)}. \end{cases} \tag{2.30}$$

Indeed, for each $n \in \mathbb{N}$, define $F_n := n\,\overline{T\big(B_X(0,1)\big)}$. Since T is surjective, we have

$$Y = \bigcup_{n\in\mathbb{N}} F_n.$$

Therefore, by Baire's Category Theorem, there exists $n_0 \in \mathbb{N}$ such that $\text{int}\big(F_{n_0}\big) \neq \emptyset$. Thus,

$$\text{int}\left(\overline{T\big(B_X(0,1)\big)}\right) \neq \emptyset.$$

So there exist $c > 0$ and $y_0 \in Y$ such that

$$B(y_0, 4c) \subset \overline{T\big(B_X(0,1)\big)}. \tag{2.31}$$

In particular, $y_0 \in \overline{T\big(B_X(0,1)\big)}$, and by symmetry,

$$-y_0 \in \overline{T\big(B_X(0,1)\big)}. \tag{2.32}$$

As a consequence of (2.31) and (2.32), and taking into account that $\overline{T\big(B_X(0,1)\big)}$ is a convex set, we obtain:

$$B(0,4c) \subset \overline{T\big(B_X(0,1)\big)} + \overline{T\big(B_X(0,1)\big)} = 2\,\overline{T\big(B_X(0,1)\big)}.$$

Finally, we prove that if $T \in \mathcal{L}(X,Y)$ satisfies (2.30), then

$$B_Y(0,c) \subset T\big(B_X(0,1)\big).$$

Indeed, given $y \in Y$ with $\|y\| < c$, we must find $x \in X$ with $\|x\| < 1$ such that $T(x) = y$. Now, from (2.30), for any $\epsilon > 0$ there exists $z \in X$ with $\|z\| < \frac{1}{2}$ and $\|y - T(z)\| < \epsilon$. Taking $\epsilon = \frac{c}{2}$, we find $z_1 \in X$ such that

$$\|z_1\| < \frac{1}{2}, \quad \text{and} \quad \|y - Tz_1\| < \frac{c}{2}.$$

Applying the same procedure to $y - T(z_1)$ and $\epsilon = \frac{c}{4}$, we find $z_2 \in X$ such that

$$\|z_2\| < \frac{1}{4}, \quad \text{and} \quad \|(y - Tz_1) - Tz_2\| < \frac{c}{4}.$$

In this way, we construct a sequence $z_n \in X$ satisfying:

$$\|z_n\| < \frac{1}{2^n}, \quad \text{and} \quad \|y - T(z_1 + z_2 + \cdots + z_n)\| < \frac{c}{2^n}, \quad \forall\, n \in \mathbb{N}.$$

Taking $x_n := \sum_{j=1}^{n} z_j$, the sequence $\{x_n\}$ is Cauchy in X, so there exists $x \in X$ with $\|x\| < 1$, such that $T(x) = y$. □

Corollary 2.10 *Let X and Y be Banach spaces and let $T \in \mathcal{L}(X, Y)$ be a bijective operator. Then, T is a topological isomorphism.*

Proof By the Open Mapping Theorem, there exists a constant $c > 0$ such that

$$B_Y(0, c) \subset T\big(B_X(0, 1)\big).$$

Then, since T is injective, we have that if $\|Tx\| < c$, then $\|x\| < 1$. This implies that

$$\|x\| \leq \frac{1}{c}\|Tx\| \qquad \forall\, x \in X.$$

Thus, since T is injective, we obtain

$$\|T^{-1}(y)\| \leq \frac{1}{c}\|y\| \qquad \forall\, y \in Y.$$

Hence, T^{-1} is continuous. □

Corollary 2.11 *Let $\|\ \|_1$ and $\|\ \|_2$ be two norms on X that give it the structure of a Banach space. If one of them is finer than the other (i.e., if for example there exists $M > 0$ such that $\|x\|_2 \leq M\|x\|_1$ for all $x \in X$), then these norms are equivalent.*

Proof By hypothesis, the identity operator

$$I : (X, \|\ \|_1) \to (X, \|\ \|_2)$$

is continuous. Therefore, applying the previous corollary, I is a topological isomorphism, and hence the norms are equivalent. □

Theorem 2.17 (Closed Graph Theorem) *Let X and Y be Banach spaces, and let $T \in L(X, Y)$ be a linear operator. If the graph of T*

$$\mathcal{G}(T) := \{(x, Tx) \ : \ x \in X\}$$

is a closed subset of $X \times Y$, then T is continuous.

Proof Consider on X the graph norm

$$\|x\|_1 := \|x\| + \|Tx\|.$$

Since by hypothesis $\mathcal{G}(T)$ is a closed subset of $X \times Y$, it follows that $(X, \| \ \|_1)$ is a Banach space. Moreover,

$$\|x\| \leq \|x\|_1 \qquad \forall\, x \in X.$$

Then, applying the previous corollary, there exists $M > 0$ such that

$$\|x\|_1 \leq M\|x\| \qquad \forall\, x \in X.$$

Therefore,

$$\|Tx\| \leq (M-1)\|x\| \qquad \forall\, x \in X.$$

Thus, T is continuous. □

2.8 The Weak Topology

We begin by recalling some results about the coarsest topology that makes a family of functions continuous. Let X be a set and let $\{Y_i \ : \ i \in I\}$ be a family of Hausdorff topological spaces. For each $i \in I$, let $f_i : X \to Y_i$. The coarsest topology on X that makes the functions f_i continuous is the topology $\mathcal{T}$ on X whose neighborhood base at a point $x \in X$ consists of the sets of the form

$$\bigcap_{j=1}^{m} f_{i_j}^{-1}(V_{i_j}),$$

where V_{i_j} is a neighborhood in Y_{i_j} of the point $f_{i_j}(x)$. Some properties of this topology that are of interest to us are the following:

(i) $x_n \to x$ with respect to the topology $\mathcal{T}$ if and only if $f_i(x_n) \to f_i(x)$ in Y_i for all $i \in I$.

(ii) If Z is a topological space and $g : Z \to X$, then g is continuous if and only if $f_i \circ g : Z \to Y_i$ is continuous for all $i \in I$.

Definition 2.7 Let X be a normed space. The *weak topology* $\sigma(X, X')$ on X is the coarsest topology on X that makes all the mappings $u \in X'$ continuous. From its very definition it follows that the weak topology is coarser than the norm topology.

Proposition 2.4 *The weak topology $\sigma(X, X')$ is Hausdorff.*

Proof Let $x_1, x_2 \in X$, with $x_1 \neq x_2$. We must find W_1, W_2, $\sigma(X, X')$-neighborhoods of x_i, such that $W_1 \cap W_2 = \emptyset$. For convenience, we assume that the space is real. By Theorem 2.13, there exists $u \in X'$ and $\alpha \in \mathbb{R}$ such that

$$\langle u, x_1 \rangle < \alpha < \langle u, x_2 \rangle.$$

Then, if we take

$$W_1 := \{x \in X \ : \ \langle u, x \rangle < \alpha\} = u^{-1}(]-\infty, \alpha[)$$

$$W_2 := \{x \in X \ : \ \langle u, x \rangle > \alpha\} = u^{-1}(]\alpha, +\infty[)$$

we have that W_1, W_2 are $\sigma(X, X')$-open sets, $x_i \in W_i$, and $W_1 \cap W_2 = \emptyset$. □

Proposition 2.5 *Let X be a normed space, and $x_0 \in X$. The sets of the form*

$$V(x_0, u_1, \ldots, u_n, \epsilon) := \{x \in X \ : \ |\langle u_i, x - x_0 \rangle| < \epsilon, \ i = 1, \ldots, n\}$$

with $u_i \in X'$ and $\epsilon > 0$ form a neighborhood base of x_0 for the weak topology $\sigma(X, X')$.

Proof We have

$$V(x_0, u_1, \ldots, u_n, \epsilon) = \bigcap_{i=1}^{n} u_i^{-1}\big(B(\langle u_i, x_0 \rangle, \epsilon)\big),$$

so each set $V(x_0, u_1, \ldots, u_n, \epsilon)$ is a $\sigma(X, X')$-neighborhood of x_0. On the other hand, if U is a $\sigma(X, X')$-neighborhood of x_0, there exist $u_i \in X'$, and W_i neighborhoods in $\mathbb{K}$ of $\langle u_i, x_0 \rangle$, such that

$$W \subset \bigcap_{i=1}^{n} u_i^{-1}(W_i) \subset U.$$

Now, we can find $\epsilon > 0$ such that

$$B(\langle u_i, x_0 \rangle, \epsilon) \subset W_i, \quad i = 1, \ldots, n.$$

Therefore, $V(x_0, u_1, \ldots, u_n, \epsilon) \subset W \subset U$. □

As an immediate consequence of the previous proposition, we have that:

U is a $\sigma(X, X')$-neighborhood of x_0 $\iff$ there exists a $\sigma(X, X')$-neighborhood W of 0 such that $U = x_0 + W$.

When a sequence $\{x_n\}$ converges to x with respect to the $\sigma(X, X')$ topology (i.e., when $\langle u, x_n\rangle \to \langle u, x\rangle$ for every $u \in X'$), we say that the sequence *converges weakly*, and we write $x_n \rightharpoonup x$ in $\sigma(X, X')$. Convergence with respect to the norm is usually referred to as *strong convergence*, i.e., $x_n \to x$ strongly when $\|x_n - x\| \to 0$. Clearly, if $x_n \to x$ strongly, then $x_n \rightharpoonup x$ in $\sigma(X, X')$. However, the converse is not true in general. In fact, consider $e_n := (0, \ldots, 0, 1, 0, \ldots)$. Since $\|e_n\|_2 = 1$ for all $n \in \mathbb{N}$, the sequence $\{e_n\}$ does not converge strongly to zero in ℓ^2. However, for each $x = (x_n) \in \ell^2$, we have

$$\langle e_n, x\rangle = x_n \to 0,$$

so $e_n \rightharpoonup 0$ in $\sigma(\ell^2, \ell^2)$.

Proposition 2.6 *Let X be a normed space and $\{x_n\} \subset X$. Then:*

(i) If $x_n \rightharpoonup x$ in $\sigma(X, X')$, then the sequence $\{x_n\}$ is bounded and

$$\|x\| \le \liminf_{n\to\infty} \|x_n\|.$$

(ii) If $x_n \rightharpoonup x$ in $\sigma(X, X')$ and $u_n \to u$ strongly in X', then

$$\langle u_n, x_n\rangle \to \langle u, x\rangle.$$

Proof

(i) By Corollary 2.8, it suffices to prove that for each $u \in X'$, the set $\{\langle u, x_n\rangle : n \in \mathbb{N}\}$ is bounded in $\mathbb{K}$. This is indeed the case, since for each $u \in X'$,

$$\langle u, x_n\rangle \to \langle u, x\rangle.$$

Moreover,

$$|\langle u, x_n\rangle| \le \|u\|\|x_n\| \quad \forall\, n \in \mathbb{N}.$$

Taking limits, we get

$$|\langle u, x\rangle| \le \|u\| \liminf_{n\to\infty} \|x_n\|.$$

Thus,

$$\|x\| = \sup_{\|u\|\le 1} |\langle u, x\rangle| \le \liminf_{n\to\infty} \|x_n\|.$$

(ii) Since

$$|\langle u_n, x_n\rangle - \langle u, x\rangle| \le |\langle u_n - u, x_n\rangle| + |\langle u, x_n - x\rangle|$$

$$\le \|u_n - u\|\,\|x_n\| + |\langle u, x_n - x\rangle|,$$

it suffices to apply (i) to obtain (ii).

□

It is not difficult to see that in finite-dimensional spaces the weak topology coincides with the norm topology. However, this is not true in infinite dimensions, as shown by the following result.

Proposition 2.7 *Let X be a normed space of infinite dimension. Then:*

(i) If W is a $\sigma(X, X')$-neighborhood of a point $x_0 \in X$, then there exists $y_0 \neq 0$ such that $x_0 + ty_0 \in W$ for every $t \in \mathbb{R}$.

(ii) The unit sphere $S(X) = \{x \in X : \|x\| = 1\}$ is not $\sigma(X, X')$-closed. More precisely,

$$\overline{S(X)}^{\sigma(X, X')} = U(X) = \{x \in X : \|x\| \le 1\}.$$

(iii) The open unit ball $B(0, 1)$ is not $\sigma(X, X')$-open. More precisely, its interior with respect to the weak topology is empty.

Proof

(i) Given the $\sigma(X, X')$-neighborhood of the point x_0, there exist $u_1, \ldots, u_n \in X'$ and $\epsilon > 0$ such that

$$V := V(x_0, u_1, \ldots, u_n, \epsilon) \subset W.$$

Since X is infinite-dimensional, there exists $y_0 \neq 0$ such that $\langle u_i, y_0\rangle = 0$ for all $i = 1, \ldots, n$, because otherwise the map $\phi : X \to \mathbb{R}^n$, defined by $\phi(x) := \big(\langle u_i, x\rangle\big)_{i=1}^n$, would be injective, and thus X would have dimension less than or equal to n. Then,

$$|\langle u_i, x_0 - (x_0 + ty_0)\rangle| = |\langle u_i, ty_0\rangle| = 0,$$

so $x_0 + ty_0 \in V \subset W$ for every $t \in \mathbb{R}$.

(ii) It suffices to show that $\{x \in X : \|x\| < 1\} \subset \overline{S(X)}^{\sigma(X, X')}$. Let $x_0 \in X$ with $\|x_0\| < 1$, and let W be a $\sigma(X, X')$-neighborhood of x_0. We show that $W \cap S(X) \neq \emptyset$. Indeed, by (i), there exists $y_0 \neq 0$ such that $x_0 + ty_0 \in W$ for all $t \in \mathbb{R}$. Then, since the function $f(t) := \|x_0 + ty_0\|$ is continuous, $f(0) = \|x_0\| < 1$ and $\lim_{t\to+\infty} f(t) = +\infty$, there exists t_0 such that $f(t_0) = 1$, that is, $\|x_0 + t_0y_0\| = 1$, so $x_0 + t_0y_0 \in W \cap S(X)$.

(iii) Suppose there exists a point x_0 in the $\sigma(X, X')$-interior of $B(0, 1)$. Then there exists $x_0 \in B(0, 1)$ and a $\sigma(X, X')$-neighborhood W of x_0 such that $W \subset B(0, 1)$. But by (i) this leads to a contradiction, since $B(0, 1)$ would contain a whole line.

□

Obviously, every weakly closed set is closed in the norm topology. As a consequence of the previous proposition, we have that the converse is not true in infinite-dimensional spaces. However, it is true for convex sets. More precisely, we have the following result.

Proposition 2.8 *Let X be a normed space and let $A \subset X$ be a convex set. Then,*

$$\overline{A}^{\|\ \|} = \overline{A}^{\sigma(X, X')}.$$

Proof For simplicity, we will assume that the space is real. Since the norm topology is stronger than the weak topology, it suffices to show that

$$\overline{A}^{\sigma(X, X')} \subset \overline{A}^{\|\ \|}.$$

Let $x_0 \in X \setminus \overline{A}^{\|\ \|}$. By Theorem 2.13, there exist $u \in X'$, $\epsilon > 0$, and $\alpha \in \mathbb{R}$ such that

$$\langle u, x\rangle \leq \alpha < \alpha + \epsilon \leq \langle u, x_0\rangle \quad \forall x \in \overline{A}^{\|\ \|}.$$

Therefore,

$$\overline{A}^{\|\ \|} \subset B := \{x \in X \ : \ \langle u, x\rangle \leq \alpha\}.$$

Now, since u is weakly continuous, B is weakly closed, so $\overline{A}^{\sigma(X,X')} \subset B$, and since $x_0 \notin B$, we conclude that $x_0 \notin \overline{A}^{\sigma(X,X')}$. □

Definition 2.8 Let X be a normed space and $A \subset X$. We say that A is $\sigma(X, X')$-*bounded* if for every $u \in X'$, the set $\{\langle u, x\rangle \ : \ x \in A\}$ is bounded in $\mathbb{K}$.

As a consequence of the Banach–Steinhaus Theorem (more specifically, as a consequence of Corollary 2.8), we have:

A is $\sigma(X, X')$-bounded if and only if A is $\|\ \|$-bounded.

Theorem 2.18 *Let X and Y be Banach spaces and $T \in L(X, Y)$. Then, $T \in \mathcal{L}(X, Y)$ if and only if T is continuous with respect to the topologies $\sigma(X, X')$ and $\sigma(Y, Y')$.*

Proof Suppose $T \in \mathcal{L}(X, Y)$. By the definition of the weak topology, to prove that T is weakly continuous it suffices to show that for every $u \in Y'$, the map $x \mapsto \langle u, Tx\rangle$ from $X[\sigma(X, X')]$ to $\mathbb{K}$ is continuous. But this map is a continuous linear form on X, hence it is continuous with respect to the topology $\sigma(X, X')$.

Conversely, suppose T is weakly continuous. Then its graph $\mathcal{G}(T)$ is $\sigma(X \times Y, X' \times Y')$-closed. Hence, by Proposition 2.8, $\mathcal{G}(T)$ is closed with respect to the norm topology. Therefore, by the Closed Graph Theorem, $T \in \mathcal{L}(X, Y)$. □

Remark 2.3 An argument similar to the one used in the proof of the previous theorem shows that if $T \in L(X, Y)$, with X and Y Banach spaces, and $T : (X, \|\ \|) \to (Y, \sigma(Y, Y'))$ is continuous, then $T : (X, \|\ \|) \to (Y, \|\ \|)$ is also continuous.

Definition 2.9 Let X be a normed space. Then X' is a Banach space with respect to the norm

$$\|u\| = \sup\{|\langle u, x\rangle| \ : \ \|x\| \leq 1\}.$$

The dual of this Banach space will be denoted by X'' and is called the *bidual* of X. Note that X'' is a Banach space with respect to the norm

$$\|\xi\| = \sup\{|\langle \xi, u\rangle| \ : \ u \in X', \ \|u\| \leq 1\}.$$

For each $x \in X$, we define $\hat{x} : X' \to \mathbb{K}$ by

$$\langle \hat{x}, u\rangle := \langle u, x\rangle.$$

Then we have

$$|\langle \hat{x}, u\rangle| = |\langle u, x\rangle| \leq \|u\|\ \|x\|,$$

which shows that $\hat{x} \in X''$. We can thus define the mapping $J : X \to X''$ as: $J(x) := \hat{x}$. Moreover,

$$\|J(x)\| = \sup\{|\langle J(x), u\rangle| \ : \ u \in X', \ \|u\| \leq 1\}$$

$$= \sup\{|\langle u, x\rangle| \ : \ u \in X', \ \|u\| \leq 1\} = \|x\|.$$

Therefore, J is an isometry. Through this isometry, we identify X as a subspace of its bidual X''.

On the space X', we have two defined topologies: the strong topology (i.e., the norm topology) and the weak topology $\sigma(X', X'')$. We can consider a third topology on X', called the *weak** topology, denoted by $\sigma(X', X)$, which is the coarsest topology on X' that makes all maps $\hat{x} : X' \to \mathbb{K}$ continuous for all $x \in X$.

Since $\hat{x} \in X''$ for all $x \in X$, it follows that the topology $\sigma(X', X)$ is coarser than the topology $\sigma(X', X'')$. Considering this topology is not arbitrary; its importance lies in the fact that, as we will see, with respect to this topology the unit ball of the dual is compact, and such compactness plays a major role in many applications.

With a proof similar to that of Propositions 2.4 and 2.5, we obtain the following result.

Proposition 2.9 *The topology $\sigma(X', X)$ is Hausdorff, and a neighborhood basis at a point $u_0 \in X'$ for this topology is formed by sets of the form:*

$$W(u_0, x_1, \ldots, x_n, \epsilon) := \{u \in X' \,:\, |\langle u - u_0, x_i\rangle| < \epsilon,\ i = 1, \ldots, n\},$$

with $x_i \in X$ and $\epsilon > 0$.

When a sequence $\{u_n\}$ converges with respect to the topology $\sigma(X', X)$ to some $u \in X'$ (i.e., when $\langle u_n, x\rangle \to \langle u, x\rangle$ for every $x \in X$), we will write: $u_n \rightharpoonup u$ in $\sigma(X', X)$. Evidently, we have that
$u_n \to u$ *strongly* $\Rightarrow u_n \rightharpoonup u$ *in* $\sigma(X', X'')$ $\Rightarrow u_n \rightharpoonup u$ *in* $\sigma(X', X)$.

A proof similar to that of Proposition 2.6 gives us the following result.

Proposition 2.10 *Let X be a normed space and $u_n \in X'$. Then:*

(i) If $u_n \rightharpoonup u$ in $\sigma(X', X)$, then the set $\{\|u_n\| : n \in \mathbb{N}\}$ is bounded and

$$\|u\| \le \liminf_{n\to\infty} \|u_n\|.$$

(ii) If $u_n \rightharpoonup u$ in $\sigma(X', X)$, and $x_n \to x$ strongly in X, then

$$\langle u_n, x_n\rangle \to \langle u, x\rangle.$$

Remark 2.4 In (ii) of the previous proposition, strong convergence is necessary, i.e., if $u_n \rightharpoonup u$ in $\sigma(X', X)$ (or even if $u_n \rightharpoonup u$ in $\sigma(X', X''))$ and $x_n \rightharpoonup x$ in $\sigma(X, X')$, we cannot conclude that $\langle u_n, x_n\rangle \to \langle u, x\rangle$. To see this, just consider $X = \ell^2$ (so that $X' = X'' = \ell^2$) and take $u_n = x_n = e_n = (0, \ldots, 0, 1, 0, \ldots)$. Then we have $e_n \rightharpoonup 0$ in $\sigma(\ell^2, \ell^2)$, yet $\langle e_n, e_n\rangle = 1$ for every $n \in \mathbb{N}$.

To prove that the topological dual of the space $X'[\sigma(X', X)]$ (i.e., the set of all continuous linear functionals on $X'[\sigma(X', X)]$) coincides with X, we need the following algebraic result.

Lemma 2.8 *Let X be a vector space and let $u, u_i \in X^*$, $i = 1, \ldots, n$, such that*

$$\langle u_i, x\rangle = 0,\ \forall\, i = 1, \ldots, n \ \Rightarrow\ \langle u, x\rangle = 0. \tag{2.33}$$

Then, there exist $\lambda_i \in \mathbb{K}$ such that $u = \sum_{i=1}^n \lambda_i u_i$.

Proof For convenience, we will assume that X is real. Let $\Psi : X \to \mathbb{R}^{n+1}$ be the map defined by

$$\Psi(x) := (\langle u, x\rangle, \langle u_1, x\rangle, \ldots, \langle u_n, x\rangle).$$

As a consequence of (2.33), we have that $y = (1, 0, \ldots, 0) \notin \Psi(X)$. Now, since $\Psi(X)$ is closed (being a finite-dimensional vector subspace), applying Theorem 2.13, we get that $\{y\}$ and $\Psi(X)$ are strictly separated. Therefore, there exist $\alpha, \alpha_1, \alpha_2, \ldots, \alpha_n \in \mathbb{R}$, and a $\beta \in \mathbb{R}$ such that

$$\alpha < \beta < \alpha \langle u, x \rangle + \sum_{i=1}^{n} \alpha_i \langle u_i, x \rangle \quad \forall\, x \in X.$$

Hence,

$$\alpha \langle u, x \rangle + \sum_{i=1}^{n} \alpha_i \langle u_i, x \rangle = 0 \quad \forall\, x \in X \quad \text{and} \quad \alpha < 0.$$

Thus,

$$u = \sum_{i=1}^{n} \left(-\frac{\alpha_i}{\alpha} \right) u_i.$$

□

Theorem 2.19 *Let $\Phi : X' \to \mathbb{K}$ be a linear map that is continuous with respect to the topology $\sigma(X', X)$. Then, there exists an element $x \in X$ such that $\Phi = \hat{x}$. Therefore,*

$$\left(X'[\sigma(X', X)] \right)' = X.$$

Proof Since Φ is linear and $\sigma(X', X)$-continuous, there exists a $\sigma(X', X)$-neighborhood W of 0 such that

$$|\Phi(u)| < 1 \quad \forall\, u \in W.$$

Now, given W, there are $x_1, \ldots, x_n \in X$ and some $\epsilon > 0$ such that

$$W(0, x_1, \ldots, x_n, \epsilon) \subset W.$$

Therefore, if $\langle u, x_i \rangle = 0$ for all $i = 1, \ldots, n$, then $\Phi(u) = 0$. Hence, by applying the previous lemma, there exist $\alpha_1, \alpha_2, \ldots, \alpha_n \in \mathbb{K}$ such that

$$\Phi(u) = \sum_{i=1} \alpha_i \langle u, x_i \rangle = \left\langle u, \sum_{i=1}^{n} \alpha_i x_i \right\rangle \quad \forall\, u \in X'.$$

Therefore, it suffices to take $x = \sum_{i=1}^{n} \alpha_i x_i$. □

Corollary 2.12 *If H is a hyperplane of X' that is closed in the topology $\sigma(X', X)$, then there exists $x \in X$, $x \neq 0$, and $\alpha \in \mathbb{K}$ such that*

$$H = \{u \in X' \ : \ \langle u, x\rangle = \alpha\}.$$

Proof We know there exists a non-zero linear map $\Phi : X' \to \mathbb{K}$ and some $\alpha \in \mathbb{R}$ such that

$$H = \{u \in X' \ : \ \Phi(u) = \alpha\}.$$

Then, by the previous theorem, it will be enough to show that Φ is $\sigma(X', X)$-continuous.

Let $u_0 \notin H$. Since H is $\sigma(X', X)$-closed, there exists a $\sigma(X', X)$-neighborhood W of u_0 such that $W \subset X' \setminus H$. We can assume that there exist $x_1, \ldots, x_n \in X$ and $\epsilon > 0$ such that

$$W = \{u \in X' \ : \ |\langle u - u_0, x_i\rangle| < \epsilon\}.$$

Now, since W is convex, we must have either:

$$\Phi(u) < \alpha \qquad \forall\, u \in W, \tag{2.34}$$

or

$$\Phi(u) > \alpha \qquad \forall\, u \in W. \tag{2.35}$$

If (2.34) holds, then we have

$$\Phi(v) < \alpha - \Phi(u_0) \qquad \forall\, v \in V := W - u_0,$$

and since $-V = V$, we will have

$$|\Phi(v)| \leq |\alpha - \Phi(u_0)| \qquad \forall\, v \in V. \tag{2.36}$$

We would reach the same conclusion under assumption (2.35). Now, from (2.36) it follows that, given $r > 0$,

$$\{u \in X' \ : \ |\Phi(u)| \leq r\} =$$

$$\frac{r}{|\alpha - \Phi(u_0)|}\{u \in X' \ : \ |\Phi(u)| \leq |\alpha - \Phi(u_0)|\} \supset \frac{r}{|\alpha - \Phi(u_0)|}V,$$

and since $\dfrac{r}{|\alpha - \Phi(u_0)|} V$ is a $\sigma(X', X)$-neighborhood of zero, we conclude that Φ is $\sigma(X', X)$-continuous at zero. Since Φ is linear, it is therefore $\sigma(X', X)$-continuous on all of X'. □

As a consequence of Theorem 2.3, we have that in a normed space of infinite dimension, its closed unit ball is not strongly compact (i.e., compact with respect to the norm topology). However, with respect to the weak* topology $\sigma(X', X)$, we have the following result.

Theorem 2.20 (Alaoglu's Theorem) *Let X be a normed space. The unit ball of the dual*

$$U(X') = \{u \in X' \ : \ \|u\| \leq 1\}$$

is $\sigma(X', X)$-compact.

Proof For each $x \in U(X)$, let $D_x := \{\alpha \in \mathbb{K} \ : \ |\alpha| \leq 1\}$ and let D be the product space

$$D := \prod_{x \in U(X)} D_x.$$

By Tychonoff's Theorem, D is compact. We define $\Phi : U(X') \to D$ as

$$\Phi(u)(x) := \langle u, x \rangle, \qquad x \in U(X).$$

We will show that Φ is a homeomorphism from $U(X')[\sigma(X', X)]$ onto $\Phi(U(X'))$, and that $\Phi(U(X'))$ is closed, so $U(X')[\sigma(X', X)]$ will be compact.

Clearly, Φ is injective. Let $\{u_i\}$ be a net in $U(X')$ such that $u_i \rightharpoonup u$ in $\sigma(X', X)$. Then, for each $x \in U(X)$,

$$\Phi(u_i)(x) = \langle u_i, x \rangle \to \langle u, x \rangle = \Phi(u)(x),$$

so $\Phi(u_i) \to \Phi(u)$ in D, and thus Φ is continuous. Finally, we show that $\Phi(U(X'))$ is closed in D. Let $u_i \in U(X')$, and suppose that $\Phi(u_i) \to \hat{u}$ in D. Then, for each $x \in U(X)$ there exists

$$u(x) := \lim_i \Phi(u_i)(x) = \lim_i \langle u_i, x \rangle.$$

Given any $x \in X$, let $\alpha > 0$ such that $\|\alpha x\| \leq 1$. We define $u(x) := \frac{1}{\alpha} u(\alpha x)$. Now, if $\beta > 0$ is such that $\|\beta x\| \leq 1$, we have

$$\frac{1}{\alpha} u(\alpha x) = \frac{1}{\alpha} \lim_i \langle u_i, \alpha x \rangle = \frac{1}{\beta} \lim_i \langle u_i, \beta x \rangle = \frac{1}{\beta} u(\beta x).$$

Therefore, u is well defined on all of X. It is easy to see that u is continuous. Moreover, if $\|x\| \leq 1$, then $u(x) \in D_x$, so $u \in U(X')$, and we have $\Phi(u) = \hat{u}$. □

2.9 Reflexive Spaces

Let X be a normed space and $J : X \to X''$ the natural embedding, i.e., $J(x) = \hat{x}$, with $\langle \hat{x}, u \rangle = \langle u, x \rangle$ for each $u \in X'$.

The space X is said to be *reflexive* if $J(X) = X''$.

Remark 2.5 Since a reflexive space is isometrically isomorphic to its bidual, it will be a Banach space. However, for a space to be reflexive, it is not enough that it be isometrically isomorphic to its bidual—it must be so via the natural embedding J. In fact, in 1951, R. C. James ([30]) constructed a non-reflexive Banach space that is isometrically isomorphic to its bidual.

Theorem 2.21 *Let $(\Omega, \mathcal{B}, \mu)$ be a σ-finite measure space. For each $1 < p < \infty$, the space $L^p(\Omega, \mu)$ is reflexive. In particular, every ℓ^p space is reflexive.*

Proof Given an element $\psi \in \left(L^p(\Omega, \mu)\right)''$, we must find a function $f \in L^p(\Omega, \mu)$ such that $J(f) = \psi$. Indeed, by the Riesz Representation Theorem, the following maps Φ_i are isometries:

$$\Phi_1 : L^{p'}(\Omega, \mu) \to \left(L^p(\Omega, \mu)\right)',$$
$$\text{defined by} \quad \langle \Phi_1(g), f \rangle := \int_\Omega fg \, d\mu \quad \forall f \in L^p(\Omega, \mu),$$

$$\Phi_2 : L^p(\Omega, \mu) \to \left(L^{p'}(\Omega, \mu)\right)',$$
$$\text{defined by} \quad \langle \Phi_2(f), g \rangle := \int_\Omega fg \, d\mu \quad \forall g \in L^{p'}(\Omega, \mu).$$

Then, since $\psi \circ \Phi_1 \in \left(L^{p'}(\Omega, \mu)\right)'$, there exists a function $f \in L^p(\Omega, \mu)$ such that $\psi \circ \Phi_1 = \Phi_2(f)$. Therefore, if $h \in \left(L^p(\Omega, \mu)\right)'$, there exists $g \in L^{p'}(\Omega, \mu)$ such that $h = \Phi_1(g)$, and hence

$$\langle \psi, h \rangle = \langle \psi, \Phi_1(g) \rangle = \langle \Phi_2(f), g \rangle = \int_\Omega fg \, d\mu = \langle f, \Phi_1(g) \rangle = \langle J(f), h \rangle.$$

□

The space $L^1(\Omega, \mu)$ is not reflexive, unless it is finite-dimensional (see Problem 2.48). Another typical example of a non-reflexive Banach space is the space c_0 (see Problem 2.46).

To provide a characterization of reflexivity, we need the following result.

Lemma 2.9 *Let X be a normed space. Then $J(U(X))$ is $\sigma(X'', X')$-dense in $U(X'')$.*

Proof Let B be the $\sigma(X'', X')$-closure of $J(U(X))$. Clearly, $U(X'') \subset B$. Suppose there exists $\phi \in U(X'') \setminus B$. By applying the Hahn–Banach Separation Theorem (in the Banach space X''), there exist $f \in (X'')'$, $\alpha \in \mathbb{R}$, and $\epsilon > 0$ such that

$$\Re\langle f, x\rangle < \alpha < \alpha + \epsilon < \Re\langle f, \phi\rangle \quad \forall x \in U(X).$$

Since $0 \in U(X)$, it follows that $\alpha > 0$. Then, dividing by α and replacing f with $\frac{1}{\alpha} f$, we may assume that there exists $f \in (X'')'$ and $\epsilon > 0$ such that

$$\Re\langle f, x\rangle < 1 < 1 + \epsilon < \Re\langle f, \phi\rangle \quad \forall x \in U(X).$$

Now, since $e^{i\theta} x \in U(X)$ for all $x \in U(X)$, we get

$$|\langle f, x\rangle| \leq 1 \quad \forall x \in U(X),$$

so $f \in U(X')$. Therefore,

$$1 + \epsilon < \Re\langle f, \phi\rangle \leq |\langle f, \phi\rangle| \leq \|\phi\| \leq 1,$$

which gives a contradiction. □

We have the following characterization of reflexive spaces.

Theorem 2.22 *Let X be a Banach space. The following statements are equivalent:*

(i) *X is reflexive.*
(ii) *X' is reflexive.*
(iii) *$\sigma(X', X) = \sigma(X', X'')$.*
(iv) *$U(X)$ is $\sigma(X, X')$-compact.*

Proof Clearly, (i) implies (iii).
(iv) implies (i): Since the restriction of the topology $\sigma(X'', X')$ is the topology $\sigma(X, X')$, from (iv) we have that $J(U(X))$ is $\sigma(X'', X')$-closed in $U(X'')$. Then, as a consequence of the previous lemma, we have that $J(U(X)) = U(X'')$, hence $J(X) = X''$. Therefore, X is reflexive.
(iii) implies (ii): By Alaoglu's theorem, $U(X')$ is $\sigma(X', X)$-compact, and since (iii) holds, $U(X')$ is also $\sigma(X', X'')$-compact. Now, as we have already shown that (iv) implies (i), we conclude that X' is reflexive.
(ii) implies (i): Since J is an isometry, we have that $J(U(X))$ is strongly closed in X'', hence it is $\sigma(X'', X''')$-closed. Then, by (ii), it is $\sigma(X'', X')$-closed. Therefore, as a consequence of the previous lemma, we have that $J(U(X)) = U(X'')$, and thus X is reflexive.

(i) implies (iv): As a consequence of Alaoglu's Theorem, we have that $U(X'')$ is $\sigma(X'', X')$-compact, and then by (i), $U(X)$ is $\sigma(X, X')$-compact. □

Corollary 2.13 *Every convex, closed, and bounded subset A of a reflexive Banach space X is $\sigma(X, X')$-compact.*

Proof Since A is convex, by Theorem 2.8, A is $\sigma(X, X')$-closed. On the other hand, since A is bounded, there exists $\lambda > 0$ such that $A \subset \lambda U(X)$, and since by the previous theorem $\lambda U(X)$ is $\sigma(X, X')$-compact, we conclude that A is also $\sigma(X, X')$-compact. □

Corollary 2.14 *Every closed subspace of a reflexive Banach space is reflexive.*

Proof Let X be a reflexive Banach space and Y a closed subspace of X. It is not difficult to see that the topology $\sigma(Y, Y')$ coincides with the restriction to Y of the topology $\sigma(X, X')$. Therefore, by the previous theorem, $U(Y) = U(X) \cap Y$ will be $\sigma(Y, Y')$-compact. Again, applying the previous theorem, we conclude that Y is reflexive. □

Definition 2.10 Let $\{x_n\}$ be a sequence in the normed space X. It is said to be a *weak Cauchy sequence* in X if for every $u \in X'$ the sequence $\{\langle u, x_n\rangle\}$ is a Cauchy sequence in $\mathbb{K}$. A normed space in which every weak Cauchy sequence is weakly convergent is called a *weakly sequentially complete space*.

Corollary 2.15 *Every reflexive Banach space is weakly sequentially complete.*

Proof Let X be a reflexive space, and let $\{x_n\}$ be a weak Cauchy sequence in X. Since $\{x_n : n \in \mathbb{N}\}$ is weakly bounded, by the Banach–Steinhaus Theorem we have that $\{x_n : n \in \mathbb{N}\}$ is strongly bounded. Then, since X is reflexive, this set is $\sigma(X, X')$-compact. Hence, there exists $x \in X$ that is a weak limit point of the sequence $\{x_n\}$. Therefore, since $\{x_n\}$ is a weak Cauchy sequence in X, it follows that $x_n \rightharpoonup x$. □

Remark 2.6 In general, the converse of the previous result is not true. For example, the space ℓ^1, which is not reflexive, is weakly sequentially complete. For a proof, see for instance [36], p. 281.

Definition 2.11 Let X be a topological space and $\varphi : X \to \mathbb{R}$. We say that φ is *lower semicontinuous* on X if for every $x \in X$,

$$\varphi(x) \leq \liminf_{y \to x} \varphi(y).$$

Proposition 2.11 *Let X be a compact space and $\varphi : X \to \mathbb{R}$ a lower semicontinuous function on X. Then φ attains its infimum on X, i.e., there exists $x_0 \in X$ such that*

$$\varphi(x_0) = \inf_{x \in X} \varphi(x).$$

Proof Let $m := \inf_{x \in X} \varphi(x)$. For $\lambda > 0$, define

$$X_\lambda := \{x \in X \ : \ \varphi(x) \leq \lambda\}.$$

Since φ is lower semicontinuous, it is easy to see that X_λ is closed in X, and hence compact. Moreover, $X_\lambda \subset X_\mu$ if $m < \lambda \leq \mu$. Therefore, the family of compact sets $\{X_\lambda \ : \ \lambda > m\}$ has the finite intersection property. Thus, there exists $x_0 \in X_\lambda$ for every $\lambda > m$, and hence, $\varphi(x_0) \leq \lambda > m$, and therefore,

$$\varphi(x_0) = \inf_{x \in X} \varphi(x).$$

□

Theorem 2.23 *Let X be a reflexive Banach space and let Y be a closed subspace of X. Then, given any $x_0 \in X \setminus Y$, there exists $y_0 \in Y$ such that*

$$\|x_0 - y_0\| = dist(x_0, Y) = \inf\{\|x_0 - y\| \ : \ y \in Y\}.$$

Proof As a consequence of Proposition 2.6, we have that the function $\varphi : X \to \mathbb{R}$ defined by

$$\varphi(x) := \|x_0 - x\|,$$

is lower semicontinuous with respect to the $\sigma(X, X')$ topology. Now, if we let $d := \text{dist}(x_0, Y)$, then, since X is reflexive,

$$K := Y \cap \{x \in X \ : \ \|x_0 - x\| \leq 2d\}$$

is weakly compact. Therefore, as a consequence of the previous proposition, there exists $y_0 \in K$ such that

$$\varphi(y_0) = \inf_{y \in K} \varphi(y).$$

Hence, $y_0 \in Y$, and moreover,

$$\|x_0 - y_0\| = \text{dist}(x_0, Y) = \inf\{\|x_0 - y\| \ : \ y \in Y\}.$$

□

Remark 2.7 The previous result is not true if X is not reflexive. In fact, a result of R. C. James [31] states that:

A Banach space X is reflexive if and only if, for every closed hyperplane H of X and every $x \in X$, there exists $y \in H$ such that

$$\|x - y\| = \text{dist}(x, H).$$

2.10 Separable Spaces

Recall that a metric space is *separable* if it has a countable dense subset, or equivalently, if it has a countable basis of open sets. In the case of normed spaces, we have the following characterization.

Proposition 2.12 *A normed space X is separable if and only if there exists a countable set whose linear span is dense in X.*

Proof It suffices to prove that if there exists $\{x_n : n \in \mathbb{N}\} \subset X$ such that $\overline{Y} = X$, with $Y := \text{LIN}\{x_n : n \in \mathbb{N}\}$, then X is separable. Indeed, we may assume, without loss of generality, that X is real. Let D be the set of elements of Y of the form $\lambda_1 x_1 + \cdots + \lambda_n x_n$, with $\lambda_i \in \mathbb{Q}$. Then D is countable. We now show that D is dense in X, hence X is separable.

Given $x \in X$ and $\epsilon > 0$, since Y is dense in X, there exist $\alpha_i \in \mathbb{R}$ such that

$$\|x - (\alpha_1 x_1 + \cdots + \alpha_n x_n)\| \leq \frac{\epsilon}{2}.$$

Now, for each $i = 1, 2, \ldots, n$, there exists $\lambda_i \in \mathbb{Q}$ such that

$$|\lambda_i - \alpha_i| \leq \frac{\epsilon}{2n\|x_i\|}.$$

Therefore,

$$\|(\lambda_1 x_1 + \cdots + \lambda_n x_n) - (\alpha_1 x_1 + \cdots + \alpha_n x_n)\| \leq \frac{\epsilon}{2},$$

so that $\|x - v\| \leq \epsilon$, and thus D is dense in X. □

Example 2.4

(i) For each $1 \leq p < \infty$, the space ℓ^p is separable. Indeed, let

$$\varphi := \text{LIN}\{e_n : n \in \mathbb{N}\}.$$

Given $x = (x_n) \in \ell^p$, define $x^n := \sum_{i=1}^n x_i e_i \in \varphi$, and note that

$$\lim_{n\to\infty} \|x - x^n\|_p^p = \sum_{i=n+1}^{\infty} |x_i|^p = 0,$$

hence φ is dense in ℓ^p, and by the previous proposition, this space is separable.

(ii) The space ℓ^∞ is not separable. In fact,, let $\mathcal{A} := \{A \ : \ A \subset \mathbb{N}\}$. For each $A \in \mathcal{A}$, define

$$\chi_A(n) := \begin{cases} 1 & \text{if } n \in A \\ \\ 0 & \text{if } n \notin A \end{cases}$$

Since $\chi_A \in \ell^\infty$, and if $A, B \in \mathcal{A}$ with $A \neq B$, then

$$\|\chi_A - \chi_B\|_\infty = 1,$$

we conclude that ℓ^∞ does not have a countable base of open sets, and thus is not separable.

(iii) The space $C([a, b])$ is separable. This follows from the classical Weierstrass Approximation Theorem, which states that polynomials are dense in $C([a, b])$.

Proposition 2.13 *Let X be a normed space. If X' is separable, then X is separable.*

Proof By hypothesis, there exist $u_n \in X'$ such that $\overline{\{u_n \ : \ n \in \mathbb{N}\}} = X'$. Now, since

$$\|u_n\| = \sup\{\langle u_n, x\rangle \ : \ x \in U(X)\},$$

there exist $x_n \in X$, with $\|x_n\| = 1$, such that

$$\langle u_n, x_n\rangle \geq \frac{1}{2}\|u_n\| \quad \forall\, n \in \mathbb{N}.$$

Let us show that $Y := \text{LIN}\{x_n \ : \ n \in \mathbb{N}\}$ is dense in X, which implies that X is separable. By Corollary 2.5, it suffices to show that if $u \in X'$ is such that $\langle u, x\rangle = 0$ for every $x \in Y$, then $u = 0$. Indeed: given $\epsilon > 0$, there exists $n \in \mathbb{N}$ such that $\|u - u_n\| < \frac{\epsilon}{3}$. Then, since $\langle u, x_n\rangle = 0$, we have

$$\frac{1}{2}\|u_n\| \leq \langle u_n, x_n\rangle = \langle u_n - u, x_n\rangle + \langle u, x_n\rangle = \langle u_n - u, x_n\rangle \leq \|u_n - u\| < \frac{\epsilon}{3}.$$

Thus,

$$\|u\| \leq \|u - u_n\| + \|u_n\| < \epsilon.$$

Consequently, $u = 0$. □

In general, the converse of the previous result is not true, since ℓ^∞ is not separable, whereas ℓ^1 is.

The separability of a space is closely related to the metrizability of the weak* topology, as the following result shows.

Theorem 2.24 *Let X be a Banach space. Then X is separable if and only if $U(X')$ is $\sigma(X', X)$-metrizable.*

Proof Suppose X is separable. Then, there exist $x_n \in U(X)$ such that $\overline{\{x_n : n \in \mathbb{N}\}} = X$. We define a metric d on $U(X')$ as:

$$d(u, v) := \sum_{n=1}^{\infty} \frac{1}{2^n} |\langle u - v, x_n \rangle|.$$

Let us show that the topology defined by this metric coincides on $U(X')$ with the $\sigma(X', X)$ topology. Let $u_0 \in U(X')$ and let U be a $\sigma(X', X)$-neighborhood of u_0. We will show that there exists $\delta > 0$ such that

$$V := \{u \in U(X') : d(u, u_0) < \delta\} \subset U.$$

Since U is a $\sigma(X', X)$-neighborhood of u_0, there exist $y_1, \ldots, y_k \in U(X)$ and $\epsilon > 0$ such that

$$W := \{u \in U(X') : |\langle u - u_0, y_i \rangle| < \epsilon,\ i = 1, \ldots, k\} \subset U.$$

Since $\{x_n : n \in \mathbb{N}\}$ is dense in $U(X)$, there exist $n_i \in \mathbb{N}$ such that

$$\|y_i - x_{n_i}\| < \frac{\epsilon}{4}, \quad \text{for all } i = 1, \ldots, k.$$

Choose $\delta > 0$ such that

$$\delta < \frac{\epsilon}{2^{n_i+1}}, \quad \text{for all } i = 1, \ldots, k.$$

Then, if $d(u, u_0) < \delta$, we have

$$\frac{1}{2^{n_i}} |\langle u - u_0, x_{n_i} \rangle| < \delta, \quad \text{for all } i = 1, \ldots, k.$$

So, for each $i = 1, \ldots, k$,

$$\begin{aligned} |\langle u - u_0, y_i \rangle| &\le |\langle u - u_0, y_i - x_{n_i} \rangle| + |\langle u - u_0, x_{n_i} \rangle| \\ &\le \|u - u_0\| \cdot \|y_i - x_{n_i}\| + 2^{n_i} \delta \\ &\le 2 \cdot \frac{\epsilon}{4} + \frac{\epsilon}{2} = \epsilon. \end{aligned}$$

Hence, $u \in W$, and therefore $V \subset U$.

Now, let $u_0 \in U(X')$, $\delta > 0$, and let us show that there exists a $\sigma(X', X)$-neighborhood U of u_0 in $U(X')$ such that

$$U \subset V := \{u \in U(X') \,:\, d(u, u_0) < \delta\}.$$

Choose $\epsilon > 0$ with $\epsilon < \delta/2$ and $k \in \mathbb{N}$ such that $1/2^{k-1} < \delta/2$. Then, if

$$u \in U := \{u \in U(X') \,:\, |\langle u - u_0, x_i\rangle| < \epsilon,\ i = 1, \ldots, k\},$$

we have

$$\begin{aligned} d(u, u_0) &= \sum_{n=1}^{\infty} \frac{1}{2^n} |\langle u - u_0, x_n\rangle| \\ &= \sum_{n=1}^{k} \frac{1}{2^n} |\langle u - u_0, x_n\rangle| + \sum_{n=k+1}^{\infty} \frac{1}{2^n} |\langle u - u_0, x_n\rangle| \\ &< \epsilon + 2 \sum_{n=k+1}^{\infty} \frac{1}{2^n} = \epsilon + \frac{1}{2^{k-1}} < \frac{\delta}{2} + \frac{\delta}{2} = \delta. \end{aligned}$$

Hence, $u \in V$.

Conversely, suppose now that $U(X')$ is $\sigma(X', X)$-metrizable, and let us show that X is separable. Let d be the metric defining the $\sigma(X', X)$ topology on $U(X')$. Then, for each $n \in \mathbb{N}$,

$$U_n = \left\{u \in U(X') \,:\, d(u, 0) < \frac{1}{n}\right\},$$

there exists a finite set $A_n \subset X$ and $\epsilon_n > 0$ such that

$$V_n := \{u \in U(X') \,:\, |\langle u, x\rangle| < \epsilon_n,\ \forall\, x \in A_n\} \subset U_n.$$

Now, the set $A := \cup_{n\in\mathbb{N}} A_n$ is countable, and moreover,

$$\bigcap_{n\in\mathbb{N}} V_n \subset \bigcap_{n\in\mathbb{N}} U_n = \{0\}.$$

Therefore, if $u \in X'$ is such that $\langle u, x\rangle = 0$ for all $x \in A$, then $u = 0$. Consequently, A is dense in X, and hence X is separable. □

Corollary 2.16 *Let X be a separable Banach space. Then, every bounded sequence in X' has a $\sigma(X', X)$-convergent subsequence.*

Proof Let $\{u_n\} \subset X'$ be a bounded sequence. Then there exists $\rho > 0$ such that $\rho u_n \in U(X')$ for every $n \in \mathbb{N}$. Now, by Alaoglu's Theorem, $U(X')$ is $\sigma(X', X)$-

compact, and since by the previous theorem $U(X')$ is $\sigma(X', X)$-metrizable, we have that $U(X')$ is $\sigma(X', X)$-sequentially compact. Consequently, the sequence $\{u_n\}$ has a $\sigma(X', X)$-convergent subsequence. □

Theorem 2.25 *Let X be a reflexive Banach space. Then, every bounded sequence in X has a $\sigma(X, X')$-convergent subsequence.*

Proof Let $\{x_n\}$ be a bounded sequence in X. Define $Y := \overline{\text{LIN}\{x_n : n \in \mathbb{N}\}}$. Then Y is a reflexive and separable Banach space, and by Proposition 2.13, Y' is also separable. Hence, as a consequence of Theorem 2.24, we have that $U(Y)$ is $\sigma(Y, Y')$-metrizable. Therefore, since $U(Y)$ is also $\sigma(Y, Y')$-compact (because Y is reflexive), we can extract a $\sigma(Y, Y')$-convergent subsequence from $\{x_n\}$, which is evidently also $\sigma(X, X')$-convergent. □

In Theorem 2.24, we saw that the separability of a Banach space implies the metrizability of the unit ball of its dual with respect to the weak* topology. It is then natural to ask whether the separability of a space implies the metrizability of its unit ball with respect to the weak topology. The answer to this question is negative. To provide a counterexample, we need the following result, which is interesting in itself.

Theorem 2.26 (Schur's Theorem) *Every weakly convergent sequence in ℓ^1 is norm convergent.*

Proof Let $\{x^n\}$ be a sequence in ℓ^1 such that $x^n \rightharpoonup 0$ in $\sigma(\ell^1, \ell^\infty)$. Given $\epsilon > 0$, we need to find n_0 such that $\|x^n\|_1 \leq \epsilon$ for all $n \geq n_0$. For each $m \in \mathbb{N}$, define

$$F_m := \left\{\phi \in U(\ell^\infty) : \left|\langle x^n, \phi\rangle\right| \leq \frac{\epsilon}{3} \quad \forall n \geq m\right\}.$$

It is easy to see that each F_m is $\sigma(\ell^\infty, \ell^1)$-closed, and since $x^n \rightharpoonup 0$ in $\sigma(\ell^1, \ell^\infty)$, we have

$$U(\ell^\infty) = \bigcup_{m\in\mathbb{N}} F_m.$$

Now, since ℓ^1 is separable, by Theorem 2.24, $U(\ell^\infty)$ is $\sigma(\ell^\infty, \ell^1)$-metrizable. Moreover, this metric is given by

$$d(\phi, \psi) = \sum_{j=1}^{\infty} \frac{1}{2^j}|\phi_j - \psi_j|.$$

Then, by Alaoglu's Theorem, $U(\ell^\infty)$ is $\sigma(\ell^\infty, \ell^1)$-compact, so it is a complete metric space with respect to the topology $\sigma(\ell^\infty, \ell^1)$. Therefore, applying Baire's Theorem, there exists $m_0 \in \mathbb{N}$ such that $\text{int}(F_{m_0}) \neq \emptyset$, where the interior is taken

with respect to the topology $\sigma(\ell^\infty, \ell^1)$. Thus, there exist $\phi \in F_{m_0}$ and $\delta > 0$ such that

$$\{\psi \in U(\ell^\infty) : d(\phi, \psi) < \delta\} \subset F_{m_0}.$$

Let $j_0 \in \mathbb{N}$ such that $\frac{1}{2^{j_0-1}} < \delta$. For $n \geq m_0$, define $\psi^n \in \ell^\infty$ by

$$\psi_j^n := \begin{cases} \phi_j & \text{if } 1 \leq j < j_0, \\ \\ \text{sign}(x_j^n) & \text{if } j \geq j_0. \end{cases}$$

We have that $\psi^n \in U(\ell^\infty)$, $\psi_j^n x_j^n = |x_j^n|$ if $j \geq j_0$, and

$$d(\phi, \psi^n) = \sum_{j=j_0+1}^{\infty} \frac{1}{2^j} |\phi_j - \psi_j^n| \leq 2 \cdot \frac{1}{2^{j_0}} < \delta.$$

So $\psi^n \in F_{m_0}$, hence

$$|\langle \psi^n, x^n \rangle| \leq \frac{\epsilon}{3} \quad \forall n \geq m_0.$$

This is equivalent to

$$\left| \sum_{j=1}^{j_0} \psi_j^n x_j^n + \sum_{j=j_0+1}^{\infty} |x_j^n| \right| \leq \frac{\epsilon}{3} \quad \forall n \geq m_0. \tag{2.37}$$

On the other hand, since $x^n \rightharpoonup 0$ in $\sigma(\ell^1, \ell^\infty)$, taking functionals $e_n = (0, \ldots, 0, 1, 0, \ldots)$, it is easy to see that there exists $n_0 \geq m_0$ such that

$$\sum_{j=1}^{j_0} |x_j^n| < \frac{\epsilon}{3} \quad \forall n \geq n_0. \tag{2.38}$$

Finally, as a consequence of (2.37) and (2.38), we have $\|x^n\|_1 < \epsilon$ for all $n \geq n_0$.

□

Example 2.5 The space ℓ^1 is separable. However, $U(\ell^1)$ is not $\sigma(\ell^1, \ell^\infty)$-metrizable, because if it were, then by Schur's Theorem, the strong and weak topologies would coincide on $U(\ell^1)$, which contradicts Proposition 2.7.

Problems

2.1 Prove that the closure of a vector subspace of a normed space is a vector subspace.

2.2 Prove that the subspaces of ℓ^∞

$$c := \{(a_n),\ a_n \in \mathbb{K}\ :\ (a_n) \text{ is convergent}\}$$

and

$$c_0 := \{(a_n),\ a_n \in \mathbb{K}\ :\ \lim_{n\to\infty} a_n = 0\}$$

are $\|\ \|_\infty$-closed, and therefore they are Banach spaces with respect to the norm $\|\ \|_\infty$.

2.3 Let Ω be a locally compact topological space. Let $f : \Omega \to \mathbb{K}$ be a continuous function. We say that f *vanishes at infinity* if for every $\epsilon > 0$, the set $\{x \in \Omega\ :\ |f(x)| \geq \epsilon\}$ is compact. Prove that the space $C_\infty(\Omega)$ of all continuous functions that vanish at infinity is a Banach space endowed with the uniform norm $\|\ \|_\infty$.

2.4 Let $(X, \|\ \|)$ be a Banach space and let Y be a closed vector subspace of X. Prove that the quotient space $\frac{X}{Y}$ endowed with the norm

$$\|\hat{x}\| := \inf\{\|x\|\ :\ x \in \hat{x}\}$$

is a Banach space.

2.5 Let X be a normed space and Y and Z two closed subspaces of X. Prove that if Z is finite-dimensional, then $Y + Z$ is closed. Is the result still true if Z is not finite-dimensional?

2.6 Prove that $C^1([-1, 1])$, endowed with the uniform norm $\|\ \|_\infty$, is a normed space but not a Banach space, *i.e.*, it is not complete. Prove that the space $C^1([-1, 1])$ is a Banach space with respect to the norm

$$\|f\|^*_\infty := \sup_{t\in[a,b]} |f(t)| + \sup_{t\in[a,b]} |f'(t)|.$$

2.7 Riesz's Lemma tells us that if $(X, \|\ \|)$ is a normed space and X_0 is a closed proper subspace of X, then for every $0 < \theta < 1$, there exists an element $x_\theta \in X$ such that $\|x_\theta\| = 1$ and $\|x - x_\theta\| \geq \theta$ for all $x \in X_0$. Prove that if X is finite-dimensional, then one can take $\theta = 1$, but that this is not true in general when the space is infinite-dimensional.

2.8 Let X be a Banach space and $\{u_n\} \subset X'$. Suppose there exists a constant $M > 0$ such that

$$|\langle u_n, x\rangle| \leq M\|x\| \quad \forall\, n \in \mathbb{N}, \ \ \forall\, x \in X.$$

Prove that the set

$$S := \left\{x \in X \ : \ \exists \lim_{n\to\infty} \langle u_n, x\rangle \in X\right\}$$

is closed in X.

2.9 Let $f \in C([a, b])$, and let $T : C([a, b]) \to \mathbb{K}$ be defined by

$$\langle T, g\rangle := \int_a^b f(t)g(t)\, dt.$$

Prove that T is an element of the topological dual of $C([a, b])$, and that

$$\|T\| = \int_a^b |f(t)|\, dt.$$

2.10 Prove that ℓ^∞ is isometrically isomorphic to the topological dual of the space ℓ^1.

2.11 Prove that ℓ^1 is isometrically isomorphic to the topological dual of the space c_0.

2.12 Let X be a Banach space. Let $\{u_n\} \subset X'$ be a sequence such that there exists $M > 0$ satisfying

$$|\langle u_n, x\rangle| \leq M\|x\|, \quad \forall\, n \in \mathbb{N} \ \text{ and } \ \forall\, x \in X.$$

Prove that the set

$$S := \{x \in X \ : \ \exists \lim_{n\to\infty} \langle u_n, x\rangle\}$$

is closed.

2.13 Let $1 \leq p \leq q \leq \infty$. Prove that if $f \in L^p(\Omega) \cap L^q(\Omega)$, then $f \in L^r(\Omega)$ for every $p \leq r \leq q$, and the *interpolation inequality*

$$\|f\|_r \leq \|f\|_p^{\alpha}\, \|_q^{1-\alpha}$$

holds, where $\frac{1}{r} = \frac{\alpha}{p} + \frac{1-\alpha}{q}$ $(0 \leq \alpha \leq 1)$.

2.14 Let $1 \le p < \infty$. Prove that the set of simple functions is dense in $L^p(\Omega)$.

2.15 Let $1 \le p < \infty$. Prove that for $f \in L^p(\Omega, \mu)$ we have

$$\|f\| = \sup \left\{ \left| \int_\Omega fg \, d\mu \right| \; : \; \|g\|_{p'} = 1 \right\}.$$

2.16 Prove that for every $f \in L^1(\Omega) \cap L^\infty(\Omega)$, we have

$$\lim_{p \to \infty} \|f\|_p = \|f\|_\infty.$$

2.17 Prove that if $f, f_n \in L^p(\Omega, \mu)$ and $\lim_{n\to\infty} \|f_n - f\|_p = 0$, then there exists a subsequence $\{f_{n_k}\}$ such that

(i) $f_{n_k}(x) \to f(x)$ for μ-almost every $x \in \Omega$.
(ii) $|f_{n_k}(x)| \le g(x)$ for all $k \in \mathbb{N}$ and for μ-almost every $x \in \Omega$, with $g \in L^p(\Omega, \mu)$.

2.18 Let $1 \le p < \infty$ and let $\{f_n\}$ be a sequence in $L^p(\Omega, \mu)$ that converges μ-a.e. to a function $f \in L^p(\Omega, \mu)$. Prove that $\|f_n - f\|_p \to 0$ as $n \to \infty$ if and only if $\|f_n\|_p \to \|f\|_p$ as $n \to \infty$.

2.19 Let $f, g \in L^p(\Omega)$, with $1 \le p \le \infty$. Prove that $f \vee g \in L^p(\Omega)$, where

$$f \vee g(x) := \max\{f(x), g(x)\}.$$

Show that if $f_n \to f$ in $L^p(\Omega)$ and $g_n \to g$ in $L^p(\Omega)$, then

$$f_n \vee g_n \to f \vee g \quad \text{in } L^p(\Omega).$$

2.20 Prove that there exists a continuous linear functional $\mathrm{LIM} : \ell^\infty \to \mathbb{K}$ satisfying:

(i) $\|\mathrm{LIM}\| = 1$.
(ii) If $x = (x_n) \in c$, then $\mathrm{LIM}(x) = \lim_{n\to\infty} x_n$.
(iii) If $x = (x_n) \in \ell^\infty$ and $x_n \ge 0$ for all $n \in \mathbb{N}$, then $\mathrm{LIM}(x) \ge 0$.

The functional LIM is called the *Banach limit*.

2.21 Prove that there exist x and y in ℓ^∞ such that

$$\mathrm{LIM}(xy) \ne \mathrm{LIM}(x)\mathrm{LIM}(y).$$

2.22 Let m be the subspace of ℓ^∞ consisting of bounded sequences that take only finitely many values. Prove that m is dense in ℓ^∞.

2.23 Let X be a normed space and $A \subset X$ a convex set. Prove that

(i) $\overline{A}$ is convex.
(ii) If $a \in \text{int}(A)$ and $b \in \overline{A}$, then

$$[a, b[= \{tb + (1-t)a \ : \ t \in [0, 1[\} \subset \text{int}(A).$$

(iii) Prove that if $\text{int}(A) \neq \emptyset$, then $\overline{A} = \overline{\text{int}(A)}$.

2.24 Let $\phi : X \to]-\infty, +\infty]$. Its effective domain is defined as

$$D(\phi) := \{x \in X \ : \ \phi(x) < +\infty\}$$

and its epigraph as

$$\text{epi}(\phi) := \{(x, \alpha) \in X \times \mathbb{R} \ : \ \phi(x) \leq \alpha\}.$$

We say that ϕ is *lower semicontinuous* if for every $x \in X$ we have

$$\phi(x) \leq \liminf_{y \to x} \phi(y),$$

and that ϕ is *convex* if

$$\phi(tx + (1-t)y) \leq t\phi(x) + (1-t)\phi(y) \quad \forall\, x, y \in X, \ \ \forall\, t \in]0, 1[.$$

Prove that:

(i) ϕ is lower semicontinuous if and only if $\text{epi}(\phi)$ is closed in $X \times \mathbb{R}$.
(ii) ϕ is convex if and only if $\text{epi}(\phi)$ is convex in $X \times \mathbb{R}$.

2.25 Let $\phi : X \to]-\infty, +\infty]$ be a lower semicontinuous convex function such that $D(\phi) \neq \emptyset$. Its *conjugate* is defined as the function $\phi^* : X' \to]-\infty, +\infty]$ given by

$$\phi^*(u) := \sup_{x \in X} \{\langle u, x\rangle - \phi(x)\}.$$

Prove that ϕ^* is convex, lower semicontinuous, and that $D(\phi^*) \neq \emptyset$.

2.26 (*Jensen's Inequality*) Let $(\Omega, \mathcal{B}, \mu)$ be a finite measure space (*i.e.*, $\mu(\Omega) < \infty$). Let $\phi : \mathbb{R} \to]-\infty, +\infty]$ be a lower semicontinuous convex function such that $D(\phi) \neq \emptyset$. Let $f \in L^1(\Omega, \mu)$ such that $f(x) \in D(\phi)$ μ-almost everywhere, and $\phi(f) \in L^1(\Omega, \mu)$. Prove that

$$\phi\left(\frac{1}{\mu(\Omega)} \int_\Omega f \, d\mu\right) \leq \frac{1}{\mu(\Omega)} \int_\Omega \phi(f) \, d\mu.$$

2.27 Let $(\Omega, \mathcal{B}, \mu)$ be a finite measure space and let $1 \leq p < \infty$. Let $\phi : \mathbb{R} \to \mathbb{R}$ be a continuous convex function. Consider the function $J : L^p(\Omega, \mu) \to]-\infty, +\infty]$ defined by

$$J(f) := \begin{cases} \displaystyle\int_\Omega \phi(f)\, d\mu \text{ if } \phi(f) \in L^1(\Omega, \mu) \\ \\ +\infty \qquad\qquad \text{if } \phi(f) \notin L^1(\Omega, \mu). \end{cases}$$

Prove that J is convex and lower semicontinuous.

2.28 Let $(X, \| \|)$ be a normed space. Prove that if C is a bounded open convex set such that $0 \in C$, and C is also symmetric (*i.e.*, $-C = C$), then its *Minkowski functional* P_C is a norm on X which is equivalent to $\| \|$.

Consider the Banach space $(C([0, 1]), \| \|_\infty)$ and the set

$$C := \left\{ f \in C([0, 1]) \ : \ \int_0^1 |f(t)|^2\, dt < 1 \right\}.$$

Prove that C is an open, convex and symmetric subset of $(C([0, 1]), \| \|_\infty)$. Is C bounded in $(C([0, 1]), \| \|_\infty)$? Prove that the Minkowski functional P_C is a norm on $C([0, 1])$. Is P_C equivalent to $\| \|_\infty$?

2.29 Let $(X, \| \|)$ be a normed space and let P be a continuous seminorm on X. Prove that if $x_n \rightharpoonup x$ with respect to the weak topology $\sigma(X, X')$, then

$$P(x) \leq \liminf_{n \to \infty} P(x_n).$$

2.30 Let X be a Banach space and let $\phi : X \to]-\infty, +\infty]$ be a convex, lower semicontinuous function. Prove that if $x_0 \in \text{int}(D(\phi))$, then there exist constants $R > 0$ and M such that

$$\phi(x) \leq M \quad \forall\, x \in X \text{ with } \|x - x_0\| \leq R.$$

Moreover, given any $0 < r < R$, there exists $L \geq 0$ such that

$$|\phi(x_1) - \phi(x_2)| \leq L\|x_1 - x_2\|, \qquad \text{if } \|x_i - x_0\| \leq r \ \ i = 1, 2.$$

2.31 Let $T \in \mathcal{L}(c, c)$. Prove that there exists a double sequence $\{a_{ij}\}$, $i = 1, 2, \ldots$, $j = 0, 1, \ldots$, such that:

(i) If $(\alpha_n) \in c$, then $(\beta_n) := (T(\alpha_n))$ satisfies

$$\beta_i = a_{i0} \lim_{j \to \infty} \alpha_j + \sum_{j=1}^{\infty} a_{ij}\alpha_j.$$

(ii) $M := \sup_{i\in\mathbb{N}} \sum_{j=0}^{\infty} |a_{ij}| < \infty$.
(iii) For every $j = 1, 2, \ldots$, the limit $\lim_{i\to\infty} a_{ij}$ exists.
(iv) The limit $\lim_{i\to\infty} \sum_{j=0}^{\infty} a_{ij}$ exists.
(v) $\|T\| = M$.

2.32 Let $\{a_{ij}\}$ be a double sequence. Prove that if it satisfies conditions (ii), (iii), and (iv) of Problem (2,31), then the map T defined by (i) belongs to $\mathcal{L}(c, c)$, and moreover $\|T\| = M$.

2.33 Let $\{a_{ij}\}$ be a double sequence such that $\lim_{i\to\infty} a_{ij} = 0$ for every $j = 1, 2, \ldots$, and $M := \sup_{i\in\mathbb{N}} \sum_{j=1}^{\infty} |a_{ij}| < \infty$. Prove that if for each $(\alpha_j) \in c_0$ we define $T(\alpha_i) = \sum_{j=1}^{\infty} a_{ij}\alpha_j$, then $T \in \mathcal{L}(c_0, c_0)$ and $\|T\| = M$.

2.34 Let X be a Banach space and $\phi \in \mathcal{L}(X, \ell^1)$ a surjective map. Prove that there exists a closed subspace H of X such that

$$H \cap \operatorname{Ker} \phi = \{0\} \quad \text{and} \quad X = H + \operatorname{Ker} \phi.$$

2.35 Let X be a Banach space and $T : X \to X'$ a linear operator such that $\langle Tx, x\rangle \geq 0$ for all $x \in X$. Prove that T is continuous.

2.36 Let X be a normed space and $A \subset X$ a subset with the following property:

There exists $\lambda > 1$ such that for all $x \in X$ with $\|x\| \leq 1$, there exists $a \in A$ such that $\|a - x\| \leq \frac{1}{\lambda}$.

Prove that for each $x \in X$ with $\|x\| \leq 1$, there exists a sequence $\{a_n\} \subset A$ such that

$$x = \sum_{n=0}^{\infty} \lambda^{-n} a_n.$$

2.37 Using Problem 2.36, prove the Closed Graph Theorem.

2.38 Let X and Y be Banach spaces and $T \in \mathcal{L}(X, Y)$. Prove that $T(X)$ is closed in Y if and only if there exists a constant M such that

$$d(x, \operatorname{Ker} T) \leq M\|Tx\| \quad \forall\, x \in X.$$

2.39 Prove that in finite-dimensional spaces the weak topology coincides with the norm topology.

2.40 Let X be a normed space and $\phi : X \to]-\infty, +\infty]$ a convex and lower semicontinuous function with respect to the norm topology. Prove that ϕ is also lower semicontinuous with respect to the weak topology.

2.41 Let $x^n = (x_k^n) \in \ell^1$. Prove that $x^n \rightharpoonup x$ in $\sigma(\ell^1, \ell^\infty)$ if and only if $\sup_{n\in\mathbb{N}} \|x^n\|_1 < +\infty$ and

$$\lim_{n\to\infty} \sum_{k\in A} x_k^n = \sum_{k\in A} x_k \quad \text{for each } A \subset \mathbb{N}.$$

2.42 Let X be a normed space and let $\{x_n\}$ be a sequence in X such that $x_n \rightharpoonup x$ in $\sigma(X, X')$. Prove that if

$$\sigma_n := \frac{x_1 + \cdots + x_n}{n},$$

then $\sigma_n \rightharpoonup x$ in $\sigma(X, X')$.

2.43 Let X be a Banach space. Let $A, B \subset X$, with A $\sigma(X, X')$-closed and B $\sigma(X, X')$-compact. Prove that $A + B$ is $\sigma(X, X')$-closed. Furthermore, if A and B are disjoint nonempty convex sets, prove that they are strictly separated.

2.44 Let X be a Banach space. Prove that there exists a compact space K such that X is isometrically isomorphic to a subspace of $C(K)$.

2.45 A Banach space X is said to be *uniformly convex* if for every $\epsilon > 0$, there exists $\delta > 0$ such that if $x, y \in U(X)$ and $\|x - y\| > \epsilon$, then

$$\left\| \frac{x+y}{2} \right\| < 1 - \delta.$$

Prove that every uniformly convex Banach space is reflexive.

2.46 Prove that the space c_0 is not reflexive.

2.47 Let K be a compact topological space. Prove that $C(K)$ is reflexive if and only if K is finite.

2.48 Let $(\Omega, \mathcal{B}, \mu)$ be a σ-finite measure space. Prove that the space $L^1(\Omega, \mu)$ is reflexive if and only if it is finite-dimensional.

2.49 Prove that the space $C([0, 1])$ is not weakly sequentially complete.

2.50 Let X be a Banach space with separable dual. Prove that X is reflexive if and only if X is $\sigma(X, X')$-sequentially complete.

Chapter 3
Hilbert Spaces



3.1 Introduction

The pioneer in using integral equations in the study of partial differential equations was Liouville in 1832,[1] followed, among others, by C. Neumann, H. Poincaré, and I. Fredholm. Between 1904 and 1906, D. Hilbert gave new impetus to the study of integral equations by considering them as infinite systems of linear equations. This shift in perspective led to the consideration of sequences of numbers whose squares are summable, thus introducing for the first time the space ℓ^2.

The emergence of the concept of structure in Mathematics was fundamental for the birth, from the ideas developed by Hilbert, of what we now call the Theory of Hilbert Spaces. One of the intermediate steps was taken by M. Fréchet, who defined the concept of metric space in his 1906 thesis.

Hilbert's work invited the application of Fréchet's ideas and the transfer of Euclidean geometry to infinite-dimensional spaces. This was done by Fréchet himself[2] and by one of Hilbert's students, Erhard Schmidt,[3] in 1908. Schmidt defined the space ℓ^2, endowed it with the inner product and the norm, and used the concepts of orthogonality and closed set. He also obtained Bessel's inequality for this space and what is now known as the *"Gram–Schmidt orthogonalization method"*. His most important contribution was the introduction of the geometric language commonly used ever since. Using geometric ideas, he proved the Projection Theorem. This geometric perspective was adopted by the young mathematicians E. Fischer and F. Riesz, who independently in 1906–1907 proved the theorem that now bears their name, introduced the Lebesgue integration theory into the context of Hilbert spaces. With this new and wonderful tool, they defined the space $L^2([a, b])$,

[1] For a historical study, see the article by J. Lützen [40].

[2] Nouvelles Annales de Mathématiques, 8, 1908, 97–116, 289–317.

[3] Rendiconti del Circolo matematico di Palermo, 25, 1908, 53–77.

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J. M. Mazón, *Elements of Functional Analysis*, Universitext,
https://doi.org/10.1007/978-3-032-29210-0_3

and as a consequence of their theorem, they proved that it is complete, separable, and isomorphic to ℓ^2.

In 1907, M. Fréchet and F. Riesz independently obtained the representation

$$T(f) = \int_a^b f(t)\, g(t)\, dt$$

for every bounded linear functional T on the space $L^2([a, b])$. A generalization of this representation theorem, due to Lax and Milgram, will be extremely useful to us in the study of elliptic linear partial differential equations (see Chap. 7).

The abstract theory of Hilbert spaces began with the axiomatic definition of a Hilbert space formulated by J. von Neumann in 1927. Among the famous problems proposed by Hilbert at the International Congress of Mathematicians in Paris in 1900, the sixth concerns the axiomatic treatment of physics. Specifically, Hilbert stated:

> *Investigations of the foundations of geometry suggest the following problem: treat in the same manner, by means of axioms, those physical sciences in which mathematics plays an important role; in the first rank are the theories of probability and mechanics.*

During the years 1910–1920, Hilbert himself made progress on this problem and led a seminar at the Mathematical Institute of the University of Göttingen on mathematical problems in physics. One of the topics discussed in this seminar was the recent theory of Quantum Mechanics developed by Heisenberg, Schrödinger, Dirac, Born, and Jordan. Research in Quantum Mechanics showed that the observables of a physical system can be represented by symmetric linear operators on a Hilbert space, and that the eigenvalues and eigenvectors of the particular operator representing energy are the energy levels of an electron in the atom and the corresponding stationary quantum states of the system. The difference between two eigenvalues gives the frequencies of the emitted light quanta and thus defines the radiation spectrum of the substance. In 1926, Erwin Schrödinger presented his quantum theory based on his famous partial differential equation and demonstrated the equivalence of his theory with Werner Heisenberg's 1925 theory based on infinite matrices.

The use of operators in quantum theory stimulated work in the abstract theory of Hilbert spaces and the operators between them. This task was undertaken by a young Hungarian mathematician, John von Neumann, who had arrived in Göttingen from Budapest in 1926 to work as Hilbert's assistant. In three articles published in 1927 in the *Göttinger Nachrichten*, J. von Neumann presented his axiomatization of Quantum Mechanics. In these works, he emphasized that Hilbert's spectral theory of bounded operators and the extensions of his students were inadequate for Quantum Mechanics, since even the simplest cases require unbounded operators. As a preliminary step for his theory, J. von Neumann gave the first axiomatic definition of Hilbert spaces, pointing out that the sequence space ℓ^2 and the function space $L^2(\Omega)$ are models of these new abstract spaces. The characterization of Hilbert

spaces by means of the parallelogram identity was obtained by Jordan and von Neumann in 1935.[4]

In two extensive articles[5] published in 1929, J. von Neumann presented his axiomatic treatment of Hilbert spaces and the operators on these spaces. He completed all this work in 1932 with the publication of his famous monograph *Mathematische Grundlagen der Quantenmechanik*, where he describes in more detail than in his articles the connection of his ideas with Quantum Mechanics, and contrasts his formalism based on Hilbert spaces and the operators between them with those used by Born, Heisenberg, Jordan, and Dirac.

In 1932, M. Stone published his book *Linear Transformations in Hilbert Spaces*. As Stone himself acknowledged,[6] his textbook is strongly inspired by von Neumann's, although it contains many original results not considered by von Neumann, such as his treatment of ordinary differential equations, his theory of one-parameter unitary groups, etc. It also includes original and different proofs of known results, such as the spectral theorem for self-adjoint operators. This book is so well structured and written that it remained for a long time the standard reference for the theory of Hilbert spaces and the operators between them.

3.2 Hilbert Spaces: Generalities

Let H be a vector space over the field $\mathbb{K} = (\mathbb{R}$ or $\mathbb{C})$. A map $(.|.) : H \times H \longrightarrow \mathbb{K}$ is called an *inner product* on H if for all $x, y, z \in H$, and $\alpha, \beta \in \mathbb{K}$, the following properties hold:

$$(x|x) \geq 0 \tag{3.1}$$

$$(x|x) = 0 \iff x = 0 \tag{3.2}$$

$$(\alpha x + \beta y|z) = \alpha(x|z) + \beta(y|z) \tag{3.3}$$

$$(x|y) = \overline{(y|x)} \tag{3.4}$$

A vector space endowed with an inner product is called a *pre-Hilbert space*. Two important inequalities in pre-Hilbert spaces are the following:

Proposition 3.1 *For two elements x, y in a pre-Hilbert space $(H, (\cdot|\cdot))$, the following inequalities hold:*

[4] Ann. of Math. **36**, 719–723.

[5] Math. Ann. **102** (1929), 49–131; 370–427.

[6] See Stone's letter reproduced in [5], p. 311.

$$|(x|y)|^2 \leq (x|x)\ (y|y) \qquad \textit{(Cauchy–Schwarz Inequality)}$$

$$\sqrt{(x+y|x+y)} \leq \sqrt{(x|x)} + \sqrt{(y|y)} \qquad \textit{(Minkowski Inequality)}.$$

Proof We first prove the Cauchy–Schwarz inequality. If $(y|y) = 0$, then $y = 0$ and the inequality is obvious. Suppose $(y|y) \neq 0$. For every $\alpha \in \mathbb{K}$, we have

$$0 \leq (x - \alpha y|x - \alpha y) = (x|x) - \alpha(y|x) - \overline{\alpha}(x|y) + \alpha\overline{\alpha}(y|y).$$

Taking

$$\alpha = \frac{(x|y)}{(y|y)}$$

we obtain the inequality.

On the other hand, using the Cauchy–Schwarz inequality, we get

$$(x+y|x+y) = (x|x) + (y|y) + 2\mathfrak{R}(x|y) \leq$$

$$(x|x) + (y|y) + 2\sqrt{(x|x)}\ \sqrt{(y|y)} = \left(\sqrt{(x|x)} + \sqrt{(y|y)}\right)^2,$$

from which the Minkowski inequality follows. □

The Minkowski inequality ensures that $\|x\| := \sqrt{(x|x)}$ defines a norm on H. Therefore, every pre-Hilbert space is a normed space. Moreover, the inner product is continuous with respect to this norm. Indeed, let $x_n \to x$ and $y_n \to y$ with respect to the norm induced by the inner product. We want to show that $(x_n|y_n) \to (x|y)$; for this, it suffices to observe that using the Cauchy–Schwarz inequality we have:

$$|(x_n|y_n) - (x|y)| \leq |(x_n - x|y_n - y)| + |(x_n - x|y)| + |(x|y_n - y)|$$

$$\leq \|x_n - x\|\|y_n - y\| + \|x_n - x\|\|y\| + \|y_n - y\|\|x\|.$$

The relation between the norm and the inner product is given by the so-called *polarization identity*:

$$4(x|y) = \|x+y\|^2 - \|x-y\|^2 + i\|x+\imath y\|^2 - i\|x-\imath y\|^2.$$

Since every pre-Hilbert space is a normed space, it is natural to ask when a normed space is a pre-Hilbert space. A satisfactory answer is given by the following result.

Theorem 3.1 *Let $(H, \|\cdot\|)$ be a normed space. The following are equivalent:*

(i) H satisfies the parallelogram law, i.e.,

$$\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2, \quad \forall x, y \in H. \tag{3.5}$$

(ii) H is a pre-Hilbert space, i.e., there exists an inner product on H such that $\|x\| = \sqrt{(x|x)}$ for all $x \in H$.

Proof (ii) implies (i): As in the proof of Minkowski's inequality, we have

$$\|x + y\|^2 = \|x\|^2 + 2\Re(x|y) + \|y\|^2,$$

$$\|x - y\|^2 = \|x\|^2 - 2\Re(x|y) + \|y\|^2.$$

Adding the two equalities gives (3.5).
(i) implies (ii): For simplicity, assume the space is real. The polarization identity suggests the candidate for the inner product:

$$(x|y) := \frac{1}{4}\left(\|x + y\|^2 - \|x - y\|^2\right). \tag{3.6}$$

We verify that the function defined by (3.6) satisfies the axioms of an inner product. Clearly, $(x|y) = (y|x)$, and since $(x|x) = \|x\|^2$, we have properties (3.1) and (3.2). Let's verify property (3.3). First, we show that $(x + y|z) = (x|z) + (y|z)$. Consider the function

$$F(x, y, z) := 4\left[(x + y|z) - (x|z) - (y|z)\right].$$

We need to show that $F \equiv 0$. We can write F as

$$\begin{aligned} F(x, y, z) &= \|x + y + z\|^2 - \|x + y - z\|^2 \\ &\quad -\|x + y\|^2 + \|x - y\|^2 - \|y + z\|^2 + \|y - z\|^2. \end{aligned} \tag{3.7}$$

As a consequence of (3.5), we have

$$\|x + y \pm z\|^2 = 2\|x \pm z\|^2 + 2\|y\|^2 - \|x \pm z - y\|^2.$$

Substituting into (3.7), we get

$$\begin{aligned} F(x, y, z) &= -\|x + z - y\|^2 + \|x - z - y\|^2 + \|x + z\|^2 \\ &\quad -\|x - z\|^2 - \|y + z\|^2 + \|y - z\|^2. \end{aligned} \tag{3.8}$$

Adding (3.7) and (3.8), and applying (3.5), we obtain:

$$F(x, y, z) = \frac{1}{2}\left(\|x + y + z\|^2 + \|y + z - x\|^2\right)$$

$$-\frac{1}{2}\left(\|y - z + x\|^2 + \|y - z - x\|^2\right) - \|y + z\|^2 + \|y - z\|^2$$

$$= \left(\|y+z\|^2 + \|x\|^2\right) - \left(\|y-z\|^2 + \|x\|^2\right) - \|y+z\|^2 + \|y-z\|^2 = 0.$$

Finally, we show that $(\alpha x|y) = \alpha(x|y)$. To do this, consider the function $f(\alpha) := (\alpha x|y) - \alpha(x|y)$. We want to prove that $f \equiv 0$. We compute:

$$f(0) = \frac{1}{4}\left(\|y\|^2 - \|y\|^2\right) = 0,$$

and since $(-x|y) = -(x|y)$, we have $f(-1) = 0$. Therefore, for any integer n, we get $f(n) = 0$, because

$$\begin{aligned}(nx|y) &= (\text{sign}(n)(x + \cdots + x)|y) = \text{sign}(n)[(x|y) + \cdots + (x|y)] \\ &= |n|\text{sign}(n)(x|y) = n(x|y).\end{aligned}$$

For $p, q \in \mathbb{Z}$, with $q \neq 0$, the above implies:

$$\left(\frac{p}{q}x\middle|y\right) = p\left(\frac{1}{q}x\middle|y\right) = \frac{p}{q}q\left(\frac{1}{q}x\middle|y\right) = \frac{p}{q}(x|y),$$

hence $f(\alpha) = 0$ for every rational α, and since f is continuous, it follows that $f \equiv 0$. □

Pre-Hilbert spaces that are complete with respect to the norm induced by their inner product are called *Hilbert spaces*. Obviously, every Hilbert space is a Banach space. Important examples of Hilbert spaces include the following:

Example 3.1

(1) It is easy to see that in $\mathbb{K}^n$,

$$(x|y) := \sum_{i=1}^{n} x_i \overline{y}_i$$

is an inner product. Moreover, the norm induced by this inner product is the Euclidean norm, i.e.,

$$\sqrt{(x|x)} = \sqrt{|x_1|^2 + \cdots + |x_n|^2}.$$

Thus, $\mathbb{K}^n$, endowed with this inner product, is a Hilbert space.

(2) Let $(\Omega, \mathcal{B}, \mu)$ be a measure space. Consider the Banach space $(L^2(\Omega, \mu), \| \|_2)$. For $f, g \in L^2(\Omega, \mu)$, define

$$(f|g) := \int_\Omega f\overline{g}\, d\mu.$$

It is easy to see that this defines an inner product. Moreover,

$$(f|f) := \int_\Omega |f|^2 \, d\mu,$$

so this inner product gives the space the structure of a Hilbert space. In particular, the sequence space ℓ^2 is a Hilbert space with respect to the inner product

$$(x|y) := \sum_{n=1}^{\infty} x_n \overline{y_n}.$$

Two important particular cases, in connection with Fourier series, are the following: Consider the measure space $(\mathbb{Z}, 2^{\mathbb{Z}}, \mu)$, where $\mu(n) = 1$ for each $n \in \mathbb{Z}$. We denote by $\ell^2(\mathbb{Z})$ the space $L^2(\mathbb{Z}, \mu)$, i.e.,

$$\ell^2(\mathbb{Z}) = \left\{ (x_n)_{n\in\mathbb{Z}} \, : \, \sum_{n=-\infty}^{+\infty} |x_n|^2 = \lim_{n\to+\infty} \sum_{k=-n}^{n} |x_k|^2 < \infty \right\}.$$

In this case, the inner product is given by

$$(x|y) = \sum_{n=-\infty}^{+\infty} x_n \overline{y_n},$$

and the norm by

$$\|x\|_2 = \left(\sum_{n=-\infty}^{+\infty} |x_n|^2 \right)^{\frac{1}{2}}.$$

We denote by $\mathbb{T}$ the unit circle in the complex plane $\mathbb{C}$, i.e.,

$$\mathbb{T} := \{z \in \mathbb{C} \, : \, |z| = 1\} = \{e^{it} \, : \, t \in \mathbb{R}\}.$$

Consider on $\mathbb{T}$ the normalized Lebesgue measure m, defined by

$$m(\{e^{it} \, : \, t \in A\}) := \int_A \frac{dt}{2\pi}, \qquad \forall\, A \subset [-\pi, \pi[, \quad A \text{ Lebesgue measurable.}$$

Using the group homomorphism $\phi : (\mathbb{R}, +) \to (\mathbb{T}, \cdot)$, defined by $\phi(t) := e^{it}$, it is easy to see that the space $L^2(\mathbb{T}, m)$ can be identified with the space of all 2π-periodic functions $f : \mathbb{R} \to \mathbb{C}$ such that

$$\int_{-\pi}^{\pi} |f(t)|^2 \, dt < \infty.$$

Then $L^2(\mathbb{T})$ is a Hilbert space if we consider the inner product

$$(f|g) := \int_{\mathbb{T}} f(z)\overline{g(z)}\, dm(z).$$

We will identify a function $f \in L^2(\mathbb{T})$ with its 2π-periodic counterpart (denoted by the same symbol), $f(t) = f(e^{it})$. With this identification, the inner product becomes:

$$(f|g) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)\overline{g(t)}\, dt.$$

(3) (*Bergman spaces*) Let Ω be an open subset of the complex plane $\mathbb{C}$. The *Bergman space* $\mathcal{B}(\Omega)$ is defined as the set of all holomorphic functions $f : \Omega \to \mathbb{C}$ such that

$$\int_{\Omega} |f(x+iy)|^2\, dxdy < \infty.$$

We have that $\mathcal{B}(\Omega) \subset L^2(\Omega, \mathcal{L}^2)$, where $\mathcal{L}^2$ is the Lebesgue measure on the plane. Therefore, we can consider on $\mathcal{B}(\Omega)$ the inner product from $L^2(\Omega, \mathcal{L}^2)$, i.e., for $f, g \in \mathcal{B}(\Omega)$

$$(f|g) = \int_{\Omega} f\overline{g}.$$

To prove that this inner product gives the Bergman space the structure of a Hilbert space, it suffices to show that $\mathcal{B}(\Omega)$ is closed in $L^2(\Omega, \mathcal{L}^2)$. Let $f_n \in \mathcal{B}(\Omega)$ such that $f_n \to f \in L^2(\Omega, \mathcal{L}^2)$ with respect to the norm $\|\cdot\|_2$. We need to prove that $f \in \mathcal{B}(\Omega)$, that is, f is holomorphic in Ω.

Observe first that if $g \in \mathcal{B}(\Omega)$, then for any $z_0 \in \Omega$ and $0 < r < \operatorname{dist}(z_0, \partial\Omega)$, we have

$$|g(z_0)| \le \frac{1}{r\sqrt{\pi}}\|g\|_2. \tag{3.9}$$

Indeed: since g is holomorphic in $\overline{B(z_0, r)}$, by the mean value property we have

$$g(z_0) = \frac{1}{\pi r^2}\int_{B(z_0,r)} g(z)\, dz.$$

Then, applying the Cauchy–Schwarz inequality, we get

$$|g(z_0)| = \frac{1}{\pi r^2}\left|\int_{B(z_0,r)} g(z)\, dz\right|$$

$$\leq \frac{1}{\pi r^2}\left(\int_{B(z_0,r)} |g(z)|^2\, dz\right)^{\frac{1}{2}} \left(\int_{B(z_0,r)} \chi_\Omega\, dz\right)^{\frac{1}{2}} \leq \frac{1}{r\sqrt{\pi}}\|g\|_2.$$

Let us now show that f is holomorphic in Ω. Given $z_0 \in \Omega$, let $\rho > 0$ be such that $\overline{B(z_0,\rho)} \subset \Omega$. Take $0 < r < \operatorname{dist}(B(z_0,\rho), \partial\Omega)$. Then, as a consequence of (3.9), there exists a constant $C > 0$ such that

$$|f_n(z) - f_m(z)| \leq C\|f_n - f_m\|_2, \quad \forall\, n, m \in \mathbb{N}, \quad \text{and} \quad |z - z_0| \leq r.$$

Therefore, the sequence $\{f_n\}$ is uniformly Cauchy in every closed ball contained in Ω. Then, as a consequence of Weierstrass's Theorem, there exists a holomorphic function g in Ω such that $f_n \to g$ uniformly on compact subsets of Ω. Consequently, since $f_n \to f$ in $L^2(\Omega, \mathcal{L}^2)$, we have that $f = g$ a.e. in Ω, and thus $f \in \mathcal{B}(\Omega)$.

3.3 Geometry of Pre-Hilbert Spaces

As we saw in the previous section, pre-Hilbert spaces are those normed spaces in which two-dimensional subspaces behave like Euclidean spaces. For this reason, pre-Hilbert spaces possess an inherent geometry, a geometry that is a generalization to infinite dimensions of the geometry of Euclidean spaces. This geometry begins with the following definition of orthogonality.

Definition 3.1 Two vectors x and y in a pre-Hilbert space H are said to be *orthogonal*, denoted by $x \perp y$, if $(x|y) = 0$. If $A, B \subset H$, we write $A \perp B$ when $x \perp y$ for every $x \in A$, $y \in B$.

For orthogonal vectors x and y, the *Pythagorean identity* holds:

$$\|x + y\|^2 = \|x\|^2 + \|y\|^2.$$

To prove this, it is enough to observe that:

$$\|x + y\|^2 = (x + y|x + y) = \|x\|^2 + 2\Re(x|y) + \|y\|^2 = \|x\|^2 + \|y\|^2.$$

Definition 3.2 If M is a subset of a pre-Hilbert space H, its *orthogonal complement* is defined as

$$M^\perp := \{\, x \in H \;:\; (x|y) = 0 \;\; \forall y \in M \,\}.$$

This orthogonal complement is a closed subspace of H.

Let M be a convex subset of a pre-Hilbert space H and let $x \in H$. An element $y_0 \in M$ is said to be the *orthogonal projection* of x onto M, denoted by $y_0 = P_M(x)$, if $\|x - y_0\| = \operatorname{dist}(x, M)$.

The following result guarantees the existence and uniqueness of orthogonal projections.

Theorem 3.2 (Optimal Approximation Theorem) *Let H be a pre-Hilbert space and let M be a convex subset of H. Suppose that M, endowed with the induced metric, is complete. Then, for every $x \in H$, there exists a unique $y_0 \in M$ such that $y_0 = P_M(x)$. Moreover, y_0 is characterized as the unique element of M satisfying*

$$\Re(x - y_0 | y - y_0) \leq 0 \quad \forall\, y \in M. \tag{3.10}$$

Proof Let $y_n \in M$ be such that

$$d_n := \|x - y_n\| \to d := \operatorname{dist}(x, M) = \inf_{y \in M} \|x - y\|.$$

Applying the parallelogram law (3.5) to the vectors $x - y_n$ and $x - y_m$, we obtain

$$\left\| x - \frac{y_n + y_m}{2} \right\|^2 + \left\| \frac{y_n - y_m}{2} \right\|^2 = \frac{1}{2}\left(d_n^2 + d_m^2\right).$$

Now, since M is convex, we have $\dfrac{y_n + y_m}{2} \in M$, which implies

$$\left\| x - \frac{y_n + y_m}{2} \right\| \leq d.$$

Therefore,

$$\left\| \frac{y_n - y_m}{2} \right\|^2 \leq \frac{1}{2}\left(d_n^2 + d_m^2\right) - d^2,$$

from which it follows that the sequence $\{y_n\}$ is Cauchy, and since M is complete, there exists $y_0 \in M$ such that $y_n \to y_0$ and moreover $d = \|x - y_0\|$.

Let us now prove that y_0 is characterized by (3.10). Let $y_0 \in M$ such that $y_0 = P_M(x)$. Given any $y \in M$, the convexity of M implies that

$$z = (1 - \alpha) y_0 + \alpha y \in M \qquad \forall\, \alpha \in]0, 1].$$

Then,

$$\|x - y_0\| \leq \|x - \left[(1 - \alpha) y_0 + \alpha y\right]\| = \|(x - y_0) - \alpha(y - y_0)\|.$$

Therefore,

$$\|x - y_0\|^2 \leq \|x - y_0\|^2 - 2\alpha \Re(x - y_0|y - y_0) + \alpha^2 \|y - y_0\|^2,$$

and thus,

$$2\Re(x - y_0|y - y_0) \leq \alpha \|y - y_0\|^2.$$

Letting $\alpha \to 0^+$ yields (3.10).

Conversely, suppose that y_0 satisfies (3.10). Then,

$$\|x - y_0\|^2 - \|x - y\|^2 = 2\Re(x - y_0|y - y_0) - \|y_0 - y\|^2 \leq 0 \quad \forall\, y \in M,$$

from which it follows that $y_0 = P_M(x)$.

Finally, let us prove uniqueness. Suppose $y_i = P_M(x)$ for $i = 1, 2$. Then, since (3.10) holds, we have

$$\Re(x - y_1|y - y_1) \leq 0 \quad \forall\, y \in M, \tag{3.11}$$

$$\Re(x - y_2|y - y_2) \leq 0 \quad \forall\, y \in M. \tag{3.12}$$

Taking $y = y_2$ in (3.11) and $y = y_1$ in (3.12), and adding, we obtain $\|y_1 - y_2\|^2 \leq 0$, hence $y_1 = y_2$. □

For the case of vector subspaces, we have the following result:

Theorem 3.3 (Projection Theorem) *Let H be a pre-Hilbert space, $x \in H$, and let M be a subspace of H that is complete with respect to the induced metric (i.e., M is a Hilbert space with the induced inner product). Then an element $y_0 \in M$ satisfies $y_0 = P_M(x)$ if and only if $x - y_0 \in M^\perp$. Moreover, the map $P_M : H \longrightarrow M$ is linear and satisfies:*

(i) $\|P_M(x)\| \leq \|x\| \quad \forall x \in H$,
(ii) $(P_M)^2 = P_M$, *(i.e., P_M is a projection),*
(iii) $Ker(P_M) = M^\perp$ *and* $P_M(H) = M$.

Proof If $y_0 = P_M(x)$, then by the previous theorem, we have

$$\Re(x - y_0|z - y_0) \leq 0 \quad \forall\, z \in M.$$

Fix $y \in M$, and let $(x - y_0|y) = re^{i\theta}$, with $r = |(x - y_0|y)|$. Then, since M is a vector subspace, we have that $z = y_0 + e^{i\theta} y \in M$, so that

$$\Re(x - y_0|z - y_0) = \Re(x - y_0|e^{i\theta} y) = r \leq 0.$$

Therefore, $r = 0$, hence $(x - y_0|y) = 0$.

Conversely, suppose that $(x - y_0|y) = 0$ for every $y \in M$. Then, since $y_0 \in M$, we have $(x - y_0|y - y_0) = 0$. Applying the Optimal Approximation Theorem, we obtain that $y_0 = P_M(x)$.

Thus, the map $P_M : H \longrightarrow M$ is well defined, where $P_M(x)$ is the unique element of M such that $x - P_M(x) \in M^\perp$ and $\|x - P_M(x)\| = \text{dist}(x, M)$. Let $x_1, x_2 \in H$ and $\alpha_1, \alpha_2 \in \mathbb{C}$. Then, for any $y \in M$, we have

$$([\alpha_1 x_1 + \alpha_2 x_2] - [\alpha_1 P_M(x_1) + \alpha_2 P_M(x_2)]|y)$$

$$= \alpha_1(x_1 - P_M(x_1)|y) + \alpha_2(x_2 - P_M(x_2)|y) = 0,$$

so that $P_M(\alpha_1 x_1 + \alpha_2 x_2) = \alpha_1 P_M(x_1) + \alpha_2 P_M(x_2)$, and thus P_M is linear.

On the other hand, if $x \in H$, then $x = (x - P_M(x)) + P_M(x)$, with $P_M(x) \in M$ and $x - P_M(x) \in M^\perp$. Therefore, by the Pythagorean identity, we have

$$\|x\|^2 = \|x - P_M(x)\|^2 + \|P_M(x)\|^2 \geq \|P_M(x)\|^2,$$

which proves (i). To see that (ii) holds, just observe that if $x \in M$, then $P_M(x) = x$, so for every $x \in H$, since $P_M(x) \in M$, we have $(P_M)^2(x) = P_M(P_M(x)) = P_M(x)$, thus proving (ii).

Finally, if $P_M(x) = 0$, then $x = x - P_M(x) \in M^\perp$, and conversely, if $x \in M^\perp$, then 0 is the only vector in M such that $x - 0 = x \in M^\perp$. Therefore, $P_M(x) = 0$. Hence, $\text{Ker}(P_M) = M^\perp$, and clearly $P_M(H) = M$, so (iii) holds. □

3.4 The Dual of a Hilbert Space

As a consequence of the Projection Theorem, we obtain the following characterization of the dual of a Hilbert space.

Theorem 3.4 (Riesz–Fréchet Theorem) *In a Hilbert space H, for every linear and continuous functional u, there exists a unique $x_0 \in H$ such that*

$$\langle u, x\rangle = (x|x_0) \quad \forall x \in H$$

and moreover $\|u\| = \|x_0\|$.

Proof Let $M = \text{Ker}(u)$. Since u is continuous, M is a closed subspace of H. Assuming u is non-zero, we have $M \neq H$, so $M^\perp \neq \{0\}$. Therefore, there exists a vector $y_0 \in M^\perp$ such that $\langle u, y_0\rangle = 1$. Now, for any $x \in H$, if $\alpha = \langle u, x\rangle$, then

$$\langle u, x - \alpha y_0\rangle = \langle u, x\rangle - \alpha = 0,$$

so $x - \langle u, x\rangle y_0 \in M$. Therefore,

$$0 = (x - \langle u, x\rangle y_0 | y_0) = (x|y_0) - \langle u, x\rangle \|y_0\|^2.$$

Hence, if $x_0 := \|y_0\|^{-2} y_0$, then $\langle u, x\rangle = (x|x_0)$ for every $x \in H$. Let us prove uniqueness. If $z_0 \in H$ satisfies $(x|x_0) = (x|z_0)$ for all $x \in H$, then $x_0 - z_0 \perp H$, which implies $z_0 = x_0$. Finally, let us show that $\|u\| = \|x_0\|$. By the Cauchy–Schwarz inequality, we have

$$|\langle u, x\rangle| = |(x|x_0)| \leq \|x_0\| \|x\|,$$

which gives $\|u\| \leq \|x_0\|$. Furthermore,

$$\left\langle u, \frac{x_0}{\|x_0\|} \right\rangle = \left(x_0 \middle| \frac{x_0}{\|x_0\|} \right) = \|x_0\|,$$

so we conclude that $\|u\| = \|x_0\|$. □

As a consequence of the previous theorem, every Hilbert space is isometrically isomorphic to its dual. More precisely, if for $x \in H$, we define $x^* \in H'$ by $\langle x^*, y\rangle := (y|x)$ for every $y \in H$, then the map $C : H \to H'$ defined by $C(x) := x^*$ is an isometric isomorphism. Moreover, we have

$$(x^*|y^*) = (y|x) \quad \forall\, x, y \in H.$$

Corollary 3.1 *Every Hilbert space is reflexive.*

Proof Let $x \in H$. Then, for any $u = y^* \in H'$, we have

$$\langle x^{**}, u\rangle = (u|x^*) = (y^*|x^*) = (x|y) = \langle y^*, x\rangle = \langle u, x\rangle = \langle \hat{x}, u\rangle,$$

so $\hat{x} = x^{**}$. Therefore, the natural embedding $J : H \to H''$, $J(x) = \hat{x}$ is surjective, and thus H is reflexive. □

3.5 Orthonormal Bases and Fourier Series in Hilbert Spaces

In the n-dimensional Euclidean space $\mathbb{R}^n$, if $\{e_1, \ldots, e_n\}$ is the canonical basis, every vector $x \in \mathbb{R}^n$ can be represented in the form

$$x = \sum_{i=1}^{n} (x|e_i)\, e_i\,.$$

In this section, we will study the extension of this fact to infinite-dimensional Hilbert spaces.

Definition 3.3 Given a system of nonzero vectors $\{ e_i \ , \ \ i \in I \}$ in a pre-Hilbert space, the system is said to be *orthogonal* if $(e_i|e_j) = 0$ whenever $i \neq j$. If in addition $\|e_i\| = 1$ for all $i \in I$, the system is called *orthonormal*.

From any countable system of linearly independent vectors, one can obtain an orthonormal system by applying the *Gram–Schmidt orthonormalization process*.

Theorem 3.5 (Gram–Schmidt Orthonormalization Process) *If* $\{x_n \ , \ \ n \in \mathbb{N}\}$ *is a sequence of linearly independent vectors in a pre-Hilbert space* H*, then there exists an orthonormal system* $\{e_n \ , \ \ n \in \mathbb{N}\}$ *such that for every* $n \in \mathbb{N}$*,*

$$LIN\{e_1, \ldots, e_n\} = LIN\{x_1, \ldots, x_n\}.$$

Proof The proof proceeds by induction. Let

$$e_1 := \frac{x_1}{\|x_1\|},$$

$$e_2 := \frac{x_2 - (x_2|e_1)e_1}{\|x_2 - (x_2|e_1)e_1\|}.$$

Assuming $e_1, e_2, \ldots, e_n$ have already been defined, we define

$$e_{n+1} := \frac{x_{n+1} - \sum_{k=1}^{n}(x_{n+1}|e_k)e_k}{\|x_{n+1} - \sum_{k=1}^{n}(x_{n+1}|e_k)e_k\|}.$$

It is easy to see that the sequence $\{e_n \ , \ \ n \in \mathbb{N}\}$ satisfies the conditions required in the theorem. □

As an example of the previous method, if we consider in the real Hilbert space $L^2([-1, 1])$ the sequence of linearly independent vectors $\{t^n \ : \ n = 0, 1, \ldots\}$, and apply the Gram–Schmidt method to it, we obtain the orthonormal system of the classical *Legendre polynomials* $\{p_n(t) \ : \ n = 0, 1, 2, \ldots\}$, where:

$$p_0(t) = \frac{1}{\sqrt{2}}, \quad p_1(t) = \sqrt{\frac{3}{2}}t, \quad p_2(t) = \frac{3}{2}\sqrt{\frac{5}{2}}\left(t^2 - \frac{1}{3}\right), \ldots.$$

In general,

$$p_n(t) = \sqrt{\frac{2n+1}{2}}P_n(t),$$

where P_n are the classical Legendre polynomials, given by

$$P_n(t) = \frac{1}{2^n n!}\frac{d^n}{dt}((t^2 - 1)^n),$$

and which are the solutions of the *Legendre differential equation*

$$\frac{d}{dt}((1-t^2)\frac{du}{dt}) + n(n+1)u = 0, \quad t \in [-1,1],\ n \geq 0.$$

Definition 3.4 A system $\{e_i\ ,\ \ i \in I\}$ in the pre-Hilbert space H is said to be *complete* if

$$H = \overline{LIN\{e_i\ ,\ \ i \in I\}}.$$

A complete orthonormal system in H is called an *orthonormal basis* of H.

Given an orthonormal system $\{e_n\ ,\ n \in \mathbb{N}\}$ in H and $x \in H$, the numbers $(x|e_n)$ are called the *Fourier coefficients* of x with respect to the system $\{e_n\ ,\ \ n \in \mathbb{N}\}$, and the series

$$\sum_{n=1}^{\infty} (x|e_n)\ e_n,$$

is called the *Fourier series* of x with respect to the system $\{e_n\ ,\ \ n \in \mathbb{N}\}$.

The following questions naturally arise: Does the Fourier series $\sum_{n=1}^{\infty}(x|e_n)\ e_n$ converge in the pre-Hilbert space H? And if it converges, does its sum coincide with the element x? To answer these two questions, let us first consider the following problem: How should the coefficients α_k be chosen so that the distance between x and $S_n = \sum_{k=1}^{n} \alpha_k e_k$ is minimized? Since the system $\{e_n\ ,\ n \in \mathbb{N}\}$ is orthonormal, we have

$$\begin{aligned}
&\|x - S_n\|^2 \\
&= (x|x) - \left(x\middle|\sum_{k=1}^{n}\alpha_k e_k\right) - \left(\sum_{k=1}^{n}\alpha_k e_k\middle|x\right) + \left(\sum_{k=1}^{n}\alpha_k e_k\middle|\sum_{k=1}^{n}\alpha_k e_k\right) \\
&= \|x\|^2 - \sum_{k=1}^{n}\overline{\alpha_k}(x|e_k) - \sum_{k=1}^{n}\alpha_k\overline{(x|e_k)} + \sum_{k=1}^{n}\alpha_k\overline{\alpha_k} \\
&= \|x\|^2 - \sum_{k=1}^{n}|(x|e_k)|^2 + \sum_{k=1}^{n}|\alpha_k - (x|e_k)|^2.
\end{aligned}$$

Hence, $\|x - S_n\|^2$ is minimized when $\sum_{k=1}^{n}|\alpha_k - (x|e_k)|^2 = 0$, i.e., when $\alpha_k = (x|e_k)$. In this case, we have

$$0 \leq \|x - S_n\|^2 = \|x\|^2 - \sum_{k=1}^{n}|(x|e_k)|^2, \tag{3.13}$$

so that

$$\sum_{k=1}^{n} |(x|e_k)|^2 \leq \|x\|^2 \quad \forall\, n \in \mathbb{N}.$$

Therefore,

$$\sum_{k=1}^{\infty} |(x|e_k)|^2 \leq \|x\|^2.$$

We can thus state the following result.

Theorem 3.6 *If $\{e_n\ ,\ n \in \mathbb{N}\}$ is an orthonormal system in a pre-Hilbert space H, then for every $x \in H$, we have that $(x|e_n) \in \ell_2$, and moreover,*

$$\|(x|e_n)\|_2^2 = \sum_{n=1}^{\infty} |(x|e_n)|^2 \leq \|x\|^2. \tag{3.14}$$

Inequality (3.14) is known as *Bessel's inequality*. In the case of Hilbert spaces, we have the following result.

Theorem 3.7 (Riesz–Fischer Theorem) *Let H be a Hilbert space, and let $\{e_n\ ,\ n \in \mathbb{N}\}$ be an orthonormal system in H. Then, given $(\alpha_n) \in \ell_2$, there exists an element $x \in H$ such that $(x|e_n) = \alpha_n$ for all $n \in \mathbb{N}$, i.e., the α_n are the Fourier coefficients of x with respect to the system $\{e_n\ ,\ n \in \mathbb{N}\}$; moreover,*

$$\sum_{n=1}^{\infty} |\alpha_n|^2 = \|x\|^2.$$

Proof Given $(\alpha_n) \in \ell_2$, for each $n \in \mathbb{N}$, let $y_n := \sum_{k=1}^{n} \alpha_k e_k$. Then

$$\|y_{n+p} - y_n\|^2 = \|\alpha_{n+1}e_{n+1} + \cdots + \alpha_{n+p}e_{n+p}\|^2 = \sum_{k=n+1}^{n+p} |\alpha_k|^2,$$

and since $(\alpha_n) \in \ell_2$, the sequence $\{y_n\}$ is Cauchy in H. Therefore, there exists $x \in H$ such that $x = \lim_{n\to\infty} y_n$. Moreover,

$$(x|e_n) = (y_m|e_n) + (x - y_m|e_n). \tag{3.15}$$

Now, $(y_m|e_n) = \alpha_n$ for each $m \geq n$, and by the Cauchy–Schwarz inequality,

$$|(x - y_m|e_n)| \leq \|x - y_m\|\|e_n\| = \|x - y_m\|.$$

Therefore, taking the limit in (3.15) as $m \to \infty$, we obtain $(x|e_n) = \alpha_n$ for each $n \in \mathbb{N}$. Moreover, since $\|x - y_n\| \to 0$ as $n \to \infty$, and

$$\|x - y_n\|^2 = \|x\|^2 - \sum_{k=1}^{n} |\alpha_k|^2,$$

we have that

$$\sum_{n=1}^{\infty} |\alpha_n|^2 = \|x\|^2,$$

which completes the proof. □

To answer the second question, we give the following definition.

Definition 3.5 An orthonormal system $\{e_n\ ,\ \ n \in \mathbb{N}\}$ in a pre-Hilbert space H is said to be *closed* if for every $x \in H$ the *Parseval identity*

$$\sum_{n=1}^{\infty} |(x|e_n)|^2 = \|x\|^2$$

holds, which is equivalent, as a consequence of (3.13), to

$$\sum_{n=1}^{\infty} (x|e_n)e_n = x.$$

The following result, which is a consequence of the Riesz–Fischer Theorem, answers the second question.

Theorem 3.8 *Let H be a Hilbert space, and let $\{e_n\ ,\ n \in \mathbb{N}\}$ be an orthonormal system in H. The following are equivalent:*

(i) $\{e_n\ ,\ n \in \mathbb{N}\}$ *is a basis (i.e., it is complete).*
(ii) If $x \in H$ *and* $(x|e_n) = 0\ \ \forall n \in \mathbb{N}$, *then* $x = 0$ *(i.e., it is maximal orthonormal).*
(iii) $\{e_n\ ,\ n \in \mathbb{N}\}$ *is closed.*

Proof (i) implies (ii): Let $x \in H$ such that $(x|e_n) = 0$ for all $n \in \mathbb{N}$. Since, by hypothesis, $\{e_n,\ n \in \mathbb{N}\}$ is a basis, there exist $x_n \in \text{LIN}\{e_n\ : n \in \mathbb{N}\}$ such that $x = \lim_{n\to\infty} x_n$. Now, $(x|x_n) = 0$ for every $n \in \mathbb{N}$, hence

$$\|x\|^2 = (x|x) = \lim_{n\to\infty} (x|x_n) = 0,$$

and consequently, $x = 0$.

(ii) implies (iii): Suppose that $\{e_n\ ,\ n \in \mathbb{N}\}$ is not closed. Then there exists $x \in H$, $x \neq 0$, such that

$$(x|x) = \|x\|^2 > \sum_{n=1}^{\infty} |(x|e_n)|^2.$$

Now, since $\{(x|e_n)\}_{n\in\mathbb{N}} \in \ell^2$, by the Riesz–Fischer Theorem, there exists $y \in H$ such that

$$(x|e_n) = (y|e_n) \quad \forall\, n \in \mathbb{N}, \quad \|y\|^2 = \sum_{n=1}^{\infty} |(x|e_n)|^2.$$

Thus,

$$(x - y|e_n) = 0 \quad \forall\, n \in \mathbb{N},$$

and yet $x \neq y$, since

$$(y|y) = \|y\|^2 = \sum_{n=1}^{\infty} |(x|e_n)|^2 < (x|x).$$

(iii) implies (i): Since $\{e_n\ ,\ n \in \mathbb{N}\}$ is closed, for each $x \in H$ we have

$$x = \sum_{n=1}^{\infty} (x|e_n)e_n,$$

which implies $x \in \overline{\text{LIN}\{e_n\ ,\ n \in \mathbb{N}\}}$, and consequently the system $\{e_n\ ,\ n \in \mathbb{N}\}$ is a basis of H. □

Theorem 3.9 *A Hilbert space H is separable if and only if it has a countable orthonormal basis. Moreover, every separable Hilbert space is isometrically isomorphic to the space ℓ^2.*

Proof If H is separable, then as a consequence of Proposition 2.12, there exists a sequence $\{x_n\}$ such that

$$H = \overline{LIN\{x_n\ :\ n \in \mathbb{N}\}}.$$

Then, by the Gram–Schmidt orthogonalization process, there exists an orthonormal system $\{e_n\ ,\ n \in \mathbb{N}\}$ such that for every $n \in \mathbb{N}$,

$$LIN\{e_1, \ldots, e_n\} = LIN\{x_1, \ldots, x_n\},$$

so that $\{e_n\ ,\ n \in \mathbb{N}\}$ is an orthonormal basis of H. The converse is obvious.

Now suppose that H is a separable Hilbert space with basis $\{e_n\ ,\ n \in \mathbb{N}\}$. As a consequence of the Riesz–Fischer Theorem, the map $\mathcal{F} : H \to \ell^2$, defined by $\mathcal{F}(x) := \{(x|e_n)\}$, is an isometric isomorphism. □

Remark 3.1 For Hilbert spaces with uncountable bases, i.e., in the non-separable case, there are results similar to those we have seen. To address this, one must work with the space $\ell^2(I)$, where the index set I is uncountable, and use the concept of a summable family. This concept and the corresponding results can be found, for instance, in the textbook by J. B. Conway [11].

In a Hilbert space, all bases have the same cardinality. This leads to the definition of the *Hilbert dimension* of a Hilbert space as the cardinality of one of its bases. This concept fully characterizes Hilbert spaces, as illustrated by the following result.

Every Hilbert space H is norm isomorphic to $\ell^2(I)$, where I is an index set whose cardinality is the Hilbert dimension of H. Moreover, H is separable if and only if its Hilbert dimension is countable.

3.6 Classical Fourier Series

The general results above allow us to quickly study the classical Fourier series in $L^2(\mathbb{T})$. Since this space is separable, we know it has a countable basis. We are going to construct one that gives rise to the classical theory of Fourier series.

For each $n \in \mathbb{Z}$ we define $e_n : [-\pi, \pi] \longrightarrow \mathbb{C}$ as

$$e_n(t) := e^{int}.$$

The functions e_n are called *trigonometric functions*, and the set $\{e_n\ ,\ n \in \mathbb{Z}\}$ is called the *trigonometric system*. Since

$$(e_n|e_m) = \frac{1}{2\pi}\int_{-\pi}^{\pi} e_n(t)\overline{e_m(t)}\,dt = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{it(n-m)}\,dt = \delta_{n,m},$$

we have that $\{e_n\ ,\ n \in \mathbb{Z}\}$ is an orthonormal system in $L^2(\mathbb{T})$. We denote

$$PT(\mathbb{T}) := LIN\{e_n\ ,\ n \in \mathbb{Z}\}$$

and its elements are called *trigonometric polynomials*. As a consequence of the Stone-Weierstrass Theorem (see, for instance, [11]), we have that $PT(\mathbb{T})$ is dense in $C(\mathbb{T})$, and since $C(\mathbb{T})$ is dense in $L^2(\mathbb{T})$, it follows that $PT(\mathbb{T})$ is dense in $L^2(\mathbb{T})$. Consequently, we have that $\{e_n,\ n \in \mathbb{Z}\}$ is a basis of $L^2(\mathbb{T})$. Given $f \in L^2(\mathbb{T})$, we denote its Fourier coefficients with respect to the trigonometric system by $\hat{f}(n)$, i.e.,

$$\hat{f}(n) = (f|e_n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)e^{-int}\,dt.$$

Thus, the Fourier series generated by the function f is the trigonometric series

$$\sum_{n\in\mathbb{Z}} \hat{f}(n)e^{int}.$$

In this particular case, the *Bessel inequality* tells us that $\{\hat{f}(n)\}_{n\in\mathbb{Z}} \in \ell^2(\mathbb{Z})$, and that

$$\sum_{n\in\mathbb{Z}} |\hat{f}(n)|^2 \leq \|f\|_2.$$

Moreover, since the trigonometric system is an orthonormal basis, we have the *Parseval identity*

$$\|f\|_2 = \|\{\hat{f}(n)\}_{n\in\mathbb{Z}}\|_2 \quad \forall\, f \in L^2(\mathbb{T}).$$

Furthermore, as a consequence of Theorem 3.9, the map $\mathcal{F} : L^2(\mathbb{T}) \to \ell^2(\mathbb{Z})$, defined by $\mathcal{F}(f)(n) = \hat{f}(n)$ for each $n \in \mathbb{Z}$, is an isometric isomorphism. Finally, since the trigonometric system is closed, we see that for each $f \in L^2(\mathbb{T})$, its Fourier series converges to f in $L^2(\mathbb{T})$.

Since the space $\mathbb{T}$ has a finite measure, we have $L^2(\mathbb{T}) \subset L^1(\mathbb{T})$. Now, the Fourier coefficients and series we have considered for functions in $L^2(\mathbb{T})$ can also be defined in the broader context of the space $L^1(\mathbb{T})$, since, if $f \in L^1(\mathbb{T})$, the function $t \mapsto f(t)e^{-int}$ is measurable and $|f(t)e^{-int}| \leq |f(t)|$, so we can define

$$\hat{f}(n) := \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)e^{-int}\,dt, \qquad \forall\, n \in \mathbb{Z}.$$

As in the case of functions in $L^2(\mathbb{T})$, we call $\hat{f}(n)$ the n-th Fourier coefficient of the function f, and the trigonometric series

$$\sum_{n\in\mathbb{Z}} \hat{f}(n)e^{int},$$

the *Fourier series* of f. The partial sums of the Fourier series of f are denoted by

$$S_n(f,t) := \sum_{k=-n}^{n} \hat{f}(k)e^{ikt}, \qquad n \in \mathbb{N}.$$

Taking into account Euler's formulas:

$$\cos(kt) = \frac{e^{ikt} + e^{-ikt}}{2}, \quad \sin(kt) = \frac{e^{ikt} - e^{-ikt}}{2i},$$

we have that

$$S_n(f,t) := \frac{a_0}{2} + \sum_{k=1}^{n} a_k \cos(kt) + b_k \sin(kt),$$

where

$$a_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos(kt)\,dt, \quad b_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\sin(kt)\,dt.$$

Thus, the Fourier series of f can be written in the so-called real form:

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nt) + b_n \sin(nt).$$

If $f \in L^2(\mathbb{T})$, since $\{\hat{f}(n)\} \in \ell^2(\mathbb{Z})$, it follows that $\{\hat{f}(n)\} \in c_0(\mathbb{Z})$. Now, for $f \in L^1(\mathbb{T})$, as a consequence of the classical Riemann–Lebesgue Lemma (see for instance [56]), which states that if $f \in L^1(\mathbb{R})$, then

$$\lim_{|x|\to\infty} \int_{-\infty}^{+\infty} f(t)\, e^{-ixt}\,dt = 0,$$

we also have that if $f \in L^1(\mathbb{T})$, then $\{\hat{f}(n)\} \in c_0(\mathbb{Z})$.

Theorem 3.10 *The mapping $\mathcal{F}: L^1(\mathbb{T}) \to c_0(\mathbb{Z})$, defined by $\mathcal{F}(f)(n) = \hat{f}(n)$ for each $n \in \mathbb{Z}$, is linear, continuous, and injective, but not surjective.*

Proof As we have previously seen, $\mathcal{F}$ is well-defined. Moreover, since it is linear and

$$|\hat{f}(n)| \le \frac{1}{2\pi}\int_{-\pi}^{\pi} |f(t)e^{-int}|\,dt = \|f\|_1,$$

we have that $\|(\hat{f}(n))\|_\infty \le \|f\|_1$, so $\mathcal{F}$ is continuous. Let us show that it is injective. Let $f \in L^1(\mathbb{T})$ be such that $\hat{f}(n) = 0$ for every $n \in \mathbb{Z}$. We must show that $f = 0$. Indeed, if we consider the function

$$H(x) := \int_0^x f(t)\,dt,$$

then, since

$$H(x+2\pi) - H(x) = \int_x^{x+2\pi} f(t)\,dt = 2\pi\,\hat{f}(0) = 0,$$

we have that $H \in C(\mathbb{T})$. Therefore, the function $F(x) := H(x) - \hat{H}(0)$ is also in $C(\mathbb{T})$. Now,

$$\hat{F}(0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} (H(x) - \hat{H}(0))\, dx = \hat{H}(0) - \hat{H}(0) = 0.$$

Furthermore, integrating by parts, we have that for $n \in \mathbb{Z}$, $n \neq 0$,

$$\hat{F}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} F(t)\, e^{-int}\, dt$$

$$= \frac{1}{2\pi} \left(\left[\frac{-1}{in} e^{-int} F(t) \right]_{t=-\pi}^{t=\pi} - \int_{-\pi}^{\pi} \frac{-1}{in}\, e^{-int} f(t)\, dt \right)$$

$$= \frac{1}{2\pi in} \int_{-\pi}^{\pi} f(t) e^{-int}\, dt = \frac{1}{in} \hat{f}(n) = 0.$$

So for every trigonometric polynomial

$$P(t) = \sum_{k=-n}^{k=n} c_k\, e^{ikt},$$

$$\int_{-\pi}^{\pi} F(t) \overline{P(t)}\, dt = 2\pi \sum_{k=-n}^{k=n} c_k \hat{F}(k) = 0.$$

From this, using the density of $PT(\mathbb{T})$ in $C(\mathbb{T})$, we obtain that $F(t) = 0$ for every $t \in [-\pi, \pi]$. Consequently, $f(x) = F'(x) = 0$ for almost every $x \in [-\pi, \pi]$.

To prove that $\mathcal{F}$ is not surjective, we need to recall the following result, which can be found, for instance, in [56]:

If

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nt) + b_n \sin(nt)$$

is the Fourier series of a function $f \in L^1(\mathbb{T})$, then the series

$$\sum_{n=1}^{\infty} \frac{b_n}{n}$$

is convergent.

By Cauchy's condensation test, it is easy to see that the series

$$\sum_{n=2}^{\infty} \frac{1}{n \log n}$$

is not convergent. Then, from the above result, we have that the trigonometric series

$$\sum_{n=2}^{\infty} \frac{\sin(nt)}{\log n}$$

is not the Fourier series of any function in $L^1(\mathbb{T})$, from which we deduce that the mapping $\mathcal{F}$ is not surjective. □

The following result was established in 1912 by W. H. Young in the particular case when $p = \frac{2k}{2k-1}$ $(k = 1, 2, \ldots)$, and in its general version by F. Hausdorff in 1923. Here we present a proof due to A. P. Calderón and A. Zygmund [8] based on the Riesz–Thorin interpolation theorem.

Theorem 3.11 (Hausdorff–Young Theorem) *Let $1 \leq p \leq 2$. If $f \in L^p(\mathbb{T})$, then $(\hat{f}(n)) \in \ell^{p'}(\mathbb{Z})$, and moreover,*

$$\|(\hat{f}(n))\|_{p'} \leq \|f\|_p.$$

Proof Let $\mathcal{F}(f)(n) = \hat{f}(n)$ for each $n \in \mathbb{Z}$. As a consequence of Theorem 3.10, we have that $\mathcal{F} : L^1(\mathbb{T}) \to \ell^\infty(\mathbb{Z})$ is a bounded linear operator with norm $\|\mathcal{F}\|_{1,\infty} \leq 1$. On the other hand, by Parseval's identity, $\mathcal{F} : L^2(\mathbb{T}) \to \ell^2(\mathbb{Z})$ is a bounded linear operator with norm $\|\mathcal{F}\|_{2,2} \leq 1$. Therefore, applying the Riesz–Thorin Theorem (Theorem 2.8), we get that $\mathcal{F} : L^p(\mathbb{T}) \to \ell^q(\mathbb{Z})$ is a bounded linear operator with norm $\|\mathcal{F}\|_{p,q} \leq 1$, if $\frac{1}{p} = \frac{1-\theta}{1} + \frac{\theta}{2}$ and $\frac{1}{q} = \frac{1-\theta}{\infty} + \frac{\theta}{2}$, with $0 < \theta < 1$. Thus, fixing p, we have $\theta = 2 - \frac{2}{p}$, which gives $\frac{1}{q} = \frac{\theta}{2} = \frac{p-1}{p}$, hence $q = p'$. □

Nineteenth-century mathematicians believed that the continuity of a function implied the pointwise convergence of its Fourier series. This conjecture was disproven in 1876 by du Bois Reymond, who constructed a function $f \in C(\mathbb{T})$ whose Fourier series diverges at a point. Other examples of this kind were given by Fejér and by Lebesgue. We will see this result here using methods from Functional Analysis, more specifically, by using the Banach–Steinhaus Theorem. We need to recall from the classical theory of Fourier series (see, for instance, [56]) that if $f \in L^1(\mathbb{T})$, then

$$S_n(f,t) := \sum_{k=-n}^{n} \hat{f}(k) e^{ikt} = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(s) D_n(t-s)\, ds,$$

where

$$D_n(t) := \sum_{k=-n}^{n} e^{ikt} = 1 + 2\sum_{k=1}^{n} \cos(kt) = \frac{\sin((n+\frac{1}{2})t)}{\sin(\frac{t}{2})},$$

where the last equality holds for $t \in \mathbb{R} \setminus 2\pi\mathbb{Z}$. The sequence $\{D_n\}$ is called the *Dirichlet kernel*. Clearly,

$$D_n(t) = D_n(-t) \qquad \forall\, t \in [-\pi, \pi[, \quad n \in \mathbb{N},$$

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} D_n(t)\, dt = 1 \qquad \forall\, n \in \mathbb{N}.$$

Theorem 3.12 *There exists a function $f \in C(\mathbb{T})$ whose Fourier series diverges at a point.*

Proof By making the corresponding change of variables, we have

$$\|D_n\|_1 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |D_n(t)|\, dt = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{|\sin\left((n+\frac{1}{2})t\right)|}{|\sin(\frac{t}{2})|}\, dt$$

$$= \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} \frac{|\sin\left((2n+1)t\right)|}{|\sin t|}\, dt \geq \frac{2}{\pi} \int_{0}^{\pi/2} \frac{|\sin\left((2n+1)t\right)|}{t}\, dt$$

$$= \frac{2}{\pi} \int_{0}^{(2n+1)\pi/2} \frac{|\sin t|}{t}\, dt = \frac{2}{\pi} \sum_{k=0}^{2n} \int_{k\pi/2}^{(k+1)\pi/2} \frac{|\sin t|}{t}\, dt$$

$$\geq \frac{2}{\pi} \sum_{k=0}^{2n} \int_{k\pi/2}^{(k+1)\pi/2} \frac{|\sin t|}{(k+1)\pi/2}\, dt$$

$$= \frac{4}{\pi^2} \sum_{k=0}^{2n} \int_{k\pi/2}^{(k+1)\pi/2} \frac{|\sin t|}{k+1}\, dt = \frac{4}{\pi^2} \sum_{k=0}^{2n} \frac{1}{k+1}.$$

Hence, we have

$$\sup_{n\in\mathbb{N}} \|D_n\|_1 = +\infty. \tag{3.16}$$

For each $n \in \mathbb{N}$, consider the linear functional $\Phi_n : C(\mathbb{T}) \to \mathbb{C}$, defined by

$$\Phi_n(f) := \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) D_n(t)\, dt = S_n(f, 0).$$

Since

$$|\Phi_n(f)| \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} |f(t)|\, |D_n(t)|\, dt$$

$$\leq \|f\|_\infty \frac{1}{2\pi} \int_{-\pi}^{\pi} |D_n(t)|\, dt = \|f\|_\infty \|D_n\|_1,$$

we have that $\Phi_n \in \big(C(\mathbb{T})\big)'$, and $\|\Phi_n\| \leq \|D_n\|_1$. Let us show that $\|\Phi_n\| = \|D_n\|_1$. To this end, consider the functional $\Psi_n : L^1(\mathbb{T}) \to \mathbb{C}$ defined by

$$\Psi_n(f) := \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) D_n(t)\, dt.$$

Since

$$|\Psi_n(f)| \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} |f(t)|\, |D_n(t)|\, dt$$

$$\leq (2n+1) \frac{1}{2\pi} \int_{-\pi}^{\pi} |f(t)|\, dt = (2n+1) \|f\|_1,$$

we see that Ψ_n is continuous. For each $n \in \mathbb{N}$, consider the function

$$g_n(x) = \begin{cases} 1 & \text{if } D_n(t) \geq 0 \\ \\ -1 & \text{if } D_n(t) < 0. \end{cases}$$

Then $g_n \in L^1(\mathbb{T})$, and moreover

$$|\Psi_n(g_n)| = \frac{1}{2\pi} \left| \int_{-\pi}^{\pi} g_n(t) D_n(t)\, dt \right| = \frac{1}{2\pi} \int_{-\pi}^{\pi} |D_n(t)|\, dt = \|D_n\|_1.$$

Now, since Φ_n agrees with Ψ_n on $C(\mathbb{T})$ and $C(\mathbb{T})$ is dense in $L^1(\mathbb{T})$, there exists a sequence $f_m \in C(\mathbb{T})$ with $\|f_m\|_\infty \leq 1$, such that $f_m \to g_n$ in $L^1(\mathbb{T})$. Therefore, $\Phi_n(f_m) \to \Psi_n(g_n)$. Then, given $\epsilon > 0$, there exists $m_0 \in \mathbb{N}$ such that

$$|\Phi_n(f_m) - \Psi_n(g_n)| \leq \epsilon \quad \forall\, m \geq m_0.$$

Hence,

$$|\Phi_n(f_m)| \geq |\Psi_n(g_n)| - \epsilon = \|D_n\|_1 - \epsilon \quad \forall\, m \geq m_0.$$

Since ϵ is arbitrary, we conclude that

$$\|\Phi_n\| = \sup\{|\Phi_n(f)| \,:\, \|f\|_\infty \leq 1\} \geq \|D_n\|_1.$$

Therefore, $\|\Phi_n\| = \|D_n\|_1$, and as a consequence of (3.16), we have

$$\sup_{n \in \mathbb{N}} \|\Phi_n\| = +\infty.$$

Then, applying the Banach-Steinhaus Theorem, we conclude that there exists a function $f \in C(\mathbb{T})$ such that

$$\sup_{n\in\mathbb{N}} |\Phi_n(f)| = \sup_{n\in\mathbb{N}} |S_n(f,0)| = +\infty.$$

Consequently, the Fourier series of f diverges at $x = 0$. □

In 1915, the Russian mathematician Lusin conjectured that for $f \in L^2(\mathbb{T})$, its Fourier series converged to f almost everywhere. This conjecture was proven in 1966 by L. Carleson [9]. In 1967, R. A. Hunt [29] showed that the result also holds if $f \in L^p(\mathbb{T})$ with $1 < p \leq \infty$. The fact that the result is not true for $p = 1$ had been proven in 1926 by Kolmogoroff, who provided an example of a function $f \in L^1(\mathbb{T})$ whose Fourier series diverges at every point. We will now prove, using again the Banach–Steinhaus Theorem, that there exists a function in $L^1(\mathbb{T})$ whose Fourier series does not converge in the L^1-norm. To do this, let us first state the following result.

Lemma 3.1 *Let $1 \leq p < \infty$. Then, $S_n(f) \to f$ in $L^p(\mathbb{T})$ for every $f \in L^p(\mathbb{T})$ if and only if there exists a constant $C_p > 0$ such that*

$$\|S_n(f)\|_p \leq C_p \|f\|_p \quad \forall\, n \in \mathbb{N},\ \forall\, f \in L^p(\mathbb{T}).$$

Proof Suppose that $S_n(f) \to f$ in $L^p(\mathbb{T})$ for every $f \in L^p(\mathbb{T})$. Then, since $\{S_n(f)\}$ is bounded, by the Banach–Steinhaus Theorem, we have that $\{S_n\}$ is uniformly bounded, which is precisely what we are looking for. Conversely, suppose that there exists a constant $C_p > 0$ such that

$$\|S_n(f)\|_p \leq C_p \|f\|_p \quad \forall\, n \in \mathbb{N},\ \forall\, f \in L^p(\mathbb{T}).$$

If $f \in PT(\mathbb{T})$, there exists $n_0 \in \mathbb{N}$ such that $S_n(f) = f$ for every $n \geq n_0$, hence $S_n(f) \to f$ in $L^p(\mathbb{T})$. Given a function $f \in L^p(\mathbb{T})$, since trigonometric polynomials are dense in $L^p(\mathbb{T})$, for $\epsilon > 0$, there exists $g \in PT(\mathbb{T})$ such that

$$\|f - g\|_p \leq \frac{\epsilon}{4C_p}.$$

Now, given g, there exists $n_0 \in \mathbb{N}$ such that

$$\|S_n(g) - g\|_p \leq \frac{\epsilon}{4} \quad \forall\, n \geq n_0.$$

Therefore, for $n \geq n_0$, we have

$$\|S_n(f) - f\|_p \leq \|S_n(f-g)\|_p + \|S_n(g) - g\|_p + \|f - g\|_p$$

$$\leq C_p \|f - g\|_p + \frac{\epsilon}{2} + \frac{\epsilon}{4C_p} \leq \epsilon.$$

□

Theorem 3.13 *There exists a function $f \in L^1(\mathbb{T})$ whose Fourier series does not converge to f in $L^1(\mathbb{T})$.*

Proof By the previous lemma, it is enough to prove that if $S_n : L^1(\mathbb{T}) \to L^1(\mathbb{T})$ is the map $S_n(f)(x) := S_n(f, x)$, then the sequence $\{S_n\}$ is not uniformly bounded. Suppose it were. Then there exists $K > 0$ such that

$$\|S_n(f)\|_1 \leq K \quad \forall\, f \in L^1(\mathbb{T}), \quad \text{with } \|f\|_1 \leq 1.$$

We take the same g_n as in the proof of the previous theorem, i.e., $g_n(t) := \operatorname{sign}(D_n(t))$. Then, if $\|f\|_1 \leq 1$, we have

$$\left|\frac{1}{2\pi}\int_{-\pi}^{\pi} S_n(f) g_n\right| \leq \frac{1}{2\pi}\int_{-\pi}^{\pi} |S_n(f)| = \|S_n(f)\|_1 \leq K.$$

From here, applying Tonelli–Hobson's Theorem, we obtain

$$K \geq \left|\frac{1}{2\pi}\int_{-\pi}^{\pi} S_n(f)(x) g_n(x)\, dx\right|$$

$$= \frac{1}{2\pi}\left|\int_{-\pi}^{\pi} \frac{1}{2\pi}\left(\int_{-\pi}^{\pi} f(z) D_n(x-z)\, dz\right) g_n(x)\, dx\right|$$

$$= \frac{1}{2\pi}\left|\int_{-\pi}^{\pi} \frac{1}{2\pi}\left(\int_{-\pi}^{\pi} D_n(x-z) g_n(x)\, dx\right) f(z)\, dz\right|.$$

Now, defining

$$M_n(z) := \frac{1}{2\pi}\int_{-\pi}^{\pi} D_n(x-z) g_n(x)\, dx,$$

we have that

$$\frac{1}{2\pi}\left|\int_{-\pi}^{\pi} M_n(z) f(z)\, dz\right| \leq K \quad \forall\, f \in L^1(\mathbb{T}), \quad \|f\|_1 \leq 1. \tag{3.17}$$

For each $0 < \epsilon < \pi$, consider the function:

$$f_\epsilon(x) = \begin{cases} \dfrac{\pi}{\epsilon} \text{ if } |x| \leq \epsilon \\ \\ 0 \quad \text{if } |x| > \epsilon. \end{cases}$$

Since $\|f_\epsilon\|_1 = 1$, as a consequence of (3.17), we obtain

$$K \geq \frac{1}{2\pi}\left|\int_{-\pi}^{\pi} M_n(z) f_\epsilon(z)\, dz\right| = \frac{1}{2\pi}\left|\int_{-\epsilon}^{\epsilon} \frac{\pi}{\epsilon} M_n(z)\, dz\right|.$$

Now, since M_n is continuous, applying the Mean Value Theorem, for each $\epsilon > 0$ there exists a z_ϵ with $|z_\epsilon| \leq \epsilon$ such that

$$K \geq \frac{1}{2}|M_n(z_\epsilon)|\left|\int_{-\epsilon}^{\epsilon} \frac{1}{\epsilon}\, dz\right| = |M_n(z_\epsilon)|.$$

Then, letting $\epsilon \to 0^+$, we obtain

$$M_n(0) \leq K \quad \forall\, n \in \mathbb{N}.$$

This is a contradiction, since

$$M_n(0) = \frac{1}{2\pi}\int_{-\pi}^{\pi} D_n(x) g_n(x)\, dx = \frac{1}{2\pi}\int_{-\pi}^{\pi} |D_n(x)|\, dx = \|D_n\|_1,$$

and we know from the proof of the previous theorem that $\sup_{n\in\mathbb{N}} \|D_n\|_1 = +\infty$. □

Problems

3.1 Prove the polarization identity.

3.2 Let H be a Hilbert space. Prove that

$$\|x - z\| = \|x - y\| + \|y - z\|$$

if and only if $y = \alpha x + (1 - \alpha)z$ for some $\alpha \in [0, 1]$.

3.3 Let $1 \leq p \leq \infty$. Prove that if $p \neq 2$, then $L^p([0, 1])$ does not satisfy the parallelogram law, and hence is not a Hilbert space.

3.4 Prove that the space $C([a, b])$, with respect to

$$(f|g) := \int_a^b f(t)\overline{g(t)}\, dt$$

is a pre-Hilbert space but not a Hilbert space.

3.5 Prove that every Hilbert space is uniformly convex, and therefore reflexive (see Problem (2.45)).

3.6 Let H be a Hilbert space. Prove:

(i) If M is a closed subspace of H, then $(M^\perp)^\perp = M$.

(ii) If $A \subset H$, then $(A^{\perp})^{\perp} = \overline{LIN(A)}$.
(iii) If X is a vector subspace of H, then X is dense in H if and only if $X^{\perp} = \{0\}$.

3.7 Let $\{K_n\}$ be a sequence of closed convex subsets of a Hilbert space H.

(i) Suppose that $K_1 \supset K_2 \supset \cdots \supset K_n \supset \cdots$ and $\bigcap_{n\in\mathbb{N}} K_n \neq \emptyset$. Prove that for each $x \in H$, the sequence $y_n := P_{K_n}(x)$ converges in H.
(ii) Suppose that $K_1 \subset K_2 \subset \cdots \subset K_n \subset \cdots$. Prove that for each $x \in H$, the sequence $y_n := P_{K_n}(x)$ converges in H.

3.8 Let H be the vector space of real polynomials defined on $[0, 1]$ with the inner product $(f|g) := \int_0^1 f(t)g(t)dt$. Give an example of a continuous linear functional on H for which the Riesz–Fréchet Theorem does not hold.

3.9 Let Ω be a measurable subset of $\mathbb{R}^n$, and let $A \subset \Omega$ be measurable. Prove that $P_A(f) := \chi_A f$ is an orthogonal projection in the space $L^2(\Omega)$. What conditions must A and B satisfy for $P_A + P_B$ to also be a projection?

3.10 Let K be a nonempty closed convex subset of a Hilbert space H. Let $x \in H$ and $y = P_K(x)$. Prove that

$$\|z - y\|^2 \leq \|z - x\|^2 - \|y - x\|^2 \quad \forall\, z \in K,$$

and thus

$$\|z - y\| \leq \|z - x\| \quad \forall\, z \in K.$$

3.11 Let $X := \{x = (x_n) \in \ell^2 \ : \ x_1 = x_2\}$ be a subspace of ℓ^2. Show that X is closed in ℓ^2 and compute its distance to the point $x = (\frac{1}{n})$.

3.12 Let $(\Omega, \mathcal{B}, \mu)$ be a σ-finite measure space, and let $f_0 : \Omega \to [0, +\infty[$ be a μ-measurable function. Prove that the set

$$K := \{f \in L^2(\Omega, \mu) \ : \ |f(x)| \leq f_0(x) \ \text{ a.e. in } \ \Omega\}$$

is a closed convex subset of $L^2(\Omega, \mu)$, and determine $P_K(f)$ for each $f \in L^2(\Omega, \mu)$.

3.13 Let E and F be two closed vector subspaces of a Hilbert space H. Prove that if $E \perp F$, then $E + F$ is closed.

3.14 Let H be a Hilbert space and let M and N be closed subspaces of H. Prove that if $M \perp N$, then $P_M + P_N$ is the orthogonal projection onto $M \bigoplus N$.

3.15 Let P and Q be orthogonal projections in a Hilbert space. Prove that if $PQ = QP$, then $P + Q - PQ$ is an orthogonal projection.

3.16 Let H be a Hilbert space, X a vector space, and $T : H \to X$ a surjective linear map such that $\operatorname{Ker} T$ is a closed subspace of H.

(i) Prove that for each $x \in X$, there exists a unique element $h_x \in H$ such that

$$\|h_x\| = \min\{\|h\| \ : \ T(h) = x\}.$$

(ii) Show that h_x is the unique element $h \in H$ such that $T(h) = x$ and $(h|y) = 0$ for all $y \in \operatorname{Ker} T$.

(iii) rove that X is a Hilbert space with respect to the inner product $(x|y) := (h_x|h_y)$, whose associated norm is given by

$$\|x\| := \inf\{\|h\| \ : \ T(h) = x\}.$$

3.17 Let $\{x_n\}$ be a sequence in a Hilbert space H. Suppose that $x_n \rightharpoonup x$ weakly. Prove that if

$$\limsup_{n\to\infty} \|x_n\| \leq \|x\|,$$

then $x_n \to x$ strongly.

3.18 Let M be a closed convex subset of the Hilbert space H, and let $T : M \to H$ be a contraction, i.e., a map satisfying:

$$\|Tx - Ty\| \leq \|x - y\| \quad \forall\, x, y \in M.$$

Prove:

(i) If a sequence $\{x_n\} \subset M$ is such that

$$x_n \rightharpoonup x \ \text{ weakly and } \ x_n - Tx_n \to y \ \text{ strongly},$$

then $x - Tx = y$.

(ii) Deduce from (i) that if M is bounded and $T(M) \subset M$, then T has a fixed point, i.e., there exists $x \in M$ such that $Tx = x$.

3.19

(i) Show that if $(\alpha_n) \in \ell^2$, then the power series

$$\sum_{n=0}^{\infty} \alpha_n z^n$$

has radius of convergence greater than or equal to 1.

(ii) For $|\lambda| < 1$, consider the linear functional $T_\lambda : \ell^2 \to \mathbb{C}$ defined by

$$T_\lambda(\{\alpha_n\}) := \sum_{n=0}^{\infty} \alpha_n \lambda^n.$$

Find a vector $x_0 \in \ell^2$ such that $T_\lambda(x) = (x|x_0)$ for every $x \in \ell^2$. Compute $\|T_\lambda\|$.

3.20 For $f, g \in L^1(\mathbb{T})$, their convolution product is defined as

$$(f \star g)(s) := \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) g(s-t)\, dt.$$

Show that $f \star g \in L^1(\mathbb{T})$,

$$\|f \star g\|_1 \leq \|f\|_1 \, \|g\|_1,$$

and

$$\mathcal{F}(f \star g)(n) = \hat{f}(n)\hat{g}(n) \quad \forall\, n \in \mathbb{Z}.$$

Moreover, in the case where $f \in C(\mathbb{T})$, show that $f \star g \in C(\mathbb{T})$, and

$$\|f \star g\|_\infty \leq \|f\|_\infty \, \|g\|_1.$$

3.21 Let

$$A(\mathbb{T}) := \{f \in L^1(\mathbb{T}) \ : \ (\hat{f}(n)) \in \ell^1(\mathbb{Z})\}.$$

Show that $A(\mathbb{T})$ is a Banach space with respect to the norm

$$\|f\|_{A(\mathbb{T})} := \sum_{n=-\infty}^{+\infty} |\hat{f}(n)|.$$

Also show that if $f \in L^1(\mathbb{T})$ and $g \in A(\mathbb{T})$, then $f \star g \in A(\mathbb{T})$, and

$$\|f \star g\|_{A(\mathbb{T})} \leq \|f\|_1 \, \|g\|_{A(\mathbb{T})}.$$

3.22 For each $i = 0, 1, 2, \ldots$, the *Rademacher functions* r_i on $]0, 1[$ are defined by

$$r_i(t) := (-1)^{[2^i t]}, \quad \text{where } [x] = \text{the integer part of } x.$$

Show that $\{r_i\}$ is an orthonormal system in $L^2(]0, 1[)$. Is it a basis?

3.23 Given a non-negative integer $n \geq 0$, we consider it written in base 2, i.e., $n = \sum_{i=0}^{l} \alpha_i 2^i$, with $\alpha_i \in \{0, 1\}$. Then the function W_n on $]0, 1[$ is defined as

$$W_n(t) := \prod_{i=0}^{l} r_{i+1}(t)^{\alpha_i},$$

where r_i are the Rademacher functions. Show that the system $\{W_n\}_{n\geq 0}$, known as the *Walsh system*, is an orthonormal basis of $L^2(]0, 1[)$.

3.24 Consider in $L^2([0, 1])$ the sequence of *Haar functions* $\{h_n\}$ defined by

$$h_1(t) := 1$$

and

$$h_{2^m+k}(t) := \sqrt{2^m} \quad \text{for} \quad \frac{k-1}{2^m} \leq t \leq \frac{2k-1}{2^{m+1}},$$

$$h_{2^m+k}(t) := -\sqrt{2^m} \quad \text{for} \quad \frac{2k-1}{2^{m+1}} \leq t \leq \frac{k}{2^m},$$

$$h_{2^m+k}(t) := 0 \quad \text{otherwise},$$

where $k = 1, 2, \ldots, 2^m$ and $m = 0, 1, 2, \ldots$. Show that the sequence $\{h_n\}$ is an orthonormal basis of $L^2([0, 1])$.

Chapter 4
Spectral Theory of Operators



4.1 Introduction

The historical precedent of spectral theorems is the Principal Axis Theorem from Analytic Geometry. This theorem was already known, for the two-dimensional space, around the first half of the seventeenth century by P. Fermat and R. Descartes, and for three-dimensional space by L. Euler (1748). The general form of the Principal Axis Theorem states that every symmetric quadratic form $(Ax, x) = \sum a_{ij} x_i x_j$ in $\mathbb{R}^n$ can be rewritten, via an orthogonal transformation, in the form $\sum \lambda_i x_i^2$. The generalization from $\mathbb{R}^3$ to $\mathbb{R}^n$ was discovered by J. L. Lagrange in 1759. In 1830, A. L. Cauchy proved that the coefficients λ_i of the normal form must be real.

The next important step was made by J. J. Sylvester and A. Cayley in the mid-nineteenth century, using matrix notation to systematize the algebraic study of finite-dimensional spaces. In 1852, Sylvester proved that the coefficients λ_i of the normal form of (Ax, x) are the roots of the characteristic polynomial $det(\lambda I - A) = 0$. In 1858, Cayley provided the following matrix formulation of the Principal Axis Theorem:

> *If A is a real symmetric matrix, then there exists an orthogonal matrix T such that $D = T^{-1}AT$, where D is the diagonal matrix formed by the eigenvalues of A.*

Modern spectral theory is founded on the fact that certain linear operators in infinite-dimensional spaces can also be represented in a "diagonal" form. Following this idea, at the beginning of the twentieth century, the theory evolved from the finite matrix case to the infinite-dimensional case (infinite matrices and integral operators).

The study of integral equations began in 1837 when J. Liouville introduced the iteration method to solve a specific type of integral equation. This method was extended in 1877 by C. Neumann to a more general setting in his study of the Dirichlet problem for Laplace's equation. The entry of integral equations into the

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J. M. Mazón, *Elements of Functional Analysis*, Universitext,
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history of spectral theory is due to I. Fredholm, with the establishment in 1900 of his famous *Theorem of the Alternative* concerning the solutions ϕ of the integral equation

$$\phi(x) + \int_0^1 K(x, y)\phi(y)\, dy = \psi(x), \quad (\, 0 \leq x \leq 1\,).$$

Motivated by Fredholm's results, it was D. Hilbert who, in a series of six works published in *Göttingen Nachrichten* between 1904 and 1906, began the study of spectral theory in infinite-dimensional spaces.

Hilbert worked with integral equations

$$\phi(x) - \lambda \int_0^1 K(x, y)\phi(y)\, dy = \psi(x),$$

along with their analogous infinite-dimensional matrix equations

$$\phi(x_i) - \lambda \sum_j K(x_i, y_j)\phi(y_j)\, dy = \psi(x_i).$$

In the process of building the necessary tools to solve these equations, Hilbert defined the spectrum of the quadratic form K, distinguished the point spectrum from the continuous spectrum, and introduced the concept of a completely continuous form, a concept that gave rise to that of a compact operator.

Beginning in 1905 with his doctoral thesis,[1] written under Hilbert's supervision, E. Schmidt generalized and simplified Hilbert's work by introducing the suggestive language of Euclidean Geometry. The Hilbert–Schmidt spectral theory extended the Principal Axis Theorem to bounded symmetric linear forms defined on the space ℓ^2, proving that when there is no continuous spectrum—as in the case of completely continuous transformations—they behave as in the finite-dimensional case, that is, up to an orthogonal transformation, they can be written as the sum of a series of finite-rank projections. Schmidt's method does not rely on the results of Fredholm and Hilbert. He uses successive approximations and the Schwarz and Bessel inequalities as his only tools to prove convergence; moreover, he introduces what is now called the "Gram–Schmidt orthogonalization method."

At the same time Hilbert was developing his spectral theory, H. Lebesgue was developing his theory of integration. In 1907, F. Riesz and E. Fischer, combining the two theories, showed that Hilbert space ℓ^2 and the space L^2 of Lebesgue square-integrable functions are isomorphic. This fact allowed Riesz himself, in 1910, to construct a spectral theory for the space L^2 similar to that of Hilbert–Schmidt for ℓ^2.

In 1913, Riesz published his textbook *Les systèmes d'équations linéaires à une infinité d'inconnues* (Gauthier–Villars, Paris), where he presented Hilbert's results

[1] Which was published in: Math. Ann. **63** (1907), 433–476.

using a completely different method: he replaced Hilbert's bilinear forms with bounded linear operators on ℓ^2, gave the definition of the spectrum of an arbitrary operator, and emphasized the importance of complex Hilbert spaces for spectral theory. In the following years, spectral theorems were obtained for increasingly broader classes of operators.

In 1918, Riesz published a work[2] in which he created his famous theory of compact operators. Although he restricts himself to spaces of continuous functions, his method is valid for any Banach space, as he himself states in the introduction:

> *The restriction to continuous functions made in this work is not essential. A reader familiar with recent investigations on function spaces will immediately recognize the broader applicability of the method; they will also recognize that certain of these spaces, such as those of square-integrable functions and infinite-dimensional Hilbert spaces, even allow for simplifications.*

Riesz defined a continuous linear operator as *compact* if it maps every bounded sequence into a compact sequence. He obtained almost all the results of spectral theory for compact operators on Banach spaces, deriving the Fredholm alternative theorem as a particular case. The only result from this theory that was not obtained by Riesz is the one concerning the compactness of the adjoint operator, which was proved by J. Schauder in 1930;[3] for this reason, the theory is nowadays referred to as the Riesz–Schauder theory.

In 1925, W. Heisenberg and E. Schrödinger created Quantum Mechanics. In Heisenberg's theory, the physical fact that certain atomic observations cannot be made simultaneously was interpreted mathematically as the non-commutativity of the operations representing those observations. Since matrix algebra is non-commutative, Heisenberg, together with M. Born and P. Jordan, represented each physical quantity by an appropriate (finite or infinite) matrix; the set of possible values of the physical quantity is the spectrum of that matrix. Thus, the spectrum of the matrix representing the energy of an atom is precisely the spectrum of the atom.

Schrödinger's approach was different and based on his famous wave equation. Schrödinger unified the two theories by proving that the spectrum of the differential operator in his wave equation determines the corresponding Heisenberg matrix.

Hilbert himself was pleasantly surprised that the spectrum of his quadratic forms ended up having an interpretation as an atomic spectrum.

> *I developed my theory of infinite variables from a purely mathematical interest, and even called it 'spectral analysis' without having the slightest premonition that later it would find an application to the real spectrum of physicists,*

Hermann Weyl later said:

> *It was no merit of ours, but a stroke of fortune that, starting in 1923, it was discovered that the spectral theory of Hilbert space was the appropriate mathematical tool for quantum mechanics.*

[2] Acta Mathematica **41** (1918), 71–98.

[3] Studia Mathematica **2** (1930), 183–196.

Although Hilbert's original spectral theorem was for real symmetric forms, the theorem was soon extended to the complex case where the transformation is Hermitian (i.e., $(Ax, y) = (x, Ay)$ for all x, y). This is another fortunate coincidence between mathematics and physics, since physical quantities are measured using real numbers, and such transformations have a real spectrum.

Although each observable in the new mechanics was represented by a Hermitian transformation, not every Hermitian transformation represents an observable. In 1930, P. Dirac added the crucial hypothesis that a Hermitian transformation represents an observable if and only if its eigenvectors form a complete orthonormal system. Although the completely continuous transformations in the spectral theory of Hilbert–Schmidt have this property, that theory was not entirely satisfactory for the study of Quantum Mechanics, because the most important transformations in physics are unbounded in L^2 when dealing with differentiation. The work of generalizing spectral theory to an abstract Hilbert space and addressing the unbounded case was fundamentally carried out by J. von Neumann, who in 1927 provided the mathematical foundation of Quantum Mechanics in his famous textbook *Mathematische Begründung der Quantenmechanik*. Between 1927 and 1930, J. von Neumann published three important articles that revolutionized spectral theory. During this same period, and independently, M. H. Stone obtained results similar to those of J. von Neumann and compiled them in his 1932 textbook *Linear Transformations in Hilbert Spaces*.

The von Neumann–Stone spectral theorem states that if T is a Hermitian operator then

$$T = \int_{\sigma(T)} \lambda \, dE_\lambda,$$

where the operators E_λ are projections with the following properties:

(i) The range of E_λ is contained in the range of E_μ if $\lambda < \mu$.
(ii) $E_{\lambda+\epsilon} \to E_\lambda$ as $\epsilon \to 0$.
(iii) $E_\lambda \to 0$ as $\lambda \to -\infty$.
(iv) $E_\lambda \to I$ as $\lambda \to +\infty$.

A family of projections with the above properties is called a *resolution of the identity*, as named by Stone.

Due to the existence of important operators that are not Hermitian, von Neumann and Stone extended their spectral theorem to a more general class of operators, the *normal operators*.

The collection of all bounded linear operators on a Hilbert space forms a ring. J. von Neumann and F. J. Murray, between 1936 and 1940, extensively studied this ring under different topologies. In the same period, W. P. Steen created an axiomatic theory of operators. However, the most significant work of this period was carried out by the Russian mathematicians I. M. Gelfand, M. A. Naimark, and G. E. Silov, who developed the theory of normed algebras, which C. E. Rickart named in 1946 as *Banach Algebras*. A Banach algebra B is a complete normed algebra (over the

field $\mathbb{C}$) whose norm satisfies the inequality $\|xy\| \leq \|x\| \, \|y\|$. The space $\mathcal{L}(H)$ of bounded linear operators on the Hilbert space H (or more generally, on a Banach space) is a Banach algebra with identity given by the identity operator. By analogy with the case of $\mathcal{L}(H)$, Gelfand defined the spectrum of an element $x \in B$ as the set

$$\sigma(x) := \{ \lambda \in \mathbb{C} \ : \ x - \lambda e \ \text{ is not invertible in } B \},$$

where e is the identity element of the algebra. The set $\sigma(x)$ is compact, non-empty, and contained in the ball of radius $\|x\|$.

The theory of commutative Banach algebras developed by Gelfand relies on two fundamental concepts: homomorphisms and maximal ideals. A homomorphism of a commutative Banach algebra B is a nonzero multiplicative linear functional; its kernel is a maximal ideal. Moreover, every maximal ideal J is the kernel of a homomorphism (in this case, according to a theorem due to Gelfand and Mazur, $\frac{B}{J}$ is the field $\mathbb{C}$). The space K_B of all homomorphisms, called the structure space of the algebra B, is endowed with a topology that gives it the structure of a compact space. Furthermore, with this topology, the functions $\hat{x} : h \ \rightarrow \ h(x)$ are continuous for each $x \in B$. This allows for the definition of one of the main tools of the theory, the Gelfand transform, defined by

$$\Gamma : B \rightarrow C(K_B)$$

given by $\Gamma(x) := \hat{x}$. A fundamental result of the theory is that

$$\sigma(x) = \{ \, h(x) \ : \ h \in K_B \, \}.$$

In 1949, Gelfand and Naimark proved that the commutative Banach algebra $C(K_B)$ is characterized by the presence of an involution, namely complex conjugation. Specifically, they showed that every Banach algebra with involution (what is now called a C^*-*algebra*) is isometrically isomorphic to the algebra $C(K_B)$ for some Banach algebra B.

The impact of the Gelfand–Naimark Theorem on spectral theory lies in the fact that the spectral theorem for bounded normal operators can be proven as a consequence of the isomorphism between the algebra $B(T)$ (generated by the operator T) and the algebra $C(\sigma(T))$, using the classical result that every continuous function on $\sigma(T)$ can be uniformly approximated by functions of the form $\sum f(\lambda_i)\chi_{A_i}$, where χ_{A_i} is the characteristic function of a measurable set A_i. Thus, the theory of Banach algebras reveals that the spectral theorem is fundamentally equivalent to a simple result from function theory.

4.2 The Algebra $\mathcal{L}(X)$. Spectrum of an Operator

Throughout this chapter, X denotes a complex normed space. We denote by $\mathcal{L}(X)$ the normed space $\mathcal{L}(X, X)$ of all bounded linear operators from X to X, endowed with the norm

$$\|T\| = \sup\{\|Tx\| \ : \ \|x\| \leq 1\}, \qquad T \in \mathcal{L}(X).$$

If we define the multiplication of $T, S \in \mathcal{L}(X)$ as:

$$(ST)(x) := S(Tx), \qquad \forall\, x \in X,$$

then $\mathcal{L}(X)$ is an algebra with identity, with the identity operator I given by $I(x) := x$ for each $x \in X$. Moreover, we have

$$\|ST\| \leq \|S\|\,\|T\|,$$

since

$$\|ST\| = \sup_{\|x\|\leq 1} \|(ST)x\| = \sup_{\|x\|\leq 1} \|S(Tx)\| \leq \|S\| \sup_{\|x\|\leq 1} \|Tx\| = \|S\|\,\|T\|.$$

Remark 4.1 A Banach space X that has a multiplication operation giving it an algebra structure, and in addition satisfies

$$\|ef\| \leq \|e\|\,\|f\| \qquad \forall\, e, f \in X,$$

is called a *Banach algebra*. Thus, if X is a Banach space, then $\mathcal{L}(X)$ is a Banach algebra. Another example of a Banach algebra is provided by the space $C(K)$, where K is a compact space, equipped with the uniform norm:

$$\|f\|_\infty = \sup_{x\in K} |f(x)|,$$

since it is trivially true that

$$\|fg\|_\infty \leq \|f\|_\infty\,\|g\|_\infty \quad \forall\, f, g \in C(K).$$

Much of the spectral theory that we are going to develop in $\mathcal{L}(X)$ can also be developed in the more general context of a Banach algebra (see, for example, [11], [57]).

We are going to introduce a type of operator between Lebesgue spaces that is very useful: the *multiplication operators*.

Proposition 4.1 *Let $\Omega \subset \mathbb{R}^N$, with $0 < \mathcal{L}^N(\Omega) < +\infty$, where $\mathcal{L}^N$ denotes the Lebesgue measure on $\mathbb{R}^N$. Let $1 \leq p < \infty$, and let $\phi \in L^\infty(\Omega)$. If we define the operator $M_\phi : L^p(\Omega) \to L^p(\Omega)$ by*

$$M_\phi f(x) := \phi(x) f(x) \quad \forall\, x \in \Omega,$$

then $M_\phi \in \mathcal{L}(L^p(\Omega))$, and moreover, $\|M_\phi\| = \|\phi\|_\infty$. The operator M_ϕ is called a multiplication operator.

Proof Clearly, M_ϕ is linear. Moreover,

$$\|M_\phi f\|_p = \left(\int_\Omega |\phi(x) f(x)|^p \, dx \right)^{\frac{1}{p}} \leq \|\phi\|_\infty \, \|f\|_p,$$

which implies that $M_\phi \in \mathcal{L}(L^p(\Omega))$, and $\|M_\phi\| \leq \|\phi\|_\infty$. Now we prove the reverse inequality.

Given $\epsilon > 0$, let

$$B_\epsilon := \{x \in \Omega \,:\, |\phi(x)| > \|\phi\|_\infty - \epsilon\}.$$

Then $\mathcal{L}^N(B_\epsilon) > 0$, and since $B_\epsilon \subset \Omega$, we have $\mathcal{L}^N(B_\epsilon) < +\infty$. Therefore, $\chi_{B_\epsilon} \in L^p(\Omega)$. Now, define

$$f := \frac{\chi_{B_\epsilon}}{\left(\mathcal{L}^N(B_\epsilon)\right)^{\frac{1}{p}}},$$

so that $\|f\|_p = 1$. Then

$$\|M_\phi\| \geq \|M_\phi(f)\|_p = \left(\int_\Omega |\phi(x) f(x)|^p \, dx \right)^{\frac{1}{p}}$$

$$= \frac{1}{\left(\mathcal{L}^N(B_\epsilon)\right)^{\frac{1}{p}}} \left(\int_{B_\epsilon} |\phi(x)|^p \, dx \right)^{\frac{1}{p}}$$

$$\geq \frac{1}{\mu(B_\epsilon)^{\frac{1}{p}}} (\|\phi\|_\infty - \epsilon) \mu(B_\epsilon)^{\frac{1}{p}} = \|\phi\|_\infty - \epsilon.$$

Since this holds for every $\epsilon > 0$, we conclude that $\|M_\phi\| \geq \|\phi\|_\infty$. □

Definition 4.1 Let $T \in \mathcal{L}(X)$. We say that T is *Invertible* (in the algebra $\mathcal{L}(X)$) if T is bijective and $T^{-1} \in \mathcal{L}(X)$. Note that when X is a Banach space, as a consequence of the Open Mapping Theorem, it suffices for T to be bijective in order to be invertible.

The following result is known as the *General Invertibility Criterion*.

Theorem 4.1 *Let X be a Banach space, and $T \in \mathcal{L}(X)$. If the series*

$$\sum_{n=0}^{\infty} T^n$$

converges in $\mathcal{L}(X)$, then $I - T$ is invertible and moreover

$$(I - T)^{-1} = \sum_{n=0}^{\infty} T^n. \tag{4.1}$$

A sufficient condition for (4.1) to hold is that $\|T\| < 1$. In this case, we have

$$\|(I - T)^{-1}\| \leq \frac{1}{1 - \|T\|}.$$

If we denote by $\mathcal{G}(X)$ the set of invertible elements of $\mathcal{L}(X)$, then $\mathcal{G}(X)$ is an open set in $\mathcal{L}(X)$.

Proof If

$$S := \sum_{n=0}^{\infty} T^n \in \mathcal{L}(X),$$

then, since $S(I - T) = (I - T)S = I$, it follows that $S = (I - T)^{-1}$.

Now suppose that $\|T\| < 1$. Since for every $n \in \mathbb{N}$, $\|T^n\| \leq \|T\|^n$, it follows that the series $\sum_{n=0}^{\infty} \|T^n\|$ is convergent. Therefore, since $\mathcal{L}(X)$ is a Banach space, the series $\sum_{n=0}^{\infty} T^n$ converges in $\mathcal{L}(X)$. Moreover,

$$\|(I - T)^{-1}\| = \left\| \sum_{n=0}^{\infty} T^n \right\| \leq \sum_{n=0}^{\infty} \|T^n\| \leq \sum_{n=0}^{\infty} \|T\|^n = \frac{1}{1 - \|T\|}.$$

Finally, let us show that $\mathcal{G}(X)$ is open in $\mathcal{L}(X)$. Let $T \in \mathcal{G}(X)$, then

$$B\left(T, \frac{1}{\|T^{-1}\|}\right) \subset \mathcal{G}(X).$$

Indeed, let $S \in \mathcal{L}(X)$ with $\|T - S\| < \frac{1}{\|T^{-1}\|}$. Then

$$\|T^{-1}S - I\| = \|T^{-1}(S - T)\| \leq \|T^{-1}\| \, \|S - T\| < 1.$$

Therefore, $T^{-1}S = I - (I - T^{-1}S) \in \mathcal{G}(X)$, which implies that $S \in \mathcal{G}(X)$. □

Due to the connection of formula (4.1) with a result of C. Neumann on integral equations, the series in (4.1) is usually called the *Neumann series*.

Definition 4.2 Given an operator $T \in \mathcal{L}(X)$, the *resolvent set* of T is defined as

$$\rho(T) := \{\lambda \in \mathbb{C} \ : \ (\lambda I - T) \in \mathcal{G}(X)\}.$$

The complement of the resolvent set of T is called the *spectrum* of T, and is denoted by $\sigma(T)$, i.e.,

$$\sigma(T) := \mathbb{C} \setminus \rho(T) = \{\lambda \in \mathbb{C} \ : \ (\lambda I - T) \ \text{ is not invertible in } \mathcal{L}(X)\}.$$

For each $\lambda \in \rho(T)$, the operator

$$R(\lambda, T) := (\lambda I - T)^{-1}$$

is called the *resolvent operator* of T.

One possibility for $\lambda \in \sigma(T)$ is that the operator $(\lambda I - T)$ is not injective. In this case, λ is called an *eigenvalue* of T. The set of all eigenvalues of T is called the *point spectrum* of T, and is denoted by $\sigma_p(T)$, i.e.,

$$\sigma_p(T) := \{\lambda \in \mathbb{C} \ : \ (\lambda I - T) \ \text{ is not injective}\}.$$

When $\lambda \in \sigma_p(T)$, then $\text{Ker}(\lambda I - T) \neq \{0\}$. In this case, $\text{Ker}(\lambda I - T)$ is called the *eigenspace* associated with the eigenvalue λ, and its nonzero elements are called *eigenvectors* of T associated with λ, i.e., an element $x \in X$, $x \neq 0$, is an eigenvector of T if there exists $\lambda \in \mathbb{C}$ such that $Tx = \lambda x$.

In the finite-dimensional case, i.e., for $X = \mathbb{K}^N$, we have that $\sigma_p(T) = \sigma(T)$ for every $T \in \mathcal{L}(\mathbb{K}^N)$. However, this is not true in infinite dimension, as the following example shows.

Example 4.1 Let $S : \ell^p \to \ell^p$ be the *right shift* operator, i.e., the operator defined by

$$S(x_1, x_2, \ldots, x_n, \ldots) := (0, x_1, x_2, \ldots, x_n, \ldots).$$

Since S is not surjective, we have $0 \in \sigma(S)$. However, since S is injective, we have $0 \notin \sigma_p(S)$.

Theorem 4.2 *Let X be a Banach space and $T \in \mathcal{L}(X)$. Then, $\sigma(T)$ is a compact subset of $\mathbb{C}$. Moreover, for each $\lambda \in \mathbb{C}$ such that $|\lambda| > \|T\|$, we have $\lambda \in \rho(T)$, and*

$$R(\lambda, T) = \sum_{n=0}^{\infty} \frac{T^n}{\lambda^{n+1}}.$$

Proof Let $\lambda \in \rho(T)$. Then $\lambda I - T \in \mathcal{G}(X)$, and since $\mathcal{G}(X)$ is an open subset of $\mathcal{L}(X)$, there exists $\epsilon > 0$ such that, if $|\lambda - \mu| < \epsilon$, then $\mu I - T \in \mathcal{G}(X)$. Therefore, $D(\lambda, \epsilon) \subset \rho(T)$, which implies that $\rho(T)$ is open in $\mathbb{C}$, and consequently, $\sigma(T)$ is closed. Let us show that $\sigma(T) \subset D(0, \|T\|)$, so that $\sigma(T)$ is bounded, and thus compact. Indeed, let $\lambda \in \mathbb{C}$ with $|\lambda| > \|T\|$. Then,

$$\lambda I - T = \lambda \left(I - \frac{1}{\lambda} T \right),$$

and since

$$\left\| \frac{1}{\lambda} T \right\| = \frac{\|T\|}{|\lambda|} < 1,$$

applying the General Invertibility Criterion, we obtain that the operator $\lambda I - T$ is invertible, hence $\lambda \in \rho(T)$, and furthermore,

$$(\lambda I - T)^{-1} = \frac{1}{\lambda} \left(I - \frac{1}{\lambda} T \right)^{-1} = \frac{1}{\lambda} \sum_{n=0}^{\infty} \frac{T^n}{\lambda^{n+1}}.$$

□

Theorem 4.3 *Let X be a Banach space and $T \in \mathcal{L}(X)$. Then,*

(i) The resolvent identity holds:

$$R(\mu, T) - R(\lambda, T) = (\lambda - \mu) R(\lambda, T) R(\mu, T) \quad \text{if } \mu, \lambda \in \rho(T).$$

(ii) For each $x \in X$ and $u \in X'$, the function $\varphi : \rho(T) \to \mathbb{C}$ defined by

$$\varphi(\lambda) := \langle u, R(\lambda, T)x \rangle$$

is holomorphic in $\rho(T)$ and $\lim_{|\lambda| \to \infty} \varphi(\lambda) = 0$.

(iii) The spectrum $\sigma(T)$ is nonempty.

Proof (i): Let $\mu, \lambda \in \rho(T)$. Then,

$$R(\mu, T) - R(\lambda, T) = R(\mu, T)\big[(\lambda I - T) - (\mu I - T)\big] R(\lambda, T)$$

$$= (\lambda - \mu) R(\lambda, T) R(\mu, T).$$

(ii): First, let us show that the map $\lambda \in \rho(T) \mapsto R(\lambda, T) \in \mathcal{L}(X)$ is continuous. Let $\lambda \in \rho(T)$ and $\mu \in \mathbb{C}$. Then we have:

$$(\lambda + \mu) I - T = (\lambda I - T) + \mu I = (\lambda I - T)\big[I + \mu R(\lambda, T)\big]. \tag{4.2}$$

Now, since $\rho(T)$ is open, for μ small enough, $\lambda + \mu \in \rho(T)$, hence $(\lambda + \mu)I - T$ has an inverse $R(\lambda + \mu, T)$. Moreover, from (4.2), we have:

$$R(\lambda + \mu, T) = \big[I + \mu R(\lambda, T)\big]^{-1} R(\lambda, T).$$

Therefore,

$$\lim_{\mu \to 0} R(\lambda + \mu, T) = \lim_{\mu \to 0} \big[I + \mu R(\lambda, T)\big]^{-1} R(\lambda, T) = R(\lambda, T),$$

which proves that the map $\lambda \mapsto R(\lambda, T)$ is continuous from $\rho(T)$ into $\mathcal{L}(X)$. Consequently, using the resolvent identity, we obtain:

$$\lim_{\mu \to \lambda} \frac{R(\mu, T) - R(\lambda, T)}{\mu - \lambda} = -\lim_{\mu \to \lambda} R(\mu, T) R(\lambda, T) = -R(\lambda, T)^2. \tag{4.3}$$

For $x \in X$ and $u \in X'$, consider the function

$$\varphi(\lambda) := \langle u, R(\lambda, T)x \rangle.$$

Taking into account (4.3), we obtain:

$$\varphi'(\lambda) = \lim_{\mu \to \lambda} \frac{\varphi(\mu) - \varphi(\lambda)}{\mu - \lambda}$$

$$= \lim_{\mu \to \lambda} \left\langle u, \frac{R(\mu, T)x - R(\lambda, T)x}{\mu - \lambda} \right\rangle = -\langle u, R(\lambda, T)^2 x \rangle.$$

Therefore, φ is holomorphic in $\rho(T)$. Moreover, since

$$R(\lambda, T) = \frac{1}{\lambda} \sum_{n=0}^{\infty} \frac{T^n}{\lambda^n} \quad \text{for } |\lambda| > \|T\|,$$

we have

$$\lim_{|\lambda| \to \infty} \|R(\lambda, T)\| = 0,$$

and thus

$$\lim_{|\lambda| \to \infty} \varphi(\lambda) = 0.$$

(iii): Suppose that $\sigma(T) = \emptyset$. Then the function φ from (ii) is holomorphic on all of $\mathbb{C}$, and since $\lim_{|\lambda| \to \infty} \varphi(\lambda) = 0$, by Liouville's Theorem we get that $\varphi \equiv 0$. Therefore,

$$\langle u, R(\lambda, T)x\rangle = 0 \quad \forall\, x \in X, \text{ and } \forall\, u \in X'.$$

Hence $R(\lambda, T) = 0$, which is a contradiction. □

Remark 4.2 If the Banach space is real, there exist operators with empty spectrum. For example, if $T \in \mathcal{L}(\mathbb{R}^2)$ is the operator whose associated matrix is

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

we have that $\sigma(T) = \emptyset$, since $\lambda \in \rho(T) \iff \det(\lambda I - A) \neq 0$, and since

$$\det(\lambda I - A) = \begin{vmatrix} \lambda & 1 \\ -1 & \lambda \end{vmatrix} = \lambda^2 + 1 \neq 0$$

for every $\lambda \in \mathbb{R}$, we obtain that $\rho(T) = \mathbb{R}$.

From the previous theorems we know that if $T \in \mathcal{L}(X)$, then $\sigma(T)$ is a non-empty compact subset of $\mathbb{C}$ contained in $D(0, \|T\|)$. The *spectral radius* of T is defined as

$$r(T) := \sup\{|\lambda| \,:\, \lambda \in \sigma(T)\}.$$

Thus, we have that $0 \leq r(T) \leq \|T\|$.

We will now provide an explicit formula for the computation of the spectral radius of an operator. To that end, we first establish the following result.

Lemma 4.1 *For every $T \in \mathcal{L}(X)$, we have*

$$c(T) := \inf_{n\in\mathbb{N}} \sqrt[n]{\|T^n\|} = \lim_{n\to\infty} \sqrt[n]{\|T^n\|}.$$

Moreover, the series $\sum_{n=0}^{\infty} T^n$ is convergent if $c(T) < 1$, and divergent if $c(T) > 1$.

Proof Given $\epsilon > 0$, choose $m \in \mathbb{N}$ such that

$$\sqrt[m]{\|T^m\|} < c(T) + \epsilon.$$

Let $M := \max\{1, \|T\|, \ldots, \|T^m\|\}$. Given $n \in \mathbb{N}$, write $n = k_n m + l_n$, with $0 \leq l_n \leq m - 1$. Then, since $\|T^k\| \leq \|T\|^k$ for each $k \in \mathbb{N}$, we have

$$\sqrt[n]{\|T^n\|} = \sqrt[n]{\|T^{k_n m} T^{l_n}\|} \leq \sqrt[n]{\|T^m\|^{k_n} \|T^{l_n}\|}$$

$$\leq \sqrt[n]{M}\, \|T^m\|^{\frac{k_n}{n}} < \sqrt[n]{M}\, (c(T) + \epsilon)^{\frac{n-l_n}{n}}.$$

Now, since

$$\lim_{n\to\infty} \sqrt[n]{M}(c(T)+\epsilon)^{\frac{n-l_n}{n}} = c(T)+\epsilon,$$

there exists $N_\epsilon \in \mathbb{N}$, such that

$$\sqrt[n]{M}(c(T)+\epsilon)^{\frac{n-l_n}{n}} < c(T)+2\epsilon \quad \forall\, n \geq N_\epsilon.$$

Therefore, for each $n \geq N_\epsilon$, we have

$$c(T) \leq \sqrt[n]{\|T^n\|} < c(T)+2\epsilon.$$

Consequently,

$$c(T) = \lim_{n\to\infty} \sqrt[n]{\|T^n\|}.$$

Finally, by the Cauchy criterion, the series $\sum_{n=0}^{\infty} \|T^n\|$ is convergent if $c(T) < 1$, and divergent if $c(T) > 1$. □

Theorem 4.4 *For every $T \in \mathcal{L}(X)$ we have*

$$r(T) = \inf_{n\in\mathbb{N}} \sqrt[n]{\|T^n\|} = \lim_{n\to\infty} \sqrt[n]{\|T^n\|}. \tag{4.4}$$

Proof Consider the series

$$\sum_{n=0}^{\infty} \frac{T^n}{\lambda^{n+1}} = \frac{1}{\lambda} \sum_{n=0}^{\infty} \left(\frac{T}{\lambda}\right)^n.$$

By the previous lemma, we know that this series converges in $\mathcal{L}(X)$ if

$$c\left(\frac{T}{\lambda}\right) = \frac{1}{|\lambda|} c(T) < 1,$$

i.e., if $|\lambda| > c(T)$.

On the other hand, by Theorem 4.3, if $u \in X'$, and $x \in X$, the function $\varphi(\lambda) := \langle u, R(\lambda, T)x\rangle$ is holomorphic on $\rho(T)$. Moreover, for $|\lambda| > c(T)$, we have

$$\varphi(\lambda) = \frac{1}{\lambda} \sum_{n=0}^{\infty} \langle u, T^n x\rangle\, \lambda^{-n}. \tag{4.5}$$

Then (4.5) is the Laurent series of φ, and by uniqueness of Laurent series, it must be the Laurent series of φ in the annulus $\{\lambda \in \mathbb{C} : |\lambda| > r(T)\}$. Therefore, $r(T) \leq c(T)$. Suppose that $r(T) < c(T)$. Then, if we take λ with $r(T) < |\lambda| < c(T)$, we have that for such λ, (4.5) holds. So,

$$\lim_{n\to\infty} \langle u, T^n x\rangle \lambda^{-n} = 0 \quad \forall\, x \in X,\ \forall\, u \in X'.$$

Consequently,

$$\lambda^{-n} T^n x \rightharpoonup 0 \quad \text{in } \sigma(X, X') \quad \forall\, x \in X.$$

Therefore, as a consequence of the Banach–Steinhaus Theorem, there exists $M > 0$, such that

$$\|\lambda^{-n} T^n\| \leq M \qquad \forall\, n \in \mathbb{N}.$$

Then,

$$\frac{1}{|\lambda|} \sqrt[n]{\|T^n\|} \leq \sqrt[n]{M}.$$

so that,

$$c(T) = \lim_{n\to\infty} \sqrt[n]{\|T^n\|} \leq |\lambda|,$$

which is a contradiction. □

Formula (4.4) is called the *Gelfand–Beurling formula*.

Even in finite-dimensional spaces, there exist nonzero elements with spectral radius zero. Indeed, let $T \in \mathcal{L}(\mathbb{C}^2)$, with associated matrix

$$A = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Then, since

$$A^2 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix},$$

we have that $T^2 = 0$, and as a consequence of the Gelfand–Beurling formula, $r(T) = 0$.

4.3 Integral Operators

One of the most common types of operators that appear in applications are the so-called *integral operators*, i.e., operators of the form

$$(Tf)(x) = \int_\Omega K(x, y) f(y)\, dy,$$

where $\Omega \subset \mathbb{R}^N$, and $K : \Omega \times \Omega \to \mathbb{C}$ is a function called the *kernel* of the integral operator. For example, if $f \in C(\partial B_r)$, where B_r is the ball in $\mathbb{R}^N$ centered at 0 with radius r, then the solution to the Dirichlet problem

$$\begin{cases} \Delta u = 0 & \text{in } B_r \\ \\ u = f & \text{on } \partial B_r \end{cases}$$

is given by the *Poisson formula*

$$u(x) = \frac{r^2 - \|x\|^2}{\sigma_N r} \int_{\partial B_r} \frac{f(y)}{\|x - y\|^N}\, ds(y).$$

Now, if we define

$$K_r(x, y) := \frac{r^2 - \|x\|^2}{\sigma_N r \|x - y\|^N}, \quad x \in B_r,\ y \in \partial B_r,$$

i.e., K_r is the *Poisson kernel* of B_r, then the solution to the Dirichlet problem is given by

$$(Tf)(x) = \int_{\partial B_r} K_r(x, y) f(y)\, ds(y).$$

Another example is provided by the representation of the n-th partial sum of the Fourier series of a function $f \in L^1(\mathbb{T})$ using the Dirichlet kernel

$$D_n(t) = \frac{\sin[(n + \frac{1}{2})t]}{\sin(\frac{t}{2})}.$$

We have that

$$S_n(f, x) := \sum_{k=-n}^{n} \hat{f}(k) e^{ikx} = \frac{1}{2\pi} \int_{-\pi}^{\pi} D_n(x - t) f(t)\, dt.$$

Then, if

$$K_n(x, y) := \frac{1}{2\pi} D_n(x - y),$$

we obtain

$$S_n(f, x) = \int_{-\pi}^{\pi} K_n(x, y) f(y)\, dy.$$

Theorem 4.5 *Let $K : [a, b] \times [a, b] \to \mathbb{C}$ be a continuous function. Then, the integral operator $T : C([a, b]) \to C([a, b])$ defined by*

$$(Tf)(x) := \int_a^b K(x, y) f(y)\, dy$$

is linear and continuous. Moreover,

$$\|T\| = \max_{x\in[a,b]} \int_a^b |K(x, y)|\, dy. \tag{4.6}$$

Proof Clearly, T is linear. Let

$$M := \max_{x\in[a,b]} \int_a^b |K(x, y)|\, dy.$$

Then,

$$\|Tf\|_\infty = \sup_{x\in[a,b]} |Tf(x)| \le \sup_{x\in[a,b]} \int_a^b |K(x, y)|\, |f(y)|\, dy \le M\, \|f\|_\infty.$$

Therefore, $T \in \mathcal{L}(C([a, b]))$, and $\|T\| \le M$. Let us now show the reverse inequality. Since the function

$$x \mapsto \int_a^b |K(x, y)|\, dy$$

is continuous on $[a, b]$, there exists $x_0 \in [a, b]$ such that

$$M = \int_a^b |K(x_0, y)|\, dy.$$

Define $\phi : C([a, b]) \to \mathbb{C}$ by

$$\phi(f) := \int_a^b K(x_0, y) f(y)\, dy.$$

Clearly, ϕ is linear, and since

$$|\phi(f)| \le \int_a^b |K(x_0, y)|\, |f(y)|\, dy \le M \|f\|_\infty,$$

we have that $\phi \in \big(C([a, b])\big)'$, and $\|\phi\| \le M$. Moreover, it is not difficult to show that $\|\phi\| = M$ (see Problem 2.9). Therefore, given any $\epsilon > 0$, there exists a function $f_0 \in C([a, b])$ with $\|f_0\|_\infty \le 1$, such that

$$\phi(f_0) \geq \|\phi\| - \epsilon = M - \epsilon.$$

Then,

$$\|T\| \geq \|T(f_0)\|_\infty \geq T(f_0)(x_0) = \int_a^b K(x_0, y) f_0(y)\, dy = \phi(f_0) \geq M - \epsilon.$$

And since this holds for every $\epsilon > 0$, we conclude that $\|T\| = M$. □

An important type of integral operators are the so-called *Volterra integral operators*, named after Vito Volterra, who introduced them to solve ordinary differential equations via integral equations. These are integral operators of the form

$$Vf(x) = \int_0^x K(x, y) f(y)\, dy.$$

Consider the problem

$$\begin{cases} u''(x) + a_1(x)u'(x) + a_2(x)u(x) = F(x) \\ \\ \qquad\qquad u(0) = c_0, \quad u'(0) = c_1 \end{cases} \tag{4.7}$$

Then, if u is a solution of (4.7), and we set $\phi(x) := u''(x)$, it is not difficult to see that ϕ satisfies the so-called *Volterra integral equation*

$$\phi(x) + \int_0^x K(x, y)\phi(y)\, dy = f(x),$$

with

$$K(x, y) = a_2(x)(x - y) + a_1(x), \qquad f(x) := F(x) - c_1 a_1(x) - (c_1 x + c_0)a_2(x).$$

Using the Gelfand-Beurling formula, we will show that Volterra integral operators have spectral radius zero.

Proposition 4.2 *Let $K : [a, b] \times [a, b] \to \mathbb{C}$ be a continuous kernel, and let $V : C([a, b]) \to C([a, b])$ be the Volterra integral operator*

$$Vf(x) = \int_a^x K(x, y) f(y)\, dy.$$

Then, $r(V) = 0$.

Proof Let $M := \max\{|K(x, y)| : (x, y) \in [a, b] \times [a, b]\}$. Then,

$$|Vf(x)| \leq \int_a^x M\|f\|_\infty\, dy = M\|f\|_\infty x.$$

Thus,

$$|V^2(f)(x)| \leq \int_a^x |K(x,y)|\,|Vf(y)|\,dy \leq \int_a^x M^2\|f\|_\infty y\,dy$$

$$\leq M^2\|f\|_\infty \frac{x^2}{2}.$$

Repeating this process, we obtain

$$|V^n(f)(x)| \leq M^n\|f\|_\infty \frac{x^n}{n!} \quad \forall\, x \in [a,b].$$

Therefore, if $\alpha := \max\{|a|, |b|\}$, we have

$$\|V^n(f)\|_\infty \leq M^n\|f\|_\infty \frac{\alpha^n}{n!}.$$

Consequently,

$$\|V^n\| \leq M^n \frac{\alpha^n}{n!}.$$

From this, applying the Gelfand-Beurling formula, it follows that

$$r(V) = \lim_{n\to\infty} \sqrt[n]{\|V^n\|} \leq \lim_{n\to\infty} \sqrt[n]{M^n \frac{\alpha^n}{n!}} = M\alpha \lim_{n\to\infty} \sqrt[n]{n!} = 0.$$

□

In many applications, it is more natural to consider integral operators between Lebesgue spaces rather than between spaces of continuous functions. For these operators, we have the following result.

Theorem 4.6 *Let $K : [a,b] \times [a,b] \to \mathbb{C}$ be a measurable kernel such that there exist constants a_1 and a_2 for which:*

$$\int_a^b |K(x,y)|\,dy \leq a_1 \quad \textit{for almost every } x \in [a,b], \tag{4.8}$$

$$\int_a^b |K(x,y)|\,dx \leq a_2 \quad \textit{for almost every } y \in [a,b]. \tag{4.9}$$

Let $1 < p < \infty$, and let $T : L^p([a,b]) \to L^p([a,b])$ be the integral operator defined by

$$(Tf)(x) := \int_a^b K(x, y) f(y)\, dy.$$

Then $T \in \mathcal{L}(L^p([a, b]))$*, and moreover*

$$\|T\| \leq \sqrt[p']{a_1}\, \sqrt[p]{a_2}.$$

Proof The fact that T is well defined follows from the argument we will use to show that T is bounded. Given $f \in L^p([a, b])$, applying Hölder's inequality, we have

$$|Tf(x)| \leq \int_a^b |K(x, y)|\, |f(y)|\, dy = \int_a^b |K(x, y)|^{\frac{1}{p'}} |K(x, y)|^{\frac{1}{p}}\, |f(y)|\, dy$$

$$\leq \left(\int_a^b |K(x, y)|\, dy \right)^{\frac{1}{p'}} \left(\int_a^b |K(x, y)|\, |f(y)|^p\, dy \right)^{\frac{1}{p}}$$

$$\leq \sqrt[p']{a_1} \left(\int_a^b |K(x, y)|\, |f(y)|^p\, dy \right)^{\frac{1}{p}}.$$

From here, applying the Tonelli-Hobson Theorem, we obtain

$$\int_a^b |Tf(x)|^p\, dx \leq a_1^{\frac{p}{p'}} \int_a^b \left(\int_a^b |K(x, y)|\, |f(y)|^p\, dy \right) dx$$

$$= a_1^{\frac{p}{p'}} \int_a^b |f(y)|^p \left(\int_a^b |K(x, y)|\, dx \right) dx = a_1^{\frac{p}{p'}} a_2 \|f\|_p^p.$$

Thus, $Tf \in L^p([a, b])$, and moreover

$$\|Tf\|_p \leq \sqrt[p']{a_1}\, \sqrt[p]{a_2}\, \|f\|_p.$$

Hence, we conclude that $\|T\| \leq \sqrt[p']{a_1}\, \sqrt[p]{a_2}$. □

4.4 Adjoint of an Operator

Let X and Y be vector spaces and let $T \in L(X, Y)$. The *transpose* of T is defined as the linear operator $T^t \in L(Y^*, X^*)$ given by

$$\langle T^t u, x \rangle = \langle u, Tx \rangle \qquad \forall\, x \in X, \ \ \forall\, u \in Y^*.$$

Note that in the finite-dimensional case, i.e., when $X = \mathbb{K}^n$ and $Y = \mathbb{K}^m$, if A is the matrix associated to $T \in L(X, Y)$, then T^t is the linear operator whose associated matrix is the transpose of A.

In the case where $T \in \mathcal{L}(X, Y)$, the *adjoint* of T is defined as $T' := (T^t)|_{Y'}$, i.e.,

$$\langle T'u, x\rangle = \langle u, Tx\rangle \qquad \forall\, x \in X,\ \ \forall\, u \in Y'.$$

Now, since

$$|\langle T'u, x\rangle| = |\langle u, Tx\rangle| \le \|u\|\,\|T\|\,\|x\|,$$

we have that $T'u \in X'$ for each $u \in Y'$. Moreover, we have the following:

Theorem 4.7 *The mapping $T \mapsto T'$ is an isometric isomorphism from $\mathcal{L}(X, Y)$ to $\mathcal{L}(Y', X')$.*

Proof Clearly, the mapping $T \mapsto T'$ is linear. We show that $\|T\| = \|T'\|$. Indeed, since

$$\|Tx\| = \sup\{|\langle u, Tx\rangle| \ :\ \|u\| = 1,\ u \in Y'\},$$

we have

$$\|T'\| = \sup\{\|Tx\| \ :\ \ \|u\| = 1,\ u \in Y'\}$$

$$= \sup_{\|u\|=1} \sup\{|\langle T'u, x\rangle| \ :\ x \in X,\ \|x\| = 1\}$$

$$= \sup_{\|u\|=1} \sup\{|\langle u, Tx\rangle| \ :\ x \in X,\ \|x\| = 1\}$$

$$= \sup_{\|x\|=1} \sup\{|\langle u, Tx\rangle| \ :\ u \in Y',\ \|u\| = 1\} = \sup_{\|x\|=1} \|Tx\| = \|T\|.$$

□

Example 4.2 Let $K : [a, b] \times [a, b] \to \mathbb{R}$ be an integral kernel, $1 < p < \infty$, and consider the integral operator $T : L^p([a, b]) \to L^p([a, b])$ associated to the kernel K, i.e.,

$$(Tf)(x) = \int_a^b K(x, y) f(y)\, dy.$$

Identifying $(L^p([a, b]))'$ with $L^{p'}([a, b])$, it is easy to see that $T' : L^{p'}([a, b]) \to L^{p'}([a, b])$ is the integral operator associated to the kernel $K^*(x, y) = K(y, x)$, i.e.,

$$(T'g)(x) = \int_a^b K(y,x)g(y)\,dy.$$

Proposition 4.3 *Let X, Y, and Z be normed spaces. Then:*

(i) If $T, S \in \mathcal{L}(X,Y)$, then $(T+S)' = T' + S'$.
(ii) If $T \in \mathcal{L}(X,Y)$ and $S \in \mathcal{L}(Y,Z)$, then $(ST)' = T'S'$.

Proof (i) is a straightforward verification. (ii): For $u \in Z'$ and $x \in X$, we have

$$\langle (ST)'u, x\rangle = \langle u, (ST)x\rangle = \langle u, S(Tx)\rangle = \langle S'u, Tx\rangle$$

$$= \langle T'(S'u), x\rangle = \langle (T'S')u, x\rangle.$$

Hence, we deduce that $(ST)' = T'S'$. □

Proposition 4.4 *Let X and Y be normed spaces, and let $T \in \mathcal{L}(X,Y)$. Then the second adjoint $T'' : X'' \to Y''$ is an extension of T, and if X is reflexive, then $T'' = T$.*

Proof It suffices to observe that if $x \in X$ and $u \in Y'$, we have

$$\langle T''\hat{x}, u\rangle = \langle \hat{x}, T'u\rangle = \langle Tu, x\rangle = \langle u, \widehat{Tx}\rangle.$$

□

Regarding invertibility, we have the following result.

Theorem 4.8 *Let X be a Banach space and $T \in \mathcal{L}(X)$. Then, T is invertible in $\mathcal{L}(X)$ if and only if T' is invertible in $\mathcal{L}(X')$. In particular, we have $\sigma(T) = \sigma(T')$.*

Proof Suppose that T is invertible. Then, since I' is the identity operator on X', we have

$$I = (TT^{-1})' = (T^{-1})'T', \quad I = (T^{-1}T)' = T'(T^{-1})'.$$

Thus, T' is invertible in $\mathcal{L}(X')$, and moreover, $(T')^{-1} = (T^{-1})'$.

Now suppose that T' is invertible in $\mathcal{L}(X')$. Then by the above, there exists $(T'')^{-1} \in \mathcal{L}(X'')$. Since T'' is an extension of T, it follows that T is injective. It remains to show that $T(X) = X$. Indeed: since $J(X)$ is a closed subspace of X'', and $(T'')^{-1}$ is continuous, it follows that $T''(J(X))$ is closed, hence $T(X)$ is closed. Suppose that $T(X) \neq X$. Then, by the Hahn–Banach Theorem, there exists $u \in X'$ such that $\langle u, Tx\rangle = 0$ for every $x \in X$. Then

$$\langle T'u, x\rangle = \langle u, Tx\rangle = 0 \quad \forall\, x \in X.$$

Hence, $T'u = 0$, which contradicts the injectivity of T. □

Definition 4.3 Let X be a normed space and let $M \subset X$. The *orthogonal complement* of M is defined as:

$$M^{\perp} := \{u \in X' \,:\, \langle u, x\rangle = 0 \quad \forall\, x \in M\}.$$

Similarly, for $A \subset X'$, its *orthogonal complement* is defined as

$$A^{\perp} := \{x \in X \,:\, \langle u, x\rangle = 0 \quad \forall\, u \in A\}.$$

Given an operator $T \in \mathcal{L}(X, Y)$, we denote its *range* by $\operatorname{Ran} T$, i.e., $\operatorname{Ran} T = T(X)$.

Theorem 4.9 *Let X and Y be normed spaces and $T \in \mathcal{L}(X, Y)$. Then,*

$$\overline{\operatorname{Ran} T} = \big(\operatorname{Ker} T'\big)^{\perp}.$$

Proof We first show that $\big(\operatorname{Ker} T'\big)^{\perp} \subset \overline{\operatorname{Ran} T}$. To this end, we prove that if $y_0 \notin \overline{\operatorname{Ran} T}$, then $y_0 \notin \big(\operatorname{Ker} T'\big)^{\perp}$. Indeed, by the Hahn–Banach Theorem, there exists $u \in Y'$ such that

$$\langle u, y_0\rangle \neq 0 \quad \text{and} \quad \langle u, Tx\rangle = 0 \;\; \forall\, x \in X.$$

Then,

$$\langle T'u, x\rangle = \langle u, Tx\rangle = 0 \quad \forall\, x \in X,$$

which implies that $T'u = 0$. Thus, $y_0 \notin \big(\operatorname{Ker} T'\big)^{\perp}$.

Now we prove that $\overline{\operatorname{Ran} T} \subset \big(\operatorname{Ker} T'\big)^{\perp}$. Given $y \in \overline{\operatorname{Ran} T}$, there exist $y_n \in \operatorname{Ran} T$ such that $y_n \to y$. Let $x_n \in X$ be such that $Tx_n = y_n$. Then, if $f \in \operatorname{Ker} T'$, we have

$$\langle f, y\rangle = \lim_{n\to\infty} \langle f, y_n\rangle = \lim_{n\to\infty} \langle f, Tx_n\rangle = \lim_{n\to\infty} \langle T'f, x_n\rangle = 0.$$

Therefore, $y \in \big(\operatorname{Ker} T'\big)^{\perp}$. □

The dual of the previous result, i.e.,

$$\overline{\operatorname{Ran} T'} = \big(\operatorname{Ker} T\big)^{\perp},$$

is not true in general (see Problem 4.10). However, a weaker result does hold, and to prove it we first need to establish the following lemma.

Lemma 4.2 *Let X and Y be Banach spaces and $T \in \mathcal{L}(X, Y)$. If* $\operatorname{Ran} T$ *is closed, then there exists a constant K such that, for each $y \in \operatorname{Ran} T$, there exists $x \in X$ such that $Tx = y$ and $\|x\| \leq K\|y\|$.*

Proof Since $\operatorname{Ran} T$ is closed, by the Open Mapping Theorem, there exists $\delta > 0$ such that

$$\{y \in \operatorname{Ran} T \ : \ \|y\| < \delta\} \subset T(U(X)).$$

Therefore, given $y \in \operatorname{Ran} T$, $y \neq 0$, there exists $z \in U(X)$ such that

$$Tz = \frac{\delta y}{2\|y\|}.$$

Let $x := \frac{2\|y\|z}{\delta}$. Then $Tx = y$, and moreover,

$$\|x\| < \frac{2}{\delta}\|y\|.$$

□

Given an operator $T \in \mathcal{L}(X, Y)$, its *range* is denoted by $\operatorname{Ran} T$, i.e., $\operatorname{Ran} T = T(X)$.

Theorem 4.10 *Let X and Y be Banach spaces and $T \in \mathcal{L}(X, Y)$. If* $\operatorname{Ran} T$ *is closed, then* $\operatorname{Ran} T'$ *is also closed and, moreover,*

$$\operatorname{Ran} T' = \big(\operatorname{Ker} T\big)^{\perp}.$$

Proof We first show that $\big(\operatorname{Ker} T\big)^{\perp} \subset \operatorname{Ran} T'$. Given $u \in \big(\operatorname{Ker} T\big)^{\perp}$, we define a functional w on $\operatorname{Ran} T$ as:

$$\langle w, Tx\rangle := \langle u, x\rangle \quad \forall\, x \in X.$$

The functional w is well defined because if $Tx_1 = Tx_2$, then since $x_1 - x_2 \in \operatorname{Ker} T$, we have

$$\langle w, Tx_1\rangle - \langle w, Tx_2\rangle = \langle u, x_1\rangle - \langle u, x_2\rangle = 0.$$

Now, by the previous lemma, there exists $K > 0$ such that for any $y \in \operatorname{Ran} T$, there exists $x \in X$ with $Tx = y$ and $\|x\| \leq K\|y\|$. Therefore,

$$|\langle w, y\rangle| = |\langle u, x\rangle| \leq K\|u\|\,\|y\|.$$

Thus, $w \in \big(\text{Ran}\, T\big)'$. Then, by the Hahn-Banach Theorem, w can be extended to a functional $\tilde{w} \in Y'$, and we have

$$\langle T'\tilde{w}, x\rangle = \langle \tilde{w}, Tx\rangle = \langle w, Tx\rangle = \langle u, x\rangle.$$

Hence, $T'\tilde{w} = u$, so $u \in \text{Ran}\, T'$.

Now suppose that $u \in \text{Ran}\, T'$, and let us show that $u \in \big(\text{Ker}\, T\big)^{\perp}$. Let $w \in Y'$ such that $T'w = u$. Then, if $x \in \text{Ker}\, T$, we have

$$\langle u, x\rangle = \langle T'w, x\rangle = \langle w, Tx\rangle = \langle w, 0\rangle = 0.$$

Hence, $u \in \big(\text{Ker}\, T\big)^{\perp}$.

The fact that $\text{Ran}\, T'$ is closed follows from the equality $\text{Ran}\, T' = \big(\text{Ker}\, T\big)^{\perp}$, since $\big(\text{Ker}\, T\big)^{\perp}$ is closed. □

To conclude this section, we will make a more refined study of the spectrum of an operator.

Definition 4.4 Let $T \in \mathcal{L}(X)$. The *approximate point spectrum* is defined as

$$\sigma_{ap}(T) := \{\lambda \in \mathbb{C} \,:\, (\lambda I - T) \text{ is not injective or } \text{Ran}\,(\lambda I - T) \text{ is not closed}\}.$$

Clearly, $\sigma_p(T) \subset \sigma_{ap}(T)$.

The *residual spectrum* of T is defined as

$$\sigma_r(T) := \{\lambda \in \mathbb{C} \,:\, \text{Ran}\,(\lambda I - T) \text{ is not dense in } X\}.$$

Clearly,

$$\sigma(T) = \sigma_{ap}(T) \cup \sigma_r(T).$$

Proposition 4.5 *Let X be a Banach space and let $T \in \mathcal{L}(X)$. Then*

$$\sigma_r(T) = \sigma_p(T').$$

Proof As a consequence of Theorem 4.9, we have

$$\overline{\text{Ran}\,(\lambda I - T)} = [\text{Ker}\,(\lambda I - T')]^{\perp}.$$

Consequently,

$$\lambda \in \sigma_p(T') \iff \text{Ker}\,(\lambda I - T') \neq \{0\}$$

$$\iff X \neq [\text{Ker}\,(\lambda I - T')]^{\perp} = \overline{\text{Ran}\,(\lambda I - T)} \iff \lambda \in \sigma_r(T).$$

□

We now have the following characterization of the approximate point spectrum.

Proposition 4.6 *Let X be a Banach space and let $T \in \mathcal{L}(X)$. The following are equivalent:*

(i) $\lambda \in \sigma_{ap}(T)$
(ii) There exist $x_n \in X$ *with* $\|x_n\| = 1$ *such that*

$$\lim_{n\to\infty} \|Tx_n - \lambda x_n\| = 0.$$

Proof (i) implies (ii): By contradiction, suppose there does not exist a sequence with the required properties. Then, for any $x \neq 0$, we have that $(\lambda I - T)x \neq 0$, so $(\lambda I - T)$ is injective. Let us now show that Ran $(\lambda I - T)$ is closed, which would imply that $\lambda \notin \sigma_{ap}(T)$. First, note that:

$$\text{there exists a } C > 0 \text{ such that } \|(\lambda I - T)x\| \geq C\|x\| \quad \forall\, x \in X. \tag{4.10}$$

If this were not the case, we could find $y_n \neq 0$ such that

$$\|(\lambda I - T)y_n\| \leq \frac{1}{n}\|y_n\| \quad \forall\, n \in \mathbb{N}.$$

Then, defining $x_n := \frac{y_n}{\|y_n\|}$, we get

$$\|(\lambda I - T)x_n\| \leq \frac{1}{n} \quad \forall\, n \in \mathbb{N},$$

which contradicts our assumption. Now let us prove that Ran $(\lambda I - T)$ is closed. Suppose $x_n \in X$ and that

$$y = \lim_{n\to\infty} (\lambda I - T)x_n.$$

From (4.10), it follows that

$$\|x_n - x_m\| \leq \frac{1}{C}\|(\lambda I - T)x_n - (\lambda I - T)x_m\|,$$

which shows that the sequence $\{x_n\}$ is Cauchy. Hence, $x_n \to x \in X$, and thus $y = (\lambda I - T)x$, i.e., $y \in$ Ran $(\lambda I - T)$.
(ii) implies (i): Suppose that $\lambda \notin \sigma_{ap}(T)$. Then, $(\lambda I - T) : X \to$ Ran $(\lambda I - T)$ is bijective and continuous, and since Ran $(\lambda I - T)$ is a Banach space, by the Open Mapping Theorem, there exists a continuous linear map $R :$ Ran $(\lambda I - T) \to X$ such that $R(\lambda I - T) = (\lambda I - T)R = I$. Therefore, if $\|x\| = 1$, we have

$$1 = \|x\| = \|R(\lambda I - T)x\| \leq \|R\|\,\|(\lambda I - T)x\|,$$

and so

$$\|(\lambda I - T)x\| \geq \frac{1}{\|R\|},$$

which contradicts condition (ii). □

Proposition 4.7 *Let X be a Banach space and let $T \in \mathcal{L}(X)$. Then, if $\partial\sigma(T)$ is the topological boundary of T, we have that $\partial\sigma(T) \subset \sigma_{ap}(T)$.*

Proof Let $\lambda \in \partial\sigma(T)$. Then there exists a sequence $\{\lambda_n\} \subset \rho(T)$ such that $\lambda_n \to \lambda$. Since

$$\|R(\lambda_n, T)\| \geq \frac{1}{d(\lambda_n, \sigma(T))},$$

by the Banach–Steinhaus Theorem, there exists an $x \in X$ such that

$$\lim_{n\to\infty} \|R(\lambda_n, T)x\| = +\infty.$$

Let us define

$$x_n := \frac{1}{\|R(\lambda_n, T)x\|} R(\lambda_n, T)x.$$

Then, since

$$(\lambda - T)x_n = (\lambda - \lambda_n)x_n + (\lambda_n - T)x_n = (\lambda - \lambda_n)x_n + \frac{1}{\|R(\lambda_n, T)x\|},$$

we obtain

$$\lim_{n\to\infty} \|(\lambda - T)x_n\| = 0.$$

Therefore, by the previous proposition, it follows that $\lambda \in \sigma_{ap}(T)$. □

4.5 Compact Operators

In this section, we will study a special class of bounded linear operators between Banach spaces that arise in many applications and possess properties similar to those of operators between finite-dimensional spaces. These are the *compact operators*, also known as *completely continuous operators*.

Definition 4.5 Let X and Y be two Banach spaces, and let $T \in \mathcal{L}(X, Y)$. We say that T is a *compact operator* if $T(U(X))$ is relatively compact in Y. The set of all compact operators from X to Y is denoted by $\mathcal{K}(X, Y)$.

Proposition 4.8 *Let X and Y be Banach spaces. Then the following hold:*

(i) $\mathcal{K}(X, Y)$ is a vector subspace of $\mathcal{L}(X, Y)$.
(ii) If $T_n \in \mathcal{K}(X, Y)$, $T \in \mathcal{L}(X, Y)$ and $T_n \to T$ in $\mathcal{L}(X, Y)$, then $T \in \mathcal{K}(X, Y)$, i.e., $\mathcal{K}(X, Y)$ is closed in $\mathcal{L}(X, Y)$.
(iii) If $T \in \mathcal{L}(X)$, $S \in \mathcal{L}(X)$, and $R \in \mathcal{K}(X, Y)$, then both RT and SR belong to $\mathcal{K}(X, Y)$.

Proof (i) is a straightforward verification.
(ii): Let us show that $T(U(X))$ is precompact. Since Y is a Banach space, this will imply that $\overline{T(U(X))}$ is compact. Given $\epsilon > 0$, let $n_0 \in \mathbb{N}$ such that $\|T_n - T\| \leq \frac{\epsilon}{3}$ for all $n \geq n_0$. Since T_{n_0} is compact, there exist $x_1, x_2, \ldots, x_m \in U(X)$ such that

$$T_{n_0}(U(X)) \subset \bigcup_{j=1}^{m} B_j,$$

where

$$B_j := \left\{ y \in Y \ : \ \|T_{n_0}(x_j) - y\| < \frac{\epsilon}{3} \right\}.$$

Thus, if $x \in U(X)$, there exists $1 \leq j \leq m$ such that $T_{n_0}(x) \in B_j$, and hence

$$\|T_{n_0}(x) - T_{n_0}(x_j)\| < \frac{\epsilon}{3}.$$

Therefore,

$$\|Tx - Tx_j\| \leq \|Tx_j - T_{n_0}(x_j)\| + \|T_{n_0}(x_j) - T_{n_0}(x)\| + \|T_{n_0}(x) - Tx\| \leq \epsilon.$$

Consequently,

$$T(U(X)) \subset \bigcup_{j=1}^{m} (Tx_j + \epsilon U(Y)).$$

Hence, $T(U(X))$ is precompact in Y.
(iii): Since $T \in \mathcal{L}(X)$, there exists a constant $C > 0$ such that $T(U(X)) \subset CU(X)$. Therefore,

$$\overline{(RT)(U(X))} \subset C\overline{R(U(X))}.$$

Thus, $\overline{(RT)(U(X))}$ is compact, and hence $RT \in \mathcal{K}(X, Y)$. On the other hand, since R is compact, $\overline{R(U(X))}$ is compact in Y, and since S is continuous, $S[\overline{R(U(X))}]$ is compact. Now, $\overline{SR(U(X))} \subset S[\overline{R(U(X))}]$, so $SR \in \mathcal{K}(X, Y)$. □

Definition 4.6 An operator $T \in \mathcal{L}(X, Y)$ is said to be of *finite rank* if $\operatorname{Ran} T$ is finite dimensional. The vector space of all finite rank operators from X to Y will be denoted by $\mathcal{F}(X, Y)$.

Clearly, every finite rank operator is compact. Moreover, as a consequence of the previous result, we have the following.

If $T_n \in \mathcal{F}(X, Y)$ and $T_n \to T$ in $\mathcal{L}(X, Y)$, then $T \in \mathcal{K}(X, Y)$, i.e.,

$$\overline{\mathcal{F}(X, Y)} \subset \mathcal{K}(X, Y).$$

The famous *approximation problem* of Banach–Grothendieck refers to the reverse of this last result. More precisely, a Banach space X is said to have the *approximation property* if $\overline{\mathcal{F}(X, Y)} = \mathcal{K}(X, Y)$ for every Banach space Y. In 1973, P. Enflo [14] gave the first example of a Banach space without the approximation property. Using orthogonal projections, it is easy to see that every Hilbert space has the approximation property (see Problem 4.18).

A very useful result when studying compact operators is the classical Ascoli–Arzelà Theorem, which characterizes the compact subsets of $C(K)$. Recall that if K is a compact topological space and $\mathcal{G} \subset C(K)$, we say that $\mathcal{G}$ is *equicontinuous* if for every $\epsilon > 0$ and every $x_0 \in K$, there exists a neighborhood U of x_0 such that

$$|f(x_0) - f(x)| < \epsilon \quad \forall\, x \in U, \quad \text{and } \forall\, f \in \mathcal{G}.$$

Theorem 4.11 (Ascoli–Arzelà Theorem) *Let K be a compact topological space and $\mathcal{G} \subset C(K)$. Then, $\mathcal{G}$ is relatively compact in $C(K)$ if and only if $\mathcal{G}$ is equicontinuous and bounded.*

Proof Suppose that $\mathcal{G}$ is relatively compact in $C(K)$. Clearly, $\mathcal{G}$ is bounded in $C(K)$. Let us prove that it is equicontinuous. Given $\epsilon > 0$, since $\mathcal{G}$ is relatively compact, there exist $f_1, \ldots, f_n \in \mathcal{G}$ such that

$$\mathcal{G} \subset \bigcup_{j=1}^{n} \left\{ f \in C(K) \,:\, \|f - f_j\|_\infty < \frac{\epsilon}{3} \right\}. \tag{4.11}$$

Given $x_0 \in K$, let U be an open neighborhood of x_0 in K such that

$$|f_j(x) - f_j(x_0)| < \frac{\epsilon}{3} \quad \forall\, j = 1, \ldots, n. \tag{4.12}$$

If $f \in \mathcal{G}$, then by (4.11) there exists f_j such that

$$\|f - f_j\|_\infty < \frac{\epsilon}{3}.$$

Then, taking (4.12) into account, if $x \in U$, we have

$$|f(x) - f(x_0)| \le |f(x) - f_j(x)| + |f_j(x) - f_j(x_0)| + |f_j(x_0) - f(x_0)| < \epsilon.$$

Therefore, $\mathcal{G}$ is equicontinuous.

Now suppose that $\mathcal{G}$ is bounded and equicontinuous. Given $\epsilon > 0$, we need to find a finite number of functions in $\mathcal{G}$ such that the union of the balls of radius ϵ centered at these functions covers $\mathcal{G}$. Since $\mathcal{G}$ is bounded, there exists $M > 0$ such that

$$\|f\|_\infty \le M \qquad \forall\, f \in \mathcal{G}.$$

On the other hand, since $\mathcal{G}$ is equicontinuous, for each $x \in K$ there exists a neighborhood U_x of x in K such that

$$|f(x) - f(y)| < \frac{\epsilon}{2} \quad \forall\, y \in U_x, \text{ and } \forall\, f \in \mathcal{G}. \tag{4.13}$$

Now, since K is compact, there exist $x_1, \ldots, x_n \in K$ such that

$$K = \bigcup_{j=1}^{n} U_{x_j}.$$

Let $a_1, a_2, \ldots, a_m \in \mathbb{K}$, with $|a_i| < M$, such that

$$\{a \in \mathbb{K} \,:\, |a| \le M\} \subset \bigcup_{i=1}^{m} \left\{a \in \mathbb{K} \,:\, |a - a_i| < \frac{\epsilon}{2}\right\}.$$

Let

$$B := \{(b_1, \ldots, b_n) \,:\, b_j \in \{a_1, a_2, \ldots, a_m\},\ \forall\, j = 1, \ldots, n\}.$$

Subordinate to the cover $\{U_{x_1}, \ldots, U_{x_n}\}$, we take a partition of unity $\{\psi_1, \ldots, \psi_n\}$, i.e., $0 \le \psi_i \in C(K)$, $\operatorname{sop}(\psi_i) \subset U_{x_i}$, $i = 1, \ldots, n$, and such that

$$\sum_{i=1}^{n} \psi_i(x) = 1, \qquad \forall\, x \in K.$$

For each $b = (b_1, \ldots, b_n) \in B$, we define the continuous function

$$f_b(x) := \sum_{i=1}^{n} b_i \psi_i(x).$$

We claim that

$$\mathcal{G} \subset \bigcup_{b \in B} \{f \in C(K);\ \|f - f_b\|_\infty < \epsilon\},$$

which concludes the proof. If $f \in \mathcal{G}$, since $|f(x_i)| \leq M$ for all $i = 1, \ldots, n$, we can find $b = (b_1, \ldots, b_n) \in B$ such that

$$|b_i - f(x_i)| < \frac{\epsilon}{2}, \quad \forall\, i = 1, \ldots, n.$$

Given $x \in K$, since $\sum_{i=1}^n \psi_i(x) = 1$, we have

$$|f(x) - f_b(x)| = |f(x) - \sum_{i=1}^n b_i \psi_i(x)| = \left| \sum_{i=1}^n \big(f(x) - b_i\big)\psi_i(x) \right|$$

$$\leq \sum_{i=1}^n |f(x) - b_i| \psi_i(x).$$

Now, since $\operatorname{sop}(\psi_i) \subset U_{x_i}$, we have

$$|f(x) - b_i| \leq |f(x) - f(x_i)| + |f(x_i) - b_i| < \epsilon,$$

thus $|f(x) - f_b(x)| < \epsilon$ for all $x \in K$ and consequently,

$$\|f - f_b\|_\infty < \epsilon.$$

□

Example 4.3 In Theorem 4.5 we saw that if $K : [a, b] \times [a, b] \to \mathbb{C}$ is a continuous function, then the integral operator $T : C([a, b]) \to C([a, b])$ defined by

$$(Tf)(x) := \int_a^b K(x, y) f(y)\, dy$$

is linear and continuous. Let us now show that this operator is compact. By the Ascoli–Arzelà Theorem, it suffices to show that if B is the unit ball of $C([a, b])$, then $T(B)$ is equicontinuous. Indeed: given $\epsilon > 0$, since the kernel K is uniformly continuous, there exists $\delta > 0$ such that, if $x, x', y \in [a, b]$ and $|x - x'| \leq \delta$, then

$$|K(x, y) - K(x', y)| \leq \frac{\epsilon}{b - a}.$$

Therefore, for each $f \in C([a, b])$, with $\|f\| \leq 1$, we have

$$|Tf(x) - T(f)(x')| = \left| \int_a^b \big(K(x, y) - K(x', y)\big)\, f(y)\, dy \right|$$

$$\leq \int_a^b |K(x, y) - K(x', y)|\, |f(y)|\, dy \leq (b - a)\frac{\epsilon}{b - a} = \epsilon.$$

Hence, $T(B)$ is equicontinuous.

We now study the compactness of integral operators on $L^2([a, b])$. To do so, we first need to establish the following result.

Lemma 4.3 *Let $\{e_n : n \in \mathbb{N}\}$ be an orthonormal basis of the Hilbert space $L^2([a, b])$. Then, if we define $\phi_{n,m} : [a, b] \times [a, b] \to \mathbb{C}$ as*

$$\phi_{n,m}(x, y) := e_n(x)\overline{e_m(y)},$$

it follows that $\{\phi_{n,m} : n, m \in \mathbb{N}\}$ is an orthonormal basis of $L^2([a, b] \times [a, b])$.

Proof Using the Tonelli–Hobson Theorem (Theorem B.5) we obtain

$$\int_{[a,b]\times[a,b]} |\phi_{n,m}(x, y)|^2\, dxdy = \int_{[a,b]\times[a,b]} |e_n(x)\overline{e_m(y)}|^2\, dxdy$$

$$= \left(\int_a^b |e_n(x)|^2\, dx \right)\left(\int_a^b |e_m(y)|^2\, dy \right) = \|e_n\|_2^2\, \|e_m\|_2^2 = 1.$$

Thus, $\phi_{n,m} \in L^2([a, b] \times [a, b])$ and $\|\phi_{n,m}\|_2 = 1$.

Let $(n, m) \neq (i, j)$. Then,

$$(\phi_{n,m}|\phi_{i,j}) = \int_{[a,b]\times[a,b]} e_n(x)\overline{e_m(y)e_i(x)}e_j(y)\, dxdy = (e_n|e_i)\overline{(e_m|e_j)} = 0.$$

Therefore, $\{\phi_{n,m} : n, m \in \mathbb{N}\}$ is an orthonormal system. To show that it is a basis, we will prove it is a maximal orthonormal system. Let $\phi \in L^2([a, b] \times [a, b])$ such that $(\phi|\phi_{n,j}) = 0$ for every $n, j \in \mathbb{N}$. We must show that $\phi = 0$. Indeed: for $y \in [a, b]$, let $\phi_y(x) := \phi(x, y)$. Then, since $\phi \in L^2([a, b] \times [a, b])$, we have that for almost every $y \in [a, b]$, the function $\phi_y \in L^2([a, b])$. Therefore, for each $n \in \mathbb{N}$ we can define the function f_n as

$$f_n(y) := (e_n|\phi_y) = \int_a^b e_n(x)\overline{\phi(x, y)}\, dx.$$

Now, since f_n is measurable and also

$$\int_a^b |f_n(y)|^2\, dy = \int_a^b |(e_n|\phi_y)|^2\, dy \leq \int_a^b \|e_n\|_2^2\, \|\phi_y\|_2^2\, dy$$

$$= \int_a^b \left(\int_a^b |\phi(x,y)|^2 \, dx \right) dy = \int_{[a,b]\times[a,b]} |\phi(x,y)|^2 \, dxdy = \|\phi\|_2^2,$$

we conclude that $f_n \in L^2([a,b])$ for each $n \in \mathbb{N}$. Then, since by hypothesis $\{e_n : n \in \mathbb{N}\}$ is an orthonormal basis of $L^2([a,b])$, applying Parseval's identity, we obtain

$$\|f_n\|_2^2 = \sum_{j=1}^{\infty} |(e_j|f_n)|^2 = \sum_{j=1}^{\infty} \left| \int_a^b e_j(y)\overline{f_n(y)} \, dy \right|^2$$

$$= \sum_{j=1}^{\infty} \left| \int_a^b e_j(y) \left(\int_a^b \overline{e_n(x)}\phi(x,y) \, dx \right) dy \right|^2$$

$$= \sum_{j=1}^{\infty} \left| \int_{[a,b]\times[a,b]} \phi(x,y)\overline{e_n(x)}e_j(y) \, dxdy \right|^2$$

$$= \sum_{j=1}^{\infty} \left| \int_{[a,b]\times[a,b]} \phi(x,y)\overline{\phi_{n,j}(x,y)} \, dxdy \right|^2 = \sum_{j=1}^{\infty} |(\phi|\phi_{n,j})|^2 = 0.$$

Consequently, $f_n = 0$ for all $n \in \mathbb{N}$, from which it follows that $\phi_y = 0$ for almost every $y \in [a,b]$ and thus, $\phi = 0$. □

Theorem 4.12 *If $K \in L^2([a,b] \times [a,b])$, then the integral operator $T : L^2([a,b]) \to L^2([a,b])$ defined by*

$$(Tf)(x) := \int_a^b K(x,y) f(y) \, dy$$

is a compact operator, and moreover $\|T\| \leq \|K\|_2$.

Proof Let $\{e_n : n \in \mathbb{N}\}$ and $\{\phi_{n,m} : n, m \in \mathbb{N}\}$ be as in the previous lemma. Since $K \in L^2([a,b] \times [a,b])$, by applying Parseval's identity we have:

$$\|K\|_2^2 = \sum_{n,m=1}^{\infty} |(K|\phi_{n,m})|^2 = \sum_{n,m=1}^{\infty} \left| \int_{[a,b]\times[a,b]} K(x,y)\overline{e_n(x)}e_m(y) \, dxdy \right|^2$$

$$= \sum_{n,m=1}^{\infty} \left| \int_a^b \left(\int_a^b K(x,y)e_m(y) \, dy \right) \overline{e_n(x)} \, dx \right|^2 = \sum_{n,m=1}^{\infty} |(Te_m|e_n)|^2.$$

Given a function $f \in L^2([a,b])$, we can write $f = \sum_{m=1}^{\infty} \alpha_m e_m$, where $\alpha_m = (f|e_m)$, and moreover:

$$\|f\|_2^2 = \sum_{m=1}^{\infty} |\alpha_m|^2.$$

Then,

$$|(Tf|e_n)|^2 = \left| \sum_{m=1}^{\infty} \alpha_m (Te_m|e_n) \right|^2 \leq \left(\sum_{m=1}^{\infty} |\alpha_m|^2 \right) \left(\sum_{m=1}^{\infty} |(Te_m|e_n)|^2 \right).$$

Hence, by Parseval's identity, we obtain:

$$\|Tf\|_2^2 = \sum_{n=1}^{\infty} |(Tf|e_n)|^2 \leq \|f\|_2^2 \sum_{n=1}^{\infty} \left(\sum_{m=1}^{\infty} |(Te_m|e_n)|^2 \right) = \|K\|_2^2 \, \|f\|_2^2.$$

Therefore, $T \in \mathcal{L}(L^2([a,b]))$ and $\|T\| \leq \|K\|_2$.

Given $n \in \mathbb{N}$ and $\alpha_{i,j} \in \mathbb{C}$, consider the kernel

$$K_n(x,y) := \sum_{i,j=1}^{n} \alpha_{i,j} \phi_{i,j}(x,y),$$

and let T_n be the integral operator on $L^2([a,b])$ associated with the kernel K_n. Then,

$$(T_n f)(x) = \int_a^b K_n(x,y) f(y)\, dy$$

$$= \sum_{i,j=1}^{n} \alpha_{i,j} \int_a^b e_i(x) \overline{e_j(y)} f(y)\, dy = \sum_{i,j=1}^{n} \beta_{i,j} e_i(x).$$

Hence, $T_n f \in \operatorname{span}\{e_1, \ldots, e_n\}$ and thus T_n is a finite-rank operator, i.e., $T_n \in \mathcal{F}(L^2([a,b]))$ for each $n \in \mathbb{N}$.

Since $K \in L^2([a,b] \times [a,b])$, and by the previous lemma $\{\phi_{n,m} : n, m \in \mathbb{N}\}$ is a basis of $L^2([a,b] \times [a,b])$, we have that:

$$K = \sum_{n,m=1}^{\infty} \alpha_{n,m} \phi_{n,m},$$

and thus, if

$$K_n = \sum_{i,j=1}^{n} \alpha_{i,j} \phi_{i,j},$$

we obtain that $\|K_n - K\|_2 \to 0$ as $n \to \infty$.

Then, if T_n is the integral operator on $L^2([a,b])$ associated with the kernel K_n, we have that $T_n \in \mathcal{F}(L^2([a,b]))$, and since $\|T_n - T\| \leq \|K_n - K\|_2$, it follows that $T_n \to T$ in $\mathcal{L}(L^2([a,b]))$, and consequently, $T \in \mathcal{K}(L^2([a,b]))$. □

Theorem 4.13 (Schauder's Theorem) *Let $T \in \mathcal{L}(X,Y)$. Then,*

$$T \in \mathcal{K}(X,Y) \iff T' \in \mathcal{K}(Y',X').$$

Proof Suppose that $T \in \mathcal{K}(X,Y)$ and let us show that $T'(U(Y'))$ is relatively compact in X'. To that end, take a sequence $\{v_n\} \subset U(Y')$, and let us show that the sequence $\{T'(v_n)\}$ has a convergent subsequence. By hypothesis, we have that $K := \overline{T(U(X))}$ is a compact subset of Y. For each $n \in \mathbb{N}$, define the map $\phi_n : K \to \mathbb{K}$ by

$$\phi_n(x) := \langle v_n, x \rangle.$$

Then, $\mathcal{A} := \{\phi_n \ : \ n \in \mathbb{N}\}$ is a bounded subset of $C(K)$. Moreover, since

$$|\phi_n(x) - \phi_n(x')| = |\langle v_n, x - x' \rangle| \leq \|x - x'\|,$$

we have that $\mathcal{A}$ is equicontinuous. Therefore, by the Ascoli–Arzelà Theorem, $\mathcal{A}$ is relatively compact in $C(K)$. Consequently, there exists a subsequence $\{\phi_{n_k}\}$ and $\phi \in C(K)$ such that $\phi_{n_k} \to \phi$ in $C(K)$ as $k \to +\infty$. Thus,

$$\lim_{k\to\infty} \sup_{x\in K} |\phi_{n_k}(x) - \phi(x)| = 0.$$

Then,

$$\lim_{k,l\to\infty} \sup_{u\in U(X)} |\langle v_{n_k}, Tu \rangle - \langle v_{n_l}, Tu \rangle| = \lim_{k,l\to\infty} \sup_{u\in U(X)} |\langle T'v_{n_k} - T'v_{n_l}, u \rangle| = 0.$$

From this, it follows that the sequence $\{T'v_{n_k}\}$ is Cauchy in X', and therefore convergent.

Now suppose that $T' \in \mathcal{K}(Y',X')$. Then, from the above, we have that $T''(U(X))$ is relatively compact in Y''. Now, $T''(U(X)) = T(U(X))$ and Y is closed in Y''. Therefore, $T(U(X))$ is relatively compact in Y, so $T \in \mathcal{K}(X,Y)$. □

4.6 Spectral Theory for Compact Operators

In this section, we will see that compact linear operators behave spectrally in a way that is quite similar to that of linear operators on finite-dimensional spaces.

We will need the following result.

Lemma 4.4 *Let X be a vector space and let $u_i \in X^*$, $i = 1, 2, \ldots, n$ be linearly independent. Then, there exist $x_i \in X$ such that*

$$\langle u_i, x_j \rangle = \delta_{i,j}.$$

Proof Define

$$M_j := \ker u_1 \cap \ldots \cap \ker u_{j-1} \cap \ker u_{j+1} \cap \ldots \cap \ker u_n.$$

If there exists j such that $M_j \subset \ker u_j$, then we would have that if $u_i(x) = 0$ for all $i \neq j$, then $u_j(x) = 0$ as well. Thus, applying the Lemma 2.8, we would find that u_j is a linear combination of the remaining u_i. Therefore, $M_j \not\subset \ker u_j$ for each $j = 1, \ldots, n$. Hence, for each $j = 1, \ldots, n$ there exists $y_j \in M_j$ such that $u_j(y_j) \neq 0$. It suffices to take

$$x_j := \frac{y_j}{u_j(y_j)}.$$

□

One of the fundamental results on the spectral behavior of compact operators is the following result, originally obtained by I. Fredholm for integral operators.

Theorem 4.14 (Fredholm Alternative Theorem) *Let X be a Banach space and $T \in \mathcal{K}(X)$. Then:*

(i) $\ker(I - T)$ *is finite-dimensional.*
(ii) $\mathrm{Ran}(I - T) = [\ker(I - T')]^{\perp}$, *hence* $\mathrm{Ran}(I - T)$ *is closed.*
(iii) $\ker(I - T) = \{0\} \iff \mathrm{Ran}(I - T) = X$.
(iv) $\dim(\ker(I - T)) = \dim(\ker(I - T'))$.

Proof (i): Let $Y := \ker(I - T)$. Then $U(Y) \subset T(U(X))$, so $U(Y)$ is compact. Hence, by Theorem 2.3, Y is finite-dimensional.
(ii): It suffices to show that $\mathrm{Ran}(I - T)$ is closed, since by Theorem 4.9 we have

$$[\ker(I - T')]^{\perp} = \overline{\mathrm{Ran}(I - T)}.$$

Let $y_n = x_n - Tx_n \to x$. We want to prove that $x \in \mathrm{Ran}(I - T)$. Let $d_n := \mathrm{dist}(x_n, \ker(I - T))$. Since $\ker(I - T)$ is finite-dimensional by (i), there exist $z_n \in \ker(I - T)$ such that $d_n = \|x_n - z_n\|$. We write:

$$y_n = (x_n - z_n) - T(x_n - z_n). \tag{4.14}$$

Let us show that $\{d_n\}$ is bounded. If not, there would exist a subsequence $\{d_{n_j}\}$ with $\lim_{j\to\infty} d_{n_j} = +\infty$. Then define

$$\xi_n := \frac{x_n - z_n}{\|x_n - z_n\|}.$$

From (4.14), we get

$$\lim_{j\to\infty}(\xi_{n_j} - T\xi_{n_j}) = 0.$$

Since T is compact, there exists a further subsequence (still denoted equal) such that $\xi := \lim_{j\to\infty} T\xi_{n_j}$ exists, hence $\xi = \lim_{j\to\infty}\xi_{n_j}$. Thus $\xi \in \ker(I - T)$. On the other hand,

$$\operatorname{dist}(\xi_n, \ker(I - T)) = \frac{\operatorname{dist}(x_n, \ker(I - T))}{\|x_n - z_n\|} = 1,$$

so $\operatorname{dist}(\xi, \ker(I-T)) = 1$, a contradiction. Hence $\{d_n\}$ is bounded. By compactness of T, there exists a subsequence such that

$$w := \lim_{k\to\infty} T(x_{n_k} - z_{n_k}).$$

From (4.14) it follows that

$$\lim_{k\to\infty}(x_{n_k} - z_{n_k}) = x + w.$$

So if $y := x + w$, then $(I - T)y = x$, hence $x \in \operatorname{Ran}(I - T)$.
(iii): Suppose $\ker(I - T) = \{0\}$ and suppose, by contradiction, that $X_1 := \operatorname{Ran}(I - T) \neq X$. By (ii), X_1 is a Banach space and since $T(X_1) \subset X_1$, then $T|_{X_1} \in \mathcal{K}(X_1)$. Applying (ii) again, $X_2 := (I-T)(X_1)$ is a closed subspace of X_1. Also, $X_2 \neq X_1$, for otherwise, given $y \in X \setminus X_1$ and assuming $X_2 = X_1$, there exists $x \in X$ such that $(I-T)(x-Tx) = y-Ty$, hence $(I-T)(x-Tx-y) = 0$ and since $(I-T)$ is injective, it follows that $y = x - Tx$, contradicting $y \notin X_1$. Proceeding recursively, define $X_n := (I - T)(X_{n-1})$, each of which is closed and

$$X_1 \supset X_2 \supset X_3 \supset \ldots$$

By Riesz's Lemma (Lemma 2.1), there exist $x_n \in X_n$, $\|x_n\| = 1$ such that

$$\operatorname{dist}(x_n, X_{n+1}) \geq \frac{1}{2}, \quad \forall n \in \mathbb{N}.$$

We write

$$Tx_n - Tx_m = -(x_n - Tx_n) + (x_m - Tx_m) + (x_n - x_m).$$

For $n > m$, since $X_{n+1} \subset X_n \subset X_{m+1} \subset X_m$, we have

$$y_{n,m} := -(x_n - Tx_n) + (x_m - Tx_m) + x_n \in X_{m+1}.$$

Hence

$$\|Tx_n - Tx_m\| = \|y_{n,m} - x_m\| \geq \operatorname{dist}(x_m, X_{m+1}) \geq \frac{1}{2},$$

contradicting the compactness of T. Therefore,

$$\operatorname{Ran}(I - T) = X.$$

Now suppose $\operatorname{Ran}(I - T) = X$. Let us show that

$$\ker(I - T) = \{0\}.$$

By Theorem 4.9,

$$\ker(I - T') = [\operatorname{Ran}(I - T)]^{\perp} = X^{\perp} = \{0\}.$$

By Schauder's Theorem, $T' \in \mathcal{K}(X')$, so applying the previous implication to T' yields $\operatorname{Ran}(I - T') = X'$. Hence,

$$\ker(I - T) = [\operatorname{Ran}(I - T')]^{\perp} = \{0\}.$$

(iv): From (i), we have that

$$\dim(\operatorname{Ker}(I - T)) = n \quad \text{and} \quad \dim(\operatorname{Ker}(I - T')) = m.$$

Let $\{x_1, \ldots, x_n\}$ be a basis of $\operatorname{Ker}(I - T)$ and $\{u_1, \ldots, u_m\}$ a basis of $\operatorname{Ker}(I - T')$. By the Hahn–Banach Theorem, there exist $v_1, \ldots, v_n \in X'$ such that

$$\langle v_i, x_j \rangle = \delta_{i,j} \quad \text{for } i, j = 1, \ldots, n.$$

By Lemma 4.4, there exist $y_1, \ldots, y_m \in X$ such that

$$\langle u_i, y_j \rangle = \delta_{i,j} \quad \text{for } i, j = 1, \ldots, m.$$

Suppose that $n < m$. Define the operator S on X by

$$Sx := Tx + \sum_{i=1}^{n} \langle v_i, x \rangle y_i.$$

Then, since S is the sum of a compact operator and a finite-rank operator, it is compact. Let us show that

$$\operatorname{Ker}(I - S) = \{0\}.$$

Indeed, let $x \in \text{Ker}\,(I - S)$. Then,

$$x - Tx = \sum_{i=1}^{n} \langle v_i, x \rangle y_i.$$

Thus,

$$0 = \langle (I - T')u_j, x \rangle = \langle u_j, x - Tx \rangle = \langle v_j, x \rangle.$$

Hence, $x - Tx = 0$, and therefore $x = \sum_{i=1}^{n} \alpha_i x_i$. Then,

$$0 = \langle v_j, x \rangle = \alpha_j, \quad j = 1, \ldots, n,$$

and consequently, $x = 0$. Thus, we have $\text{Ker}\,(I - S) = \{0\}$. Then, applying (iii), we obtain that $\text{Ran}\,(I - S) = X$. In particular, there exists $x \in X$ such that $x - Sx = y_{n+1}$. Then,

$$1 = \langle u_{n+1}, y_{n+1} \rangle = \langle u_{n+1}, x - Sx \rangle = \langle u_{n+1}, x - Tx - \sum_{i=1}^{n} \langle v_i, x \rangle y_i \rangle = 0,$$

since

$$\langle u_{n+1}, y_i \rangle = 0 \quad (i = 1, \ldots, n), \quad \text{and } u_{n+1} \in \text{Ker}\,(I - T').$$

We thus arrive at a contradiction, which shows that $n \geq m$. Now, applying the above argument to the operator T', we obtain that if $k = \dim(\text{Ker}\,(I - T''))$, then $m \geq k$. Since obviously $n \leq k$, we obtain $n \leq m$ and consequently, $n = m$.

□

The Fredholm alternative theorem concerns the solution of the equation

$$x - Tx = y,$$

and states the following alternative: either, for every y, the equation $x - Tx = y$ has a unique solution, or the homogeneous equation

$$x - Tx = 0$$

admits n linearly independent solutions, and in this case, the non-homogeneous equation $x - Tx = y$ has a solution if and only if

$$y \in [\text{Ker}\,T]^{\perp}.$$

Definition 4.7 Let $T \in \mathcal{L}(X)$ and $\lambda \in \sigma_p(T)$. Then,

$$\text{Ker}\,(\lambda I - T) \neq \{0\}.$$

The *dimension* of Ker $(\lambda I - T)$ is called the *geometric multiplicity* of λ.

Lemma 4.5 *Let $T \in \mathcal{K}(X)$. Every nonzero element of $\sigma_p(T)$ is an isolated point of $\sigma_p(T)$.*

Proof We must show that if $\lambda_n \in \sigma_p(T)$, with $\lambda_n \neq \lambda_m$, and $\lambda_n \to \lambda$, then $\lambda = 0$. Since $\lambda_n \in \sigma_p(T)$, there exist $e_n \neq 0$ such that $Te_n = \lambda_n e_n$. For each $n \in \mathbb{N}$, let $X_n := \text{LIN}\{e_1, \ldots, e_n\}$. We will show by induction that for each $n \in \mathbb{N}$, the set $\{e_1, \ldots, e_n\}$ is linearly independent. Assume the result holds for n and suppose

$$e_{n+1} = \sum_{i=1}^{n} \alpha_i e_i.$$

Then

$$\sum_{i=1}^{n} \alpha_i \lambda_{n+1} e_i = \lambda_{n+1} e_{n+1} = T e_{n+1} = \sum_{i=1}^{n} \alpha_i \lambda_i e_i,$$

so

$$\alpha_i (\lambda_i - \lambda_{n+1}) = 0, \quad \text{for } i = 1, \ldots, n.$$

From which we deduce that $\alpha_i = 0$ for all $i = 1, \ldots, n$.

Therefore, we have that $X_n \subset X_{n+1}$, $X_n \neq X_{n+1}$ for every $n \in \mathbb{N}$. On the other hand, we have that $(\lambda_n I - T)X_n \subset X_{n-1}$ for all $n \in \mathbb{N}$. Then, applying Riesz's Lemma (Lemma 2.1), there exist $x_n \in X_n$, with $\|x_n\| = 1$, such that

$$\text{dist}(x_n, X_{n-1}) \geq \frac{1}{2}, \quad \forall\, n \geq 2.$$

For $2 \leq m < n$, we have

$$X_{m-1} \subset X_m \subset X_{n-1} \subset X_n.$$

Hence,

$$\left\| \frac{Tx_n}{\lambda_n} - \frac{Tx_m}{\lambda_m} \right\| = \left\| \frac{Tx_n - \lambda_n x_n}{\lambda_n} - \frac{Tx_m - \lambda_m x_m}{\lambda_m} + x_n - x_m \right\|$$

$$\geq \text{dist}(x_n, X_{n-1}) \geq \frac{1}{2}.$$

Now, if $\lambda \neq 0$, we reach a contradiction with the compactness of the operator T, since compactness implies that the sequence $\left\{\frac{Tx_n}{\lambda_n}\right\}$ has a convergent subsequence. □

Using the previous lemma and Fredholm's Alternative Theorem, we obtain the following characterization of the spectrum of a compact operator.

Theorem 4.15 (Riesz-Schauder Theorem) *Let X be an infinite-dimensional Banach space and let $T \in \mathcal{K}(X)$. Then the following hold:*

(i) $0 \in \sigma(T)$.
(ii) $\sigma(T) \setminus \{0\} = \sigma_p(T) \setminus \{0\}$, *and if* $\lambda \in \sigma_p(T) \setminus \{0\}$, *then* λ *has finite geometric multiplicity.*
(iii) One of the following situations occurs:

(a) $\sigma(T) = \{0\}$,
(b) $\sigma(T)$ *is finite,*
(c) $\sigma(T) \setminus \{0\} = \{\lambda_n\}$ *with* $\lim_{n\to\infty} \lambda_n = 0$.

Proof (i): If $0 \notin \sigma(T)$, then T is bijective, so that $I = TT^{-1} \in \mathcal{K}(X)$, which is a contradiction since this would imply that $U(X)$ is compact and, consequently, X would be finite-dimensional.
(ii): Let $\lambda \in \sigma(T) \setminus \{0\}$. If $\lambda \notin \sigma_p(T)$, then $\text{Ker}\,(\lambda I - T) = \{0\}$. Hence, $\text{Ker}\,(I - \frac{1}{\lambda}T) = \{0\}$. Therefore, since $\frac{1}{\lambda}T$ is a compact operator, by applying Fredholm's Alternative Theorem, we obtain

$$\text{Ran}\,(\lambda I - T) = \text{Ran}\left(I - \frac{1}{\lambda}T\right) = X,$$

which is a contradiction, since this would imply that $(\lambda I - T)$ is bijective and, consequently, $\lambda \notin \sigma(T)$. Moreover, by Fredholm's Alternative Theorem, we also have

$$\dim \text{Ker}\,(\lambda I - T) = \dim \text{Ker}\left(I - \frac{1}{\lambda}T\right) < \infty.$$

Thus, λ has finite geometric multiplicity.
(iii): For each $n \in \mathbb{N}$, the set

$$\sigma(T) \cap \left\{\lambda \in \mathbb{C} \,:\, |\lambda| \geq \frac{1}{n}\right\}$$

is empty or finite, since if it were infinite, then $\sigma(T)$, being compact, would have an accumulation point, which would contradict Lemma 4.5. Therefore, when $\sigma(T)\setminus\{0\}$ has infinitely many points, we can order them as a sequence tending to zero. □

To show that all the situations in part (iii) of the previous theorem can occur, we introduce the notion of *diagonalizable operators*.

Proposition 4.9 *Let H be a separable Hilbert space with orthonormal basis $\{e_n : n \in \mathbb{N}\}$. Let $\{\alpha_n : n \in \mathbb{N}\} \subset \mathbb{K}$, with*

$$M = \sup\{|\alpha_n| : n \in \mathbb{N}\} < \infty.$$

If $A(e_n) := \alpha_n e_n$ for all $n \in \mathbb{N}$, then A can be extended to an operator $A \in \mathcal{L}(H)$ with $\|A\| = M$, satisfying:

(i) A is compact $\iff \lim_{n\to\infty} \alpha_n = 0$.
(ii) $\sigma_p(A) = \{\alpha_n : n \in \mathbb{N}\}$, and if $\alpha \in \sigma_p(A)$ and $J_\alpha := \{j \in \mathbb{N} : \alpha = \alpha_j\}$, then the geometric multiplicity of α equals the cardinality of J_α.

Proof Given $x \in H$, since $\{e_n : n \in \mathbb{N}\}$ is an orthonormal basis of H, we have

$$x = \sum_{n=1}^{\infty} (x|e_n)e_n.$$

We define

$$Ax := \sum_{n=1}^{\infty} (x|e_n)Ae_n = \sum_{n=1}^{\infty} (x|e_n)\alpha_n e_n.$$

By Parseval's identity, we obtain

$$\|Ax\|^2 = \sum_{n=1}^{\infty} |(x|e_n)|^2|\alpha_n|^2 \le M^2 \sum_{n=1}^{\infty} |(x|e_n)|^2 = M^2\|x\|^2.$$

Hence, $A \in \mathcal{L}(H)$ and $\|A\| \le M$.

Moreover,

$$\|A\| \ge \|Ae_n\| = |\alpha_n|\|e_n\| = |\alpha_n|, \quad \forall\, n \in \mathbb{N}.$$

Therefore, $\|A\| = M$.

(i): Let P_n be the orthogonal projection onto $LIN\{e_1, \ldots, e_n\}$, that is,

$$P_n(x) = \sum_{i=1}^{n} (x|e_i)e_i, \quad \forall\, x \in H.$$

Define $A_n := AP_n$. Then,

$$A_n(e_j) = \begin{cases} 0, & \text{if } j > n, \\ \alpha_j e_j, & \text{if } 1 \le j \le n. \end{cases}$$

Clearly, $A_n \in \mathcal{F}(H)$ and, as shown above,

$$\|A_n - A\| = \sup\{|\alpha_j| : j > n\}.$$

Therefore, if $\alpha_n \to 0$, we have that $A_n \to A$ in $\mathcal{L}(H)$ and consequently $A \in \mathcal{K}(H)$. Conversely, if A is compact, it is easy to see that $\|A_n - A\| \to 0$, so $\alpha_n \to 0$.
(ii): Since $Ae_n = \alpha_n e_n$, we have that $\alpha_n \in \sigma_p(A)$. Let us show the reverse inclusion. Suppose there exists $\alpha \in \sigma_p(A)$, with $\alpha \neq \alpha_n$ for all $n \in \mathbb{N}$. Let $x \in H$, $x \neq 0$, such that $Ax = \alpha x$. Since

$$x = \sum_{n=1}^{\infty} (x|e_n)e_n,$$

we get

$$0 = \alpha x - Ax = \sum_{n=1}^{\infty} (\alpha - \alpha_n)(x|e_n)e_n.$$

Using Parseval's identity, we have

$$0 = \|\alpha x - Ax\|^2 = \sum_{n=1}^{\infty} |\alpha - \alpha_n|^2 |(x|e_n)|^2.$$

Then, since $|\alpha - \alpha_n|^2 \neq 0$, it follows that $(x|e_n) = 0$ for all $n \in \mathbb{N}$, hence $x = 0$, which contradicts the assumption. Therefore, $\sigma_p(A) = \{\alpha_n : n \in \mathbb{N}\}$.

Let $\alpha \in \sigma_p(A)$, and let $J_\alpha = \{j \in \mathbb{N} : \alpha_j = \alpha\}$. We claim that

$$\mathrm{Ker}(\alpha I - A) = LIN\{e_j : j \in J_\alpha\}.$$

Indeed, if $x \in LIN\{e_j : j \in J_\alpha\}$, then $x = \sum_{j \in J_\alpha} \beta_j e_j$. Therefore,

$$\alpha x - Ax = \sum_{j \in J_\alpha} \beta_j (\alpha - \alpha_j) e_j = 0,$$

so $x \in \mathrm{Ker}(\alpha I - A)$. Suppose that $x \notin LIN\{e_j : j \in J_\alpha\}$. Then $x = \sum_{n=1}^{\infty} (x|e_n)e_n$, with $(x|e_{n_0}) \neq 0$ for some $n_0 \notin J_\alpha$. But then, since

$$0 = \|\alpha x - Ax\|^2 = \sum_{n=1}^{\infty} |\alpha - \alpha_n|^2 |(x|e_n)|^2,$$

it follows that $(x|e_{n_0}) = 0$, which is a contradiction. □

Example 4.4 Let us consider the Volterra integral operator $V : L^2([0,1]) \to L^2([0,1])$,

$$Vf(x) = \int_0^x f(t)\,dt.$$

By Theorem 4.12, we know that V is a compact operator. Let us show that $\sigma(V) = \{0\}$ and that it has no eigenvalues. By Proposition 4.2 we know that $r(V) = 0$, so it suffices to show that there are no nonzero elements in $\sigma_p(V)$. Suppose that $\lambda \in \sigma_p(V)$. Then, there exists $f \in L^2([0,1])$, $f \neq 0$, such that $Vf = \lambda f$, and therefore

$$\lambda f(x) = \int_0^x f(t)\,dt \qquad \text{a.e. on } [0,1].$$

Define

$$g(x) := \int_0^x |f(t)|^2\,dt.$$

Then, for almost every $x \in [0,1]$, we have that $g'(x) = |f(x)|^2$. Moreover, g is increasing, $g(0) = 0$, and $g(1) = \|f\|_2^2 > 0$. Therefore, there exists $a \in [0,1[$ such that $g(a) = 0$ and $g(s) > 0$ for all $s \in]a,1]$. On the other hand, applying the Cauchy–Schwarz inequality, we obtain:

$$|\lambda|^2 g'(x) = |\lambda|^2 |f(x)|^2 = \left| \int_0^x f(t)\,dt \right|^2 \leq x \int_0^x |f(t)|^2\,dt = x g(x).$$

Hence,

$$|\lambda|^2 \frac{g'(x)}{g(x)} \leq x, \quad \text{a.e. on }]a,1].$$

Therefore,

$$|\lambda|^2 (\log g(1) - \log g(a)) = |\lambda|^2 \int_a^1 \frac{g'(s)}{g(s)}\,ds \leq \int_a^1 s\,ds = \frac{1}{2} - \frac{a^2}{2},$$

from which we conclude that $\lambda = 0$.

4.7 The Spectrum of a Self-adjoint Operator

Let $(H, (\cdot|\cdot))$ be a Hilbert space. By the Riesz–Fréchet Theorem, we can identify H with its dual H'. More precisely, the map $\Psi : H \to H'$ defined by

$$\langle \Psi(x), y\rangle := (x|y) \qquad \forall y \in H,$$

is an isometric isomorphism. Consequently, given an operator $T \in \mathcal{L}(H)$, its adjoint $T' \in \mathcal{L}(H')$ can be uniquely identified with an operator in H, specifically with the operator $\Psi^{-1}T'\Psi \in \mathcal{L}(H)$, since we have

$$(\Psi^{-1}T'\Psi(x)|y) = \langle T'\Psi(x), y\rangle = \langle \Psi(x), Ty\rangle = (x|Ty).$$

The operator $\Psi^{-1}T'\Psi$ is denoted by T^* and is called the *adjoint* of T. Thus, T^* is characterized as the unique element of $\mathcal{L}(H)$ satisfying:

$$(Tx|y) = (x|T^*y) \quad \forall x, y \in H.$$

From the known properties of the adjoint, we obtain the following result.

Proposition 4.10 *Let H be a Hilbert space and $T, S \in \mathcal{L}(H)$. Then:*

(i) $(T + S)^* = T^* + S^*$.
(ii) $(TS)^* = S^*T^*$.
(iii) *T is invertible if and only if T^* is invertible, and in that case,*

$$(T^*)^{-1} = (T^{-1})^*.$$

(iv) $\sigma(T) = \sigma(T^*)$.
(v) $\|T\| = \|T^*\| = \sqrt{\|T^*T\|}$

Proof We only need to prove (v). Let $x \in H$ with $\|x\| \leq 1$. Then,

$$\|Tx\|^2 = (Tx|Tx) = (x|T^*Tx) \leq \|T^*Tx\|\,\|x\| \leq \|T^*T\| \leq \|T^*\|\,\|T\| = \|T\|^2.$$

Taking the supremum over all such x gives the result. □

Definition 4.8 Let $T \in \mathcal{L}(H)$. We say that T is a *self-adjoint operator* (or *Hermitian*) if $T = T^*$, i.e., if

$$(Tx|y) = (x|Ty) \qquad \forall\, y \in H.$$

Example 4.5

(1) Let $\phi \in L^\infty(\Omega)$ and consider the multiplication operator $M_\phi : L^2(\Omega) \to L^2(\Omega)$, defined by $M_\phi(f)(x) = \phi(x)f(x)$. Then, $(M_\phi)^* = M_{\overline{\phi}}$, since:

$$(M_\phi(f)|g) = \int_\Omega M_\phi(f)(x)\overline{g(x)}\,dx = \int_\Omega \phi(x) f(x)\overline{g(x)}\,dx$$

$$= \int_\Omega f(x)\overline{g(x)\overline{\phi(x)}}\,dx = \int_\Omega f(x)\overline{M_{\overline{\phi}}(g)(x)}\,dx = (f|M_{\overline{\phi}}(g)).$$

Thus, M_ϕ is self-adjoint if and only if $\phi(\Omega) \subset \mathbb{R}$.

(2) Now consider the integral operator $T : L^2([a,b]) \to L^2([a,b])$ with kernel K,

$$Tf(x) = \int_a^b K(x,y)\, f(y)\, dy.$$

Then its adjoint T^* is the integral operator with kernel $K^*(x,y) := \overline{K(y,x)}$, since

$$(Tf|g) = \int_a^b (Tf)(x)\overline{g(x)}\,dx = \int_a^b \left(\int_a^b K(x,y)\, f(y)\, dy \right)\overline{g(x)}\,dx$$

$$= \int_a^b f(y)\overline{\left(\int_a^b K^*(x,y) g(x)\, dx \right)}\, dy = (f|T^*g).$$

Therefore, T is self-adjoint if and only if the kernel K is symmetric and real.

There is the following characterization of self-adjoint operators.

Theorem 4.16 *Let H be a complex Hilbert space and $T \in \mathcal{L}(H)$. Then,*

$$T \text{ is self-adjoint} \iff (Tx|x) \in \mathbb{R} \ \ \forall\, x \in H.$$

Proof ($\Rightarrow$) By hypothesis, $T = T^*$, hence

$$(Tx|x) = (x|Tx) = \overline{(Tx|x)},$$

so $(Tx|x) \in \mathbb{R}$ for each $x \in H$.

($\Leftarrow$) By hypothesis, we have that for every $x, y \in H$ and $\alpha \in \mathbb{C}$,

$$(T(x+\alpha y)|x+\alpha y) = (Tx|x) + \overline{\alpha}(Tx|y) + \alpha(Ty|x) + |\alpha|^2(Ty|y) \in \mathbb{R}.$$

Then this expression equals its complex conjugate, and since

$$(Tx|x), (Ty|y) \in \mathbb{R},$$

we obtain that

$$\alpha(Ty|x) + \overline{\alpha}(Tx|y) = \overline{\alpha}(T^*x|y) + \alpha(T^*y|x).$$

Taking $\alpha = 1$ and $\alpha = i$, we obtain

$$(Ty|x) + (Tx|y) = (T^*x|y) + (T^*y|x), \tag{4.15}$$

$$i(Ty|x) - i(Tx|y) = -i(T^*x|y) + i(T^*y|x). \tag{4.16}$$

Then, multiplying (4.15) by i, and adding it to (4.16), we get

$$2i(Ty|x) = 2i(T^*y|x),$$

and since this holds for all $x, y \in H$, it follows that $T = T^*$. □

The previous theorem is not valid if the space is real. Let us look at an example. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the operator associated with the matrix

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Then,

$$(Tx|x) = ((x_2, -x_1)|(x_1, x_2)) = x_1x_2 - x_1x_2 = 0.$$

However, T is not self-adjoint, since

$$A^t = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \neq A.$$

The following result provides a formula for computing the norm of a self-adjoint operator.

Theorem 4.17 *Let $T \in \mathcal{L}(H)$ be a self-adjoint operator. Then,*

$$\|T\| = \sup\{|(Tx|x)| \ : \ \|x\| = 1\}.$$

Proof Let

$$M := \sup\{|(Tx|x)| \ : \ \|x\| = 1\}.$$

If $\|x\| = 1$, we have

$$|(Tx|x)| \leq \|Tx\| \, \|x\| \leq \|T\|,$$

so that $M \leq \|T\|$. Let us now prove the reverse inequality. By the Riesz–Fréchet Theorem, we have

$$\|Tx\| = \sup\{|(Tx|y)| \ : \ \|y\| = 1\}.$$

Thus, we must show that $|(Tx|y)| \leq M$ for all $x, y \in H$ with $\|x\| = \|y\| = 1$. Indeed, if $\|x\| = \|y\| = 1$, we have

$$(T(x \pm y)|x \pm y) = (Tx|x) \pm (Tx|y) \pm (Ty|x) + (Ty|y)$$

$$= (Tx|x) \pm (Tx|y) \pm (y|Tx) + (Ty|y) = (Tx|x) \pm 2\Re(Tx|y) + (Ty|y),$$

so that

$$4\Re(Tx|y) = (T(x+y)|x+y) - (T(x-y)|x-y).$$

Then, by the parallelogram law we obtain

$$\begin{aligned} 4|\Re(Tx|y)| &\leq |(T(x+y)|x+y)| + |(T(x-y)|x-y)| \\ &\leq M(\|x+y\|^2 + \|x-y\|^2) \leq 2M(\|x\|^2 + \|y\|^2) = 4M. \end{aligned} \tag{4.17}$$

Now, $(Tx|y) = e^{i\theta}|(Tx|y)|$, so replacing x by $e^{-i\theta}x$ in (4.17), we conclude that $|(Tx|y)| \leq M$. □

Corollary 4.1 *If $T \in \mathcal{L}(H)$ is self-adjoint and $(Tx|x) = 0$ for every $x \in H$, then $T = 0$.*

Definition 4.9 Let X and Y be normed spaces and let $T : X \to Y$ be a linear operator. We say that T is 0-*bounded* if there exists a constant $M > 0$ such that

$$\|Tx\| \geq M\|x\| \qquad \forall\, x \in X.$$

The following invertibility criterion holds in $\mathcal{L}(H)$.

Theorem 4.18 *Let H be a Hilbert space and let $T \in \mathcal{L}(H)$. The following are equivalent:*

(i) T is invertible in $\mathcal{L}(H)$.
(ii) T and T^* are 0-bounded.
(iii) T is 0-bounded and T^* is injective.
(iv) T is 0-bounded and Ran T *is dense.*

Proof (i) ⇒ (ii): By the Inverse Mapping Theorem, T is 0-bounded. Moreover, since T^* is also invertible, it follows that T^* is 0-bounded as well.
(ii) ⇒ (iii): This is obvious.
(iii) ⇒ (iv): We know that $\text{Ker}\, T^* = (\overline{\text{Ran}\, T})^\perp$, so since by hypothesis $\text{Ker}\, T^* = 0$, we have $\overline{\text{Ran}\, T} = X$.
(iv) ⇒ (i): Since T is 0-bounded, it is injective. We now show that it is surjective. Let $\{Tx_n\}$ be a Cauchy sequence. Since T is 0-bounded, it follows that $\{x_n\}$ is a Cauchy sequence in H. Therefore, $x_n \to x$ in H, and since T is continuous, $Tx_n \to Tx$.

Hence, Ran T is complete and thus closed in H. Then, since by hypothesis Ran T is dense in H, it follows that Ran $T = H$. □

Definition 4.10 An operator $T \in \mathcal{L}(H)$ is said to be *positive*, and denoted by $T \geq 0$, if

$$(Tx|x) \geq 0 \quad \forall\, x \in H.$$

Lemma 4.6 *Let H be a complex Hilbert space. The following hold:*

(i) If $T \in \mathcal{L}(H)$ is positive, then $T + I$ is invertible.
(ii) For every $T \in \mathcal{L}(H)$, $T^*T + I$ is invertible.

Proof (i): Since T is positive, it follows from Theorem 4.16 that T is self-adjoint. On the other hand,

$$\|(T + I)x\|^2 = (Tx + x|Tx + x) = \|Tx\|^2 + 2(Tx|x) + \|x\|^2 \geq \|x\|^2.$$

Hence, $T + I$ is bounded below by 0. Therefore, by the previous theorem, $T + I$ is invertible.
(ii): Since

$$(T^*Tx|x) = (Tx|Tx) = \|Tx\|^2 \geq 0,$$

we have that T^*T is positive. Consequently, (ii) follows from (i). □

Theorem 4.19 *Let H be a complex Hilbert space and let $T \in \mathcal{L}(H)$ be a self-adjoint operator. Then, $\sigma(T) \subset \mathbb{R}$.*

Proof We need to show that if $\lambda = \alpha + i\beta$ with $\beta \neq 0$, then $(\lambda I - T)$ is invertible. Now, note that

$$\lambda I - T = (\alpha I - T) + i\beta I = \beta \left[\frac{1}{\beta}(\alpha I - T) + iI \right]$$

and $\frac{1}{\beta}(\alpha I - T)$ is self-adjoint, so it will suffice to show that if T is self-adjoint, then $T + iI$ is invertible. Indeed, by the previous lemma, $T^2 + I$ is invertible, and since

$$(T + iI)^*(T + iI) = (T + iI)(T + iI)^* = T^2 + I,$$

we conclude that $T + iI$ is invertible. □

Proposition 4.11 *Let H be a complex Hilbert space and let $T \in \mathcal{L}(H)$ be a self-adjoint operator. Then, for every $\lambda \in \mathbb{C}$, we have*

$$H = \operatorname{Ker}(\lambda I - T) \bigoplus \overline{\operatorname{Ran}(\lambda I - T)}.$$

Proof By the previous theorem, if λ is not a real number, then

$$\operatorname{Ker}(\lambda I - T) = \operatorname{Ker}(\bar{\lambda} I - T) = \{0\}.$$

Therefore, it suffices to show that

$$H = \operatorname{Ker}(\bar{\lambda} I - T) \bigoplus \overline{\operatorname{Ran}(\lambda I - T)}.$$

If $x \in \operatorname{Ker}(\bar{\lambda} I - T)$ and $y \in \operatorname{Ran}(\lambda I - T)$, then $(\bar{\lambda} I - T)x = 0$, and there exists $z \in H$ such that $y = (\lambda I - T)z$. Thus,

$$(x|y) = (x|(\lambda I - T)z) = ((\bar{\lambda} I - T)x|z) = 0.$$

Hence, $(x|y) = 0$ for every $x \in \operatorname{Ker}(\bar{\lambda} I - T)$ and $y \in \overline{\operatorname{Ran}(\lambda I - T)}$, because given $y \in \overline{\operatorname{Ran}(\lambda I - T)}$, there exist $y_n \in \operatorname{Ran}(\lambda I - T)$ such that $y_n \to y$, and therefore, $(x|y) = \lim_{n\to\infty}(x|y_n) = 0$. We thus have that

$$\operatorname{Ker}(\bar{\lambda} I - T) \perp \overline{\operatorname{Ran}(\lambda I - T)}.$$

Finally, let us show that if

$$x \perp \left(\operatorname{Ker}(\bar{\lambda} I - T) + \operatorname{Ran}(\lambda I - T)\right),$$

then $x = 0$. Indeed, for every $z \in H$ we have

$$((\bar{\lambda} I - T)x|z) = (x|(\lambda I - T)z) = 0,$$

so $x \in \operatorname{Ker}(\bar{\lambda} I - T)$, and since $x \perp \operatorname{Ker}(\bar{\lambda} I - T)$, we conclude that $x = 0$. □

Corollary 4.2 *Let H be a complex Hilbert space and let $T \in \mathcal{L}(H)$ be a self-adjoint operator. Then, $\sigma_r(T) \subset \sigma_p(T)$ and, consequently, $\sigma(T) = \sigma_{ap}(T)$.*

Theorem 4.20 *Let $T \in \mathcal{L}(H)$ be a self-adjoint operator, and define*

$$m(T) := \inf\{(Tx|x) \ : \ \|x\| = 1\},$$

$$M(T) := \sup\{(Tx|x) \ : \ \|x\| = 1\}.$$

Then, the following hold:

(i) $\sigma(T) \subset [m(T), M(T)]$.
(ii) $m(T), M(T) \in \sigma(T)$.
(iii) $r(T) = \|T\|$.

Proof First observe that, as a consequence of Theorem 4.17, we have

$$\|T\| = \max\{|m(T)|, |M(T)|\}. \tag{4.18}$$

(i): Suppose that $\lambda > M(T)$. Then,

$$((\lambda I - T)x|x) = \lambda(x|x) - (Tx|x) \geq (\lambda - M(T))\,(x|x).$$

Hence,

$$(\lambda - M(T))\|x\|^2 \leq ((\lambda I - T)x|x) \leq \|(\lambda I - T)x\|\,\|x\|,$$

which implies

$$\|(\lambda I - T)x\| \geq (\lambda - M(T))\|x\|.$$

Therefore, by Theorem 4.18, we have $\lambda \in \rho(T)$. Similarly, it can be shown that if $\lambda < m(T)$, then $\lambda \in \rho(T)$.
(ii): By the definition of $M(T)$, there exists a sequence $x_n \in H$ with $\|x_n\| = 1$ such that $(Tx_n|x_n) = M(T) - \delta_n$, where $0 \leq \delta_n \to 0$. Let $\gamma > \max\{|m(T)|, |M(T)|\}$. Then,

$$\sup_{\|x\|=1}((T+\gamma)x|x) = \sup_{\|x\|=1}(Tx|x) + \gamma(x|x) = M(T) + \gamma,$$

$$\inf_{\|x\|=1}((T+\gamma)x|x) = \inf_{\|x\|=1}(Tx|x) + \gamma(x|x) = m(T) + \gamma.$$

Since $M(T) + \gamma \geq m(T) + \gamma > 0$, we have $\|T + \gamma I\| = M(T) + \gamma$. Hence,

$$\|(T + \gamma I)x_n\| \leq (M(T) + \gamma)\|x_n\| = M(T) + \gamma.$$

Now,

$$\|(T - M(T))x_n\|^2 = \|(T + \gamma I)x_n - (M(T) + \gamma)x_n\|^2$$

$$= \|(T + \gamma I)x_n\|^2 + (M(T) + \gamma)^2\|x_n\|^2 - 2(M(T) + \gamma)((T + \gamma I)x_n|x_n)$$

$$\leq (M(T) + \gamma)^2 + (M(T) + \gamma)^2 - 2(M(T) + \gamma)(M(T) - \delta_n + \gamma) \to 0.$$

Consequently, $M(T) \in \sigma_{ap}(T) = \sigma(T)$. The proof that $m(T) \in \sigma(T)$ is similar. Indeed, take a sequence $y_n \in H$, with $\|y_n\| = 1$, such that $(Ty_n|y_n) = m(T) + \epsilon_n$, where $\epsilon_n \to 0$, and show that

$$\|(T - m(T))y_n\|^2 = \|(T - \gamma I)y_n - (m(T) - \gamma)y_n\|^2 \to 0.$$

(iii): We know that $r(T) \leq \|T\|$. Now, since $m(T), M(T) \in \sigma(T)$, taking into account (4.18), we have

$$r(T) \geq \max\{|m(T)|, |M(T)|\} = \|T\|,$$

hence $r(T) = \|T\|$. □

Corollary 4.3 *Let H be a complex Hilbert space and $T \in \mathcal{L}(H)$. The following are equivalent:*

(i) T is positive.
(ii) T is self-adjoint and $\sigma(T) \subset [0, +\infty[$.

Proof (i) $\Rightarrow$ (ii): Since T is positive, it must be self-adjoint, so we know that $\sigma(T) \subset \mathbb{R}$. Let us show that if $\beta \in \mathbb{R}$ with $\beta < 0$, then $\beta \in \rho(T)$. Indeed, since

$$\beta I - T = \beta \left(I + \frac{1}{(-\beta)} T \right),$$

and $\frac{1}{(-\beta)} T \geq 0$, by applying Lemma 4.6, we conclude that $\beta I - T$ is invertible.
(ii) $\Rightarrow$ (i): By the previous theorem, we have that $m(T) \in \sigma(T) \subset [0, +\infty[$, hence T is positive. □

Remark 4.3 If T is positive, then $r(T) \in \sigma(T)$, since $r(T) = \|T\| = M(T) \in \sigma(T)$. However, this is not true in general for self-adjoint operators. For example, if T is the operator on $\mathbb{C}^2$ associated to the matrix

$$A = \begin{pmatrix} -2 & 0 \\ 0 & 1 \end{pmatrix},$$

then $\sigma(T) = \{-2, 1\}$, and $r(T) = 2 \notin \sigma(T)$.

4.8 Spectral Decomposition of an Operator

In this section, we will see that every compact self-adjoint operator is diagonalizable. We need the following result.

Lemma 4.7 *Let H be a Hilbert space and $T \in \mathcal{L}(H)$. Then:*

(i) If M is a closed subspace of H that is invariant under T (i.e., $T(M) \subset M$), then $M^{\perp}$ is invariant under T^*.
(ii) If T is self-adjoint and $\lambda, \mu \in \sigma_p(T)$ with $\lambda \neq \mu$, then

$$\text{Ker}\,(\lambda I - T) \perp \text{Ker}\,(\mu I - T).$$

Proof (i): Let $x \in M^{\perp}$ and $y \in M$. Since $Ty \in M$, we have that $0 = (Ty|x) = (y|T^*x)$, hence $T^*x \in M^{\perp}$.

(ii): Let $x \in \text{Ker}\,(\lambda I - T)$ and $y \in \text{Ker}\,(\mu I - T)$. Then $Tx = \lambda x$ and $Ty = \mu y$, so that

$$0 = (Tx|y) - (x|Ty) = (\lambda x|y) - (x|\mu y) = (\lambda - \overline{\mu})(x|y).$$

Now, since $\lambda, \mu \in \mathbb{R}$, we have $\lambda - \overline{\mu} = \lambda - \mu \neq 0$, hence $(x|y) = 0$. □

Theorem 4.21 (The Spectral Theorem for Compact Self-Adjoint Operators) *Let H be a Hilbert space, and let $T \in \mathcal{L}(H)$ be a compact self-adjoint operator. There exists an orthonormal sequence $\{x_n\}$ of eigenvectors of T, with corresponding eigenvalues $\{\lambda_n\}$, such that $|\lambda_1| \geq |\lambda_2| \geq \ldots$, and*

$$Tx = \sum_{n=1}^{\infty} \lambda_n (x|x_n) x_n.$$

Moreover, if there are infinitely many distinct λ_n, then $\lim_{n\to\infty} \lambda_n = 0$.

Proof We will obtain the eigenvectors x_n inductively. As a consequence of Theorem 4.20, we know that $\lambda_1 = \|T\|$ or $\lambda_1 = -\|T\|$ is an eigenvalue of T. Take a unit norm eigenfunction x_1 corresponding to the eigenvalue λ_1. Since $LIN\{x_1\}$ is invariant under T, and $T^* = T$, as a consequence of the previous lemma we have that $H_2 := LIN\{x_1\}^{\perp}$ is invariant under T. Thus, we can define $T_2 := T_{|H_2}$, and we have that $T_2 \in \mathcal{K}(H_2)$ and it is self-adjoint. Applying Theorem 4.20 to T_2, we obtain that $\lambda_2 = \|T_2\|$ or $\lambda_2 = -\|T_2\|$ is an eigenvalue of T_2. Let x_2 be a unit eigenfunction corresponding to the eigenvalue λ_2. Note that $(x_1|x_2) = 0$.

Inductively, suppose we have constructed unit, mutually orthogonal eigenvectors $x_1, x_2, \ldots, x_n$ of T, with corresponding eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$, such that $|\lambda_i| = \|T_i\|$, for $i = 1, 2, \ldots, n$, where $T_1 = T$ and $T_j = T_{|H_j}$, with

$$H_j = LIN\{x_1, \ldots, x_{j-1}\}^{\perp}, \quad \text{for } 1 \leq j \leq n.$$

Then let $H_{n+1} = LIN\{x_1, \ldots, x_n\}^{\perp}$, and define $T_{n+1} := T_{|H_{n+1}}$. Again, $\lambda_{n+1} = \pm\|T_{n+1}\|$ is an eigenvalue of T_{n+1}; thus, we can choose a unit vector $x_{n+1} \in H_{n+1}$ that is an eigenfunction associated with the eigenvalue λ_{n+1}. Then $x_1, \ldots, x_{n+1}$ are mutually orthogonal unit eigenvectors of T, and their corresponding eigenvalues $\lambda_1, \ldots, \lambda_{n+1}$ satisfy $|\lambda_i| = \|T_i\|$ for $1 \leq i \leq n+1$. We continue inductively as long as $T_n \neq 0$.

If there exists m such that $T_m = 0$, we stop the construction, and for any $x \in H$, we have

$$y = x - \sum_{j=1}^{m-1} (x|x_j) x_j \in H_m.$$

Then,

$$0 = T_m y = Ty = Tx - \sum_{j=1}^{m-1} (x|x_j) Tx_j,$$

so that

$$Tx = \sum_{j=1}^{m-1} \lambda_j (x|x_j) x_j.$$

Consequently, in this case the statement of the theorem holds.

Finally, suppose that $T_n \neq 0$ for all $n \in \mathbb{N}$. Given $x \in H$, let

$$y_n := x - \sum_{j=1}^{n-1} (x|x_j) x_j \in H_n.$$

Then, applying the Pythagorean Theorem to the relation

$$x = y_n + \sum_{j=1}^{n-1} (x|x_j) x_j,$$

we obtain

$$\|x\|^2 = \|y_n\|^2 + \sum_{j=1}^{n-1} |(x|x_j)|^2,$$

so $\|y_n\| \leq \|x\|$ for every $n \in \mathbb{N}$. Now,

$$\|Ty_n\| = \|T_n y_n\| \leq \|T_n\| \, \|y_n\| \leq |\lambda_n| \, \|x\|.$$

Hence,

$$\left\| Tx - \sum_{j=1}^{n-1} \lambda_j (x|x_j) x_j \right\| = \|Ty_n\| \leq |\lambda_n| \, \|x\|.$$

By the Riesz–Schauder Theorem, we have that $|\lambda_n| \to 0$, and thus

$$Tx = \sum_{n=1}^{\infty} \lambda_n (x|x_n) x_n.$$

□

In general, the orthogonal sequence $\{x_n\}$ from the previous theorem is not necessarily an orthonormal basis of the space H if the space is not separable. Now, in the separable case, we have the following result.

Corollary 4.4 *Let H be a separable Hilbert space of infinite dimension. Then, given $T \in \mathcal{L}(H)$ a compact and self-adjoint operator, there exists an orthonormal basis $\{e_n\}$ of H consisting of eigenvectors of T. Moreover, for each $x \in H$,*

$$Tx = \sum_{n=1}^{\infty} \lambda_n (x|e_n) e_n,$$

where λ_n is the eigenvalue corresponding to e_n.

Proof By the previous theorem, there exists an orthonormal sequence $\{x_n\}$ of eigenvectors of T, with eigenvalues $\lambda_n \neq 0$, such that

$$Tx = \sum_{n=1}^{\infty} \lambda_n (x|x_n) x_n. \tag{4.19}$$

Now, since $\operatorname{Ker} T$ is a separable Hilbert space, there exists a countable orthonormal basis $\{y_n \ : \ n \in I \subset \mathbb{N}\}$ of $\operatorname{Ker} T$. Then, each y_n is an eigenvector corresponding to the eigenvalue 0, so y_n is orthogonal to x_m for each $n \in I$, $m \in \mathbb{N}$. We then consider the countable orthonormal system

$$\mathcal{S} := \{x_m \ : \ m \in \mathbb{N}\} \cup \{y_n \ : \ n \in I\}.$$

By (4.19), for each $x \in H$,

$$x - \sum_{n=1}^{\infty} (x|x_n) x_n \in \operatorname{Ker} T,$$

and since $\{y_n \ : \ n \in I \subset \mathbb{N}\}$ is a basis of $\operatorname{Ker} T$, we have that

$$x - \sum_{n=1}^{\infty} (x|x_n) x_n = \sum_{m \in I} (x|y_m) y_m.$$

Therefore, $\mathcal{S}$ is a basis of H. Moreover, if we write

$$\mathcal{S} = \{e_n \ : \ n \in \mathbb{N}\},$$

we have

$$Tx = \sum_{n=1}^{\infty} \lambda_n (x|e_n) e_n.$$

□

4.9 The Fredholm Integral Equation

A *Fredholm integral equation* (of the second kind) is an equation of the form:

$$f(s) - \lambda \int_a^b K(s,t)\, f(t)\, dt = g(s), \tag{4.20}$$

where $\lambda \neq 0$, g is given, and f is the unknown function.

Theorem 4.22 (Fredholm Alternative Theorem) *Let $K \in L^2([a,b] \times [a,b])$ and $\lambda \neq 0$, and consider the integral equation*

$$f(s) - \lambda \int_a^b K(s,t)\, f(t)\, dt = g(s). \tag{4.21}$$

Then, one of the following alternatives holds:

(i) For each $g \in L^2([a,b])$, there exists a unique solution $f \in L^2([a,b])$ to Eq. (4.21).

(ii) The homogeneous equation

$$f(s) - \lambda \int_a^b K(s,t)\, f(t)\, dt = 0$$

has a finite number of linearly independent solutions, and in this case, Eq. (4.21) has a solution if and only if

$$g \in \left[\operatorname{Ker} \left(\frac{1}{\lambda} I - T^* \right) \right]^{\perp}.$$

Proof To study this equation with the help of spectral theory, let us consider the integral operator associated with the kernel K, i.e., the operator $T : L^2([a,b]) \to L^2([a,b])$, defined by

$$Tf(s) := \int_a^b K(s,t)\, f(t)\, dt.$$

By Theorem 4.12, if $K \in L^2([a,b] \times [a,b])$, then T is a compact operator.

Equation (4.20) can be written in the form:

$$(I - \lambda T) f = g. \tag{4.22}$$

Since $\lambda \neq 0$, if we set $\mu := \frac{1}{\lambda}$, we can rewrite (4.22) as:

$$(\mu I - T) f = \mu g. \tag{4.23}$$

Then, applying the Fredholm Alternative Theorem, we obtain the result. □

Corollary 4.5 *For every $g \in L^2([a, b])$ and $\lambda \neq 0$, the Volterra integral equation*

$$f(s) - \lambda \int_a^s f(t)\, dt = g(s)$$

has a unique solution $f \in L^2([a, b])$.

Proof As we saw in Example 4.4, if V is the Volterra integral operator associated with the integral equation, i.e., if

$$Vf(s) = \int_a^s f(t)\, dt,$$

we have that $\sigma(V) = \{0\}$. Hence, $\frac{1}{\lambda} \notin \sigma(V)$, and by applying the previous theorem, we obtain the result. □

Now suppose that the kernel $K \in L^2([a, b] \times [a, b])$ is symmetric, i.e., $K(s, t) = \overline{K(t, s)}$. Then the integral operator T associated with the kernel K is self-adjoint. Thus, we can apply the spectral theorem to T, and since $L^2([a, b])$ is separable, we can apply Corollary 4.4, which ensures that there exists an orthonormal basis $\{e_n\}$ of $L^2([a, b])$ such that

$$Tf = \sum_{n=1}^{\infty} \mu_n (f|e_n) e_n,$$

with $\{\mu_n\} = \sigma_p(T) \setminus \{0\}$. Therefore,

$$f = \lambda T f + g = \lambda \sum_{n=1}^{\infty} \mu_n (f|e_n) e_n + g.$$

Hence, for each $n \in \mathbb{N}$,

$$(f|e_n) = \lambda \mu_n (f|e_n) + (g|e_n).$$

So, if $\frac{1}{\lambda} \notin \sigma(T)$, we have

$$(f|e_n) = \frac{(g|e_n)}{1-\lambda\mu_n}.$$

Thus, in this case, the unique solution to the Fredholm integral equation (4.20) is given by

$$f = \lambda \sum_{n=1}^{\infty} \frac{\mu_n}{1-\lambda\mu_n} (g|e_n)e_n + g. \tag{4.24}$$

Now suppose that $\frac{1}{\lambda} \in \sigma(T) \setminus \{0\}$. Let $\lambda = \mu_{n_0}$. Then, applying the previous reasoning to the operator T restricted to the subspace $[\mathrm{Ker}\,(\mu_{n_0} I - T)]^{\perp}$, we have that if $g \in [\mathrm{Ker}\,(\mu_{n_0} I - T)]^{\perp}$, then

$$f_1 = \lambda \sum_{n=1, n\neq n_0}^{\infty} \frac{\mu_n}{1-\lambda\mu_n} (g|e_n)e_n + g$$

is a solution of Eq. (4.20). Now, if $f_2 \in \mathrm{Ker}\,(\mu_{n_0} I - T)$, then $f_1 + f_2$ is also a solution of (4.20). Hence, in this case, the solutions of (4.20) are of the form:

$$f = \lambda \sum_{n=1}^{\infty} c_n (g|e_n)e_n + g,$$

with

$$c_n = \begin{cases} \dfrac{\mu_n}{1-\lambda\mu_n} & \text{if } n \neq n_0 \\ \\ \text{arbitrary} & \text{if } n = n_0. \end{cases}$$

Problems

4.1 Let $T_n, T \in \mathcal{L}(X)$ with $T_n \to T$ in $\mathcal{L}(X)$. Prove that if $\alpha_n \in \sigma(T_n)$ and $\alpha_n \to \alpha$ in $\mathbb{C}$, then $\alpha \in \sigma(T)$.

4.2 Let $T, S \in \mathcal{L}(X)$. Prove that $r(TS) = r(ST)$. What is the relationship between $\sigma(TS)$ and $\sigma(ST)$?

4.3 Let $T \in \mathcal{L}(\ell^1)$, defined by $T[(a_n)] := \left(\sum_{n=2}^{\infty} a_n, a_1, a_2, \ldots\right)$ for every $(a_n) \in \ell^1$. Prove that

$$\sigma(T) = \{z \in \mathbb{C} \,:\, |z| \leq 1\} \cup \left\{\frac{1+\sqrt{5}}{2}\right\}.$$

4.4 Let H be a Hilbert space with orthonormal basis $\{e_n \,:\, n \in \mathbb{N}\}$. Define the operator $T : H \to H$ by

$$T(e_1) = 0, \quad T(e_2) = \frac{e_1}{2}, \ \ldots, \ T(e_n) = \frac{e_{n-1}}{n}.$$

Find the spectrum of T.

4.5 Let $V : C([0,1]) \to C([0,1])$ be the Volterra integral operator

$$Vf(x) := \int_0^x f(t)\, dt \qquad \forall\ f \in C([0,1]).$$

Prove that $\sigma(V) = \{0\}$.

4.6 Compute the spectrum of the integral operator on $L^2([0,1])$ with kernel $K(x,y) := \min\{x,y\}$.

4.7 Let $K \subset \mathbb{C}$ be a compact set. Find an operator $T \in \mathcal{L}(C([0,1]))$ such that $\sigma(T) = K$.

4.8 Let X be a Banach space and $T \in \mathcal{L}(X)$. Prove that if T is an isometry (i.e., $\|Tx\| = \|x\|, \forall\, x \in X$), then $\sigma(T) \subset \mathbb{T}$ or $\sigma(T) = \{z \in \mathbb{C} \ :\ |z| \le 1\}$.

4.9 Let $T \in \mathcal{L}(\ell^1)$, defined by $T[(a_n)] := (0, -a_1, -a_2, \ldots)$ for every $(a_n) \in \ell^1$. Prove that:

(i) $\rho(T) = \{z \in \mathbb{C} \ :\ |z| > 1\}$, and

$$\|R(z,T)\| = \frac{1}{|z| - 1} \quad \forall\, z \in \mathbb{C}.$$

(ii) If $|z| \le 1$, then the range of the operator $zI - T$ is not dense in ℓ^1. Consequently, $\sigma(T) = \sigma_r(T)$.

4.10 Give an example of an operator $T \in \mathcal{L}(X,Y)$ for which

$$\overline{\operatorname{Ran} T'} \neq \big(\operatorname{Ker} T\big)^{\perp}.$$

4.11 Let $T : \ell^1 \to \ell^1$ be the operator defined by the infinite matrix $A = (a_{i,j})$, with

$$\begin{cases} a_{i,j} = 1 \ \text{ if } \ |i-j| = 1, \\ \\ a_{i,j} = 0 \ \text{ if } \ |i-j| \neq 1. \end{cases}$$

Prove that $\sigma(T) = \{z \in \mathbb{R} \ :\ -2 \le z \le 2\}$.

4.12 Let $K \subset \mathbb{C}$ be a compact set, and let $\{a_n\}$ be a dense sequence in K. Let $T : \ell^2 \to \ell^2$ be the operator defined by $T[(b_n)] := (a_n b_n)$. Prove that:

(i) $\sigma(T) = K$.

(ii) Each a_n is an eigenvalue of T.
(iii) If $z \in K$ with $z \neq a_n$ for every $n \in \mathbb{N}$, then z is not an eigenvalue and $z \in \sigma_r(T)$.

4.13 Let $T \in \mathcal{L}(X)$. Prove that $\sigma_{ap}(T)$ is closed.

4.14 Find an operator $T \in \mathcal{L}(\ell^2)$ such that $\sigma_p(T) = \emptyset$ and $\sigma_{ap}(T) = \{0\}$.

4.15 Let $P(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_0$ be a polynomial with complex coefficients. Prove that if for $T \in \mathcal{L}(X)$ we define $P(T) := a_n T^n + a_{n-1} T^{n-1} + \cdots + a_0 I$, then

$$\sigma(P(T)) = \{\mu = P(\lambda) \ : \ \lambda \in \sigma(T)\}.$$

4.16 Let H be a Hilbert space with orthonormal basis $\{e_n \ : \ n \in \mathbb{N}\}$. Let $\{\alpha_n \ : \ n \in \mathbb{N}\} \subset \mathbb{K}$ such that

$$M = \sup\{|\alpha_n| \ : \ n \in \mathbb{N}\} < \infty,$$

and let T be the diagonal operator associated with $\{\alpha_n \ : \ n \in \mathbb{N}\}$, i.e., the one defined by $T(e_n) := \alpha_n e_n$ for all $n \in \mathbb{N}$. Prove that

$$\sigma(T) = \sigma_{ap}(T) = \overline{\{\alpha_n \ : \ n \in \mathbb{N}\}}.$$

4.17 Prove that every compact operator maps weakly convergent sequences to norm convergent sequences.

4.18 Prove that every Hilbert space has the approximation property.

4.19 Let S be the unilateral shift operator on ℓ^2, i.e., $S[(a_n)] := (0, a_1, a_2, \ldots)$. Prove that

$$\inf\{\|S - K\| \ : \ K \in \mathcal{K}(\ell^2)\} = 1.$$

4.20 Let X be a Banach space. Prove that every non-zero ideal of $\mathcal{L}(X)$ contains the finite-rank operators.

4.21 Let X be a Banach space with a basis (i.e., there exists a sequence $\{x_n \ : \ n \in \mathbb{N}\}$ such that every $x \in X$ can be uniquely written as $x = \sum_{n=1}^{\infty} a_n x_n$). Prove that $\mathcal{F}(X)$ is dense in $\mathcal{K}(X)$.

4.22 Let X be a reflexive Banach space. Prove that $\mathcal{L}(X, \ell^1) = \mathcal{K}(X, \ell^1)$ and that $\mathcal{L}(c_0, X) = \mathcal{K}(c_0, X)$.

4.23 Let H_1 and H_2 be two Hilbert spaces, and let $T \in \mathcal{L}(H_1, H_2)$. We say that T is a *Hilbert-Schmidt operator* if there exists an orthonormal basis $\{e_n \ : \ n \in \mathbb{N}\}$ of H_1 such that $\sum_{n=1}^{\infty} \|Te_n\|^2 < +\infty$. Prove that every Hilbert-Schmidt operator is compact.

4.24 Give an example of a compact operator that is not Hilbert-Schmidt.

4.25 Let H be a Hilbert space and $T \in \mathcal{L}(H)$. Prove that $x \neq 0$ is an eigenvector of T if and only if $|(x|Tx)| = \|Tx\|\,\|x\|$.

4.26 Let H be a Hilbert space with orthonormal basis $\{e_n : n \in \mathbb{N}\}$. We denote by $\mathcal{HS}(H)$ the vector space of all Hilbert-Schmidt operators on H. Prove that $\mathcal{HS}(H)$ is a Hilbert space with respect to the inner product

$$(T|S) := \sum_{n=1}^{\infty} (Te_n|Se_n).$$

4.27 Let H be a Hilbert space and $T \in \mathcal{L}(H)$. We say that T is a *normal operator* if $TT^* = T^*T$. Prove that:

(i) T is normal $\iff$ $\|Tx\| = \|T^*x\|$ for all $x \in H$.
(ii) If T is normal, then T is injective if and only if T has dense range.
(iii) Give an example of an injective operator with non-dense range, and of a surjective but non-injective operator.

4.28 Let H be a complex Hilbert space and $T \in \mathcal{L}(H)$ such that $\operatorname{Ker} T$ has finite dimension. We define:

$$\alpha(T) := \inf\{\|T - S\| : \dim T < \dim S\}$$

$$\gamma(T) := \inf\{\|Tx\| : x \perp \operatorname{Ker} T \text{ and } \|x\| = 1\}.$$

Prove that $\alpha(T) = \gamma(T)$.

4.29 Let H be a complex Hilbert space and $T \in \mathcal{L}(H)$. A subspace M of H is said to be *reducing* for T if both M and its orthogonal complement are invariant under T (i.e., $T(M) \subset M$ and $T(M^\perp) \subset M^\perp$). Give an example of an operator T that has nontrivial invariant subspaces, but whose only reducing subspaces are $\{0\}$ and the whole space.

4.30 Let H be a separable Hilbert space. Prove that the Hilbert exponential matrix $(a_{i,j})$, with $a_{i,j} := \dfrac{1}{2^{i+j+1}}$, $i, j = 0, 1, \ldots$, defines a self-adjoint operator $T \in \mathcal{L}(H)$ such that $\|T\| = \frac{2}{3}$, and $\sigma_p(T) = \{0, \frac{2}{3}\}$. Moreover, 0 has infinite multiplicity and $\frac{2}{3}$ has multiplicity one.

4.31 Let H be a finite-dimensional Hilbert space. Let $T \in \mathcal{L}(H)$ such that $\|T\| = 1$ and $\sigma(T) = \{1\}$. Prove that $T = I$.

4.32 Prove that the result from the previous problem does not hold in infinite dimensions, i.e., find an operator $T \in \mathcal{L}(H)$, with H infinite-dimensional, such that $\|T\| = 1$, $\sigma(T) = \{1\}$, and $T \neq I$.

4.33 Let H be a Hilbert space and $T \in \mathcal{L}(H)$ a positive operator. Prove that there exists a unique positive operator $D \in \mathcal{L}(H)$ such that $D^2 = T$. This operator D is called the square root of T, and is denoted $D = \sqrt{T}$.

4.34 Let T be a compact self-adjoint operator on the separable Hilbert space H. By the spectral theorem, there exists an orthonormal basis $\{e_n : n \in \mathbb{N}\}$ consisting of eigenvectors of T. Let λ_n be the corresponding eigenvalues. Suppose that $f : \sigma(T) \to \mathbb{C}$ is a function such that $\lim_{n\to\infty} f(\lambda_n)$ exists. Prove that the operator

$$f(T)x := \sum_{n=1}^{\infty} f(\lambda_n)(x|e_n)e_n$$

is well defined and that $f(T) \in \mathcal{L}(H)$. Also prove that if f is real-valued, then $f(T)$ is self-adjoint, and if $\lim_{n\to\infty} f(\lambda_n) = 0$, then $f(T)$ is compact.

Chapter 5
Theory of Distributions



5.1 Introduction

The need to generalize the concept of function, which gives rise to the theory of distributions or generalized functions, stems from the physicists' necessity to use derivatives of non-differentiable functions. In 1912, Peano wrote:

> *I am sure that something must be found. There must exist a notion of generalized function that is to functions what real numbers are to rational numbers.*

This necessity arises from the fact that some physical solutions to partial differential equations, which represent the fundamental equations of Physics, correspond to functions that are not differentiable or have discontinuous derivatives. One of the first to point out this need was B. Riemann in 1858;[1] in his study of the propagation of compressible gases, he concluded that the mathematical model (a system of nonlinear partial differential equations), understood in the sense of classical solutions, was contradictory, as it predicted intersecting characteristic lines along which the physical variables simultaneously took on different values. However, Riemann ventured that the theory was correct if one radically changed the point of view and accepted that the solutions of a differential equation might not be differentiable, or even continuous. One of the simplest examples where this situation occurs is the *Burgers' equation*, i.e., the equation

$$u_t + uu_x = 0 \tag{5.1}$$

Suppose that u is a classical solution to Eq. (5.1) with initial data

$$u(0, x) = u_0(x), \qquad \forall\, x \in \mathbb{R}. \tag{5.2}$$

[1] Abh. Ges. Wiss. Göttingen Math, Kl. **18** (1858/59), 43–65.

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Let us consider the so-called characteristic curves of the equation, i.e., the curves $x(t)$ solving the differential equation

$$\frac{dx(t)}{dt} = u(t, x(t)). \tag{5.3}$$

Along these curves, if $f(t) = u(t, x(t))$, we have

$$f'(t) = \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x}\frac{\partial x}{\partial t} = 0 \quad \forall t > 0.$$

Therefore, f is constant. Thus, u is constant along the characteristics. Consequently, it follows from (5.3) that these curves are straight lines. The characteristic curve passing through $(0, x_0)$ is given by $x(t) = x_0 + u_0(x_0)t$, and along this curve u takes the constant value $u(t, x(t)) = u_0(x_0)$. Hence, if two characteristic lines intersect at a point (t, x), the value of $u(t, x)$ is not well defined. Consequently, except for small times, we cannot expect u to be continuous. Therefore, the problem (5.1)–(5.2) does not have a classical solution. However, since this problem originates from a physical context, it presents a dilemma. The resolution of this dilemma is to introduce a notion of solution weaker than the classical solution (which we will call a weak solution), which allows for the possibility that u is discontinuous.

We can rewrite Eq. (5.1) in the form

$$u_t + \left(\frac{u^2}{2}\right)_x = 0,$$

which is then a particular case of an equation of the form

$$u_t + \big(f(u)\big)_x = 0. \tag{5.4}$$

Such equations are called *conservation laws*, due to their analogy with the form taken by conservation laws in Physics.

Suppose that u is a classical solution of (5.4) with initial data $u(0, x) = u_0(x)$. Then, if $\phi \in C_0^\infty(]0, +\infty[\times\mathbb{R})$, by multiplying the equation by ϕ and integrating by parts, we obtain

$$\int_0^\infty \int_{\mathbb{R}} u\phi_t + f(u)\phi_x + \int_{\mathbb{R}} u_0(x)\phi(0, x)\, dx = 0. \tag{5.5}$$

A measurable and bounded function u that satisfies (5.5) for every test function $\phi \in C_0^\infty(]0, +\infty[\times\mathbb{R})$ is called a *weak solution* of (5.4).

This type of issue arises even in more classical linear equations than Burgers' equation. Consider the *vibrating string equation*

$$\frac{1}{v^2}\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, \tag{5.6}$$

where v is the wave propagation speed. In 1747, D'Alembert, and a year later Euler, showed that the general solution of this equation is of the form $u(t, x) = f(x+tv)+g(x-tv)$, where f and g are functions of one variable. Naturally, this assumes that f and g are of class C^2. What happens if f and g are only of class C^1, or continuous, or even not continuous at all? Is u still a wave?

The historical development of the mathematical tools that rigorously address the above problem is very interesting. This development is what we aim to sketch in this introduction. It is not our intention to discuss here the question of who invented distributions. A good textbook on this topic is that of J. Lützen [41], from which we take the following comment:

> *If one asks who was the first person to use distributions in mathematics, the answer is Fourier in 1822, Kirchhoff in 1882, and Heaviside in 1898. If one asks for a rigorous theory, which possibly only uses distributions implicitly, the answer is Bochner in 1932. If one wants to know who first rigorously defined distributions as functionals, the answer is Sobolev in 1935. Finally, if one wants to name the person who saw the great applications of distributions and created a broad and comprehensive theory of these objects, Schwartz is the one to cite.*

In 1882, G. Kirchhoff introduced a so-called "function" δ with the strange property of vanishing everywhere on the real line except at the origin, and such that $\int_a^b \delta(x)\,dx = 1$ if $a < 0 < b$. Although such δ-functions have a physical meaning, like point masses or charges, many nineteenth-century mathematical physicists avoided using them due to the mathematical issues they entailed. For example, Maxwell in his *Treatise on Electricity and Magnetism* of 1873, comments on the formula for the potential of a point charge:

$$V = \frac{e}{r}, \quad r = \sqrt{(x-a)^2 + (y-b)^2 + (z-c)^2} \tag{5.7}$$

in the following way:

> *The value of V given by Eq. (5.7) may be the actual value of the potential in the space outside a closed surface surrounding the point (a, b, c), but we cannot, except for purely mathematical purposes, assume that this is the form of the function at the point (a, b, c) itself.*

Prior to Kirchhoff's definition, a mathematical description of δ-functions appeared in connection with Fourier series. Even in Fourier's own 1822 monograph *Théorie Analytique de la Chaleur*, Fourier showed how the expansion into a Fourier series can give rise to an integral expression of the form

$$f(x) = \int \delta(\alpha - x)\,d\alpha, \tag{5.8}$$

with

$$\delta(x) = \frac{1}{\pi}\left(\frac{1}{2} + \sum_{n=1}^{\infty} \cos nx\right) \quad \text{for } x \in [-\pi, \pi]. \tag{5.9}$$

In his proof, Fourier unscrupulously interchanges integration and summation. This lack of rigor was corrected within the framework of distribution theory, as Schwartz showed in [52, p. 225] that

$$\frac{1}{2} + \sum_{i=1}^{p} \cos\, i(x-\alpha) \to \pi \sum_{n=-\infty}^{+\infty} \delta(x + 2\pi n),$$

in the sense of distributions. In his proof of what is now known as the Fourier Integral Theorem, Fourier used an argument which, expressed in more modern language, can be written as

$$\lim_{n\to\infty} \frac{\sin\, n(\alpha - x)}{\pi(\alpha - x)} = \delta(\alpha - x), \tag{5.10}$$

a result that is indeed correct within the framework of distribution theory. This idea of proving the Fourier Integral Theorem or results on the convergence of Fourier series by showing that the Dirichlet kernel or similar kernels as in (5.10) converge to a δ-function was also used by Cauchy in 1823.

It was the electrical engineer Oliver Heaviside who was among the first to point out the connection between Fourier series and δ-functions. He began his work with the δ-function in a 1893 article[2] where he introduced his *symbolic calculus*, which he later applied and extended in his textbook on Electromagnetism. Heaviside introduced the function he called *"the unit-step"*, which is now known as the *Heaviside function*, i.e., the real function H defined by $H(t) = 0$ for $t < 0$ and $H(t) = 1$ for $t \geq 0$. This function is not differentiable at the origin. Heaviside called its derivative $H'(t)$ *"the unit-impulse"*; this "function" is equal to zero except at the origin, where it equals $+\infty$, but its integral must be equal to 1, i.e.,

$$\int_{-\epsilon}^{\epsilon} H'(t)\, dt = H(\epsilon) - H(-\epsilon) = 1,$$

for each $\epsilon > 0$, so that

$$\int_{-\infty}^{\infty} H'(t)\, dt = 1.$$

This function was rediscovered by Paul Dirac in 1926,[3] introducing it as a convenient notation in the mathematical formulation of Quantum Mechanics, in the following form:

> *We need a notation for the function which is zero except when x is very small, and whose integral over a range containing $x = 0$ is equal to one. We shall use the symbol $\delta(x)$ to*

[2] Proc. Roy. Soc. London **52** (1893), 504–529.

[3] Proc. Roy. Soc., A **113** (1927), 621-641.

denote this function, i.e., $\delta(x)$ is defined as

$$\delta(x) = 0 \quad \text{when } x \neq 0$$

and

$$\int_{-\infty}^{+\infty} dx\, \delta(x) = 1.$$

Strictly speaking, of course, $\delta(x)$ is not an actual function of x, but may only be regarded as the limit of a certain sequence of functions. At the same time, $\delta(x)$ can be used as if it were a proper function for practically all purposes of Quantum Mechanics without giving wrong results.

Nowadays, this function is denoted by δ and is called the Dirac measure (or the Dirac delta), since it is actually a measure; there are no functions that are zero almost everywhere and yet have an integral equal to one. Heaviside, in his symbolic calculus, also introduced the successive derivatives of the function δ, as well as formulas of the type:

$$\delta \star f = f, \quad \delta' \star f = f'.$$

Heaviside's calculus had a great impact among engineers, but it was strongly criticized by the mathematical community for its lack of rigor. The first justification of Heaviside's symbolic calculus, via the Laplace transform, was given by N. Wiener in 1926;[4] although the function δ and its derivatives were still unjustified; that justification had to wait until the introduction of Schwartz's theory of distributions.

Weak solutions of partial differential equations originated in a work by Du Bois-Reymond on the Calculus of Variations. Although such solutions appear sporadically in the works of Poincaré, their systematic study began between 1930 and 1940. The first important work where the concept of a weak solution appears is in the famous article by Jean Leray from 1934[5] on the Navier–Stokes equations, in which he introduced the concept of *quasi-derivative*.

Leray also introduced the concept of *regularization* of a locally integrable function f by a sequence $\{\rho_n\}$ of positive infinitely differentiable functions with compact support, which tend to zero and satisfy $\int \rho_n = 1$. This type of regularization had already been used by Wiener in his 1926 work we discussed earlier. He showed that if f is continuous, then $\rho_n \star f$ is infinitely differentiable, $\rho_n \star f$ converges uniformly to f on compact sets, and that if h is the "generalized derivative" of f, then $\rho_n \star h$ is the (usual) derivative of $\rho_n \star f$. L. Schwartz learned this theory from lectures that

[4] Math. Ann. **95** (1926), 557–584.

[5] Acta Math. **63** (1934),193–248.

Leray himself gave at the Collège de France. In his autobiography [53] Schwartz writes:

> *I followed Leray's Peccot lectures at the Collège de France in 1934–35, with great enthusiasm, since it was something very modern compared to everything I had learned at the ENS.*[6]

Generalized functions were implicitly used by S. Bochner in a 1927 paper[7] and later in his monograph *Vorlesungen über Fouriersche Integrale* (Leipzig, 1932). Bochner, building on the work of H. Hahn[8] and N. Wiener,[9] developed a general theory of the Fourier transform for functions with polynomial growth. Since he did not consider the concept of distributional derivative, he restricted himself to tempered distributions that are ordinary functions, which gave his theory a certain asymmetry. In 1952, Bochner wrote a very harsh review[10] of Schwartz's book *Théorie des Distributions* (1950), in which he essentially claimed that almost all the concepts in it were not new. Lützen's book [41] contains a very detailed study of this review, where Bochner's views on the originality of Schwartz's work are refuted. Schwartz himself, in his autobiography [53], refers to Bochner's opinions on his Theory of Distributions:

> *Bochner was quite a sensitive person and was not particularly pleased with the distributions I had introduced; he even went so far as to write a review of them somewhere, in which he described them as merely "not bad". He also told people close to him that I had found some obvious things and some difficult ones, but that he had already found all the difficult ones himself... When I discovered distributions, I was not aware of Bochner's two publications (the second actually appeared in 1946,*[11] *after the discovery of distributions in 1944). I only became aware of them later, and of course I cited them in my book. However, they did not help me solve any of my problems.*

In the direction of extending the Fourier transform, the work of T. Carleman in 1944[12] on analytic Fourier transforms and those of Beurling in 1947[13] on harmonic Fourier transforms are also noteworthy. The connection of these theories with Schwartz's theory was later established with the development of the theory of hyperfunctions in the 1950s.

In 1936,[14] S. L. Sobolev, in order to obtain an existence theorem for Cauchy problems for second-order hyperbolic equations, was the first to introduce generalized functions as continuous linear functionals in spaces of differentiable functions

[6] Ecole Normale Supérieure.

[7] Math. Ann. **97** (1927), 635–662.

[8] Acta Math. **49** (1926), 301–353.

[9] Math. Z. **24** (1925), 575–616.

[10] Bull. Amer. Math. Soc. **58** (1952), 78–85.

[11] Ann. Math. **47** (1946), 202–212.

[12] *"L'Intégral de Fourier et Questions qui s'y Rattachent"*, Uppsala, 1944.

[13] Colloque Analyse Harmonique Nancy. CNRS. Paris, 1947, 9–30.

[14] Mat. Sb. **1** (1936), 39–71.

with compact support. He also differentiated these functionals by transposing operators and realized that they are generalized solutions of certain Cauchy problems.

Sobolev invented distributions, but used them exclusively in relation to Cauchy problems. The creation of an extensive and rigorous theory of distributions is the work of L. Schwartz, who also foresaw the great applications of this theory. His sources of inspiration were the theory of generalized solutions of partial differential equations, which, as we mentioned, he learned from Leray himself, the symbolic calculus of Heaviside and the theory of Radon measures from N. Bourbaki and H. Cartan, the finite parts of divergent integrals by Hadamard, and de Rham's currents. In his autobiography [53] Schwartz writes:

> *One of the major influences in the creation of distributions came from Jacques Hadamard's theory of the finite parts of divergent integrals... I was also influenced by my meeting with George de Rham in the autumn of 1942. I was in Clermont–Ferrand, in a period of mathematical expansion, learning about differential forms and their integrals, when de Rham arrived to speak about harmonic differential forms. It was a very beautiful and coherent theory... Cartan was in Clermont for a Bourbaki group meeting and came with us. De Rham told us that he was waiting for a more general theory, since his present theory of currents was visibly flawed.*

Functional Analysis was the foundation for the formalism of his theory, especially the theory of duality; one of the basic tools used by Schwartz to extend concepts from functions to distributions is the transposition of operators.

Schwartz's starting point was a problem about polyharmonic functions proposed by G. Choquet and J. Deny.[15] Schwartz first introduced certain operators, which he called convolution operators, defined as operators $T : \mathcal{D} \to \mathcal{E}$ satisfying the condition.

$$T(\phi \star \psi) = T(\phi) \star \psi \quad \forall\, \phi, \psi \in \mathcal{D}.$$

After intensively working with this type of operator, he encountered serious difficulties when trying to develop the Fourier transform. In 1944, suddenly,[16] he realized that the ideal objects for the development of his theory were not convolution operators, but rather the functionals he called distributions. One of the most important contributions of the theory developed by Schwartz, in contrast to that of Sobolev, was that it provided a satisfactory generalization of the Fourier transform that included all previous versions. Schwartz had the idea of considering as test functions the space $\mathcal{S}$—now called the Schwartz space—consisting of all infinitely differentiable functions whose derivatives multiplied by polynomials are integrable; he endowed this space with a topology that gives it the structure of a Fréchet space and such that the Fourier transform is a topological isomorphism of

[15] Bull. Soc. Math. France **77** (194), 118–141.

[16] In his autobiography [53], Schwartz says: *"The invention of the theory of distributions occurred in Paris, at the beginning of November 1944, while my identity card and ration book were under the illegal name Sèlimartin. The discovery was very sudden—it happened in a single night"*.

$\mathcal{S}$ onto $\mathcal{S}$. This allowed him to extend the Fourier transform to the topological dual of $\mathcal{S}$, whose elements he called tempered distributions.

Although at first there were some criticisms, such as those mentioned by Bochner, the success and acceptance of Schwartz's distributions did not take long to arrive. In 1950, Schwartz was awarded the Fields Medal, the highest honor a mathematician can receive, for his theory of distributions. Since then, distributions have been accepted among mathematicians as one of the essential tools of modern calculus and have provided a unified language for the general treatment of solutions to partial differential equations. On the impact of the theory of distributions, F. Treves wrote in 1984:[17]

> *Schwartz gave a strong perspective from the point of view of Functional Analysis to the theory, extending Banach's 1932 book. But the great merit of the Theory of Distributions has been that the basic operations of analysis, differentiation and convolution, and the Fourier/Laplace transforms and their inversion, which had previously required so much care in the classical context, could now be handled unscrupulously following purely algebraic rules.*

To conclude, let us recall what J. Dieudonné said in 1964 about distributions in an article[18] on recent developments in mathematics:

> *The application of these new ideas [the theory of distributions], and in particular the extension given to the convolution product and the Fourier transform, was soon felt; I will only mention here the work of Gårding, Hörmander, Malgrange, Ehrenpreis, Łojasiewicz, Calderín and many others, who have taught us much about the general properties of linear partial differential equations, especially regarding existence and uniqueness problems.*

5.2 The Space of Test Functions

For the study of partial differential equations, it is very convenient to use the notation of multi-indices (due to Laurent Schwartz). A *multi-index* is a vector $\alpha = (\alpha_1, \ldots, \alpha_N)$ where the α_k are non-negative integers, i.e., $\alpha \in \mathbb{N}^N$. Given $x = (x_1, \ldots, x_N) \in \mathbb{R}^N$ and $\alpha = (\alpha_1, \ldots, \alpha_N) \in \mathbb{N}^N$ we define:

$$x^\alpha := x_1^{\alpha_1} x_2^{\alpha_2} \ldots x_N^{\alpha_N}$$

$$|\alpha| := \alpha_1 + \cdots + \alpha_N$$

$$\alpha! := \alpha_1!\, \alpha_2! \ldots \alpha_N!.$$

[17] Bull. Amer. Math. Soc. **10** (1984), 337–340.

[18] Amer. Math. Monthly **79** (1964), 239–248.

On $\mathbb{N}^N$ we consider the partial order

$$\alpha \geq \beta \iff \alpha_i \geq \beta_i \quad \text{for } i = 1, 2, \ldots, N.$$

A polynomial of degree k in $\mathbb{R}^N$ is a function of the form

$$P(x) = \sum_{|\alpha| \leq k} c_\alpha x^\alpha, \qquad \text{with } c_\alpha \in \mathbb{C}.$$

The following formulas with multi-indices are useful:

Binomial Theorem

$$(x + y)^\alpha = \sum_{\beta+\gamma=\alpha} \frac{\alpha!}{\beta!\,\gamma!}\, x^\beta\, y^\gamma.$$

Multinomial Theorem

$$(x_1 + \cdots + x_N)^m = \sum_{|\alpha|=m} \frac{m!}{\alpha!} x^\alpha.$$

For partial derivatives, we use the notation

$$D_i u = \frac{\partial u}{\partial x_i} = u_{x_i}, \qquad D_{ij} u = \frac{\partial^2 u}{\partial x_i \partial x_j} = u_{x_i x_j}.$$

The differential operator of order m is given by

$$D^\alpha = D_1^{\alpha_1}\, D_2^{\alpha_2} \ldots D_N^{\alpha_N} = \frac{\partial^m}{\partial x_1^{\alpha_1} \ldots \partial x_N^{\alpha_N}},$$

with $|\alpha| = m$. We agree that $D^0 u = u$. Given a polynomial of degree k in $\mathbb{R}^N$,

$$P(x) = \sum_{|\alpha| \leq k} c_\alpha x^\alpha,$$

its associated differential polynomial is defined as

$$P(D)(f) = \sum_{|\alpha| \leq k} c_\alpha D^\alpha f.$$

We will use the following function spaces: Let $\Omega \subset \mathbb{R}^N$,

$$C(\Omega) := \{\, u : \Omega \to \mathbb{C},\ u \text{ continuous}\}.$$

If Ω is open and $k \in \mathbb{N}$,

$$C^k(\Omega) := \{\, u : \Omega \to \mathbb{C} \;:\; \exists D^\alpha u \in C(\Omega),\ \forall |\alpha| \le k\}.$$

$C^k(\overline{\Omega})$ denotes the space of functions $u \in C^k(\Omega)$ such that each $D^\alpha u$ extends continuously to the closure of Ω for every $|\alpha| \le k$.

$$C^\infty(\Omega) := \bigcap_{k=1}^{\infty} C^k(\Omega), \quad \text{and} \quad C^\infty(\overline{\Omega}) := \bigcap_{k=1}^{\infty} C^k(\overline{\Omega}).$$

Using multi-index notation, the Leibniz and Taylor formulas become:

Leibniz Formula. If $u, v \in C^k(\Omega)$ and $|\alpha| \le k$, then

$$D^\alpha(uv) = \sum_{\beta+\gamma=\alpha} \frac{\alpha!}{\beta!\,\gamma!}(D^\beta u)(D^\gamma v).$$

Taylor Formula. Let $u \in C^\infty(\Omega)$ and $x_0 \in \Omega$. Then, for every $k \in \mathbb{N}$ and x in a neighborhood of x_0, one can write

$$u(x) = \sum_{|\alpha|\le k} \frac{1}{\alpha!}(x - x_0)^\alpha D^\alpha u(x_0) + \frac{1}{k!}\int_0^1 (1-t)^k \times \sum_{|\alpha|=k+1} (x - x_0)^\alpha D^\alpha u(tx + (1-t)x_0)\, dt.$$

To give the concept of the support of a measurable function, first observe the following. Let $\Omega \subset \mathbb{R}^N$ be an open set and $u : \Omega \to \mathbb{C}$ a measurable function. Let $\{\Omega_i : i \in I\}$ be the family of open subsets of Ω where u vanishes almost everywhere. Then, $u = 0$ almost everywhere on $\Gamma = \bigcup_{i\in I} \Omega_i$. Indeed: For each $m \in \mathbb{N}$, define

$$K_m = \left\{ x \in \Gamma : \operatorname{dist}(x, \mathbb{R}^N \setminus \Gamma) \ge \frac{1}{m},\ \|x\| \le m \right\}.$$

Each K_m is compact and $\Gamma = \bigcup_{m\in\mathbb{N}} K_m$. Therefore, for each $m \in \mathbb{N}$, there exists a finite subset $I_m \subset I$ such that

$$K_m \subset \bigcup_{i\in I_m} \Omega_i.$$

We then have that $J = \bigcup_{m\in\mathbb{N}} I_m$ is countable and also $\Gamma = \bigcup_{j\in J} \Omega_j$. Consequently, $u = 0$ almost everywhere on Γ.

The *support* of a function u is defined as the complement of the largest open set where u vanishes almost everywhere. From the previous observation, we have

$$\text{supp(u)} = \Omega \setminus \Gamma.$$

It is easy to see that if $u : \Omega \to \mathbb{C}$ is continuous on Ω, then

$$\text{supp(u)} = \overline{\{x \in \Omega \ : \ u(x) \neq 0\}}.$$

We define the following spaces of functions:

$$C_0(\Omega) := \{u \in C(\Omega) \ : \ \text{supp(u)} \ \text{ is } \ \text{compact}\}$$

$$C_0^k(\Omega) := \{u \in C^k(\Omega) \ : \ \text{supp(u)} \ \text{ is } \ \text{compact}\}.$$

$$C_0^\infty(\Omega) := \{u \in C^\infty(\Omega) \ : \ \text{supp(u)} \ \text{ is compact}\}.$$

The space $C_0^\infty(\Omega)$ is also denoted by $\mathcal{D}(\Omega)$, and is called the *space of test functions*.

In order to show that the space $\mathcal{D}(\Omega)$ is rich in functions, we need the following result.

Lemma 5.1 *For every $x_0 \in \Omega$ and every $r > 0$, such that the closed ball $\overline{B}(x_0, r)$ is contained in Ω, there exists a function $0 \leq \phi \in \mathcal{D}(\Omega)$ satisfying:*

$$\text{supp}(\phi) \subset \overline{B}(x_0, r), \qquad \int_\Omega \phi(x)\, dx = 1.$$

Proof Consider the function

$$\psi(x) = \begin{cases} \exp\left(\frac{-r^2}{r^2 - \|x - x_0\|^2}\right) & \text{if } \|x - x_0\| < r \\ \\ 0 & \text{if } \|x - x_0\| \geq r. \end{cases}$$

It is easy to see that the desired function is

$$\phi(x) = \frac{\psi(x)}{\int_\Omega \psi(x)\, dx}.$$

□

To give a systematic process for constructing functions in $\mathcal{D}(\Omega)$, we need to recall the notion of convolution product, as well as some of its properties.

The space of *locally integrable* functions in $\mathbb{R}^N$ is denoted by $L^1_{loc}(\mathbb{R}^N)$, i.e.,

$$L^1_{loc}(\mathbb{R}^N) = \{f : \mathbb{R}^N \to \mathbb{C} \text{ measurable } : \ f \in L^1(K), \ \forall\, K \subset \mathbb{R}^N \text{ compact}\}.$$

Given two functions $f, g \in L^1_{loc}(\mathbb{R}^N)$, their *convolution product* $f \star g$ is defined as the function

$$f \star g(x) = \int_{\mathbb{R}^N} f(x-y)g(y)\,dy$$

when the integral exists. In Chap. 2 we saw cases where the convolution product of two functions is defined. We will now see that convolution is useful to regularize functions. More specifically, we have the following result.

Proposition 5.1 *The following results hold:*

(i) If $f \in C_0(\mathbb{R}^N)$ and $g \in L^1_{loc}(\mathbb{R}^N)$, then $f \star g \in C(\mathbb{R}^N)$.
(ii) If $f \in C^k_0(\mathbb{R}^N)$ and $g \in L^1_{loc}(\mathbb{R}^N)$, then $f \star g \in C^k(\mathbb{R}^N)$, and we have

$$D^\alpha(f \star g) = (D^\alpha f) \star g.$$

(iii) If $f \in L^1_{loc}(\mathbb{R}^N)$, $g \in L^1_{loc}(\mathbb{R}^N)$ and $f \star g$ is defined, then

$$\operatorname{supp}(f \star g) \subset \overline{\operatorname{supp(f)} + \operatorname{supp}\,(g)}.$$

Proof

(i): First observe that for each $x \in \mathbb{R}^N$, the function $y \mapsto f(x-y)g(y)$ is integrable, so $(f \star g)(x)$ is well defined. We will show that it is continuous. Let $x_n \to x$, we want to prove that $(f \star g)(x_n) \to (f \star g)(x)$. Define the functions

$$F_n(y) := f(x_n - y)g(y), \quad F(y) := f(x-y)g(y).$$

We have that $F_n \to F$ a.e. in $\mathbb{R}^N$. Moreover, since supp(f) is compact, we can find a compact set $K \subset \mathbb{R}^N$ such that $x_n - \operatorname{supp(f)} \subset K$ for all $n \in \mathbb{N}$. Hence, $|F_n(y)| \le \|f\|_\infty \chi_K(y)\,|g(y)|$. Therefore, by the Dominated Convergence Theorem (Theorem B.1),

$$(f \star g)(x_n) = \int_{\mathbb{R}^N} F_n(y)\,dy \to \int_{\mathbb{R}^N} F(y)\,dy = (f \star g)(x).$$

(ii): By induction, it suffices to prove it for $k = 1$. Let $x \in \mathbb{R}^N$. We show that

$$\frac{\partial}{\partial x_i}(f \star g)(x) = \left(\frac{\partial f}{\partial x_i} \star g\right)(x), \quad i = 1, 2, \ldots, n.$$

Indeed:

$$\left| f(x + te_i - y) - f(x - y) - t\frac{\partial f}{\partial x_i}(x - y) \right|$$

$$= \left| \int_0^1 [t\frac{\partial f}{\partial x_i}(x + ste_i - y) - t\frac{\partial f}{\partial x_i}(x - y)]\, ds \right| \leq |t|\,\epsilon(|t|),$$

with $\epsilon(|t|) \to 0$ as $t \to 0$, since $\frac{\partial f}{\partial x_i}$ is uniformly continuous. Let $K \subset \mathbb{R}^N$ be a compact set such that

$$x + te_i - \text{supp(f)} \subset K \quad \text{if } |t| \leq 1.$$

Then

$$\left| f(x + te_i - y) - f(x - y) - t\frac{\partial f}{\partial x_i}(x - y) \right|$$

$$\leq |t|\epsilon(|t|)\chi_K(y), \quad \forall\, y \in \mathbb{R}^N,\ |t| \leq 1.$$

Hence,

$$\left| \frac{(f \star g)(x + te_i) - (f \star g)(x)}{t} - \left(\frac{\partial f}{\partial x_i} \star g\right)(x) \right|$$

$$\leq \int_{\mathbb{R}^N} \left| \frac{f(x + te_i - y)g(y) - f(x - y)g(y)}{t} - \frac{\partial f}{\partial x_i}(x - y)g(y) \right| dy$$

$$\leq \epsilon(|t|) \int_K |g(y)|\, dy.$$

Therefore, $\frac{\partial}{\partial x_i}(f \star g)(x) = (\frac{\partial f}{\partial x_i} \star g)(x)$. Moreover, since $\frac{\partial f}{\partial x_i}$ is continuous, from (i) we conclude that $\frac{\partial f}{\partial x_i} \star g$ is continuous.

(iii): Since

$$(f \star g)(x) = \int_{\mathbb{R}^N} f(x - y)g(y)\, dy = \int_{(x - \text{supp(f)}) \cap \text{supp(g)}} f(x - y)g(y)\, dy,$$

we have that if $x \notin \text{supp(f)} + \text{supp(g)}$, then $(f \star g)(x) = 0$. Therefore, $f \star g = 0$ almost everywhere in $\mathbb{R}^N \setminus (\text{supp(f)} + \text{supp(g)})$. In particular, $f \star g = 0$ almost everywhere in $\text{Int}(\mathbb{R}^N \setminus (\text{supp(f)} + \text{supp(g)}))$, hence

$$\text{supp(f} \star \text{g)} \subset \mathbb{R}^N \setminus \text{Int}(\mathbb{R}^N \setminus (\text{supp(f)} + \text{supp(g)})) = \overline{\text{supp(f)} + \text{supp(g)}}.$$

□

Let ϕ_0 be the function obtained from Lemma 5.1, taking $x_0 = 0$ and $r = 1$, i.e., $0 \leq \phi_0 \in \mathcal{D}(\mathbb{R}^N)$ is a function satisfying:

$$\text{supp}(\phi_0) \subset \overline{B}(0,1), \quad \text{and} \quad \int_{\mathbb{R}^N} \phi_0(x)\, dx = 1.$$

For each $\epsilon > 0$ we define the function

$$\rho_\epsilon(x) := \frac{1}{\epsilon^N}\, \phi_0\left(\frac{x}{\epsilon}\right).$$

We then have that

$$0 \leq \rho_\epsilon \in \mathcal{D}(\mathbb{R}^N), \quad \text{supp}(\rho_\epsilon) \subset \overline{B}(0,\epsilon), \quad \text{and} \quad \int_{\mathbb{R}^N} \rho_\epsilon(x)\, dx = 1.$$

We will denote by ϕ_n the function ρ_ϵ with $\epsilon = \frac{1}{n}$, i.e.,

$$\phi_n(x) = n^N \phi_0(nx).$$

The functions ϕ_n satisfy:

$$0 \leq \phi_n \in \mathcal{D}(\mathbb{R}^N), \quad \text{supp}(\phi_n) \subset \overline{B}\left(0, \frac{1}{n}\right), \quad \int_{\mathbb{R}^N} \phi_n(x)\, dx = 1. \tag{5.11}$$

A sequence satisfying (5.11) is called a *mollifier sequence*.

Lemma 5.2 *Let $f \in L^1_{loc}(\Omega)$ have compact support. Then, for sufficiently small $\epsilon > 0$, we have that*

$$\rho_\epsilon \star f \in \mathcal{D}(\Omega).$$

Proof Since $\text{supp}(f)$ is compact in Ω and

$$\text{supp}(\rho_\epsilon \star f) \subset \overline{\text{supp}(\rho_\epsilon) + \text{supp(f)}} \subset \overline{B}(0,\epsilon) + \text{supp(f)},$$

for sufficiently small ϵ, we have that $\text{supp}(\rho_\epsilon \star f)$ is a compact subset of Ω. Moreover, since

$$D^\alpha(\rho_\epsilon \star f) = D^\alpha \rho_\epsilon \star f$$

is continuous, we conclude that $\rho_\epsilon \star f \in \mathcal{D}(\Omega)$ for sufficiently small ϵ. □

The previous result ensures that the space $\mathcal{D}(\Omega)$ is rich in functions.

Proposition 5.2 *Let K be a compact subset of $\mathbb{R}^N$ and let $\epsilon > 0$. Define*

$$K_\epsilon := \bigcup_{x \in K} B(x, \epsilon).$$

Then there exists a function $0 \leq \psi_\epsilon \in \mathcal{D}(\mathbb{R}^N)$ such that

$$\psi_\epsilon = 1 \quad \forall\, x \in K, \quad \textit{and} \quad \operatorname{supp}(\psi_\epsilon) \subset K_\epsilon.$$

Proof By the previous lemma, we have that

$$\psi_\epsilon := \chi_{K_{\epsilon/3}} \star \rho_{\epsilon/2} \in \mathcal{D}(\mathbb{R}^N),$$

and furthermore,

$$\operatorname{supp}(\psi_\epsilon) \subset K_{\epsilon/2} + B\left(0, \frac{\epsilon}{2}\right) = K_\epsilon.$$

Finally, for each $x \in K$,

$$\psi_\epsilon(x) = \int_{\mathbb{R}^N} \rho_{\epsilon/3}(y) \chi_{K_{\epsilon/3}}(x - y)\, dy = \int_{\mathbb{R}^N} \rho_{\epsilon/3}(y)\, dy = 1.$$

□

Theorem 5.1 ***(Partition of Unity)*** *If $\mathcal{U}$ is a collection of open subsets of $\mathbb{R}^N$ whose union is Ω, then there exists a sequence $\{\psi_i\}_{i \in \mathbb{N}} \subset \mathcal{D}(\Omega)$ of positive functions satisfying:*

(i) Each ψ_i has its support contained in some member of $\mathcal{U}$.
(ii) $\sum_{i=1}^{\infty} \psi_i(x) = 1$ for every $x \in \Omega$.
(iii) If $K \subset \Omega$ is compact, there exists $m \in \mathbb{N}$, and an open set $W \supset K$ such that

$$\psi_1(x) + \cdots + \psi_m(x) = 1 \quad \forall\, x \in W.$$

Proof Let $A \subset \Omega$ be a countable dense subset, and let $\{\overline{B}(p_i, r_i)\}_{i=1}^{\infty}$ be the sequence of all closed balls with center $p_i \in A$ and rational radius, contained in some member of $\mathcal{U}$. It is easy to see that

$$\Omega = \bigcup_{i \in \mathbb{N}} B\left(p_i, \frac{r_i}{2}\right).$$

As a consequence of the previous proposition, we can find functions $\varphi_i \in \mathcal{D}(\Omega)$, $0 \leq \varphi_i \leq 1$, such that $\varphi_i(x) = 1$ for every $x \in B(p_i, \frac{r_i}{2})$ and $\varphi_i(x) = 0$ for every $x \in \Omega \setminus \overline{B}(p_i, r_i)$. We define:

$$\psi_1 := \varphi_1,$$

$$\psi_{i+1} := (1 - \varphi_1) \cdots (1 - \varphi_i)\varphi_{i+1}.$$

Since $\psi_i(x) = 0$ for each $x \in \Omega \setminus \overline{B}(p_i, r_i)$, there exists $U \in \mathcal{U}$ such that $\text{supp}(\psi_i) \subset U$. Thus, condition (i) is satisfied. By induction, we have:

$$\psi_1 + \cdots + \psi_i = 1 - (1 - \varphi_1) \cdots (1 - \varphi_i). \tag{5.12}$$

Since $\varphi_i(x) = 1$ for every $x \in B(p_i, \frac{r_i}{2})$, from (5.12) we deduce that

$$\psi_1(x) + \cdots + \psi_j(x) = 1 \quad \text{if } x \in \bigcup_{i=1}^{j} B\left(p_i, \frac{r_i}{2}\right). \tag{5.13}$$

From this, we deduce (ii). Moreover, if $K \subset \Omega$ is compact, there exists $m \in \mathbb{N}$ such that

$$K \subset \bigcup_{i=1}^{m} B\left(p_i, \frac{r_i}{2}\right). \tag{5.14}$$

From (5.13)and (5.14), we deduce (iii). □

A sequence $\{\psi_i\}_{i \in \mathbb{N}}$ satisfying the properties of the previous theorem is called a *partition of unity* in Ω subordinate to the covering $\mathcal{U}$.

Using the regularizing sequence $\{\phi_n\}$, we obtain the following approximation results.

Proposition 5.3 *If $f \in C(\mathbb{R}^N)$, then*

$$\phi_n \star f \to f \quad \textit{uniformly on compact subsets of } \mathbb{R}^N.$$

Proof Let $K \subset \mathbb{R}^N$ be a compact set. Since f is uniformly continuous on K, given $\epsilon > 0$, there exists $\delta > 0$ such that

$$|f(x - y) - f(x)| \leq \epsilon \quad \forall x \in K, \ \forall y \in B(0, \delta).$$

Then,

$$(\phi_n \star f)(x) - f(x) = \int_{\mathbb{R}^N} [f(x - y) - f(x)]\phi_n(y)\, dy$$

$$= \int_{\overline{B}(0,1/n)} [f(x-y) - f(x)]\phi_n(y)\, dy.$$

Thus, taking $n \in \mathbb{N}$ such that $n > \frac{1}{\delta}$, we have that for all $x \in K$,

$$|(\phi_n \star f)(x) - f(x)| \leq \epsilon \int_{\mathbb{R}^N} \phi_n(y)\, dy = \epsilon.$$

□

Proposition 5.4 *If $f \in L^p(\mathbb{R}^N)$ with $1 \leq p < \infty$, then*

$$\phi_n \star f \to f \quad \text{in } L^p(\mathbb{R}^N).$$

Proof Given $\epsilon > 0$, since $C_0(\mathbb{R}^N)$ is dense in $L^p(\mathbb{R}^N)$, there exists $g \in C_0(\mathbb{R}^N)$ such that $\|f - g\|_p \leq \frac{\epsilon}{4}$. Now, by Proposition 5.3,

$$\phi_n \star g \to g \quad \text{uniformly on compact subsets of } \mathbb{R}^N.$$

Moreover,

$$\text{supp}(\phi_n \star g) \subset \text{supp(g)} + \overline{B}\left(0, \frac{1}{n}\right) \subset \text{supp(g)} + \overline{B}(0,1) \quad \forall n \in \mathbb{N},$$

so by the Dominated Convergence Theorem (Theorem B.1), we have $\phi_n \star g \to g$ in $L^p(\mathbb{R}^N)$. Then, using Young's inequality, for $n \in \mathbb{N}$ large enough, we obtain

$$\|\phi_n \star f - f\|_p = \|\phi_n \star (f-g) + (\phi_n \star g - g) + (f-g)\|_p$$

$$\leq \|\phi_n\|_1 \|f-g\|_p + \|\phi_n \star g - g\|_p + \|f-g\|_p \leq \epsilon.$$

□

Theorem 5.2 *For every open set $\Omega \subset \mathbb{R}^N$ and every $1 \leq p < \infty$, the space $\mathcal{D}(\Omega)$ is dense in $L^p(\Omega)$.*

Proof Let $f \in L^p(\Omega)$ and $\epsilon > 0$. Since $C_0(\Omega)$ is dense in $L^p(\Omega)$, there exists $g \in C_0(\Omega)$ such that $\|f - g\|_p \leq \frac{\epsilon}{2}$. Let $\overline{g}$ be the function

$$\overline{g}(x) := \begin{cases} g(x) \text{ if } x \in \Omega \\ \\ 0 \quad \text{if } x \in \mathbb{R}^N \setminus \Omega. \end{cases}$$

Then, $\overline{g} \in L^p(\mathbb{R}^N)$, and hence, by Proposition 5.4,

$$\overline{g} \star \phi_n \to \overline{g} \quad \text{in } L^p(\mathbb{R}^N).$$

Moreover, for $n \in \mathbb{N}$ sufficiently large, $\operatorname{supp}(\overline{g} \star \phi_n)$ is a compact subset of Ω. Consequently, if u_n is the restriction to Ω of $\overline{g} \star \phi_n$, we have that $u_n \in \mathcal{D}(\Omega)$, and $u_n \to g$ in $L^p(\Omega)$. Therefore, for $n \in \mathbb{N}$ large enough,

$$\|u_n - f\|_p \leq \|u_n - g\|_p + \|f - g\|_p \leq \epsilon.$$

□

5.3 The Space of Distributions

The space $\mathcal{D}(\Omega)$ can be endowed with a topology compatible with its vector space structure (see, for example, [4, 50]). However, for our purposes, we do not need to describe this topology in detail; it is sufficient for us to know which sequences converge with respect to this topology.

Definition 5.1 Let $\varphi_n, \varphi \in \mathcal{D}(\Omega)$. We say that the sequence $\{\varphi_n\}$ converges to φ in $\mathcal{D}(\Omega)$, and we write

$$\varphi_n \to \varphi \quad \text{in } \mathcal{D}(\Omega),$$

if there exists a compact set $K \subset \Omega$ such that $\operatorname{supp}(\varphi_n) \subset K$ for all $n \in \mathbb{N}$, and for every multi-index $\alpha \in \mathbb{N}^N$

$$D^\alpha \varphi_n \to D^\alpha \varphi \quad \text{uniformly on } K.$$

A *distribution* on Ω is a linear functional T on $\mathcal{D}(\Omega)$ that is continuous with respect to the convergence defined in $\mathcal{D}(\Omega)$, i.e., it satisfies:

$$\text{if } \varphi_n \to \varphi \text{ in } \mathcal{D}(\Omega) \Rightarrow T(\varphi_n) \to T(\varphi).$$

The space of all distributions on Ω is denoted by $\mathcal{D}'(\Omega)$.

Note that, due to linearity, the continuity condition only needs to be checked at 0, i.e., for a linear functional $T : \mathcal{D}(\Omega) \to \mathbb{C}$ to be a distribution on Ω, it is enough that

$$\text{if } \varphi_n \to 0 \text{ in } \mathcal{D}(\Omega) \Rightarrow T(\varphi_n) \to 0.$$

From now on, for $T \in \mathcal{D}'(\Omega)$ and $\varphi \in \mathcal{D}(\Omega)$, we will use either $T(\varphi)$ or $\langle T, \varphi \rangle$ to denote the action of T on φ.

The simplest examples of distributions are locally integrable functions. More precisely, to each function $f \in L^1_{loc}(\Omega)$ one can associate the linear functional T_f on $\mathcal{D}(\Omega)$ defined by

$$\langle T_f, \varphi \rangle := \int_\Omega f\varphi \quad \forall\, \varphi \in \mathcal{D}(\Omega).$$

As a direct consequence of the Dominated Convergence Theorem (Theorem B.1), we have that $T_f \in \mathcal{D}'(\Omega)$. Moreover, different functions give rise to different distributions. More precisely, we have the following result.

Proposition 5.5 *Let $f, g \in L^1_{loc}(\Omega)$. Then,*

$$T_f = T_g \iff f = g \quad \text{a.e. on } \Omega.$$

Proof It is enough to show that if $T_f = 0$, then $f = 0$ a.e. on Ω, which is equivalent to proving that if $K \subset \Omega$ is a compact set, then $\int_K f = 0$.

Now, given a compact set K, by Proposition 5.2, we can find functions $\varphi_n \in \mathcal{D}(\Omega)$ such that $0 \le \varphi_n \le 1$, $\text{supp}(\varphi_n) \subset K_{1/n}$, and $\varphi_n(x) = 1$ for every $x \in K$.

Therefore, applying the Dominated Convergence Theorem, we get:

$$0 = \langle T_f, \varphi_n \rangle = \int_\Omega f\varphi_n \to \int_K f.$$

Hence, $\int_K f = 0$. □

From now on, we will not distinguish between a locally integrable function and the distribution it generates. We will say that a distribution T is a function if there exists $f \in L^1_{loc}(\Omega)$ such that $T = T_f$.

Note that since $C(\Omega) \subset L^1_{loc}(\Omega)$ and $L^p(\Omega) \subset L^1_{loc}(\Omega)$ $(1 \le p \le \infty)$, it follows that all continuous functions and all functions in the Lebesgue spaces are distributions.

We now present a characterization of distributions. For this purpose, we need to introduce the following norms. Given $m \in \mathbb{N}$, for $f \in \mathcal{D}(\Omega)$, we define:

$$P_m(f) := \sup\{|D^\alpha f(x)| \ : \ x \in \Omega, \ \ |\alpha| \le m\}.$$

Given a compact set $K \subset \Omega$, we denote:

$$\mathcal{D}_K(\Omega) := \{f \in C^\infty(\Omega) \ : \ \text{supp(f)} \subset K\}.$$

If we denote by $\mathcal{K}(\Omega)$ the family of all compact subsets of Ω, then we have:

$$\mathcal{D}(\Omega) = \bigcup_{K \in \mathcal{K}(\Omega)} \mathcal{D}_K(\Omega).$$

Theorem 5.3 *Let $\Omega \subset \mathbb{R}^N$ be an open set, and let $T : \mathcal{D}(\Omega) \to \mathbb{C}$ be a linear map. The following are equivalent:*

(i) $T \in \mathcal{D}'(\Omega)$.
(ii) For every $K \in \mathcal{K}(\Omega)$, there exist a constant C_K and an $m_K \in \mathbb{N}$ such that

$$|\langle T, \varphi\rangle| \leq C_K P_{m_K}(\varphi) \quad \forall\, \varphi \in \mathcal{D}_K(\Omega).$$

Proof (ii) $\Rightarrow$ (i): This follows easily from the definition of convergence in $\mathcal{D}(\Omega)$. Let us show that (i) $\Rightarrow$ (ii): Suppose that (ii) does not hold. Then there exists $K \in \mathcal{K}(\Omega)$ such that, for every $m \in \mathbb{N}$, we can find $\varphi_m \in \mathcal{D}_K(\Omega)$ such that $|\langle T, \varphi_m\rangle| > m P_m(\varphi_m)$. For each $m \in \mathbb{N}$, define

$$\psi_m := \frac{\varphi_m}{\langle T, \varphi_m\rangle} \in \mathcal{D}_K(\Omega).$$

Then, given $\alpha \in \mathbb{N}^n$, if we take $m > |\alpha|$, we have

$$\sup_{x\in\Omega} |D^\alpha \psi_m(x)| \leq \frac{P_{|\alpha|}(\varphi_m)}{\langle T, \varphi_m\rangle} \leq \frac{P_m(\varphi_m)}{\langle T, \varphi_m\rangle} \leq \frac{1}{m}.$$

Therefore, $\psi_m \to 0$ in $\mathcal{D}(\Omega)$. However, $|\langle T, \psi_m\rangle| = 1$ for all $m \in \mathbb{N}$, which leads to a contradiction. □

The previous result leads to the following definition.

Definition 5.2 Let $T \in \mathcal{D}'(\Omega)$. Suppose that there exists $m \in \mathbb{N}$, independent of the compact set K, such that

$$|\langle T, \varphi\rangle| \leq C_K P_m(\varphi) \quad \forall\, \varphi \in \mathcal{D}_K(\Omega), \tag{5.15}$$

Then, the smallest such $m \in \mathbb{N}$ for which (5.15) holds is called the *order of the distribution* T. If no such integer exists, we say that the distribution is of *infinite order*.

5.4 Radon Measures

A similar process to the one used to define distributions in Ω allows us to define a special type of measure, the so-called Radon measures. We begin by defining convergence in $C_0(\Omega)$.

Definition 5.3 Let $f_n, f \in C_0(\Omega)$. We say that

$$f_n \to f \quad \text{in } C_0(\Omega),$$

if there exists a compact set $K \in \mathcal{K}(\Omega)$ such that $\text{supp}(\text{f}_\text{n}) \subset K$ for every $n \in \mathbb{N}$, and

$$f_n \to f \quad \text{uniformly on } K.$$

A *Radon measure* on Ω is a linear functional μ on $C_0(\Omega)$ that is continuous with respect to the convergence just defined on $C_0(\Omega)$, i.e., it satisfies:

$$\text{if } \ f_n \to f \text{ in } C_0(\Omega) \ \Rightarrow \ \mu(f_n) \to \mu(f).$$

We denote by $\mathcal{M}(\Omega)$ the space of all Radon measures on Ω. Given $\mu \in \mathcal{M}(\Omega)$ and $f \in C_0(\Omega)$, we use the notation

$$\mu(f) = \int_\Omega f \, d\mu.$$

Since $\varphi_n \to \varphi$ in $\mathcal{D}(\Omega)$ implies that $\varphi_n \to \varphi$ in $C_0(\Omega)$, it follows that if $\mu \in \mathcal{M}(\Omega)$ and we define T_μ by

$$\langle T_\mu, \varphi \rangle := \int_\Omega \varphi \, d\mu \quad \forall \, \varphi \in \mathcal{D}(\Omega),$$

then $T_\mu \in \mathcal{D}'(\Omega)$, i.e., every Radon measure is a distribution.

A completely similar proof to that of Theorem 5.3 yields the following characterization of Radon measures.

Theorem 5.4 *Let $\Omega \subset \mathbb{R}^N$ be an open set, and let $\mu : C_0(\Omega) \to \mathbb{C}$ be a linear map. The following are equivalent:*

(i) $\mu \in \mathcal{M}(\Omega)$.
(ii) For every $K \in \mathcal{K}(\Omega)$, there exists a constant C_K such that

$$\left| \int_\Omega f \, d\mu \right| \leq C_K \, P_0(f) \quad \forall \, f \in C_K(\Omega),$$

where

$$C_K(\Omega) := \{ f \in C(\Omega) \ : \ \text{supp(f)} \subset K \}.$$

As a consequence of this result, every Radon measure is a distribution of order zero. The converse is also true, i.e., distributions of order zero are Radon measures. This follows from the next result.

Theorem 5.5 *Let $T \in \mathcal{D}'(\Omega)$. Then T is defined by a Radon measure, i.e., there exists a Radon measure $\mu \in \mathcal{M}(\Omega)$ such that $T = T_\mu$, if and only if T is continuous on $\mathcal{D}_K(\Omega)$ with respect to the uniform convergence on K, for every $K \in \mathcal{K}(\Omega)$.*

Proof Clearly, the condition is necessary. Let us prove the sufficiency: Let $T \in \mathcal{D}'(\Omega)$ be P_0-continuous on $\mathcal{D}_K(\Omega)$. Then T can be uniquely extended to a P_0-continuous functional T_K on the closure of $\mathcal{D}_K(\Omega)$ in $C_K(\Omega)$. Moreover, if $K_1, K_2 \in \mathcal{K}(\Omega)$ with $K_1 \subset K_2$, then T_{K_2} is an extension of T_{K_1}. Consequently, we can define a continuous linear extension μ of T on the space

$$\bigcup_{K \in \mathcal{K}(\Omega)} \overline{\mathcal{D}_K(\Omega)}^{C_K(\Omega)}.$$

Now, this space is $C_0(\Omega)$, since if $f \in C_0(\Omega)$ and $K = \text{supp(f)}$, given $\epsilon > 0$ such that $K_\epsilon \subset \Omega$, and choosing $m \in \mathbb{N}$ with $\frac{1}{m} \leq \epsilon$, we have $\phi_m \star f \in \mathcal{D}_{K_\epsilon}(\Omega)$ and $\phi_m \star f \to f$ uniformly on K_ϵ, hence

$$f \in \overline{\mathcal{D}_{K_\epsilon}(\Omega)}^{C_{K_\epsilon}(\Omega)}.$$

□

Example 5.1 As a consequence of Theorem 5.4, it is easy to provide examples of distributions that are not Radon measures. Let us consider the functional $T : \mathcal{D}(\mathbb{R}) \to \mathbb{C}$ defined by

$$\langle T, \varphi \rangle := \varphi'(0).$$

Clearly, $T \in \mathcal{D}'(\mathbb{R})$. However, T is not a Radon measure, since if we take $\varphi \in \mathcal{D}(\mathbb{R})$, with $0 \leq \varphi \leq 1$, $\text{supp}(\varphi) \subset [-1, 1]$, $\varphi(x) = 1$ for every $x \in [-\frac{1}{2}, \frac{1}{2}]$, and define

$$\psi_k(x) := \sin(kx)\varphi(x),$$

we get

$$\langle T, \psi_k \rangle = \psi_k'(0) = k \quad \forall\, k \in \mathbb{N}.$$

Now, if there existed a Radon measure $\mu \in \mathcal{M}(\mathbb{R})$ such that $T = T_\mu$, we would have

$$k = \langle T_\mu, \psi_k \rangle = \left| \int_{-1}^{1} \sin(kx)\varphi(x)\, d\mu \right| \leq C \quad \forall\, k \in \mathbb{N},$$

which is a contradiction.

In mathematical physics, the distribution T is called the *dipole distribution*. More generally, for each $m \in \mathbb{N}$, the distribution defined by

$$\langle T, \varphi \rangle := \varphi^{(m)}(0) \quad \forall\, \varphi \in \mathcal{D}(\mathbb{R}),$$

is a distribution that is not a Radon measure. It is not difficult to verify that such a distribution is of order m.

Let $f \in L^1_{loc}(\Omega)$. As a consequence of the Dominated Convergence Theorem, one can associate to f the Radon measure μ_f defined by

$$\mu_f(\phi) := \int_\Omega f(x)\phi(x)\,dx \quad \forall\, \phi \in C_0(\Omega).$$

In this way, $L^1_{loc}(\Omega)$ is embedded in $\mathcal{M}(\Omega)$. Moreover, we have that $T_f = T_{\mu_f}$. Thus, every function distribution is of order zero. Radon measures of the form μ_f are called *absolutely continuous measures*. In this case, the function f is called the *density* of the measure μ_f with respect to the Lebesgue measure, which is usually denoted by $d\mu_f = f\,dx$.

Not all Radon measures are absolutely continuous. One of the most interesting examples of a non-absolutely continuous measure is the *Dirac measure*, defined as follows: let $x \in \Omega$, and define δ_x by

$$\delta_x(f) := f(x) \qquad \forall\, f \in C_0(\Omega).$$

Clearly, $\delta_x \in \mathcal{M}(\Omega)$. In the particular case where $\Omega = \mathbb{R}^N$, we denote δ_0 by δ, i.e., δ is the Radon measure on $\mathbb{R}^N$ defined by

$$\delta(f) := f(0) \qquad \forall\, f \in C_0(\mathbb{R}^N).$$

Let us show that δ is not absolutely continuous. If it were, there would exist a function $f \in L^1_{loc}(\mathbb{R}^N)$ such that $\mu_f = \delta$. Then, taking a sequence of functions $\psi_m \in \mathcal{D}(\mathbb{R}^N)$, with $0 \le \psi_m \le 1$, $\psi_m(0) = 1$ and $\mathrm{supp}(\psi_\mathrm{m}) \subset B(0, \frac{1}{\mathrm{m}})$, we would have $\delta(\psi_m) = \psi_m(0) = 1$ for all $m \in \mathbb{N}$, whereas

$$\mu_f(\psi_m) = \int_{\mathbb{R}^N} f(x)\psi_m(x)\,dx \le \int_{B(0,\frac{1}{m})} |f|,$$

so that

$$\lim_{m\to\infty} \mu_f(\psi_m) = 0,$$

leading to a contradiction.

Definition 5.4 A distribution $T \in \mathcal{D}'(\Omega)$ is said to be *positive*, and we denote this by $T \ge 0$, if

$$\langle T, \varphi\rangle \ge 0 \quad \text{for every } 0 \le \varphi \in \mathcal{D}(\Omega).$$

Similarly, a Radon measure μ is said to be *positive* if

$$\mu(f) \geq 0 \quad \text{for every } 0 \leq f \in C_0(\Omega).$$

Clearly, every positive Radon measure is a positive distribution. In the next result we will see that the converse is also true.

Theorem 5.6 *Every positive distribution is a positive Radon measure.*

Proof Let $0 \leq T \in \mathcal{D}'(\Omega)$. By Theorem 5.5, to prove that T is defined by a Radon measure, we must show that for every compact set $K \subset \Omega$, the distribution T is a continuous linear functional on $\mathcal{D}_K(\Omega)$ with respect to uniform convergence on K.

Indeed, let $f_m \in \mathcal{D}_K(\Omega)$ be such that

$$f_m \to 0 \quad \text{uniformly on } K.$$

Then, taking $0 \leq \varphi \in \mathcal{D}(\Omega)$ such that $\varphi \geq 1$ on K, there exist constants $\epsilon_m > 0$, with $\epsilon_m \to 0$ as $m \to \infty$, satisfying:

$$|f_m(x)| \leq \epsilon_m \varphi(x) \qquad \forall\, x \in \Omega.$$

Hence,

$$-\epsilon_m \varphi \leq \Re f_m \leq \epsilon_m \varphi \quad \text{and} \quad -\epsilon_m \varphi \leq \Im f_m \leq \epsilon_m \varphi,$$

and since T is positive, it follows that

$$-\epsilon_m \langle T, \varphi\rangle \leq \langle T, \Re f_m\rangle \leq \epsilon_m \langle T, \varphi\rangle$$

$$-\epsilon_m \langle T, \varphi\rangle \leq \langle T, \Im f_m\rangle \leq \epsilon_m \langle T, \varphi\rangle.$$

Consequently,

$$|\langle T, f_m\rangle| \leq 2\epsilon_m \langle T, \varphi\rangle,$$

and thus $\langle T, f_m\rangle \to 0$. Therefore, there exists $\mu \in \mathcal{M}(\Omega)$ such that $T = T_\mu$. Finally, let us show that μ is positive. We already know that $\mu(\varphi) \geq 0$ for every $0 \leq \varphi \in \mathcal{D}(\Omega)$. Now, if $0 \leq f \in C_0(\Omega)$, then $0 \leq f \star \phi_m \in \mathcal{D}(\Omega)$ and $f \star \phi_m \to f$ uniformly, hence

$$\mu(f) = \lim_{m\to\infty} \mu(f \star \phi_m) \geq 0.$$

□

If $\mu \in \mathcal{M}(\Omega)$, it is immediate that the mapping $\overline{\mu}(f) := \overline{\mu(\overline{f})}$ $(f \in C_0(\Omega))$, is also a Radon measure on Ω, and is called the *conjugate* of μ. One has: $\overline{\overline{\mu}} = \mu$, and $\overline{a\mu_1 + b\mu_2} = \overline{a}\,\overline{\mu_1} + \overline{b}\,\overline{\mu_2}$.

A Radon measure $\mu \in \mathcal{M}(\Omega)$ is said to be *real* if $\overline{\mu} = \mu$; this is equivalent to saying that $\mu(f) \in \mathbb{R}$ for every $f \in C_0(\Omega, \mathbb{R})$. The vector space of all real Radon measures on Ω is denoted by $\mathcal{M}_{\mathbb{R}}(\Omega)$. Clearly, every positive Radon measure on Ω is an element of $\mathcal{M}_{\mathbb{R}}(\Omega)$. The notion of a positive measure allows us to define an order relation on the vector space $\mathcal{M}_{\mathbb{R}}(\Omega)$. For $\mu, \nu \in \mathcal{M}_{\mathbb{R}}(\Omega)$, we define:

$$\mu \leq \nu \iff \nu - \mu \geq 0.$$

Clearly, this relation is an order relation on $\mathcal{M}_{\mathbb{R}}(\Omega)$, and it satisfies that if $\mu \leq \nu$, then $\lambda + \mu \leq \lambda + \nu$ for all $\lambda \in \mathcal{M}_{\mathbb{R}}(\Omega)$, and $a\mu \leq a\nu$ for every scalar $a \geq 0$.

In the following result, we will see that it is possible to define the modulus of a Radon measure.

Theorem 5.7 *Let $\mu \in \mathcal{M}(\Omega)$. There exists a minimal (with respect to the previous order) positive Radon measure ν on Ω such that $|\mu(f)| \leq \nu(|f|)$ for all $f \in C_0(\Omega)$.*

Proof We first define the measure ν on the positive functions of $C_0(\Omega)$ by the formula

$$\nu(f) := \sup\{|\mu(g)| \ : \ g \in C_0(\Omega),\ |g| \leq f\},\ \ 0 \leq f \in C_0(\Omega). \tag{5.16}$$

Note that the above supremum is finite, since if $|g| \leq f$, then supp(g) $\subset K :=$ supp(f), and as a consequence of Theorem 5.4 we obtain:

$$|\mu(g)| \leq C_K \|g\|_\infty \leq C_K \|f\|_\infty.$$

It is clear that for every scalar $a \geq 0$, we have $\nu(af) = a\nu(f)$. Let us show that for $f_1, f_2 \in C_0(\Omega)$ positive, the following holds:

$$\nu(f_1 + f_2) = \nu(f_1) + \nu(f_2) \tag{5.17}$$

Indeed: given $\epsilon > 0$, there exist functions $g_i \in C_0(\Omega)$ such that

$$|g_i| \leq f_i, \quad |\mu(g_i)| \geq \nu(f_i) - \epsilon, \quad i = 1, 2.$$

Moreover, by multiplying g_i by a complex number of modulus one, we may assume that $\mu(g_i) = |\mu(g_i)|$. Then we have:

$$\mu(g_1 + g_2) = |\mu(g_1)| + |\mu(g_2)| \geq \nu(f_1) + \nu(f_2) - 2\epsilon.$$

From this, since $|g_1 + g_2| \leq f_1 + f_2$, we have:

$$\nu(f_1 + f_2) \geq \nu(f_1) + \nu(f_2) - 2\epsilon.$$

Since ϵ is arbitrary, we deduce that $\nu(f_1 + f_2) \geq \nu(f_1) + \nu(f_2)$. Let us now show the reverse inequality. Let $h \in C_0(\Omega)$ such that $|h| \leq f_1 + f_2$. For each $i = 1, 2$, define the function h_i by:

$$h_i(x) := \begin{cases} \dfrac{h(x) f_i(x)}{f_1(x) + f_2(x)} & \text{if } f_1(x) + f_2(x) \neq 0 \\ \\ 0 & \text{if } f_1(x) + f_2(x) = 0. \end{cases}$$

It is easy to verify that h_i is continuous on Ω. Moreover, since $|h_i| \leq f_i$ and $h = h_1 + h_2$, we obtain:

$$|\mu(h)| \leq |\mu(h_1)| + |\mu(h_2)| \leq \nu(f_1) + \nu(f_2),$$

and since h is arbitrary, we conclude that $\nu(f_1 + f_2) \leq \nu(f_1) + \nu(f_2)$, which proves (5.17).

We now extend the definition of $\nu(f)$ to any function $f \in C_0(\Omega, \mathbb{R})$ by writing: $\nu(f) := \nu(f_1) - \nu(f_2)$ for any decomposition $f = f_1 - f_2$ as the difference of two positive functions in $C_0(\Omega, \mathbb{R})$. As a consequence of (5.17), this definition is independent of the chosen decomposition, and moreover, the formula (5.17) also holds for functions $f_1, f_2 \in C_0(\Omega, \mathbb{R})$. Finally, the previous definition gives us that if $a \geq 0$, then $\nu(af) = a\nu(f)$; on the other hand, if $a < 0$, then:

$$\nu(af) = \nu(af^+ - af^-) = \nu(-af^- - (-af^+)) = \nu(-af^-) - \nu(-af^+)$$

$$= -a\nu(f^-) - (-a)\nu(f^+) = a(\nu(f^+) - \nu(f^-)) = a\nu(f).$$

Thus, ν is a (real) linear form on $C_0(\Omega, \mathbb{R})$, and since $\nu(f) \geq 0$ for every $0 \leq f \in C_0(\Omega, \mathbb{R})$, by virtue of Theorem 5.6, we conclude that ν is a positive Radon measure on Ω. Let us finally prove the minimality of ν. Let $0 \leq \lambda \in \mathcal{M}(\Omega)$ such that $|\mu(f)| \leq \lambda(|f|)$ for all $f \in C_0(\Omega)$. Then, given $0 \leq g \in C_0(\Omega)$, if $h \in C_0(\Omega)$ is such that $|h| \leq g$, then $|\mu(h)| \leq \lambda(|h|) \leq \lambda(g)$, and by (5.16) we obtain that $\nu(g) \leq \lambda(g)$. Consequently, $\nu \leq \lambda$, from which the minimality of ν follows. □

The measure ν from the previous theorem is denoted by $|\mu|$ and is called the *total variation* or the *modulus* of the measure μ. Thus, the measure $|\mu|$ satisfies the following:

$$|\mu(f)| \leq |\mu|(|f|) \qquad \forall\, f \in C_0(\Omega) \tag{5.18}$$

and

$$|\mu|(f) := \sup\{|\mu(g)| \,:\, g \in C_0(\Omega),\ |g| \leq f\},\ \ 0 \leq f \in C_0(\Omega). \tag{5.19}$$

It is immediate that for all $a \in \mathbb{C}$ and all $\mu \in \mathcal{M}(\Omega)$, we have

$$|a\mu| = |a|\,|\mu|, \tag{5.20}$$

and as a consequence of (5.18) we obtain

$$\mu \le |\mu| \quad \forall\, \mu \in \mathcal{M}_{\mathbb{R}}(\Omega). \tag{5.21}$$

Moreover, if $\mu \in \mathcal{M}(\Omega)$ is positive, then $|\mu| = \mu$. Indeed, taking into account (5.19), it suffices to prove that $|\mu(f)| \le \mu(|f|)$ for all $f \in C_0(\Omega)$. Now, given $f \in C_0(\Omega)$, there exists a $z \in \mathbb{C}$ with $|z| = 1$ such that $|\mu(f)| = z\mu(f) = \mu(zf)$; since μ is real, this implies $\mu(zf) = \mu(\Re(zf))$, and since μ is positive and $\Re(zf) \le |f|$, we have

$$|\mu(f)| = \mu(\Re(zf)) \le \mu(|f|).$$

Remark 5.1 By the classical Riesz Representation Theorem (see for example [17] or [3]), every real Radon measure $\mu \in \mathcal{M}_{\mathbb{R}}(\Omega)$ is uniquely associated with a regular Borel measure. More precisely, let $\mathcal{B}(\Omega)$ be the family of Borel subsets of Ω, i.e., the smallest σ-algebra containing the open sets of Ω. Recall that a set function $\nu : \mathcal{B}(\Omega) \to]-\infty, +\infty]$ is a *regular Borel measure* on Ω, if $\nu(\emptyset) = 0$, for every sequence $\{A_n\}$ of disjoint elements of $\mathcal{B}(\Omega)$,

$$\nu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \nu(A_n),$$

$$\nu(K) < \infty \quad \text{for every compact } K \subset \Omega,$$

and for every $E \in \mathcal{B}(\Omega)$, we have

$$\nu(E) = \sup\{\nu(K) \,:\, K \subset E,\ K \text{ compact}\}$$

$$= \inf\{\nu(A) \,:\, E \subset A,\ A \text{ open}\}.$$

The Riesz Representation Theorem guarantees that for every $\mu \in \mathcal{M}_{\mathbb{R}}(\Omega)$, there exists a unique regular Borel measure ν on Ω such that

$$\mu(f) = \int_{\Omega} \varphi \, d\nu \qquad \forall\, \varphi \in C_0(\Omega).$$

Moreover,

$$\|\mu\| := \sup\{\mu(f) \,:\, f \in C_0(\Omega),\ |f| \le 1\} = |\nu|(\Omega),$$

where $|\nu|$ denotes the total variation of the measure ν, i.e., the positive measure defined on $\mathcal{B}(\Omega)$ by

$$|\nu|(E) := \sup\left\{\sum_{n=1}^{\infty}|\nu(E_n)| \ : \ E_n \text{ disjoint and } E=\bigcup_{n=1}^{\infty}E_n\right\}.$$

5.5 Calculus with Distributions

In this section we will define the derivative of a distribution and study some of its properties. To motivate the definition of the derivative, observe that if f is a continuously differentiable function on $]a,b[$ (or more generally, absolutely continuous), then by applying integration by parts we have

$$\int_a^b f'\,\phi = -\int_a^b f\,\phi', \quad \forall\,\phi\in\mathcal{D}(]a,b[).$$

Thus, f' as a distribution on $]a,b[$ satisfies

$$\langle f',\phi\rangle = -\langle f,\phi'\rangle, \quad \forall\,\phi\in\mathcal{D}(]a,b[).$$

This motivates the following definition:

Definition 5.5 Let T be a distribution on an open set $\Omega\subset\mathbb{R}^N$ and let α be a multi-index. The *α-th derivative* of T is defined by the formula

$$\langle D^\alpha T,\varphi\rangle := (-1)^{|\alpha|}\,\langle T, D^\alpha\varphi\rangle \quad \forall\,\varphi\in\mathcal{D}(\Omega).$$

It is immediate that $D^\alpha T$ is a distribution in Ω.

Example 5.2

(1) Let δ be the Dirac distribution on $\mathbb{R}$. Then, for each $\varphi\in\mathcal{D}(\mathbb{R})$,

$$\langle\delta',\varphi\rangle = -\langle\delta,\varphi'\rangle = -\varphi'(0).$$

Hence, the derivative δ' is minus the dipole distribution.

(2) Let H be the *Heaviside function*, defined by

$$H(x) := \begin{cases} 0 \ \text{ if } \ x<0 \\ \\ 1 \ \text{ if } \ x\geq 0. \end{cases}$$

Since $H \in L^1_{loc}(\mathbb{R})$, it follows that $T_H \in \mathcal{D}'(\mathbb{R})$. Then, if $\varphi \in \mathcal{D}(\mathbb{R})$, we have

$$\langle (T_H)', \varphi \rangle = -\langle T_H, \varphi' \rangle = -\int_{\mathbb{R}} H(x)\varphi'(x)\, dx$$

$$= -\int_0^{\infty} \varphi'(x)\, dx = \varphi(0) = \langle \delta, \varphi \rangle.$$

Therefore, $(T_H)' = \delta$, i.e., the distributional derivative of the Heaviside function is the Dirac delta.

(3) Let sign be the sign function, i.e.,

$$\operatorname{sign}(x) := \begin{cases} -1 & \text{if } x < 0 \\ \\ 1 & \text{if } x \geq 0. \end{cases}$$

This function is the distributional derivative of the function $x \mapsto |x|$. Indeed, for $\varphi \in \mathcal{D}(\mathbb{R})$, we have

$$\langle D|x|, \varphi \rangle = -\int_{\mathbb{R}} |x|\varphi'(x)\, dx = \int_{-\infty}^{0} x\varphi'(x)\, dx - \int_0^{\infty} x\varphi'(x)\, dx$$

$$= \int_0^{\infty} \varphi(x)\, dx - \int_{-\infty}^{0} \varphi(x)\, dx = \int_{\mathbb{R}} \operatorname{sign}(x)\varphi(x)\, dx.$$

Consequently, $D|x| = \operatorname{sign}(x)$.

(4) Consider on $\mathbb{R}$ the distribution defined by the locally integrable function $x \mapsto \log|x|$. To compute its distributional derivative, take $\varphi \in \mathcal{D}(\mathbb{R})$. Then,

$$\langle D\log|x|, \varphi \rangle = -\int_{\mathbb{R}} \log|x|\varphi'(x)\, dx = -\lim_{\epsilon \to 0^+} \int_{|x| \geq \epsilon} \log|x|\varphi'(x)\, dx$$

$$= \lim_{\epsilon \to 0^+} \left\{ \log\epsilon[\varphi(\epsilon) - \varphi(-\epsilon)] + \int_{\epsilon}^{+\infty} \frac{\varphi(x) - \varphi(-x)}{x}\, dx \right\}$$

$$= \lim_{\epsilon \to 0^+} \int_{\epsilon}^{+\infty} \frac{\varphi(x) - \varphi(-x)}{x}\, dx,$$

where the last equality follows easily from the Mean Value Theorem. We introduce the notation:

$$\operatorname{Pf} \int_{-\infty}^{+\infty} \frac{\varphi(x)}{x}\, dx := \lim_{\epsilon \to 0^+} \int_{\epsilon}^{+\infty} \frac{\varphi(x) - \varphi(-x)}{x}\, dx.$$

The symbol Pf denotes the *finite part* of the divergent integral $\int_{-\infty}^{+\infty} \frac{\varphi(x)}{x}\, dx$, a concept introduced by J. Hadamard in his study of hyperbolic partial differential equations. Since

$$\lim_{\epsilon\to 0^+} \int_{\epsilon}^{+\infty} \frac{\varphi(x)-\varphi(-x)}{x}\, dx = \lim_{\epsilon\to 0^+} \int_{|x|\geq\epsilon} \frac{\varphi(x)}{x}\, dx = \mathrm{PV} \int_{-\infty}^{+\infty} \frac{\varphi(x)}{x}\, dx,$$

where the latter expression represents the *Cauchy principal value* of the integral $\int_{-\infty}^{+\infty} \frac{\varphi(x)}{x}\, dx$, we have that

$$\mathrm{Pf} \int_{-\infty}^{+\infty} \frac{\varphi(x)}{x}\, dx = \mathrm{PV} \int_{-\infty}^{+\infty} \frac{\varphi(x)}{x}\, dx.$$

It is easy to verify that the mapping

$$\varphi \mapsto \mathrm{Pf} \int_{-\infty}^{+\infty} \frac{\varphi(x)}{x}\, dx$$

is linear and continuous on $\mathcal{D}(\Omega)$, so it defines a distribution denoted by $\mathrm{Pf}\frac{1}{x}$, which is called the pseudofunction $\frac{1}{x}$. Therefore, we have that

$$D \log|x| = \mathrm{Pf}\frac{1}{x}.$$

Definition 5.6 Let $S \in \mathcal{D}'(\mathbb{R})$. A distribution $T \in \mathcal{D}'(\mathbb{R})$ is said to be a *primitive* of S if $T' = S$.

Theorem 5.8 *Every distribution on $\mathbb{R}$ has infinitely many primitives, with the difference between any two of them being a constant.*

Proof Let

$$X := \left\{ \varphi \in \mathcal{D}(\mathbb{R}) \ : \ \int_{\mathbb{R}} \varphi(x)\, dx = 0 \right\}.$$

It is immediate to verify that a function $\varphi \in X$ if and only if there exists $\psi \in \mathcal{D}(\mathbb{R})$ such that $\psi' = \varphi$. It is enough to take

$$\psi(x) = \int_{-\infty}^{x} \varphi(y)\, dy.$$

Moreover, X is a hyperplane of $\mathcal{D}(\mathbb{R})$, since if $\theta \in \mathcal{D}(\mathbb{R})$ with

$$\int_{\mathbb{R}} \theta(y)\, dy = 1,$$

then, given $\varphi \in \mathcal{D}(\mathbb{R})$, we have that $\varphi = \lambda\theta + \psi$, with

$$\lambda = \int_{\mathbb{R}} \varphi(y)\, dy$$

and $\psi = \varphi - \lambda\theta \in X$.

Given the distribution $S \in \mathcal{D}'(\mathbb{R})$, we define T as follows: Given $\varphi \in \mathcal{D}(\mathbb{R})$, we know that there exists a unique $\psi \in \mathcal{D}(\mathbb{R})$ such that $\varphi = \lambda\theta + \psi'$. We then define

$$\langle T, \varphi\rangle := a\lambda - \langle S, \psi\rangle,$$

where a is an arbitrary constant. Let us show that T is a primitive of S. Let $\varphi \in \mathcal{D}(\mathbb{R})$, then, since $\varphi' \in X$, we have

$$\langle T', \varphi\rangle = -\langle T, \varphi'\rangle = \langle S, \varphi\rangle,$$

so that $T' = S$.

Finally, suppose that T_1 and T_2 are two primitives of S. Then, for each $\varphi \in \mathcal{D}(\Omega)$,

$$0 = \langle T_1' - T_2', \varphi\rangle = -\langle T_1 - T_2, \varphi'\rangle.$$

Consequently, $\langle T_1 - T_2, \psi\rangle = 0$ for every $\psi \in X$. Therefore, given $\varphi \in \mathcal{D}(\Omega)$,

$$\varphi = \left(\int_{\mathbb{R}} \varphi\right)\theta + \psi, \quad \text{with } \psi \in X,$$

we have that

$$\langle T_1 - T_2, \varphi\rangle = \langle T_1, \left(\int_{\mathbb{R}} \varphi\right)\theta\rangle - \langle T_2, \left(\int_{\mathbb{R}} \varphi\right)\theta\rangle = \int_{\mathbb{R}} \langle T_1 - T_2, \theta\rangle\varphi.$$

Consequently,

$$T_1 - T_2 = \langle T_1 - T_2, \theta\rangle.$$

□

Corollary 5.1 *If $T \in \mathcal{D}'(\mathbb{R})$ has vanishing derivative, then T is a constant.*

Suppose that $f \in L^1_{loc}(\Omega)$ is such that $D^\alpha f \in L^1_{loc}(\Omega)$. Then we may consider the distributions T_f and $T_{D^\alpha f}$. The question of consistency arises as to whether $D^\alpha T_f = T_{D^\alpha f}$. This is equivalent to the identity

$$(-1)^{|\alpha|} \int_\Omega f\ D^\alpha\phi = \int_\Omega D^\alpha f\ \phi \quad \forall\, \phi \in \mathcal{D}(\Omega). \tag{5.22}$$

By integration by parts, (5.22) holds if f admits continuous partial derivatives up to order $|\alpha|$. Now, as we shall see, (5.22) does not hold in general. To this end, we are going to characterize when (5.22) holds in the particular case $\Omega = \mathbb{R}$. Recall that a function f is *absolutely continuous* on $\mathbb{R}$ if and only if $f \in L^1_{loc}(\mathbb{R})$, there exists f' a.e. in $\mathbb{R}$, with $f' \in L^1_{loc}(\mathbb{R})$, and for every $a, x \in \mathbb{R}$,

$$f(x) - f(a) = \int_a^x f'(y)\, dy.$$

We have the following characterization theorem.

Theorem 5.9 (Nikodym's Theorem) *Let $f \in L^1_{loc}(\mathbb{R})$. If f is absolutely continuous on $\mathbb{R}$, then $(T_f)' = T_{f'}$. Conversely, if $(T_f)' \in L^1_{loc}(\mathbb{R})$, then f is absolutely continuous on $\mathbb{R}$.*

Proof Suppose that f is absolutely continuous on $\mathbb{R}$. Let $\varphi \in \mathcal{D}(\mathbb{R})$ with $\operatorname{supp}(\varphi) \subset [-a, a]$, and let $A = \{(x, y) \ : -a \le x < y \le a\}$. Then, applying Fubini's Theorem, we have

$$\langle (T_f)', \varphi \rangle = -\langle T_f, \varphi' \rangle = -\int_{-a}^{a} f(x)\varphi'(x)\, dx$$

$$= \int_{-a}^{a} \varphi'(x)\left(\int_x^a f'(y)\, dy - f(a) \right) dx = \int_{-a}^{a} \varphi'(x)\left(\int_x^a f'(y)\, dy \right) dx$$

$$= \int_A \varphi'(x) f'(y)\, dx dy = \int_{-a}^{a} f'(y)\left(\int_{-a}^{y} \varphi'(x)\, dx \right) dy$$

$$= \int_{-a}^{a} f'(y)\varphi(y)\, dy = \langle T_{f'}, \varphi \rangle.$$

Consequently, $(T_f)' = T_{f'}$.

Now suppose that $f \in L^1_{loc}(\mathbb{R})$ and that there exists a function $g \in L^1_{loc}(\mathbb{R})$ such that $(T_f)' = T_g$. Consider the function $h(x) := \int_0^x g(y)\, dy$. We have that h is absolutely continuous and $h'(x) = g(x)$ almost everywhere on $\mathbb{R}$. Then, by what was proved in the first part, we have that $(T_h)' = T_{h'} = T_g = (T_f)'$. Therefore, by Theorem 5.8, there exists a constant k such that $T_h - T_f = T_k$, hence $f = h - k$ almost everywhere on $\mathbb{R}$, and consequently, f is absolutely continuous on $\mathbb{R}$. □

If f is a function of bounded variation on every compact interval of $\mathbb{R}$ (i.e., if on every compact interval f is the difference of two monotone functions), then by Lebesgue's Theorem, there exists f' almost everywhere on $\mathbb{R}$, and $f' \in L^1_{loc}(\mathbb{R})$.

Now, in the case where f is not absolutely continuous, as happens for example with the Cantor staircase function (see e.g. [37]), by Nikodym's Theorem we have that $(T_f)' \neq T_{f'}$.

In general, the following result holds, whose proof can be found in the book by L. Schwartz [52]:

In order for a distribution on $\mathbb{R}$ to have a Radon measure as its distributional derivative, it is necessary and sufficient that it be of bounded variation on every compact subinterval of $\mathbb{R}$.

Various topologies can be defined on the space $\mathcal{D}'(\Omega)$ (see for instance [52]), but for our purposes the following notion of convergence will suffice.

Definition 5.7 We say that a sequence of distributions $T_n \in \mathcal{D}'(\Omega)$ converges to a distribution $T \in \mathcal{D}'(\Omega)$, and we write $T_n \to T$ in $\mathcal{D}'(\Omega)$, if

$$\langle T_n, \varphi \rangle \to \langle T, \varphi \rangle \quad \forall\, \varphi \in \mathcal{D}(\Omega).$$

It is evident that if $T_n \to T$ in $\mathcal{D}'(\Omega)$, then $D^\alpha T_n \to D^\alpha T$ in $\mathcal{D}'(\Omega)$ for every multi-index $\alpha \in \mathbb{N}^N$.

Example 5.3 Let $\{\phi_n\}$ be a regularizing sequence in $\mathbb{R}$. Let us show that $T_{\phi_n} \to \delta$ in $\mathcal{D}'(\mathbb{R})$. Indeed, given $\varphi \in \mathcal{D}(\mathbb{R})$, we have

$$\langle T_{\phi_n}, \varphi \rangle = \int_{\mathbb{R}} \phi_n(x)\varphi(x)\, dx = \varphi(0) + \int_{\mathbb{R}} \phi_n(x)(\varphi(x) - \varphi(0))\, dx$$

$$= \varphi(0) + kn \int_{-1/n}^{1/n} \exp\left(\frac{-1}{1 - |nx|^2}\right)(\varphi(x) - \varphi(0))\, dx$$

$$= \varphi(0) + k \int_{-1}^{1} \exp\left(\frac{-1}{1 - |x|^2}\right)(\varphi(x/n) - \varphi(0))\, dx \to \varphi(0) = \langle \delta, \varphi \rangle,$$

by the Dominated Convergence Theorem. Therefore, $T_{\phi_n} \to \delta$ in $\mathcal{D}'(\mathbb{R})$.

Let us now define the product of a function with a distribution.

Definition 5.8 Let $\psi \in C^\infty(\Omega)$ and $T \in \mathcal{D}'(\Omega)$. The product distribution ψT is defined by the formula

$$\langle \psi T, \varphi \rangle := \langle T, \psi\varphi \rangle \quad \forall\, \varphi \in \mathcal{D}(\Omega).$$

The standard calculus rules still apply. For example, we have

$$\langle (\psi T)', \varphi \rangle = -\langle \psi T, \varphi' \rangle = -\langle T, \psi\varphi' \rangle = -\langle T, (\psi\varphi)' - \psi'\varphi \rangle$$

$$= \langle T', \psi\varphi \rangle + \langle T, \psi'\varphi \rangle = \langle \psi T' + \psi' T, \varphi \rangle,$$

which implies that

$$(\psi T)' = \psi T' + \psi' T.$$

It can also be easily shown by induction that the *Leibniz formula* holds, i.e., *If* $\psi \in C^\infty(\Omega)$ and $T \in \mathcal{D}'(\Omega)$, *then for every multi-index* $\alpha \in \mathbb{N}^n$, *we have*

$$D^\alpha(\psi T) = \sum_{\beta \leq \alpha} \frac{\alpha!}{\beta!(\alpha! - \beta!)} D^\beta \psi D^{\alpha-\beta} T.$$

Although we will study the wave equation in more detail later, in the following example we will compute the distributional solutions of the one-dimensional wave equation.

Example 5.4 The wave equation arose in the study of the vibration of a string. If a string with uniform linear density ρ is under a uniform tension T, and if in the initial position the string coincides with the x-axis, then when its equilibrium position is perturbed, the transverse displacement $u(x, t)$ satisfies the one-dimensional wave equation

$$\frac{\partial^2 u(t,x)}{\partial t^2} = k^2 \frac{\partial^2 u(t,x)}{\partial x^2} \tag{5.23}$$

where $k^2 = \frac{T}{\rho}$.

If u is a distribution on $\mathbb{R}^2$ that satisfies (5.23) in the distributional sense, we say that u is a weak (or distributional) solution of that equation. It is easy to see that if u is a classical solution of (5.23), then u is also a weak solution.

For $f \in L^1_{loc}(\mathbb{R})$, consider the traveling wave

$$u(t, x) := f(x - kt).$$

Since $u \in L^1_{loc}(\mathbb{R}^2)$, it defines a distribution on $\mathbb{R}^2$, which we will also denote by u. Let us show that this distribution is a weak solution of (5.23). For $\varphi \in \mathcal{D}(\mathbb{R}^2)$, we have

$$\left\langle \frac{\partial^2 u}{\partial t^2}, \varphi \right\rangle = \left\langle u, \frac{\partial^2 \varphi}{\partial t^2} \right\rangle = \int_{\mathbb{R}^2} f(x - kt) \frac{\partial^2 \varphi}{\partial t^2}(t, x)\, dtdx.$$

$$\left\langle \frac{\partial^2 u}{\partial x^2}, \varphi \right\rangle = \left\langle u, \frac{\partial^2 \varphi}{\partial x^2} \right\rangle = \int_{\mathbb{R}^2} f(x - kt) \frac{\partial^2 \varphi}{\partial x^2}(t, x)\, dtdx.$$

Thus,

$$\left\langle \frac{\partial^2 u}{\partial t^2} - k^2 \frac{\partial^2 u}{\partial x^2}, \varphi \right\rangle = \int_{\mathbb{R}^2} f(x - kt) \left(\frac{\partial^2 \varphi}{\partial t^2}(t, x) - k^2 \frac{\partial^2 \varphi}{\partial x^2}(t, x) \right) dtdx.$$

We want to show that this expression is zero. To do so, we perform the change of variables: $y = x - kt$, $z = x + kt$, for which

$$\frac{\partial(y,z)}{\partial(t,x)} = \begin{vmatrix} -k & 1 \\ k & 1 \end{vmatrix} = -2k.$$

Moreover,

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial y} + \frac{\partial}{\partial z}, \quad \frac{\partial}{\partial t} = -k\frac{\partial}{\partial y} + k\frac{\partial}{\partial z},$$

so that

$$\frac{\partial^2}{\partial t^2} - k^2\frac{\partial^2}{\partial x^2} = -4k^2\frac{\partial^2}{\partial y \partial z}.$$

Therefore, we obtain

$$\int_{\mathbb{R}^2} f(x-kt)\left(\frac{\partial^2\varphi}{\partial t^2}(t,x) - k^2\frac{\partial^2\varphi}{\partial x^2}(t,x)\right) dtdx$$

$$= -2k\int_{\mathbb{R}^2} f(y)\frac{\partial^2\varphi}{\partial y\partial z}(y,z)\, dydz$$

$$= -2k\int_{\mathbb{R}} f(y)\left(\int_{\mathbb{R}} \frac{\partial^2\varphi}{\partial y\partial z}(y,z)\, dz\right) dy = 0,$$

since $\varphi \in \mathcal{D}(\mathbb{R}^2)$.

5.6 Distributions with Compact Support

We have defined the support of a measurable function as the complement of the largest open set where the function vanishes almost everywhere. We want to extend this notion to distributions, and to do so, we need to give meaning to the phrase "largest open set where a distribution vanishes."

Let Ω and Λ be two open subsets of $\mathbb{R}^n$, with $\Lambda \subset \Omega$. Let T_1 and T_2 be two distributions on Ω. By the expression

$$T_1 = T_2 \quad \text{in } \Lambda$$

we mean, by definition, that

$$\langle T_1, \varphi\rangle = \langle T_2, \varphi\rangle \quad \forall\, \varphi \in \mathcal{D}(\Lambda).$$

Theorem 5.10 (Localization Principle) *Let $\mathcal{U}$ be an open covering of Ω, and suppose that for each $U \in \mathcal{U}$ there exists a distribution $T_U \in \mathcal{D}'(U)$ such that*

$$T_{U_1} = T_{U_2} \quad \text{in } U_1 \cap U_2 \quad (\text{when } U_1 \cap U_2 \neq \emptyset).$$

Then there exists a unique distribution $T \in \mathcal{D}'(\Omega)$ such that

$$T = T_U \quad \text{in } U \quad \forall\, U \in \mathcal{U}.$$

Proof Let $\{\psi_i\}_{i=1}^{\infty}$ be a partition of unity subordinate to the covering $\mathcal{U}$. For each $i \in \mathbb{N}$, let $U_i \in \mathcal{U}$ be such that $\operatorname{supp}(\psi_i) \subset U_i$. Given $\varphi \in \mathcal{D}(\Omega)$, we have that $\varphi = \sum_{i=1}^{\infty} \varphi\psi_i$, and since φ has compact support, we know that only a finite number of summands are nonzero. We define $T : \mathcal{D}(\Omega) \to \mathbb{C}$ as

$$\langle T, \varphi \rangle := \sum_{i=1}^{\infty} \langle T_{U_i}, \varphi\psi_i \rangle.$$

Clearly, T is linear. Let us show that it is continuous. Let $\varphi_m \to 0$ in $\mathcal{D}(\Omega)$. Then there exists a compact set $K \subset \Omega$ such that $\operatorname{supp}(\varphi_{\mathrm{m}}) \subset \mathrm{K}$ for each $m \in \mathbb{N}$. Then there exists $i_0 \in \mathbb{N}$ such that

$$\langle T, \varphi_m \rangle := \sum_{i=1}^{i_0} \langle T_{U_i}, \varphi_m\psi_i \rangle \quad \forall\, m \in \mathbb{N}. \tag{5.24}$$

Now, since $\varphi_m\psi_i \to 0$ in $\mathcal{D}(U_i)$, it follows from (5.24) that $\langle T, \varphi_m \rangle \to 0$. Consequently, $T \in \mathcal{D}'(\Omega)$. Let $U \in \mathcal{U}$; we now show that $T = T_U$ in U. Given $\varphi \in \mathcal{D}(U)$, we have that $\varphi\psi_i \in \mathcal{D}(U \cap U_i)$, and thus

$$\langle T_{U_i}, \varphi\psi_i \rangle = \langle T_U, \varphi\psi_i \rangle.$$

Therefore,

$$\langle T, \varphi \rangle = \sum_{i=1}^{\infty} \langle T_{U_i}, \varphi\psi_i \rangle = \sum_{i=1}^{\infty} \langle T_U, \varphi\psi_i \rangle = \left\langle T_U, \sum_{i=1}^{\infty} \varphi\psi_i \right\rangle = \langle T_U, \varphi \rangle.$$

Finally, suppose there exists another $S \in \mathcal{D}'(\Omega)$ such that $S = T_U$ for each $U \in \mathcal{U}$. In particular, $S = T_{U_i}$ for each $i \in \mathbb{N}$, and thus, by the definition of T, we have $T = S$. □

As a consequence of the localization principle, we have that if $\{U_i\}_{i \in I}$ is a collection of open subsets of Ω, and $T \in \mathcal{D}'(\Omega)$ is such that $T = 0$ on each U_i for all $i \in I$, then $T = 0$ on $\bigcup_{i \in I} U_i$. This allows us to give the following definition.

Definition 5.9 Let $T \in \mathcal{D}'(\Omega)$. Its *support* is defined as the complement in Ω of the largest open set on which T vanishes. The support of T is denoted by supp(T).

Clearly, if $f \in L^1_{loc}(\Omega)$, then $\mathrm{supp}(\mathrm{T_f}) = \mathrm{supp(f)}$. On the other hand, it is easy to see that for each $x \in \mathbb{R}^N$, we have $\mathrm{supp}(\delta_x) = \{x\}$.

The support of a distribution is a closed set. It is useful to know when a distribution has compact support. In order to characterize distributions with compact support, we need to introduce a topology on the space $C^\infty(\Omega)$, a space which, following L. Schwartz, we denote by $\mathcal{E}(\Omega)$. As in the case of the space $\mathcal{D}(\Omega)$, we introduce this topology by defining its convergent sequences.

Definition 5.10 Let $\varphi_n, \varphi \in \mathcal{E}(\Omega)$. We say that the sequence $\{\varphi_n\}$ converges to φ in $\mathcal{E}(\Omega)$ if for each multi-index $\alpha \in \mathbb{N}^N$ and each compact set $K \in \mathcal{K}(\Omega)$, we have $D^\alpha \varphi_n \to D^\alpha \varphi$ uniformly on K. To denote this convergence, we write

$$\varphi_n \to \varphi \quad \text{in } \mathcal{E}(\Omega).$$

Remark 5.2 A metric can be defined on the space $\mathcal{E}(\Omega)$ whose convergent sequences coincide with those defined above. To do this, given the open set Ω, we set

$$K_m := \left\{ x \in \Omega \,:\, \|x\| \le m, \;\; d(x, \mathbb{R}^N \setminus \Omega) \ge \frac{1}{m} \right\},$$

so that each K_m is compact,

$$\Omega = \bigcup_{m \in \mathbb{N}} K_m,$$

and for any compact $K \subset \Omega$, there exists some $m_0 \in \mathbb{N}$ such that $K \subset K_{m_0}$. Then, for each $m \in \mathbb{N}$ and $\phi \in \mathcal{E}(\Omega)$, we define

$$q_m(\phi) := \sup\{|D^\alpha \phi(x)| \,:\, x \in K_m, \;\; |\alpha| \le m\}.$$

Each q_m is a seminorm on $\mathcal{E}(\Omega)$. Then we have

$$d(\phi, \varphi) := \sum_{m=1}^{\infty} \frac{1}{2^m} \, \frac{q_m(\phi - \varphi)}{1 + q_m(\phi - \varphi)}$$

is a metric on $\mathcal{E}(\Omega)$, and

$$\varphi_m \to \varphi \;\; \text{in } \mathcal{E}(\Omega) \iff \lim_{m \to \infty} d(\varphi_m, \varphi) = 0.$$

Moreover, the space $\mathcal{E}(\Omega)$ is complete with respect to this metric (see [52]).

We denote the topological dual of the space $\mathcal{E}(\Omega)$ by $\mathcal{E}'(\Omega)$, i.e., given a linear form $T : \mathcal{E}(\Omega) \to \mathbb{C}$, we have that

$$T \in \mathcal{E}'(\Omega) \iff \langle T, \varphi_m \rangle \to 0 \text{ as } \varphi_m \to 0 \text{ in } \mathcal{E}(\Omega).$$

We are going to identify $\mathcal{E}'(\Omega)$ with the space of distributions in Ω with compact support. For this, we first need to establish the following characterization of the elements of $\mathcal{E}'(\Omega)$.

Lemma 5.3 *Let $T : \mathcal{E}(\Omega) \to \mathbb{C}$ be a linear form. Then, $T \in \mathcal{E}'(\Omega)$ if and only if there exist $m \in \mathbb{N}$ and a constant $C > 0$ such that*

$$|\langle T, \phi \rangle| \leq C q_m(\phi) \quad \forall\, \phi \in \mathcal{E}(\Omega). \tag{5.25}$$

Proof Clearly, if T satisfies (5.25), then $T \in \mathcal{E}'(\Omega)$. To prove the converse, we proceed by contradiction. Let $T \in \mathcal{E}'(\Omega)$, and suppose that it does not satisfy (5.25). Then, for each $m \in \mathbb{N}$, there exists $\phi_m \in \mathcal{E}(\Omega)$ such that

$$|\langle T, \phi_m \rangle| > m q_m(\phi_m).$$

Now, define

$$\psi_m := \frac{\phi_m}{\langle T, \phi_m \rangle},$$

so that

$$|\langle T, \psi_m \rangle| = 1 \quad \text{and} \quad q_m(\psi_m) < \frac{1}{m}.$$

Let us show that $\psi_m \to 0$ in $\mathcal{E}(\Omega)$, which will lead to a contradiction. Given a compact set $K \subset \Omega$ and $\alpha \in \mathbb{N}^n$, there exists $m_0 \in \mathbb{N}$ such that $K \subset K_{m_0}$ and $|\alpha| \leq m_0$. Therefore, for $m \geq m_0$, we have

$$\sup_{x \in K} |D^\alpha \psi_m(x)| \leq q_m(\psi_m) < \frac{1}{m},$$

which concludes the proof. □

Theorem 5.11 *Let $T \in \mathcal{D}'(\Omega)$ be a distribution with compact support. Then there exists a unique $\tilde{T} \in \mathcal{E}'(\Omega)$ whose restriction to $\mathcal{D}(\Omega)$ is T.*

Conversely, if $S \in \mathcal{E}'(\Omega)$, its restriction to $\mathcal{D}(\Omega)$ is a distribution in Ω with compact support.

Proof Let $T \in \mathcal{D}'(\Omega)$, with $K = \text{supp(T)}$ compact. Take a function $\varphi \in \mathcal{D}(\Omega)$ that is identically equal to one on a neighborhood of K. Then, if $\phi \in \mathcal{D}(\Omega)$, since

$\text{supp}((1-\varphi)\phi) \subset \Omega \setminus K$, we have

$$0 = \langle T, (1-\varphi)\phi\rangle = \langle T, \phi\rangle - \langle T, \varphi\phi\rangle.$$

Thus, $T = \varphi T$. We then define $\tilde{T} : \mathcal{E}(\Omega) \to \mathbb{C}$ as

$$\langle \tilde{T}, \psi\rangle := \langle T, \varphi\psi\rangle.$$

Clearly, $\tilde{T}$ is linear; moreover, if $\psi_m \to 0$ in $\mathcal{E}(\Omega)$, then $\varphi\psi_m \to 0$ in $\mathcal{D}(\Omega)$, and hence

$$\langle \tilde{T}, \psi_m\rangle = \langle T, \varphi\psi_m\rangle \to 0.$$

Therefore, $\tilde{T} \in \mathcal{E}'(\Omega)$, and furthermore, for $\phi \in \mathcal{D}(\Omega)$,

$$\langle \tilde{T}, \phi\rangle = \langle T, \varphi\phi\rangle = \langle T, \phi\rangle.$$

Uniqueness follows from the density of $\mathcal{D}(\Omega)$ in $\mathcal{E}(\Omega)$.

Conversely, let $S \in \mathcal{E}'(\Omega)$. Clearly, if T is the restriction of S to $\mathcal{D}(\Omega)$, then $T \in \mathcal{D}'(\Omega)$, and as a consequence of Lemma 5.3, there exists a constant $C > 0$ and an $m \in \mathbb{N}$ such that

$$|\langle T, \phi\rangle| \le C q_m(\phi) \quad \forall\, \phi \in \mathcal{D}(\Omega).$$

Thus, supp(T) $\subset K_m$, and consequently, it is compact. □

Theorem 5.12 *Every distribution with compact support is of finite order.*

Proof Let $T \in \mathcal{D}'(\Omega)$, with $K = \text{supp(T)}$ compact. As we saw in the proof of the previous theorem, if $\varphi \in \mathcal{D}(\Omega)$ is a function identically equal to 1 in a neighborhood of K, then $T = \varphi T$. Let $\tilde{K} = \text{supp}(\varphi)$. Then, given a compact set $Q \in \mathcal{K}(\Omega)$, since $\text{supp}(\varphi\phi) \subset \tilde{K}$ for every $\phi \in \mathcal{D}_Q(\Omega)$, there exist constants $C_{\tilde{K}} > 0$ and $m_{\tilde{K}} \in \mathbb{N}$ such that

$$|\langle T, \phi\rangle| = |\langle T, \varphi\phi\rangle| \le C_{\tilde{K}} P_{m_{\tilde{K}}}(\varphi\phi) \quad \forall\, \phi \in \mathcal{D}_Q(\Omega).$$

Now, by the Leibniz formula,

$$P_{m_{\tilde{K}}}(\varphi\phi) \le C P_{m_{\tilde{K}}}(\phi).$$

Consequently,

$$|\langle T, \phi\rangle| \le C C_{\tilde{K}} P_{m_{\tilde{K}}}(\phi) \quad \forall\, \phi \in \mathcal{D}_Q(\Omega),$$

which shows that the order of T is less than or equal to $m_{\tilde{K}}$, and therefore finite. □

Clearly, the converse is not true, i.e., not every distribution of finite order has compact support. It suffices to consider a Radon measure (which by Theorem 5.4 has order zero) that does not have compact support; for example, the Lebesgue measure on $\mathbb{R}^N$.

Proposition 5.6 *If $T \in \mathcal{D}'(\Omega)$, with $K = \mathrm{supp(T)}$ compact, then for every multi-index $\alpha \in \mathbb{N}^n$, the distribution $D^\alpha T$ has compact support, with* $\mathrm{supp(D^\alpha T)} \subset \mathrm{K}$.

Proof Let $\omega := \Omega \setminus K$. Then,

$$\langle T, \varphi \rangle = 0 \quad \forall\, \varphi \in \mathcal{D}(\omega).$$

Hence,

$$\langle D^\alpha T, \varphi \rangle = (-1)^{|\alpha|} \langle T, D^\alpha \varphi \rangle = 0 \quad \forall\, \varphi \in \mathcal{D}(\omega).$$

Therefore, $\mathrm{supp(D^\alpha T)} \subset K$. □

The following result characterizes distributions whose support is a single point.

Theorem 5.13 *Let $T \in \mathcal{D}'(\Omega)$ with $\mathrm{supp(T)} = \{\mathrm{x}\}$ and order m. Then there exist constants $c_\alpha \in \mathbb{C}$ such that*

$$T = \sum_{|\alpha| \le m} c_\alpha D^\alpha \delta_x.$$

Conversely, every distribution of the above form has support equal to $\{x\}$.

Proof Since the support of $D^\alpha \delta_x$ is $\{x\}$, the second part is proven. To prove the first part, without loss of generality, assume $x = 0$. By Lemma 2.8, if

$$X = \bigcap_{|\alpha| \le m} \mathrm{Ker}(D^\alpha \delta) = \{\phi \in \mathcal{D}(\Omega) \ : \ D^\alpha \phi(0) = 0 \ \forall\, |\alpha| \le m\},$$

it suffices to prove that $X \subset \mathrm{Ker}(T)$, i.e.,

$$\langle T, \phi \rangle = 0 \quad \forall\, \phi \in X. \tag{5.26}$$

Let $\phi \in X$ and $\epsilon > 0$. There exists $\delta > 0$ such that $\overline{B}(0, \delta) \subset \Omega$ and

$$|D^\alpha \phi(x)| \le \epsilon \quad \forall\, x \in \overline{B}(0, \delta), \ \ |\alpha| = m. \tag{5.27}$$

We claim that

$$|D^\alpha \phi(x)| \le \epsilon n^{m-|\alpha|} \|x\|^{m-|\alpha|} \quad \forall\, x \in \overline{B}(0, \delta), \ |\alpha| \le m. \tag{5.28}$$

When $|\alpha| = m$, (5.28) is (5.27). Suppose $1 \leq k \leq m$, and assume that (5.28) holds for every α with $|\alpha| = k$, and suppose $|\beta| = k - 1$. Then, since

$$\nabla D^\beta \phi = (D_1 D^\beta \phi, \ldots, D_n D^\beta \phi),$$

the induction hypothesis yields

$$|\nabla D^\beta \phi(x)| \leq n\epsilon n^{m-k} \|x\|^{m-k} \quad \forall\, x \in \overline{B}(0, \delta). \tag{5.29}$$

Now, since $D^\beta \phi(0) = 0$, by the Mean Value Theorem (Theorem A.1) and using (5.29), we obtain (5.28) with α replaced by β, and thus we have proven (5.28).

Let $\varphi \in \mathcal{D}(\mathbb{R}^N)$ with $\text{supp}(\varphi) \subset \text{B}(0, 1)$, and $\varphi = 1$ in a neighborhood of 0. For each $\eta > 0$, consider the function

$$\varphi_\eta(x) := \varphi\left(\frac{x}{\eta}\right) \quad x \in \mathbb{R}^N.$$

For η small enough, we have $\text{supp}(\varphi_\eta) \subset \overline{B}(0, \delta)$. By the Leibniz formula,

$$D^\alpha(\varphi_\eta \phi)(x) = \sum_{\beta \leq \alpha} C_{\alpha\beta} \left(D^{\alpha-\beta}\varphi\right)\left(\frac{x}{\eta}\right)\left(D^\beta \phi\right)(x) \eta^{|\beta|-|\alpha|}. \tag{5.30}$$

It follows from (5.28) and (5.30) that

$$P_m(\varphi_\eta \phi) \leq \epsilon C_1 P_m(\varphi) \tag{5.31}$$

for η small enough, with constant C depending only on n and m.

Since T has order m, there exists a constant C_2 such that

$$|\langle T, \psi \rangle| \leq C_2 P_m(\psi) \;\; \forall\, \psi \in \mathcal{D}_{\overline{B}(0,\delta)}. \tag{5.32}$$

Now, since $\varphi_\eta = 1$ in a neighborhood of the support of T, we have that $T = \varphi_\eta T$. Then, as a consequence of (5.31) and (5.32), we obtain

$$|\langle T, \phi \rangle| = |\langle \varphi_\eta T, \phi \rangle| \leq \epsilon C_1 C_2 P_m(\psi).$$

From this, since ϵ is arbitrary, we obtain (5.26). □

5.7 Convolution of Distributions

In this section, we will generalize the notion of convolution of functions to certain classes of distributions. There are two ways to introduce the convolution product of distributions: one using the tensor product, as done by L. Schwartz in [52], and the other, which we will follow here, is the one described in the textbooks by Hörmander [28] and Rudin [50]. We begin by introducing some notation.

Let $f : \mathbb{R}^N \to \mathbb{C}$ and $x \in \mathbb{R}^N$. We define the functions $\tau_x f$ and $\check{f}$ as follows:

$$(\tau_x f)(y) := f(y - x), \qquad \check{f}(y) := f(-y), \quad \forall\, y \in \mathbb{R}^N.$$

If $T \in \mathcal{D}'(\mathbb{R}^N)$ and $x \in \mathbb{R}^N$, we define $\tau_x T \in \mathcal{D}'(\mathbb{R}^N)$ as

$$\langle \tau_x T, \varphi \rangle := \langle T, \tau_{-x}\varphi \rangle \quad \forall\, \varphi \in \mathcal{D}(\mathbb{R}^N).$$

Note that if $f, g \in L^1_{loc}(\mathbb{R}^N)$ are two convolvable functions, then

$$(f \star g)(x) = \int_{\mathbb{R}^N} f(y)g(x - y)\, dy = \int_{\mathbb{R}^N} f(y)(\tau_x \check{g})(y)\, dy.$$

This motivates the following definition.

Definition 5.11 Let $T \in \mathcal{D}'(\mathbb{R}^N)$ and $\phi \in \mathcal{D}(\mathbb{R}^N)$. The convolution of T and ϕ is defined as the function $T \star \phi$ given by

$$(T \star \phi)(x) := \langle T, \tau_x \check{\phi} \rangle.$$

The basic properties of this convolution product are collected in the following result.

Theorem 5.14 *Let $T \in \mathcal{D}'(\mathbb{R}^n)$ and $\phi \in \mathcal{D}(\mathbb{R}^n)$. Then:*

(i) For each $x \in \mathbb{R}^n$,

$$\tau_x(T \star \phi) = (\tau_x T) \star \phi = T \star (\tau_x \phi).$$

(ii) For each multi-index α,

$$D^\alpha(T \star \phi) = (D^\alpha T) \star \phi = T \star (D^\alpha \phi).$$

In particular, $T \star \phi \in \mathcal{E}(\mathbb{R}^n)$.

(iii) If $\psi \in \mathcal{D}(\mathbb{R}^n)$, then

$$T \star (\phi \star \psi) = (T \star \phi) \star \psi.$$

(iv) *If* $T \star \phi = 0$ *for all* $\phi \in \mathcal{D}(\mathbb{R}^n)$, *then* $T = 0$.
(v) $\operatorname{supp}(\mathrm{T} \star \phi) \subset \operatorname{supp}(\mathrm{T}) + \operatorname{supp}(\phi)$.

Proof

(i) Just observe that

$$\tau_x(T \star \phi)(y) = (T \star \phi)(y - x) = \langle T, \tau_{y-x}\check{\phi}\rangle = \langle \tau_x T, \tau_y \check{\phi}\rangle$$

$$= \big((\tau_x T) \star \phi\big)(y) = \langle T, \tau_y(\tau_x \phi)^\vee\rangle = (T \star \tau_x \phi)(y).$$

(ii) It suffices to prove it for $\alpha = (0, \ldots, 0, 1, 0, \ldots, 0)$, i.e., for $D^\alpha = \frac{\partial}{\partial x_i}$. Let $\mathbf{e}_i = (0, \ldots, 0, 1, 0, \ldots, 0)$, then, using (i), we get

$$\frac{\partial}{\partial x_i}(T \star \phi)(x) = \lim_{h\to 0} \frac{1}{h}\big[(T \star \phi)(x) - (T \star \phi)(x - h\mathbf{e}_i)\big]$$

$$= \lim_{h\to 0} \frac{1}{h}\big[(T\star\phi)(x) - \tau_{h\mathbf{e}_i}(T\star\phi)(x)\big] = \lim_{h\to 0} \frac{1}{h}\big[(T\star\phi)(x) - (T\star(\tau_{h\mathbf{e}_i}\phi))(x)\big]$$

$$= \lim_{h\to 0} \frac{1}{h}\left[T \star (\phi - \tau_{h\mathbf{e}_i}\phi)\right](x) = \lim_{h\to 0} \left\langle T, \tau_x \left(\frac{\phi - \tau_{h\mathbf{e}_i}\phi}{h}\right)^\vee \right\rangle$$

$$= \left\langle T, \tau_x \left(\frac{\partial \phi}{\partial x_i}\right)^\vee \right\rangle = \left(T \star \frac{\partial \phi}{\partial x_i}\right)(x),$$

since

$$\frac{\phi - \tau_{h\mathbf{e}_i}\phi}{h} \to \frac{\partial \phi}{\partial x_i} \quad \text{in } \mathcal{D}(\mathbb{R}^N).$$

Let us now prove the other equality. It is easy to see that

$$\lim_{h\to 0} \frac{T - \tau_{h\mathbf{e}_i} T}{h} = \frac{\partial T}{\partial x_i} \quad \text{in } \mathcal{D}'(\mathbb{R}^N).$$

Hence,

$$\frac{\partial}{\partial x_i}(T \star \phi)(x) = \lim_{h\to 0} \left(\left(\frac{T - \tau_{h\mathbf{e}_i} T}{h}\right) \star \phi\right)(x)$$

$$= \lim_{h\to 0} \left\langle \frac{T - \tau_{h\mathbf{e}_i} T}{h}, \tau_x \check{\phi} \right\rangle = \left\langle \frac{\partial T}{\partial x_i}, \tau_x \check{\phi} \right\rangle = \left(\frac{\partial T}{\partial x_i} \star \phi\right)(x),$$

(iii) It is enough to prove that

$$(T \star (\phi \star \psi))(0) = ((T \star \phi) \star \psi)(0), \tag{5.33}$$

since by taking $\tau_{-x}\psi$ instead of ψ in (5.33), we get the desired result. Now,

$$(\phi\star\psi)^{\vee}(x) = (\phi\star\psi)(-x) = \int_{\mathbb{R}^N} \phi(-x-y)\psi(y)\,dy = \int_{\mathbb{R}^N} \tau_y\check{\phi}(x)\check{\psi}(y)\,dy.$$

Since $\text{supp}(\check{\psi})$ is compact, the last integral is the limit as $\epsilon \to 0^+$ of Riemann sums of the form

$$\sum_p \tau_{\epsilon p}\check{\phi}(x)\check{\psi}(\epsilon p).$$

Hence,

$$(\phi \star \psi)^{\vee}(\cdot) = \lim_{\epsilon\to 0^+} \sum_p \tau_{\epsilon p}\check{\phi}(\cdot)\check{\psi}(\epsilon p)$$

in the topology of $\mathcal{D}(\mathbb{R}^N)$. Consequently,

$$(T \star (\phi \star \psi))(0) = \langle T, (\phi \star \psi)^{\vee}\rangle = \lim_{\epsilon\to 0^+} \sum_p \left\langle T, \tau_{\epsilon p}\check{\phi}(\cdot)\check{\psi}(\epsilon p)\right\rangle$$

$$= \lim_{\epsilon\to 0^+} \sum_p \langle T, \tau_{\epsilon p}\check{\phi}\rangle\check{\psi}(\epsilon p) = \lim_{\epsilon\to 0^+} \sum_p T \star \phi(\epsilon p)\check{\psi}(\epsilon p)$$

$$= \int_{\mathbb{R}^N} (T \star \phi)(y)\psi(-y)\,dy = ((T \star \phi) \star \psi)(0),$$

which proves (5.33).

(iv) If $T \star \phi = 0$ for each $\phi \in \mathcal{D}(\Omega)$, then since $\check{\phi} \in \mathcal{D}(\Omega)$, we have

$$0 = (T \star \check{\phi})(0) = \langle T, \phi\rangle,$$

hence $T = 0$.

(v) Let $K = \text{supp(T)}$ and $Q = \text{supp}(\phi)$. Since

$$\text{supp}(\tau_x\check{\phi}) = x - Q$$

and

$$(T \star \phi)(x) = \langle T, \tau_x\check{\phi}\rangle,$$

this function vanishes unless $(x - Q) \cap K \neq \emptyset$, i.e., $x \in K + Q$. Therefore,

$$\operatorname{supp}(\mathrm{T} \star \phi) \subset \mathrm{K} + \mathrm{Q} \subset \operatorname{supp}(\mathrm{T}) + \operatorname{supp}(\phi).$$

□

The above definition of convolution can be extended to functions in $\mathcal{E}(\mathbb{R}^N)$, provided that the distribution has compact support. More precisely, we have the following definition.

Definition 5.12 Let $T \in \mathcal{E}'(\mathbb{R}^N)$ and $\psi \in \mathcal{E}(\mathbb{R}^N)$. The convolution of T and ψ is defined as the function $T \star \psi$ given by

$$(T \star \psi)(x) := \langle T, \tau_x \check{\psi} \rangle.$$

For this convolution product, properties analogous to those established in Theorem 5.14 hold.

Theorem 5.15 *Let $T \in \mathcal{E}'(\mathbb{R}^N)$ and $\psi \in \mathcal{E}(\mathbb{R}^N)$. Then:*

(i) For each $x \in \mathbb{R}^N$,

$$\tau_x(T \star \psi) = (\tau_x T) \star \psi = T \star (\tau_x \psi).$$

(ii) For each multi-index α,

$$D^\alpha(T \star \psi) = (D^\alpha T) \star \psi = T \star (D^\alpha \psi).$$

In particular, $T \star \psi \in \mathcal{E}(\mathbb{R}^N)$. Moreover, if $\phi \in \mathcal{D}(\mathbb{R}^N)$, then

$$T \star (\psi \star \phi) = (T \star \psi) \star \phi = (T \star \phi) \star \psi. \tag{5.34}$$

Proof It is enough to prove (5.34); the other proofs are analogous to those of Theorem 5.14. Given $\phi \in \mathcal{D}(\mathbb{R}^N)$, by (v) of Theorem 5.14, we have that $\operatorname{supp}(\mathrm{T} \star \phi)$ is compact, so $T \star \phi \in \mathcal{D}(\mathbb{R}^N)$. As in the proof of the previous theorem, to prove (5.34), it suffices to show that the three functions coincide at the origin. Since $K := \operatorname{supp}(\mathrm{T}) + \operatorname{supp}(\phi)$ is compact, we can find a function $\psi_0 \in \mathcal{D}(\mathbb{R}^n)$ such that $\check{\psi} = \check{\psi}_0$ in a neighborhood of K. Then, since $\operatorname{supp}(\mathrm{T} \star \psi) \subset K$, we have

$$\begin{aligned}\big((T \star \phi) \star \psi\big)(0) &= \int_K (T \star \phi)(y)\psi(-y)\,dy = \int_K (T \star \phi)(y)\psi_0(-y)\,dy \\ &= \big((T \star \phi) \star \psi_0\big)(0) = \big((T \star \psi_0) \star \phi\big)(0).\end{aligned}$$

Now, if $-x \in \operatorname{supp}(\phi)$, then $\tau_x \check{\psi} = \tau_x \check{\psi}_0$ in a neighborhood of $\operatorname{supp}(\mathrm{T})$, and thus

$$T \star \psi = T \star \psi_0 \quad \text{on} \quad -\operatorname{supp}(\phi).$$

Consequently,

$$\big((T \star \psi) \star \phi\big)(0) = \big((T \star \psi_0) \star \phi\big)(0),$$

and therefore,

$$(T \star \psi) \star \phi = (T \star \phi) \star \psi. \tag{5.35}$$

To prove the other equality, let $\varphi \in \mathcal{D}(\mathbb{R}^N)$ be an arbitrary function. Then,

$$\big((T \star \phi) \star \psi\big) \star \varphi = \big((T \star \phi) \star \varphi\big) \star \psi = (T \star \varphi) \star (\phi \star \psi).$$

Now, since $\varphi \in \mathcal{D}(\mathbb{R}^N)$ and $\phi \star \psi \in \mathcal{E}(\mathbb{R}^N)$, by (5.35) we have

$$\big((T \star \phi) \star \psi\big) \star \varphi = \big(T \star (\phi \star \psi)\big) \star \varphi.$$

From here, since φ is arbitrary, applying (iv) of Theorem 5.15, we obtain

$$(T \star \phi) \star \psi = T \star (\phi \star \psi).$$

□

Just as with the convolution of functions we were able to regularize locally integrable functions, we will now see that with the convolution product we have just defined, we can regularize distributions, i.e., approximate them by smooth functions.

Theorem 5.16 *Given a distribution $T \in \mathcal{D}'(\Omega)$, there exists a sequence $\{\varphi_n\} \subset \mathcal{D}(\Omega)$ such that*

$$T_{\varphi_n} \to T \qquad \text{in } \mathcal{D}'(\Omega).$$

Proof Let $K_1 \subset K_2 \subset \cdots \subset K_m \subset \cdots$ be a sequence of compact sets such that

$$\Omega = \bigcup_{m \in \mathbb{N}} K_m.$$

Let $\psi_m \in \mathcal{D}(\Omega)$ be such that $\psi_m = 1$ on K_m. Then we have $\psi_m T \to T$ in $\mathcal{D}'(\Omega)$. Consequently, we may assume that T has compact support. Let $\{\phi_n\}$ be a regularizing sequence. Since

$$\operatorname{supp}(T \star \phi_n) \subset \operatorname{supp}(\mathrm{T}) + \operatorname{supp}(\phi_{\mathrm{n}}),$$

for n sufficiently large, we have that $\varphi_n := T \star \phi_n \in \mathcal{D}(\Omega)$. Let us show that

$$T_{\varphi_n} \to T \qquad \text{in } \mathcal{D}'(\Omega). \tag{5.36}$$

For $\psi \in \mathcal{D}(\Omega)$, we have

$$\langle T \star \phi_n, \psi \rangle = \big((T \star \phi_n) \star \check{\psi}\big)(0) = \big(T \star (\phi_n \star \check{\psi})\big)(0)$$

$$= \langle T, (\phi_n \star \check{\psi})^{\vee} \rangle \to \langle T, \psi \rangle,$$

since

$$\phi_n \star \check{\psi} \to \check{\psi} \qquad \text{in } \mathcal{D}(\Omega),$$

which proves (5.36). □

We will now define the convolution product of two distributions, when at least one of them has compact support. The basis for this definition is the following proposition.

Proposition 5.7 *Let $L : \mathcal{D}(\mathbb{R}^N) \to \mathcal{E}(\mathbb{R}^N)$ be a linear and continuous mapping that commutes with translations (i.e., such that $\tau_x L = L\tau_x$ for all $x \in \mathbb{R}^N$). Then there exists a unique distribution $T \in \mathcal{D}'(\mathbb{R}^N)$ such that*

$$L(\phi) = T \star \phi \qquad \forall\, \phi \in \mathcal{D}(\mathbb{R}^N).$$

Proof We define $T : \mathcal{D}(\mathbb{R}^N) \to \mathbb{C}$ by

$$\langle T, \phi \rangle := L(\check{\phi})(0).$$

Clearly, $T \in \mathcal{D}'(\mathbb{R}^N)$, and we have

$$L(\phi)(0) = \langle T, \check{\phi} \rangle = (T \star \phi)(0).$$

Then, applying the translation τ_{-x} to both sides, and using that L commutes with translations, we obtain

$$L(\phi)(x) = (\tau_{-x} L(\phi))(0) = L(\tau_{-x}\phi)(0) = (T \star \tau_{-x}\phi)(0) = (T \star \phi)(x).$$

Hence,

$$L(\phi) = T \star \phi \qquad \forall\, \phi \in \mathcal{D}(\mathbb{R}^N).$$

Moreover, if $S \in \mathcal{D}'(\mathbb{R}^N)$ satisfies

$$L(\phi) = S \star \phi \qquad \forall\, \phi \in \mathcal{D}(\mathbb{R}^N),$$

then

$$(T - S) \star \phi = 0 \qquad \forall\, \phi \in \mathcal{D}(\mathbb{R}^N),$$

which, by (iv) of Theorem 5.14, implies that $T = S$. □

Let $T \in \mathcal{D}'(\mathbb{R}^N)$ and $S \in \mathcal{E}'(\mathbb{R}^N)$. Then, as a consequence of Theorems 5.14 and 5.15, the mapping $L : \mathcal{D}(\mathbb{R}^N) \to \mathcal{E}(\mathbb{R}^N)$ defined by

$$L(\phi) := T \star (S \star \phi)$$

commutes with translations. Therefore, as a consequence of Proposition 5.7, there exists a unique distribution $R \in \mathcal{D}'(\mathbb{R}^N)$ such that

$$R \star \phi = L(\phi) = T \star (S \star \phi) \qquad \forall\, \phi \in \mathcal{D}(\mathbb{R}^N).$$

An analogous condition is obtained if we consider the mapping

$$\phi \mapsto S \star (T \star \phi).$$

Therefore, the following definition makes sense.

Definition 5.13 Let T and S be two distributions on $\mathbb{R}^N$, one of which has compact support. The *convolution product* $T \star S$ of T and S is defined as the unique distribution satisfying the following equivalent conditions:

$$(T \star S) \star \phi = T \star (S \star \phi) \qquad \forall\, \phi \in \mathcal{D}(\mathbb{R}^N).$$

$$\langle T \star S, \phi \rangle = (T \star (S \star \check{\phi}))(0).$$

The following result summarizes the main properties of the convolution product of distributions.

Theorem 5.17 *Let T, S, R be three distributions on $\mathbb{R}^N$, and assume that at least two of them have compact support. Then the following hold:*

(i) $T \star S = S \star T$.
(ii) $\mathrm{supp}(\mathrm{T} \star \mathrm{S}) \subset \mathrm{supp}(\mathrm{T}) + \mathrm{supp}(\mathrm{S})$.
(iii) $D^\alpha(T \star S) = D^\alpha T \star S = T \star D^\alpha S$.
(iv) $(T \star S) \star R = T \star (S \star R)$.
(v) For every $T \in \mathcal{D}'(\mathbb{R}^N)$,

$$T = \delta \star T = T \star \delta.$$

Moreover, for every multi-index $\alpha \in \mathbb{N}^N$,

$$D^\alpha T = (D^\alpha \delta) \star T.$$

Proof

(i) Let $\phi_1, \phi_2 \in \mathcal{D}(\mathbb{R}^N)$. Then,

$$(T \star S) \star (\phi_1 \star \phi_2) = T \star (S \star (\phi_1 \star \phi_2)) = T \star ((S \star \phi_1) \star \phi_2) = T \star (\phi_2 \star (S \star \phi_1)).$$

Thus, if $T \in \mathcal{E}'(\mathbb{R}^N)$, by applying (iii) of Theorem 5.15, and if $S \in \mathcal{E}'(\mathbb{R}^N)$, by applying (iii) of Theorem 5.14 (since in that case $S \star \phi_2 \in \mathcal{D}(\mathbb{R}^N)$), we have

$$(T \star S) \star (\phi_1 \star \phi_2) = (T \star \phi_2) \star (S \star \phi_1) \tag{5.37}$$

Similarly, we obtain

$$(S \star T) \star (\phi_1 \star \phi_2) = (S \star \phi_1) \star (T \star \phi_2) \tag{5.38}$$

Now, since

$$(T \star \phi_2) \star (S \star \phi_1) = (S \star \phi_1) \star (T \star \phi_2), \tag{5.39}$$

being functions, from (5.37), (5.38), and (5.39) it follows that

$$((T \star S) \star \phi_1) \star \phi_2 = ((S \star T) \star \phi_1) \star \phi_2.$$

Hence, since ϕ_2 is arbitrary, by (iv) of Theorem 5.14, it follows that

$$(T \star S) \star \phi_1 = (S \star T) \star \phi_1.$$

Since ϕ_1 is arbitrary, applying the same result again we deduce (i).

(ii) As a consequence of (i), we may assume, without loss of generality, that $S \in \mathcal{E}'(\mathbb{R}^N)$, so that supp(S) is compact. Then we have:

$$\langle T \star S, \phi \rangle = (T \star (S \star \check{\phi}))(0) = \langle T, (S \star \check{\phi})^{\vee} \rangle.$$

Now,

$$\text{supp}(\text{S} \star \check{\phi}) \subset \text{supp}(\text{S}) - \text{supp}(\phi).$$

Therefore, $\langle T \star S, \phi \rangle = 0$ unless

$$\text{supp}(\text{T}) \cap \Big(\text{supp}(\phi) - \text{supp}(\text{S}) \Big) \neq \emptyset.$$

Which proves (ii).

(iii) Let $\alpha \in \mathbb{N}^N$ and $\phi \in \mathcal{D}(\mathbb{R}^N)$. Then,

$$D^\alpha(T\star S)\star\phi = (T\star S)\star D^\alpha\phi = T\star(S\star D^\alpha\phi) = T\star(D^\alpha S\star\phi) = (T\star D^\alpha S)\star\phi.$$

And since this holds for every $\phi \in \mathcal{D}(\mathbb{R}^N)$, we have proved (iii).

(iv) By (ii), if two distributions have compact support, their convolution product also has compact support. Therefore, since at least two of the distributions T, S, R have compact support, we can define both $(T \star S) \star R$ and $T \star (S \star R)$. Suppose that $R \in \mathcal{E}'(\mathbb{R}^N)$. Then, given $\phi \in \mathcal{D}(\mathbb{R}^N)$, since $R \star \phi \in \mathcal{D}(\mathbb{R}^N)$, we have

$$(T \star (S \star R)) \star \phi = T \star ((S \star R) \star \phi) = T \star (S \star (R \star \phi))$$

and

$$((T \star S) \star R) \star \phi = (T \star S) \star (R \star \phi) = T \star (S \star (R \star \phi)),$$

so, since this holds for every $\phi \in \mathcal{D}(\mathbb{R}^N)$, we have

$$T \star (S \star R) = (T \star S) \star R.$$

In the case where $R \notin \mathcal{E}'(\mathbb{R}^N)$, by hypothesis we have $T, S \in \mathcal{E}'(\mathbb{R}^N)$. Therefore, using the above, we obtain

$$T\star(S\star R) = T\star(R\star S) = (R\star S)\star T = R\star(S\star T) = (S\star T)\star R = (T\star S)\star R.$$

(v) Given $\phi \in \mathcal{D}(\mathbb{R}^N)$, we have

$$(\delta \star \phi)(x) = \langle \delta, \tau_x \check{\phi} \rangle = (\tau_x \check{\phi})(0) = \check{\phi}(-x) = \phi(x).$$

Consequently, $\delta \star \phi = \phi$. Therefore, for $T \in \mathcal{D}'(\mathbb{R}^N)$, we have

$$(\delta \star T) \star \phi = (T \star \delta) \star \phi = T \star (\delta \star \phi) = T \star \phi,$$

so that $T = \delta \star T = T \star \delta$. Moreover,

$$D^\alpha T = D^\alpha(T \star \delta) = D^\alpha T \star \delta = T \star D^\alpha \delta = D^\alpha \delta \star T.$$

□

Example 5.5 As we will see in this example, in order to define the convolution product of three distributions, it is not necessary for two of them to have compact support. However, this condition is necessary for associativity to hold. Indeed: Let us consider the distributions: $T = T_{\mathbf{1}}$ (i.e., the distribution generated by the locally

integrable function $\mathbf{1}(x) = 1$ for all $x \in \mathbb{R}$), $S = \delta'$, and $R = T_H$ (i.e., the distribution generated by the Heaviside function). Then,

$$\delta' \star H = \delta \star H' = \delta \star \delta = \delta$$

and

$$\mathbf{1} \star \delta' = \mathbf{1}' \star \delta = 0.$$

Therefore,

$$\mathbf{1} \star (\delta' \star H) = \mathbf{1} \star \delta = \mathbf{1}$$

$$(\mathbf{1} \star \delta') \star H = 0 \star H = 0.$$

5.8 The Fourier Transform

In this section we will study the Fourier transform, which, together with convolution, is one of the fundamental tools in the study of partial differential equations. We will begin by studying the Fourier transform in $L^1(\mathbb{R}^N)$, and then extend it to a class of distributions, the so-called *tempered distributions*.

Definition 5.14 For $f \in L^1(\mathbb{R}^N)$, its *Fourier transform*, denoted by $\hat{f}$ or by $\mathcal{F}f$, is defined by the formula

$$\hat{f}(y) := \int_{\mathbb{R}^N} e^{-i<x,y>} f(x)\, dx \qquad y \in \mathbb{R}^N,$$

where $< x, y >:= \sum_{i=1}^N x_i y_i$.

Since $f \in L^1(\mathbb{R}^N)$, it follows immediately that

$$|\hat{f}(y)| \leq \|f\|_1 \qquad \forall\, y \in \mathbb{R}^N. \tag{5.40}$$

A straightforward application of the Dominated Convergence Theorem (Theorem B.1) gives the continuity of $\hat{f}$. Furthermore, we have:

Lemma 5.4 *For each $f \in L^1(\mathbb{R}^N)$, the Fourier transform $\hat{f}$ is uniformly continuous on $\mathbb{R}^N$.*

Proof Given $\epsilon > 0$, since $f \in L^1(\mathbb{R}^N)$, there exists $R > 0$ such that

$$\int_{\|x\| \geq R} |f(x)|\, dx < \frac{\epsilon}{4}.$$

Let $\delta > 0$ be such that

$$R\delta \int_{\|x\| \le R} |f(x)|\, dx < \frac{\epsilon}{2}.$$

Then, if we take $\|h\| \le \delta$, we have

$$|\hat{f}(y+h) - \hat{f}(y)| = \left| \int_{\mathbb{R}^N} \left(e^{-i\langle x, y+h\rangle} - e^{-i\langle x, y\rangle} \right) f(x)\, dx \right|$$

$$\le \int_{\mathbb{R}^N} |f(x)| \left| e^{-i\langle x, y\rangle} (e^{-i\langle x, h\rangle} - 1) \right| dx = \int_{\mathbb{R}^N} |f(x)| \left| e^{-i\langle x, h\rangle} - 1 \right| dx$$

$$= \int_{\mathbb{R}^N} |f(x)| \cdot 2 \left| \sin\left(\frac{\langle x, h\rangle}{2} \right) \right| dx$$

$$\le 2 \int_{\|x\| \ge R} |f(x)|\, dx + 2 \int_{\|x\| \le R} |f(x)| \left| \sin\left(\frac{\langle x, h\rangle}{2} \right) \right| dx$$

$$\le \frac{\epsilon}{2} + \int_{\|x\| \le R} |f(x)| \cdot |\langle x, h\rangle|\, dx \le \frac{\epsilon}{2} + R\delta \int_{\|x\| \le R} |f(x)|\, dx \le \epsilon.$$

□

Lemma 5.5 (Riemann–Lebesgue Lemma) *For every $f \in L^1(\mathbb{R}^N)$, we have*

$$\lim_{\|y\| \to \infty} \hat{f}(y) = 0.$$

Proof Given $\epsilon > 0$ and $f \in L^1(\mathbb{R}^N)$, since $\mathcal{D}(\mathbb{R}^N)$ is dense in $L^1(\mathbb{R}^N)$, there exists $\varphi \in \mathcal{D}(\mathbb{R}^N)$ such that

$$\|f - \varphi\|_1 \le \frac{\epsilon}{2}.$$

Now, given φ, integrating by parts with respect to the variable x_j, we have

$$\hat{\varphi}(y) = \int_{\mathbb{R}^N} e^{-i\langle x, y\rangle}\, \varphi(x)\, dx$$

$$= \int_{\mathbb{R}^{N-1}} \left(\int_{\mathbb{R}} -\frac{1}{iy_j} D_j(e^{-i\langle x, y\rangle}) \varphi(x)\, dx_j \right) d\hat{x}_j = \frac{1}{iy_j} \int_{\mathbb{R}^N} e^{-i\langle x, y\rangle}\, D_j\varphi(x)\, dx.$$

From which it follows that

$$|\hat{\varphi}(y)| \le \frac{1}{|y_j|} \|D_j\varphi\|_1 \to 0 \quad \text{as} \quad \|y\| \to \infty.$$

Then there exists $R > 0$ such that

$$|\hat{\varphi}(y)| \leq \frac{\epsilon}{2} \quad \forall y \text{ such that } \|y\| \geq R.$$

Consequently, when $\|y\| \geq R$, we have

$$|\hat{f}(y)| \leq |\widehat{\varphi - f}(y)| + |\hat{\varphi}(y)| \leq \|f - \varphi\|_1 + |\hat{\varphi}(y)| \leq \epsilon.$$

□

We denote by

$$C_\infty(\mathbb{R}^N) := \{\, f : \mathbb{R}^N \to \mathbb{C} \text{ continuous such that } \lim_{\|x\| \to \infty} f(x) = 0 \,\},$$

and we endow it with the norm

$$\|f\|_\infty := \sup_{x \in \mathbb{R}^N} |f(x)|,$$

so that it has the structure of a Banach space (see Problem (2.3)).

As a consequence of the Riemann-Lebesgue Lemma, (5.40), and Lemma 5.4, we obtain the following result.

Corollary 5.2 *The Fourier transform $\mathcal{F}$ is a linear and continuous mapping from $L^1(\mathbb{R}^N)$ to $C_\infty(\mathbb{R}^N)$, with $\|\mathcal{F}\| \leq 1$.*

The following result summarizes the most useful properties of the Fourier transform.

Theorem 5.18 *Let $f, g \in L^1(\mathbb{R}^N)$. The following properties hold:*

(i) $\displaystyle\int_{\mathbb{R}^n} \hat{f}(x) g(x)\, dx = \int_{\mathbb{R}^n} f(x) \hat{g}(x)\, dx.$

(ii) $\widehat{\tau_a f}(y) = e^{-i<a,y>}\, \hat{f}(y), \quad a \in \mathbb{R}^N, \quad y \in \mathbb{R}^N.$

(iii) $\mathcal{F}(e^{-i<a,x>}\, f) = \tau_{-a} \hat{f}.$

(iv) $\mathcal{F}(f \star g) = \mathcal{F}(f)\mathcal{F}(g).$

(v) If $x^\alpha f \in L^1(\mathbb{R}^N)$, $|\alpha| \leq k$, then $\hat{f} \in C^k$ and

$$D^\alpha(\hat{f}) = (-i)^{|\alpha|} \widehat{x^\alpha f}.$$

(vi) If $f \in C^k$ with $D^\beta f \in L^1(\mathbb{R}^N)$ for every β with $|\beta| \leq k$, then $y^\beta \hat{f} \in L^\infty(\mathbb{R}^N)$ and

$$i^{|\beta|}\, y^\beta \hat{f} = \mathcal{F}(D^\beta f).$$

Proof (i), (ii), and (iii) follow by direct computation.

(iv) It suffices to apply Fubini's Theorem (Theorem B.4), since

$$\mathcal{F}(f \star g)(y) = \int_{\mathbb{R}^N} e^{-i<x,y>}\ (f \star g)(x)\ dx$$

$$= \int_{\mathbb{R}^N} e^{-i<x,y>} \left(\int_{\mathbb{R}^N} f(x-z) g(z)\ dz \right) dx$$

$$= \int_{\mathbb{R}^N} g(z) \left(\int_{\mathbb{R}^N} e^{-i<x,y>}\ f(x-z)\ dx \right) dz$$

$$= \int_{\mathbb{R}^N} g(z) \left(\int_{\mathbb{R}^N} e^{-i<z+\eta,y>}\ f(\eta)\ d\eta \right) dz$$

$$= \left(\int_{\mathbb{R}^N} e^{-i<z,y>}\ g(z)\ dz \right) \left(\int_{\mathbb{R}^N} e^{-i<\eta,y>}\ f(\eta)\ d\eta \right) = \hat{f}(y)\hat{g}(y).$$

(v) It is enough to prove it for $\alpha = (0, \ldots, 0, 1, 0, \ldots, 0)$, so that $x^\alpha = x_j$. Assume that $x_j f \in L^1(\mathbb{R}^N)$. Then,

$$\frac{\hat{f}(y + t\mathbf{e}_j) - \hat{f}(y)}{t} = \frac{1}{t} \int_{\mathbb{R}^N} \left(e^{-i<x,y+t\mathbf{e}_j>} - e^{-i<x,y>} \right) f(x)\ dx$$

$$= \frac{1}{t} \int_{\mathbb{R}^N} e^{-i<x,y>} \left(e^{-i<x,t\mathbf{e}_j>} - 1 \right) f(x)\ dx$$

$$= \int_{\mathbb{R}^N} x_j f(x) e^{-i<x,y>} \left(\frac{e^{-itx_j} - 1}{tx_j} \right) dx.$$

Now, by the Dominated Convergence Theorem, as $t \to 0$, the last integral tends to

$$-i \int_{\mathbb{R}^N} x_j f(x) e^{-i<x,y>}\ dx.$$

Therefore, we obtain

$$D_j \hat{f}(y) = -i\widehat{x_j f}(y).$$

(vi) As in (v), we may assume that $\beta = (0, \ldots, 0, 1, 0, \ldots, 0)$. Then, applying Fubini's Theorem and integrating by parts, we get

$$\mathcal{F}(D^\beta f)(y) = \mathcal{F}(D_j f)(y) = \int_{\mathbb{R}^N} D_j f(x) e^{-i<x,y>}\ dx$$

$$= iy_j \int_{\mathbb{R}^{N-1}} e^{-i(x_1y_1+\cdots+x_{j-1}y_{j-1}+x_{j+1}y_{j+1}+\cdots+x_ny_n)} \left(\int_{\mathbb{R}} f(x) e^{-ix_jy_j}\, dx_j \right) d\hat{x}_j$$

$$= iy_j \int_{\mathbb{R}^N} e^{-i<x,y>}\ f(x)\, dx = iy_j \hat{f}(y).$$

□

In the following example, we will compute the Fourier transform of the Gaussian function.

Example 5.6 Let us consider the function $f(x) = e^{-\|x\|^2}$, where $x \in \mathbb{R}^N$. It is well known that $f \in L^1(\mathbb{R}^N)$ and that

$$\|f\|_1 = \int_{\mathbb{R}^N} e^{-\|x\|^2}\, dx = \pi^{\frac{N}{2}}. \tag{5.41}$$

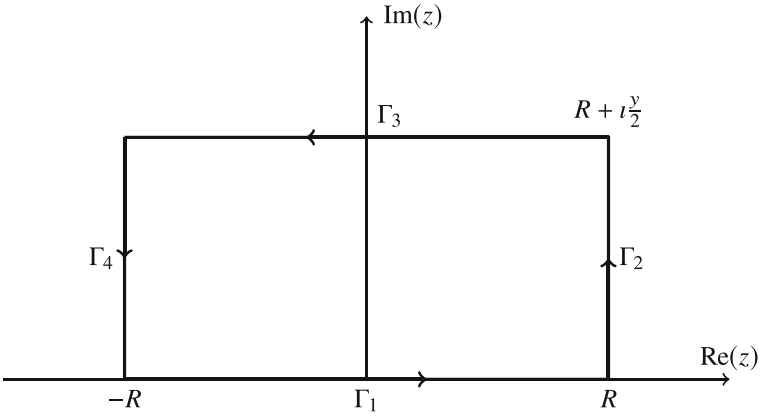
We will first assume that $N = 1$. Then,

$$\hat{f}(y) = \int_{\mathbb{R}} e^{-i\langle x,y\rangle} f(x)\, dx = e^{-\frac{y^2}{4}} \int_{\mathbb{R}} e^{-(x+\frac{1}{2}iy)^2}\, dx.$$

We will evaluate this integral using Cauchy's Theorem. To this end, we consider the path $\Gamma = \bigcup_{j=1}^4 \Gamma_j$, where Γ_j are the segments:

$$\Gamma_1 = [-R, R],\ \ \Gamma_2 = [R, R+i\tfrac{y}{2}],\ \ \Gamma_3 = [R+i\tfrac{y}{2}, -R+i\tfrac{y}{2}],\ \ \Gamma_4 = [-R+i\tfrac{y}{2}, -R],$$

as shown in the figure:



Now,

$$\left|\int_{\Gamma_2} e^{-z^2}\,dz\right| = \left|\int_0^{\frac{y}{2}} e^{-(R+i\xi)^2}\,d\xi\right| = \left|\int_0^{\frac{y}{2}} e^{-R^2}e^{-iR\xi}e^{\xi^2}\,d\xi\right|$$

$$\leq e^{-R^2}\int_0^{\frac{y}{2}} e^{\xi^2}\,d\xi \leq Ce^{-R^2},$$

so that

$$\int_{\Gamma_2} e^{-z^2}\,dz \to 0 \quad \text{as} \quad R \to +\infty. \tag{5.42}$$

Similarly, we have

$$\int_{\Gamma_4} e^{-z^2}\,dz \to 0 \quad \text{as} \quad R \to +\infty. \tag{5.43}$$

Moreover,

$$\int_{\Gamma_1} e^{-z^2}\,dz = \int_{-R}^{R} e^{-x^2}\,dx \to \int_{\mathbb{R}} e^{-x^2}\,dx = \sqrt{\pi}, \quad \text{as} \quad R \to +\infty. \tag{5.44}$$

$$\int_{\Gamma_3} e^{-z^2}\,dz = -\int_{-R}^{R} e^{-(x+i\frac{y}{2})^2}\,dx. \tag{5.45}$$

Letting $R \to +\infty$, as a consequence of (5.42)–(5.45), we obtain

$$\hat{f}(y) = e^{-\frac{y^2}{4}}\int_{-\infty}^{+\infty} e^{-(x+i\frac{y}{2})^2}\,dx = \sqrt{\pi}e^{-\frac{y^2}{4}}.$$

In the general case, applying the above and Fubini's Theorem, we obtain

$$\hat{f}(y) = \int_{\mathbb{R}^N} e^{-i\langle x,y\rangle}\,e^{-\|x\|^2}\,dx = \prod_{j=1}^{N}\int_{\mathbb{R}} e^{-(ix_jy_j+x_j^2)}\,dx$$

$$= \pi^{\frac{N}{2}}\prod_{j=1}^{n} e^{-\frac{y_j^2}{4}} = \pi^{\frac{N}{2}}e^{-\frac{\|y\|^2}{4}}.$$

Similarly, we obtain that if $f(x) = e^{-a\|x\|^2}$, with $a > 0$, $x \in \mathbb{R}^N$, then

$$\hat{f}(y) = \left(\frac{\pi}{a}\right)^{\frac{N}{2}} e^{-\frac{\|y\|^2}{4a}}.$$

Properties (v) and (vi) of Theorem 5.18 suggest the introduction of the following subspace of $L^1(\mathbb{R}^N)$ where (v) and (vi) hold for every $k \in \mathbb{N}$.

Definition 5.15 The space of *rapidly decreasing functions*, also called the *Schwartz space*, is defined as

$$\mathcal{S}(\mathbb{R}^N) := \left\{ f \in C^\infty(\mathbb{R}^N) \ : \lim_{\|x\| \to +\infty} x^\alpha D^\beta f(x) \to 0, \ \forall\, \alpha, \ \beta \in \mathbb{N}^N \right\}.$$

It is easy to check that if $f \in C^\infty(\mathbb{R}^N)$, then:

(1) $f \in \mathcal{S}(\mathbb{R}^N)$ if and only if, for every polynomial P and every multi-index α, the function $P(x)D^\alpha f(x)$ is bounded in $\mathbb{R}^N$.
(2) $f \in \mathcal{S}(\mathbb{R}^N)$ if and only if, for every integer k and every multi-index α, the function

$$(1 + \|x\|^2)^k \ D^\alpha f(x)$$

is bounded in $\mathbb{R}^N$.

Example 5.7 Clearly, $\mathcal{D}(\mathbb{R}^N) \subset \mathcal{S}(\mathbb{R}^N)$. However, the inclusion is strict, since for example, the function

$$f(x) := e^{-\|x\|^2} \in \mathcal{S}(\mathbb{R}^N) \setminus \mathcal{D}(\mathbb{R}^N),$$

because

$$D^\alpha f(x) = Q(x) e^{-\|x\|^2},$$

with Q a polynomial.

As we did with the spaces $\mathcal{D}(\mathbb{R}^N)$ and $\mathcal{E}(\mathbb{R}^N)$, we will describe the topology of the space $\mathcal{S}(\mathbb{R}^N)$ by means of its convergent sequences.

Definition 5.16 We say that a sequence $\{f_n\}$ converges to a function f in $\mathcal{S}(\mathbb{R}^N)$, and we write

$$f_n \to f \quad \text{in } \mathcal{S}(\mathbb{R}^N),$$

if for every polynomial P and every multi-index $\alpha \in \mathbb{N}^N$,

$$P D^\alpha f_n \to P D^\alpha f \quad \text{uniformly on } \mathbb{R}^N.$$

Remark 5.3 As was the case with the space $\mathcal{E}(\mathbb{R}^N)$, the topology of $\mathcal{S}(\mathbb{R}^N)$ can be described by a countable family of seminorms. To this end, it suffices to consider, for each pair of multi-indices $\alpha, \beta \in \mathbb{N}^N$, the seminorm:

$$P_{\alpha,\beta}(f) := \sup_{x\in\mathbb{R}^N} |x^\alpha D^\beta f(x)|, \qquad f \in \mathcal{S}(\mathbb{R}^N).$$

Another family of seminorms that define the same topology is given by the sequence $\{S_m\}$ with

$$S_m(f) := \sup_{x\in\mathbb{R}^N} \left\{ (1+\|x\|^2)^m |D^\alpha f(x)| \ : \ |\alpha| \le m \right\}.$$

It then follows that the topology of the space $\mathcal{S}(\mathbb{R}^N)$ is the one defined by the metric:

$$d(f,g) := \sum_{m=0}^{\infty} \frac{1}{2^m} \frac{S_m(f-g)}{1+S_m(f-g)}.$$

It can be shown without much difficulty (see, for instance, [4]) that with respect to this metric, $\mathcal{S}(\mathbb{R}^N)$ is a Fréchet space (i.e., a complete metric space whose topology is compatible with the vector space structure).

Proposition 5.8 *For every $1 \le p < \infty$, the space $\mathcal{S}(\mathbb{R}^N)$ is a dense subspace of $L^p(\mathbb{R}^N)$, and the inclusion is continuous.*

Proof For convenience, we will prove the case $p = 1$. Since $\mathcal{D}(\mathbb{R}^N) \subset \mathcal{S}(\mathbb{R}^N)$ and $\mathcal{D}(\mathbb{R}^N)$ is dense in $L^1(\mathbb{R}^N)$, it suffices to show that $\mathcal{S}(\mathbb{R}^N) \subset L^1(\mathbb{R}^N)$ and that the inclusion is continuous. Indeed, let $f \in \mathcal{S}(\mathbb{R}^N)$. Then, for each $k \in \mathbb{N}$, there exists $M_k > 0$ such that

$$\sup_{x\in\mathbb{R}^N} \left| f(x) \left(1+\|x\|^2\right)^k \right| \le M_k. \tag{5.46}$$

Now, since for $k > \frac{N}{2}$ the function $x \mapsto (1+\|x\|^2)^{-k}$ belongs to $L^1(\mathbb{R}^N)$, using (5.46), we get:

$$\begin{aligned} \|f\|_1 &= \int_{\mathbb{R}^N} |f(x)| \left(1+\|x\|^2\right)^k \left(1+\|x\|^2\right)^{-k} dx \\ &\le M_k \int_{\mathbb{R}^N} \left(1+\|x\|^2\right)^{-k} dx < +\infty. \end{aligned}$$

Therefore, $f \in L^1(\mathbb{R}^N)$, and if

$$C := \int_{\mathbb{R}^N} \left(1 + \|x\|^2\right)^{-k} \, dx,$$

then $\|f\|_1 \leq C S_k(f)$, from which the continuity of the inclusion follows. □

The usefulness of the Schwartz space lies in its good behavior with respect to the Fourier transform. Specifically, we have the following result.

Theorem 5.19 (Fourier Inversion Theorem) *The Fourier transform $\mathcal{F}$ is a topological isomorphism from $\mathcal{S}(\mathbb{R}^N)$ onto $\mathcal{S}(\mathbb{R}^N)$, and its inverse is given by the inversion formula:*

$$f(x) = \mathcal{F}^{-1}(\hat{f})(x) = \frac{1}{(2\pi)^N} \int_{\mathbb{R}^N} e^{i<x,y>} \, \hat{f}(y) \, dy.$$

Proof Let $f \in \mathcal{S}(\mathbb{R}^N)$. Then, for each $\alpha \in \mathbb{N}^N$, we have $x^\alpha f \in L^1(\mathbb{R}^N)$, and hence, by part (v) of Theorem 5.18, we have $\hat{f} \in C^\infty(\mathbb{R}^N)$ and

$$\mathcal{F}(D^\alpha(x^\beta f))(y) = i^{|\alpha|} y^\alpha \widehat{x^\beta f}(y) = i^{|\alpha|} y^\alpha \frac{1}{(-i)^{|\beta|}} D^\beta(\hat{f})(y).$$

Thus, since $D^\alpha(x^\beta f) \in L^1(\mathbb{R}^N)$, it follows that $y^\alpha D^\beta(\hat{f})$ is bounded, and therefore $\hat{f} \in \mathcal{S}(\mathbb{R}^N)$. Moreover, for each $y \in \mathbb{R}^N$,

$$\left|y^\alpha D^\beta(\hat{f})(y)\right| \leq C_1 \int_{\mathbb{R}^N} \left|D^\alpha\left(x^\beta f(x)\right)\right| \, dx$$

$$= \int_{\mathbb{R}^N} \left(|D^\alpha(x^\beta f(x))|(1 + \|x\|^2)^k\right) \left(1 + \|x\|^2\right)^{-k} \, dx.$$

Then if $k > \frac{N}{2}$, we have:

$$\sup_{y \in \mathbb{R}^N} \left|y^\alpha D^\beta(\hat{f})(y)\right| \leq C_2 \sup_{x \in \mathbb{R}^N} \left[\left|D^\alpha(x^\beta f(x))\right| \left(1 + \|x\|^2\right)^k\right].$$

This last inequality implies the continuity of the Fourier transform from $\mathcal{S}(\mathbb{R}^N)$ into itself.

Let us now show that the inversion formula holds for $f \in \mathcal{S}(\mathbb{R}^N)$. Let $\phi \in \mathcal{S}(\mathbb{R}^N)$. For $\lambda > 0$, define $g(x) := \phi\left(\frac{x}{\lambda}\right)$. Then $g \in \mathcal{S}(\mathbb{R}^N)$ and $\hat{g}(y) = \lambda^N \hat{\phi}(\lambda y)$. Applying part (i) of Theorem 5.18, we obtain:

$$\int_{\mathbb{R}^N} f\left(\frac{y}{\lambda}\right) \hat{\phi}(y) \, dy = \int_{\mathbb{R}^N} f(y) \lambda^N \hat{\phi}(\lambda y) \, dy = \int_{\mathbb{R}^N} f(y) \hat{g}(y) \, dy$$

$$= \int_{\mathbb{R}^N} \hat{f}(y) g(y)\, dy = \int_{\mathbb{R}^N} \hat{f}(y) \phi\left(\frac{y}{\lambda}\right) dy.$$

Letting $\lambda \to +\infty$ and applying the Dominated Convergence Theorem (Theorem B.1), we obtain:

$$f(0) \int_{\mathbb{R}^N} \hat{\phi}(y)\, dy = \phi(0) \int_{\mathbb{R}^N} \hat{f}(y)\, dy. \tag{5.47}$$

If $\phi(x) = e^{-\|x\|^2}$, then from Example 5.6, we know that

$$\hat{\phi}(y) = \pi^{\frac{N}{2}} e^{-\frac{\|y\|^2}{4}}.$$

Hence, from (5.47), we obtain:

$$f(0) = \frac{1}{(2\pi)^N} \int_{\mathbb{R}^N} \hat{f}(y)\, dy,$$

so the inversion formula holds at $x = 0$. Applying it to the function $g := \tau_{-x} f$, we get:

$$f(x) = \tau_{-x} f(0) = g(0) = \frac{1}{(2\pi)^N} \int_{\mathbb{R}^N} \hat{g}(y)\, dy = \frac{1}{(2\pi)^N} \int_{\mathbb{R}^N} \widehat{\tau_{-x} f}(y)\, dy$$

$$= \frac{1}{(2\pi)^N} \int_{\mathbb{R}^N} e^{i<x,y>}\, \hat{f}(y)\, dy.$$

Thus we have established the inversion formula, which implies that $\mathcal{F}$ is injective on $\mathcal{S}(\mathbb{R}^N)$. Moreover, it is also surjective. Indeed, for $f \in \mathcal{S}(\mathbb{R}^N)$, we have:

$$\check{f}(x) = f(-x) = \frac{1}{(2\pi)^N} \int_{\mathbb{R}^N} e^{-i<x,y>}\, \hat{f}(y)\, dy = \frac{1}{(2\pi)^N} \mathcal{F}^2(f)(x). \tag{5.48}$$

Therefore,

$$\mathcal{F}^4(f)(x) = \mathcal{F}^2\left(\mathcal{F}^2(f)\right)(x) = (2\pi)^N \left(\mathcal{F}^2(f)\right)^{\vee}(x)$$

$$= (2\pi)^N \left((2\pi)^N \check{f}\right)^{\vee}(x) = (2\pi)^{2N} f(x).$$

Hence,

$$f(x) = \frac{1}{(2\pi)^{2N}} \mathcal{F}^4(f)(x) = \mathcal{F}\left(\frac{1}{(2\pi)^{2N}} \mathcal{F}^3(f)\right)(x).$$

Consequently, $\mathcal{F}\colon \mathcal{S}(\mathbb{R}^N) \to \mathcal{S}(\mathbb{R}^N)$ is surjective, and moreover,

$$\mathcal{F}^{-1} = \frac{1}{(2\pi)^{2N}}\mathcal{F}^3, \tag{5.49}$$

which shows that $\mathcal{F}^{-1} : \mathcal{S}(\mathbb{R}^N) \to \mathcal{S}(\mathbb{R}^N)$ is continuous. □

Corollary 5.3 *If $f, g \in \mathcal{S}(\mathbb{R}^N)$, then $f \star g \in \mathcal{S}(\mathbb{R}^N)$ and moreover*

$$\mathcal{F}(fg) = \frac{1}{(2\pi)^N}\hat{f} \star \hat{g}.$$

Proof Let $f, g \in \mathcal{S}(\mathbb{R}^N)$. Since $\hat{f}, \hat{g} \in L^1(\mathbb{R}^N)$, applying item (iv) of Theorem 5.18, we get

$$\mathcal{F}(\hat{f} \star \hat{g}) = \mathcal{F}^2(f)\mathcal{F}^2(g).$$

Now, by (5.48), $\mathcal{F}^2(f) = (2\pi)^N \check{f}$. Therefore, since $fg \in \mathcal{S}(\mathbb{R}^N)$, we obtain

$$\mathcal{F}(\hat{f} \star \hat{g}) = (2\pi)^{2N}\check{f}\check{g} = (2\pi)^N\mathcal{F}^2(fg).$$

Applying $\mathcal{F}^{-1}$, we get

$$\hat{f} \star \hat{g} = (2\pi)^N\mathcal{F}(fg).$$

Consequently, $\hat{f} \star \hat{g} \in \mathcal{S}(\mathbb{R}^N)$, and since $\mathcal{F} : \mathcal{S}(\mathbb{R}^N) \to \mathcal{S}(\mathbb{R}^N)$ is surjective, we conclude that $f \star g \in \mathcal{S}(\mathbb{R}^N)$. □

As a consequence of the properties of the Fourier transform on the Schwartz space, we will see that it is possible to define the Fourier transform for functions in $L^2(\mathbb{R}^N)$. More precisely, we have the following result.

Theorem 5.20 (Plancherel's Theorem) *There exists a unique linear bijection*

$$\mathcal{P} : L^2(\mathbb{R}^N) \to L^2(\mathbb{R}^N)$$

such that $\mathcal{P}(f) = \hat{f}$ for every $f \in \mathcal{S}(\mathbb{R}^N)$, and in addition, it satisfies Parseval's identity

$$\|f\|_2 = (2\pi)^{-\frac{N}{2}}\|\hat{f}\|_2, \quad \forall f \in L^2(\mathbb{R}^n).$$

Proof By the inversion formula, we have that $\mathcal{F}^{-1} = (2\pi)^{-N}\overline{\mathcal{F}}$, where

$$\overline{\mathcal{F}}(\phi)(x) := \int_{\mathbb{R}^N} e^{i\langle x,y\rangle}\,\phi(y)\,dy.$$

Given $f, g \in \mathcal{S}(\mathbb{R}^N)$, since $\overline{g} \in \mathcal{S}(\mathbb{R}^N)$, there exists $h \in \mathcal{S}(\mathbb{R}^N)$ such that $\hat{h} = \overline{g}$, and thus, applying item (i) of Theorem 5.18, we obtain

$$\int_{\mathbb{R}^N} f\overline{g} = \int_{\mathbb{R}^N} f\hat{h} = \int_{\mathbb{R}^N} \hat{f}h.$$

Now,

$$h(x) = \mathcal{F}^{-1}(\overline{g})(x) = (2\pi)^{-N}\overline{\mathcal{F}(\overline{g})}(x) = (2\pi)^{-N}\int_{\mathbb{R}^N} e^{i\langle x,y\rangle}\overline{g}(y)\,dy$$

$$= (2\pi)^{-N}\overline{\int_{\mathbb{R}^N} e^{-i\langle x,y\rangle}g(y)\,dy} = (2\pi)^{-N}\overline{\hat{g}(y)}.$$

Therefore,

$$\int_{\mathbb{R}^N} f\overline{g} = (2\pi)^{-N}\int_{\mathbb{R}^N} \hat{f}\overline{\hat{g}}.$$

Taking $f = g$ in the previous equality, we obtain

$$\|f\|_2^2 = (2\pi)^{-\frac{N}{2}}\|\hat{f}\|_2^2 \qquad \forall\, f \in \mathcal{S}(\mathbb{R}^N). \tag{5.50}$$

Since $\mathcal{S}(\mathbb{R}^N)$ is dense in $L^2(\mathbb{R}^N)$, from (5.50) we conclude that $\mathcal{F}$ extends uniquely to a linear operator $\mathcal{P} : L^2(\mathbb{R}^N) \to L^2(\mathbb{R}^N)$.

Let $f \in L^2(\mathbb{R}^N)$, and let $f_m \in \mathcal{S}(\mathbb{R}^N)$ such that $f_m \to f$ in $L^2(\mathbb{R}^N)$. Then,

$$\|\mathcal{P}(f)\|_2 = \lim_{m\to\infty}\|\mathcal{P}(f_m)\|_2 = \lim_{m\to\infty}\|\hat{f}_m\|_2$$

$$= \lim_{m\to\infty}(2\pi)^{-\frac{N}{2}}\|f_m\|_2 = (2\pi)^{-\frac{N}{2}}\|f\|_2.$$

Therefore, Parseval's identity holds, from which we deduce that $\mathcal{P}$ has closed range. Then, since $\mathcal{S}(\mathbb{R}^N)$ is contained in the range and dense in $L^2(\mathbb{R}^N)$, we have that $\mathcal{P}$ is surjective. □

The operator $\mathcal{P}$ from the previous theorem is called the *Fourier-Plancherel transform*.

Theorem 5.21 (The Hausdorff–Young Inequality) *If $1 \le p \le 2$, there exists a bounded linear operator $\mathcal{T}_p : L^p(\mathbb{R}^N) \to L^{p'}(\mathbb{R}^N)$ such that $\mathcal{T}_p(f) = \mathcal{F}(f)$ for every $f \in \mathcal{S}(\mathbb{R}^N)$, and*

$$\|\mathcal{T}_p(f)\|_{p'} \le (2\pi)^{\frac{N}{p'}}\|f\|_p, \quad \forall\, f \in L^p(\mathbb{R}^N).$$

Moreover, $\mathcal{T}_1 = \mathcal{F}$ and $\mathcal{T}_2 = \mathcal{P}$.

Proof We know that $\mathcal{F} : L^1(\mathbb{R}^N) \to L^\infty(\mathbb{R}^N)$ is a bounded operator with $\|\mathcal{F}\|_{1,\infty} \leq 1$. By Plancherel's Theorem, we have that $\mathcal{P} : L^2(\mathbb{R}^N) \to L^2(\mathbb{R}^N)$ is a bounded linear operator with $\|\mathcal{P}\|_{2,2} \leq (2\pi)^{\frac{N}{2}}$. Moreover, $\mathcal{F}(f) = \mathcal{P}(f)$ if $f \in \mathcal{S}(\mathbb{R}^N)$. Then, applying the Riesz–Thorin Interpolation Theorem (see Remark 2.2), we conclude that there exists a bounded linear operator $\mathcal{T}_p : L^p(\mathbb{R}^N) \to L^{p'}(\mathbb{R}^N)$, provided that

$$\frac{1}{p} = \frac{1-\theta}{1} + \frac{\theta}{2}, \quad \frac{1}{q} = \frac{1-\theta}{\infty} + \frac{\theta}{2}, \quad 0 < \theta < 1,$$

with operator norm $\|\mathcal{T}_p\|_{p,q} \leq (2\pi)^{\frac{N\theta}{2}}$, and such that $\mathcal{T}_p(f) = \mathcal{F}(f)$ for every $f \in \mathcal{S}(\mathbb{R}^N)$. Eliminating θ gives $q = p'$ and $\theta = \frac{2}{p'}$, which completes the proof. □

To define the Fourier transform of certain distributions, we first need to establish the following result.

Theorem 5.22 *The space $\mathcal{D}(\mathbb{R}^N)$ is dense in $\mathcal{S}(\mathbb{R}^N)$, and the inclusion $\mathcal{D}(\mathbb{R}^N) \subset \mathcal{S}(\mathbb{R}^N)$ is continuous.*

Proof We prove the density by multiplying with a sequence of cut-off functions. Let $\varphi \in \mathcal{D}(\mathbb{R}^N)$, $0 \leq \varphi \leq 1$, $\text{supp}(\varphi) \subset \text{B}(0, 2)$, and such that $\varphi(x) = 1$ for every $x \in \overline{B}(0, 1)$. Given $f \in \mathcal{S}(\mathbb{R}^N)$, we consider the functions $f_m(x) := f(x)\varphi\left(\frac{x}{m}\right)$. Then $f_m \in \mathcal{D}(\mathbb{R}^N)$, $f_m(x) = f(x)$ if $\|x\| \leq m$, and $\text{supp}(\text{f}_\text{m}) \subset B(0, 2m)$. Let us show that $f_m \to f$ in $\mathcal{S}(\mathbb{R}^N)$. For this, we need to prove that for each $\alpha, \beta \in \mathbb{N}^N$,

$$x^\alpha D^\beta f_m(x) \to x^\alpha D^\beta f(x) \quad \text{uniformly on } \mathbb{R}^N. \tag{5.51}$$

We prove (5.51) by induction on $|\beta|$. For $|\beta| = 0$, we have

$$\begin{aligned}
\sup_{x\in\mathbb{R}^N} \left|x^\alpha D^\beta f_m(x) - x^\alpha D^\beta f(x)\right| &= \sup_{x\in\mathbb{R}^N} \left|x^\alpha f(x)\left(\varphi\left(\frac{x}{m}\right) - 1\right)\right| \\
&= \sup_{\|x\|>m} \left|x^\alpha f(x)\left(\varphi\left(\frac{x}{m}\right) - 1\right)\right| \\
&\leq \sup_{\|x\|>m} |x^\alpha f(x)| \to 0, \quad \text{as } m \to \infty,
\end{aligned}$$

since $f \in \mathcal{S}(\mathbb{R}^N)$.

Assume it holds for $|\beta| = k$, and we prove it for $|\beta| = k+1$. Given β with $|\beta| = k+1$, there exists $1 \le i \le n$ such that $\beta_i \neq 0$. Then $\beta = \tilde{\beta} + \mathbf{e}_i$, and so,

$$x^\alpha D^\beta f_m(x) - x^\alpha D^\beta f(x) = x^\alpha D^{\tilde{\beta}} D_i \left[f(x) \left(\varphi\left(\frac{x}{m}\right) - 1 \right) \right]$$
$$= x^\alpha D^{\tilde{\beta}} \left[D_i f(x) \left(\varphi\left(\frac{x}{m}\right) - 1 \right) + f(x) \frac{1}{m} D_i \varphi\left(\frac{x}{m}\right) \right].$$

Consequently,

$$\sup_{x\in\mathbb{R}^N} \left| x^\alpha D^\beta f_m(x) - x^\alpha D^\beta f(x) \right| \le \sup_{x\in\mathbb{R}^N} \left| x^\alpha D^{\tilde{\beta}} \left[D_i f(x) \left(\varphi\left(\frac{x}{m}\right) - 1 \right) \right] \right|$$
$$+ \sup_{x\in\mathbb{R}^N} \left| x^\alpha D^{\tilde{\beta}} \left[f(x) \frac{1}{m} D_i \varphi\left(\frac{x}{m}\right) \right] \right|.$$

Now, since $D_i f \in \mathcal{S}(\mathbb{R}^N)$ and $|\tilde{\beta}| = k$, the first term tends to zero as $m \to \infty$, by the induction hypothesis. As for the second term, $x^\alpha D^{\tilde{\beta}} \left[f(x) D_i \varphi\left(\frac{x}{m}\right) \right]$ is a finite sum of powers of x times derivatives of f and φ. Hence, since $f, \varphi \in \mathcal{S}(\mathbb{R}^N)$,

$$\sup_{x\in\mathbb{R}^N} \left| x^\alpha D^{\tilde{\beta}} \left[f(x) \frac{1}{m} D_i \varphi\left(\frac{x}{m}\right) \right] \right| \le \frac{M}{m} \to 0, \quad \text{as } m \to \infty.$$

Now we prove that the inclusion is continuous. Let $\varphi_m \to 0$ in $\mathcal{D}(\mathbb{R}^N)$. Then there exists a compact set $K \subset \mathbb{R}^N$ such that $\mathrm{supp}(\varphi_m) \subset K$ for all $m \in \mathbb{N}$, and

$$D^\beta \varphi_m \to 0 \quad \text{uniformly on } \mathbb{R}^N \quad \forall \beta \in \mathbb{N}^N.$$

Therefore, given $\alpha, \beta \in \mathbb{N}^N$, if $M_\alpha := \sup\{|x^\alpha| \,:\, x \in K\}$, then

$$\sup_{x\in\mathbb{R}^N} \left| x^\alpha D^\beta \varphi_m(x) \right| \le M_\alpha \sup_{x\in K} \left| D^\beta \varphi_m(x) \right| \to 0, \quad \text{as } m \to \infty.$$

Consequently, the inclusion of $\mathcal{D}(\mathbb{R}^N)$ into $\mathcal{S}(\mathbb{R}^N)$ is continuous. □

The space $\mathcal{D}(\mathbb{R}^N)$ is not stable under the Fourier transform, i.e., if $f \in \mathcal{D}(\mathbb{R}^N)$, then $\hat{f} \in \mathcal{S}(\mathbb{R}^N)$ but we cannot expect $\hat{f} \in \mathcal{D}(\mathbb{R}^N)$, since the Fourier transform of a compactly supported function does not have compact support, being analytic. Indeed: suppose that $f \in \mathcal{D}(\mathbb{R})$, and that supp(f) $\subset [a, b]$. Then,

$$\hat{f}(y) = \int_a^b e^{-i\langle x, y\rangle} \, f(x) \, dx,$$

and this function can be analytically continued to an entire function, defined by:

$$g(\alpha + i\beta) := \int_a^b e^{-i\langle x, \alpha+i\beta\rangle} f(x)\, dx.$$

Clearly, the function g is defined on the entire complex plane. Moreover,

$$\frac{d}{dz} g(z) = \int_a^b \frac{d}{dz}\left(e^{-i\langle x,z\rangle} f(x)\right) dx = -ix \int_a^b e^{-i\langle x,z\rangle} f(x)\, dx.$$

Therefore, g is an analytic function on $\mathbb{C}$. By Liouville's Theorem, if g had compact support, it would be identically zero, and hence $\hat{f} = 0$, which implies $f = 0$.

What we have just seen is a particular case of the following characterization of the Fourier transform of functions in $\mathcal{D}(\mathbb{R}^N)$, whose proof can be found in Rudin's book [50].

Paley-Wiener Theorem *Let $a > 0$. The following are equivalent:*

(i) $\psi = \hat{\varphi}$ *with* $\varphi \in \mathcal{D}(\mathbb{R}^N)$ *and* $\text{supp}(\varphi) \subset \text{B}(0, \text{a})$.
(ii) ψ *can be extended to an entire function* $\tilde{\psi}$ *in* $\mathbb{C}^n$*, with the property:*

$$\text{given } k \in \mathbb{N},\ \exists C_k > 0\ :\ |\tilde{\psi}(z)| \leq C_k\big(1 + |z|^2\big)^{-\frac{k}{2}}\, e^{a|\Im z|}\ \ \forall\, z \in \mathbb{C}^N.$$

Definition 5.17 The topological dual of the space $\mathcal{S}(\mathbb{R}^N)$ is denoted by $\mathcal{S}'(\mathbb{R}^N)$, and its elements are called *tempered distributions*, i.e., a tempered distribution is a linear form $T : \mathcal{S}(\mathbb{R}^N) \to \mathbb{C}$ such that, if $f_n \to 0$ in $\mathcal{S}(\mathbb{R}^N)$, then $\langle T, f_n\rangle \to 0$.

As a consequence of Theorem 5.22, each element of $\mathcal{S}'(\mathbb{R}^N)$ determines a unique distribution. Consequently, $\mathcal{S}'(\mathbb{R}^N)$ can be identified with a subspace of $\mathcal{D}'(\mathbb{R}^N)$.

Example 5.8

(i) Every distribution with compact support is a tempered distribution. This is an immediate consequence of Theorem 5.11.
(ii) As a consequence of Proposition 5.8, every function in $L^p(\mathbb{R}^N)$, with $1 \leq p < \infty$, defines a tempered distribution.
(iii) A Radon measure $\mu \in \mathcal{M}(\mathbb{R}^N)$ is said to have *slow growth* if there exists an integer $k \geq 0$ such that

$$\int_{\mathbb{R}^N} \frac{d\mu(x)}{(1 + \|x\|^2)^k} < \infty.$$

Viewed as distributions, Radon measures of slow growth are tempered distributions. Indeed: given a Radon measure $\mu \in \mathcal{M}(\mathbb{R}^N)$ of slow growth, define $T_\mu : \mathcal{S}(\mathbb{R}^N) \to \mathbb{C}$ by

$$T_\mu(f) := \int_{\mathbb{R}^N} f(x)\, d\mu(x).$$

Then, since

$$|T_\mu(f)| \le \sup_{x\in\mathbb{R}^N}\Big(|f(x)(1+\|x\|^2)^k|\Big)\int_{\mathbb{R}^N}\frac{d\mu(x)}{(1+\|x\|^2)^k},$$

we have that T_μ is continuous on $\mathcal{S}(\mathbb{R}^N)$ and, consequently, $T_\mu \in \mathcal{S}'(\mathbb{R}^N)$.

(iv) The functions in $L^\infty(\mathbb{R}^N)$ are tempered distributions. Indeed: by (iii), it suffices to show that for every $f \in L^\infty(\mathbb{R}^N)$, the associated Radon measure T_f has slow growth, which holds since, for $k > \frac{n}{2}$,

$$\int_{\mathbb{R}^N}\frac{f(x)\,dx}{(1+\|x\|^2)^k} < \infty.$$

(v) Let us give an example of a non-tempered distribution. Consider the distribution T_f associated to the function $f(x) := e^x$, and let us show that $T_f \notin \mathcal{S}'(\mathbb{R})$. Let $\varphi \in \mathcal{D}(\mathbb{R})$ with $0 \le \varphi \le 1$, $\varphi(x) = 1$ for all $x \in [-1,1]$, and $\operatorname{supp}(\varphi) \subset [-2,2]$. Given $\phi \in \mathcal{S}(\mathbb{R}^N)$, define $\phi_m(x) := \phi(x)\varphi\big(\frac{x}{m}\big)$, then $\phi_m \to \phi$ in $\mathcal{S}(\mathbb{R}^N)$ (see the proof of Theorem 5.22). Therefore, if $T_f \in \mathcal{S}'(\mathbb{R})$, then

$$\langle T_f,\phi\rangle = \lim_{m\to\infty}\langle T_f,\phi_m\rangle = \lim_{m\to\infty}\int_{\mathbb{R}} e^x\phi_m(x)\,dx.$$

Consider the function $\phi \in \mathcal{S}(\mathbb{R}^N)$ defined by

$$\phi(x) := \begin{cases} e^{-x}e^{-\frac{1}{x}} & \text{if } x > 0 \\ \\ 0 & \text{if } x \le 0. \end{cases}$$

Then,

$$\langle T_f,\phi\rangle = \lim_{m\to\infty}\int_{\mathbb{R}} e^x\phi_m(x)\,dx = \lim_{m\to\infty}\int_0^\infty e^{-\frac{1}{x}}\varphi\left(\frac{x}{m}\right)dx$$

$$\ge \lim_{m\to\infty}\int_0^m e^{-\frac{1}{x}}\,dx \ge \lim_{m\to\infty} e^{-1}\int_1^m dx = \lim_{m\to\infty} e^{-1}(m-1) = +\infty.$$

Consequently, $e^x \notin \mathcal{S}'(\mathbb{R})$. Similarly, one can show that $e^{\|x\|^2} \notin \mathcal{S}'(\mathbb{R}^N)$.

Definition 5.18 Let $T \in \mathcal{S}'(\mathbb{R}^N)$ be a tempered distribution. The *Fourier transform* $\hat{T}$ of T is defined by the formula

$$\langle \hat{T},\phi\rangle := \langle T,\hat{\phi}\rangle \qquad \forall\,\phi\in\mathcal{S}(\mathbb{R}^N).$$

As a consequence of Theorem 5.19, we have that $\hat{T} \in \mathcal{S}'(\mathbb{R}^N)$. Therefore, we can define the map $\mathcal{F}: \mathcal{S}'(\mathbb{R}^N) \to \mathcal{S}'(\mathbb{R}^N)$ by $\mathcal{F}(T) := \hat{T}$.

Since $L^1(\mathbb{R}^N) \subset \mathcal{S}'(\mathbb{R}^N)$, for elements of $L^1(\mathbb{R}^N)$ we have two definitions of the Fourier transform. Now, if $f \in L^1(\mathbb{R}^N)$, for each $\phi \in \mathcal{S}(\mathbb{R}^N)$, we have

$$\langle \widehat{T_f}, \phi \rangle = \langle T_f, \hat{\phi} \rangle = \int_{\mathbb{R}^N} f\hat{\phi} = \int_{\mathbb{R}^N} \hat{f}\phi = \langle T_{\hat{f}}, \phi \rangle.$$

Hence, $\widehat{T_f} = T_{\hat{f}}$, and the two definitions of the Fourier transform agree on $L^1(\mathbb{R}^N)$.

Since $L^2(\mathbb{R}^N) \subset \mathcal{S}'(\mathbb{R}^N)$, the same consistency issue arises for the Fourier-Plancherel transform. Now, since the identity

$$\int_{\mathbb{R}^N} f\hat{\phi} = \int_{\mathbb{R}^N} \mathcal{P}(f)\phi, \qquad f \in L^2(\mathbb{R}^N), \quad \phi \in \mathcal{S}(\mathbb{R}^N),$$

holds, the same argument as before shows that: if $f \in L^2(\mathbb{R}^N)$, then its Fourier–Plancherel transform $\mathcal{P}(f)$ and its transform $\widehat{T_f}$, as an element of $\mathcal{S}'(\mathbb{R}^N)$, coincide.

Definition 5.19 Just as in the space $\mathcal{D}'(\mathbb{R}^N)$, in the space $\mathcal{S}'(\mathbb{R}^N)$ we consider the weak topology, i.e., if $T_n, T \in \mathcal{S}'(\mathbb{R}^N)$, we say that

$$T_n \to T \quad \text{in } \mathcal{S}'(\mathbb{R}^N),$$

if

$$\langle T_n, \phi \rangle \to \langle T, \phi \rangle \qquad \forall\, \phi \in \mathcal{S}(\mathbb{R}^N).$$

The following theorem gathers some of the properties of the Fourier transform of tempered distributions.

Theorem 5.23

(i) The Fourier transform $\mathcal{F}: \mathcal{S}'(\mathbb{R}^N) \to \mathcal{S}'(\mathbb{R}^N)$ is a bijective continuous map with a continuous inverse.

(ii) For each $a \in \mathbb{R}^N$ and $T \in \mathcal{S}'(\mathbb{R}^N)$, we have:

$$\mathcal{F}(\tau_a T) = e^{-i<a,y>}\, \mathcal{F}(T). \tag{5.52}$$

$$\mathcal{F}(e^{-i<a,x>}\, T) = \tau_{-a}\, \mathcal{F}(T). \tag{5.53}$$

(iii) For every multi-index $\alpha \in \mathbb{N}^N$, we have:

$$\mathcal{F}(D^\alpha T) = (iy)^\alpha \mathcal{F}(T). \tag{5.54}$$

$$D^\alpha \mathcal{F}(T) = \mathcal{F}\big((-ix)^\alpha T\big). \tag{5.55}$$

Proof

(i) Since $\mathcal{F}: \mathcal{S}(\mathbb{R}^N) \to \mathcal{S}(\mathbb{R}^N)$ is a surjective map, from the definition it follows easily that $\mathcal{F}: \mathcal{S}'(\mathbb{R}^N) \to \mathcal{S}'(\mathbb{R}^N)$ is injective. On the other hand, given $S \in \mathcal{S}'(\mathbb{R}^N)$, define $T: \mathcal{S}(\mathbb{R}^N) \to \mathbb{C}$ by $\langle T, \hat{\phi}\rangle := \langle S, \phi\rangle$; this is well-defined and $T \in \mathcal{S}'(\mathbb{R}^N)$. Moreover,

$$\langle \mathcal{F}(T), \phi\rangle = \langle T, \hat{\phi}\rangle = \langle S, \phi\rangle \quad \forall\, \phi \in \mathcal{S}(\mathbb{R}^N).$$

Consequently, $\mathcal{F}$ is bijective. Suppose that $T_m \to T$ in $\mathcal{S}'(\mathbb{R}^N)$, then

$$\langle \widehat{T_m}, \phi\rangle = \langle T_m, \hat{\phi}\rangle \to \langle T, \hat{\phi}\rangle = \langle \hat{T}, \phi\rangle.$$

Hence, $\mathcal{F}: \mathcal{S}'(\mathbb{R}^N) \to \mathcal{S}'(\mathbb{R}^N)$ is continuous. On the other hand,

$$\langle \mathcal{F}^{-1}T, \phi\rangle = \langle \mathcal{F}(\mathcal{F}^{-1}T), \mathcal{F}^{-1}(\phi)\rangle = \langle T, \mathcal{F}^{-1}(\phi)\rangle,$$

from which it easily follows that $\mathcal{F}^{-1}: \mathcal{S}'(\mathbb{R}^N) \to \mathcal{S}'(\mathbb{R}^N)$ is continuous.

(ii) is a straightforward verification. Let us now prove (iii). For every $\phi \in \mathcal{S}(\mathbb{R}^N)$, we have:

$$\langle \mathcal{F}(D^\alpha T), \phi\rangle = \langle D^\alpha T, \hat{\phi}\rangle = (-1)^{|\alpha|}\langle T, D^\alpha(\hat{\phi})\rangle$$

$$= (-1)^{|\alpha|}\langle T, (-i)^{|\alpha|}\widehat{y^\alpha \phi}\rangle = \langle (iy)^\alpha \hat{T}, \phi\rangle.$$

Therefore,

$$\mathcal{F}(D^\alpha T) = (iy)^\alpha \mathcal{F}(T).$$

Finally, for every $\phi \in \mathcal{S}(\mathbb{R}^N)$, we have:

$$\langle \mathcal{F}\big((-ix)^\alpha T\big), \phi\rangle = \langle (-ix)^\alpha T, \hat{\phi}\rangle = (-1)^{|\alpha|}\langle T, i^{|\alpha|}x^\alpha \hat{\phi}\rangle$$

$$= (-1)^{|\alpha|}\langle T, \mathcal{F}(D^\alpha \phi)\rangle = (-1)^{|\alpha|}\langle \hat{T}, D^\alpha \phi\rangle = \langle D^\alpha \hat{T}, \phi\rangle.$$

Consequently, (5.55) holds.

□

Definition 5.20 Given $T \in \mathcal{S}'(\mathbb{R}^N)$ and a polynomial $P(x)$, we define $P(x)T: \mathcal{S}(\mathbb{R}^N) \to \mathbb{C}$ by the formula:

$$\langle P(x)T, \phi\rangle := \langle T, P(x)\phi\rangle \quad \forall\, \phi \in \mathcal{S}(\mathbb{R}^N).$$

Since $P(x)\phi \in \mathcal{S}(\mathbb{R}^N)$ and, moreover, if $\phi_m \to 0$ in $\mathcal{S}(\mathbb{R}^N)$, then $P(x)\phi_m \to 0$ in $\mathcal{S}(\mathbb{R}^N)$, we have that $P(x)T \in \mathcal{S}'(\mathbb{R}^N)$.

Given the polynomial $P(x) = \sum_{|\alpha|\le k} c_\alpha x^\alpha$, we denote by P^* the polynomial

$$P^*(x) := \sum_{|\alpha|\le k} i^\alpha c_\alpha x^\alpha = \sum_{|\alpha|\le k} c_\alpha (ix)^\alpha = P(ix).$$

With this notation, as a consequence of the previous theorem, we obtain the following result.

Corollary 5.4 *For each polynomial $P(x)$ and each $T \in \mathcal{S}'(\mathbb{R}^N)$, the following identities hold:*

$$\mathcal{F}\big(P(D)T\big) = P^*(x)\hat{T}. \tag{5.56}$$

$$\mathcal{F}\big(P(x)T\big) = P^*(D)\hat{T}. \tag{5.57}$$

Example 5.9

(i) If δ is the Dirac measure on $\mathbb{R}^N$, then

$$\hat{\delta}(y) = 1 \quad \forall\, y \in \mathbb{R}^N.$$

Moreover, we have

$$\mathcal{F}(D^\alpha \delta) = (iy)^\alpha,$$

and

$$\mathcal{F}(\mathbf{1}) = (2\pi)^N \delta.$$

Indeed: Let $\phi \in \mathcal{S}(\mathbb{R}^N)$, then

$$\langle \hat{\delta}, \phi\rangle = \langle \delta, \hat{\phi}\rangle = \hat{\phi}(0) = \int_{\mathbb{R}^N} \phi(x)\, dx = \langle \mathbf{1}, \phi\rangle.$$

Hence, $\hat{\delta} = \mathbf{1}$. Then, by the previous theorem we get

$$\mathcal{F}(D^\alpha \delta) = (iy)^\alpha \hat{\delta} = (iy)^\alpha \mathbf{1}.$$

Finally, taking into account the inversion formula, we obtain:

$$\langle \mathcal{F}(\mathbf{1}), \phi\rangle = \langle \mathbf{1}, \hat{\phi}\rangle = \int_{\mathbb{R}^N} \hat{\phi}(x)\, dx = (2\pi)^N \phi(0) = (2\pi)^N \langle \delta, \phi\rangle.$$

(ii) Let χ_a be the characteristic function of the interval $[-a, a]$. We show that

$$\widehat{\chi_a} = 2\,\frac{\sin ay}{y}.$$

Indeed: For $\phi \in \mathcal{S}(\mathbb{R})$, we have

$$\langle \widehat{\chi_a}, \phi\rangle = \langle \chi_a, \hat{\phi}\rangle = \int_{-a}^{a} \hat{\phi}(t)\,dt = \int_{-a}^{a}\left(\int_{\mathbb{R}} e^{-ixt}\phi(x)\,dx\right)dt$$

$$= \int_{\mathbb{R}} \phi(x)\left(\int_{-a}^{a} e^{-ixt}\,dt\right)dx = \int_{\mathbb{R}} \phi(x)\left(\frac{1}{-ix}(e^{-iax} - e^{iax})\right)dx$$

$$= \int_{\mathbb{R}} \phi(x) 2\frac{\sin ax}{x}\,dx = \langle 2\frac{\sin ax}{x}, \phi\rangle.$$

(iii) Given $T \in \mathcal{S}'(\mathbb{R}^N)$, we define $\check{T}$ as

$$\langle \check{T}, \phi\rangle := \langle T, \check{\phi}\rangle \qquad \forall\, \phi \in \mathcal{S}(\mathbb{R}^N).$$

Evidently, $\check{T} \in \mathcal{S}'(\mathbb{R}^N)$ and we have

$$\langle \mathcal{F}^2(T), \phi\rangle = \langle T, \mathcal{F}^2(\phi)\rangle = \langle T, (2\pi)^N \check{\phi}\rangle = (2\pi)^N \langle \check{T}, \phi\rangle.$$

Therefore,

$$\mathcal{F}^2(T) = (2\pi)^N \check{T}. \tag{5.58}$$

Just as we have defined the convolution product for $T \in \mathcal{D}'(\mathbb{R}^N)$ and $\phi \in \mathcal{D}(\mathbb{R}^N)$, and for $T \in \mathcal{E}'(\mathbb{R}^N)$ and $\phi \in \mathcal{E}(\mathbb{R}^N)$, we define the convolution product of a distribution $T \in \mathcal{S}'(\mathbb{R}^N)$ and a function $\phi \in \mathcal{S}(\mathbb{R}^N)$ as

$$(T \star \phi)(x) := \langle T, \tau_x \check{\phi}\rangle \qquad \forall\, x \in \mathbb{R}^N.$$

To prove the properties of this convolution product, we need the following result.

Lemma 5.6 *Let $\phi \in \mathcal{S}(\mathbb{R}^N)$ and $\epsilon > 0$. Then, if*

$$\phi_\epsilon(x) := \frac{\phi(x + \epsilon \mathbf{e}_j) - \phi(x)}{\epsilon} \quad (x \in \mathbb{R}^N),$$

we have that

$$\phi_\epsilon \to D_j\phi \quad \text{in } \mathcal{S}(\mathbb{R}^N) \text{ as } \epsilon \to 0.$$

Proof Since $\mathcal{F}: \mathcal{S}(\mathbb{R}^N) \to \mathcal{S}(\mathbb{R}^N)$ is an isomorphism, it will suffice to prove that

$$\hat{\phi}_\epsilon \to \widehat{D_j \phi} \quad \text{in } \mathcal{S}(\mathbb{R}^N) \quad \text{as } \epsilon \to 0.$$

Now,

$$\hat{\phi}_\epsilon(y) = \frac{1}{\epsilon}\left[\mathcal{F}(\tau_{-\epsilon \mathbf{e}_j}\phi)(y) - \hat{\phi}(y)\right]$$

$$= \frac{1}{\epsilon}\left(e^{i\epsilon <\mathbf{e}_j, y>} - 1\right)\hat{\phi}(y) = \frac{1}{\epsilon}\left(e^{i\epsilon y_j} - 1\right)\hat{\phi}(y),$$

and

$$\widehat{D_j \phi}(y) = i y_j \hat{\phi}(y).$$

Therefore, we need to prove that

$$\psi_\epsilon \hat{\phi} \to 0 \quad \text{in } \mathcal{S}(\mathbb{R}^N) \quad \text{as } \epsilon \to 0, \tag{5.59}$$

where

$$\psi_\epsilon(y) := \frac{1}{\epsilon}\left(e^{i\epsilon y_j} - 1\right) - i y_j.$$

To prove (5.59), we need to show that for every $\alpha, \gamma \in \mathbb{N}^N$,

$$x^\gamma D^\alpha(\psi_\epsilon \hat{\phi})(x) = x^\gamma \sum_{\beta \le \alpha} \frac{\alpha!}{\beta!(\alpha-\beta)!} D^{\alpha-\beta}\hat{\phi}(x) D^\beta \psi_\epsilon(x) \to 0 \tag{5.60}$$

uniformly in $x \in \mathbb{R}^N$ as $\epsilon \to 0$. Now, it is easy to verify that

$$|D^\beta \psi_\epsilon(y)| \le \begin{cases} \epsilon y_j^2 & \text{if } |\beta| = 0 \\ \epsilon |y_j| & \text{if } |\beta| = 1 \\ \epsilon^{|\beta|-1} & \text{if } |\beta| > 1. \end{cases}$$

From this, since $\hat{\phi} \in \mathcal{S}(\mathbb{R}^N)$, it follows that (5.60) holds. □

Theorem 5.24 *For $T \in \mathcal{S}'(\mathbb{R}^N)$ and $\phi \in \mathcal{S}(\mathbb{R}^N)$, the following properties hold:*

(i) $T \star \phi \in C^\infty(\mathbb{R}^N)$, and

$$D^\alpha(T \star \phi) = (D^\alpha T) \star \phi = T \star (D^\alpha \phi).$$

(ii) $T \star \phi \in \mathcal{S}'(\mathbb{R}^N)$.
(iii) $\mathcal{F}(T \star \phi) = \hat{\phi}\hat{T}$.
(iv) $(T \star \phi) \star \psi = T \star (\phi \star \psi) \quad \forall\, \psi \in \mathcal{S}(\mathbb{R}^N)$.
(v) $\mathcal{F}(\phi T) = \frac{1}{(2\pi)^N}\, \hat{T} \star \hat{\phi}$.

Proof

(i) It is easy to verify that convolution commutes with translations, i.e.,

$$\tau_x(T \star \phi) = (\tau_x T) \star \phi = T \star (\tau_x \phi).$$

Thus, (i) is proved similarly to (ii) in Theorem 5.14, using the previous lemma.

(ii) It suffices to show that $T \star \phi$ has slow growth, i.e., there exists $k \in \mathbb{N}$ such that

$$\int_{\mathbb{R}^N} \frac{(T \star \phi)(x)}{(1 + \|x\|^2)^k}\, dx < \infty. \tag{5.61}$$

Now, since $T \in \mathcal{S}'(\mathbb{R}^N)$, for each $m \in \mathbb{N}$ we have

$$|(T \star \phi)(x)| = |\langle T, \tau_x \check{\phi}\rangle| \leq C_m S_m(\tau_x \check{\phi})$$

$$= C_m \sup\left\{(1 + \|y\|^2)^m |D^\alpha(\tau_x \check{\phi})(y)| : y \in \mathbb{R}^N,\ |\alpha| \leq m\right\}$$

$$= C_m \sup\left\{(1 + \|y\|^2)^m |D^\alpha \phi(x - y)| : y \in \mathbb{R}^N,\ |\alpha| \leq m\right\}$$

$$= C_m \sup\left\{(1 + \|x - z\|^2)^m |D^\alpha \phi(z)| : z \in \mathbb{R}^N,\ |\alpha| \leq m\right\}$$

$$\leq C_m \sup\left\{\left(2(1 + \|x\|^2)(1 + \|z\|^2)\right)^m |D^\alpha \phi(z)| : z \in \mathbb{R}^N,\ |\alpha| \leq m\right\}$$

$$\leq C_m 2^m (1 + \|x\|^2)^m \sup\left\{(1 + \|z\|^2)^m |D^\alpha \phi(z)| : z \in \mathbb{R}^N,\ |\alpha| \leq m\right\}$$

$$\leq C_m 2^m (1 + \|x\|^2)^m S_m(\phi).$$

Therefore,

$$\frac{|(T \star \phi)(x)|}{(1 + \|x\|^2)^{2m}} \leq S_m(\phi) C_m 2^m (1 + \|x\|^2)^{-m}.$$

Hence, taking $k = 2m$, with $m > \frac{N}{2}$, we get (5.61).

(iii) From (ii), we have that $T \star \phi \in \mathcal{S}'(\mathbb{R}^N)$, thus $\mathcal{F}(T \star \phi) \in \mathcal{S}'(\mathbb{R}^N)$. Let $\psi \in \mathcal{D}(\mathbb{R}^N)$ and let $K = \operatorname{supp}(\psi)$. Then,

$$\langle \mathcal{F}(T \star \phi), \hat{\psi} \rangle = \langle T \star \phi, \mathcal{F}^2(\psi) \rangle = \langle T \star \phi, (2\pi)^N \check{\psi} \rangle$$

$$= (2\pi)^N \int_{-K} (T \star \phi)(x)\psi(-x)\, dx = (2\pi)^N \int_{-K} \langle T, \tau_x \check{\phi} \rangle \psi(-x)\, dx$$

$$= (2\pi)^N \int_{-K} \langle T, \psi(-x)\tau_x \check{\phi} \rangle\, dx = (2\pi)^N \langle T, \int_{-K} \psi(-x)\tau_x \check{\phi}\, dx \rangle$$

$$= (2\pi)^N \langle T, (\phi \star \psi)^\vee \rangle = \langle T, \mathcal{F}^2(\phi \star \psi) \rangle$$

$$= \langle \hat{T}, \mathcal{F}(\phi \star \psi) \rangle = \langle \hat{T}, \hat{\phi}\hat{\psi} \rangle = \langle \hat{\phi}\hat{T}, \hat{\psi} \rangle.$$

Now, since the set $\{\hat{\psi} : \psi \in \mathcal{D}(\mathbb{R}^N)\}$ is dense in $\mathcal{S}(\mathbb{R}^N)$, it follows that $\mathcal{F}(T \star \phi) = \hat{\phi}\hat{T}$.

(iv) In the proof of (iii), we saw that

$$\langle (T \star \phi), \check{\psi} \rangle = \langle T, (\phi \star \psi)^\vee \rangle,$$

for each $\psi \in \mathcal{D}(\mathbb{R}^N)$; hence, by density, it also holds for each $\psi \in \mathcal{S}(\mathbb{R}^N)$. Therefore, we have that

$$\big((T \star \phi) \star \psi\big)(0) = \big(T \star (\phi \star \psi)\big)(0).$$

Replacing ψ by $\tau_{-x}\psi$, we obtain

$$\big((T \star \phi) \star \psi\big)(x) = \big(T \star (\phi \star \psi)\big)(x).$$

(v) From (iii), we get

$$\frac{1}{(2\pi)^N} \mathcal{F}(\hat{T} \star \hat{\phi}) = \frac{1}{(2\pi)^N} \mathcal{F}^2(T)\mathcal{F}^2(\phi) = (2\pi)^N \check{T}\check{\phi} =$$

$$= (2\pi)^N (\phi T)^\vee = \mathcal{F}^2(\phi T).$$

Therefore, $\mathcal{F}(\phi T) = \frac{1}{(2\pi)^N} \hat{T} \star \hat{\phi}$.

□

The Fourier transform of a distribution with compact support is a C^∞ function. More precisely, we have the following result.

Theorem 5.25 *If $T \in \mathcal{E}'(\mathbb{R}^N)$, then its Fourier transform $\hat{T}$ is a C^∞ function given by*

$$\hat{T}(y) = \langle T, e^{-i\langle x,y\rangle}\rangle \qquad \forall\, y \in \mathbb{R}^N.$$

Moreover, for every $y \in \mathbb{R}^N$ and every multi-index $\alpha \in \mathbb{N}^N$, we have

$$D^\alpha(\hat{T})(y) = \langle (-ix)^\alpha T, e^{-i\langle x,y\rangle}\rangle.$$

Proof Let $T \in \mathcal{E}'(\mathbb{R}^N)$ and set $K := \text{supp(T)}$. Then, if we take a function $\varphi \in \mathcal{D}(\mathbb{R}^N)$ such that $\varphi(x) = 1$ if $x \in K_1 = \bigcup_{x\in K} B(x,1)$, we know that $T = \varphi T$. Therefore, by item (v) of the previous theorem we have

$$\hat{T} = \widehat{\varphi T} = \frac{1}{(2\pi)^N}\big(\hat{\varphi} \star \hat{T}\big).$$

Hence $\hat{T} \in \mathcal{E}(\mathbb{R}^N)$. Moreover, if we take $\phi \in \mathcal{E}(\mathbb{R}^N)$ such that $\hat{\phi} = \varphi$, then

$$\big(\hat{T} \star \hat{\varphi}\big)(x) = \big(\hat{T} \star \mathcal{F}^2(\phi)\big)(x) = \big(\hat{T} \star (2\pi)^N \check{\phi}\big)(x)$$

$$= \langle \hat{T}, (2\pi)^N \tau_x \phi\rangle = \langle T, (2\pi)^N \widehat{\tau_x \phi}\rangle = \langle T, (2\pi)^N e^{-i\langle x,y\rangle}\hat{\phi}\rangle$$

$$= \langle T, (2\pi)^N e^{-i\langle x,y\rangle}\varphi\rangle = (2\pi)^N \langle \varphi T, e^{-i\langle x,y\rangle}\rangle = (2\pi)^N \langle T, e^{-i\langle x,y\rangle}\rangle.$$

Thus,

$$\hat{T}(x) = \frac{1}{(2\pi)^N}\big(\hat{\varphi} \star \hat{T}\big)(x) = \langle T, e^{-i\langle x,y\rangle}\rangle.$$

Finally, given $\alpha \in \mathbb{N}^N$, since $D^\alpha \hat{T} = \mathcal{F}\big[(-ix)^\alpha T\big]$, we obtain

$$(D^\alpha \hat{T})(y) = \mathcal{F}\big[(-ix)^\alpha T\big](y) = \langle (-ix)^\alpha T, e^{-i\langle x,y\rangle}\rangle.$$

□

If $T \in \mathcal{E}'(\mathbb{R}^N)$ and $S \in \mathcal{S}'(\mathbb{R}^N)$, then the convolution product $T \star S$ is defined. On the other hand, by the previous theorem, $\hat{T}$ is a C^∞ function, so the distribution $\hat{T}\hat{S}$ is also defined. We are going to show that a similar formula to the one for functions holds.

Theorem 5.26 *If $T \in \mathcal{E}'(\mathbb{R}^N)$ and $S \in \mathcal{S}'(\mathbb{R}^N)$, then $T \star S \in \mathcal{S}'(\mathbb{R}^N)$ and*

$$\mathcal{F}(T \star S) = \hat{T}\hat{S}.$$

Proof Given $\varphi \in \mathcal{D}(\mathbb{R}^N)$, we have

$$\langle T \star S, \varphi \rangle = (T \star (S \star \check{\varphi}))(0) = \langle \check{T}, S \star \check{\varphi} \rangle.$$

Therefore, since $\check{T} \in \mathcal{E}'(\mathbb{R}^N)$, there exist $C \geq 0$, $K \in \mathcal{K}(\mathbb{R}^N)$, and $m \in \mathbb{N}$ such that

$$|\langle T \star S, \varphi \rangle| \leq C \sum_{|\alpha| \leq m} \sup_{x \in K} |D^\alpha (S \star \check{\varphi})(x)| = C \sum_{|\alpha| \leq m} \sup_{x \in K} |(D^\alpha S \star \check{\varphi})(x)|$$

$$= C \sum_{|\alpha| \leq m} \sup_{x \in K} |\langle D^\alpha S, \tau_x \varphi \rangle| \leq M P_{\gamma\beta}(\varphi),$$

since $D^\alpha S \in \mathcal{S}'(\mathbb{R}^N)$. Then, as $\mathcal{D}(\mathbb{R}^N)$ is dense in $\mathcal{S}(\mathbb{R}^N)$, we conclude that $T \star S$ extends uniquely to an element of $\mathcal{S}'(\mathbb{R}^N)$, i.e., $T \star S \in \mathcal{S}'(\mathbb{R}^N)$.

Now suppose $T \in \mathcal{D}(\mathbb{R}^N) \subset \mathcal{E}'(\mathbb{R}^N)$. Let $\psi \in \mathcal{D}(\mathbb{R}^N)$ such that $\hat{\psi} \in \mathcal{S}(\mathbb{R}^N)$. Then,

$$(\check{T} \star \hat{\psi})(x) = \int_{\mathbb{R}^N} \check{T}(y) \hat{\psi}(x - y)\, dy = \int_{\mathbb{R}^N} T(-y) \hat{\psi}(x - y)\, dy$$

$$= \int_{\mathbb{R}^N} T(z) \hat{\psi}(x + z)\, dz = \int_{\mathbb{R}^N} T(z) \left(\int_{\mathbb{R}^N} e^{-i\langle x+z, \xi \rangle} \psi(\xi)\, d\xi \right) dz$$

$$= \int_{\mathbb{R}^N} \left(\int_{\mathbb{R}^N} T(z) e^{-i\langle z, \xi \rangle}\, dz \right) e^{-i\langle x, \xi \rangle} \psi(\xi)\, d\xi$$

$$= \int_{\mathbb{R}^N} \hat{T}(\xi) e^{-i\langle x, \xi \rangle} \psi(\xi)\, d\xi = \mathcal{F}(\hat{T}\psi)(x).$$

Therefore,

$$\langle \mathcal{F}(T \star S), \psi \rangle = \langle T \star S, \hat{\psi} \rangle = \langle S, \check{T} \star \hat{\psi} \rangle$$

$$= \langle S, \mathcal{F}(\hat{T}\psi) \rangle = \langle \hat{S}, \hat{T}\psi \rangle = \langle \hat{T}\hat{S}, \psi \rangle.$$

Now, since $\mathcal{D}(\mathbb{R}^N)$ is dense in $\mathcal{S}(\mathbb{R}^N)$, from the above we have $\mathcal{F}(T \star S) = \hat{T}\hat{S}$ when $T \in \mathcal{D}(\mathbb{R}^N)$ and $S \in \mathcal{S}'(\mathbb{R}^N)$. Finally, suppose $T \in \mathcal{E}'(\mathbb{R}^N)$ and $S \in \mathcal{S}'(\mathbb{R}^N)$. Take $\varphi \in \mathcal{D}(\mathbb{R}^N)$ such that $\hat{\varphi}(x) \neq 0$ for every $x \in \mathbb{R}^N$. Then,

$$(\varphi \star T) \star S = \varphi \star (T \star S).$$

Now, $\varphi \star T \in \mathcal{D}(\mathbb{R}^N)$, so applying the above we obtain

$$\hat{\varphi}\mathcal{F}(T \star S) = \mathcal{F}[\varphi \star (T \star S)] = \mathcal{F}[(\varphi \star T) \star S] = \mathcal{F}(\varphi \star T)\hat{S} = \hat{\varphi}\hat{T}\hat{S},$$

and since $\hat{\varphi}(x) \neq 0$ for every $x \in \mathbb{R}^N$, we conclude that

$$\mathcal{F}(T \star S) = \hat{T}\hat{S}.$$

□

5.9 Fundamental Solutions

In this section, we study some problems of linear partial differential equations with constant coefficients using the theory of distributions. We obtain the fundamental solutions of the classical operators in Mathematical Physics and establish the fundamental result of Malgrange and Ehrenpreis regarding the existence of fundamental solutions for every linear differential operator with constant coefficients.

Definition 5.21 A *linear differential operator* of order m in $\mathbb{R}^N$ is an expression of the form

$$P(x, D) = \sum_{|\alpha| \leq m} a_\alpha(x) D^\alpha,$$

where a_α are continuous functions valued in $\mathbb{C}$ and defined on some open subset of $\mathbb{R}^N$. The expression

$$P_m(x, D) = \sum_{|\alpha| = m} a_\alpha(x) D^\alpha$$

is called the *principal part* of the operator $P(x, D)$. When the functions are constant, we refer to it as a *linear differential operator with constant coefficients*, and we denote it by $P(D)$.

Let $\Omega \subset \mathbb{R}^N$ be open. Consider the linear partial differential equation in Ω

$$P(x, D)u = f, \tag{5.62}$$

where $P(x, D)$ is the differential operator

$$P(x, D) = \sum_{|\alpha| \leq m} a_\alpha(x) D^\alpha.$$

A *classical solution* of (5.62) in Ω is a function u of class $C^m(\Omega)$ that satisfies (5.62) pointwise in Ω. This concept of solution is too restrictive for our purposes, so we introduce the following more general concept of solution.

A distribution $T \in \mathcal{D}'(\Omega)$ that satisfies Eq. (5.62) in the sense of distributions is called a *distributional solution* of Eq. (5.62), i.e., $T \in \mathcal{D}'(\Omega)$ is a distributional

solution of (5.62) if

$$\langle P(x, D)T, \phi\rangle = \langle f, \phi\rangle \quad \forall\, \phi \in \mathcal{D}(\Omega),$$

or equivalently,

$$\langle T, \sum_{|\alpha|\le m} (-1)^{\alpha} a_{\alpha} D^{\alpha}\phi\rangle = \langle f, \phi\rangle \quad \forall\, \phi \in \mathcal{D}(\Omega).$$

Distributional solutions are also commonly referred to as *weak* or *generalized* solutions. Clearly, every classical solution is a distributional solution. As we saw in the introduction, the converse is not true in general. Let us now look at some additional examples.

Example 5.10

(i) Let us consider the equation

$$xu' = 0 \tag{5.63}$$

in $\mathbb{R}$. Clearly, every constant function is a classical solution of Eq. (5.63). Now, the Heaviside function H is a distributional solution of Eq. (5.63), since $H' = \delta$ and, for each $\varphi \in \mathcal{D}(\mathbb{R})$, we have

$$\langle x\delta, \varphi\rangle = \langle \delta, x\varphi\rangle = (x\varphi)(0) = 0 \cdot \varphi(0) = 0.$$

However, since H is not of class C^1, it is not a classical solution of Eq. (5.63).

(ii) Let us consider the equation

$$D_1 u = 0 \tag{5.64}$$

in $\mathbb{R}^2$. The function

$$u(x, y) := \begin{cases} 1 \text{ if } y > 0 \\ \\ 0 \text{ if } y \le 0 \end{cases}$$

is a distributional solution of Eq. (5.64), since, for each $\varphi \in \mathcal{D}(\mathbb{R}^2)$, we have

$$\langle D_1 u, \varphi\rangle = -\langle u, D_1\varphi\rangle = -\int_{\mathbb{R}^2} u(x, y) D_1\varphi(x, y)\, dxdy$$

$$= -\int_0^{\infty} \left(\int_{\mathbb{R}} D_1\varphi(x, y)\, dx\right) dy = 0$$

since $\text{supp}(\varphi)$ is a compact subset of $\mathbb{R}^2$.

We now compute a distributional solution of Eq. (5.64) that is not a function distribution. Define $T : \mathcal{D}(\mathbb{R}^2) \to \mathbb{C}$ by

$$\langle T, \varphi \rangle := \int_{\mathbb{R}} \varphi(x, 0)\, dx.$$

It is easy to verify that $T \in \mathcal{D}'(\mathbb{R}^2)$, and also,

$$\langle D_1 T, \varphi \rangle = -\langle T, D_1 \varphi \rangle = -\int_{\mathbb{R}} D_1 \varphi(x, 0)\, dx = 0.$$

Therefore, T is a distributional solution of Eq. (5.64). Let us show that it is not a function distribution. If it were, there would exist $f \in L^1_{loc}(\mathbb{R}^2)$ such that $T = T_f$. Let $\phi \in \mathcal{D}(\mathbb{R})$, with $\text{supp}(\phi) = [-1, 1]$ and $\int_{\mathbb{R}} \phi(x)\, dx = 1$, and let $\psi_m \in \mathcal{D}(\mathbb{R})$, with $0 \leq \psi_m \leq 1$, such that

$$\text{supp}(\psi_{\text{m}}) \subset]-\frac{1}{m}, \frac{1}{m}[$$

and $\psi_m(0) = 1$ for all $m \in \mathbb{N}$. Then consider the functions $\varphi_m \in \mathcal{D}(\mathbb{R}^2)$, defined by $\varphi_m(x, y) := \phi(x)\psi_m(y)$. Then we have

$$\langle T, \varphi_m \rangle = \int_{\mathbb{R}} \varphi_m(x, 0)\, dx = \int_{\mathbb{R}} \phi(x) = 1.$$

On the other hand,

$$\langle T_f, \varphi_m \rangle = \int_{\mathbb{R}^2} f(x, y)\varphi_m(x, y)\, dxdy$$

$$= \int_{\text{supp}(\phi)\times[-1,1]} f(x, y)\phi(x)\psi_m(y)\, dxdy.$$

Now, if $y \neq 0$, and $m_0 \in \mathbb{N}$ is such that $|y| > \frac{1}{m_0}$, then

$$f(x, y)\phi(x)\psi_m(y) = 0 \qquad \forall\, x \in \mathbb{R}, \ \ \forall\, m \geq m_0.$$

Hence,

$$f(x, y)\phi(x)\psi_m(y) \to 0 \qquad \text{a.e. in } \mathbb{R}^2.$$

Moreover,

$$|f(x, y)\phi(x)\psi_m(x, y)| \leq |f(x, y)\phi(x)| \in L^1(\text{supp}(\phi) \times [-1, 1]).$$

Consequently, applying the Dominated Convergence Theorem (Theorem B.1), we obtain

$$\langle T_f, \varphi_m \rangle \to 0 \quad \text{as } m \to \infty.$$

Therefore, $T \neq T_f$.

The following result tells us when a distributional solution is classical.

Proposition 5.9 *Let u be a distributional solution of the equation*

$$P(x, D)u = f \quad \text{in } \Omega.$$

Then, if $u \in C^m(\Omega)$, where m is the order of the differential operator, and $f \in C(\Omega)$, it follows that u is a classical solution.

Proof Since $u \in C^m(\Omega)$, for each multi-index $|\alpha| \leq m$, we have that $D^\alpha u = T_{D^\alpha u}$. Therefore, $P(x, D)u$ is a continuous function. Thus, since

$$T_{P(x,D)u} = T_f,$$

we have that $P(x, D)u = f$ almost everywhere in Ω, and by continuity, the equality holds at every point. □

In the following result we obtain a generalization of the classical Liouville Theorem for harmonic functions.

Theorem 5.27 (Generalized Liouville Theorem) *Let $P(D)$ be a linear differential operator with constant coefficients. Suppose that $P(ix) \neq 0$ for every $x \neq 0$. Then, if $T \in \mathcal{S}'(\mathbb{R}^N)$ satisfies $P(D)T = 0$ in the distributional sense, it follows that T is a polynomial in x.*

Proof Applying the Fourier transform we get

$$0 = \mathcal{F}(P(D)T) = P(i\xi)\hat{T}.$$

Hence, $\operatorname{supp}(\hat{T}) \subset \{0\}$. As a consequence of Theorem 5.13, we have

$$\hat{T} = \sum_{|\alpha| \leq m} c_\alpha D^\alpha \delta.$$

Consequently,

$$T = \mathcal{F}^{-1}\left(\sum_{|\alpha| \leq m} c_\alpha D^\alpha \delta \right) = \sum_{|\alpha| \leq m} c_\alpha \mathcal{F}^{-1}\left(D^\alpha \delta\right)$$

$$= \sum_{|\alpha|\leq m} c_\alpha (-ix)^\alpha \mathcal{F}^{-1}(\delta) = \sum_{|\alpha|\leq m} c_\alpha (-ix)^\alpha \frac{1}{(2\pi)^N}.$$

□

Corollary 5.5 (Liouville's Theorem) *Every bounded harmonic function in* $\mathbb{R}^N$ *is constant.*

Proof Let u be harmonic and bounded in $\mathbb{R}^N$. Then $u \in C^2(\mathbb{R}^N)$ and $\Delta u = 0$. Hence, by the previous result, u is a polynomial, and since it is bounded, it must be constant. □

Definition 5.22 Let $P(D)$ be a linear differential operator with constant coefficients. A distribution $E \in \mathcal{D}'(\mathbb{R}^N)$ that satisfies $P(D)E = \delta$ is called a *fundamental solution* of $P(D)$.

The importance of fundamental solutions lies in the fact that, from them, we can construct distributional solutions of the non-homogeneous equations associated with the operator $P(D)$. Indeed, suppose that E is a fundamental solution of the operator $P(D)$. Then, if $T \in \mathcal{D}'(\mathbb{R}^N)$ is a distribution for which the convolution $E \star T$ is defined, we have

$$P(D)(E \star T) = (P(D)E) \star T = \delta \star T = T.$$

Thus, $E \star T$ is a distributional solution of the equation $P(D)U = T$. Moreover, this solution is unique in the class of distributions that are convolutionable with E, since if S is a distribution in this class such that $P(D)S = T$, then

$$S = S \star \delta = S \star P(D)E = P(D)S \star E = T \star E.$$

In general, fundamental solutions are not unique. Indeed, consider the first-order differential operator $\frac{d}{dx}$. Then, if H is the Heaviside function and C is a constant, we have

$$\frac{d}{dx}(H + C) = \frac{d}{dx}H = \delta.$$

The fundamental solutions of classical differential operators such as Laplace's, the heat operator, the Cauchy-Riemann operator, and the wave operator have been known for a long time. The conjecture that every linear differential operator with constant coefficients has a fundamental solution was independently proved by Malgrange and Ehrenpreis in [15, 42]. Before stating the Malgrange–Ehrenpreis Theorem, we will compute the fundamental solutions of classical differential operators. We begin with the Laplace operator. Recall that if $\Gamma(s) := \int_0^\infty t^{s-1}e^{-t}\,dt$

(for $s > 0$) is the Gamma function, then the $(N-1)$-dimensional measure of the unit sphere S_1 in $\mathbb{R}^N$ is given by

$$\sigma_N = \frac{2\pi^{\frac{N}{2}}}{\Gamma\left(\frac{N}{2}\right)}.$$

Theorem 5.28 *The locally integrable functions*

$$E_N(x) := \begin{cases} \frac{1}{2\pi}\log\|x\| & \text{if } N = 2 \\ \\ \frac{1}{(2-N)\sigma_N}\|x\|^{2-N} & \text{if } N \geq 3, \end{cases}$$

are fundamental solutions of the Laplace operator.

Proof We will assume that $N \geq 3$, as the case $N = 2$ is handled similarly. Making the change to polar coordinates, we have:

$$\int_{B(0,a)} E_N(x)\,dx = \frac{1}{(2-N)\sigma_N}\int_{B(0,a)} \|x\|^{2-N}$$

$$= \frac{1}{(2-N)\sigma_N}\int_0^a r^{2-N} r^{N-1}\sigma_N\,dr$$

$$= \frac{1}{(2-N)}\int_0^a r\,dr = \frac{a^2}{2(2-N)}.$$

Therefore, $E_N \in L^1_{loc}(\mathbb{R}^N)$. Let us now show that $\Delta E_N = \delta$. To do this, we must prove that

$$\langle \Delta E_N, \phi\rangle = \phi(0) \qquad \forall\, \phi \in \mathcal{D}(\mathbb{R}^N). \tag{5.65}$$

Given $\phi \in \mathcal{D}(\mathbb{R}^N)$, let $R > 0$ such that $\text{supp}(\phi) \subset B(0,R)$. For $0 < r < R$, let $\Omega_r := \{x \in \mathbb{R}^N : r \leq \|x\| < R\}$. Then, by the Dominated Convergence Theorem, we have

$$\langle \Delta E_N, \phi\rangle = \int_{\mathbb{R}^N} E_N \Delta\phi = \int_{B(0,R)} E_N\Delta\phi = \lim_{r\to 0^+}\int_{\Omega_r} E_N\Delta\phi.$$

Now, since $\Delta E_N = 0$ in Ω_r, applying Green's Formula (Theorem A.4), we get

$$\int_{\Omega_r} E_N\Delta\phi = \int_{\Omega_r}(E_N\Delta\phi - \phi\Delta E_N) = \int_{\partial\Omega_r}\left(E_N\frac{\partial\phi}{\partial\eta} - \phi\frac{\partial E_N}{\partial\eta}\right)ds$$

$$= \int_{S_r} \phi \frac{\partial E_N}{\partial \eta}\, ds - \int_{S_r} E_N \frac{\partial \phi}{\partial \eta}\, ds.$$

Now, since

$$\eta(x) = \frac{-x}{\|x\|} \quad \text{and} \quad \nabla E_N(x) = \frac{x}{\sigma_N} \|x\|^{-N},$$

we have

$$\frac{\partial E_N}{\partial \eta}(x) = \langle \nabla E_N(x), \eta(x) \rangle = \frac{1}{\sigma_N} \|x\|^{-N+1}.$$

Thus,

$$\int_{\Omega_r} E_N \Delta \phi = \int_{S_r} \phi(s) \frac{1}{\sigma_N} r^{-N+1}\, ds - \frac{1}{(2-N)\sigma_N} \int_{S_r} r^{2-N} \frac{\partial \phi}{\partial \eta}\, ds.$$

Since

$$\left| \frac{\partial \phi}{\partial \eta}(x) \right| = \left| \left\langle \nabla \phi(x), \frac{x}{\|x\|} \right\rangle \right| \le M$$

and

$$\int_{S_r} r^{2-N}\, ds = \int_{S_1} r\, ds = r\sigma_N,$$

we have

$$\lim_{r \to 0^+} \frac{1}{(2-N)\sigma_N} \int_{S_r} r^{2-N} \frac{\partial \phi}{\partial \eta}\, ds = 0.$$

Therefore,

$$\langle \Delta E_N, \phi \rangle = \lim_{r \to 0^+} \int_{S_r} \phi(s) \frac{1}{\sigma_N} r^{-N+1}\, ds$$

$$= \lim_{r \to 0^+} \frac{1}{\sigma_N} \int_{S_1} \phi(ry) r^{-N+1} r^{N-1} ds(y)$$

$$= \lim_{r \to 0^+} \frac{1}{\sigma_N} \int_{S_1} \big(\phi(ry) - \phi(0)\big)\, ds(y) + \phi(0) = \phi(0),$$

since ϕ is uniformly continuous. □

Remark 5.4 An alternative proof of the previous result can be given using the Fourier transform. Let us see such a proof in the case $N \geq 3$. We have $\Delta E_N = P(D)E_N$, where $P(x) := x_1^2 + \cdots + x_N^2$. Then, if $E_N \in \mathcal{S}'(\mathbb{R}^N)$ satisfies $\Delta E_N = \delta$, applying the Fourier transform we obtain:

$$\mathbf{1} = \hat{\delta} = \mathcal{F}(P(D)E_N) = P^*(x)\widehat{E_N} = -(x_1^2 + \cdots + x_N^2)\hat{E_N}.$$

Hence,

$$E_N = -\mathcal{F}^{-1}(\|x\|^{-2}).$$

To compute $\mathcal{F}^{-1}(\|x\|^{-2})$, note first that for $\alpha > -N$, we have:

$$\int_0^\infty s^{-\frac{\alpha}{2}-1} e^{-s\|x\|^2}\, ds = \|x\|^\alpha \int_0^\infty t^{-\frac{\alpha}{2}-1} e^{-t}\, dt = \|x\|^\alpha \Gamma\left(-\frac{\alpha}{2}\right).$$

Therefore,

$$\|x\|^\alpha = \frac{1}{\Gamma\left(-\frac{\alpha}{2}\right)} \int_0^\infty s^{-\frac{\alpha}{2}-1} e^{-s\|x\|^2}\, ds.$$

From this, applying Fubini's Theorem, we get

$$\mathcal{F}\left(\|x\|^\alpha\right)(y) = \frac{1}{\Gamma\left(-\frac{\alpha}{2}\right)} \int_0^\infty s^{-\frac{\alpha}{2}-1} \mathcal{F}\left(e^{-s\|x\|^2}\right)(y)\, ds$$

$$= \frac{1}{\Gamma\left(-\frac{\alpha}{2}\right)} \int_0^\infty s^{-\frac{\alpha}{2}-1} \left(\frac{\pi}{s}\right)^{\frac{N}{2}} e^{-\frac{\|y\|^2}{4s}}\, ds$$

$$= \frac{\pi^{\frac{N}{2}}}{\Gamma\left(-\frac{\alpha}{2}\right)} \int_0^\infty s^{-\frac{\alpha}{2}-\frac{N}{2}-1} e^{-\frac{\|y\|^2}{4s}}\, ds.$$

Then, making the change of variable $t = \frac{\|y\|^2}{4s}$, we obtain

$$\mathcal{F}\left(\|x\|^\alpha\right)(y) = \frac{\pi^{\frac{N}{2}}}{\Gamma\left(-\frac{\alpha}{2}\right)} 2^{\alpha+N} \|y\|^{-\alpha-N} \int_0^\infty t^{\frac{\alpha}{2}+\frac{N}{2}-1} e^{-t}\, dt$$

$$= \frac{\pi^{\frac{n}{2}}}{\Gamma\left(-\frac{\alpha}{2}\right)} 2^{\alpha+N} \|y\|^{-\alpha-N} \Gamma\left(\frac{\alpha}{2} + \frac{N}{2}\right).$$

Taking $\alpha = 2 - N$, we obtain

$$\mathcal{F}\left(\|x\|^{2-N}\right)(y) = 4\frac{\pi^{\frac{N}{2}}}{\Gamma\left(\frac{N-2}{2}\right)}\|y\|^{-2}.$$

Therefore,

$$E_N(x) = -\mathcal{F}^{-1}\left(\|y\|^{-2}\right)(x) = -\frac{\Gamma\left(\frac{N-2}{2}\right)}{4\pi^{\frac{N}{2}}}\|x\|^{2-N},$$

and since

$$\Gamma\left(\frac{N}{2}\right) = \Gamma\left(\frac{N}{2} - 1 + 1\right)$$

$$= \left(\frac{N}{2} - 1\right)\Gamma\left(\frac{N}{2} - 1\right) = \frac{N-2}{2}\Gamma\left(\frac{N-2}{2}\right),$$

we obtain

$$E_N(x) = \frac{\Gamma\left(\frac{N}{2}\right)}{(2-N)2\pi^{\frac{N}{2}}}\|x\|^{2-N} = \frac{1}{(2-N)\sigma_N}\|x\|^{2-N}.$$

Remark 5.5 As a consequence of the previous theorem, we have that if $T \in \mathcal{D}'(\mathbb{R}^N)$ is a distribution that can be convolved with E_N, then $E_N \star T$ is a distributional solution of the equation

$$\Delta u = T \qquad \text{in } \mathbb{R}^N.$$

Let Ω be a bounded open subset of $\mathbb{R}^N$ ($N \geq 3$). Suppose that $f \in C(\Omega)$ is a bounded function. Define

$$F(x) = \begin{cases} f(x) \text{ if } x \in \Omega \\ \\ 0 \qquad \text{if } x \notin \Omega. \end{cases}$$

Then, since $T_F \in \mathcal{E}'(\mathbb{R}^N)$, T_F can be convolved with E_N, and thus

$$\Delta(E_N \star T_F) = T_F \qquad \text{in } \mathcal{D}'(\mathbb{R}^N).$$

Hence,

$$\Delta(E_N \star T_f) = T_f \qquad \text{in } \mathcal{D}'(\Omega).$$

It follows that the function

$$(E_N \star f)(x) = \frac{1}{(2-N)\sigma_N} \int_{\mathbb{R}^N} \frac{f(x)}{\|x-y\|^{N-2}}\, dy$$

is harmonic in $\mathbb{R}^N \setminus \overline{\Omega}$ (i.e., $\Delta(E_N \star f) = 0$ in $\mathbb{R}^N \setminus \overline{\Omega}$) and satisfies in Ω the *Poisson equation*

$$\Delta(E_N \star f) = f \qquad \text{in } \Omega.$$

The function $E_N \star f$ is called the *Newtonian potential* generated by the mass distribution $f(y)dy$ in Ω. In the case $N = 3$, it coincides with the gravitational potential created by a mass distribution $f(y)dy$ in Ω. Note that in this case, if

$$V(x) := \frac{1}{\|x\|},$$

then

$$(E_3 \star f)(x) = \Big(-\frac{1}{\sigma_3}\|y\|^{-1} \star f\Big)(x) = -\frac{1}{4\pi}(V \star f)(x),$$

we have

$$\Delta V = -4\pi f \qquad \text{in } \Omega,$$

which is the classical equation established by Poisson in 1813.

The *heat equation* is the partial differential equation

$$\rho(x)\frac{\partial u}{\partial t}(x,t) - \operatorname{div}_x(k(x)\nabla_x u(x,t)) = f(x,t), \tag{5.66}$$

where ρ is the heat capacity of the medium, k the thermal conductivity coefficient, $u(x,t)$ represents the temperature of the medium at point x at time t, and $f(x,t)$ is the heat source. Assuming that the heat capacity and the conductivity coefficient are constant, and also, for simplicity, that $\rho = k = 1$, Eq. (5.66) becomes:

$$\frac{\partial u}{\partial t}(x,t) - \Delta_x u(x,t) = f(x,t).$$

The differential operator in $\mathbb{R}^{N+1}$

$$\frac{\partial}{\partial t} - \Delta_x$$

is called the *heat operator*. We are going to compute its fundamental solutions.

Theorem 5.29 *The locally integrable function*

$$G(x,t) := (4\pi t)^{-\frac{N}{2}} \exp\left(-\frac{\|x\|^2}{4t}\right) H(t),$$

called the Gaussian kernel, is a fundamental solution of the heat operator.

Proof For each $t > 0$, let

$$G_t(x) := (4\pi t)^{-\frac{N}{2}} \exp\left(-\frac{\|x\|^2}{4t}\right) H(t).$$

Then we have

$$\int_{\mathbb{R}^N} G_t(x)\, dx = 1.$$

Therefore, $G \in L^1_{loc}(\mathbb{R}^{N+1})$. Moreover,

$$\int_{\mathbb{R}^{N+1}} \frac{G(x,t)}{(1+t^2)}\, dxdt = \int_0^\infty \frac{dt}{(1+t^2)} \int_{\mathbb{R}^N} G_t(x)\, dx = \int_0^\infty \frac{dt}{(1+t^2)} < +\infty.$$

Hence, $G \in \mathcal{S}'(\mathbb{R}^{N+1})$. For each $\epsilon > 0$, let

$$G^\epsilon(x,t) := \begin{cases} (4\pi t)^{-\frac{N}{2}} \exp\left(-\frac{\|x\|^2}{4t}\right) & \text{if } t > \epsilon \\ \\ 0 & \text{if } t \le \epsilon \end{cases}$$

By the Dominated Convergence Theorem, we have that $G^\epsilon \to G$ in $\mathcal{D}'(\mathbb{R}^{N+1})$ as $\epsilon \to 0^+$, hence

$$\partial_t G^\epsilon - \Delta_x G^\epsilon \to \partial_t G - \Delta_x G \quad \text{in } \mathcal{D}'(\mathbb{R}^{N+1}) \quad \text{as } \epsilon \to 0^+.$$

Therefore, it suffices to prove that

$$\partial_t G^\epsilon - \Delta_x G^\epsilon \to \delta \quad \text{in } \mathcal{D}'(\mathbb{R}^{N+1}) \quad \text{as } \epsilon \to 0^+.$$

To this end, we need to show that for every $\phi \in \mathcal{D}(\mathbb{R}^{N+1})$, we have

$$\langle G^\epsilon, -\partial_t \phi - \Delta_x \phi \rangle \to \phi(0,0) \quad \text{as } \epsilon \to 0^+. \tag{5.67}$$

Integrating by parts yields

$$\langle G^\epsilon, -\partial_t\phi - \Delta_x\phi\rangle = \int_\epsilon^\infty \int_{\mathbb{R}^N} G(x,t)\big(-\partial_t\phi(x,t) - \Delta_x\phi(x,t)\big)\,dtdx$$

$$= \int_\epsilon^\infty \int_{\mathbb{R}^N} (\partial_t G(x,t) - \Delta_x G(x,t))\phi(x,t)\,dtdx + \int_{\mathbb{R}^N} G(x,\epsilon)\phi(x,\epsilon)\,dx.$$

Then, since $G(x,\epsilon) = G(-x,\epsilon)$ and $\partial_t G - \Delta_x G = 0$ in $]\epsilon,\infty[\times\mathbb{R}^N$, it follows that

$$\langle G^\epsilon, -\partial_t\phi - \Delta_x\phi\rangle = \int_{\mathbb{R}^N} G(x,\epsilon)\phi(x,\epsilon)\,dx$$

$$= [G(.,\epsilon)\star\phi(\cdot,\epsilon)](0) = [G(.,\epsilon)\star\phi(\cdot,0)](0) + [G(.,\epsilon)\star\big(\phi(\cdot,\epsilon) - \phi(\cdot,0)\big)](0).$$

Now, by the Dominated Convergence Theorem (Theorem B.1),

$$[G(.,\epsilon)\star\phi(\cdot,0)](0) = \int_{\mathbb{R}^N} G(-x,\epsilon)\phi(x,0)\,dx \to \phi(0,0).$$

On the other hand,

$$\big|[G(.,\epsilon)\star\big(\phi(\cdot,\epsilon) - \phi(\cdot,0)\big)](0)\big| = \left|\int_{\mathbb{R}^N} G(-x,\epsilon)\big(\phi(x,\epsilon) - \phi(x,0)\big)\,dx\right|$$

$$\le \sup_{x\in\mathbb{R}^N} |\phi(x,\epsilon) - \phi(x,0)| \cdot \|G(.,\epsilon)\|_1 \to 0 \quad \text{as } \epsilon \to 0^+.$$

This proves (5.67) and, consequently, the theorem. □

As a consequence of the previous theorem, we have that if $T \in \mathcal{D}'(\mathbb{R}^{N+1})$ is a distribution that can be convolved with G, then $G\star T$ is a distributional solution of the equation

$$\frac{\partial u}{\partial t} - \Delta_x u = T.$$

An important example of this situation is given by the following result.

Theorem 5.30 *For each $f \in L^1(\mathbb{R}^{N+1})$, the function $u := G\star f$ is a distributional solution of the heat equation*

$$\frac{\partial u}{\partial t} - \Delta_x u = f.$$

Proof It suffices to show that $u := G \star f$ is well defined. Indeed:

$$u(t,x) = (G \star f)(t,x) = \int_{\mathbb{R}^{N+1}} G(t-s, x-y) f(s,y)\, ds dy$$

$$= \int_{-\infty}^{t} \left(\int_{\mathbb{R}^N} G_{t-s}(y-x) f(s,y)\, dy \right) ds = \int_{-\infty}^{t} \big(G_{t-s} \star f(s,\cdot) \big)(x)\, ds.$$

Now, since $f \in L^1(\mathbb{R}^{N+1})$, for almost every $s \in \mathbb{R}$, we have $f(s,\cdot) \in L^1(\mathbb{R}^N)$, and thus, by Young's inequality, we get

$$\sup_{t\in\mathbb{R}} \int_{\mathbb{R}^N} |u(t,x)|\, dx \le \sup_{t\in\mathbb{R}} \int_{-\infty}^{t} \int_{\mathbb{R}^N} |G_{t-s} \star f(s,\cdot)(x)|\, dx ds$$

$$\le \sup_{t\in\mathbb{R}} \int_{-\infty}^{t} \|G_{t-s}\|_{L^1(\mathbb{R}^N)} \|f(s,\cdot)\|_{L^1(\mathbb{R}^N)}\, ds \le \|f\|_{L^1(\mathbb{R}^{n+1})} < +\infty.$$

□

In Example 5.4, we saw that traveling waves were distributional solutions of the one-dimensional wave equation. In addition to the vibrating string problem, the modeling of phenomena such as sound or light waves involves the partial differential equation

$$\frac{1}{c^2} \frac{\partial^2 u}{\partial t^2}(x,t) - \Delta_x u(x,t) = f(x,t),$$

where the term $f(x,t)$ corresponds to the sources of perturbation in the medium. The symbol $\Box$ is often used to denote the wave operator

$$\Box u := \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2} - \Delta_x u,$$

which is also known as the *D'Alembert operator*.

We will now compute the fundamental solutions of the one-dimensional and two-dimensional wave operators. We begin with the one-dimensional case.

Theorem 5.31 *The locally integrable function*

$$O_1(x,t) := \frac{c}{2} H(ct+x) H(ct-x)$$

is a fundamental solution of the one-dimensional wave operator.

Proof Let $\phi \in \mathcal{D}(\mathbb{R}^2)$. As in Example 5.4, using the change of variables $y = ct + x$, $s = ct - x$, we obtain

$$\frac{1}{c^2}\frac{\partial^2 \phi}{\partial t^2}(x,t) - \frac{\partial^2 \phi}{\partial x^2}(x,t) = 4\frac{\partial^2 \psi}{\partial y \partial s}(y,s),$$

where $\psi(y,s) = \phi(x,t)$. Therefore,

$$\langle \frac{1}{c^2}\frac{\partial^2 O_1}{\partial t^2} - \frac{\partial^2 O_1}{\partial x^2}, \phi\rangle = \frac{c}{2}\int_{\mathbb{R}^2} H(ct+x)H(ct-x)\left(\frac{1}{c^2}\frac{\partial^2 \phi}{\partial t^2} - \frac{\partial^2 \phi}{\partial x^2}\right)(x,t)\, dxdt$$

$$= \frac{1}{4}\int_0^\infty \int_0^\infty \left(\frac{1}{c^2}\frac{\partial^2 \phi}{\partial t^2} - \frac{\partial^2 \phi}{\partial x^2}\right)\left(\frac{y-s}{2}, \frac{y+s}{2c}\right) dyds$$

$$= \int_0^\infty \int_0^\infty \frac{\partial^2 \psi}{\partial y \partial s}(y,s)\, dyds = -\int_0^\infty \frac{\partial \psi}{\partial s}(0,s)\, ds = \psi(0,0) = \phi(0,0) = \langle \delta, \phi\rangle.$$

Therefore, O_1 is a fundamental solution of the one-dimensional wave equation. □

Theorem 5.32 *The locally integrable function on* $\mathbb{R}^2 \times \mathbb{R}$

$$O_2(x,t) := \frac{cH(ct - \|x\|)}{2\pi\sqrt{c^2t^2 - \|x\|^2}}$$

is a fundamental solution of the two-dimensional wave operator.

Proof Given $\phi \in \mathcal{D}(\mathbb{R}^3)$, we must show that $\langle \Box O_2, \phi\rangle = \phi(0,0,0)$.

$$\langle \Box O_2, \phi\rangle = \langle O_2, \Box\phi\rangle = \frac{c}{2\pi}\int_{\mathbb{R}^3} \frac{H(ct - \|x\|)}{\sqrt{c^2t^2 - \|x\|^2}} \Box\phi(x,t)\, dxdt$$

$$= \frac{c}{2\pi}\int_0^\infty \int_{B(0,ct)} \frac{\Box\phi(x,t)}{\sqrt{c^2t^2 - \|x\|^2}}\, dxdt.$$

Switching to polar coordinates in the spatial variable $x = (r\cos\,\theta, r\sin\,\theta)$, define

$$\psi(r,\theta,t) = \phi(r\cos\,\theta, r\sin\,\theta, t),$$

so that

$$\Box\phi(r\sin\,\theta, r\cos\,\theta, t) = \left(\frac{1}{c^2}\frac{\partial^2 \psi}{\partial t^2} - \frac{\partial^2 \psi}{\partial r^2} - \frac{1}{r}\frac{\partial \psi}{\partial r} - \frac{1}{r^2}\frac{\partial^2 \psi}{\partial \theta^2}\right)(r,\theta,t),$$

and thus

$$\langle \Box O_2, \phi \rangle$$

$$= \frac{c}{2\pi} \int_0^\infty \int_0^{ct} \int_0^{2\pi} \frac{r}{\sqrt{c^2t^2 - r^2}} \left(\frac{1}{c^2} \frac{\partial^2 \psi}{\partial t^2} - \frac{\partial^2 \psi}{\partial r^2} - \frac{1}{r} \frac{\partial \psi}{\partial r} - \frac{1}{r^2} \frac{\partial^2 \psi}{\partial \theta^2} \right) d\theta dr dt$$

where

$$\int_0^{2\pi} \frac{\partial^2 \psi}{\partial \theta^2}(r, \theta, t)\, d\theta = \frac{\partial \psi}{\partial \theta}(r, 2\pi, t) - \frac{\partial \psi}{\partial \theta}(r, 0, t)$$

$$= r \frac{\partial \phi}{\partial x_1}(0, r, t) - r \frac{\partial \phi}{\partial x_1}(0, r, t) = 0,$$

so

$$\langle \Box O_2, \phi \rangle$$

$$= \frac{c}{2\pi} \int_0^\infty \int_0^{ct} \int_0^{2\pi} \frac{r}{\sqrt{c^2t^2 - r^2}} \left(\frac{1}{c^2} \frac{\partial^2 \psi}{\partial t^2} - \frac{\partial^2 \psi}{\partial r^2} - \frac{1}{r} \frac{\partial \psi}{\partial r} \right)(r, \theta, t)\, d\theta dr dt.$$

Let us consider the function

$$\eta(r, t) := \int_0^{2\pi} \psi(r, \theta, t)\, d\theta$$

and make the change of variable $r = cts$, from which it follows that

$$\langle \Box O_2, \phi \rangle = \frac{c}{2\pi} \int_0^\infty \int_0^{ct} \frac{r}{\sqrt{c^2t^2 - r^2}} \left(\frac{1}{c^2} \frac{\partial^2 \eta}{\partial t^2} - \frac{\partial^2 \eta}{\partial r^2} - \frac{1}{r} \frac{\partial \eta}{\partial r} \right)(r, t)\, dr dt$$

$$= \frac{c}{2\pi} \int_0^\infty \int_0^1 \frac{cts}{\sqrt{1 - s^2}} \left(\frac{1}{c^2} \frac{\partial^2 \eta}{\partial t^2} - \frac{\partial^2 \eta}{\partial r^2} - \frac{1}{cts} \frac{\partial \eta}{\partial r} \right)(cts, t)\, ds dt.$$

Let $f(s, t) := \eta(cst, t)$, then

$$\left(\frac{1}{c^2} \frac{\partial^2 \eta}{\partial t^2} - \frac{\partial^2 \eta}{\partial r^2} - \frac{1}{cts} \frac{\partial \eta}{\partial r} \right)(cts, t) =$$

$$\left(\frac{1}{c^2} \frac{\partial^2 f}{\partial t^2} + \frac{1}{c^2 t^2}(s^2 - 1) \frac{\partial^2 f}{\partial s^2} - \frac{2s}{c^2 t} \frac{\partial^2 f}{\partial s \partial t} + \frac{1}{c^2 t^2} \left(2s - \frac{1}{s}\right) \frac{\partial f}{\partial s} \right)(s, t),$$

thus

$$\langle \Box O_2, \phi \rangle = \frac{1}{2\pi} \int_0^\infty \int_0^1 \frac{-s}{t} \sqrt{1-s^2}\, \frac{\partial^2 f}{\partial s^2}\, ds dt$$

$$= \frac{1}{2\pi} \int_0^\infty \int_0^1 \frac{1}{\sqrt{1-s^2}} \left(ts \frac{\partial^2 f}{\partial t^2} - 2s^2 \frac{\partial^2 f}{\partial s \partial t} + \frac{2s^2-1}{t} \frac{\partial f}{\partial s} \right) ds dt.$$

Now, integrating by parts, we obtain

$$\int_0^1 \frac{-s}{t} \sqrt{1-s^2}\, \frac{\partial^2 f}{\partial s^2}\, ds = -\int_0^1 \frac{2s^2-1}{t\sqrt{1-s^2}} \frac{\partial f}{\partial s}\, ds$$

and

$$\int_0^\infty \frac{\partial^2 f}{\partial s \partial t}\, dt = -\frac{\partial f}{\partial s}(s,0) = 0.$$

Therefore,

$$\langle \Box O_2, \phi \rangle = \frac{1}{2\pi} \int_0^\infty \int_0^1 \frac{st}{\sqrt{1-s^2}} \frac{\partial^2 f}{\partial t^2}\, ds dt$$

$$= \frac{1}{2\pi} \int_0^1 \frac{s}{\sqrt{1-s^2}} \left(\int_0^\infty t \frac{\partial^2 f}{\partial t^2}\, dt \right) ds$$

$$= -\frac{1}{2\pi} \int_0^1 \frac{s}{\sqrt{1-s^2}} \left(\int_0^\infty \frac{\partial f}{\partial t}\, dt \right) ds$$

$$= \frac{1}{2\pi} \int_0^1 \frac{s f(s,0)}{\sqrt{1-s^2}}\, ds = \frac{1}{2\pi} \int_0^1 \frac{s \eta(0,0)}{\sqrt{1-s^2}}\, ds$$

$$= \frac{\eta(0,0)}{2\pi} = \frac{1}{2\pi} \int_0^{2\pi} \psi(0,\theta,0)\, d\theta = \phi(0,0,0).$$

□

We now state the Malgrange–Ehrenpreis Theorem. We will present an elementary proof due to J. P. Rosay [47]. For a shorter but more complicated proof, see for example [50].

We denote the inner product in $L^2(\mathbb{R}^N)$ by

$$(f/g) = \int_{\mathbb{R}^N} f(x) \overline{g}(x)\, dx.$$

Then, if $P(D) = \sum_{|\alpha|\leq m} a_\alpha D^\alpha$, for $P(D)u \in L^2(\mathbb{R}^N)$, we have that

$$(P(D)u/\phi) = (u/\overline{P}(D)\phi) \qquad \forall\, \phi \in \mathcal{D}(\mathbb{R}^N), \tag{5.68}$$

where $\overline{P}(D) := \sum_{|\alpha|\leq m}(-1)^{|\alpha|}\overline{a_\alpha} D^\alpha$. In fact:

$$(P(D)u/\phi) = \int_{\mathbb{R}^N} P(D)u\ \overline{\phi} = \langle P(D)u, \overline{\phi}\rangle = \sum_{|\alpha|\leq m} (-1)^{|\alpha|} \int_{\mathbb{R}^N} u a_\alpha D^\alpha \overline{\phi}$$

$$= \int_{\mathbb{R}^N} u\ \overline{\sum_{|\alpha|\leq m} (-1)^{|\alpha|}\overline{a_\alpha} D^\alpha \phi} = \int_{\mathbb{R}^N} u\ \overline{\overline{P}(D)\phi} = (u/\overline{P}(D)\phi).$$

As a consequence of (5.68), taking into account that linear differential operators with constant coefficients commute, we obtain that for each $\phi \in \mathcal{D}(\mathbb{R}^N)$,

$$\|P(D)\phi\|_2^2 = (P(D)\phi/P(D)\phi) = (\phi/\overline{P}(D)P(D)\phi)$$

$$= (\phi/P(D)\overline{P}(D)\phi) = (\overline{P}(D)\phi/\overline{P}(D)\phi) = \|\overline{P}(D)\phi\|_2^2.$$

Therefore, we have that

$$\|P(D)\phi\|_2 = \|\overline{P}(D)\phi\|_2 \qquad \forall\, \phi \in \mathcal{D}(\mathbb{R}^N). \tag{5.69}$$

First, we will show that for each $g \in L^2_{loc}(\mathbb{R}^N)$, the equation $P(D)u = g$ has a distributional solution $u \in L^2_{loc}(\mathbb{R}^N)$. To do so, we need the following inequality due to Hörmander.

Theorem 5.33 (Hörmander's Inequality) *Let $\Omega \subset \mathbb{R}^N$ be a bounded open set and $P(D) = \sum_{|\alpha|\leq m} a_\alpha D^\alpha$. Then there exists a constant $C > 0$ such that*

$$\|P(D)\phi\|_2 \geq C\|\phi\|_2 \qquad \forall\, \phi \in \mathcal{D}(\Omega).$$

Moreover, $C = |P|_m K(m, \Omega)$, with $|P|_m = \max\{|a_\alpha| \ :\ |\alpha| = m\}$ and $K(m, \Omega)$ depending only on m and the diameter of Ω.

Proof Before proving the general case, let us illustrate the idea with the particular case where $n = 1$, $\Omega =]0, 1[$ and $P = \frac{d}{dx}$. In this case, we want to show that $\|\phi'\|_2 \geq C\|\phi\|_2$ for $\phi \in \mathcal{D}(]0, 1[)$. In fact, we have:

$$\big((x\phi)'/\phi\big) = (x\phi'/\phi) + (\phi/\phi).$$

Also, integrating by parts,

$$\big((x\phi)'/\phi\big) = -(x\phi/\phi'),$$

thus we obtain

$$(\phi/\phi) = -(x\phi'/\phi) - (x\phi'/\phi).$$

Now, since $|x| \leq 1$, applying the Cauchy–Schwarz inequality, we get

$$\|\phi\|_2^2 \leq 2\|\phi'\|_2\|\phi\|_2.$$

We now prove the general statement by induction on the degree of P. For $j = 1, 2, \ldots, N$, we define $P_j(D)$ by the relation:

$$P(D)(x_j\phi) = x_j P(D)\phi + P_j(D)\phi.$$

The operator $P_j(D)$ is zero if and only if $P(D)$ involves no differentiation with respect to x_j, and if $P_j(D) \neq 0$, its degree is less than m. Let $R := \sup_{x\in\Omega} \|x\|$. We will prove by induction on m that

$$\|P_j(D)\phi\|_2 \leq 2mR\|P(D)\phi\|_2 \qquad \forall\, \phi \in \mathcal{D}(\Omega). \tag{5.70}$$

Note first that from (5.70) and the definition of P_j it follows that

$$\|P(D)(x_j\phi)\|_2 \leq (2m+1)R\|P(D)\phi\|_2 \qquad \forall\, \phi \in \mathcal{D}(\Omega). \tag{5.71}$$

The inequality (5.70) is trivial for $m = 0$, since in that case $P_j(D) = 0$. Suppose now that (5.70) holds for operators of order $m-1$, and let us prove it for $P(D)$ of order m. From the definition of $P_j(D)$ we have

$$\big(P(D)(x_j\phi)/P_j(D)\phi\big) = \big(x_j P(D)\phi/P_j(D)\phi\big) + \|P_j(D)\phi\|_2^2.$$

On the other hand,

$$\big(P(D)(x_j\phi)/P_j(D)\phi\big) = \big(x_j\phi/\overline{P}(D)P_j(D)\phi\big) = \big(\overline{P_j}(D)(x_j\phi)/\overline{P}(D)\phi\big).$$

Consequently,

$$\|P_j(D)\phi\|_2^2 = \big(\overline{P_j}(D)(x_j\phi)/\overline{P}(D)\phi\big) - \big(x_j P(D)\phi/P_j(D)\phi\big).$$

Now, since the degree of P_j is less than m, by the induction hypothesis we have

$$\|\overline{P_j}(D)(x_j\phi)\|_2 \leq (2m-1)R\|P_j(D)\phi\|_2.$$

On the other hand, by the Cauchy–Schwarz inequality,

$$\big|\big(x_j P(D)\phi/P_j(D)\phi\big)\big| \leq R\|P(D)\phi\|_2\|P_j(D)\phi\|_2.$$

Thus, we obtain

$$\|P_j(D)\phi\|_2^2 \leq 2mR\|P_j(D)\phi\|_2\|P(D)\phi\|_2,$$

which proves (5.70). Finally, we prove, by induction on m, that Hörmander's inequality holds. Clearly it is true for $m = 0$. Assume it holds for operators of order $m-1$, and let us prove it for an operator $P(D)$ of order m. There exists some $j \in \{1, 2, \ldots, N\}$ such that $P_j(D)$ has order $m-1$. Hence, by the induction hypothesis,

$$\|P_j(D)\phi\|_2 \geq C_1 \|\phi\|_2 \quad \forall\, \phi \in \mathcal{D}(\Omega). \tag{5.72}$$

Finally, from (5.70) and (5.72), we deduce

$$\|P(D)\phi\|_2 \geq \frac{1}{2mR}\|P_j(D)\phi\|_2 \geq \frac{C_1}{2mR}\|\phi\|_2,$$

which concludes the proof of the Hörmander inequality. □

Corollary 5.6 *Let $\Omega \subset \mathbb{R}^N$ be a bounded open set. For each $g \in L^2(\Omega)$, there exists $u \in L^2(\Omega)$ such that*

$$P(D)u = g \quad \text{in} \quad \mathcal{D}'(\Omega).$$

Moreover, there exists a constant $C > 0$ such that

$$\|u\|_2 \leq C\|g\|_2.$$

Proof By Hörmander's inequality we have that

$$\|\overline{P}(D)\phi\|_2 \geq C\|\phi\|_2 \qquad \forall\, \phi \in \mathcal{D}(\Omega). \tag{5.73}$$

On the other hand,

$$P(D)u = g \ \text{ in } \ \mathcal{D}'(\Omega) \iff (g/\phi) = (u/\overline{P}(D)\phi) \qquad \forall\, \phi \in \mathcal{D}(\Omega). \tag{5.74}$$

Let

$$X := \{\psi \in \mathcal{D}(\Omega) \ : \ \psi = \overline{P}(D)\phi \ \text{ for some } \ \phi \in \mathcal{D}(\Omega)\}.$$

We define $T : X \to \mathbb{C}$ by

$$T(\overline{P}(D)\phi) := (\phi/g).$$

Then, by (5.73) we have that

$$|T(\overline{P}(D)\phi)| \leq \|g\|_2\|\phi\|_2 \leq \frac{1}{C}\|g\|_2\|\overline{P}(D)\phi\|_2.$$

Therefore, T is continuous with respect to the $\|\ \|_2$-norm. Hence, it can be extended to a continuous functional on $\overline{X}^{L^2(\Omega)}$. Then, applying the Riesz Theorem in the Hilbert space $\overline{X}^{L^2(\Omega)}$, we obtain that there exists $u \in \overline{X}^{L^2(\Omega)}$ such that

$$(\overline{P}(D)\phi/u) = T(\overline{P}(D)\phi) := (\phi/g) \qquad \forall\, \phi \in \mathcal{D}(\Omega).$$

Thus, (5.74) holds, and the result is proved. □

We now proceed to establish a weighted Hörmander inequality.

Theorem 5.34 *Let $\Omega \subset \mathbb{R}^N$ be a bounded open set and $P(D) = \sum_{|\alpha|\leq m} a_\alpha D^\alpha$. There exists a constant $C > 0$ such that*

$$\int_\Omega e^{\eta x_1}|P(D)\phi|^2 \geq C\int_\Omega e^{\eta x_1}|\phi|^2 \qquad \forall\, \phi \in \mathcal{D}(\Omega),\ \ \eta \in \mathbb{R}.$$

Proof Let $Q(D)$ be the differential operator defined by

$$Q(D)\left(\exp\left(\frac{\eta x_1}{2}\right)\phi\right) = \exp\left(\frac{\eta x_1}{2}\right)P(D)\phi \qquad \forall\, \phi \in \mathcal{D}(\Omega).$$

It is easy to see that $Q(D)$ is a differential operator with constant coefficients and of the same order as $P(D)$. Then, applying Hörmander's inequality to $Q(D)$, we obtain

$$\int_\Omega e^{\eta x_1}|P(D)\phi|^2 = \left\|\exp\left(\frac{\eta x_1}{2}\right)P(D)\phi\right\|_2^2 = \left\|Q(D)\left(\exp\left(\frac{\eta x_1}{2}\right)\phi\right)\right\|_2^2$$

$$\geq C^2\left\|\exp\left(\frac{\eta x_1}{2}\right)\phi\right\|_2^2 = C^2\int_\Omega e^{\eta x_1}|\phi|^2.$$

□

Corollary 5.7 *If $\phi \in \mathcal{D}(\mathbb{R}^N)$ and $P(D)\phi = 0$ in the half-space $H = \{(x_1, x_2, \ldots, x_N) : x_1 > 0\}$, then $\phi = 0$ in H.*

Proof If the result were not true, we would have $\text{supp}(\phi) \cap \text{H} \neq \emptyset$. Let $\Omega \subset \mathbb{R}^N$ be a bounded open set such that $\text{supp}(\phi) \subset \Omega$. Now, since $\text{supp}(\text{P(D)}\phi) \cap \text{H} = \emptyset$, we have that for $\eta > 0$,

$$e^{\eta x_1}|P(D)\phi|^2 \leq |P(D)\phi|^2.$$

Therefore, applying the previous theorem, we get

$$\int_{\mathbb{R}^N} |P(D)\phi|^2 \geq \int_{\Omega} e^{\eta x_1} |P(D)\phi|^2 \geq C \int_{\Omega} e^{\eta x_1} |\phi|^2,$$

which leads to a contradiction since

$$\lim_{\eta \to +\infty} \int_{\Omega} e^{\eta x_1} |\phi|^2 = +\infty.$$

□

Corollary 5.8 *Let $\phi \in L^2(\mathbb{R}^N)$ with compact support. If*

$$\mathrm{supp(P(D)}\phi) \subset \mathrm{B(0, r)},$$

then

$$\mathrm{supp}(\phi) \subset \mathrm{B(0, r)}.$$

Proof First suppose that $\phi \in \mathcal{D}(\mathbb{R}^N)$. Using rotations and translations, the result follows from the previous corollary by expressing the ball as an intersection of half-spaces. Now suppose that $\phi \in L^2(\mathbb{R}^N)$. Let $\{\phi_m\}$ be a regularizing sequence, and let $\varphi_m := \phi_m \star \phi \in \mathcal{D}(\mathbb{R}^N)$. Then,

$$P(D)\varphi_m = (P(D)\phi) \star \phi_m,$$

so that

$$\mathrm{supp(P(D)}\varphi_{\mathrm{m}}) \subset \mathrm{B}\left(0, \mathrm{r} + \frac{1}{\mathrm{m}}\right).$$

Therefore, by the previous result,

$$\mathrm{supp}(\varphi_{\mathrm{m}}) \subset \mathrm{B}\left(0, \mathrm{r} + \frac{1}{\mathrm{m}}\right) \qquad \forall\, \mathrm{m} \in \mathbb{N}.$$

From this, since $\varphi_m \to \phi$ in $L^2(\mathbb{R}^N)$, we conclude that $\mathrm{supp}(\phi) \subset \mathrm{B(0, r)}$. □

For convenience, we will denote by B_r the ball $B(0, r)$ in $\mathbb{R}^N$. We have the following approximation theorem.

Theorem 5.35 *Let $0 < r < r' < R$. If $v \in L^2(B_{r'})$ satisfies $P(D)v = 0$ in $\mathcal{D}'(B_{r'})$, then there exists a sequence $\{v_m\} \subset L^2(B_R)$ such that $P(D)v_m = 0$ in $\mathcal{D}'(B_R)$ and $v_m \to v$ in $L^2(B_r)$.*

Proof By convolving with a mollifying sequence, we can assume that v is smooth.

Define

$$E := \left\{ u \in L^2(B_R) \ : \ P(D)u = 0 \text{ in } \mathcal{D}'(B_R) \right\}.$$

We must show that E is dense in the $L^2(B_r)$-norm in

$$F := \left\{ v \in L^2(B_{r'}) \ : \ P(D)v = 0 \text{ in } \mathcal{D}'(B_{r'}) \right\}.$$

Since E is convex, it suffices to prove weak density, or equivalently, that if $g \in L^2(B_r)$ satisfies $(u, g)_{L^2(B_r)} = 0$ for every $u \in E$, then $(v, g)_{L^2(B_r)} = 0$ for every $v \in F$.

To this end, we first prove that:

(*a*) If $g \in L^2(B_r)$ satisfies $(u, g)_{L^2(B_r)} = 0 \, \forall u \in E$, then $\exists w \in L^2(B_R)$ such that

$$(\varphi, g)_{L^2(B_r)} = (P(D)\varphi, w)_{L^2(B_R)} \quad \forall \varphi \in \mathcal{D}(\mathbb{R}^N).$$

By the Riesz Representation Theorem, (a) is equivalent to the existence of $C > 0$ such that

$$|(\varphi, g)_{L^2(B_r)}| \leq C \|P(D)\varphi\|_{L^2(B_R)} \quad \forall \varphi \in \mathcal{D}(\mathbb{R}^N), \tag{5.75}$$

because defining the operator T by $T(P(D)\varphi) := (\varphi, g)_{L^2(B_r)}$, inequality (5.75) implies the continuity of T on the closure

$$\overline{\{P(D)\varphi : \varphi \in \mathcal{D}(\mathbb{R}^N)\}}^{L^2(B_R)}.$$

We now prove (5.75). Note that if $P(D)\varphi = 0$, then $\varphi \in E$ and hence $(\varphi, g)_{L^2(B_r)} = 0$. Thus we may assume $P(D)\varphi \neq 0$. Then, by Corollary 5.7, there exists $\psi \in L^2(B_R)$ such that $P(D)\psi = P(D)\varphi$ and

$$\|\psi\|_{L^2(B_R)} \leq C_1 \|P(D)\varphi\|_{L^2(B_R)}.$$

Since $P(D)(\varphi - \psi) = 0$ in B_R, we have $\varphi - \psi \in E$ and thus

$$(\varphi - \psi, g)_{L^2(B_r)} = 0.$$

Consequently,

$$(\varphi, g)_{L^2(B_r)} = (\psi, g)_{L^2(B_r)}.$$

Therefore,

$$|(\varphi, g)_{L^2(B_r)}| = |(\psi, g)_{L^2(B_r)}| \le \|g\|_{L^2(B_r)} \|\psi\|_{L^2(B_r)}$$

$$\le C_1 \|g\|_{L^2(B_r)} \|P(D)\varphi\|_{L^2(B_r)},$$

which proves (5.75) and thus (a).

Define

$$\tilde{g}(x) := \begin{cases} g(x), & x \in B_r \\ 0, & x \in \mathbb{R}^N \setminus B_r \end{cases} \quad \text{and} \quad \tilde{w}(x) := \begin{cases} w(x), & x \in B_R \\ 0, & x \in \mathbb{R}^N \setminus B_R. \end{cases}$$

Then, for every $\varphi \in \mathcal{D}(\mathbb{R}^N)$, by (a) we have

$$(\varphi, \overline{P}(D)\tilde{w})_{L^2(\mathbb{R}^N)} = (P(D)\varphi, \tilde{w})_{L^2(\mathbb{R}^N)}$$

$$= (P(D)\varphi, w)_{L^2(B_R)} = (\varphi, g)_{L^2(B_r)} = (\varphi, \tilde{g})_{L^2(\mathbb{R}^N)}.$$

Hence, $\overline{P}(D)\tilde{w} = \tilde{g}$ in $\mathbb{R}^N$. Since $\operatorname{supp}(\tilde{w})$ is compact and

$$\operatorname{supp}(\overline{P}(D)\tilde{w}) = \operatorname{supp}(\tilde{g}) \subset B_r,$$

applying Corollary 5.8 we get $\operatorname{supp}(\tilde{w}) \subset B_r$. Thus, $w = 0$ in $B_R \setminus B_r$.

Finally, let $v \in \mathcal{D}(B_{r'})$ with $P(D)v = 0$ in $B_{r'}$. We show $(v, g)_{L^2(B_r)} = 0$. Indeed, let

$$\tilde{v}(x) := \begin{cases} v(x), & x \in B_{r'} \\ 0, & x \in \mathbb{R}^N \setminus B_{r'}. \end{cases}$$

Then, $P(D)\tilde{v} = 0$ in B_r. Hence, by (a),

$$(v, g)_{L^2(B_r)} = (\tilde{v}, g)_{L^2(B_r)} = (P(D)\tilde{v}, w)_{L^2(B_R)} = 0.$$

□

Let us now see that for each $g \in L^2_{\mathrm{loc}}(\mathbb{R}^N)$, the equation

$$P(D)u = g$$

is solvable in $L^2_{\mathrm{loc}}(\mathbb{R}^N)$.

Theorem 5.36 *Let $P(D)$ be a linear differential operator with constant coefficients. Then, for every $g \in L^2_{loc}(\mathbb{R}^N)$, there exists a function $u \in L^2_{loc}(\mathbb{R}^N)$ such that*

$$P(D)u = g \quad \text{in } \mathcal{D}'(\mathbb{R}^N).$$

Proof By Corollary 5.74, there exists $u_1 \in L^2(B_2)$ such that

$$P(D)u_1 = g \quad \text{in } \mathcal{D}'(B_2).$$

Assume inductively that we have chosen $u_m \in L^2(B_{m+1})$ such that

$$P(D)u_m = g \quad \text{in } \mathcal{D}'(B_{m+1}).$$

We then define u_{m+1} as follows: again, by Corollary 5.74, there exists $w \in L^2(B_{m+2})$ such that

$$P(D)w = g \quad \text{in } \mathcal{D}'(B_{m+2}),$$

so that

$$P(D)(u_m - w) = g \quad \text{in } \mathcal{D}'(B_{m+1}).$$

Then, by Theorem 5.35, there exists $v \in L^2(B_{m+2})$ such that

$$P(D)v = 0 \quad \text{in } \mathcal{D}'(B_{m+2}) \quad \text{and} \quad \|v - (u_m - w)\|_{L^2(B_m)} \leq \frac{1}{2^m}.$$

We set $u_{m+1} := v + w$. Then we have

$$P(D)u_{m+1} = g \quad \text{in } \mathcal{D}'(B_{m+2}) \quad \text{and} \quad \|u_{m+1} - u_m\|_{L^2(B_m)} \leq \frac{1}{2^m}.$$

Therefore, the sequence $\{u_m\}$ converges in $L^2_{\text{loc}}(\mathbb{R}^N)$ to a function $u \in L^2_{\text{loc}}(\mathbb{R}^N)$, which clearly satisfies

$$P(D)u = g \quad \text{in } \mathcal{D}'(\mathbb{R}^N).$$

□

Theorem 5.37 (Malgrange–Ehrenpreis Theorem) *Every linear differential operator with constant coefficients has a fundamental solution.*

Proof Let H be the Heaviside function on $\mathbb{R}^N$, i.e.,

$$H(x_1, \ldots, x_n) := \begin{cases} 1 \text{ if } x_j > 0 \ \forall\, j = 1, \ldots, N \\ \\ 0 \text{ otherwise.} \end{cases}$$

Then, if $\alpha = (1, \ldots, 1)$, we have that $D^\alpha H = \delta$. Now, since $H \in L^2_{loc}(\mathbb{R}^N)$, by the previous theorem, there exists $u \in L^2_{loc}(\mathbb{R}^N)$ such that $P(D)u = H$ in $\mathcal{D}'(\mathbb{R}^N)$. Let $E \in \mathcal{D}'(\mathbb{R}^N)$ be the distribution $E := D^\alpha u$. Then,

$$P(D)E = P(D)D^\alpha u = D^\alpha(P(D)u) = D^\alpha H = \delta.$$

□

Remark 5.6 There are other versions of the Malgrange–Ehrenpreis Theorem, such as the one given by Hörmander in [28], which, using more sophisticated proofs, provide more precise information about the fundamental solution. The idea behind Hörmander's proof is the following: one wants to solve the problem

$$P(D)E = \delta \qquad \text{in } \mathcal{D}'(\mathbb{R}^N).$$

Taking the Fourier transform, we obtain

$$P^*(\xi)\hat{E} = \mathbf{1}.$$

Therefore, if $P^*(\xi) \neq 0$ for every $\xi \in \mathbb{R}^N$, it is natural to try to define E as the inverse Fourier transform of the function $\frac{1}{P^*(\xi)}$, i.e.,

$$\langle E, \phi \rangle = \int_{\mathbb{R}^N} \frac{1}{P^*(\xi)} \mathcal{F}^{-1}\phi(\xi)\, d\xi.$$

In the general case, the idea is to integrate in the complex domain to avoid the zeros of $P^*(\xi)$.

L. Schwartz proposed the following *"division problem for distributions"*: Given a polynomial $P(x)$ and a tempered distribution $S \in \mathcal{S}'(\mathbb{R}^N)$, does there exist a tempered distribution $X \in \mathcal{S}'(\mathbb{R}^N)$ such that $P(x)X = S$? If this problem has a solution, then the linear partial differential equation $P(D)T = S$ has a solution $T \in \mathcal{S}'(\mathbb{R}^N)$. Indeed, applying the Fourier transform to the equation yields

$$P^*(\xi)\hat{T} = \hat{S}.$$

Hence, if $X \in \mathcal{S}'(\mathbb{R}^N)$ is the solution to this equation, then $T := \mathcal{F}^{-1}(X)$ is the solution to the equation $P(D)T = S$.

The division problem of Schwartz was solved independently by L. Hörmander in 1958 [27], and by S. Lojasiewicz in 1959 [38].

In the following example, we will see that if the differential operator does not have constant coefficients, the corresponding equation does not necessarily have a distributional solution.

Example 5.11 Let $f \in \mathcal{D}(\mathbb{R})$ such that

$$\int_0^\infty t f^2(t^2)\, dt \neq 0.$$

Define $g(x, y) := f(x^2 + y^2)$. Let us show that the equation

$$xu_y - yu_x = g \tag{5.76}$$

has no distributional solution in $\mathbb{R}^2$. Indeed, if $u \in \mathcal{D}'(\mathbb{R}^2)$ were a distributional solution of (5.76), then

$$\langle u, -(x\varphi)_y + (y\varphi)_x \rangle = \langle g, \varphi \rangle \qquad \forall\, \varphi \in \mathcal{D}(\mathbb{R}^2).$$

In particular, taking $\varphi = g$, we would obtain

$$\langle u, -(xg)_y + (yg)_x \rangle = \int_{\mathbb{R}^2} g^2 = 2\pi \int_0^\infty t f^2(t^2)\, dt \neq 0.$$

Now,

$$-(xg)_y + (yg)_x = -(xf(x^2 + y^2))_y + (yf(x^2 + y^2))_x = 0.$$

Hence, Eq. (5.76) has no distributional solution.

Remark 5.7 For a long time, it was believed that a linear partial differential equation should have plenty of solutions. In particular, suppose we have a linear equation

$$\sum_{|\alpha| \leq m} a_\alpha D^\alpha u = f, \tag{5.77}$$

with the functions a_α and f being of class C^∞. Given a point $x_0 \in \mathbb{R}^N$, does there exist a solution (not necessarily of class C^∞) that satisfies (5.77) in a neighborhood of x_0? According to the classical Cauchy–Kowalewski Theorem, the answer is positive when f and the a_α are analytic functions. It seemed reasonable to think that in the linear case, the analyticity assumption was not essential. However, in

1959, Hans Lewy gave a simple example that proved the conjecture false. Lewy's example is the differential operator in $\mathbb{R}^3$ defined by

$$Lu := u_x + iu_y - 2i(x + iy)u_z.$$

For a study of this operator, see for instance [32] or [54].

5.10 Hypoelliptic Operators

In this section, we study the regularity of distributional solutions. More specifically, we focus on linear differential operators with constant coefficients $P(D)$ such that the distributional solutions of the equation $P(D)u = f$ are smooth whenever f is smooth. One of the first results in this direction is the well-known *Weyl's Lemma*, obtained by H. Weyl in 1940, which states that if $u \in L^1_{loc}(\Omega)$ is a weak solution of the equation $\Delta u = 0$, then $u \in C^\infty(\Omega)$.

Definition 5.23 Let $\Omega \subset \mathbb{R}^N$ be an open set. A linear differential operator with constant coefficients $P(D)$ is said to be *hypoelliptic* in Ω if for every open set $U \subset \Omega$ and every distribution $T \in \mathcal{D}'(U)$, the implication

$$P(D)T \in C^\infty(U) \quad \Rightarrow \quad T \in C^\infty(U)$$

holds.

The following result characterizes the hypoellipticity of an operator in terms of its fundamental solutions.

Theorem 5.38 (Schwartz's Theorem) *Let $P(D)$ be a linear differential operator with constant coefficients. The following statements are equivalent:*

(i) $P(D)$ is hypoelliptic in $\mathbb{R}^N$.
(ii) Every fundamental solution of $P(D)$ is of class C^∞ in $\mathbb{R}^N \setminus \{0\}$.
(iii) $P(D)$ admits a fundamental solution of class C^∞ in $\mathbb{R}^N \setminus \{0\}$.

Proof (i) implies (ii): Let E be a fundamental solution of $P(D)$. Since $P(D)E = 0$ in $\mathcal{D}'(\mathbb{R}^N \setminus \{0\})$, and by hypothesis $P(D)$ is hypoelliptic, it follows that $E \in C^\infty(\mathbb{R}^N \setminus \{0\})$.

(ii) implies (iii): By the Malgrange–Ehrenpreis Theorem, there exists a fundamental solution of $P(D)$, and by (ii), this fundamental solution must be of class C^∞ in $\mathbb{R}^N \setminus \{0\}$.

(iii) implies (i): Let $U \subset \mathbb{R}^N$ be open, and $T \in \mathcal{D}'(U)$ such that $f := P(D)T \in C^\infty(U)$. We need to prove that $T \in C^\infty(U)$. It suffices to show that for any $x_0 \in U$, T is C^∞ in a neighborhood of x_0.

Let W be a relatively compact neighborhood of x_0 with $W \subset U$, and let $g \in \mathcal{D}(U)$ such that $g = 1$ in W. Applying the Leibniz rule, we have:

$$P(D)(gT) = gP(D)T + v = gf + v,$$

where v is a linear combination of derivatives of g of nonzero order. Therefore, $v = 0$ where the derivatives of g vanish. In particular,

$$v = 0 \quad \text{in } W \quad \text{and in } \mathbb{R}^N \setminus \text{supp(g)}.$$

Now, since by hypothesis there exists $E \in C^\infty(\mathbb{R}^N \setminus \{0\})$ such that $P(D)E = \delta$, we have:

$$E \star P(D)(gT) = (P(D)E) \star (gT) = \delta \star (gT) = gT.$$

Hence,

$$gT = E \star (gf) + E \star v.$$

Since $gf \in \mathcal{D}(\mathbb{R}^N)$, we have $E \star (gf) \in C^\infty(\mathbb{R}^N)$. Therefore, it suffices to show that $E \star v$ is of class C^∞ in a neighborhood of x_0; because then gT will be C^∞ in a neighborhood of x_0, and since $gT = T$ in W, it will follow that T is C^∞ in a neighborhood of x_0.

We now show that $E \star v$ is of class C^∞ near x_0. Let $\epsilon > 0$ be such that

$$V := \{x \in \mathbb{R}^N : d(x, \mathbb{R}^N \setminus W) > \epsilon\}$$

is a neighborhood of x_0 (e.g., $\epsilon = \frac{1}{2}d(x_0, \mathbb{R}^N \setminus W)$). Let $\phi \in \mathcal{D}(\mathbb{R}^N)$ such that $\phi(x) = 1$ if $\|x\| < \frac{\epsilon}{2}$ and $\phi(x) = 0$ if $\|x\| > \epsilon$. We write:

$$E \star v = (\phi E) \star v + [(\mathbf{1} - \phi)E] \star v.$$

By hypothesis, $(\mathbf{1}-\phi)E \in C^\infty(\mathbb{R}^N)$. Then, since v is a distribution with compact support, $[(\mathbf{1} - \phi)E] \star v \in C^\infty(\mathbb{R}^N)$. On the other hand, since

$$\text{supp}((\phi\text{E}) \star \text{v}) \subset \text{supp}(\phi\text{E}) + \text{supp(v)},$$

we get

$$\text{supp}((\phi\text{E}) \star \text{v}) \subset \bigcup_{\text{x}\in\text{supp(v)}} B(x, \epsilon).$$

Now, $\text{supp(v)} \subset \mathbb{R}^\text{N} \setminus \text{W}$. Therefore, $(\phi E) \star v = 0$ in V. Hence, $E \star v$ is of class C^∞ in V. □

As a consequence of Theorem 5.28, we know that there exist fundamental solutions of the Laplace operator that are of class C^∞ on $\mathbb{R}^N \setminus \{0\}$. Therefore, as a consequence of the previous theorem, we have that the Laplace operator is hypoelliptic, and consequently, the classical Weyl's Lemma:

Corollary 5.9 (Weyl's Lemma) *Let Ω be an open subset of $\mathbb{R}^N$, and let u be a distributional solution of $\Delta u = f$ in Ω. Then, if $f \in C^\infty(\Omega)$, it follows that $u \in C^\infty(\Omega)$, and consequently u is a classical solution.*

The problem of finding algebraic conditions on the polynomial $P(x)$ that characterize the hypoellipticity of the differential operator $P(D)$ was solved by L. Hörmander in 1957 in his thesis. In his textbook [28] there is a good exposition of these results (see also [12]). For example, we have the following result.

Theorem 5.39 (Hörmander's Theorem) *Let $P(D)$ be a linear differential operator in $\mathbb{R}^N$ with constant coefficients. The following are equivalent:*

(i) $P(D)$ is hypoelliptic.
(ii) Given a constant C_1, there exists a constant C_2 such that

$$P(\xi) = 0, \quad |\Re \xi| \leq C_1 \;\Rightarrow\; |\xi| \leq C_2.$$

(iii) There exists a function $C : \mathbb{R}^N \to \mathbb{R}^+$ with $C(\xi) \to 0$ as $\|\xi\| \to +\infty$, such that

$$|D^\alpha P^*(\xi)| \leq C(\xi)|P^*(\xi)| \quad \forall\, \xi \in \mathbb{R}^N, \;\; \forall \alpha \in \mathbb{N}^n, \; |\alpha| > 0.$$

Remark 5.8 In Theorem 5.31 we saw that the function

$$O_1(x,t) = \frac{c}{2} H(ct + x) H(ct - x)$$

is a fundamental solution of the one-dimensional wave operator. Since this function is not of class C^∞ in $\mathbb{R}^2 \setminus \{(0,0)\}$, by Schwartz's Theorem, we conclude that the operator is not hypoelliptic in $\mathbb{R}^2$. Similarly, in Theorem 5.32, we proved that the function

$$O_2(x,t) = \frac{c H(ct - \|x\|)}{2\pi \sqrt{c^2 t^2 - \|x\|^2}}$$

is a fundamental solution of the two-dimensional wave operator. By the previous argument, we also deduce that this operator is not hypoelliptic in $\mathbb{R}^3$. More generally, with the help of Hörmander's Theorem, we have that the wave operator in $\mathbb{R}^{N+1}$, i.e., the operator $\Box u = u_{tt} - \Delta_x u$, is not hypoelliptic, since $\Box = P(D)$ with $P(\xi) = \xi_1^2 - (\xi_2^2 + \cdots + \xi_{N+1}^2)$, and thus $P^*(\xi) = \xi_2^2 + \cdots + \xi_{N+1}^2 - \xi_1^2$. Hence, for $\alpha = (2, 0, \ldots, 0)$, we have $D^\alpha P^*(\xi) = -2$ for all $\xi \in \mathbb{R}^{N+1}$, and yet, $P^*(m, m, 0, \ldots, 0) = 0$ for all $m \in \mathbb{N}$.

Definition 5.24 A differential operator $P(D)$ is said to be *elliptic* if its principal part P_m satisfies

$$P_m(\xi) \neq 0 \qquad \forall\, \xi \in \mathbb{R}^N \setminus \{0\}.$$

Proposition 5.10 *Let $P(D)$ be a linear differential operator in $\mathbb{R}^N$ with constant coefficients. The following are equivalent:*

(i) $P(D)$ is elliptic.
(ii) There exists $C_1 > 0$ such that $|P_m(\xi)| \geq C_1 \|\xi\|^m \quad \forall\, \xi \in \mathbb{R}^N$.
(iii) There exist $C_2 > 0$ and $A > 0$ such that

$$|P^*(\xi)| \geq C_2 \|\xi\|^m \quad \forall\, \xi \in \mathbb{R}^N, \text{ with } \|\xi\| \geq A.$$

Proof (i) implies (ii): Since the map $\xi \mapsto |P_m(\xi)|$ is continuous, we have

$$\min_{\|\xi\|=1} |P_m(\xi)| = C_1 > 0.$$

Therefore, given $\xi \neq 0$, we get

$$C_1 \leq \left| P_m\left(\frac{\xi}{\|\xi\|}\right)\right| = \frac{1}{\|\xi\|^m} |P_m(\xi)|.$$

(ii) implies (iii): We write

$$P^*(\xi) = P_m(i\xi) + Q(i\xi).$$

Then,

$$|Q(i\xi)| \leq \sum_{|\alpha| \leq m-1} |c_\alpha| |(i\xi)^\alpha| \leq B\|\xi\|^{m-1} \quad \text{if} \quad \|\xi\| \geq 1.$$

Therefore, if $A := \max\{1, \frac{2B}{C_1}\}$, for $\xi \in \mathbb{R}^N$ with $\|\xi\| \geq A$, we have

$$|P^*(\xi)| \geq |P_m(i\xi)| - |Q(i\xi)| \geq C_1\|\xi\|^m - B\|\xi\|^{m-1} \geq C_2\|\xi\|^m.$$

(iii) implies (i): Using the above decomposition, for $\xi \in \mathbb{R}^N$ with $\|\xi\| \geq A$, we have

$$|P_m(\xi)| = |P^*(i\xi) - Q(i\xi)| \geq C_2\|\xi\|^m - B\|\xi\|^{m-1} \geq C_3\|\xi\|^m.$$

From this, since P_m is homogeneous, we conclude that $P_m(\xi) \neq 0$ for every $\xi \in \mathbb{R}^N \setminus \{0\}$. □

The most characteristic example of an elliptic operator is the Laplacian. The terminology *"hypoelliptic"* is semantically correct, since as a consequence of Hörmander's Theorem, we have that every elliptic operator is hypoelliptic.

Proposition 5.11 *Every elliptic operator is hypoelliptic.*

Proof If $P(D)$ is elliptic, then by the previous proposition, there exists $C_1 > 0$ such that $|P_m(\xi)|^2 \geq C_1\|\xi\|^{2m}$ for all $\xi \in \mathbb{R}^N$. Then, since $P_m = P + (P_m - P)$, we have

$$C_1\|\xi\|^{2m} \leq |P_m(\xi)|^2 = |P(\xi)|^2 + |P_m(\xi) - P(\xi)|^2 + 2|P_m(\xi) - P(\xi)||P(\xi)|$$

$$\leq |P(\xi)|^2 + C_2\big(\|\xi\|^{2m-1} + 1\big).$$

Then, if $\|\xi\| \geq \frac{2C_2}{C_1}$, we have

$$C_1\|\xi\|^{2m} \leq 2|P(\xi)|^2 + 2C_2.$$

From this it follows that there exists $C > 0$ such that if $\|\xi\| \geq \frac{2C_2}{C_1}$, then

$$1 + \|\xi\|^{2m} \leq C\big(|P(\xi)|^2 + 1\big).$$

From here, by Hörmander's Theorem, we deduce that $P(D)$ is hypoelliptic. □

The converse of the previous result is not true in general. An example is provided by the heat operator $P(D) = \frac{\partial}{\partial t} - \Delta_x$. By Schwartz's Theorem, we have that this operator is hypoelliptic, since its fundamental solution

$$G(x,t) := (4\pi t)^{-\frac{N}{2}} \exp\left(-\frac{\|x\|^2}{4t}\right) H(t)$$

is of class C^∞ on $\mathbb{R}^{N+1} \setminus \{0\}$; however, this operator is not elliptic since $P_m(\xi) = -(\xi_2^2 + \cdots + \xi_{N+1}^2)$, and so $P_m(1, 0, \ldots, 0) = 0$.

Clearly, every linear differential operator on $\mathbb{R}$ with constant coefficients is elliptic and, consequently, hypoelliptic. Therefore, we have the following result.

Proposition 5.12 *Let $\Omega \subset \mathbb{R}$ be an open set, and let u be a distributional solution of the ordinary differential equation*

$$a_0u + a_1u' + \cdots + a_mu^{(m)} = f \qquad \text{in} \quad \Omega,$$

with $f \in \mathbb{C}^\infty(\Omega)$. Then $u \in \mathbb{C}^\infty(\Omega)$, and consequently it is a classical solution.

There is the following relationship between ellipticity and hypoellipticity.

Theorem 5.40 *Let $P(D)$ be a linear differential operator with constant coefficients of order m. Then $P(D)$ is elliptic if and only if $P(D)$ is hypoelliptic and satisfies*

$$\sum_{j=1}^{n} \left| \frac{\partial P_m}{\partial \xi_j}(\xi) \right| \neq 0 \qquad \forall\, \xi \in \mathbb{R}^N \setminus \{0\}. \tag{5.78}$$

Proof Suppose that $P(D)$ is elliptic. By the previous proposition, we have that $P(D)$ is hypoelliptic. On the other hand, since P_m is a homogeneous function of degree m, by Euler's Theorem we have

$$m P_m(\xi) = \sum_{j=1}^{N} \xi_j \frac{\partial P_m}{\partial \xi_j}(\xi).$$

Then, Eq. (5.78) holds.

Conversely, suppose that (5.78) holds and that $P(D)$ is not elliptic. Then, there exists $\xi \neq 0$ such that $P_m(\xi) = 0$. If we take $y = \frac{1}{i} t\xi$, with $t \in \mathbb{R}$, we get

$$P^*(y) = P(t\xi) = O(|t|^{m-1}) \qquad \text{as } t \to \infty.$$

and

$$\sum_{j=1}^{N} \left| \frac{\partial P^*}{\partial y_j}(y) \right| = \left| D_j P(t\xi) \right| \sim |t|^m \sum_{j=1}^{N} |D_j P_m(\xi)| \quad \text{as } t \to \infty.$$

Consequently, $P(D)$ does not satisfy condition (iii) of Hörmander's Theorem and therefore is not hypoelliptic. □

5.11 The Schrödinger Equation

In this section, using the Fourier transform of tempered distributions, we will obtain distributional solutions to the Cauchy problem for the Schrödinger equation.

Suppose that a quantum particle of mass m moves in a force field with potential V. Then, if $u(x, t)$ denotes the wave function of the particle, i.e.,

$$\int_{\Omega} |u(x, t)|^2 \, dx$$

represents the probability that the particle is in Ω at time t, the function $u(x,t)$ satisfies the *Schrödinger equation*

$$i\hbar\frac{\partial u}{\partial t} = -\frac{\hbar^2}{2m}\Delta_x u + V(u), \tag{5.79}$$

where $\hbar$ is the Planck constant.

The probability density for the momentum of the particle is given by the Fourier transform of $u(\cdot,t)$. More precisely,

$$(2\pi)^{-N}\int_\Omega |\hat{u}(x,t)|^2\,dx,$$

where $\hat{u}$ denotes the Fourier transform with respect to the spatial variable x, represents the probability that the momentum of the particle is in Ω at time t.

This probabilistic interpretation requires that

$$\int_{\mathbb{R}^N} |u(x,t)|^2\,dx = 1 \quad \forall\, t \geq 0 \tag{5.80}$$

for physically relevant solutions. Note that by Plancherel's Theorem, if u satisfies (5.80), then

$$(2\pi)^{-N}\int_{\mathbb{R}^N} |\hat{u}(x,t)|^2\,dx = 1 \quad \forall\, t \geq 0,$$

so the probabilistic interpretation of momentum is also consistent.

In the case where there is no force field, i.e., for a free particle, the Schrödinger equation takes the form

$$\frac{\partial u}{\partial t}(x,t) = i\frac{\hbar}{2m}\Delta_x u(x,t). \tag{5.81}$$

The differential operator $P(D) := \frac{1}{i}\frac{\partial}{\partial t} + \Delta_x$ is called the *Schrödinger operator*. This operator is not hypoelliptic, since

$$P^*(\xi) = \xi_1 - (\xi_2^2 + \cdots + \xi_{N+1}^2),$$

so that

$$\frac{\partial P^*}{\partial \xi_1} = 1,$$

and if we take $\xi^m := (m, \sqrt{m}, 0, \ldots, 0)$, then $P^*(\xi^m) = 0$. Therefore, by Hörmander's Theorem, the Schrödinger operator is not hypoelliptic.

Let us consider the Cauchy problem for the Schrödinger equation:

$$\begin{cases} u_t = i\,\Delta_x u & \text{in } \mathbb{R}^+ \times \mathbb{R}^N \\ \\ u(x,0) = f(x) \quad x \in \mathbb{R}^N \end{cases}$$

with initial data $f \in \mathcal{S}(\mathbb{R}^N)$. We will solve this problem by applying the Fourier transform $\mathcal{F}_x$ with respect to the spatial variable $x \in \mathbb{R}^N$. Applying this transform leads to the ordinary differential equation:

$$\begin{cases} \dfrac{\partial}{\partial t}\mathcal{F}_x(u) = i\mathcal{F}_x(\Delta_x u) = -i\|\xi\|^2 \mathcal{F}_x(u) \\ \\ \hat{u}(\xi,0) = \hat{f}(\xi). \end{cases}$$

This equation has the solution

$$\mathcal{F}_x(u)(\xi,t) = \hat{f}(\xi)\exp(-it\|\xi\|^2),$$

so, since the function $h(\xi) := \hat{f}(\xi)\exp(-it\|\xi\|^2)$ belongs to $\mathcal{S}(\mathbb{R}^N)$, by the Inversion Formula, we obtain

$$u(x,t) = \mathcal{F}_x^{-1}\big(\hat{f}(\xi)\exp(-it\|\xi\|^2)\big)(x)$$

$$= (2\pi)^{-N}\int_{\mathbb{R}^N} e^{i\langle x,\xi\rangle}\exp(-it\|\xi\|^2)\hat{f}(\xi)\,d\xi.$$

We now aim to obtain another formula for the solution u. The function

$$g(\xi) := \exp(-it\|\xi\|^2)$$

does not belong to $L^1(\mathbb{R}^N)$ nor to $\mathcal{S}(\mathbb{R}^N)$. However, since g is continuous and bounded, we have $g \in \mathcal{S}'(\mathbb{R}^N)$. Consequently,

$$u(x,t) = \Big(\mathcal{F}_x^{-1}\big(\exp(-it\|\xi\|^2)\big) \star f\Big)(x). \tag{5.82}$$

To compute $\mathcal{F}_x^{-1}\big(\exp(-it\|\xi\|^2)\big)$, we will compute the Fourier transform of the tempered distribution defined by the function

$$\psi(x) := \exp(is\|x\|^2),$$

with $s \neq 0$. For $\phi \in \mathcal{S}(\mathbb{R}^N)$, we have

$$\langle \hat{\psi}, \phi\rangle = \langle \psi, \hat{\phi}\rangle.$$

Now, for $a > 0$, we have

$$\mathcal{F}\Big(\exp(-a\|x\|^2)\Big)(y) = \Big(\frac{\pi}{a}\Big)^{\frac{N}{2}} \exp\left(-\frac{\|y\|^2}{4a}\right).$$

Therefore, for $\phi \in \mathcal{S}(\mathbb{R}^N)$, we get

$$\int_{\mathbb{R}^N} \exp(-a\|x\|^2)\hat{\phi}(x)\,dx = \Big(\frac{\pi}{a}\Big)^{\frac{N}{2}} \int_{\mathbb{R}^N} \exp\left(-\frac{\|y\|^2}{4a}\right)\phi(y)\,dy. \tag{5.83}$$

We want to see whether (5.83) holds for $a = -is$ with $s \neq 0$. To this end, fix $\phi \in \mathcal{S}(\mathbb{R}^N)$ and define the complex functions, for $z \in \mathbb{C}$ with $\Re z \geq 0$:

$$F(z) := \int_{\mathbb{R}^N} \exp\left(-z\|x\|^2\right)\hat{\phi}(x)\,dx,$$

$$G(z) := \Big(\frac{\pi}{z}\Big)^{\frac{N}{2}} \int_{\mathbb{R}^N} \exp\left(-\frac{\|x\|^2}{4z}\right)\phi(x)\,dx.$$

For $\Re z > 0$, the integrals defining the functions F and G are convergent, hence F and G are holomorphic in the set $\{z \in \mathbb{C} \,:\, \Re z > 0\}$. Now, by (5.83), we have $F(a) = G(a)$ for every $a \in \mathbb{R}$, so by analytic continuation we get $F(z) = G(z)$ for all $z \in \mathbb{C}$ with $\Re z > 0$. Moreover, F and G are continuous on the set $\{z = is \,:\, s \in \mathbb{R} \setminus \{0\}\}$, since, by the Dominated Convergence Theorem,

$$\lim_{\epsilon \to 0^+} F(\epsilon + is) = \lim_{\epsilon \to 0^+} \int_{\mathbb{R}^N} \exp\big(-(\epsilon + is)\|x\|^2\big)\hat{\phi}(x)\,dx = F(is).$$

A similar argument holds for G. Consequently,

$$F(is) = G(is) \quad \forall\, s \in \mathbb{R} \setminus \{0\}.$$

Therefore,

$$\int_{\mathbb{R}^N} \exp\big(is\|x\|^2\big)\hat{\phi}(x)\,dx = \left(\frac{\pi}{-is}\right)^{\frac{N}{2}} \int_{\mathbb{R}^N} \exp\left(\frac{\|x\|^2}{4is}\right)\phi(x)\,dx,$$

from which we deduce that

$$\mathcal{F}\big(\exp(is\|x\|^2)\big)(y) = \left(\frac{\pi}{-is}\right)^{\frac{N}{2}} \exp\left(-i\,\frac{\|y\|^2}{4s}\right).$$

Now, for $s \neq 0$, we have

$$\mathcal{F}^{-1}\left(\exp\left(-i\frac{\|y\|^2}{4s}\right)\right)(x) = \left(\frac{\pi}{-is}\right)^{-\frac{N}{2}} \exp\left(is\|x\|^2\right).$$

We thus have that

$$\mathcal{F}^{-1}\left(\exp\left(-it\|x\|^2\right)\right)(y) = (4t\pi i)^{-\frac{N}{2}} \exp\left(i\frac{\|y\|^2}{4t}\right). \tag{5.84}$$

Then, as a consequence of (5.82) and (5.84), we obtain that

$$u(x,t) = (4t\pi i)^{-\frac{N}{2}} \int_{\mathbb{R}^N} \exp\left(i\frac{\|x-y\|^2}{4t}\right) f(y)\, dy.$$

We have established the following result.

Theorem 5.41 *For every $f \in \mathcal{S}(\mathbb{R}^N)$, the function*

$$u(x,t) = (4t\pi i)^{-\frac{N}{2}} \int_{\mathbb{R}^N} \exp\left(i\frac{\|x-y\|^2}{4t}\right) f(y)\, dy$$

is a distributional solution of the Cauchy problem

$$\begin{cases} u_t = i\Delta_x u & \text{in } \mathbb{R}^+ \times \mathbb{R}^N \\ u(x,0) = f(x) & x \in \mathbb{R}^N. \end{cases}$$

Problems

5.1 Prove that there does not exist a function $f \in L^1_{loc}(\mathbb{R}^N)$ such that

$$\int_{\mathbb{R}^N} f(x)\phi(x)\, dx = \phi(0), \qquad \forall\, \phi \in \mathcal{D}(\mathbb{R}^N).$$

5.2 Let $f_k \in L^1_{loc}(\mathbb{R}^N)$. The sequence $\{f_k\}$ is said to be δ-*convergent* if

$$\lim_{k\to\infty} \int_{\mathbb{R}^N} f_k(x)\phi(x)\, dx = \phi(0), \qquad \forall\, \phi \in \mathcal{D}(\mathbb{R}^N).$$

Let $0 \leq f \in L^1(\mathbb{R}^N)$, with $\int_{\mathbb{R}^N} f(x)\, dx = 1$. Prove that if $f_k(x) := kf(kx)$, $\{f_k\}$ is a δ-convergent sequence.

5.3 Prove that the following sequences are δ-convergent:

(i) $f_k(x) := \dfrac{1}{2} k e^{-k\|x\|}$.

(ii) $f_k(x) := \dfrac{\sin(kx)}{\pi x}$.

(iii) $f_k(x) := \dfrac{1}{2k} \|x\|^{\frac{1}{k}-1}$.

(iv) $f_k(x) := \dfrac{k}{2} \chi_{]-\frac{1}{k}, \frac{1}{k}[}$.

5.4 Prove that each of the following functionals is a distribution on Ω:

(i) $\langle T, \phi \rangle := \displaystyle\sum_{n=1}^{\infty} \phi^{(n)}(n)$, $\Omega = \mathbb{R}$.

(ii) $\langle T, \phi \rangle := \displaystyle\sum_{n=1}^{\infty} n^2 (\phi(\frac{1}{n}) - \phi(-\frac{1}{n}))$, $\Omega = \mathbb{R} \setminus \{0\}$.

(iii) $\langle T, \phi \rangle := \displaystyle\int_{\gamma} \phi \, ds$, $\Omega = \mathbb{R}^2$, where γ is a piecewise C^1 curve.

(iv) $\langle T, \phi \rangle := \displaystyle\int_{\mathbb{R}^2} \frac{\phi(x, y)}{x + iy} \, dxdy$, $\Omega = \mathbb{R}^2$.

(v) $\langle T, \phi \rangle := \displaystyle\int_{\mathbb{R}} \phi(x, -x) \, dx$, $\Omega = \mathbb{R}^2$.

(vi) $\langle T, \phi \rangle := \displaystyle\sum_{n=1}^{\infty} \left[\phi'\left(\frac{1}{n^2}\right) - \phi'(0) \right]$, $\Omega = \mathbb{R}$.

(vii) $\langle T, \phi \rangle := \displaystyle\int_0^{\infty} \frac{\phi(x) - \phi(0)}{\sqrt{x}} \, dx$, $\Omega = \mathbb{R}$.

5.5 Prove that

$$\left\langle \text{p.v.}\frac{1}{x}, \phi \right\rangle := \lim_{\epsilon \to 0^+} \int_{|x| \geq \epsilon} \frac{\phi(x)}{x} \, dx,$$

defines a distribution on $\mathbb{R}$ which is not a function.

5.6 Extend the concept of homogeneous function of degree m to distributions, and prove that the distribution $\text{p.v.}\frac{1}{x}$ is homogeneous of degree -1.

5.7 Let H be the Heaviside function. We define:

$$\left\langle \text{Pf}\frac{H}{x^2}, \phi \right\rangle := \lim_{\epsilon \to 0^+} \left(\int_{\epsilon}^{\infty} \frac{\phi(x)}{x^2} \, dx - \frac{\phi(0)}{\epsilon} + \phi'(0) \log \epsilon \right).$$

Prove that $\text{Pf}\frac{H}{x^2}$ is a distribution on $\mathbb{R}$.

5.8 Let f be a measurable function on $]0, 1[$ such that

$$\int_0^1 x^m |f(x)|\, dx < \infty, \quad m \in \mathbb{N}.$$

Prove that

$$\langle T, \phi \rangle := \int_0^1 f(x) \left(\phi(x) - \sum_{k=0}^{m-1} x^k \frac{\phi^{(k)}(0)}{k!} \right) dx$$

defines a distribution on $\mathbb{R}$ whose restriction to $]0, 1[$ is regular.

5.9 Let $0 \leq \phi \in \mathcal{D}(\mathbb{R})$ with $\operatorname{supp}(\phi) \subset [1, 2]$ and $\int_{\mathbb{R}} \phi(x)\, dx = 1$. Define $f_\epsilon(x) := \phi\left(\frac{x}{\epsilon}\right)$ for $0 < \epsilon < 1$. By studying the behavior of $\langle T, f_\epsilon \rangle$ as $\epsilon \to 0$, prove that there exists no distribution $T \in \mathcal{D}'(\mathbb{R})$ such that

$$\langle T, \phi \rangle = \int_0^\infty e^{\frac{1}{x}} \phi(x)\, dx,$$

for all $\phi \in \mathcal{D}(]0, +\infty[)$.

5.10 Prove that

$$\frac{\partial}{\partial x_i}(fT) = \frac{\partial f}{\partial x_i} T + f \frac{\partial T}{\partial x_i}$$

for every distribution $T \in \mathcal{D}'(\Omega)$ and every function $f \in C^\infty(\Omega)$.

5.11 Prove that

$$\frac{\partial^2 T}{\partial x_i \partial x_j} = \frac{\partial^2 T}{\partial x_j \partial x_i}$$

for every distribution $T \in \mathcal{D}'(\Omega)$.

5.12 Compute $x^m \delta^{(k)}$, where $m, k \in \mathbb{N}$.

5.13 Let $T \in \mathcal{D}'(\Omega)$ and $\phi \in \mathcal{D}(\Omega)$. Determine the relationship between the following two statements:

(i) $\langle T, \phi \rangle = 0$
(ii) $\phi T = 0$

5.14 Compute the support of the following distributions:

(i) $\langle T, \phi \rangle := \int_{\mathbb{R}} \phi(x, -x)\, dx,\ \Omega = \mathbb{R}^2.$

(ii) $\langle T, \phi \rangle := \sum_{n=1}^{\infty} \left[\phi'\left(\frac{1}{n^2}\right) - \phi'(0) \right],\ \Omega = \mathbb{R}.$

5.15

(i) Let $T \in \mathcal{D}'(\Omega)$ and $\phi \in \mathcal{D}(\Omega)$. Suppose that ϕ vanishes on the support of T. Can we conclude that $\langle T, \phi \rangle = 0$?

(ii) ° Let B be the unit ball in $\mathbb{R}^N$ and $T \in \mathcal{D}'(\mathbb{R}^N)$ such that supp(T) $\subset B$. Prove that $\langle T, \phi \rangle = 0$ for every $\phi \in \mathcal{D}(\Omega)$ that vanishes on B.

5.16 Let $\Omega =]a, b[$, $x_0 \in \Omega$, and $f \in C^1(\Omega \setminus \{x_0\})$, such that f' is bounded in a punctured neighborhood of x_0, and the one-sided limits $f(x_0+)$ and $f(x_0-)$ exist. Prove that

$$(T_f)' = T_{f'} + (f(x_0+) - f(x_0-))\delta_{x_0}.$$

5.17 Let

$$f(x) := \begin{cases} |x| & \text{if } -1 < x < 1 \\ \\ 1 & \text{if } |x| \geq 1 \end{cases}$$

Compute $(T_f)''$.

5.18 Compute the distributional derivative of the following functions:

(i) $f(x) = \frac{\sin x}{|x|}$

(ii) $f(x) = \log |x|$

(iii) $f(x) = H(x) \log |x|$

(iv) $f(x) = 0$ if $x < 0$, and $f(x) = \frac{1}{\sqrt{x}}$ if $x > 0$

5.19 Let $g(x) := \exp(ie^{\frac{1}{x}})$. Show that $T := ix^2(T_g)'$ defines a distribution on $\mathbb{R}$ whose restriction to $]0, +\infty[$ coincides with the function $f(x) = e^{\frac{1}{x}} \exp(ie^{\frac{1}{x}})$.

5.20 For $\epsilon > 0$, let $f_\epsilon(x) := \log_\pi(x + i\epsilon)$, and let f_0 be the function defined by: $f_0(x) := \log x$ when $x > 0$ and $f_0(x) := \log |x| + i\pi$ when $x < 0$. Prove that $\lim_{\epsilon \to 0^+} f_\epsilon = f_0$ in $\mathcal{D}'(\mathbb{R})$. Compute f_0', and deduce that

$$\lim_{\epsilon \to 0^+} \frac{1}{x + i\epsilon} = -i\pi\delta + \text{p.v.}\, \frac{1}{x}.$$

5.21 Prove that the Dirac delta is not the m-th derivative of any continuous function with compact support.

5.22 Let $\overline{\partial} = \frac{1}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right)$ be the Cauchy–Riemann operator, and let $f(z) = \frac{1}{z}$, $z \in \mathbb{C}$. Prove that $\overline{\partial} f = \pi\delta$.

5.23 Prove that the partial derivatives of the characteristic function of the unit ball in $\mathbb{R}^N$ are distributions supported on the unit sphere.

5.24 Let f be the characteristic function of the unit disk in $\mathbb{R}^2$. Prove that $x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y}$ is a distribution of order zero whose support is the unit circle.

5.25 Characterize all distributions on $\mathbb{R}$ whose support consists exactly of two points.

5.26 Let $T \in \mathcal{D}'(\mathbb{R})$ such that $x^m T = 0$. Prove that T is a linear combination of $\delta, \delta', \ldots, \delta^{(m-1)}$. Find the general solution of the equation $xT = 1$ in $\mathcal{D}'(\mathbb{R})$.

5.27 Let f_a be the characteristic function of the interval $]0, a[$ with $a > 0$, and let T be the distribution $T := \sum_{k=0}^{\infty} \delta'_{ka}$. Prove that $f_a \star T = \delta$.

5.28 Let

$$\phi_k(x) := \frac{k^n}{\pi^{\frac{n}{2}}}\left(1 - \frac{\|x\|^2}{k}\right)^{k^3}, \quad x \in \mathbb{R}^N.$$

Prove that the sequence $\{\phi_k\}$ is δ-convergent and deduce that every distribution with compact support is the limit of a sequence of polynomials.

5.29 Prove that the distribution generated by the function $f(x) = e^x$ is not tempered, but the one generated by $g(x) = e^x \cos e^x$ is tempered.

5.30 Let $\phi \in \mathcal{D}(\mathbb{R})$ be nonzero, and let $\phi_n(x) := \frac{1}{n}\phi(x - n)$, $n \in \mathbb{N}$. Consider:

(i) Find a distribution T such that $\lim_{n\to\infty} \langle T, \phi_n\rangle \neq 0$.
(ii) Does there exist a tempered distribution T such that $\lim_{n\to\infty} \langle T, \phi_n\rangle \neq 0$?

5.31 Compute the Fourier transform of the following functions:

(i) $f(x) = \frac{1}{1+x^2}$.
(ii) $g(x) = \frac{x}{1+x^2}$.
(iii) $h(x) = \frac{x}{x+i}$.

5.32 Let $f, g \in \mathcal{S}(\mathbb{R}^N)$ such that $f \star g = 0$. Can one deduce that $f = 0$ or $g = 0$?

5.33 Let $f \in \mathcal{S}(\mathbb{R}^2)$, and define $g(x) := f(x, 0)$. Prove that

$$\hat{g}(\xi) = \frac{1}{2\pi}\int_{\mathbb{R}^2} \hat{f}(\xi, \eta)\, d\eta.$$

5.34 Compute the Fourier transform of the following distributions on $\mathbb{R}$:

(i) $(\delta_a)^{(k)}$.
(ii) $f(x) = x^k$.
(iii) $f(x) = \cos ax$, $a \in \mathbb{R}$.
(iv) $f(x) = \sin^3 x$.
(v) $f(x) = \frac{\sin ax}{\pi x}$.
(vi) $f(x) = \operatorname{sign} x$.

5.35 Let f be the function supported on $[-1, 1]$ whose graph is the triangle with vertices $(-1, 0)$, $(0, b)$, $(1, 0)$. Compute the Fourier transform of f.

5.36 Given that $\overline{\partial}(\frac{1}{z}) = \pi\delta$, we ask:

(i) Compute the Fourier transform of $f(z) = \frac{1}{z}$.
(ii) Prove that for every test function ϕ one has

$$\phi(z) = -\frac{1}{\pi} \int_{\mathbb{R}^2} \frac{\overline{\partial}(w)}{w - z}\, dw.$$

5.37 Prove that $VP\left(\frac{1}{x}\right)$ is a tempered distribution and compute its Fourier transform.

5.38 Compute the Fourier transform of the Heaviside function and of the function $f(x) = |x|$.

5.39 Let $f \in L^2(\mathbb{R})$. Show that if f is nonzero, then f and $\hat{f}$ cannot both vanish outside a bounded interval.

5.40 Given the sequence of functions $f_k(x) := \displaystyle\int_0^k y \cos(xy)\, dy$, compute $\hat{f}_k$ and prove that the sequence $\{f_k\}$ is convergent in $\mathcal{S}'(\mathbb{R})$.

5.41 Given $f \in \mathcal{S}(\mathbb{R})$, define the 2π-periodic function

$$F(x) := \sum_{k=-\infty}^{+\infty} f(x - 2k\pi).$$

Suppose it is known that F admits a Fourier series expansion $F(x) = \sum_{k=-\infty}^{+\infty} c_k e^{ikx}$. Prove that $c_k = \frac{1}{2\pi}\hat{f}(k)$, and as a consequence, deduce the *Poisson summation formula*

$$\sum_{k=-\infty}^{+\infty} f(2k\pi) = \frac{1}{2\pi} \sum_{k=-\infty}^{+\infty} \hat{f}(k).$$

5.42 Prove that if T is a distribution on $\mathbb{R}$ with compact support such that $\langle T, x^n \rangle = 0$ for all $n \in \mathbb{N}$, then $T = 0$.

5.43 Let S, T be nonzero distributions with compact support. Prove that $S \star T$ is nonzero.

5.44 Given $f(x, y) = \log \sqrt{x^2 + y^2}$, do the following:

(i) Prove that f is harmonic in $\mathbb{R}^2 \setminus \{0\}$.
(ii) Prove that $\Delta f = 2\pi\delta$, i.e., $E(x, y) := \frac{\pi}{2} \log \sqrt{x^2 + y^2}$ is the fundamental solution of the Laplace operator in the plane.
(iii) If $u(z) = 0$ when $|z| < 1$ and $u(z) = f(z)$ when $|z| > 1$, prove that Δu is the arc length measure on the unit circle.

5.45 Let Ω be a bounded domain in $\mathbb{R}^N$ with smooth boundary, and $N \geq 2$. Let $\phi_N : [0, +\infty[\to \mathbb{R}$ be the function

$$\phi_N(r) := \begin{cases} \frac{1}{2\pi} \log(r) & \text{if } N = 2 \\ \\ \frac{1}{(2-N)\sigma_N} r^{2-N} & \text{if } N \geq 3, \end{cases}$$

i.e., $\phi_N(\|x\|) = E_N(x)$, where E_N is the fundamental solution of the Laplacian in $\mathbb{R}^N$. Prove that if $u \in C^2(\Omega) \cap C(\overline{\Omega})$ and $B(x, r) \subset \Omega$, then

$$u(x) = \int_{B(x,r)} \Delta u(y)(\phi_N(\|x - y\|) - \phi_N(r))\, dy$$

$$+ \frac{1}{\sigma_N r^{N-1}} \int_{\|x-y\|=r} u(y)\, dS(y).$$

As a consequence, if $\Delta u = 0$ in Ω, then

$$u(x) = \frac{1}{\sigma_N r^{N-1}} \int_{\|x-y\|=r} u(y)\, dS(y),$$

i.e., every harmonic function satisfies the mean value property, and if $\Delta u \leq 0$ in Ω, then

$$u(x) \geq \frac{1}{\sigma_N r^{N-1}} \int_{\|x-y\|=r} u(y)\, dS(y).$$

5.46 Let $\mathbf{a} = (a_1, \ldots, a_N) \in \mathbb{R}^N$, and $\lambda \in \mathbb{R}$. Prove that the differential operator $a_1 \frac{\partial}{\partial x_1} + \cdots + a_N \frac{\partial}{\partial x_N} - \lambda$ admits as fundamental solution the distribution E defined by

$$\langle E, \phi \rangle = \int_0^\infty \phi(t\mathbf{a}) e^{\lambda t}\, dt.$$

Chapter 6
Sobolev Spaces



6.1 Introduction

One of the most important problems in Analysis at the end of the nineteenth century, which drove the development of Analysis in the twentieth century, is the so-called *Dirichlet problem*, i.e., the problem of the existence of a harmonic function in a domain Ω of $\mathbb{R}^N$ that takes prescribed values on its boundary. In 1850, using an observation by Gauss, Dirichlet proved the existence of a solution to the problem by assuming what is now known as the *Dirichlet principle*, which states that the solution u to the Dirichlet problem is, among all the functions v that take the prescribed boundary values, the one that minimizes the *energy integral* (also called the *Dirichlet integral*)

$$\int_\Omega |\nabla v|^2.$$

Riemann accepted the validity of the Dirichlet principle in his work on potential theory, but neither Gauss nor Riemann gave any rigorous proof of this principle, so criticism regarding its validity soon began to emerge.

Towards the end of the nineteenth century, progress in abstraction and better understanding of the foundations of Calculus gave rise to an era of criticism. Weierstrass, in lectures given at the Berlin Academy in 1870, presented an example of a variational problem that has no minimum. Weierstrass's example consists of finding a continuously differentiable function $u : [-1, 1] \to \mathbb{R}$ that minimizes the integral

$$I(u) = \int_{-1}^{1} |xu'(x)|^2 \, dx,$$

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J. M. Mazón, *Elements of Functional Analysis*, Universitext,
https://doi.org/10.1007/978-3-032-29210-0_6

subject to the boundary conditions $u(\pm 1) = \pm 1$. Choosing the functions

$$u_n(x) := \frac{\operatorname{arctg}(nx)}{\operatorname{arctg}(n)},$$

Weierstrass showed that the infimum must be zero; however, the value zero is not admissible. With this criticism, the Dirichlet Principle fell into disrepute and underwent a crisis.

Due to Weierstrass's example, the principle was forgotten until 1899, when Hilbert revived it, showing that for a broad class of domains, the principle was valid, and therefore, the Dirichlet problem had a solution for such domains. For a long time, it was believed that the Dirichlet problem did have a solution, but that the limitations were inherent to the methods used to prove it rather than to the problem itself. In 1911, the Polish mathematician E. Zaremba was the first to find a domain for which the Dirichlet problem has no solution. In 1913, Lebesgue provided a more significant example of the impossibility of the Dirichlet problem (see [33], p. 334). These results gave rise to a broader concept of solution, the concept of a *weak solution*. In this generalized context, the Dirichlet problem is equivalent to a variational problem in a Hilbert space of functions. The appropriate Hilbert spaces of functions for this new formulation of the Dirichlet problem are the so-called *Sobolev spaces*. Sobolev introduced these spaces in 1936[1] as the suitable functional framework for obtaining an existence theorem for the Cauchy problem associated with a hyperbolic partial differential equation; although the first person to consider the type of functions that form the Sobolev spaces was Beppo Levi (Pag. 6.2) in 1906.[2]

In his first work on the Dirichlet Principle, where the Lebesgue integral is used instead of the Riemann integral, Levi states that he makes this change not out of *"amore di generalità"*, but out of *"necessità di cose"*. This work by Levi was continued by Otto Nikodym in 1935.[3]

One of the antecedents of the spaces defined by Sobolev are the spaces considered by J. Schauder in 1935[4] in his study of a class of nonlinear hyperbolic equations. In order to apply his fixed point theorem, he needed to find a suitable compact set. He achieved this using a method introduced in 1926 by Friedrichs and Lewy, a method that led him to consider the space of functions f of class C^m in Ω such that $D^\alpha f \in L^2(\Omega)$ for all $|\alpha| \leq m$, endowed with the norm

$$\|f\| = \sqrt{\int_\Omega \sum_{|\alpha| \leq m} |D^\alpha f(x)|^2 \, dx}.$$

[1] Mat. Sb. **1** (1936), 39–71.

[2] Rend. Circ. Mat, Palermo **22** (1906), 239–358.

[3] Math. Cluj. **9** (1935), 110–128.

[4] Fund. Math. **24** (1935), 213–246.

Unfortunately, this space is not complete. In his 1936 work, Sobolev had the idea to consider functions in $L^2(\Omega)$ whose generalized derivatives up to order m belong to $L^2(\Omega)$. The space of all such functions, denoted by $H^m(\Omega)$, is complete with the above norm. These spaces were rediscovered in 1940 by J. W. Calkin[5] and C. B. Morrey Jr.[6] in their studies on variational problems. It was later observed that the spaces $H^m(\mathbb{R}^N)$ could be defined using the Fourier transform of tempered distributions, which allowed the definition to be extended to any real number m.

In 1934, Friedrichs[7] introduced the space $H_0^1(\Omega)$, i.e., the closure in $H^1(\Omega)$ of $\mathcal{D}(\Omega)$, while studying the problem of finding a minimum of an elliptic quadratic functional that vanishes on the boundary of Ω. Another basic tool used by Friedrichs is an inequality that was used by Poincaré in 1894[8] in his study of the eigenvalues of the heat equation. Friedrichs recognized the importance of this type of inequality in the study of elliptic equations.

In his 1936 work, Sobolev observed that increasing the index m results in the functions in the corresponding spaces being more regular. More precisely, if $\Omega \subset \mathbb{R}^N$ and $m > [\frac{N}{2}] + 1$, then $H^m(\Omega) \subset C^r(\Omega)$, with $r = m - [\frac{N}{2}] - 1$. Hence, the intersection of the spaces $H^m(\Omega)$ consists of C^∞ functions whose derivatives are square integrable. Sobolev himself, in 1938,[9] obtained more general results on continuous embeddings. Results on continuous embeddings were also obtained by Morrey in 1940 and later, among others, by K. O. Friedrichs,[10] E. Gagliardo,[11] and L. Nirenberg.[12]

These continuous embedding results are very important in relation to the regularity problem, i.e., knowing when a weak solution of a partial differential equation is actually a classical solution.

Another type of important embedding theorems in the theory of Sobolev spaces are the so-called compact embedding theorems, i.e., when the embedding is a compact map. These allow the application of Riesz–Fredholm theory on compact operators, and enable the proof, among other interesting results, that the spectrum of elliptic linear operators on bounded domains is discrete. The origin of these compact embedding results is a theorem by F. Rellich from 1930,[13] which asserts that the embedding of the space $H^1(\Omega)$ into $L^2(\Omega)$ is compact, where Ω is a bounded open set of class C^1. This result was generalized by V. I. Kondrachov.[14]

[5] Duke Math. J. **6** (1940), 170–186.

[6] Duke Math. J. **6** (1940), 187–215.

[7] Math. Ann. **109** (1934), 465–487; 685–713.

[8] Rend. Circ. Mat. Palermo **8** (1894), 57–155.

[9] Mat. Sb. **46** (1938), 471–496.

[10] Trans. Amer. Math. Soc. **55** (1944), 132–151.

[11] Ricerche Mat. **7** (1958), 102–137.

[12] Ann. Scuola Norm. Sup. Pisa **20**(1966), 733–737.

[13] Göttingen Nachr (1930), 30–35.

[14] Dokl. Akad. Nauk. SSSR **48** (1945), 535–538.

In the study of boundary value problems, the issue of giving meaning to the restriction of a function to the boundary of its domain arises naturally; this is known as the trace of the function on the boundary. The problem of characterizing the trace on a domain was solved by J. L. Lions in the early 1960s using interpolation between Sobolev spaces. This theory is presented in his textbook with E. Magenes [39].

In 1964, N. Meyer and J. Serrin[15] proved that Sobolev spaces are the completion of the spaces of differentiable functions with respect to the Sobolev norm. It is surprising, especially because of the simplicity of the proof, that this result—which clarifies the relationship between these spaces—remained undiscovered for so long.

Our study of Sobolev spaces focuses almost exclusively on those properties we need to develop, in the next chapter, the study of boundary value problems for elliptic equations. Sobolev spaces are among the clearest examples of one of the most important applications of Functional Analysis to partial differential equations, namely, the selection or construction—according to the requirements of the problem to which they are applied—of specific function spaces. A thorough study of Sobolev spaces, as well as their generalizations—Sobolev spaces with weights and Sobolev–Orlicz spaces—can be found in the textbook by Adams [1].

Functions of bounded variation on the real line were introduced by C. Jordan in 1881[16] in connection with Dirichlet's criterion for the convergence of Fourier series. In that work, Jordan proved his characterization of functions of bounded variation as the difference of two monotone functions. In 1905, G. Vitali[17] introduced absolutely continuous functions of one variable and gave his famous example of a continuous function of bounded variation that is not absolutely continuous. In the same work, he introduced functions of bounded variation of two variables by means of their variation over rectangles. This same definition was later given by H. Lebesgue in 1910 and by Ch. J. De La Vallée-Poussin in 1916.

L. Tonelli, who was the first to observe the geometric aspects of functions of bounded variation in one variable—more specifically, their connection with the length of curves—noticed that Vitali's definition for functions of two variables was not an adequate generalization of the one-dimensional case. He proposed, as a definition in the two-dimensional case, to call functions of bounded variation in $\Omega \subset \mathbb{R}^2$ those continuous functions for which the surface area of the projections of their graph onto the vertical coordinate planes is finite. L. Cesari[18] generalized Tonelli's definition to the case of integrable functions, giving the definition of what is now known as functions of bounded variation in the sense of Tonelli–Cesari, which is the following: for simplicity, we assume that $\Omega = [a, b]\times[a, b]$. A function $f : \Omega \to \mathbb{R}$ is said to be of bounded variation in the sense of Tonelli–Cesari if there

[15] Proc. Nat. Acad. Sci. USA **51** (1964), 1055–1056.

[16] C. R. Acad. Sci. Paris **XCII** (1881), 228–230.

[17] Atti Accad. Sci. Torino Cl. Sci. Fis. Mat. Natur. **40** (1904–5), 1021–1034.

[18] Ann. Scuola Norm. Sup. Pisa. **5** (1936), 299–313.

exists $g = f$ almost everywhere in Ω such that

$$\int_a^b V_a^b(g(x,\cdot))\,dx < \infty, \qquad \int_a^b V_a^b(g(\cdot,y))\,dy < \infty,$$

where $V_a^b(h)$ denotes the total variation of h on $[a,b]$.

In 1954, G. Fichera and E. De Giorgi were the first to study functions of bounded variation within the framework of Distribution Theory. Fichera, for $f : \Omega \subset \mathbb{R}^N \to \mathbb{R}$ continuous, said that f is of bounded variation if its partial derivatives, in the sense of distributions, are measures with finite total variation, and he proved the equivalence of this concept with the one introduced by Tonelli. De Giorgi also considered functions whose distributional derivatives are measures. More precisely, for $f \in L^\infty(\mathbb{R}^N)$, if $\phi_\lambda(x) = (\pi\lambda)^{-N/2}\exp(-\|x\|^2/\lambda)$, he considered

$$I(f) := \lim_{\lambda \downarrow 0} \int_{\mathbb{R}^N} |\nabla(f \star \phi_\lambda)|\,dx,$$

and proved that $I(f)$ is finite if and only if the distributional gradient of f, Df, is an $\mathbb{R}^N$-valued measure with finite total variation $\|Df\|$. Moreover, it holds that

$$\|Df\|(\mathbb{R}^N) = I(f).$$

In 1952, R. Caccioppoli[19] defined the perimeter of a measurable set $E \subset \mathbb{R}^N$ as

$$\mathrm{Per}(E) :=$$

$$\inf\left\{\liminf_{h\to\infty} \mathrm{Per}(E_h) \ : \ \partial E_h \text{ polyhedral}, \ \lim_{h\to\infty} \mathcal{L}^N(E_h \Delta E) = 0\right\},$$

where $\mathrm{Per}(E_h)$ is the usual perimeter of the polyhedral approximant E_h. In 1954, De Giorgi[20] also proved that $I(\chi_E)$ coincides with the perimeter of E in the sense of Caccioppoli, that is,

$$I(\chi_E) = \|D\chi_E\|(\mathbb{R}^N) = \mathrm{Per}(E).$$

In 1964, M. Miranda,[21] in order to characterize the functions $f \in L^1_{loc}(\mathbb{R}^N)$ whose distributional gradient is a measure, introduced the variation $V(f,\Omega)$ of f on an open set $\Omega \subset \mathbb{R}^N$ as

$$V(f,\Omega) := \sup\left\{\int_\Omega f\ \mathrm{div}(\varphi)\,dx \ : \ \varphi \in (C_c^1(\Omega))^N, \ \|\varphi\|_\infty \le 1\right\}.$$

[19] Rend. Acc. Naz. Lincei **12** (1952), 3–11, 137–146.

[20] Ann. Mat. Pura Appl. **36** (1954), 191–213.

[21] Ann. Scuola Norm. Sup. Pisa Cl. Sci. (3) **18** (1964), 27–56.

In the case when f is continuously differentiable on Ω, a simple integration by parts shows that

$$V(f,\Omega)=\int_\Omega \|\nabla f\|\,dx.$$

He also used the variation $V(f,\Omega)$ to define the perimeter of a set. Later, in 1967, motivated by the Dirichlet problem for the mean curvature equation, he proved the trace theorem for functions of bounded variation.

The most important contributions on the fine properties of functions of bounded variation were obtained in the mid-1960s by H. Federer and A. I. Vol'pert. Federer's motivations were geometric in nature, and his results are compiled in his textbook [18]. Vol'pert developed his results with an eye toward applications to partial differential equations [58].

6.2 Sobolev Spaces

Let $\Omega\subset\mathbb{R}^N$ be an open set and $1\le p\le\infty$. The norm in the Lebesgue space $L^p(\Omega)$ will be denoted by $\|\ \|_p$, and the Lebesgue measure on $\mathbb{R}^N$ will be denoted by $\mathcal{L}^N$. In the previous chapter we saw that, considered as distributions, functions in $L^1_{loc}(\Omega)$ can be differentiated in the weak sense. More precisely, given $f\in L^1_{loc}(\Omega)$ and a multi-index $\alpha\in\mathbb{N}^N$, we define $D^\alpha f$ as the distribution

$$\langle D^\alpha f,\phi\rangle:=(-1)^{|\alpha|}\int_\Omega f(x)D^\alpha\phi(x)\,dx,\qquad\forall\,\phi\in\mathcal{D}(\Omega).$$

In this chapter we study spaces of integrable functions whose weak derivatives are also functions. Specifically, for $m=0,1,2,\ldots$, the *Sobolev space* of order m, p on Ω is defined as

$$W^{m,p}(\Omega):=\left\{f\in L^p(\Omega)\ :\ D^\alpha f\in L^p(\Omega)\ \ \forall\,|\alpha|\le m\right\}.$$

We equip this space with the norm

$$\|f\|_{m,p}:=\sum_{|\alpha|\le m}\|D^\alpha f\|_p.$$

Clearly, in the case $1\le p<\infty$, the norm $\|\ \|_{m,p}$ is equivalent to the norm

$$N_{m,p}(f):=\left(\sum_{|\alpha|\le m}\|D^\alpha f\|_p^p\right)^{\frac{1}{p}}.$$

Note that for $m = 0$, $W^{m,p}(\Omega)$ coincides with $L^p(\Omega)$, and that for $p = 2$, the norm is induced by the inner product

$$(f|g) := \sum_{|\alpha| \le m} \int_\Omega D^\alpha f \, \overline{D^\alpha g}.$$

In this case, the Sobolev space is usually denoted by $H^m(\Omega)$, i.e., $H^m(\Omega) := W^{m,2}(\Omega)$.

Proposition 6.1 *Let $f \in W^{m,p}(\Omega)$, $|\alpha| \le m$. Then:*

(i) $D^\alpha f \in W^{m-|\alpha|,p}(\Omega)$ and $D^\beta(D^\alpha f) = D^\alpha(D^\beta f) = D^{\alpha+\beta} f$, whenever $|\alpha| + |\beta| \le m$.

(ii) If $\phi \in \mathcal{D}(\Omega)$, then $\phi f \in W^{m,p}(\Omega)$ and the Leibniz formula holds:

$$D^\alpha(\phi f) = \sum_{\beta \le \alpha} \frac{\alpha!}{\beta!(\alpha - \beta)!} D^\beta \phi \, D^{\alpha-\beta} f. \tag{6.1}$$

Proof

(i) Let $\phi \in \mathcal{D}(\Omega)$. Then,

$$\int_\Omega D^\alpha f D^\beta \phi \, dx = (-1)^{|\alpha|} \int_\Omega f D^{\alpha+\beta} \phi \, dx$$

$$= (-1)^{|\alpha|}(-1)^{|\alpha+\beta|} \int_\Omega D^{\alpha+\beta} f \phi \, dx = (-1)^{|\beta|} \int_\Omega D^{\alpha+\beta} f \phi \, dx.$$

Therefore, $D^\beta(D^\alpha f) = D^{\alpha+\beta} f$, from which (i) follows.

(ii) We will prove (6.1) by induction on $|\alpha|$. Suppose that $|\alpha| = 1$. For $\varphi \in \mathcal{D}(\Omega)$, we have

$$\int_\Omega \phi f D^\alpha \varphi \, dx = \int_\Omega f D^\alpha(\phi\varphi) \, dx - f(D^\alpha \phi)\varphi \, dx$$

$$= -\int_\Omega (\phi D^\alpha f + f D^\alpha \phi)\varphi \, dx.$$

Therefore, $D^\alpha(\phi f) = \phi D^\alpha f + f D^\alpha \phi$, as desired.

Assume that $k < m$ and that (6.1) holds for all $|\alpha| \le k$ and every function $\phi \in \mathcal{D}(\Omega)$. Let us choose a multi-index α with $|\alpha| = k + 1$. Then, $\alpha = \beta + \gamma$, for some $|\beta| = k$, $|\gamma| = 1$. Then, for $\varphi \in \mathcal{D}(\Omega)$, using the induction hypothesis, we get

$$\int_\Omega \phi f D^\alpha \varphi \, dx = \int_\Omega \phi f D^\beta(D^\gamma \varphi) \, dx = (-1)^{|\beta|} \int_\Omega D^\beta(\phi f) D^\gamma \varphi \, dx$$

$$= (-1)^{|\beta|} \int_\Omega \sum_{\sigma \le \beta} \frac{\beta!}{\sigma!(\beta-\sigma)!} D^\sigma \phi \, D^{\beta-\sigma} f \, D^\gamma \varphi \, dx$$

$$= (-1)^{|\beta|+|\gamma|} \int_\Omega \sum_{\sigma \le \beta} \frac{\beta!}{\sigma!(\beta-\sigma)!} D^\gamma (D^\sigma \phi \, D^{\beta-\sigma} f) \varphi \, dx$$

$$= (-1)^{|\alpha|} \int_\Omega \sum_{\sigma \le \beta} \frac{\beta!}{\sigma!(\beta-\sigma)!} [D^\rho \phi D^{\alpha-\rho} f + D^\sigma \phi D^{\alpha-\sigma} f] \varphi \, dx,$$

where $\rho = \sigma + \gamma$. From here, since

$$\frac{\beta!}{\sigma!(\beta-\sigma)!} + \frac{\beta!}{(\sigma-\gamma)!(\beta-(\sigma-\gamma))!} = \frac{\alpha!}{\sigma!(\alpha-\sigma)!},$$

we obtain

$$\int_\Omega \phi f \, D^\alpha \varphi \, dx = (-1)^{|\alpha|} \int_\Omega \left(\sum_{\sigma \le \alpha} \frac{\alpha!}{\sigma!(\alpha-\sigma)!} D^\sigma \phi D^{\alpha-\sigma} f \right) \varphi \, dx,$$

which completes the proof of (6.1). □

We will use the notation $V \subset\subset U$ to indicate that V is compactly contained in U, i.e., $\overline{V}$ is a compact set and $\overline{V} \subset U$. We define $W^{m,p}_{loc}(\Omega)$ as: $f \in W^{m,p}_{loc}(\Omega)$ if and only if $f \in W^{m,p}(U)$ for every open set $U \subset\subset \Omega$. We write

$$f_n \to f \quad \text{in} \quad W^{m,p}_{loc}(\Omega)$$

to mean that

$$f_n \to f \quad \text{in} \quad W^{m,p}(U)$$

for every open set $U \subset\subset \Omega$.

Theorem 6.1 *The Sobolev spaces $W^{m,p}(\Omega)$ are Banach spaces. In particular, $H^m(\Omega)$ is a Hilbert space. Moreover, if $m \ge k$, then $W^{m,p}(\Omega) \subset W^{k,p}(\Omega)$, and the embedding is continuous.*

Proof Denote by $L^p_m(\Omega) := \prod_{|\alpha| \le m} L^p(\Omega)$ with the product norm. Then, $W^{m,p}(\Omega)$ can be embedded into $L^p_m(\Omega)$ via the map $P : W^{m,p}(\Omega) \to L^p_m(\Omega)$ defined by $P(f) := (D^\alpha f)_{|\alpha| \le m}$ for each $f \in W^{m,p}(\Omega)$. It suffices to prove that $P(W^{m,p}(\Omega))$ is closed in $L^p_m(\Omega)$. Indeed: let $\{f_n\}$ be a sequence in $W^{m,p}(\Omega)$ such that $D^\alpha f_n \to f_\alpha$ in $L^p(\Omega)$ for every $|\alpha| \le m$. Given $\phi \in \mathcal{D}(\Omega)$, since

$$\int_\Omega D^\alpha f_n \, \phi = (-1)^{|\alpha|} \int_\Omega f_n \, D^\alpha \phi,$$

taking the limit as $n \to \infty$, we obtain for every $|\alpha| \leq m$,

$$\int_\Omega f_\alpha \, \phi = (-1)^{|\alpha|} \int_\Omega f_0 \, D^\alpha \phi.$$

Thus, $D^\alpha f_0 = f_\alpha$ and $f_0 \in W^{m,p}(\Omega)$. □

Remark 6.1 Since the operator $P : W^{m,p}(\Omega) \to L^p_m(\Omega)$ from the proof of the previous theorem is an isometry, it follows that $W^{m,p}(\Omega)$ is reflexive if $1 < p < \infty$, and separable if $1 \leq p < \infty$.

6.3 Approximation by Smooth Functions

To study certain properties of Sobolev spaces, we need to develop a procedure for approximating functions in a Sobolev space by smooth functions. For this purpose, we will use the method of convolution with mollifiers, introduced in the first section of the previous chapter.

For each $\epsilon > 0$, let $0 \leq \rho_\epsilon \in \mathcal{D}(\mathbb{R}^N)$ be functions satisfying:

$$\operatorname{supp}(\rho_\epsilon) \subset B(0, \epsilon), \quad \int_{\mathbb{R}^N} \rho_\epsilon(x) \, dx = 1.$$

We know that functions with these properties exist (see Sect. 5.2), and moreover we may assume that

$$\rho_\epsilon(x) = \frac{1}{\epsilon^N} \rho\left(\frac{x}{\epsilon}\right),$$

where $0 \leq \rho \in \mathcal{D}(\mathbb{R}^N)$ with

$$\int_{\mathbb{R}^N} \rho(x) \, dx = 1, \quad \operatorname{supp}(\rho) \subset B(0, 1).$$

Let $\Omega \subset \mathbb{R}^N$ be an open set. We use the following notation:

$$\Omega_\epsilon := \{x \in \Omega \, : \, \operatorname{dist}(x, \partial\Omega) > \epsilon\}.$$

If $f \in L^1_{loc}(\Omega)$, we define its *regularization* f^ϵ in Ω_ϵ by

$$f^\epsilon := \rho_\epsilon \star f,$$

that is,

$$f^\epsilon(x) = \int_\Omega \rho_\epsilon(x - y) f(y) \, dy = \int_{B(0,\epsilon)} \rho_\epsilon(y) f(x - y) \, dy, \quad \forall \, x \in \Omega_\epsilon.$$

From the previous chapter, we know that the functions f^ϵ have the following properties:

$$f^\epsilon \in C^\infty(\Omega_\epsilon). \tag{6.2}$$

$$\text{If } f \in C(\Omega), \text{ then } f^\epsilon \to f \text{ uniformly on compact subsets of } \Omega. \tag{6.3}$$

$$\text{If } f \in L^p_{loc}(\Omega), \text{ then } f^\epsilon \to f \text{ in } L^p_{loc}(\Omega). \tag{6.4}$$

Given a function $f \in L^1_{loc}(\mathbb{R}^N)$ and a measurable set $B \subset \mathbb{R}^N$ with $0 < \mathcal{L}^N(B) < \infty$, we denote by the *average* of f over B the quantity:

$$f_B := \fint_B f(x)\,dx = \frac{1}{\mathcal{L}^N(B)} \int_B f(x)\,dx.$$

Recall that the Lebesgue Differentiation Theorem (see for instance [17]) states that if $f \in L^1_{loc}(\mathbb{R}^N)$, then for almost every point $x_0 \in \mathbb{R}^N$, we have

$$\lim_{r\to 0} \fint_{B(x_0,r)} |f(x) - f(x_0)|\,dx = 0. \tag{6.5}$$

The points x_0 that satisfy (6.5) are called *Lebesgue points* of f. Thus, almost every point is a Lebesgue point, and if x_0 is a Lebesgue point of f, then

$$\lim_{r\to 0} \fint_{B(x_0,r)} f(x)\,dx = f(x_0).$$

Lemma 6.1

(i) Let $f \in L^1_{loc}(\mathbb{R}^N)$, then if x is a Lebesgue point of f, we have

$$\lim_{\epsilon\to 0} f^\epsilon(x) = f(x).$$

In particular,

$$f^\epsilon \to f \quad \mathcal{L}^N - a.e.$$

(ii) If $f \in W^{1,p}_{loc}(\Omega)$ for some $1 \le p \le \infty$, then, in Ω_ϵ,

$$\frac{\partial f^\epsilon}{\partial x_i} = \rho_\epsilon \star \frac{\partial f}{\partial x_i}, \quad \text{for } i = 1, 2, \ldots, N. \tag{6.6}$$

In particular, if $f \in W^{1,p}_{loc}(\Omega)$ *for some* $1 \le p < \infty$, *then*

$$f^\epsilon \to f \quad in \ W^{1,p}_{loc}(\Omega).$$

Proof

(i) If x is a Lebesgue point of f, we have

$$|f^\epsilon(x) - f(x)| \le \frac{1}{\epsilon^N} \int_{B(x,\epsilon)} \rho\left(\frac{x-y}{\epsilon}\right) |f(y) - f(x)|\, dy$$

$$\le C\|\rho\|_\infty \fint_{B(x,\epsilon)} |f(y) - f(x)|\, dy \to 0, \quad \text{as } \epsilon \to 0.$$

(ii) For $x \in \Omega_\epsilon$, we have

$$\frac{\partial f^\epsilon}{\partial x_i}(x) = \int_\Omega \frac{\partial \rho_\epsilon}{\partial x_i}(x-y) f(y)\, dy = -\int_\Omega \frac{\partial \rho_\epsilon}{\partial y_i}(x-y) f(y)\, dy$$

$$= \int_\Omega \rho_\epsilon(x-y) \frac{\partial f}{\partial y_i}(y)\, dy = \rho_\epsilon \star \frac{\partial f}{\partial x_i}(x).$$

□

Before moving on to the global approximation theorems, let us observe that, as a consequence of the previous local approximation result, some of the usual rules of Calculus also hold for weak derivatives.

Theorem 6.2 *Suppose that* $1 \le p < \infty$. *Then:*

(i) If $f, g \in W^{1,p}(\Omega) \cap L^\infty(\Omega)$, *then* $fg \in W^{1,p}(\Omega) \cap L^\infty(\Omega)$ *and*

$$\frac{\partial (fg)}{\partial x_i} = f \frac{\partial g}{\partial x_i} + g \frac{\partial f}{\partial x_i}, \quad i = 1, 2, \ldots, N.$$

*(ii) (***Chain Rule***) If* $f \in W^{1,p}(\Omega)$ *and* $F \in C^1(\mathbb{R})$, $F' \in L^\infty(\Omega)$, *and* $F(0) = 0$, *then* $F(f) \in W^{1,p}(\Omega)$ *and*

$$\frac{\partial F(f)}{\partial x_i} = F'(f) \frac{\partial f}{\partial x_i}, \quad i = 1, 2, \ldots, N.$$

(iii) If $f \in W^{1,p}(\Omega)$, *then* $f^+, f^-, |f| \in W^{1,p}(\Omega)$ *and moreover:*

$$\nabla f^+ = \begin{cases} \nabla f \ \text{a.e. on } \{f > 0\} \\ \\ 0 \quad \text{a.e. on } \{f \le 0\} \end{cases}$$

$$\nabla f^- = \begin{cases} -\nabla f \ \text{a.e. on } \{f < 0\} \\ \\ 0 \qquad \text{a.e. on } \{f \geq 0\} \end{cases}$$

$$\nabla |f| = \begin{cases} \nabla f \quad \text{a.e. on } \{f > 0\} \\ \\ 0 \qquad \text{a.e. on } \{f = 0\} \\ \\ -\nabla f \ \text{a.e. on } \{f < 0\} \end{cases}$$

$$\nabla f = 0 \ \text{ a.e. on } \ \{f = 0\}. \tag{6.7}$$

Proof

(i) Let $\phi \in \mathcal{D}(\Omega)$, with $\text{supp}(\phi) \subset U \subset\subset \Omega$. For $\epsilon > 0$ sufficiently small, define $f^\epsilon := \rho_\epsilon \star f$, $g^\epsilon := \rho_\epsilon \star g$. Then, as a consequence of the previous lemma, we have:

$$\int_\Omega f g \frac{\partial \phi}{\partial x_i}\, dx = \int_U f g \frac{\partial \phi}{\partial x_i}\, dx = \lim_{\epsilon \to 0} \int_U f^\epsilon g^\epsilon \frac{\partial \phi}{\partial x_i}\, dx$$

$$= -\lim_{\epsilon \to 0} \int_U \frac{\partial (f^\epsilon g^\epsilon)}{\partial x_i}\, \phi\, dx = -\lim_{\epsilon \to 0} \int_U \left(\frac{\partial f^\epsilon}{\partial x_i} g^\epsilon + \frac{\partial g^\epsilon}{\partial x_i} f^\epsilon \right) \phi\, dx$$

$$= -\int_U \left(\frac{\partial f}{\partial x_i} g + \frac{\partial g}{\partial x_i} f \right) \phi\, dx = -\int_\Omega \left(\frac{\partial f}{\partial x_i} g + \frac{\partial g}{\partial x_i} f \right) \phi\, dx.$$

From this, statement (i) follows.

(ii) Choosing ϕ, U and f^ϵ as in (i), and taking into account the previous lemma, if we apply the Dominated Convergence Theorem, we get:

$$\int_\Omega F(f) \frac{\partial \phi}{\partial x_i}\, dx = \int_U F(f) \frac{\partial \phi}{\partial x_i}\, dx = \lim_{\epsilon \to 0} \int_U F(f^\epsilon) \frac{\partial \phi}{\partial x_i}\, dx$$

$$= -\lim_{\epsilon \to 0} \int_U F'(f^\epsilon) \frac{\partial f^\epsilon}{\partial x_i}\, \phi\, dx = -\int_U F'(f) \frac{\partial f}{\partial x_i}\, \phi\, dx = -\int_\Omega F'(f) \frac{\partial f}{\partial x_i}\, \phi\, dx.$$

From this, statement (ii) follows.

(iii) To prove the first identity of (iii), we consider smooth functions that approximate the function $r \mapsto r^+$. For instance, for $\epsilon > 0$, consider the function

$$F_\epsilon(r) := \begin{cases} \sqrt{r^2 + \epsilon^2} - \epsilon & \text{if } r \geq 0 \\ \\ 0 & \text{if } r < 0. \end{cases}$$

Then $F_\epsilon \in C^1(\Omega)$, $F'_\epsilon \in L^\infty(\Omega)$, so applying (ii) we obtain, for every $\phi \in \mathcal{D}(\Omega)$:

$$\int_\Omega F_\epsilon(f)\frac{\partial \phi}{\partial x_i}\,dx = -\int_\Omega F'_\epsilon(f)\frac{\partial f}{\partial x_i}\,\phi\,dx.$$

Now, taking the limit as $\epsilon \to 0$ and applying the Dominated Convergence Theorem, we obtain:

$$\int_\Omega f^+\frac{\partial \phi}{\partial x_i}\,dx = -\int_{\{f>0\}}\frac{\partial f}{\partial x_i}\,\phi\,dx.$$

From this, the first identity in (iii) follows. The next two follow from this and the equalities:

$$f^- = (-f)^+, \quad |f| = f^+ + f^-.$$

Finally, Eq. (6.7) follows from the previous formulas and the identity:

$$\nabla f = \nabla f^+ - \nabla f^-.$$

□

A first global approximation result is the following.

Theorem 6.3 (Meyers-Serrin Theorem) *Let $\Omega \subset \mathbb{R}^N$ be an open set, and let $1 \le p < \infty$. Given a function $f \in W^{1,p}(\Omega)$, there exists a sequence $f_n \in C^\infty(\Omega) \cap W^{1,p}(\Omega)$ such that*

$$f_n \to f \quad \text{in } W^{1,p}(\Omega).$$

Proof Let $\Omega_0 = \emptyset$, and for each $n \in \mathbb{N}$, define

$$\Omega_n := \left\{x \in \Omega \,:\, \operatorname{dist}(x, \partial\Omega) > \frac{1}{n}\right\} \cap B(0, n).$$

Consider the family of open sets

$$U_n := \Omega_{n+1} \setminus \overline{\Omega_{n-1}}, \quad n = 1, 2, \ldots.$$

By Theorem 5.1, there exists a partition of unity $\{\psi_n\}$ subordinate to the open cover $\{U_n\}$, i.e., $\psi_n \in \mathcal{D}(U_n)$, $0 \le \psi_n \le 1$ $(n = 1, 2, \ldots)$, and

$$\sum_{n=1}^{\infty} \psi_n = 1 \quad \text{in } \Omega.$$

Fix $\epsilon > 0$. For each $n \in \mathbb{N}$, the function $f\psi_n \in W^{1,p}(\Omega)$, with $\operatorname{supp}(f\psi_n) \subset U_n$. Therefore, there exists $\epsilon_n > 0$ such that

$$\begin{cases} \operatorname{supp}(\rho_{\epsilon_n} \star (f\psi_n)) \subset U_n \\ \left(\displaystyle\int_\Omega |\rho_{\epsilon_n} \star (f\psi_n) - f\psi_n|^p \, dx \right)^{\frac{1}{p}} \leq \frac{\epsilon}{2^n} \\ \left(\displaystyle\int_\Omega |\rho_{\epsilon_n} \star (\nabla(f\psi_n)) - \nabla(f\psi_n)|^p \, dx \right)^{\frac{1}{p}} \leq \frac{\epsilon}{2^n} \end{cases} \tag{6.8}$$

Define

$$f_\epsilon := \sum_{n=1}^{\infty} \rho_{\epsilon_n} \star (f\psi_n).$$

Since $\{\psi_n\}$ is a partition of unity, for each $x \in \Omega$ there exists a neighborhood where the sum defining the function f_ϵ has only finitely many nonzero terms. Consequently, $f_\epsilon \in C^\infty(\Omega)$.

Since

$$f = \sum_{n=1}^{\infty} f\psi_n,$$

and taking into account (6.8), we obtain

$$\|f_\epsilon - f\|_p \leq \sum_{n=1}^{\infty} \left(\int_\Omega |\rho_{\epsilon_n} \star (f\psi_n) - f\psi_n|^p \, dx \right)^{\frac{1}{p}} \leq \epsilon$$

and

$$\|\nabla f_\epsilon - \nabla f\|_p \leq \sum_{n=1}^{\infty} \left(\int_\Omega |\rho_{\epsilon_n} \star (\nabla(f\psi_n)) - \nabla(f\psi_n)|^p \, dx \right)^{\frac{1}{p}} \leq \epsilon.$$

Consequently, $f_\epsilon \in W^{1,p}(\Omega)$ and

$$f_\epsilon \to f \quad \text{in } W^{1,p}(\Omega) \text{ as } \epsilon \to 0.$$

□

The previous result does not guarantee that the functions f_n can be taken in $C^\infty(\overline{\Omega})$. To approximate up to the boundary, we need some smoothness of the

boundary. For that, recall the concept of open sets with smooth boundary (see Appendix A).

Let $\Omega \subset \mathbb{R}^N$ be an open set, and let $k \in \mathbb{N}$. We say that $\partial\Omega$ is of class C^k if for every point $x^0 \in \partial\Omega$ there exists an $r > 0$ and a C^k function $\gamma : \mathbb{R}^{N-1} \to \mathbb{R}$ such that

$$\Omega \cap B(x^0, r) = \{x \in B(x^0, r) \ : \ x_N > \gamma(x_1, \ldots, x_{N-1})\}.$$

When $\partial\Omega$ is of class C^k for all $k = 1, 2, \ldots$, we say that $\partial\Omega$ is of class C^∞. If the function γ is only Lipschitz continuous, we say that $\partial\Omega$ is Lipschitz.

Frequently, we need to perform a change of coordinates near a point $x^0 \in \partial\Omega$ so that, with respect to the new coordinates, the boundary becomes flat. To be more precise, if $x^0 \in \partial\Omega$ is fixed and γ is the function such that

$$\Omega \cap B(x^0, r) = \{x \in B(x^0, r) \ : \ x_N > \gamma(x_1, \ldots, x_{N-1})\},$$

we define

$$\phi^i(x) := x_i, \qquad \text{for } i = 1, 2, \ldots, N-1$$

$$\phi^N(x) := x_N - \gamma(x_1, \ldots, x_{N-1}),$$

and we write $y = \phi(x) = (\phi^1(x), \ldots, \phi^N(x))$. Similarly, we take

$$\psi^i(y) := y_i, \qquad \text{for } i = 1, 2, \ldots, N-1$$

$$\psi^N(y) := y_N + \gamma(y_1, \ldots, y_{N-1}),$$

and we write $x = \psi(y) = (\psi^1(y), \ldots, \psi^N(y))$. Then, $\phi = \psi^{-1}$, and the map $x \mapsto \phi(x) = y$ flattens $\partial\Omega$ near x^0.

Theorem 6.4 (Global Approximation by Smooth Functions) *Let $\Omega \subset \mathbb{R}^N$ be a bounded open set with boundary $\partial\Omega$ of class C^1, and let $1 \le p < \infty$. Then, given a function $f \in W^{m,p}(\Omega)$, there exist functions $f_n \in C^\infty(\overline{\Omega})$ such that*

$$f_n \to f \quad \text{in } W^{m,p}(\Omega).$$

Proof Fix a point $x^0 \in \partial\Omega$. Since $\partial\Omega$ is of class C^1, there exists $r > 0$ and a function $\gamma : \mathbb{R}^{N-1} \to \mathbb{R}$ of class C^1 such that

$$\Omega \cap B(x^0, r) = \left\{x \in B(x^0, r) \ : \ x_N > \gamma(x_1, \ldots, x_{N-1})\right\}.$$

Let $U := \Omega \cap B(x^0, \frac{r}{2})$. For $x \in U$, and $\lambda, \epsilon > 0$, set

$$x^\epsilon := x + \lambda\epsilon e_N.$$

Then, for $\lambda > 0$ large enough, we have $B(x^\epsilon, \epsilon) \subset \Omega \cap B(x^0, r)$ for all $x \in U$ and $\epsilon > 0$ small enough.

We define, for $\epsilon > 0$ sufficiently small and $x \in U$, the function $g_\epsilon(x) := f(x^\epsilon)$. Then, the function $f^\epsilon := \rho_\epsilon \star g_\epsilon$ is well defined, since for $x \in U$,

$$f^\epsilon(x) = \frac{1}{\epsilon^N} \int_\Omega \rho\left(\frac{z}{\epsilon}\right) f((x-z)^\epsilon)\, dz = \frac{1}{\epsilon^N} \int_{B(x^\epsilon,\epsilon)} \rho\left(\frac{x-w}{\epsilon} + \lambda e_N\right) f(w)\, dw.$$

Clearly, $f^\epsilon \in C^\infty(\overline{U})$. We claim that

$$\lim_{\epsilon \to 0} f^\epsilon = f \quad \text{in } W^{m,p}(U). \tag{6.9}$$

Indeed, given a multi-index α with $|\alpha| \le m$, we have

$$\|D^\alpha f^\epsilon - D^\alpha f\|_{L^p(U)} \le \|D^\alpha f^\epsilon - D^\alpha g_\epsilon\|_{L^p(U)} + \|D^\alpha g_\epsilon - D^\alpha f\|_{L^p(U)}.$$

The second term tends to zero as $\epsilon \to 0$, because translation is continuous with respect to the L^p-norm. The first term also tends to zero due to the definition of f^ϵ as the convolution of g_ϵ with smoothing functions.

Given $\delta > 0$, since $\partial\Omega$ is compact, there exist points $x_i^0 \in \partial\Omega$ and radii $r_i > 0$ such that

$$\partial\Omega \subset \bigcup_{i=1}^q B\left(x_i^0, \frac{r_i}{2}\right).$$

Now, if we take $U_i := \Omega \cap B(x_i^0, \frac{r_i}{2})$, as a consequence of (6.9), we can find functions $f^i \in C^\infty(\overline{U_i})$ such that

$$\|f^i - f\|_{W^{m,p}(U_i)} \le \delta, \qquad \text{for } i = 1, 2, \ldots, q. \tag{6.10}$$

Let $U_0 \subset\subset \Omega$ be an open set such that

$$\Omega \subset \bigcup_{i=0}^q U_i.$$

By Lemma 6.1, there exists a function $f^0 \in C^\infty(\overline{U_0})$ such that

$$\|f^0 - f\|_{W^{m,p}(U_0)} \le \delta. \tag{6.11}$$

Let $\{\psi_i\}_{i=0}^q$ be a partition of unity subordinate to the covering $\{U_i\}_{i=0}^q$ of Ω. Define

$$g := \sum_{i=0}^{q} \psi_i f^i \in C^\infty(\overline{\Omega}).$$

Then, since $f = \sum_{i=0}^q \psi_i f$, and using Proposition 6.1, from (6.10) and (6.11), we deduce that for $|\alpha| \le m$,

$$\|D^\alpha g - D^\alpha f\|_p \le \sum_{i=0}^{q} \|D^\alpha(\psi_i f^i) - D^\alpha(\psi_i f)\|_p \le C \sum_{i=0}^{q} \|f^i - f\|_{W^{m,p}(U_i)} \le Cq\delta.$$

□

The previous theorem is also valid if $\partial\Omega$ is Lipschitz. For a proof, see, for example, [17].

6.4 The Space $H^s(\mathbb{R}^N)$

Using the Fourier–Plancherel transform, we have the following characterization of Sobolev spaces $H^m(\mathbb{R}^N)$.

Theorem 6.5 *Let $m \in \mathbb{N}$. Then,*

$$H^m(\mathbb{R}^N) = \left\{ u \in \mathcal{S}'(\mathbb{R}^N) \ : \ (1 + \|x\|^2)^{\frac{m}{2}} \hat{u} \in L^2(\mathbb{R}^N) \right\}.$$

Moreover, the norm $\| \ \|_{m,2}$ is equivalent to the one defined by

$$|u|_m := \left(\int_{\mathbb{R}^N} (1 + \|x\|^2)^m |\hat{u}(x)|^2 \, dx \right)^{\frac{1}{2}}.$$

Proof Clearly, every element of $H^m(\mathbb{R}^N)$ is a tempered distribution since $H^m(\mathbb{R}^N) \subset L^2(\mathbb{R}^N) \subset \mathcal{S}'(\mathbb{R}^N)$.

We need the following inequalities: For every $\alpha \in \mathbb{N}^N$ with $0 < |\alpha| \le m$, there exists a constant $C > 0$ such that

$$\prod_{j=1}^{N} |x_j|^{2\alpha_j} \le \left(1 + \sum_{j=1}^{N} |x_j|^2 \right)^m \le C \left(1 + \sum_{0<|\alpha|\le m} \prod_{j=1}^{N} |x_j|^{2\alpha_j} \right) \tag{6.12}$$

for every $x \in \mathbb{R}^N$. The first inequality follows from the fact that $\alpha_1 + \alpha_2 + \cdots + \alpha_N \le m$, and the second is obtained by applying the binomial formula.

If $u \in H^m(\mathbb{R}^N)$, then since $D^\alpha u \in L^2(\mathbb{R}^N)$ for all $|\alpha| \le m$, by Plancherel's Theorem we have

$$\widehat{D^\alpha u} = (i\xi)^\alpha \hat{u} \in L^2(\mathbb{R}^N), \quad \text{if } |\alpha| \le m.$$

Then, applying (6.12), we get

$$\sum_{|\alpha|\le m} \int_{\mathbb{R}^N} |x^\alpha \hat{u}(x)|^2 \, dx = \int_{\mathbb{R}^N} \left(1 + \sum_{0<|\alpha|\le m} \prod_{j=1}^N |x_j|^{2\alpha_j}\right) |\hat{u}(x)|^2 \, dx$$

$$\ge \frac{1}{C} \int_{\mathbb{R}^N} \left(1 + \sum_{j=1}^N |x_j|^2\right)^m |\hat{u}(x)|^2 \, dx.$$

Hence, $(1 + \|x\|^2)^{\frac{m}{2}} \hat{u} \in L^2(\mathbb{R}^N)$. Moreover, by Parseval's identity, we have

$$|u|_m = \int_{\mathbb{R}^N} (1 + \|x\|^2)^m |\hat{u}(x)|^2 \, dx \le C \sum_{|\alpha|\le m} \int_{\mathbb{R}^N} |x^\alpha \hat{u}(x)|^2 \, dx$$

$$= C \int_{\mathbb{R}^N} \sum_{|\alpha|\le m} |\widehat{D^\alpha u}(x)|^2 \, dx = C_1 \int_{\mathbb{R}^N} \sum_{|\alpha|\le m} |D^\alpha u(x)|^2 \, dx = C_1 \|u\|_{m,2}^2.$$

Now suppose that $(1 + \|x\|^2)^{\frac{m}{2}} \hat{u} \in L^2(\mathbb{R}^N)$, then $\hat{u} \in L^2(\mathbb{R}^N)$. Applying (6.12), we get

$$\int_{\mathbb{R}^N} \prod_{j=1}^N |x_j|^{2\alpha_j} |\hat{u}(x)|^2 \, dx \le C \int_{\mathbb{R}^N} (1 + \|x\|^2)^m |\hat{u}(x)|^2 \, dx < +\infty.$$

Thus, $x^\alpha \hat{u} \in L^2(\mathbb{R}^N)$ for $|\alpha| \le m$. Then, since $\widehat{D^\alpha u} = (ix)^\alpha \hat{u}$, we have $\widehat{D^\alpha u} \in L^2(\mathbb{R}^N)$. Therefore, by Plancherel's Theorem, $D^\alpha u \in L^2(\mathbb{R}^N)$ for each $|\alpha| \le m$, and consequently $u \in H^m(\mathbb{R}^N)$. Moreover, taking into account (6.12) and Parseval's identity, we have

$$\|u\|_{m,2}^2 = \int_{\mathbb{R}^N} \sum_{|\alpha|\le m} |D^\alpha u(x)|^2 \, dx = M \int_{\mathbb{R}^N} \sum_{|\alpha|\le m} |\widehat{D^\alpha u}(x)|^2 \, dx$$

$$= M \int_{\mathbb{R}^N} \sum_{|\alpha|\le m} |x^\alpha \hat{u}(x)|^2 \, dx = M \sum_{|\alpha|\le m} \int_{\mathbb{R}^N} \prod_{j=1}^N |x_j|^{2\alpha_j} |\hat{u}(x)|^2 \, dx$$

$$\leq M_1 \int_{\mathbb{R}^N} (1 + \|x\|^2)^m |\hat{u}(x)|^2 \, dx = M_1 |u|_m,$$

which concludes the proof of the theorem. □

The characterization of the previous theorem suggests the following definition.

Definition 6.1 For $s \in \mathbb{R}$, the space $H^s(\mathbb{R}^N)$ is defined as

$$H^s(\mathbb{R}^N) := \left\{ u \in \mathcal{S}'(\mathbb{R}^N) \ : \ (1 + \|x\|^2)^{\frac{s}{2}} \hat{u} \in L^2(\mathbb{R}^N) \right\}.$$

This space is endowed with the inner product

$$(u|v) := \int_{\mathbb{R}^N} (1 + \|x\|^2)^s \hat{u}(x) \overline{\hat{v}(x)} \, dx,$$

whose associated norm is given by

$$|u|_s = \left(\int_{\mathbb{R}^N} (1 + \|x\|^2)^s |\hat{u}(x)|^2 \, dx \right)^{\frac{1}{2}}.$$

The spaces $H^s(\mathbb{R}^N)$ can also be defined via interpolation between $H^m(\mathbb{R}^N)$ and $L^2(\mathbb{R}^N)$. For this method of definition, see for example [39].

Theorem 6.6 *For every $s \in \mathbb{R}$, $H^s(\mathbb{R}^N)$ is a Hilbert space. Moreover, if $s_1 \geq s_2$, then $H^{s_1}(\mathbb{R}^N) \subset H^{s_2}(\mathbb{R}^N)$ with continuous embedding.*

Proof To prove that $H^s(\mathbb{R}^N)$ is a Hilbert space, it suffices to show that it is complete. Let $\{u_n\}$ be a Cauchy sequence in $H^s(\mathbb{R}^N)$. Then, $\{(1 + \|x\|^2)^{\frac{s}{2}} \hat{u}_n\}$ is a Cauchy sequence in $L^2(\mathbb{R}^N)$, so there exists $v \in L^2(\mathbb{R}^N)$ such that

$$\{(1 + \|x\|^2)^{\frac{s}{2}} \hat{u}_n\} \to v \quad \text{in} \quad L^2(\mathbb{R}^N).$$

Now, since $(1 + \|x\|^2)^{-\frac{s}{2}} v \in \mathcal{S}'(\mathbb{R}^N)$, there exists $u \in \mathcal{S}'(\mathbb{R}^N)$ such that $\hat{u} = (1 + \|x\|^2)^{-\frac{s}{2}} v$. Then, $(1 + \|x\|^2)^{\frac{s}{2}} \hat{u} \in L^2(\mathbb{R}^N)$, hence $u \in H^s(\mathbb{R}^N)$ and, moreover, $u_n \to u$ in $H^s(\mathbb{R}^N)$. Finally, the continuous inclusion follows from the inequality $(1 + \|x\|^2)^{s_2} \leq (1 + \|x\|^2)^{s_1}$ when $s_1 \geq s_2$. □

For $s_1 \geq s_2 \geq 0$, we have the inclusions:

$$\mathcal{S}'(\mathbb{R}^N) \subset H^{s_1}(\mathbb{R}^N) \subset H^{s_2}(\mathbb{R}^N)$$

$$\subset L^2(\mathbb{R}^N) \subset H^{-s_2}(\mathbb{R}^N) \subset H^{-s_1}(\mathbb{R}^N) \subset \mathcal{S}'(\mathbb{R}^N).$$

Theorem 6.7 *For each $s \in \mathbb{R}$, $\mathcal{D}(\mathbb{R}^N)$ is dense in $H^s(\mathbb{R}^N)$.*

Proof Let $u \in H^s(\mathbb{R}^N)$. Then, $(1+\|x\|^2)^{\frac{s}{2}}\hat{u} \in L^2(\mathbb{R}^N)$, so, since $\mathcal{D}(\mathbb{R}^N)$ is dense in $L^2(\mathbb{R}^N)$, there exist $\phi_n \in \mathcal{D}(\mathbb{R}^N)$ such that $\phi_n \to (1+\|x\|^2)^{\frac{s}{2}}\hat{u}$ in $L^2(\mathbb{R}^N)$.

We define u_n by

$$\hat{u}_n(x) := (1+\|x\|^2)^{-\frac{s}{2}}\phi_n(x).$$

Then, since $(1+\|x\|^2)^{-\frac{s}{2}}\phi_n \in \mathcal{D}(\mathbb{R}^N) \subset \mathcal{S}(\mathbb{R}^N)$, we have that $u_n \in \mathcal{S}(\mathbb{R}^N)$. Now, by construction,

$$(1+\|x\|^2)^{\frac{s}{2}}\hat{u}_n = \phi_n \to (1+\|x\|^2)^{\frac{s}{2}}\hat{u} \quad \text{in } L^2(\mathbb{R}^N),$$

so $u_n \to u$ in $H^s(\mathbb{R}^N)$. To complete the proof, it suffices to observe that, by Theorem 5.22, we know that $\mathcal{D}(\mathbb{R}^N)$ is dense in $\mathcal{S}(\mathbb{R}^N)$ with continuous inclusion. □

As a consequence of the previous theorem, we have that $(H^s(\mathbb{R}^N))' \subset \mathcal{D}'(\mathbb{R}^N)$, i.e., the elements of the dual of $H^s(\mathbb{R}^N)$ are distributions in $\mathbb{R}^N$. More precisely, we have the following characterization of the dual of $H^s(\mathbb{R}^N)$.

Theorem 6.8 *For every $s \in \mathbb{R}$, the topological dual of the space $H^s(\mathbb{R}^N)$ coincides algebraically and topologically with the space $H^{-s}(\mathbb{R}^N)$.*

Proof We must show that there exists a bijection

$$T : (H^s(\mathbb{R}^N))' \to H^{-s}(\mathbb{R}^N)$$

such that T and T^{-1} are continuous.

Given $F \in (H^s(\mathbb{R}^N))'$, by the Riesz Representation Theorem, there exists $v_F \in H^s(\mathbb{R}^N)$ such that

$$\langle F, u\rangle = (u|v_F) = \int_{\mathbb{R}^N} (1+\|x\|^2)^s \hat{u}(x)\overline{\widehat{v_F}(x)}\,dx, \quad \forall\, u \in H^s(\mathbb{R}^N) \tag{6.13}$$

and moreover $|v_F|_s = \|F\|$.

Since

$$f := (1+\|x\|^2)^{\frac{s}{2}}\widehat{v_F} \in L^2(\mathbb{R}^N) \subset \mathcal{S}(\mathbb{R}^N)',$$

we have that

$$T_{v_F} := \mathcal{F}\left[(1+\|x\|^2)^s \widehat{v_F}\right] \in \mathcal{S}(\mathbb{R}^N)'.$$

Moreover,

$$\left(1+\|x\|^2\right)^{-\frac{s}{2}} \mathcal{F}(T_{v_F}) = \left(1+\|x\|^2\right)^{-\frac{s}{2}} \mathcal{F}^2\left[(1+\|x\|^2)^{\frac{s}{2}} f\right]$$

$$= \left(1 + \|x\|^2\right)^{-\frac{s}{2}} (2\pi)^N \left[(1 + \|x\|^2)^{\frac{s}{2}} f\right]^\vee = (2\pi)^N f^\vee \in L^2(\mathbb{R}^N).$$

Therefore, $T_{v_F} \in H^{-s}(\mathbb{R}^N)$ and, moreover,

$$|T_{v_F}|_{-s} = \left\| \left(1 + \|x\|^2\right)^{-\frac{s}{2}} \mathcal{F}(T_{v_F}) \right\|_2 = \|(2\pi)^N f^\vee\|_2$$

$$= (2\pi)^N \left\| \left(1 + \|x\|^2\right)^{\frac{s}{2}} \widehat{v_F} \right\|_2 = (2\pi)^N |v_F|_s.$$

Consequently,

$$|T_{v_F}|_{-s} = (2\pi)^N \|F\|. \tag{6.14}$$

Let us now show that

$$T_{v_F} = F \qquad \text{in } \mathcal{D}'(\mathbb{R}^N). \tag{6.15}$$

For $\phi \in \mathcal{D}(\mathbb{R}^N)$, we have

$$\langle T_{v_F}, \phi \rangle = \langle T_{v_F}, \phi \rangle_{\mathcal{S}',\mathcal{S}} = \langle \mathcal{F}^{-1}(T_{v_F}), \mathcal{F}(\phi) \rangle_{\mathcal{S}',\mathcal{S}}$$

$$= \left\langle \left(1 + \|x\|^2\right)^{-\frac{s}{2}} \mathcal{F}^{-1}(T_{v_F}), \left(1 + \|x\|^2\right)^{\frac{s}{2}} \mathcal{F}(\phi) \right\rangle_{\mathcal{S}',\mathcal{S}}.$$

Now,

$$\left(1 + \|x\|^2\right)^{-\frac{s}{2}} \mathcal{F}^{-1}(T_{v_F}) = \left(1 + \|x\|^2\right)^{-\frac{s}{2}} \left(1 + \|x\|^2\right)^s \widehat{v_F}$$

$$= \left(1 + \|x\|^2\right)^{\frac{s}{2}} \widehat{v_F} \in L^2(\mathbb{R}^N).$$

Therefore, taking into account (6.13), we obtain

$$\langle T_{v_F}, \phi \rangle = \int_{\mathbb{R}^N} \left(1 + \|x\|^2\right)^{\frac{s}{2}} \hat{\phi}(x) \left(1 + \|x\|^2\right)^{\frac{s}{2}} \overline{\widehat{v_F}(x)}\, dx = \langle F, \phi \rangle,$$

which proves (6.15).

We define the mapping $T : (H^s(\mathbb{R}^N))' \to H^{-s}(\mathbb{R}^N)$ by $T(F) := T_{v_F}$. From (6.14) and (6.15) it follows that T is injective and continuous. Let us now show that it is surjective. Given $v \in H^{-s}(\mathbb{R}^N)$, let

$$u := \mathcal{F}\left[(2\pi)^{-2N} \left(1 + \|x\|^2\right)^{-s} \hat{v}\right] \in H^s(\mathbb{R}^N).$$

Consider $F \in (H^s(\mathbb{R}^N))'$ defined by

$$\langle F, \phi \rangle := (\phi | u) \quad \forall\, \phi \in H^s(\mathbb{R}^N).$$

Then we have

$$T(F) = T_u = \mathcal{F}\left[\left(1 + \|x\|^2\right)^s \hat{u}\right]$$

$$= \mathcal{F}\left[\left(1 + \|x\|^2\right)^s \mathcal{F}^2\left[(2\pi)^{-2N}\left(1 + \|x\|^2\right)^{-s} \hat{v}\right]\right]$$

$$= \mathcal{F}\left[\left(1 + \|x\|^2\right)^s (2\pi)^{-N}\left(1 + \|x\|^2\right)^{-s} (\mathcal{F}v)^\vee\right] = \mathcal{F}\left[(2\pi)^{-N}(\mathcal{F}v)^\vee\right] = v.$$

Therefore, T is linear, bijective and continuous. Then, applying the Open Mapping Theorem, T is a homeomorphism. □

When s is a non-negative integer, the space $H^{-s}(\mathbb{R}^N)$ admits the following characterization.

Theorem 6.9 *Let $m \in \mathbb{N}$. Then, $u \in H^{-m}(\mathbb{R}^N)$ if and only if there exist $v_\alpha \in L^2(\mathbb{R}^N)$ such that*

$$u = \sum_{|\alpha| \le m} D^\alpha v_\alpha.$$

Proof If $v \in L^2(\mathbb{R}^N)$, then

$$\left(1 + \|x\|^2\right)^{-\frac{m}{2}} \widehat{D^\alpha v} = \left(1 + \|x\|^2\right)^{-\frac{m}{2}} (ix)^\alpha \hat{v},$$

and

$$\left|\left(1 + \|x\|^2\right)^{-\frac{m}{2}} (ix)^\alpha\right| \le 1, \quad \text{for } |\alpha| \le m,$$

so we have that $D^\alpha v \in H^{-m}(\mathbb{R}^N)$. Thus, the condition is sufficient.

Conversely, let $u \in H^{-m}(\mathbb{R}^N)$. Then,

$$\left(1 + \|x\|^2\right)^{-\frac{m}{2}} \hat{u} \in L^2(\mathbb{R}^N).$$

Now,

$$\left(1 + \|x\|^2\right)^{\frac{m}{2}} \le \left(1 + \|x\|^2\right)^m = \left[1 + (x_1^2 + \cdots + x_N^2)^{\frac{1}{2}}\right]^m$$

$$\leq (1 + |x_1| + \cdots + |x_N|)^m = 1 + \sum_{1 \leq |\alpha| \leq m} c_\alpha |x^\alpha|.$$

As a consequence of this inequality, we have that if

$$\hat{v}(x) = \left(1 + \sum_{1 \leq |\alpha| \leq m} c_\alpha |x^\alpha|\right)^{-1} \hat{u}(x),$$

then,

$$|\hat{v}(x)| \leq \left(1 + \|x\|^2\right)^{-\frac{m}{2}} |\hat{u}(x)|.$$

Thus, $v \in L^2(\mathbb{R}^N)$. Now, we can write

$$\hat{u}(x) = \left(1 + \sum_{1 \leq |\alpha| \leq m} c_\alpha |x^\alpha|\right) \hat{v}(x) = \sum_{|\alpha| \leq m} i^{|\alpha|} x^\alpha \widehat{v_\alpha}(x), \tag{6.16}$$

with $\widehat{v_\alpha} = \hat{v}$ when $|\alpha| = 0$, and $\widehat{v_\alpha}(x) = i^{-|\alpha|} c_\alpha |x^{-\alpha} \hat{v}(x)$, for $1 \leq |\alpha| \leq m$. Since $v \in L^2(\mathbb{R}^N)$, we have that $v_\alpha \in L^2(\mathbb{R}^N)$. Taking the inverse of the Fourier-Plancherel transform in (6.16), we get

$$u = \sum_{|\alpha| \leq m} \mathcal{P}^{-1}\left(i^{|\alpha|} x^\alpha \widehat{v_\alpha}\right) = \sum_{|\alpha| \leq m} D^\alpha v_\alpha.$$

□

As a first application of the spaces $H^s(\mathbb{R}^N)$, we have the following result.

Theorem 6.10 *Let $s \in \mathbb{R}$ and $k \in \mathbb{R}$, $k \neq 0$. For every $f \in H^s(\mathbb{R}^N)$ there exists a unique $u \in H^{s+2}(\mathbb{R}^N)$, distributional solution of the equation*

$$(-\Delta + k^2)u = f \quad \text{in } \mathbb{R}^N.$$

Moreover, the operator $(-\Delta + k^2)$ is an isomorphism from $H^{s+2}(\mathbb{R}^N)$ onto $H^s(\mathbb{R}^N)$.

Proof Clearly, $(-\Delta + k^2) : H^{s+2}(\mathbb{R}^N) \to H^s(\mathbb{R}^N)$ is a linear operator. Moreover, if $u \in H^{s+2}(\mathbb{R}^N)$ is such that $(-\Delta + k^2)u = 0$, then $(\|x\|^2 + k^2)\hat{u} = 0$, and since $k \neq 0$, we have $\hat{u} = 0$, and consequently, $u = 0$. Therefore, the operator $(-\Delta + k^2)$ is injective. Let us now prove that it is surjective. Let $v \in H^s(\mathbb{R}^N)$. Then,

$$\left(1 + \|x\|^2\right)^{\frac{s}{2}} \hat{v} \in L^2(\mathbb{R}^N).$$

Define

$$u := \mathcal{F}^{-1}\left(\frac{\hat{v}}{\|x\|^2 + k^2}\right).$$

We have

$$\left|(1+\|x\|^2)^{\frac{s+2}{2}}\hat{u}\right| = \left|\left(1+\|x\|^2\right)^{\frac{s+2}{2}} \frac{\hat{v}}{\|x\|^2+k^2}\right|$$

$$\leq C\left(1+\|x\|^2\right)^{\frac{s+2}{2}} \frac{|\hat{v}|}{1+\|x\|^2} = C\left(1+\|x\|^2\right)^{\frac{s}{2}} |\hat{v}| \in L^2(\mathbb{R}^N),$$

so $u \in H^{s+2}(\mathbb{R}^N)$ and, moreover,

$$(-\Delta + k^2)u = (-\Delta + k^2)\mathcal{F}^{-1}\left(\frac{\hat{v}}{\|x\|^2+k^2}\right) = v.$$

Consequently, $(-\Delta + k^2)$ is bijective. Furthermore, its continuity also follows from the above. Finally, by the Open Mapping Theorem, its inverse is also continuous. □

In the following result, we will see that elliptic operators have good regularity properties.

Theorem 6.11 *Let $P(D) = \sum_{|\alpha|\leq m} c_\alpha D^\alpha$ be an elliptic operator of order m in $\mathbb{R}^N$, and let $u \in L^2(\mathbb{R}^N)$. If $P(D)u \in H^s(\mathbb{R}^N)$ with $s \geq 0$, then $u \in H^{s+m}(\mathbb{R}^N)$ and the following a priori estimate holds:*

$$|u|_{s+m} \leq M\big(\|u\|_2 + |P(D)u|_s\big).$$

Proof Since $P(D)$ is elliptic, there exist constants $C > 0$ and $A > 0$ such that

$$|P^*(\xi)| \geq C\|\xi\|^m \quad \text{for} \quad \|\xi\| \geq A. \tag{6.17}$$

By Parseval's identity we have

$$\int_{\|\xi\|\leq A} \left(1+\|\xi\|^2\right)^{s+m} |\hat{u}(\xi)|^2\, d\xi \leq (1+A^2)^{s+m} \int_{\|\xi\|\leq A} |\hat{u}(\xi)|^2\, dx \leq M\|u\|_2^2.$$

On the other hand, for $\|\xi\| \geq A$, we have

$$\frac{(1+\|\xi\|^2)^m}{\|\xi\|^{2m}} = \left(1+\frac{1}{\|\xi\|^2}\right)^m \leq \left(1+\frac{1}{A^2}\right)^m = K,$$

and therefore

$$(1 + \|\xi\|^2)^m \leq K\|\xi\|^{2m}.$$

From this, and using (6.17), it follows that

$$\int_{\|\xi\|\geq A} (1 + \|\xi\|^2)^{s+m} |\hat{u}(\xi)|^2 \, d\xi \leq K \int_{\|\xi\|\geq A} \|\xi\|^{2m} (1 + \|\xi\|^2)^s |\hat{u}(\xi)|^2 \, d\xi$$

$$\leq C^2 K \int_{\|\xi\|\geq A} |P^*(\xi)\hat{u}(\xi)|^2 (1 + \|\xi\|^2)^s \, d\xi$$

$$\leq C^2 K \int_{\mathbb{R}^N} |P^*(\xi)\hat{u}(\xi)|^2 (1 + \|\xi\|^2)^s \, d\xi$$

$$= C^2 K \int_{\mathbb{R}^N} |\widehat{P(D)u}|^2 (1 + \|\xi\|^2)^s \, d\xi = C^2 K |P(D)u|_s^2.$$

Consequently, we have

$$|u|_{s+m}^2 = \int_{\mathbb{R}^N} (1 + \|\xi\|^2)^{s+m} |\hat{u}(\xi)|^2 \, d\xi \leq \int_{\|\xi\|\leq A} (1 + \|\xi\|^2)^{s+m} |\hat{u}(\xi)|^2 \, d\xi$$

$$+ \int_{\|\xi\|\geq A} (1 + \|\xi\|^2)^{s+m} |\hat{u}(\xi)|^2 \, d\xi \leq M\|u\|_2^2 + C^2 K |P(D)u|_s^2.$$

Therefore, there exists $M > 0$ such that

$$|u|_{s+m} \leq M\big(\|u\|_2 + |P(D)u|_s\big).$$

□

6.5 The Space $W_0^{m,p}(\Omega)$

For $1 \leq p < \infty$ and $m \in \mathbb{N}$, we denote by $W_0^{m,p}(\Omega)$ the closure of $\mathcal{D}(\Omega)$ in $W^{m,p}(\Omega)$. Therefore, $f \in W_0^{m,p}(\Omega)$ if and only if there exist $f_n \in \mathcal{D}(\Omega)$ such that $f_n \to f$ in $W^{m,p}(\Omega)$. By definition, endowed with the norm $\|\cdot\|_{m,p}$, the space $W_0^{m,p}(\Omega)$ is a separable Banach space, which is reflexive if $1 < p < \infty$. We will also use the notation $H_0^m(\Omega)$ to denote the Hilbert space $W_0^{m,2}(\Omega)$.

It is easy to see that $W^{m,p}(\mathbb{R}^N) = W_0^{m,p}(\mathbb{R}^N)$ (see Problem 6.13). However, for Ω a bounded open set, we have $W^{1,p}(\Omega) \neq W_0^{1,p}(\Omega)$ (see Problem 6.14).

Since functions in $W^{1,p}(\Omega)$ are defined $\mathcal{L}^N$-a.e., and $\partial\Omega$ is $\mathcal{L}^N$-null, at first it makes no sense to say that a function in $W^{1,p}(\Omega)$ vanishes on $\partial\Omega$. However, as we will see in the section dedicated to the trace, when $\partial\Omega$ is of class C^1, one can give meaning to the restriction of functions in $W^{1,p}(\Omega)$ to $\partial\Omega$, and the functions in $W_0^{1,p}(\Omega)$ are precisely those that, in this sense, vanish on $\partial\Omega$. In this section, we will see characterizations of $W_0^{1,p}(\Omega)$ that suggest what we have just discussed.

Lemma 6.2 *Let $f \in W^{1,p}(\Omega)$, $1 \le p < \infty$. If $\operatorname{supp}(f)$ is a compact set contained in Ω, then $f \in W_0^{1,p}(\Omega)$.*

Proof Let U be an open set such that $\operatorname{supp}(f) \subset U \subset\subset \Omega$. Let $\phi \in \mathcal{D}(U)$ such that $\phi(x) = 1$ for every $x \in \operatorname{supp}(f)$. As a consequence of Lemma 6.1, there exists a sequence $g_n \in C^\infty(\mathbb{R}^N)$ such that $g_n \to f$ in $W^{1,p}_{loc}(\Omega)$. Therefore, if $f_n := \phi g_n$, then $f_n \in \mathcal{D}(\Omega)$ and $f_n \to f$ in $W^{1,p}(\Omega)$, hence $f \in W_0^{1,p}(\Omega)$. □

Theorem 6.12 *Let $f \in W^{1,p}(\Omega) \cap C(\overline{\Omega})$. If $f_{|\partial\Omega} = 0$, then $f \in W_0^{1,p}(\Omega)$.*

Proof First, suppose that $\operatorname{supp}(f)$ is bounded. Let $F \in C^1(\mathbb{R})$ be such that $|F(t)| \le |t|$ for all $t \in \mathbb{R}$, and

$$F(t) = \begin{cases} 0 & \text{if } |t| \le 1 \\ \\ t & \text{if } |t| \ge 2. \end{cases}$$

By the chain rule, we have that $f_n := \frac{1}{n}\big(F \circ (nf)\big) \in W^{1,p}(\Omega)$. Now,

$$f_n(x) = \frac{1}{n} F(nf(x)) = f(x), \quad \text{if } |f(x)| \ge \frac{2}{n}.$$

Moreover,

$$\frac{\partial f_n}{\partial x_i}(x) = \frac{1}{n} F'(nf(x)) n \frac{\partial f}{\partial x_i}(x) = F'(nf(x)) \frac{\partial f}{\partial x_i}(x),$$

so that

$$\frac{\partial f_n}{\partial x_i}(x) = \frac{\partial f}{\partial x_i}(x), \quad \text{if } |f(x)| \ge \frac{2}{n}.$$

Therefore, by the Dominated Convergence Theorem, we have that $f_n \to f$ in $W^{1,p}(\Omega)$. On the other hand,

$$\operatorname{supp}(f_n) \subset \left\{ x \in \Omega \,:\, |x| \ge \frac{1}{n} \right\}.$$

Hence, $\text{supp}(f_n)$ is a compact subset of Ω, and applying the previous lemma, we obtain that $f_n \in W_0^{1,p}(\Omega)$, and consequently, $f \in W_0^{1,p}(\Omega)$.

In the general case where $\text{supp}(f)$ is not bounded, we take a sequence of truncations $\{\xi_n\}$, i.e., $\xi_n(x) = \xi\left(\frac{x}{n}\right)$, with $\xi \in \mathcal{D}(\Omega)$, $0 \leq \xi \leq 1$, and

$$\xi(x) = \begin{cases} 1 & \text{if } \|x\| \leq 1 \\ \\ 0 & \text{if } \|x\| \geq 2, \end{cases}$$

and we consider $f_n := \xi_n f$. It is easy to verify that $f_n \to f$ in $W^{1,p}(\Omega)$. Now, since $\text{supp}(f_n)$ is bounded, by the above we have that $f_n \in W_0^{1,p}(\Omega)$, and consequently, $f \in W_0^{1,p}(\Omega)$. □

Given $f \in W^{1,p}(\Omega)$, we denote by f^0 the function

$$f^0(x) := \begin{cases} f(x) & \text{if } x \in \Omega \\ \\ 0 & \text{if } x \in \mathbb{R}^N \setminus \Omega. \end{cases}$$

In general, f^0 is not an element of $W^{1,p}(\mathbb{R}^N)$. For example, if $\Omega =]0, 1[$, and $f(x) = 1$ for every $x \in \Omega$, then

$$\frac{df^0}{dx} = \delta_0 - \delta_1,$$

so that $f^0 \notin W^{1,p}(\mathbb{R})$. In the following result we will see that the zero extension of functions in $W_0^{1,p}(\Omega)$ is indeed an element of $W^{1,p}(\mathbb{R}^N)$.

Theorem 6.13 *Let $1 < p < \infty$, and let $\Omega \subset \mathbb{R}^N$ be an open set. For each $f \in W_0^{1,p}(\Omega)$, there exists $f^0 \in W^{1,p}(\mathbb{R}^N)$ such that*

$$\frac{\partial f^0}{\partial x_i} = \left(\frac{\partial f}{\partial x_i}\right)^0, \quad i = 1, \ldots, N. \tag{6.18}$$

Proof Let $f_n \in \mathcal{D}(\Omega)$ be such that $f_n \to f$ in $W^{1,p}(\Omega)$. Let $\phi \in \mathcal{D}(\mathbb{R}^N)$. Then, by Hölder's inequality, we have

$$\left| \int_\Omega f_n \frac{\partial \phi}{\partial x_i} \right| = \left| \int_\Omega \frac{\partial f_n}{\partial x_i} \phi \right| \leq \left\| \frac{\partial f_n}{\partial x_i} \right\|_{L^p(\Omega)} \|\phi\|_{L^{p'}(\Omega)}$$

$$\leq \|f_n\|_{W^{1,p}(\Omega)} \, \|\phi\|_{L^{p'}(\mathbb{R}^N)}.$$

Therefore, letting $n \to \infty$, we obtain

$$\left| \int_{\mathbb{R}^N} f^0 \frac{\partial \phi}{\partial x_i} \right| = \left| \int_{\Omega} f \frac{\partial \phi}{\partial x_i} \right| \leq \|f\|_{W^{1,p}(\Omega)} \, \|\phi\|_{L^{p'}(\mathbb{R}^N)}. \tag{6.19}$$

Now, since $1 < p' < \infty$, we have that $\mathcal{D}(\mathbb{R}^N)$ is dense in $L^{p'}(\mathbb{R}^N)$, and thus from (6.19) it follows that the linear functional

$$\phi \mapsto \int_{\mathbb{R}^N} f^0 \frac{\partial \phi}{\partial x_i}$$

can be extended to a unique continuous linear functional on $L^{p'}(\mathbb{R}^N)$. Consequently, by the Riesz Representation Theorem, there exist $g_i \in L^p(\mathbb{R}^N)$ $(i = 1, \ldots, N)$ such that:

$$\|g_i\|_{L^p(\mathbb{R}^N)} \leq \|f\|_{W^{1,p}(\Omega)} \tag{6.20}$$

and

$$\int_{\mathbb{R}^N} g_i \phi = \int_{\mathbb{R}^N} f^0 \frac{\partial \phi}{\partial x_i}, \quad \forall\, \phi \in \mathcal{D}(\mathbb{R}^N). \tag{6.21}$$

As a consequence of (6.21), we have that

$$\frac{\partial f^0}{\partial x_i} = -g_i \in L^p(\mathbb{R}^N), \quad \forall\, i = 1, \ldots, N.$$

so that $f^0 \in W^{1,p}(\mathbb{R}^N)$. Now, since

$$\left.\frac{\partial f^0}{\partial x_i}\right|_{\Omega} = \frac{\partial f}{\partial x_i} \quad \text{and} \quad \left.\frac{\partial f^0}{\partial x_i}\right|_{\mathbb{R}^N \setminus \Omega} = 0,$$

taking into account that $\mathcal{L}^N(\partial\Omega) = 0$, it follows that (6.18) holds. □

In the case where $\partial\Omega$ is of class C^1, the converse of the previous result also holds (see Problem 6.17).

In the chain rule, we saw that if we compose a function in $W^{1,p}(\Omega)$ with a regular real function with bounded derivative, we obtain a function in $W^{1,p}(\Omega)$. We will now see that for functions in $W_0^{1,p}(\Omega)$, it is enough that the real function is Lipschitz continuous.

Theorem 6.14 (Stampacchia's Theorem) *Let $F : \mathbb{R} \to \mathbb{R}$ be a Lipschitz continuous function with $F(0) = 0$. Let $\Omega \subset \mathbb{R}^N$ be a bounded open set and $1 < p < \infty$. Then, if $f \in W_0^{1,p}(\Omega)$, it holds that $F \circ f \in W_0^{1,p}(\Omega)$.*

Proof Take $\phi_n \in \mathcal{D}(\Omega)$ such that $\phi_n \to f$ in $W^{1,p}(\Omega)$. Define $f_n := F \circ \phi_n$. Then, since $F(0) = 0$, we have that $\text{supp}(f_n)$ is compact for each $n \in \mathbb{N}$. Also, since F is Lipschitz continuous, by the Mean Value Theorem (Theorem A.1), we have

$$|f_n(x) - f_n(y)| = |F(\phi_n(x)) - F(\phi_n(y))| \leq K|\phi_n(x) - \phi_n(y)|$$

$$\leq K \sup_{z\in\Omega} \|\nabla\phi_n(z)\|_\infty \|x - y\|.$$

Hence, each f_n is Lipschitz continuous. Thus, $f_n \in L^p(\Omega)$ and moreover, $|\frac{\partial f_n}{\partial x_i}| \leq K_n$ for $i = 1, \ldots, N$. Since Ω is bounded, $f_n \in W^{1,p}(\Omega)$ and because $\text{supp}(f_n)$ is compact, by Lemma 6.2 we have $f_n \in W_0^{1,p}(\Omega)$.

Now, we show that $f_n \to F \circ f$ in $W^{1,p}(\Omega)$, which implies $F \circ f \in W_0^{1,p}(\Omega)$. Indeed:

$$|f_n(x) - F(f(x))| \leq K|\phi_n(x) - f(x)|,$$

so $f_n \to F \circ f$ in $L^p(\Omega)$. On the other hand,

$$\left|\frac{f_n(x + he_i) - f_n(x)}{h}\right| \leq K \frac{|\phi_n(x + he_i) - \phi_n(x)|}{|h|}.$$

Therefore,

$$\left|\frac{\partial f_n}{\partial x_i}(x)\right| \leq K \left|\frac{\partial \phi_n}{\partial x_i}(x)\right|,$$

and hence,

$$\left\|\frac{\partial f_n}{\partial x_i}\right\|_p \leq K \left\|\frac{\partial \phi_n}{\partial x_i}\right\|_p.$$

Since

$$\frac{\partial \phi_n}{\partial x_i} \to \frac{\partial f}{\partial x_i} \quad \text{in} \quad L^p(\Omega),$$

we get

$$\limsup_{n\to\infty} \left\|\frac{\partial f_n}{\partial x_i}\right\|_p \leq K \left\|\frac{\partial f}{\partial x_i}\right\|_p.$$

Therefore, for each $i = 1, \ldots, N$, the sequence $\left\{\frac{\partial f_n}{\partial x_i}\right\}$ is bounded in $L^p(\Omega)$. Consequently, since $L^p(\Omega)$ is reflexive, there exist $g_i \in L^p(\Omega)$ such that

$$\sigma(L^p(\Omega), L^{p'}(\Omega)) - \lim_{n\to\infty} \frac{\partial f_n}{\partial x_i} = g_i, \quad i = 1, \ldots, N.$$

Hence, for any $\phi \in \mathcal{D}(\Omega)$, since $\phi \in L^{p'}(\Omega)$,

$$\lim_{n\to\infty} \int_\Omega \frac{\partial f_n}{\partial x_i}\phi = \int_\Omega g_i \phi,$$

and

$$\lim_{n\to\infty} \int_\Omega \frac{\partial f_n}{\partial x_i}\phi = -\lim_{n\to\infty} \int_\Omega f_n \frac{\partial \phi}{\partial x_i} = -\int_\Omega (F \circ f)\frac{\partial \phi}{\partial x_i} = \int_\Omega \frac{\partial (F \circ f)}{\partial x_i}\phi.$$

Therefore,

$$\frac{\partial (F \circ f)}{\partial x_i} = g_i \in L^p(\Omega), \quad i = 1, \ldots, N.$$

Consequently,

$$F \circ f \in \overline{W_0^{1,p}(\Omega)}^{\|\cdot\|_p \times \sigma(L^p(\Omega), L^{p'}(\Omega))} = W_0^{1,p}(\Omega).$$

□

Remark 6.2 If $\Omega \subset \mathbb{R}^N$ is a bounded open set with $\partial\Omega$ of class C^1, then, by Theorem 6.4, given a function $f \in W^{1,p}(\Omega)$, there exist functions $f_n \in \mathcal{D}(\mathbb{R}^N)$ such that

$$f_n \to f \quad \text{in } W^{1,p}(\Omega).$$

Therefore, the same proof as in the previous result ensures that Stampacchia's Theorem also holds in this context.

We denote by $W^{-1,p'}(\Omega)$ the dual space of $W_0^{1,p}(\Omega)$, with $1 \le p < \infty$, and by $H^{-1}(\Omega)$ the dual of $H_0^1(\Omega)$. The following result characterizes the elements of $W^{-1,p'}(\Omega)$.

Theorem 6.15 *Given $T \in W^{-1,p'}(\Omega)$, there exist functions $g_0, g_1, \ldots, g_N \in L^{p'}(\Omega)$ such that*

$$\langle T, f\rangle = \int_\Omega g_0 f + \sum_{i=1}^N \int_\Omega g_i \frac{\partial f}{\partial x_i}, \quad \forall\, f \in W_0^{1,p}(\Omega), \tag{6.22}$$

and

$$\|T\| = \max_{0\le i\le N} \|g_i\|_{p'}. \tag{6.23}$$

Proof Let $E := (L^p(\Omega))^{N+1}$, and $P : W_0^{1,p}(\Omega) \to E$ be the isometry defined by

$$P(f) := \left(f, \frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_N}\right).$$

Let $Q : P(W_0^{1,p}(\Omega)) \to W_0^{1,p}(\Omega)$ be the inverse of P. Consider the continuous linear functional

$$h \in P(W_0^{1,p}(\Omega)) \mapsto \langle T, Q(h)\rangle.$$

By the Hahn-Banach Theorem, this functional can be extended to a continuous linear functional $\Phi \in E'$. Then, by applying the Riesz Representation Theorem, there exist functions $g_0, g_1, \ldots, g_N \in L^{p'}(\Omega)$ such that

$$\langle \Phi, h\rangle = \int_\Omega g_0 h_0 + \sum_{i=1}^N \int_\Omega g_i h_i, \quad h = (h_0, h_1, \ldots, h_N) \in E$$

and

$$\|\Phi\| = \|T\|.$$

Therefore, we have that $\|T\| = \max_{0\le i\le N} \|g_i\|_{p'}$, and given $f \in W_0^{1,p}(\Omega)$, if we take $h = P(f) = (f, \frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_N})$, we obtain

$$\langle T, f\rangle = \int_\Omega g_0 f + \sum_{i=1}^N \int_\Omega g_i \frac{\partial f}{\partial x_i}.$$

□

Remark 6.3 Since $\mathcal{D}(\Omega)$ is dense in $W_0^{1,p}(\Omega)$, the elements of $W^{-1,p'}(\Omega)$ are uniquely determined by their restriction to $\mathcal{D}(\Omega)$. Therefore, given $T \in W^{-1,p'}(\Omega)$, from (6.22), we can write for $\phi \in \mathcal{D}(\Omega)$,

$$\langle T, \phi\rangle = \int_\Omega g_0 \phi - \sum_{i=1}^N \int_\Omega \phi \frac{\partial g_i}{\partial x_i}.$$

Consequently, we can identify the functional T with the distribution

$$g_0 - \sum_{i=1}^{N} \frac{\partial g_i}{\partial x_i}.$$

6.6 Extension of Functions

It is often convenient to establish properties of the spaces $W^{m,p}(\Omega)$ by first studying the case $\Omega = \mathbb{R}^N$, as we will do, for example, in the section dedicated to embedding theorems. Therefore, it is useful to be able to extend functions in $W^{m,p}(\Omega)$ to functions in $W^{m,p}(\mathbb{R}^N)$. This is not true for every open set Ω; certain regularity conditions on its boundary $\partial\Omega$ are required.

Theorem 6.16 (Extension Theorem) *Assume that $1 \le p \le \infty$ and that $\Omega \subset \mathbb{R}^N$ is a bounded open set with $\partial\Omega$ of class C^1. Let $W \subset \mathbb{R}^N$ be an open set such that $\Omega \subset\subset W$. Then, there exists a bounded linear operator $E : W^{1,p}(\Omega) \to W^{1,p}(\mathbb{R}^N)$ such that for each $f \in W^{1,p}(\Omega)$, the following hold:*

(i) $E(f) = f$ *almost everywhere in* Ω,
(ii) $\operatorname{supp}(E(f)) \subset W$.

Moreover, there exists a constant C, depending only on p, Ω, and W, such that

$$\|E(f)\|_{W^{1,p}(\mathbb{R}^N)} \le C\|f\|_{W^{1,p}(\Omega)}. \tag{6.24}$$

Proof First, suppose $f \in C^\infty(\overline{\Omega})$. Fix a point $x^0 \in \partial\Omega$ and assume that $\partial\Omega$ is flat near x^0, i.e., that near x^0, $\partial\Omega$ lies in the plane $\{x_N = 0\}$. Then we can assume that there exists a ball $B = B(x^0, r)$ such that

$$B^+ = B \cap \{x_N \ge 0\} \subset \overline{\Omega},$$

$$B^- := B \cap \{x_N \le 0\} \subset \mathbb{R}^N \setminus \Omega.$$

We extend f from B^+ to B^- by reflection as follows:

$$\tilde{f}(x) := \begin{cases} f(x), & \text{if } x \in B^+ \\ -3f(x_1, \ldots, x_{N-1}, -x_N) + 4f(x_1, \ldots, x_{N-1}, -\frac{x_N}{2}), & \text{if } x \in B^-. \end{cases}$$

We claim that

$$\tilde{f} \in C^1(B) \quad \text{and} \quad \|\tilde{f}\|_{W^{1,p}(B)} \le C\|f\|_{W^{1,p}(B^+)}, \tag{6.25}$$

with the constant C independent of f. Indeed, define $f_1 := \tilde{f}_{|B^-}$ and $f_2 := \tilde{f}_{|B^+}$. Then,

$$\frac{\partial f_1}{\partial x_N}(x) = 3\frac{\partial f}{\partial x_N}(x_1, \ldots, x_{N-1}, -x_N) - 2\frac{\partial f}{\partial x_N}\left(x_1, \ldots, x_{N-1}, -\frac{x_N}{2}\right),$$

so that

$$\frac{\partial f_1}{\partial x_N}(x) = \frac{\partial f_2}{\partial x_N}(x), \quad \forall x \in \{x_N = 0\}. \tag{6.26}$$

Since $f_1(x) = f_2(x)$ for every $x \in \{x_N = 0\}$, we also have

$$\frac{\partial f_1}{\partial x_i}(x) = \frac{\partial f_2}{\partial x_i}(x), \quad \forall x \in \{x_N = 0\}, \quad i = 1, \ldots, N-1. \tag{6.27}$$

From (6.26) and (6.27), and by the definition of $\tilde{f}$, we conclude that (6.25) holds.

Now, if $\partial\Omega$ is not necessarily flat near x^0, we know that there exists a C^1 map ϕ with C^1 inverse ψ such that in the new coordinates $y = \phi(x)$, $\partial\Omega$ is flat near x^0. Then, if $g(y) := f(\psi(y))$, applying the previous result, there exists a ball B such that we can extend g from B^+ to a function $\tilde{g} \in C^1(B)$ satisfying

$$\|\tilde{g}\|_{W^{1,p}(B)} \leq C\|g\|_{W^{1,p}(B^+)}.$$

Let $U := \psi(B)$. Going back to the x-variables, we obtain an extension $\tilde{f}$ of f to U, satisfying:

$$\|\tilde{f}\|_{W^{1,p}(U)} \leq C\|f\|_{W^{1,p}(\Omega)}. \tag{6.28}$$

Now, since $\partial\Omega$ is compact, we can find finitely many points $x_i^0 \in \partial\Omega$ (for $i = 1, 2, \ldots, m$), open neighborhoods U_i of x_i^0, and extensions $\tilde{f}_i \in C^1(U_i)$ of f, such that

$$\|\tilde{f}_i\|_{W^{1,p}(U_i)} \leq C\|f\|_{W^{1,p}(\Omega)}, \quad i = 1, 2, \ldots, m, \tag{6.29}$$

and

$$\partial\Omega \subset \bigcup_{i=1}^{m} U_i.$$

Take $U_0 \subset\subset \Omega$ such that

$$\Omega \subset \bigcup_{i=0}^{m} U_i.$$

Let $\{\psi_i\}_{i=0}^m$ be a partition of unity subordinate to the cover $\{U_i\}_{i=0}^m$. Define $\tilde{f} := \sum_{i=0}^m \psi_i \tilde{f}_i$, with $\tilde{f}_0 = f$. Then $\tilde{f}$ is an extension of f to $\mathbb{R}^N$, and by (6.29), there exists a constant C, independent of f, such that

$$\|\tilde{f}\|_{W^{1,p}(\mathbb{R}^N)} \leq C\|f\|_{W^{1,p}(\Omega)}. \tag{6.30}$$

Note that we can take $\tilde{f}$ so that $\operatorname{supp}(\tilde{f}) \subset W$.

If we define $E(f) := \tilde{f}$, we already have what we want in the case where $f \in C^\infty(\overline{\Omega})$. To move to the general case, we apply the Global Approximation Theorem for smooth functions (Theorem 6.4). By this theorem, given $f \in W^{1,p}(\Omega)$, there exist $f_n \in C^\infty(\overline{\Omega})$ such that

$$f_n \to f \quad \text{in } W^{m,p}(\Omega).$$

Then, by estimate (6.30) and the linearity of E, we have

$$\|E(f_n) - E(f_k)\|_{W^{1,p}(\mathbb{R}^N)} \leq C\|f_n - f_k\|_{W^{1,p}(\Omega)}.$$

Therefore, $\{E(f_n)\}$ is a Cauchy sequence in $W^{1,p}(\mathbb{R}^N)$, and hence, there exists

$$E(f) = \lim_{n\to\infty} E(f_n) \quad \text{in } W^{1,p}(\mathbb{R}^N).$$

This extension does not depend on the sequence $\{f_n\}$, and satisfies the conclusions of the theorem. □

6.7 Continuous Embedding

In this section, we study the continuous embedding of Sobolev spaces into L^p spaces and spaces of continuous functions. We begin by studying the embeddings of the spaces $W^{1,p}(\mathbb{R}^N)$, and then we obtain the general results by means of extension operators. We analyze separately the cases: $1 \leq p < N$, $p = N$, and $p > N$.

We begin with the case $1 \leq p < N$. In this case, we define the exponent p^*, called the *Sobolev conjugate* of p, as

$$p^* := \frac{Np}{N-p}.$$

Note that

$$\frac{1}{p^*} = \frac{1}{p} - \frac{1}{N}, \qquad p^* > p.$$

We need the following result.

Lemma 6.3 (Gagliardo's Lemma) *Let $N \geq 2$, $f_1, \ldots, f_N \in L^{N-1}(\mathbb{R}^{N-1})$. For $x = (x_1, \ldots, x_N) \in \mathbb{R}^N$, we write $\hat{x}_i := (x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_N)$. Then, if $f : \mathbb{R}^N \to \mathbb{R}$ is defined by $f(x) := f_1(\hat{x}_1) f_2(\hat{x}_2) \cdots f_N(\hat{x}_N)$, we have that $f \in L^1(\mathbb{R}^N)$, and*

$$\|f\|_1 \leq \prod_{i=1}^{N} \|f_i\|_{N-1}.$$

Proof We will prove it by induction on N. For $N = 2$, we have $f_1, f_2 \in L^1(\mathbb{R})$, and $f(x_1, x_2) = f_1(x_2) f_2(x_1)$. Then, by Fubini's Theorem,

$$\|f\|_1 = \int_{\mathbb{R}^2} |f(x_1, x_2)| = \int_{\mathbb{R}^2} |f_1(x_1)|\, |f_2(x_2)| = \|f_1\|_1 \, \|f_2\|_1.$$

Assume the result holds for N, and let us prove it for $N + 1$. Indeed, let $f_1, \ldots, f_{N+1} \in L^N(\mathbb{R}^N)$. Fixing x_{N+1} and applying Hölder's inequality, we get

$$\int_{\mathbb{R}^N} |f(x)|\, dx_1 \ldots dx_N \leq \|f_{N+1}\|_{L^N(\mathbb{R}^N)} \left(\int_{\mathbb{R}^N} |f_1(\hat{x}_1) \cdots f_N(\hat{x}_N)|^m \right)^{\frac{1}{m}},$$

with

$$\frac{1}{N} + \frac{1}{m} = 1.$$

Now, since x_{N+1} is a fixed parameter and

$$|f_i|^{m(N-1)} = |f_i|^N \in L^1(\mathbb{R}^{N-1}) \quad \text{for } i = 1, \ldots, N,$$

we have $|f_i|^m \in L^{N-1}(\mathbb{R}^{N-1})$. Then, by the induction hypothesis, we have

$$g(x) := |f_1(\hat{x}_1)|^m \cdots |f_N(\hat{x}_N)|^m \in L^1(\mathbb{R}^N),$$

and

$$\int_{\mathbb{R}^N} |g(x)|\, dx \leq \prod_{i=1}^{N} \| |f_i|^m \|_{L^{N-1}(\mathbb{R}^{N-1})}.$$

Now,

$$\| |f_i|^m \|_{L^{N-1}(\mathbb{R}^{N-1})} = \left(\int_{\mathbb{R}^{N-1}} |f_i|^{m(N-1)} \right)^{\frac{1}{N-1}} = \left(\int_{\mathbb{R}^{N-1}} |f_i|^N \right)^{\frac{m}{N}} = \|f_i\|^m_{L^N(\mathbb{R}^{N-1})}.$$

Therefore,

$$\int_{\mathbb{R}^N} |g(x)|\,dx \le \prod_{i=1}^{N} \|f_i\|^m_{L^N(\mathbb{R}^{N-1})}.$$

Then,

$$\int_{\mathbb{R}^{N+1}} |f(x)|\,dx \le \|f_{N+1}\|_{L^N(\mathbb{R}^N)} \int_{\mathbb{R}} \prod_{i=1}^{N} \|f_i\|^m_{L^N(\mathbb{R}^{N-1})}\,dx_{N+1}.$$

Now, since $|f_i|^N \in L^1(\mathbb{R}^N)$ for $i = 1, \ldots, N$, we have

$$x_{N+1} \mapsto \int_{\mathbb{R}^{N-1}} |f_i(\cdot, x_{N+1})|^N\,dx_1 \ldots dx_N \in L^1(\mathbb{R}),$$

so that

$$x_{N+1} \mapsto \|f_i\|_{L^N(\mathbb{R}^{N-1})} \in L^N(\mathbb{R}).$$

Then, by Hölder's inequality,

$$x_{N+1} \mapsto \prod_{i=1}^{N} \|f_i\|_{L^N(\mathbb{R}^{N-1})} \in L^1(\mathbb{R}),$$

and moreover,

$$\int_{\mathbb{R}} \prod_{i=1}^{N} \|f_i\|_{L^N(\mathbb{R}^{N-1})}\,dx_{N+1} \le \prod_{i=1}^{N} \int_{\mathbb{R}} \|f_i\|_{L^N(\mathbb{R}^{N-1})} = \prod_{i=1}^{N} \|f_i\|_{L^N(\mathbb{R}^N)}.$$

Consequently,

$$\|f\|_{L^1(\mathbb{R}^{N+1})} \le \prod_{i=1}^{N+1} \|f_i\|_{L^N(\mathbb{R}^N)}.$$

□

Theorem 6.17 (Gagliardo-Nirenberg Inequality) *Suppose that* $1 \le p < N$. *Then there exists a constant* C, *depending only on* p *and* N, *such that*

$$\|f\|_{p^*} \le C\|\nabla f\|_p \quad \forall\, f \in W^{1,p}(\mathbb{R}^N). \tag{6.31}$$

Proof First, suppose that $p = 1$, so that $p^* = \frac{N}{N-1}$. For $f \in \mathcal{D}(\mathbb{R}^N)$, we have

$$f(x) = \int_{-\infty}^{x_i} \frac{\partial f}{\partial x_i}(x_1, \ldots, x_{i-1}, t, x_{i+1}, \ldots, x_N)\, dt,$$

which gives

$$|f(x)| \leq \int_{-\infty}^{x_i} \left| \frac{\partial f}{\partial x_i}(x_1, \ldots, x_{i-1}, t, x_{i+1}, \ldots, x_N) \right| dt$$

$$\leq \int_{\mathbb{R}} \left| \frac{\partial f}{\partial x_i}(x_1, \ldots, x_{i-1}, t, x_{i+1}, \ldots, x_N) \right| dt.$$

Therefore, if we define

$$f_i(\hat{x}_i) := \int_{\mathbb{R}} \left| \frac{\partial f}{\partial x_i}(x_1, \ldots, x_{i-1}, t, x_{i+1}, \ldots, x_N) \right| dt,$$

we get

$$|f(x)|^N \leq \prod_{i=1}^{N} f_i(\hat{x}_i),$$

so that

$$|f(x)|^{\frac{N}{N-1}} \leq \prod_{i=1}^{N} |f_i(\hat{x}_i)|^{\frac{1}{N-1}}. \tag{6.32}$$

Now, since $\frac{\partial f}{\partial x_i} \in L^1(\mathbb{R}^N)$, we have $|f_i|^{\frac{1}{N-1}} \in L^{N-1}(\mathbb{R}^{N-1})$, and thus, applying Gagliardo's Lemma and taking into account (6.31), we obtain

$$\int_{\mathbb{R}^N} |f(x)|^{\frac{N}{N-1}}\, dx \leq \prod_{i=1}^{N} \left\| |f_i|^{\frac{1}{N-1}} \right\|_{L^{N-1}(\mathbb{R}^{N-1})}$$

$$= \prod_{i=1}^{N} \|f_i\|_{L^1(\mathbb{R}^{N-1})}^{\frac{1}{N-1}} = \prod_{i=1}^{N} \left\| \frac{\partial f}{\partial x_i} \right\|_{L^1(\mathbb{R}^N)}^{\frac{1}{N-1}}.$$

Consequently, for every $f \in \mathcal{D}(\mathbb{R}^N)$, we have

$$\|f\|_{p^*} \le \prod_{i=1}^{N} \left\| \frac{\partial f}{\partial x_i} \right\|_{L^1(\mathbb{R}^N)}^{\frac{1}{N}} \le \frac{1}{N} \sum_{i=1}^{N} \left\| \frac{\partial f}{\partial x_i} \right\|_{L^1(\mathbb{R}^N)} = \frac{1}{N} \|\nabla f\|_1. \tag{6.33}$$

Let $f \in W^{1,1}(\mathbb{R}^N)$. By the density of $\mathcal{D}(\mathbb{R}^N)$ in $W^{1,1}(\mathbb{R}^N)$, there exists a sequence $f_n \in \mathcal{D}(\mathbb{R}^N)$ such that $f_n \to f$ in $W^{1,1}(\mathbb{R}^N)$. Moreover, extracting a subsequence if necessary, we may assume that $f_n \to f$ almost everywhere. Now, by (6.33), we have

$$\|f_n\|_{p^*} \le \frac{1}{N} \|\nabla f_n\|_1, \qquad \forall\, n \in \mathbb{N}.$$

Then, by Fatou's Lemma, we conclude that $f \in L^{p^*}(\mathbb{R}^N)$, and that

$$\|f\|_{p^*} \le \frac{1}{N} \|\nabla f\|_1.$$

Now consider the case $1 < p < N$. Let $f \in \mathcal{D}(\mathbb{R}^N)$. For $\gamma \ge 1$, let $g := |f|^{\gamma-1} f$. Applying (6.33) to g, we obtain

$$\|g\|_{\frac{N}{N-1}} \le \prod_{i=1}^{N} \left\| \gamma |f|^{\gamma-1} \frac{\partial f}{\partial x_i} \right\|_1^{\frac{1}{N}}.$$

Therefore, by Hölder's inequality, we get

$$\|f\|_{\frac{\gamma N}{N-1}}^{\gamma} \le \gamma \prod_{i=1}^{N} \||f|^{\gamma-1}\|_{p'}^{\frac{1}{N}} \left\| \frac{\partial f}{\partial x_i} \right\|_p^{\frac{1}{N}} = \gamma \|f\|_{p'(\gamma-1)}^{\gamma-1} \prod_{i=1}^{N} \left\| \frac{\partial f}{\partial x_i} \right\|_p^{\frac{1}{N}}.$$

Now take $\gamma := \frac{p^*(N-1)}{N} = \frac{p(N-1)}{N-p} \ge 1$. Then $\frac{\gamma N}{N-1} = p'(\gamma - 1)$, and so

$$\|f\|_{p^*}^{\gamma} \le \gamma \|f\|_{p^*}^{\gamma-1} \prod_{i=1}^{N} \left\| \frac{\partial f}{\partial x_i} \right\|_p^{\frac{1}{N}},$$

from which it follows that

$$\|f\|_{p^*} \le \frac{\gamma}{N} \sum_{i=1}^{N} \left\| \frac{\partial f}{\partial x_i} \right\|_p = \frac{\gamma}{N} \|\nabla f\|_p.$$

Again, the result follows by density. □

Remark 6.4 The value of the Sobolev conjugate p^* can be easily obtained through the following homogeneity argument. Suppose that q is an exponent satisfying the Gagliardo-Nirenberg inequality, i.e.,

$$\|f\|_q \leq C\|\nabla f\|_p \quad \forall\, f \in W^{1,p}(\mathbb{R}^N). \tag{6.34}$$

Let $f \in \mathcal{D}(\mathbb{R}^N)$, $f \neq 0$. For each $\lambda > 0$, we define the rescaled function

$$f_\lambda(x) := f(\lambda x), \qquad x \in \mathbb{R}^N.$$

As a consequence of (6.34) we have that

$$\|f_\lambda\|_q \leq C\|\nabla f_\lambda\|_p \quad \text{for each } \lambda > 0. \tag{6.35}$$

Now,

$$\int_{\mathbb{R}^N} |f_\lambda(x)|^q \, dx = \frac{1}{\lambda^N} \int_{\mathbb{R}^N} |f(x)|^q \, dx,$$

and

$$\int_{\mathbb{R}^N} \|\nabla f_\lambda(x)\|^p \, dx = \frac{\lambda^p}{\lambda^N} \int_{\mathbb{R}^N} \|\nabla f(x)\|^p \, dx.$$

Consequently, from (6.35) we deduce that

$$\frac{1}{\lambda^{N/q}} \|f\|_q \leq C \frac{\lambda}{\lambda^{N/p}} \|\nabla f\|_p,$$

so that

$$\|f\|_q \leq C\lambda^{1-\frac{N}{p}+\frac{N}{q}} \|\nabla f\|_p. \tag{6.36}$$

But then, if $1 - \frac{N}{p} + \frac{N}{q} \neq 0$, we can let $\lambda \to 0$ or $\lambda \to \infty$ in (6.36), yielding a contradiction. Therefore, we must necessarily have that $1 - \frac{N}{p} + \frac{N}{q} = 0$, from which it follows that $\frac{1}{q} = \frac{1}{p} - \frac{1}{N}$, i.e., $q = p^*$.

Corollary 6.1 *For $1 \leq p < N$, we have*

$$W^{1,p}(\mathbb{R}^N) \subset L^r(\mathbb{R}^N), \qquad \forall\, r \in [p, p^*]$$

with continuous embedding.

Proof We know that $W^{1,p}(\mathbb{R}^N) \subset L^p(\mathbb{R}^N)$ with continuous embedding, and by the Gagliardo–Nirenberg inequality, $W^{1,p}(\mathbb{R}^N) \subset L^{p^*}(\mathbb{R}^N)$ with continuous embedding. Then, if $r \in [p, p^*]$, since there exists $\alpha \in [0, 1]$ such that $\frac{1}{r} =$

$\frac{\alpha}{p} + \frac{1-\alpha}{p^*}$, applying the interpolation inequality (Problem 2.13), we obtain that if $f \in W^{1,p}(\mathbb{R}^N)$,

$$\|f\|_r \leq \|f\|_p^{\alpha} \|f\|_{p^*}^{1-\alpha} \leq C\|f\|_{1,p}^{\alpha} (C_1\|\nabla f\|_p)^{1-\alpha} \leq C_2\|f\|_{1,p}.$$

□

Let's now consider the case $p = N$.

Theorem 6.18 *If $p = N$, then for each $r \geq p$,*

$$W^{1,p}(\mathbb{R}^N) \subset L^r(\mathbb{R}^N)$$

with continuous embedding.

Proof In the proof of the Gagliardo–Nirenberg inequality, we showed that if $f \in \mathcal{D}(\mathbb{R}^N)$ and $\gamma \geq 1$, then

$$\|f\|_{\frac{\gamma N}{N-1}}^{\gamma} \leq \frac{\gamma}{N}\|f\|_{p'(\gamma-1)}^{\gamma-1}\|\nabla f\|_N. \tag{6.37}$$

Now, since $(a+b)^{\gamma} \geq \gamma a^{\gamma-1}b$ for $a, b \geq 0$, and $p' = \frac{N}{N-1}$, we obtain

$$\|f\|_{\frac{\gamma N}{N-1}}^{\gamma} \leq \frac{\gamma}{N}\|f\|_{\frac{N}{N-1}(\gamma-1)}^{\gamma-1}\|\nabla f\|_N \leq \frac{1}{N}\left(\|f\|_{\frac{N}{N-1}(\gamma-1)} + \|\nabla f\|_N\right)^{\gamma}.$$

From here, taking $\gamma = N$, we get

$$\|f\|_{\frac{N^2}{N-1}} \leq \sqrt[N]{\frac{1}{N}}\left(\|f\|_N + \|\nabla f\|_N\right) \quad \forall\, f \in \mathcal{D}(\mathbb{R}^N).$$

Therefore, by the density of $\mathcal{D}(\mathbb{R}^N)$ in $W^{1,N}(\mathbb{R}^N)$, we conclude that

$$W^{1,N}(\mathbb{R}^N) \subset L^{\frac{N^2}{N-1}}(\mathbb{R}^N),$$

with continuous embedding. Now, by interpolation, we obtain

$$W^{1,N}(\mathbb{R}^N) \subset L^r(\mathbb{R}^N)$$

with continuous embedding, for $N \leq r \leq \frac{N^2}{N-1}$. Taking $\gamma = N+1$ in (6.37), we get for $f \in \mathcal{D}(\mathbb{R}^N)$,

$$\|f\|_{\frac{N(N+1)}{N-1}} \leq \sqrt[N+1]{\frac{1}{N}}\left(\|f\|_{\frac{N^2}{N-1}} + \|\nabla f\|_N\right) \leq C\left(\|f\|_N + \|\nabla f\|_N\right).$$

Then, again by density, we obtain

$$W^{1,N}(\mathbb{R}^N) \subset L^{\frac{N(N+1)}{N-1}}(\mathbb{R}^N),$$

with continuous embedding. Then, using interpolation again, we complete the proof by induction. □

Before studying the case $N < p < \infty$, we are going to introduce the Hölder spaces. Let $\Omega \subset \mathbb{R}^N$ be an open set and $0 < \gamma \leq 1$. A function $f : \Omega \to \mathbb{R}$ is said to be *Hölder continuous* in Ω with exponent γ if there exists a constant $C > 0$ such that

$$|f(x) - f(y)| \leq C\|x - y\|^{\gamma}, \quad \forall\, x, y \in \Omega.$$

For $\gamma = 1$, we obtain the Lipschitz continuous functions.

For $f : \Omega \to \mathbb{R}$ Hölder continuous in Ω with exponent γ, we define its γ-th Hölder seminorm by

$$[f]_{\gamma,\Omega} := \sup_{x,y\in\Omega,\ x\neq y} \left\{ \frac{|f(x) - f(y)|}{\|x - y\|^{\gamma}} \right\},$$

and its γ-th Hölder norm by

$$\|f\|_{C^{0,\gamma}(\overline{\Omega})} := \|f\|_{\infty} + [f]_{\gamma,\Omega}.$$

Definition 6.2 For $k = 0, 1, 2, \ldots$, the *Hölder space* $C^{k,\gamma}(\overline{\Omega})$ consists of the functions $f \in C^k(\overline{\Omega})$ for which the norm

$$\|f\|_{C^{k,\gamma}(\overline{\Omega})} := \sum_{|\alpha|\leq k} \|D^{\alpha} f\|_{\infty} + \sum_{|\alpha|=k} [D^{\alpha} f]_{\gamma,\Omega}$$

is finite.

Thus, the space $C^{k,\gamma}(\overline{\Omega})$ consists of functions f that are k times continuously differentiable, and whose partial derivatives of order k are Hölder continuous of exponent γ. These spaces, with respect to the norm $\|\cdot\|_{C^{k,\gamma}(\overline{\Omega})}$, are Banach spaces (Problem 6.18).

Theorem 6.19 (Morrey's Inequality) *Assume that $N < p \leq \infty$. Then there exists a constant C depending only on N and p, such that*

$$\|f\|_{C^{0,\gamma}(\mathbb{R}^N)} \leq C\|f\|_{1,p} \qquad \forall\, f \in W^{1,p}(\mathbb{R}^N), \tag{6.38}$$

where $\gamma = 1 - \frac{N}{p}$.

Proof Assume $N < p < \infty$; the proof for $p = \infty$ is similar. Fix a ball $B(x, r) \subset \mathbb{R}^N$. We will show that there exists a constant $C > 0$, depending only on N, such that

$$\fint_{B(x,r)} |f(y) - f(x)|\, dy \le C \int_{B(x,r)} \frac{\|\nabla f(y)\|}{\|y - x\|^{N-1}}\, dy. \tag{6.39}$$

Indeed, fix a point $z \in \partial B(0, 1)$. Then, if $0 < s < r$, we have

$$|f(x + sz) - f(x)| = \left| \int_0^s \frac{d}{dt} f(x + tz)\, dt \right|$$

$$= \left| \int_0^s \nabla f(x + tz) \cdot z\, dt \right| \le \int_0^s \|\nabla f(x + tz)\|\, dt.$$

Therefore, using polar coordinates, we obtain

$$\int_{\partial B(0,1)} |f(x + sz) - f(x)|\, dS(z) \le \int_0^s \int_{\partial B(0,1)} \|\nabla f(x + tz)\|\, dS(z) dt$$

$$= \int_{B(x,s)} \frac{\|\nabla f(y)\|}{\|x - y\|^{N-1}}\, dy \le \int_{B(x,r)} \frac{\|\nabla f(y)\|}{\|x - y\|^{N-1}}\, dy.$$

Now, multiplying by s^{N-1} and integrating from 0 to r with respect to s, we get

$$\int_{B(x,r)} |f(y) - f(x)|\, dy \le \frac{r^N}{N} \int_{B(x,r)} \frac{\|\nabla f(y)\|}{\|x - y\|^{N-1}}\, dy,$$

thus proving (6.39).

Fix $x \in \mathbb{R}^N$. Applying (6.39) we get

$$|f(x)| \le \fint_{B(x,1)} |f(y) - f(x)|\, dy + \fint_{B(x,1)} |f(y)|\, dy$$

$$\le C \int_{B(x,1)} \frac{\|\nabla f(y)\|}{\|y - x\|^{N-1}}\, dy + C \|f\|_{L^p(B(x,1))}$$

$$\le C \left(\int_{\mathbb{R}^N} \|\nabla f(y)\|^p\, dy \right)^{\frac{1}{p}} \left(\int_{B(x,1)} \frac{dy}{\|x - y\|^{(N-1)\frac{p}{p-1}}} \right)^{\frac{p-1}{p}} + C \|f\|_{L^p(B(x,1))}.$$

Now, since $p > N$, we have $(N - 1)\frac{p}{p-1} < N$, hence

$$\int_{B(x,1)} \frac{dy}{\|x - y\|^{(N-1)\frac{p}{p-1}}} < \infty,$$

so

$$|f(x)| \leq C\|f\|_{1,p},$$

and since this holds for all $x \in \mathbb{R}^N$, we obtain

$$\|f\|_\infty \leq C\|f\|_{1,p}. \tag{6.40}$$

Let us now estimate the Hölder seminorm. Given $x, y \in \mathbb{R}^N$, let $r := \|x - y\|$ and $U := B(x, r) \cap B(y, r)$. Then,

$$|f(x) - f(y)| \leq \fint_U |f(x) - f(z)|\, dz + \fint_U |f(y) - f(z)|\, dz. \tag{6.41}$$

As a consequence of (6.39), we have

$$\fint_U |f(x) - f(z)|\, dz \leq C \fint_{B(x,r)} |f(x) - f(z)|\, dz \leq \int_{B(x,r)} \frac{\|\nabla f(z)\|}{\|z - x\|^{N-1}}\, dz$$

$$\leq \left(\int_{B(x,r)} \|\nabla f(z)\|^p\, dz\right)^{\frac{1}{p}} \left(\int_{B(x,r)} \frac{dz}{\|x - z\|^{(N-1)\frac{p}{p-1}}}\right)^{\frac{p-1}{p}}$$

$$\leq C\left(r^{N-(N-1)\frac{p}{p-1}}\right)^{\frac{p-1}{p}} \|\nabla f\|_p = Cr^{1-\frac{N}{p}} \|\nabla f\|_p.$$

Similarly, we obtain

$$\fint_U |f(y) - f(z)|\, dz \leq Cr^{1-\frac{N}{p}} \|\nabla f\|_p.$$

Therefore, substituting these estimates into (6.41), we obtain

$$|f(x) - f(y)| \leq Cr^{1-\frac{N}{p}} \|\nabla f\|_p = C\|x - y\|^{1-\frac{N}{p}} \|\nabla f\|_p.$$

Thus,

$$[f]_{1-\frac{N}{p},\mathbb{R}^N} = \sup_{x \neq y} \left\{ \frac{|f(x) - f(y)|}{\|x - y\|^{1-\frac{N}{p}}} \right\} \leq C\|\nabla f\|_p.$$

This inequality together with (6.40) gives us (6.38). □

As a consequence of the previous results and the Extension Theorem, we can obtain similar results to the ones above in the case of bounded open sets with smooth boundary. More precisely, we have the following result.

Theorem 6.20 (Sobolev Inequalities) *Let $\Omega \subset \mathbb{R}^N$ be a bounded open set with boundary $\partial\Omega$ of class C^1, and let $f \in W^{1,p}(\Omega)$. Then:*

(i) If $1 \le p < N$, then $f \in L^{p^*}(\Omega)$, and there exists a constant C, depending only on p, N, and Ω, such that

$$\|f\|_{p^*} \le C\|f\|_{1,p}. \tag{6.42}$$

(ii) If $p = N$, then $f \in L^q(\Omega)$ for every $q \in [N, \infty[$, and there exists a constant C_q, depending only on q, N, and Ω, such that

$$\|f\|_q \le C_q\|f\|_{1,p}. \tag{6.43}$$

(iii) If $N < p < \infty$, there exists $f^* \in C^{0,1-\frac{N}{p}}(\overline{\Omega})$, with $f^* = f$ almost everywhere in Ω, such that

$$\|f^*\|_{C^{0,1-\frac{N}{p}}(\overline{\Omega})} \le C\|f\|_{1,p}, \tag{6.44}$$

and the constant C depends only on p, N, and Ω.

Proof We will prove only item (i), since the others are similar. Since $\partial\Omega$ is of class C^1, by Theorem 6.16, there exists an extension $E(f) = \overline{f} \in W^{1,p}(\mathbb{R}^N)$ such that $\overline{f} = f$ in Ω, $\overline{f}$ has compact support, and there exists a constant C such that

$$\|\overline{f}\|_{W^{1,p}(\mathbb{R}^N)} \le C\|f\|_{W^{1,p}(\Omega)}. \tag{6.45}$$

Since $\overline{f}$ has compact support, by Lemma 6.1, there exist $f_n \in \mathcal{D}(\mathbb{R}^N)$ such that

$$f_n \to \overline{f} \quad \text{in } W^{1,p}(\mathbb{R}^N). \tag{6.46}$$

Now, by the Gagliardo–Nirenberg inequality, we have

$$\|f_n - f_m\|_{p^*} \le C\|\nabla f_n - \nabla f_m\|_p, \quad \forall\, n, m \in \mathbb{N}.$$

Therefore,

$$f_n \to \overline{f} \quad \text{in } L^{p^*}(\mathbb{R}^N). \tag{6.47}$$

Since by the Gagliardo–Nirenberg inequality we also have

$$\|f_n\|_{p^*} \le C\|\nabla f_n\|_p, \quad \forall\, n \in \mathbb{N},$$

from (6.46) and (6.47) we obtain

$$\|\overline{f}\|_{p^*} \leq C\|\nabla \overline{f}\|_p. \tag{6.48}$$

Finally, from (6.45) and (6.48) we obtain (6.42). □

Remark 6.5 In the case $p = N$, in general, it is not true that $W^{1,p}(\Omega) \subset L^\infty(\Omega)$ (see Problem 6.22).

In the case of the space $W_0^{1,p}(\Omega)$, using the extension

$$f^0(x) := \begin{cases} f(x) & \text{if } x \in \Omega \\ \\ 0 & \text{if } x \in \mathbb{R}^N \setminus \Omega. \end{cases}$$

and Theorem 6.13, instead of the general Extension Theorem, we can obtain a similar result as before but for arbitrary open sets. Specifically, we have the following result.

Theorem 6.21 (Sobolev Inequalities for $W_0^{1,p}(\Omega)$) *Let $\Omega \subset \mathbb{R}^N$ be an open set and suppose that $1 < p < \infty$. Let $f \in W_0^{1,p}(\Omega)$. Then:*

(i) If $1 < p < N$, then $f \in L^{p^*}(\Omega)$ and there exists a constant C, depending only on p, N, and Ω, such that

$$\|f\|_{p^*} \leq C\|f\|_{1,p}. \tag{6.49}$$

(ii) If $p = N$, then $f \in L^q(\Omega)$ for every $q \in [N, \infty[$ and there exists a constant C_q, depending only on q, N, and Ω, such that

$$\|f\|_q \leq C_q\|f\|_{1,p}. \tag{6.50}$$

(iii) If $N < p < \infty$, then there exists $f^* \in C^{0,1-\frac{N}{p}}(\overline{\Omega})$ with $f^* = f$ a.e. in Ω, such that

$$\|f^*\|_{C^{0,1-\frac{N}{p}}(\overline{\Omega})} \leq C\|f\|_{1,p}, \tag{6.51}$$

and the constant C depends only on p, N, and Ω.

Once the Sobolev inequalities for the spaces $W^{1,p}(\Omega)$ have been established, it is not difficult to extend them to the general case of the spaces $W^{m,p}(\Omega)$. Specifically, we have the following result, whose proof is left as an exercise (Problem 6.23).

Theorem 6.22 (Sobolev Inequalities for $W^{m,p}(\Omega)$) *Let $\Omega \subset \mathbb{R}^N$ be a bounded open set with boundary $\partial\Omega$ of class C^1, and let $f \in W^{m,p}(\Omega)$. Then:*

(i) If $m < \frac{N}{p}$, then $f \in L^q(\Omega)$ for $\frac{1}{q} = \frac{1}{p} - \frac{m}{N}$, and there exists a constant C, depending only on m, p, N, and Ω, such that

$$\|f\|_q \leq C\|f\|_{m,p}. \tag{6.52}$$

(ii) If $p = \frac{N}{m}$, then $f \in L^q(\Omega)$ for every $q \in [p, \infty)$, and there exists a constant C_q, depending only on m, q, N, and Ω, such that

$$\|f\|_q \leq C_q\|f\|_{m,p}. \tag{6.53}$$

(iii) If $m > \frac{N}{p}$, there exists $f^* \in C^{m-\left[\frac{N}{p}\right]-1,\gamma}(\overline{\Omega})$ with $f^* = f$ almost everywhere in Ω, where

$$\gamma = \begin{cases} \left[\frac{N}{p}\right] + 1 - \frac{N}{p}, & \text{if } \frac{N}{p} \text{ is not an integer,} \\ \text{any } 0 < \gamma < 1, & \text{if } \frac{N}{p} \text{ is an integer,} \end{cases}$$

such that

$$\|f^*\|_{C^{m-\left[\frac{N}{p}\right]-1,\gamma}(\overline{\Omega})} \leq C\|f\|_{m,p}, \tag{6.54}$$

and the constant C depends only on m, p, N, γ, and Ω.

Moreover, in the case $f \in W_0^{m,p}(\Omega)$, the same inequalities hold for any open set Ω.

To finish this section, let us consider the case $p = \infty$. More specifically, we will look at the following characterization of the elements of $W^{1,\infty}(\Omega)$.

Theorem 6.23 *Let $\Omega \subset \mathbb{R}^N$ be a bounded open set with boundary of class C^1. Then, a function $f : \Omega \to \mathbb{R}$ is Lipschitz continuous if and only if $f \in W^{1,\infty}(\Omega)$.*

Proof Since Ω is a bounded open set with C^1 boundary, we know there exists an extension operator $E : W^{1,\infty}(\Omega) \to W^{1,\infty}(\mathbb{R}^N)$ satisfying the properties of Theorem 6.16. Consequently, it suffices to prove the result for $f \in W^{1,\infty}(\mathbb{R}^N)$ with compact support.

Let $f \in W^{1,\infty}(\mathbb{R}^N)$ with $\operatorname{supp}(f)$ compact. Given a mollifier sequence $\{\phi_n\}$, if we define $f_n := f \star \phi_n$, then $f_n \to f$ uniformly as $n \to \infty$, and moreover, $\|\nabla f_n\|_\infty \leq \|\nabla f\|_\infty$. Then, for $x, y \in \mathbb{R}^N$, $x \neq y$, we have:

$$f_n(x) - f_n(y) = \int_0^1 \frac{d}{dt} f_n(tx + (1-t)y)\, dt$$

$$= \int_0^1 \nabla f_n(tx + (1-t)y) \cdot (x - y)\, dt,$$

so that

$$|f_n(x) - f_n(y)| \leq \|\nabla f_n\|_\infty \|x - y\| \leq \|\nabla f\|_\infty \|x - y\|.$$

Therefore, f is Lipschitz continuous.

Now suppose that f is Lipschitz continuous in $\mathbb{R}^N$. We must show that $\frac{\partial f}{\partial x_i} \in L^\infty(\mathbb{R}^N)$ for each $i = 1, \ldots, N$. Indeed, since f is Lipschitz continuous in $\mathbb{R}^N$, denoting the difference quotient of f in the direction of $\mathbf{e}_i$ by

$$\delta_i^h(f)(x) := \frac{f(x + h\mathbf{e}_i) - f(x)}{h},$$

we have

$$\|\delta_i^{-h}(f)\|_\infty \leq [f]_{1,\mathbb{R}^N} < \infty.$$

Therefore, for each $i = 1, \ldots, N$, there exists a function $g_i \in L^\infty(\mathbb{R}^N)$ and a sequence $h_n \to 0$ such that

$$\delta_i^{-h_n}(f) \rightharpoonup g_i \quad \text{weakly in} \quad L^2_{loc}(\mathbb{R}^N). \tag{6.55}$$

Taking (6.55) into account, given $\phi \in \mathcal{D}(\mathbb{R}^N)$, we have:

$$\int_{\mathbb{R}^N} f \frac{\partial \phi}{\partial x_i}\, dx = \lim_{n\to\infty} \int_{\mathbb{R}^N} f\, \delta_i^{h_n}(\phi)\, dx$$

$$= -\lim_{n\to\infty} \int_{\mathbb{R}^N} \delta_i^{-h_n}(f)\, \phi\, dx = -\int_{\mathbb{R}^N} g_i\, \phi\, dx.$$

Consequently, $\frac{\partial f}{\partial x_i} = g_i \in L^\infty(\mathbb{R}^N)$, which concludes the proof. □

Remark 6.6 The proof of the previous theorem easily adapts to show that if $\Omega \subset \mathbb{R}^N$ is an open set, then $f \in W^{1,\infty}_{loc}(\Omega)$ if and only if f is locally Lipschitz continuous in Ω.

6.8 Compact Embedding

In this section, we will see that some of the continuous embeddings from the previous section are compact, i.e., the unit ball of the first space is relatively compact in the second space. Let us first observe that in order to obtain this type

of result, it is necessary that Ω be bounded. To illustrate this, we will show that the embedding of $W^{1,p}(]0,\infty[)$ into $L^r(]0,\infty[)$ is not compact for any $r \geq 1$. Indeed, for $f \in \mathcal{D}(]0,1[)$, we define

$$f_n(x) := \begin{cases} f(x-n) \text{ if } n < x < n+1 \\ \\ 0 \qquad\qquad \text{otherwise.} \end{cases}$$

Then, since for each $n \in \mathbb{N}$,

$$\|f_n\|_{W^{1,p}(]0,\infty[)} = \|f\|_{W^{1,p}(]0,1[)},$$

we have that $\{f_n\}$ is bounded in $W^{1,p}(]0,\infty[)$. However,

$$\|f_n - f_m\|_{L^r(]0,\infty[)} = 2^{\frac{1}{r}} \|f\|_{L^r(]0,1[)},$$

so that no subsequence of $\{f_n\}$ converges in $L^r(]0,\infty[)$.

To establish the compact embedding theorem, we need compactness criteria in function spaces, such as the Ascoli-Arzelà Theorem, which we saw in Chap. 3, and the following compactness criterion in L^p spaces.

Theorem 6.24 (Fréchet–Kolmogorov Theorem) *Let $\Omega \subset \mathbb{R}^N$ be open and let $\tilde{\Omega} \subset\subset \Omega$. Let $\mathcal{K}$ be a bounded subset of $L^p(\Omega)$ with $1 \leq p < \infty$. Suppose that for every $\epsilon > 0$ there exists $\delta > 0$, $\delta < \text{dist}(\tilde{\Omega}, \mathbb{R}^N \setminus \Omega)$ such that*

$$\|\tau_{-h} f - f\|_{L^p(\tilde{\Omega})} < \epsilon \quad \forall\, h \in \mathbb{R}^N \quad \text{with } \|h\| < \delta \quad \text{and } \forall\, f \in \mathcal{K}. \tag{6.56}$$

Then, $\mathcal{K}|_{\tilde{\Omega}} := \{f|_{\tilde{\Omega}} \,:\, f \in \mathcal{K}\}$ is relatively compact in $L^p(\tilde{\Omega})$.

Proof Without loss of generality, we may assume that Ω is bounded. Then, for $f \in \mathcal{K}$, if $\tilde{f}$ is its extension by zero outside Ω, we have that $\tilde{\mathcal{K}} := \{\tilde{f} \,:\, f \in \mathcal{K}\}$ is bounded in both $L^p(\mathbb{R}^N)$ and $L^1(\mathbb{R}^N)$.

Let $\{\rho_n\}$ be a mollifier sequence. Given $\epsilon > 0$, there exists $\delta > 0$ satisfying (6.56). We claim that

$$\|\rho_n \star \tilde{f} - \tilde{f}\|_{L^p(\tilde{\Omega})} < \epsilon \quad \forall\, f \in \mathcal{K} \text{ and } \forall\, n > \frac{1}{\delta}. \tag{6.57}$$

Indeed, applying Hölder's inequality, and noting that $\displaystyle\int_{\mathbb{R}^N} \rho_n(y)\, dy = 1$, we have

$$|(\rho_n \star \tilde{f})(x) - \tilde{f}(x)| \leq \int_{\mathbb{R}^N} |\tilde{f}(x-y) - \tilde{f}(x)|\, \rho_n(y)\, dy$$

$$= \int_{\mathbb{R}^N} |\tilde{f}(x-y) - \tilde{f}(x)|\, \rho_n(y)^{\frac{1}{p}}\, \rho_n(y)^{\frac{1}{p'}}\, dy$$

$$\leq \left(\int_{\mathbb{R}^N} |\tilde{f}(x-y) - \tilde{f}(x)|^p\, \rho_n(y)\, dy \right)^{\frac{1}{p}},$$

so that

$$|(\rho_n \star \tilde{f})(x) - \tilde{f}(x)|^p \leq \int_{B(0,\frac{1}{n})} |\tilde{f}(x-y) - \tilde{f}(x)|^p\, \rho_n(y)\, dy.$$

Then, for $n > \frac{1}{\delta}$ and $f \in \mathcal{K}$, using (6.56), we obtain

$$\int_{\tilde{\Omega}} |(\rho_n \star \tilde{f})(x) - \tilde{f}(x)|^p\, dx \leq \int_{B(0,\frac{1}{n})} \rho_n(y)\, dy \int_{\tilde{\Omega}} |\tilde{f}(x-y) - \tilde{f}(x)|^p\, dx < \epsilon^p,$$

thus proving (6.57).

Now we show that for each $n \in \mathbb{N}$, the set $\mathcal{K}_n := \{(\rho_n \star \tilde{f})_{|\tilde{\Omega}} : f \in \mathcal{K}\}$ satisfies the hypotheses of the Ascoli-Arzelà Theorem. Indeed, since $\tilde{\mathcal{K}}$ is bounded in $L^1(\mathbb{R}^N)$, we have

$$\|\rho_n \star \tilde{f}\|_\infty \leq \|\rho_n\|_\infty \|\tilde{f}\|_1 \leq C_n \quad \forall\, f \in \mathcal{K},$$

so that $\mathcal{K}_n$ is bounded in $C(\overline{\tilde{\Omega}})$. Moreover, for $x_1, x_2 \in \mathbb{R}^N$ and $f \in \mathcal{K}$, we have

$$|(\rho_n \star \tilde{f})(x_1) - (\rho_n \star \tilde{f})(x_2)| \leq \|x_1 - x_2\|\, \|\rho_n'\|_\infty \|\tilde{f}\|_1 \leq A_n \|x_1 - x_2\|.$$

Hence $\mathcal{K}_n$ is equicontinuous in $C(\overline{\tilde{\Omega}})$. Consequently, by the Ascoli–Arzelà Theorem, $\mathcal{K}_n$ is relatively compact in $C(\overline{\tilde{\Omega}})$, and therefore also in $L^p(\tilde{\Omega})$.

To conclude the proof, given $\epsilon > 0$, let $\delta > 0$ satisfy (6.56). Then, for fixed $n > \frac{1}{\delta}$, from (6.56) we get

$$\|(\rho_n \star \tilde{f}) - f\|_{L^p(\tilde{\Omega})} < \epsilon \quad \forall\, f \in \mathcal{K}. \tag{6.58}$$

Now, since $\mathcal{K}_n$ is relatively compact in $L^p(\tilde{\Omega})$, we can cover it by a finite number of balls of radius ϵ. Then, taking into account (6.58), we deduce that $\mathcal{K}|_{\tilde{\Omega}}$ can be covered by a finite number of balls of radius 2ϵ. Hence, $\mathcal{K}|_{\tilde{\Omega}}$ is relatively compact in $L^p(\tilde{\Omega})$. □

Corollary 6.2 (M. Riesz) *Let $\Omega \subset \mathbb{R}^N$ be an open set and $\mathcal{K}$ a bounded subset of $L^p(\Omega)$ with $1 \leq p < \infty$. Suppose that*

$$\begin{cases} \forall\, \epsilon > 0 \ \text{and}\ \tilde{\Omega} \subset\subset \Omega, \ \ \exists\, 0 < \delta < \operatorname{dist}(\tilde{\Omega}, \mathbb{R}^N \setminus \Omega) \ \text{such that} \\ \\ \|\tau_{-h} f - f\|_{L^p(\tilde{\Omega})} < \epsilon \quad \forall\, h \in \mathbb{R}^N \quad \text{with} \ \ \|h\| < \delta \quad \text{and} \ \ \forall\, f \in \mathcal{K}. \end{cases} \tag{6.59}$$

$$\forall\, \epsilon > 0, \ \ \exists\, \tilde{\Omega} \subset\subset \Omega \ \text{ such that } \ \|f\|_{L^p(\Omega \setminus \tilde{\Omega})} < \epsilon \quad \forall\, f \in \mathcal{K}. \tag{6.60}$$

Then, $\mathcal{K}$ is relatively compact in $L^p(\Omega)$.

Proof Given $\epsilon > 0$, by (6.60), we can fix $\tilde{\Omega} \subset\subset \Omega$ such that

$$\|f\|_{L^p(\Omega \setminus \tilde{\Omega})} < \epsilon \quad \forall\, f \in \mathcal{K}. \tag{6.61}$$

By the Fréchet–Kolmogorov Theorem, the set $\mathcal{K}|_{\tilde{\Omega}} := \{f|_{\tilde{\Omega}} : \ f \in \mathcal{K}\}$ is relatively compact in $L^p(\tilde{\Omega})$. Therefore, there exist $f_1, \ldots, f_k \in L^p(\tilde{\Omega})$ such that

$$\mathcal{K}|_{\tilde{\Omega}} \subset \bigcup_{j=1}^{k} B(f_j, \epsilon). \tag{6.62}$$

Now, as a consequence of (6.61) and (6.62), if we define

$$h_j(x) := \begin{cases} f_j(x) & x \in \tilde{\Omega} \\ \\ 0 & x \in \Omega \setminus \tilde{\Omega}, \end{cases}$$

then

$$\mathcal{K} \subset \bigcup_{j=1}^{k} B(h_j, 2\epsilon),$$

where the balls $B(h_j, 2\epsilon)$ are in $L^p(\Omega)$. Consequently, $\mathcal{K}$ is relatively compact in $L^p(\Omega)$. □

We now have the necessary tools to state the main result on compact embedding, which is the following:

Theorem 6.25 (Rellich–Kondrachov Theorem) *Let $\Omega \subset \mathbb{R}^N$ be a bounded open set of class C^1. Then, the following embeddings are compact:*

(i) If $1 \leq p < N$, then $W^{1,p}(\Omega) \hookrightarrow L^r(\Omega)$, for all $1 \leq r < p^*$.
(ii) If $p = N$, then $W^{1,p}(\Omega) \hookrightarrow L^r(\Omega)$, for all $1 \leq r < \infty$.
(iii) If $p > N$, then $W^{1,p}(\Omega) \hookrightarrow C(\overline{\Omega})$.

Proof

(i) We begin by proving the case $r = 1$, i.e., we want to show that

$$\text{the embedding } \; W^{1,p}(\Omega) \hookrightarrow L^1(\Omega) \; \text{ is compact.} \tag{6.63}$$

Let B be the unit ball in $W^{1,p}(\Omega)$. As a consequence of the Sobolev inequality (Theorem 6.20), we know that B is bounded in $L^1(\Omega)$. Therefore, by Corollary 6.2, it suffices to verify that B satisfies (6.59) and (6.60). Indeed, we first show that

$$\|\tau_{-h} f - f\|_{L^1(\tilde{\Omega})} \le \|h\| \|f\|_{W^{1,p}(\Omega)} \tag{6.64}$$

for each $\tilde{\Omega} \subset\subset \Omega$ and each $h \in \mathbb{R}^N$ with $\|h\| < \operatorname{dist}(\tilde{\Omega}, \mathbb{R}^N \setminus \Omega)$. First, assume that $\phi \in \mathcal{D}(\mathbb{R}^N)$. For $h \in \mathbb{R}^N$ we have

$$\phi(x+h) - \phi(x) = \int_0^1 \langle h, \nabla\phi(x+th)\rangle \, dt.$$

Hence,

$$\int_{\tilde{\Omega}} |\tau_{-h}\phi(x) - \phi(x)| \, dx \le \int_{\tilde{\Omega}} \|h\| \int_0^1 \|\nabla\phi(x+th)\| \, dt dx$$

$$= \|h\| \int_0^1 \left(\int_{\tilde{\Omega}+th} \|\nabla\phi(y)\| \, dy \right) dt.$$

Fixing $h \in \mathbb{R}^N$ with $\|h\| < \operatorname{dist}(\tilde{\Omega}, \mathbb{R}^N \setminus \Omega)$, we can find a set $\omega \subset\subset \Omega$ such that $\tilde{\Omega} + th \subset \omega$ for all $t \in [0,1]$, so that

$$\|\tau_{-h}\phi - \phi\|_{L^1(\tilde{\Omega})} \le \|h\| \int_\omega \|\nabla\phi(y)\| \, dy. \tag{6.65}$$

Now, given $f \in W^{1,1}(\Omega)$, by Lemma 6.1, there exists a sequence $\phi_n \in \mathcal{D}(\mathbb{R}^N)$ such that $\phi_n \to f$ in $L^1(\Omega)$ and $\nabla\phi_n \to \nabla f$ in $L^1(\omega)^N$. Applying (6.65) to each ϕ_n and taking the limit, we obtain (6.64). Clearly, (6.64) implies (6.59). Finally, to prove (6.60), observe that since B is bounded in $L^{p^*}(\Omega)$, by Hölder's inequality we have, for $f \in B$ and $\tilde{\Omega} \subset\subset \Omega$,

$$\|f\|_{L^1(\Omega\setminus\tilde{\Omega})} = \int_\Omega |f| \chi_{\Omega\setminus\tilde{\Omega}} \le \|f\|_{L^{p^*}(\Omega)} \mathcal{L}^N(\Omega \setminus \tilde{\Omega}) \le \mathcal{L}^N(\Omega \setminus \tilde{\Omega}).$$

This proves (6.63).

Now consider the case $1 < r < p^* = \frac{Np}{N-p}$. From Theorem 6.20, we have the continuous embedding $W^{1,p}(\Omega) \hookrightarrow L^1(\Omega) \cap L^{p^*}(\Omega)$. There exists

$\alpha \in [0, 1]$ such that $\frac{1}{r} = \frac{\alpha}{1} + \frac{1-\alpha}{p^*}$, and so, using the interpolation inequality (see Problem (2. 13)), we get

$$\|f\|_r \leq \|f\|_1^{\alpha} \|f\|_{p^*}^{1-\alpha} \leq C\|f\|_1^{\alpha}\|f\|_{1,p}^{1-\alpha} \quad \forall\, f \in W^{1,p}(\Omega).$$

Thus, if B is the unit ball of $W^{1,p}(\Omega)$, then

$$\|f\|_r \leq C\|f\|_1^{\alpha} \quad \forall\, f \in B. \tag{6.66}$$

Let $\{f_n\} \subset B$. By (6.63), there exists a subsequence $\{f_{n_k}\}$ converging in $L^1(\Omega)$, and by (6.66) it follows that $\{f_{n_k}\}$ is a Cauchy sequence in $L^r(\Omega)$, hence convergent. Therefore, B is relatively compact in $L^r(\Omega)$.

(ii) Since Ω is bounded, for each $0 < \epsilon < 1$, we have the continuous embedding $W^{1,N}(\Omega) \hookrightarrow W^{1,N-\epsilon}(\Omega)$. By (i), $W^{1,N-\epsilon}(\Omega) \hookrightarrow L^r(\Omega)$ is compact. Thus, (ii) follows, since $(N-\epsilon)^* = \frac{N(N-\epsilon)}{\epsilon}$.

(iii) Let B be the unit ball of $W^{1,p}(\Omega)$. Since the embedding $W^{1,p}(\Omega) \hookrightarrow C(\overline{\Omega})$ is continuous, B is a bounded subset of $C(\overline{\Omega})$. Therefore, by the Ascoli–Arzelà Theorem, it is enough to show that B is equicontinuous. Now, from (6.44), we have

$$|f(x) - f(y)| \leq C\|x - y\|^{1-\frac{N}{p}} \quad \forall\, f \in B, \quad \forall\, x, y \in \Omega,$$

which implies that B is equicontinuous.

□

Corollary 6.3 *Let $\Omega \subset \mathbb{R}^N$ be a bounded open set. Then, the following embeddings are compact:*

(i) If $1 \leq p < N$, then $W_0^{1,p}(\Omega) \hookrightarrow L^r(\Omega)$ for $1 \leq r < p^*$.
(ii) If $p = N$, then $W_0^{1,p}(\Omega) \hookrightarrow L^r(\Omega)$ for $1 \leq r < \infty$.
(iii) If $p > N$, then $W_0^{1,p}(\Omega) \hookrightarrow C(\overline{\Omega})$.

Proof Let $R > 0$ be such that $\overline{\Omega} \subset B(0, R)$. We have the following continuous embeddings:

$$W_0^{1,p}(\Omega) \hookrightarrow W_0^{1,p}(B(0, R)) \hookrightarrow W^{1,p}(B(0, R)) \hookrightarrow L^r(B(0, R)).$$

Now, since $B(0, 1)$ is a bounded open set with boundary of class C^1, by the Rellich–Kondrachov Theorem, the last embedding is compact for suitable values of r. Consequently, for such values of r, the embedding $W_0^{1,p}(\Omega) \hookrightarrow L^r(\Omega)$ is compact.

□

In the Rellich–Kondrachov Theorem and its Corollary, we have not considered the limiting case $r = p^*$ when $1 \leq p < N$, i.e., we have not proved the compact

embedding of $W^{1,p}(\Omega)$ into $L^{p^*}(\Omega)$. In the following example, we will see that, in general, this is not true.

Example 6.1 Let $\Omega \subset \mathbb{R}^N$ be a bounded open set. Let $a_i \in \Omega$ and $0 < r_i \leq 1$ such that $B(a_i, r_i) \subset \Omega$ and $B(a_i, r_i) \cap B(a_j, r_j) = \emptyset$ for $i \neq j$. Fix a function $\phi \in \mathcal{D}(B(0,1))$, and define $\phi_i \in \mathcal{D}(B(a_i, r_i))$ by

$$\phi_i(x) := r_i^{1-\frac{N}{p}} \phi\left(\frac{x - a_i}{r_i}\right).$$

We then have

$$\begin{aligned}
\|\phi_i\|_{p^*}^{p^*} &= \int_{B(a_i,r_i)} |\phi_i(x)|^{p^*} \, dx \\
&= \int_{B(a_i,r_i)} \left| r_i^{1-\frac{N}{p}} \phi\left(\frac{x - a_i}{r_i}\right) \right|^{p^*} \, dx \\
&= \int_{B(0,1)} r_i^{(1-\frac{N}{p})p^*} |\phi(y)|^{p^*} r_i^N \, dy \\
&= r_i^{\frac{p-N}{p} \cdot \frac{Np}{N-p} + N} \|\phi\|_{p^*}^{p^*} = \|\phi\|_{p^*}^{p^*}.
\end{aligned}$$

Similarly, we have

$$\|\phi_i\|_p^p = r_i^p \, \|\phi\|_p^p \leq \|\phi\|_p^p$$

and

$$\|\nabla\phi_i\|_p^p = r_i^{p-1} \, \|\nabla\phi\|_p^p \leq \|\nabla\phi\|_p^p.$$

Therefore,

$$\|\phi_i\|_{1,p} \leq \|\phi\|_{1,p} \quad \text{for all } i,$$

so the sequence $\{\phi_i\}$ is bounded in $W^{1,p}(\Omega)$. However,

$$\|\phi_i - \phi_j\|_{p^*} = 2^{\frac{1}{p^*}} \, \|\phi\|_{p^*},$$

so the sequence $\{\phi_i\}$ has no convergent subsequence. Consequently, the embedding $W^{1,p}(\Omega) \hookrightarrow L^{p^*}(\Omega)$ is not compact.

6.9 Trace Theory

In this section, we aim to make sense of the restriction of functions $f \in W^{1,p}(\Omega)$ to the boundary $\partial\Omega$. In the case where $f \in C(\overline{\Omega})$, the meaning is clear: it is simply the usual restriction $f|_{\partial\Omega}$. However, in general, functions in $W^{1,p}(\Omega)$ are not continuous; in fact, they are defined $\mathcal{L}^N$-a.e., and $\mathcal{L}^N(\partial\Omega) = 0$. Nevertheless, we shall see in the following result that, by means of the *trace operator*, we can give meaning to the restriction of functions in $W^{1,p}(\Omega)$ to the boundary $\partial\Omega$.

Theorem 6.26 (Trace Theorem) *Let $1 \le p < \infty$, and suppose that $\Omega \subset \mathbb{R}^N$ is a bounded open set with boundary $\partial\Omega$ of class C^1. Then there exists a bounded linear operator*

$$\gamma : W^{1,p}(\Omega) \to L^p(\partial\Omega),$$

called the trace operator, such that

$$\gamma(f) = f|_{\partial\Omega} \quad \text{if} \quad f \in W^{1,p}(\Omega) \cap C(\overline{\Omega}) \tag{6.67}$$

and

$$\|\gamma(f)\|_{L^p(\partial\Omega)} \le C\|f\|_{1,p}, \quad \forall f \in W^{1,p}(\Omega), \tag{6.68}$$

with the constant C depending only on p and $\partial\Omega$.
Moreover, $\operatorname{Ker}\gamma = W_0^{1,p}(\Omega)$*, i.e.,*

$$f \in W_0^{1,p}(\Omega) \iff \gamma(f) = 0 \quad \text{on } \partial\Omega.$$

Proof Let us first assume that $f \in C^1(\overline{\Omega})$. Given $x^0 \in \partial\Omega$, since $\partial\Omega$ is of class C^1, we may assume that $\partial\Omega$ is flat near x^0, i.e., that near x^0, $\partial\Omega$ is contained in the hyperplane $\{x_N = 0\}$. Then, we can assume that there exists a ball $B = B(x^0, r)$ such that

$$B^+ = B \cap \{x_N \ge 0\} \subset \overline{\Omega}, \quad B^- := B \cap \{x_N \le 0\} \subset \mathbb{R}^N \setminus \Omega.$$

Let $\Gamma := \partial\Omega \cap B(x^0, \frac{r}{2})$, and take $\varphi \in \mathcal{D}(B)$ with $\varphi \ge 0$ and $\varphi(x) = 1$ for every $x \in B(x^0, \frac{r}{2})$. Let $x' = (x_1, \dots, x_{N-1}) \in \mathbb{R}^{N-1} = \{x_N = 0\}$. Applying Young's inequality, we obtain

$$\int_\Gamma |f|^p \, dx' \le \int_{\{x_N=0\}} \varphi |f|^p \, dx' = -\int_{B^+} \frac{\partial(\varphi|f|^p)}{\partial x_N} \, dx$$

$$= -\int_{B^+} \left(|f|^p \frac{\partial\varphi}{\partial x_N} + p|f|^{p-1}(\operatorname{sign} f)\frac{\partial f}{\partial x_N} \right) dx \le C \int_{B^+} \left(|f|^p + \|\nabla f\|^p \right) dx.$$

Therefore,

$$\|f\|_{L^p(\Gamma)} \le C\|f\|_{1,p}. \tag{6.69}$$

Now, since $\partial\Omega$ is compact, there exists a finite number of points $x_i^0 \in \partial\Omega$ and subsets $\Gamma_i \subset \partial\Omega$ such that $\partial\Omega = \cup\Gamma_i$, satisfying (6.69). Consequently, if we write $\gamma(f) := f|_{\partial\Omega}$, there exists a constant C, independent of f, such that

$$\|\gamma(f)\|_{L^p(\partial\Omega)} \le C\|f\|_{1,p}. \tag{6.70}$$

Now suppose $f \in W^{1,p}(\Omega)$. Then, by Theorem 6.4, there exist $f_n \in C^\infty(\overline{\Omega})$ such that $f_n \to f$ in $W^{1,p}(\Omega)$. As a consequence of (6.70), we have

$$\|\gamma(f_n) - \gamma(f_m)\|_{L^p(\partial\Omega)} \le C\|f_n - f_m\|_{1,p}. \tag{6.71}$$

Thus, $\{\gamma(f_n)\}$ is a Cauchy sequence in $L^p(\partial\Omega)$. We can therefore define

$$\gamma(f) := \lim_{n\to\infty} \gamma(f_n).$$

Due to (6.71), this definition is independent of the choice of approximating sequence. Moreover, taking limits in (6.71) yields (6.68). To prove (6.67), observe that when $f \in W^{1,p}(\Omega) \cap C(\overline{\Omega})$, the approximating sequence constructed in the proof of Theorem 6.4 converges uniformly to f.

Let us now show that $\operatorname{Ker}\gamma = W_0^{1,p}(\Omega)$. Suppose $f \in W_0^{1,p}(\Omega)$. Then, there exist $f_n \in \mathcal{D}(\Omega)$ such that $f_n \to f$ in $W^{1,p}(\Omega)$. Since $\gamma(f_n) = 0$ for all $n \in \mathbb{N}$ and $\gamma : W^{1,p}(\Omega) \to L^p(\partial\Omega)$ is bounded, it follows that $\gamma(f) = 0$, i.e., $f \in \operatorname{Ker}\gamma$.

Now suppose $f \in \operatorname{Ker}\gamma$. Since $\partial\Omega$ is of class C^1, by using partitions of unity and flattening the boundary, we may assume that

$$f \in W^{1,p}(\mathbb{R}^N_+), \quad f \text{ has compact support in } \overline{\mathbb{R}^N_+},$$

$$\text{and} \quad \gamma(f) = 0 \quad \text{on} \quad \partial\mathbb{R}^N_+ = \mathbb{R}^{N-1}.$$

Then, there exist $f_n \in C^1(\overline{\mathbb{R}^N_+})$ such that

$$f_n \to f \quad \text{in} \quad W^{1,p}(\mathbb{R}^N_+) \tag{6.72}$$

and

$$\gamma(f_n) = f_n|_{\mathbb{R}^{N-1}} \to 0 \quad \text{in} \quad L^p(\mathbb{R}^{N-1}). \tag{6.73}$$

Now, for $x' \in \mathbb{R}^{N-1}$, and $x_N \geq 0$, we have

$$|f_n(x', x_N)| \leq |f_n(x', 0)| + \int_0^{x_N} \left| \frac{\partial f_n}{\partial x_N}(x', t) \right| dt.$$

Hence,

$$\int_{\mathbb{R}^{N-1}} |f_n(x', x_N)|^p \, dx'$$

$$\leq C \left(\int_{\mathbb{R}^{N-1}} |f_n(x', 0)|^p \, dx' + x_N^{p-1} \int_0^{x_N} \int_{\mathbb{R}^{N-1}} |\nabla f_n(x', t)|^p \, dx' dt \right).$$

Taking into account (6.72) and (6.73), letting $n \to \infty$, we obtain

$$\int_{\mathbb{R}^{N-1}} |f(x', x_N)|^p \, dx' \leq C x_N^{p-1} \int_0^{x_N} \int_{\mathbb{R}^{N-1}} |\nabla f|^p \, dx' dt \tag{6.74}$$

for almost every $x_N > 0$.

Now take a function $\varphi \in C^\infty([0, \infty))$ such that $0 \leq \varphi \leq 1$, $\varphi(t) = 1$ for $t \in [0, 1]$, and $\varphi(t) = 0$ for $t \in [2, \infty)$. Define $\varphi_n(x) := \varphi(n x_N)$ for $x \in \mathbb{R}^N_+$, and set

$$\phi_n(x) := f(x)(1 - \varphi_n(x)), \quad x \in \mathbb{R}^N_+.$$

Then,

$$\frac{\partial \phi_n}{\partial x_N} = \frac{\partial f}{\partial x_N}(1 - \varphi_n) - n f \varphi',$$

$$\frac{\partial \phi_n}{\partial x_j} = \frac{\partial f}{\partial x_j}(1 - \varphi_n), \quad \text{for } j = 1, \ldots, N-1,$$

so

$$\begin{aligned} &\int_{\mathbb{R}^N_+} |\nabla \phi_n - \nabla f|^p \, dx \leq \\ &C \int_{\mathbb{R}^N_+} |\varphi_n|^p \, \|\nabla f\|^p \, dx + C n^p \int_0^{\frac{2}{n}} \int_{\mathbb{R}^{N-1}} |f|^p \, dx' dt. \end{aligned} \tag{6.75}$$

Since $\varphi_n(x) = 0$ when $x_N > \frac{2}{n}$, we have

$$\lim_{n \to \infty} \int_{\mathbb{R}^N_+} |\varphi_n|^p \, \|\nabla f\|^p \, dx = 0. \tag{6.76}$$

Using (6.74), we also get

$$\begin{aligned} & n^p \int_0^{\frac{2}{n}} \int_{\mathbb{R}^{N-1}} |f|^p \, dx' dt \leq \\ & n^p \left(\int_0^{\frac{2}{n}} t^{p-1} dt \right) \left(\int_0^{\frac{2}{n}} \int_{\mathbb{R}^{N-1}} |\nabla f|^p \, dx' dx_N \right) \leq \\ & C \int_0^{\frac{2}{n}} \int_{\mathbb{R}^{N-1}} |\nabla f|^p \, dx' dx_N \to 0 \quad \text{as } n \to \infty. \end{aligned} \tag{6.77}$$

As a consequence of (6.76) and (6.77), taking limits in (6.75) yields $\nabla \phi_n \to \nabla f$ in $L^p(\mathbb{R}^N_+)^N$, and since $\phi_n \to f$ in $L^p(\mathbb{R}^N_+)$, we conclude that

$$\phi_n \to f \quad \text{in } W^{1,p}(\mathbb{R}^N_+).$$

Finally, since $\phi_n(x) = 0$ when $0 < x_N < \frac{1}{n}$, it is easy to construct $f_n \in \mathcal{D}(\mathbb{R}^N_+)$ such that $f_n \to f$ in $W^{1,p}(\mathbb{R}^N_+)$, and hence $f \in W_0^{1,p}(\mathbb{R}^N_+)$. □

Definition 6.3 As a consequence of the previous theorem and Problem 3.16, we have that $\operatorname{Ran} \gamma$ is a Hilbert space with respect to the norm

$$\|g\| := \inf \left\{ \|f\|_{H^1(\Omega)} \, ; \, \gamma(f) = g \right\}.$$

We denote this Hilbert space by $H^{1/2}(\partial\Omega)$. The dual of the space $H^{1/2}(\partial\Omega)$ is denoted by $H^{-1/2}(\partial\Omega)$, and the duality pairing between these spaces is denoted by $\langle \cdot, \cdot \rangle_{\partial\Omega}$.

Remark 6.7 From (6.68), we have that

$$\|\gamma(f)\|_{L^2(\partial\Omega)} \leq C_1 \|f\|_{H^1(\Omega)} \quad \forall \, f \in H^1(\Omega),$$

which implies that

$$\|\gamma(f)\|_{L^2(\partial\Omega)} \leq C_1 \|\gamma(f)\|_{H^{1/2}(\partial\Omega)} \quad \forall \, f \in H^1(\Omega). \tag{6.78}$$

Now, by the open mapping theorem, there exists a constant $C_2 > 0$ such that

$$\|f\|_{H^1(\Omega)} \leq C_2 \|\gamma(f)\|_{L^2(\partial\Omega)} \quad \forall \, f \in H^1(\Omega),$$

which implies that

$$\|\gamma(f)\|_{H^{1/2}(\partial\Omega)} \leq C_2 \|\gamma(f)\|_{L^2(\partial\Omega)} \quad \forall \, f \in H^1(\Omega). \tag{6.79}$$

As a consequence of (6.78) and (6.79), we conclude that the norms

$$\|\cdot\|_{H^{1/2}(\partial\Omega)} \quad \text{and} \quad \|\cdot\|_{L^2(\partial\Omega)}$$

are equivalent on $H^{1/2}(\partial\Omega)$.

We now show that the classical Green's formula (Theorem A.4) for smooth functions can be extended to the context of Sobolev spaces.

Theorem 6.27 (Green's Formula) *Let $\Omega \subset \mathbb{R}^N$ be a bounded open set with boundary $\partial\Omega$ of class C^1. Then, for all $f, g \in H^1(\Omega)$ and for all $i = 1, 2, \ldots, N$, we have*

$$\int_\Omega f \frac{\partial g}{\partial x_i} = -\int_\Omega g \frac{\partial f}{\partial x_i} + \int_{\partial\Omega} \gamma(f)\gamma(g)\eta_i, \tag{6.80}$$

where $\eta = (\eta_1, \ldots, \eta_n)$ is the outward unit normal vector to the boundary of Ω.

Proof Since $\partial\Omega$ is of class C^1, by applying Theorem 6.4, there exist sequences $f_n, g_n \in C^\infty(\overline{\Omega})$ such that

$$f_n \to f, \quad g_n \to g \quad \text{in } H^1(\Omega).$$

Applying the classical Green's formula to the functions f_n, g_n, we obtain

$$\int_\Omega f_n \frac{\partial g_n}{\partial x_i} = -\int_\Omega g_n \frac{\partial f_n}{\partial x_i} + \int_{\partial\Omega} \gamma(f_n)\gamma(g_n)\eta_i.$$

Letting $n \to \infty$ and using the continuity of the trace operator (Theorem 6.26), we obtain (6.80). □

To generalize the concept of the normal derivative of smooth functions to the setting of functions in Sobolev spaces, we introduce the following space.

Definition 6.4 For an open set $\Omega \subset \mathbb{R}^N$, we define

$$\mathrm{Div}(\Omega) := \left\{ \mathbf{f} \in L^2(\Omega)^N \ : \ \mathrm{div}\,\mathbf{f} \in L^2(\Omega) \right\},$$

where by $\mathrm{div}\,\mathbf{f}$ we mean the distributional divergence of the vector field $\mathbf{f} = (f_1, \ldots, f_N)$, i.e.,

$$\langle \mathrm{div}\,\mathbf{f}, \phi \rangle = -\sum_{i=1}^N \int_\Omega f_i \frac{\partial \phi}{\partial x_i} = -\int_\Omega \mathbf{f} \cdot \nabla\phi \quad \forall\, \phi \in \mathcal{D}(\Omega).$$

Theorem 6.28 $\mathrm{Div}(\Omega)$ *is a separable Hilbert space with respect to the inner product*

$$(\mathbf{f}|\mathbf{g}) := (\mathbf{f}|\mathbf{g})_{L^2(\Omega)^N} + (\operatorname{div}\mathbf{f}|\operatorname{div}\mathbf{g})_{L^2(\Omega)}.$$

Proof Clearly, $\mathrm{Div}(\Omega)$ is a pre-Hilbert space with respect to this inner product. Let us show that it is complete.

Let $\{\mathbf{f}_n\}$ be a Cauchy sequence in $\mathrm{Div}(\Omega)$. Then, $\{\mathbf{f}_n\}$ is a Cauchy sequence in $L^2(\Omega)^N$, and $\{\operatorname{div}\mathbf{f}_n\}$ is a Cauchy sequence in $L^2(\Omega)$. Consequently, there exist $\mathbf{f} \in L^2(\Omega)^N$ and $g \in L^2(\Omega)$ such that

$$\mathbf{f}_n \to \mathbf{f} \quad \text{in } L^2(\Omega)^N \quad \text{and} \quad \operatorname{div}\mathbf{f}_n \to g \quad \text{in } L^2(\Omega).$$

Now, for $\phi \in \mathcal{D}(\Omega)$, by applying the Dominated Convergence Theorem, we have

$$\langle \operatorname{div}\mathbf{f}, \phi\rangle = -\int_\Omega \mathbf{f}\cdot\nabla\phi = -\lim_{n\to\infty}\int_\Omega \mathbf{f}_n\cdot\nabla\phi = \lim_{n\to\infty}\langle \operatorname{div}\mathbf{f}_n, \phi\rangle =$$

$$= \lim_{n\to\infty}\int_\Omega \operatorname{div}(\mathbf{f}_n)\phi = \int_\Omega g\phi.$$

Therefore, $\operatorname{div}\mathbf{f} = g$, so $\mathbf{f} \in \mathrm{Div}(\Omega)$, and moreover, $\mathbf{f}_n \to \mathbf{f}$ in $\mathrm{Div}(\Omega)$. Hence, the space is complete and, consequently, it is a Hilbert space.

To prove separability, consider the isometry

$$\mathcal{I} : \mathrm{Div}(\Omega) \to [L^2(\Omega)]^{N+1},$$

defined by: $\mathcal{I}(\mathbf{f}) := (\mathbf{f}, \operatorname{div}\mathbf{f})$. Using an argument similar to the one used to prove that the space is complete, it is easy to show that $\mathcal{I}(\mathrm{Div}(\Omega))$ is a closed subspace of $[L^2(\Omega)]^{N+1}$, and thus separable, from which the separability of $\mathrm{Div}(\Omega)$ follows. □

We now present the following characterization of the elements of the dual of $\mathrm{Div}(\Omega)$.

Lemma 6.4 *If* $T \in (\mathrm{Div}(\Omega))'$*, then there exists* $\mathbf{g} \in [L^2(\Omega)]^{N+1}$ *such that*

$$\langle T, \mathbf{f}\rangle = \int_\Omega (g_1, \ldots, g_N)\cdot\mathbf{f} + \int_\Omega g_{N+1}\operatorname{div}(\mathbf{f}) \quad \forall\, \mathbf{f} \in \mathrm{Div}(\Omega).$$

Proof Let us again consider the isometry

$$\mathcal{I} : \mathrm{Div}(\Omega) \to [L^2(\Omega)]^{N+1},$$

defined by $\mathcal{I}(\mathbf{f}) := (\mathbf{f}, \operatorname{div}\mathbf{f})$. Clearly, the linear form

$$T \circ \mathcal{I}^{-1} : T(\operatorname{Div}(\Omega)) \to \mathbb{R}$$

can be extended to a continuous linear form on $[L^2(\Omega)]^{N+1}$, from which the result follows, taking into account the identification of $[L^2(\Omega)]^{N+1}$ with its dual. □

In the following result we show that it is possible to approximate the elements of $\operatorname{Div}(\Omega)$ by infinitely differentiable vector fields.

Theorem 6.29 *The space* $\mathcal{D}(\overline{\Omega})^N$ *is dense in* $\operatorname{Div}(\Omega)$.

Proof Taking into account Corollary 2.5, it suffices to prove that if $T \in (\operatorname{Div}(\Omega))'$ satisfies

$$\langle T, \mathbf{f}\rangle = 0 \quad \forall\, \mathbf{f} \in \mathcal{D}(\overline{\Omega})^N, \tag{6.81}$$

then $T = 0$. Indeed, by the previous lemma, there exists a function $\mathbf{g} \in [L^2(\Omega)]^{N+1}$ such that

$$\langle T, \mathbf{f}\rangle = \int_\Omega (g_1, \dots, g_N) \cdot \mathbf{f} + \int_\Omega g_{N+1} \operatorname{div}(\mathbf{f}) \quad \forall\, \mathbf{f} \in \operatorname{Div}(\Omega). \tag{6.82}$$

For each $i = 1, \dots, N+1$, let $\tilde{g}_i \in L^2(\mathbb{R}^N)$ be the extension of g_i that is zero on $\mathbb{R}^N \setminus \Omega$. Now, as a consequence of (6.81) and (6.82), given $\phi \in \mathcal{D}(\mathbb{R}^N)$, we have

$$\left\langle \frac{\partial \tilde{g}_{N+1}}{\partial x_i}, \phi \right\rangle = -\left\langle \tilde{g}_{N+1}, \frac{\partial \phi}{\partial x_i} \right\rangle = -\int_\Omega g_{N+1} \frac{\partial \phi}{\partial x_i}$$

$$= \int_\Omega (g_1, \dots, g_N) \cdot (0, \dots, \phi, \dots, 0) - \langle T, (0, \dots, \phi, \dots, 0)\rangle = \int_{\mathbb{R}^N} \tilde{g}_i \phi.$$

Therefore, $\frac{\partial \tilde{g}_{N+1}}{\partial x_i} = \tilde{g}_i \in L^2(\mathbb{R}^N)$, so $\tilde{g}_{N+1} \in H^1(\mathbb{R}^N)$. Hence, since $\partial\Omega$ is of class C^1, it follows that $g_{N+1} \in H_0^1(\Omega)$ (see Problem 6.17). Consequently, there exist $\phi_n \in \mathcal{D}(\Omega)$ such that $\phi_n \to g_{N+1}$ in $H_0^1(\Omega)$. Then, for each $\mathbf{f} \in \operatorname{Div}(\Omega)$, since $\frac{\partial g_{N+1}}{\partial x_i} = g_i$, and taking into account (6.82), we have

$$\langle T, \mathbf{f}\rangle = \int_\Omega \nabla g_{N+1} \cdot \mathbf{f} + \int_\Omega g_{N+1} \operatorname{div}(\mathbf{f}) = \lim_{n\to\infty} \left(\int_\Omega \nabla \phi_n \cdot \mathbf{f} + \int_\Omega \phi_n \operatorname{div}(\mathbf{f}) \right)$$

$$= \lim_{n\to\infty} \left(-\int_\Omega \phi_n \operatorname{div}(\mathbf{f}) + \int_\Omega \phi_n \operatorname{div}(\mathbf{f}) \right) = 0.$$

□

In the following result, we will see that it is possible to define the component in the direction of the unit outward normal η on $\partial\Omega$ for each field in $\operatorname{Div}(\Omega)$.

Theorem 6.30 *There exists a unique linear and continuous map*

$$\gamma_\eta : \mathrm{Div}(\Omega) \to H^{-1/2}(\partial\Omega)$$

such that

$$\gamma_\eta(\mathbf{f}) = (\mathbf{f}\cdot\eta)_{|\partial\Omega} \quad \forall\, \mathbf{f} \in \mathcal{D}(\overline{\Omega}) \quad \text{and} \quad \|\gamma_\eta\| \le 1.$$

Moreover, for each $g \in H^1(\Omega)$ and $\mathbf{f} \in \mathrm{Div}(\Omega)$, the following integration by parts formula holds:

$$\int_\Omega \nabla g \cdot \mathbf{f}\, dx = \langle \gamma_\eta(\mathbf{f}), \gamma(g)\rangle_{\partial\Omega} - \int_\Omega g\, \mathrm{div}(\mathbf{f})\, dx \tag{6.83}$$

Proof For $g \in H^1(\Omega)$ and $\mathbf{f} \in \mathcal{D}(\overline{\Omega})$, using Green's formula (6.80), we obtain

$$\int_\Omega g\, \mathrm{div}(\mathbf{f}) = -\int_\Omega \nabla g \cdot \mathbf{f} + \int_{\partial\Omega} \gamma(g)(\mathbf{f}\cdot\eta). \tag{6.84}$$

Then, applying the Cauchy–Schwarz inequality, we get

$$\left| \int_{\partial\Omega} \gamma(g)(\mathbf{f}\cdot\eta) \right| \le \|g\|_{H^1(\Omega)}\, \|\mathbf{f}\|_{\mathrm{Div}(\Omega)}. \tag{6.85}$$

Let $h \in H^{1/2}(\partial\Omega)$ and let $g \in H^1(\Omega)$ such that $\gamma(g) = h$. As a consequence of (6.85), we obtain

$$\left| \int_{\partial\Omega} h(\mathbf{f}\cdot\eta) \right| \le \|g\|_{H^1(\Omega)}\, \|\mathbf{f}\|_{\mathrm{Div}(\Omega)}.$$

Taking the infimum over all functions $g \in H^1(\Omega)$ such that $\gamma(g) = h$, we get

$$\left| \int_{\partial\Omega} h(\mathbf{f}\cdot\eta) \right| \le \|\mathbf{f}\|_{\mathrm{Div}(\Omega)} \|h\|_{H^{1/2}(\partial\Omega)},$$

from which we deduce that $(\mathbf{f}\cdot\eta)_{|\partial\Omega} \in H^{-1/2}(\partial\Omega)$ and

$$\|(\mathbf{f}\cdot\eta)_{|\partial\Omega}\|_{H^{-1/2}(\partial\Omega)} \le \|\mathbf{f}\|_{\mathrm{Div}(\Omega)}. \tag{6.86}$$

Therefore, since $\mathcal{D}(\overline{\Omega})^N$ is dense in $\mathrm{Div}(\Omega)$, we can uniquely extend to $\mathrm{Div}(\Omega)$ the map from $\mathcal{D}(\overline{\Omega})$ into $H^{-1/2}(\partial\Omega)$ that associates to each $\mathbf{f}$ the value $(\mathbf{f}\cdot\eta)_{|\partial\Omega}$, thereby obtaining a linear and continuous map $\gamma_\eta : \mathrm{Div}(\Omega) \to H^{-1/2}(\partial\Omega)$ such that $\gamma_\eta(\mathbf{f}) = (\mathbf{f}\cdot\eta)_{|\partial\Omega}$ for every $\mathbf{f} \in \mathcal{D}(\overline{\Omega})$ and, as a consequence of (6.85), satisfying $\|\gamma_\eta\| \le 1$.

Finally, the integration by parts formula (6.83) follows from (6.84), taking into account the density of $\mathcal{D}(\overline{\Omega})^N$ in $\mathrm{Div}(\Omega)$. □

Remark 6.8 Taking $g = \chi_\Omega$ in (6.83) we obtain

$$\int_\Omega \mathrm{div}(\mathbf{f})\, dx = \langle \gamma_\eta(\mathbf{f}), \gamma(\chi_\Omega)\rangle_{\partial\Omega}.$$

Then, denoting

$$\langle \gamma_\eta(\mathbf{f}), \gamma(\chi_\Omega)\rangle_{\partial\Omega} = \int_{\partial\Omega} \mathbf{f}\cdot\eta,$$

we obtain the following generalization of the Divergence Theorem:

Theorem 6.31 (Divergence Theorem) *Let $\Omega \subset \mathbb{R}^N$ be a bounded open set of class C^1. For every $\mathbf{f} \in \mathrm{Div}(\Omega)$, we have*

$$\int_\Omega \mathrm{div}(\mathbf{f})\, dx = \int_{\partial\Omega} \mathbf{f}\cdot\eta. \tag{6.87}$$

Recall that for $g \in C^1(\overline{\Omega})$, its outer normal derivative is defined as

$$\frac{\partial g}{\partial \eta} := \nabla g \cdot \eta = \sum_{i=1}^N \frac{\partial g}{\partial x_i}\eta_i.$$

Now, given $g \in H^2(\Omega)$, since $\nabla g \in \mathrm{Div}(\Omega)$, by the theorem above, we have that $\gamma_\eta(\nabla g) \in H^{-1/2}(\partial\Omega)$ is defined. We denote this element by $\frac{\partial g}{\partial \eta}$ and call it the *outer normal derivative* of g. Moreover, as a consequence of (6.83), we obtain the following generalization of the integration by parts formula.

Corollary 6.4 (Integration by Parts Formula) *Let $\Omega \subset \mathbb{R}^N$ be a bounded open set of class C^1. For $f \in H^1(\Omega)$ and $g \in H^2(\Omega)$, we have*

$$\int_\Omega \nabla f \cdot \nabla g = \left\langle \frac{\partial g}{\partial \eta}, \gamma(f)\right\rangle_{\partial\Omega} - \int_\Omega f\, \Delta g. \tag{6.88}$$

6.10 Poincaré-Type Inequalities

In the section on continuous embedding, we discussed the so-called Sobolev inequalities, which provide bounds for the norms of functions in Sobolev spaces in terms of their norms in Lebesgue spaces. In this section, we present some of the so-called Poincaré-type inequalities, which provide bounds for the norms of functions

in Sobolev spaces in Lebesgue spaces in terms of the norms of their gradients in those spaces. This type of inequality will be very useful in the next chapter to establish the existence of weak solutions to certain elliptic boundary value problems.

Definition 6.5 Let $\xi \in \mathbb{R}^N$ with $\|\xi\| = 1$, and let $a < b$, with $d = b - a$. We denote by $B_d(\xi)$ the strip of thickness d in the direction of ξ, that is,

$$B_d(\xi) := \left\{ x \in \mathbb{R}^N \ : \ a < x \cdot \xi < b \right\}.$$

Theorem 6.32 (Poincaré Inequality) *Let $\Omega \subset \mathbb{R}^N$ be an open set and suppose there exists a strip $B_d(\xi)$ such that $\Omega \subset B_d(\xi)$. Then, for every $f \in W_0^{1,p}(\Omega)$ $(1 \leq p < \infty)$, we have*

$$\|f\|_p^p \leq \frac{d^p}{p} \|\nabla f\|_p^p. \tag{6.89}$$

Proof Since $\mathcal{D}(\Omega)$ is dense in $W_0^{1,p}(\Omega)$, it is sufficient to prove (6.89) for $f \in \mathcal{D}(\Omega)$. Given $f \in \mathcal{D}(\Omega)$, we can write

$$f(x + t\xi) = \int_a^t \nabla f(x + s\xi) \cdot \xi \, ds.$$

Therefore, using Hölder's inequality, we get

$$|f(x + t\xi)|^p \leq (t - a)^{p-1} \int_a^b \|\nabla f(x + s\xi)\|^p \, ds. \tag{6.90}$$

Let $H = \{x \in \mathbb{R}^N \ : \ x \cdot \xi = 0\}$ be the hyperplane orthogonal to ξ. If $g \in L^p(\mathbb{R}^N)$ with $\text{supp}(g) \subset \Omega$, then since $\Omega \subset B_d(\xi)$, we have

$$\int_\Omega |g(y)|^p \, dy = \int_a^b \int_H |g(x + t\xi)|^p \, dx dt. \tag{6.91}$$

Finally, as a consequence of (6.90) and (6.91), we obtain

$$\|f\|_p^p = \int_a^b \int_H |f(x + t\xi)|^p \, dx dt$$

$$\leq \int_a^b (t - a)^{p-1} \left(\int_H \left(\int_a^b \|\nabla f(x + s\xi)\|^p \, ds \right) dx \right) dt$$

$$= \int_a^b (t - a)^{p-1} \|\nabla f\|_p^p \, dt = \frac{(b - a)^p}{p} \|\nabla f\|_p^p.$$

□

Remark 6.9 Poincaré's inequality does not hold for unbounded open sets. Let us consider an example where Poincaré's inequality fails. Let $\Omega = \mathbb{R}^N$, and let $\phi \in \mathcal{D}(\mathbb{R}^N)$ be such that $0 \leq \phi \leq 1$, $\phi(x) = 1$ if $\|x\| \leq 1$, and $\phi(x) = 0$ if $\|x\| \geq 2$. Consider the sequence of functions $\phi_n \in \mathcal{D}(\mathbb{R}^N)$ defined by $\phi_n(x) := \phi(\frac{x}{n})$. Then, if $N < p$, we have that

$$\lim_{n\to\infty} \|\nabla \phi_n\|_p = 0$$

and yet,

$$\|\phi_n\|_p \geq \mathcal{L}^N(B(0,n)) \to \infty, \quad \text{as} \quad n \to \infty.$$

Let us now examine some consequences of Poincaré's inequality.

Corollary 6.5 *Let $\Omega \subset \mathbb{R}^N$ be an open set and let $1 \leq p < \infty$. If Ω is bounded in at least one direction, i.e., if $\Omega \subset]a, b[\times\mathbb{R}^{N-1}$, then for every $f \in W_0^{1,p}(\Omega)$ we have*

$$\|f\|_p^p \leq \frac{(b-a)^p}{p} \|\nabla f\|_p^p. \tag{6.92}$$

In particular, if Ω is a bounded open set with diameter d, then for every $f \in W_0^{1,p}(\Omega)$ we have

$$\|f\|_p^p \leq \frac{d^p}{p} \|\nabla f\|_p^p. \tag{6.93}$$

Proof This is an immediate consequence of the previous theorem, noting that if $\xi = (1, 0, \ldots, 0)$, then

$$\Omega \subset]a, b[\times\mathbb{R}^{N-1} \subset B_{b-a}(\xi).$$

□

Another immediate consequence of the previous theorem is the following result.

Corollary 6.6 *Let $\Omega \subset \mathbb{R}^N$ be an open set and $1 \leq p < \infty$. If there exists a strip $B_d(\xi)$ such that $\Omega \subset B_d(\xi)$, then*

$$f \mapsto \left(\sum_{i=1}^N \| \frac{\partial f}{\partial x_i} \|_p^p \right)^{\frac{1}{p}}$$

is a norm on $W_0^{1,p}(\Omega)$ equivalent to the norm $\|\cdot\|_{1,p}$.

Corollary 6.7 *Let $\Omega \subset \mathbb{R}^N$ be an open set and suppose that there exists a strip $B_d(\xi)$ such that $\Omega \subset B_d(\xi)$, and let $1 \leq p < \infty$, $m \in \mathbb{N}$. Then there exists a*

constant C, depending only on m, p, and Ω, such that

$$\|f\|_p^p \le C \sum_{|\alpha|=m} \|D^\alpha f\|_p^p \quad \forall\, f \in W_0^{m,p}(\Omega). \tag{6.94}$$

In particular, the expression

$$f \mapsto \left(\sum_{|\alpha|=m} \|D^\alpha f\|_p^p \right)^{\frac{1}{p}}$$

is a norm on $W_0^{m,p}(\Omega)$ equivalent to the norm $\|\cdot\|_{m,p}$.

Proof It follows immediately by iterating Poincaré's inequality, taking into account that if $f \in W_0^{m,p}(\Omega)$, then $\frac{\partial f}{\partial x_i} \in W_0^{m-1,p}(\Omega)$. □

Remark 6.10 Poincaré's inequality does not hold for unbounded open sets. Let us look at an example where Poincaré's inequality fails. Let $\Omega = \mathbb{R}^N$, and let $\phi \in \mathcal{D}(\mathbb{R}^N)$, with $0 \le \phi \le 1$, $\phi(x) = 1$ if $\|x\| \le 1$, and $\phi(x) = 0$ if $\|x\| \ge 2$. Consider the sequence of functions $\phi_n \in \mathcal{D}(\mathbb{R}^N)$ defined by $\phi_n(x) := \phi\left(\frac{x}{n}\right)$. Then, if $N < p$, we have

$$\lim_{n\to\infty} \|\nabla \phi_n\|_p = 0$$

and yet,

$$\|\phi_n\|_p \ge \mathcal{L}^N(B(0,n)) \to \infty \quad \text{as} \quad n \to \infty.$$

Theorem 6.33 (Generalized Poincaré Inequality) *Let $\Omega \subset \mathbb{R}^N$ be a bounded domain with C^1 boundary, and let $1 < p < \infty$. Let N be a continuous seminorm on $W^{1,p}(\Omega)$ such that $N(a\chi_\Omega) = 0$ implies $a = 0$. Then, there exists a constant C such that*

$$\|f\|_p \le C\Big(\|\nabla f\|_p + N(f) \Big) \quad \forall\, f \in W^{1,p}(\Omega). \tag{6.95}$$

Proof Suppose that (6.95) is not true. Then, we can find functions $f_n \in W^{1,p}(\Omega)$ such that

$$\|f_n\|_p \ge n\Big(\|\nabla f_n\|_p + N(f_n) \Big) \quad \forall\, n \in \mathbb{N}. \tag{6.96}$$

Moreover, by homogeneity, we may assume that $\|f_n\|_p = 1$ for all $n \in \mathbb{N}$. Therefore, from (6.96), we get

$$\|\nabla f_n\|_p \le \frac{1}{n}, \quad N(f_n) \le \frac{1}{n}, \quad \forall\, n \in \mathbb{N}. \tag{6.97}$$

Hence, the sequence $\{f_n\}$ is bounded in $W^{1,p}(\Omega)$ and, since this space is reflexive, we may assume (extracting a subsequence if necessary) that $f_n \rightharpoonup f$ weakly in $W^{1,p}(\Omega)$. Since N is a continuous seminorm on $W^{1,p}(\Omega)$, we deduce (see Problem 2.29) that

$$N(f) \leq \liminf_{n\to\infty} N(f_n) = 0. \tag{6.98}$$

On the other hand, as a consequence of the Rellich–Kondrachov Theorem, we have that $f_n \to f$ in $L^p(\Omega)$, and thus $\|f\|_p = 1$. Moreover, $\nabla f_n \rightharpoonup \nabla f$ weakly in $L^p(\Omega)^N$, from which, using (6.97), we deduce that $\nabla f = 0$. Since Ω is connected, it follows that f is constant (see Problem 6.6). Now, since $N(f) = 0$, we conclude that $f = 0$, which contradicts the fact that $\|f\|_p = 1$. □

Different choices of the continuous seminorm N in the previous theorem give rise to inequalities that are quite useful in applications. Let us see some examples:

Corollary 6.8 (Friedrichs' Inequality) *Let $\Omega \subset \mathbb{R}^N$ be a bounded domain with C^1 boundary, and let $1 < p < \infty$. If $\Gamma \subset \partial\Omega$, with $H^{N-1}(\Gamma) > 0$, then there exists a constant C such that*

$$\|f\|_p \leq C\left(\|\nabla f\|_p + \left(\int_\Gamma |\gamma(f)|^p\right)^{\frac{1}{p}}\right) \quad \forall\, f \in W^{1,p}(\Omega). \tag{6.99}$$

Proof It suffices to apply the previous result with

$$N(f) := \left(\int_\Gamma |\gamma(f)|^p\right)^{\frac{1}{p}},$$

since, by the trace theorem, N is a continuous seminorm on $W^{1,p}(\Omega)$ and also satisfies that $N(a\chi_\Omega) = 0$ implies $a = 0$. □

Another inequality, which is also a consequence of the generalized Poincaré inequality and has many applications in the study of the asymptotic behavior of solutions to partial differential equations, is the following.

Corollary 6.9 *Let $\Omega \subset \mathbb{R}^N$ be a bounded domain with boundary of class C^1 and let $1 < p < \infty$. The following inequality holds:*

$$\|f\|_p \leq C\left(\|\nabla f\|_p + \left|\int_\Omega f(x)\,dx\right|\right) \quad \forall\, f \in W^{1,p}(\Omega). \tag{6.100}$$

In particular, one has the so-called Poincaré–Wirtinger inequality:

$$\|f - f_\Omega\|_p \leq C\|\nabla f\|_p \quad \forall\, f \in W^{1,p}(\Omega), \tag{6.101}$$

where f_Ω denotes the average of f over Ω, i.e.,

$$f_\Omega := \fint_\Omega f(x)\,dx = \frac{1}{\mathcal{L}^N(\Omega)} \int_\Omega f(x)\,dx.$$

Proof To prove (6.100), it suffices to apply the generalized Poincaré inequality, choosing as a continuous seminorm

$$N(f) := \left| \int_\Omega f(x)\,dx \right|.$$

To obtain (6.101), it is enough to apply (6.100) to the function $f - f_\Omega$. □

In the textbook by W. P. Ziemer [61], one can find more general Poincaré-type inequalities.

6.11 Functions of Bounded Variation

Throughout this section, Ω will denote an open subset of $\mathbb{R}^N$. We will give an introduction to functions of bounded variation in Ω; for a more thorough study of such functions, see [3, 17, 23] or [61].

Definition 6.6 A function $u \in L^1(\Omega)$ is said to be a *function of bounded variation* in Ω if its partial derivatives in the sense of distributions are Radon measures with finite total variation, i.e., if there exist Radon measures $\mu_1, \ldots, \mu_N \in \mathcal{M}_{\mathbb{R}}(\Omega)$ such that

$$\int_\Omega u \frac{\partial \phi}{\partial x_i}\,dx = -\int_\Omega \phi\,d\mu_i \quad \forall \phi \in \mathcal{D}(\Omega), \quad i = 1, 2, \ldots, N,$$

with $|\mu_i|(\Omega) < \infty$ for $i = 1, 2, \ldots, N$.

Here we are identifying the Radon measure μ_i with the regular Borel measure whose existence is guaranteed by the Riesz Representation Theorem (see Remark 5.1). We denote $D_i u = \mu_i$, and $Du = (D_1 u, \ldots, D_N u)$. The vector space of all functions of bounded variation in Ω will be denoted by $BV(\Omega)$.

A function u is said to be *locally of bounded variation* in Ω if $u \in BV(U)$ for every open set $U \subset\subset \Omega$. The space of all locally bounded variation functions in Ω will be denoted by $BV_{loc}(\Omega)$.

Clearly, the Sobolev space $W^{1,1}(\Omega)$ is contained in $BV(\Omega)$, and moreover, for every $u \in W^{1,1}(\Omega)$, we have $Du = \nabla u\,\mathcal{L}^N$, where $\mathcal{L}^N$ denotes the N-dimensional Lebesgue measure. Furthermore, the inclusion is strict, i.e., there exist functions $u \in BV(\Omega)$ such that Du is singular with respect to the measure $\mathcal{L}^N$. For example, if $\Omega = (-1, 1)$ and we take $u := \chi_{(0,1)}$, we have $Du = \delta$, hence $u \in BV(\Omega) \setminus W^{1,1}(\Omega)$.

For every $u \in BV(\Omega)$, we define

$$\|Du\| := \sup\left\{\int_\Omega u \operatorname{div}\varphi\, dx \ : \ \varphi \in C_0^1(\Omega,\mathbb{R}^N),\ \|\varphi\|_\infty \le 1\right\}.$$

Using integration by parts, it follows that

$$\|Du\| = \int_\Omega |\nabla u|\, dx \quad \forall\, u \in W^{1,1}(\Omega).$$

By the Riesz Representation Theorem for vector-valued measures (see [17]), it can be shown that $\|Du\|$ coincides with the total variation of the vector measure Du, i.e., $\|Du\| = |Du|(\Omega)$.

Theorem 6.34 *Let $u \in L^1(\Omega)$. Then, $u \in BV(\Omega)$ if and only if $\|Du\| < \infty$. The mapping $u \mapsto \|Du\|$ is lower semicontinuous with respect to L^1-convergence, i.e., if $u_n \to u$ in $L^1(\Omega)$, then*

$$\|Du\| \le \liminf_{n\to\infty} \|Du_n\|.$$

Proof Suppose that $u \in BV(\Omega)$ and let $\varphi \in C_0^1(\Omega,\mathbb{R}^N)$ with $\|\varphi\|_\infty \le 1$. Then,

$$\left|\int_\Omega u\ \operatorname{div}\varphi\, dx\right| = \left|\sum_{i=1}^N \int_\Omega u\, \frac{\partial\varphi_i}{\partial x_i}\, dx\right| = \left|-\sum_{i=1}^N \int_\Omega \varphi_i\, dD_i u\right|$$

$$\le \sum_{i=1}^N \int_\Omega |\varphi_i|\, d|D_i u| \le \sum_{i=1}^N |D_i u|(\Omega) < \infty,$$

which implies that $\|Du\| < \infty$.

Now suppose that $u \in L^1(\Omega)$ and $\|Du\| < \infty$. Given $\phi \in \mathcal{D}(\Omega)$ with $|\phi| \le 1$, consider the vector field $\varphi = (0,\dots,0,\phi,0,\dots,0)$. Then,

$$\left|\int_\Omega \phi\, dD_i u\right| = \left|\int_\Omega u\ \operatorname{div}\varphi\, dx\right| \le \|Du\|,$$

from which we deduce by density that $|D_i u|(\Omega) \le \|Du\|$, and consequently $u \in BV(\Omega)$.

Suppose now that $u_n \to u$ in $L^1(\Omega)$. Then, for any $\varphi \in C_0^1(\Omega,\mathbb{R}^N)$ with $\|\varphi\|_\infty \le 1$, we have

$$\int_\Omega u\ \operatorname{div}\varphi\, dx = \lim_{n\to\infty} \int_\Omega u_n\ \operatorname{div}\varphi\, dx \le \liminf_{n\to\infty} \|Du_n\|,$$

from which it follows that

$$\|Du\| \leq \liminf_{n\to\infty} \|Du_n\|.$$

□

Remark 6.11 It can be shown that $BV(\Omega)$ is a Banach space with respect to the norm

$$\|u\|_{BV(\Omega)} := \|u\|_1 + \|Du\|.$$

However, this norm is too strong because it does not make $BV(\Omega)$ a separable space (see Problem 6.32). Moreover, $C^1(\Omega) \cap BV(\Omega)$ is not dense in $BV(\Omega)$ with respect to this norm, since the $\|\cdot\|_{BV(\Omega)}$-closure of $C^1(\Omega) \cap BV(\Omega)$ coincides with $W^{1,1}(\Omega)$.

Nevertheless, functions in $BV(\Omega)$ can be approximated in the L^1-sense by smooth functions. More precisely, we have the following approximation theorem.

Theorem 6.35 (Approximation by Smooth Functions) *Given $u \in BV(\Omega)$, there exist $u_n \in C^\infty(\Omega) \cap BV(\Omega)$ such that:*

(i) $u_n \to u$ *in* $L^1(\Omega)$, *and*

(ii) $\|Du\| = \lim_{n\to\infty} \int_\Omega |\nabla u_n|\, dx.$

Proof Fix $\epsilon > 0$. Due to the regularity of the measure $\|Du\|$, there exists $m \in \mathbb{N}$ such that

$$\|Du\|(\Omega \setminus \Omega_1) < \epsilon, \tag{6.102}$$

where

$$\Omega_k := \left\{ x \in \Omega \,:\, \operatorname{dist}(x, \partial\Omega) > \frac{1}{m+k} \right\} \cap B(0, k+m), \qquad k \in \mathbb{N}.$$

Let $\Omega_0 := \emptyset$, and define

$$U_k := \Omega_{k+1} \setminus \overline{\Omega_k}, \qquad k \in \mathbb{N}.$$

Let $\{\psi_k\}$ be a partition of unity subordinate to the open cover $\{U_k\}$, and for $\delta > 0$ define

$$\rho_\delta(x) = \frac{1}{\delta^N} \rho\left(\frac{x}{\delta}\right),$$

where $0 \leq \rho \in \mathcal{D}(\Omega)$, with

$$\int_{\mathbb{R}^N} \rho(x)\, dx = 1, \quad \operatorname{supp}(\rho) \subset B(0,1).$$

For each $k \in \mathbb{N}$, choose $\delta_k > 0$ sufficiently small so that the following holds:

$$\begin{cases} \operatorname{supp}(\rho_{\delta_k} \star (u\psi_k)) \subset U_k, \\ \displaystyle\int_\Omega |\rho_{\delta_k} \star (u\psi_k) - u\psi_k|\, dx < \frac{\epsilon}{2^k}, \\ \displaystyle\int_\Omega |\rho_{\delta_k} \star (u\nabla\psi_k) - u\nabla\psi_k|\, dx < \frac{\epsilon}{2^k}. \end{cases} \tag{6.103}$$

Define

$$u_\epsilon := \sum_{k=1}^{\infty} \rho_{\delta_k} \star (u\psi_k).$$

Since for each $x \in \Omega$ there exists a neighborhood where only finitely many terms in the series defining u_ϵ are non-zero, we have

$$u_\epsilon \in C^\infty(\Omega).$$

Moreover, since $\{\psi_k\}$ is a partition of unity,

$$u = \sum_{k=1}^{\infty} u\psi_k,$$

and using (6.103) we obtain

$$\|u_\epsilon - u\|_1 \le \sum_{k=1}^{\infty} \int_\Omega |\rho_{\delta_k} \star (u\psi_k) - u\psi_k|\, dx < \epsilon.$$

Therefore,

$$u_\epsilon \to u \quad \text{in } L^1(\Omega), \quad \text{as } \epsilon \to 0.$$

From this, applying the previous theorem we get

$$\|Du\| \le \liminf_{\epsilon \to 0} \|Du_\epsilon\|. \tag{6.104}$$

Let $\varphi \in C_0^1(\Omega, \mathbb{R}^N)$ with $\|\varphi\|_\infty \le 1$. Then, using the fact that $\sum_{k=1}^{\infty} \nabla\psi_k = 0$, we get

$$\int_\Omega u_\epsilon \operatorname{div}\varphi\, dx = \sum_{k=1}^{\infty} \int_\Omega \rho_{\delta_k} \star (u\psi_k) \operatorname{div}\varphi\, dx = \sum_{k=1}^{\infty} \int_\Omega u\psi_k \operatorname{div}(\rho_{\delta_k} \star \varphi)\, dx$$

$$= \sum_{k=1}^{\infty} \int_{\Omega} u \operatorname{div}(\psi_k(\rho_{\delta_k} \star \varphi))\, dx - \sum_{k=1}^{\infty} \int_{\Omega} u\, \nabla\psi_k \cdot (\rho_{\delta_k} \star \varphi)\, dx$$

$$= \sum_{k=1}^{\infty} \int_{\Omega} u \operatorname{div}(\psi_k(\rho_{\delta_k} \star \varphi))\, dx - \sum_{k=1}^{\infty} \int_{\Omega} \varphi \cdot (\rho_{\delta_k} \star (u\nabla\psi_k) - u\nabla\psi_k)\, dx.$$

Now, since $|\psi_k(\rho_{\delta_k} \star \varphi)| \leq 1$ and each point of Ω belongs to at most three sets of $\{U_k\}$, and using (6.102), we obtain

$$\left| \sum_{k=1}^{\infty} \int_{\Omega} u \operatorname{div}(\psi_k(\rho_{\delta_k} \star \varphi))\, dx \right|$$

$$= \left| \int_{\Omega} u \operatorname{div}(\psi_1(\rho_{\delta_1} \star \varphi))\, dx + \sum_{k=2}^{\infty} \int_{\Omega} u \operatorname{div}(\psi_k(\rho_{\delta_k} \star \varphi))\, dx \right|$$

$$\leq \|Du\|(\Omega) + \sum_{k=2}^{\infty} \|Du\|(U_k) \leq \|Du\|(\Omega) + 3\|Du\|(\Omega \setminus \Omega_k) \leq \|Du\| + 3\epsilon.$$

On the other hand, as a consequence of (6.104), we have

$$\left| \sum_{k=1}^{\infty} \int_{\Omega} \varphi \cdot (\rho_{\delta_k} \star (u\nabla\psi_k) - u\nabla\psi_k)\, dx \right| < \epsilon.$$

Therefore, we obtain

$$\int_{\Omega} u_\epsilon \operatorname{div} \varphi\, dx \leq \|Du\| + 4\epsilon,$$

and consequently,

$$\|Du_\epsilon\| \leq \|Du\| + 4\epsilon.$$

From this last inequality and (6.103), we obtain (ii) and conclude the proof. □

Corollary 6.10 (Compact Embedding) *Let $\Omega \subset \mathbb{R}^N$ be a bounded open set of class C^1. Let $\{u_n\}$ be a sequence in $BV(\Omega)$ such that $\|u_n\|_{BV(\Omega)} < M$ for all $n \in \mathbb{N}$. Then, there exists a subsequence $\{u_{n_k}\}$ and a function $u \in BV(\Omega)$ such that*

$$\lim_{k\to\infty} u_{n_k} = u \qquad \text{in} \quad L^1(\Omega).$$

Proof By the previous theorem, for each $n \in \mathbb{N}$ we can find a function $v_n \in C^\infty(\Omega) \cap BV(\Omega)$ such that

$$\|u_n - v_n\|_1 \leq \frac{1}{n}, \qquad \forall\, n \in \mathbb{N}, \tag{6.105}$$

$$\sup_{n\in\mathbb{N}} \int_\Omega |\nabla v_n|\, dx < \infty. \tag{6.106}$$

As a consequence of (6.105) and (6.106), we have that $\{v_n\}$ is a bounded sequence in $W^{1,1}(\Omega)$. Then, by the Rellich–Kondrachov Theorem, there exists a subsequence $\{v_{n_k}\}$ and a function $u \in L^1(\Omega)$ such that

$$\lim_{k\to\infty} v_{n_k} = u \quad \text{ in } L^1(\Omega).$$

Then, from (6.105), we obtain that

$$\lim_{k\to\infty} u_{n_k} = u \quad \text{ in } L^1(\Omega).$$

Finally, as a consequence of Theorem 6.34, we conclude that $u \in BV(\Omega)$. □

Using a technique similar to the one employed in the proof of the previous result, we can easily obtain Sobolev-type inequalities for functions of bounded variation. More precisely, we have the following result.

Theorem 6.36 *For $N \geq 2$, there exists a constant $S(N) > 0$ such that*

$$\|u\|_{L^{N/N-1}(\mathbb{R}^N)} \leq S(N)\|Du\|(\mathbb{R}^N) \quad \forall\, u \in BV(\mathbb{R}^N). \tag{6.107}$$

Proof By Theorem 6.35, there exist $u_n \in C^\infty(\mathbb{R}^N) \cap BV(\Omega)$ such that

$$\begin{cases} u_n \to u \quad \text{in } L^1(\Omega), \quad u_n \to u \quad \text{a.e.} \\ \\ \|Du\|(\mathbb{R}^N) = \lim\limits_{n\to\infty} \displaystyle\int_{\mathbb{R}^N} |\nabla u_n|\, dx. \end{cases}$$

Then, applying Fatou's Lemma and the Gagliardo–Nirenberg inequality (Theorem 6.17), we get

$$\|u\|_{L^{N/N-1}(\mathbb{R}^N)} \leq \liminf_{n\to\infty} \|u_n\|_{L^{N/N-1}(\mathbb{R}^N)}$$

$$\leq \lim_{n\to\infty} S(N) \int_{\mathbb{R}^N} |\nabla u_n| = S(N)\|Du\|(\mathbb{R}^N),$$

which proves the theorem. □

Remark 6.12 If in the proof of the previous result we use Theorem 6.18, instead of the Gagliardo–Nirenberg inequality, we obtain that

$$\|u\|_{L^{\infty}(\mathbb{R})} \leq S(1)\|Du\|(\mathbb{R}) \quad \forall\, u \in BV(\mathbb{R}). \tag{6.108}$$

Poincaré–Wirtinger-type inequalities can also be obtained for functions of bounded variation. More precisely, we have the following result.

Theorem 6.37 *Let $\Omega \subset \mathbb{R}^N$ be a bounded and connected open set with boundary of class C^1. Then there exists a constant $C > 0$, depending only on Ω, such that*

$$\int_{\Omega} |u - u_{\Omega}|\, dx \leq C\|Du\|(\Omega) \quad \forall\, u \in BV(\Omega). \tag{6.109}$$

Proof Suppose the result is not true. Then there exists a sequence $\{u_n\} \subset BV(\Omega)$ such that

$$\int_{\Omega} |u_n - (u_n)_{\Omega}|\, dx \geq n\|Du_n\|(\Omega) \quad \forall\, n \in \mathbb{N}.$$

Since both sides of the above inequality are homogeneous of degree 1, we can rescale and translate the functions u_n and obtain functions $v_n \in BV(\Omega)$ such that

$$\int_{\Omega} |v_n|\, dx = 1, \quad \int_{\Omega} v_n\, dx = 0, \quad \|Dv_n\|(\Omega) \leq \frac{1}{n}, \quad \forall\, n \in \mathbb{N}.$$

Applying Corollary 6.10, we can assume (extracting a subsequence if necessary) that $v_n \to v$ in $L^1(\Omega)$, with

$$\int_{\Omega} |v|\, dx = 1, \quad \int_{\Omega} v\, dx = 0.$$

Now, since $\|Dv_n\|(\Omega) \leq \frac{1}{n}$ for each $n \in \mathbb{N}$, by the lower semicontinuity of the variation (Theorem 6.34), we have that $\|Dv\|(\Omega) = 0$. Then, since Ω is connected, v must be constant in Ω (see Problem 6.33), which leads to a contradiction. □

Remark 6.13 If we apply the previous result to the case where $\Omega = B(x, r)$, and use the notation

$$(u)_{x,r} = u_{B(x,r)} = \fint_{B(x,r)} u(x)\, dx,$$

we obtain that there exists a constant $C > 0$, depending on $B(x, r)$, such that

$$\int_{B(x,r)} |u - (u)_{x,r}|\, dy \leq C\|Du\|(B(x, r)) \quad \forall\, u \in BV(B(x, r)).$$

Now, the optimal constant C in the above inequality does not depend on x. Indeed, it suffices to consider the transformation

$$T : BV(B(x,r)) \to BV(B(0,1)),$$

defined by $T(u)(y) := u(x + ry)$, which is surjective, to see that if $C(B(x,r))$ is the best constant associated with the ball $B(x,r)$, then $C(B(x,r)) = C(N)r$, where $C(N) = C(B(0,1))$. Therefore, as a consequence of (6.109), we have

$$\int_{B(x,r)} |u - (u)_{x,r}|\, dy \le C(N)\, r\, \|Du\|(B(x,r)) \quad \forall\, u \in BV(B(x,r)) \tag{6.110}$$

for every $x \in \mathbb{R}^N$ and every $r > 0$.

Definition 6.7 Let $E \subset \mathbb{R}^N$ be an $\mathcal{L}^N$-measurable set. We say that E is a *set of finite perimeter* in the open set Ω if $\chi_E \in BV(\Omega)$; in this case, the *perimeter* of E in Ω is defined as

$$P(E,\Omega) := \|D\chi_E\|(\Omega).$$

Thus, we have

$$P(E,\Omega) = \sup\left\{ \int_E \operatorname{div} \varphi\, dx \ :\ \varphi \in C_0^1(\Omega,\mathbb{R}^N),\ \|\varphi\|_\infty \le 1 \right\}. \tag{6.111}$$

We say that E is a *set of locally finite perimeter* if $P(E,\Omega) < \infty$ for every bounded open set $\Omega \subset \mathbb{R}^N$. Such sets are also commonly called *Caccioppoli sets*. When E is of finite perimeter in $\mathbb{R}^N$, we simply say that E is a *set of finite perimeter* and write

$$\operatorname{Per}(E) = P(E,\mathbb{R}^N).$$

Remark 6.14 If $E \subset \mathbb{R}^N$ is a bounded open set with boundary ∂E of class C^2, then E is a Caccioppoli set and, moreover, for every bounded open set $\Omega \subset \mathbb{R}^N$, we have

$$P(E,\Omega) = \mathcal{H}^{N-1}(\Omega \cap \partial E), \tag{6.112}$$

where $\mathcal{H}^{N-1}$ denotes the $(N-1)$-dimensional Hausdorff measure. Indeed, if $\varphi \in C_0^1(\Omega,\mathbb{R}^N)$ with $\|\varphi\|_\infty \le 1$, by applying the classical Green-Gauss Theorem (Theorem A.2), we obtain

$$\int_E \operatorname{div} \varphi\, dx = \int_{\partial E} \varphi \cdot \nu\, d\mathcal{H}^{N-1} \le \mathcal{H}^{N-1}(\Omega \cap \partial E) < \infty,$$

where ν is the unit outer normal to ∂E. Then, from (6.111), we get

$$P(E, \Omega) \leq \mathcal{H}^{N-1}(\Omega \cap \partial E).$$

Let us now prove the opposite inequality. Since ∂E is of class C^2, there exists an open set U containing ∂E, such that $d(x) = \operatorname{dist}(x, E)$ is of class C^1 in $U \setminus \partial E$, and

$$\nabla d(x) = \frac{x - \xi(x)}{d(x)},$$

where $\xi(x)$ is the unique point of ∂E closest to x. Therefore, the unit outer normal ν to ∂E admits an extension $\tilde{\nu} \in C_0^1(\mathbb{R}^N)$ such that $|\tilde{\nu}| \leq 1$. Consequently, if we take $\varphi = \eta\tilde{\nu}$ with $\eta \in \mathcal{D}(\Omega)$, we obtain

$$\int_E \operatorname{div} \varphi \, dx = \int_E \operatorname{div}(\eta\tilde{\nu}) \, dx = \int_{\partial E} \eta \, d\mathcal{H}^{N-1}.$$

Therefore,

$$P(E, \Omega) \geq \sup \left\{ \int_{\partial E} \eta \, d\mathcal{H}^{N-1} \ : \ \eta \in \mathcal{D}(\Omega), \ |\eta| \leq 1 \right\} = \mathcal{H}^{N-1}(\Omega \cap \partial E).$$

Let us recall that the *classical coarea formula* (see [3]) for a smooth function u ensures that

$$\int_\Omega |\nabla u(x)| \, dx = \int_{-\infty}^{+\infty} \mathcal{H}^{N-1}(\{x \in \Omega \ : \ u(x) = t\}) \, dt. \tag{6.113}$$

By Sard's Theorem (see [26]), for almost every $t \in \mathbb{R}$, the set

$$\{x \in \Omega \ : \ u(x) = t\}$$

is a smooth hypersurface which coincides with the boundary of the superlevel set

$$E_t(u) := \{x \in \Omega \ : u(x) > t\}.$$

Therefore, as a consequence of (6.112), we can rewrite the coarea formula as

$$\int_\Omega |\nabla u(x)| \, dx = \int_{-\infty}^{+\infty} \|D\chi_{E_t(u)}\|(\Omega) \, dt. \tag{6.114}$$

In 1960, W. H. Fleming and R. Rishel [19] proved that the coarea formula also holds for functions of bounded variation. Specifically, they established the following result.

Theorem 6.38 (The Coarea Formula) *Let $u \in BV(\Omega)$. Then, for almost every $t \in \mathbb{R}$, the set*

$$E_t(u) := \{x \in \Omega \ : u(x) > t\}$$

has finite perimeter, and

$$\|Du\|\, dx = \int_{-\infty}^{+\infty} \|D\chi_{E_t(u)}\|(\Omega)\, dt. \tag{6.115}$$

Conversely, if $u \in L^1(\Omega)$ and

$$\int_{-\infty}^{+\infty} \|D\chi_{E_t(u)}\|(\Omega)\, dt < \infty,$$

then $u \in BV(\Omega)$.

Proof First, observe that for every function $u \in L^1(\Omega)$,

$$u(x) = u^+(x) - u^-(x) = \int_0^{+\infty} \chi_{E_t(u)}(x)\, dt - \int_{-\infty}^{0} (1 - \chi_{E_t(u)}(x))\, dt. \tag{6.116}$$

Now, for $\varphi \in C_0^1(\Omega, \mathbb{R}^N)$ with $\|\varphi\|_\infty \le 1$, we have

$$\int_\Omega u \operatorname{div} \varphi\, dx = \int_{-\infty}^{+\infty} \left(\int_{E_t(u)} \operatorname{div} \varphi\, dx \right) dt. \tag{6.117}$$

Indeed, using (6.116) and the fact that $\int_\Omega \operatorname{div} \varphi\, dx = 0$, we get:

$$\int_\Omega u \operatorname{div} \varphi\, dx = \int_\Omega \left(\int_0^{+\infty} \chi_{E_t(u)}(x)\, dt - \int_{-\infty}^{0} (1 - \chi_{E_t(u)}(x))\, dt \right) \operatorname{div} \varphi\, dx$$

$$= \int_0^{+\infty} \left(\int_\Omega \chi_{E_t(u)}(x) \operatorname{div} \varphi\, dx \right) dt - \int_{-\infty}^{0} \left(\int_\Omega (1 - \chi_{E_t(u)}(x)) \operatorname{div} \varphi\, dx \right) dt$$

$$= \int_{-\infty}^{+\infty} \left(\int_\Omega \chi_{E_t(u)}(x) \operatorname{div} \varphi\, dx \right) dt = \int_{-\infty}^{+\infty} \left(\int_{E_t(u)} \operatorname{div} \varphi\, dx \right) dt.$$

As a consequence of (6.117), we obtain

$$\|Du\|(\Omega) \le \int_{-\infty}^{+\infty} \|D\chi_{E_t(u)}\|(\Omega)\, dt. \tag{6.118}$$

Then, if

$$\int_{-\infty}^{+\infty} \|D\chi_{E_t(u)}\|(\Omega)\,dt < \infty,$$

we have $\|Du\|(\Omega) < \infty$, and hence $u \in BV(\Omega)$ by Theorem 6.34.

Now suppose $u \in BV(\Omega)$. First, let us prove that the function $t \mapsto \|D\chi_{E_t(u)}\|$ is $\mathcal{L}^1$-measurable. Indeed, since the function

$$(x,t) \mapsto \chi_{E_t(u)}(x)$$

is $(\mathcal{L}^N \times \mathcal{L}^1)$-measurable, it follows that for each $\varphi \in C_0^1(\Omega,\mathbb{R}^N)$, the function

$$t \mapsto \int_{E_t(u)} \operatorname{div}\varphi\,dx = \int_\Omega \chi_{E_t(u)} \operatorname{div}\varphi\,dx$$

is $\mathcal{L}^1$-measurable. Now, let D be a countable dense subset of $C_0^1(\Omega,\mathbb{R}^N)$. Then

$$t \mapsto \|D\chi_{E_t(u)}\| = \sup\left\{\int_{E_t(u)} \operatorname{div}\varphi\,dx \;:\; \varphi \in D,\ \|\varphi\|_\infty \le 1\right\},$$

so the function $t \mapsto \|D\chi_{E_t(u)}\|$ is $\mathcal{L}^1$-measurable.

Let us now prove that (6.115) holds for $u \in BV(\Omega) \cap C^\infty(\Omega)$. Define

$$f(t) := \int_{\Omega\setminus E_t(u)} |\nabla u|\,dx = \int_{\{u\le t\}} |\nabla u|\,dx.$$

Since the function f is increasing, $f'(t)$ exists for almost every $t \in \mathbb{R}$, and

$$\int_{-\infty}^{+\infty} f'(t)\,dt \le \int_\Omega |\nabla u|\,dx. \tag{6.119}$$

Fix $t \in \mathbb{R}$ and $r > 0$, and define $\eta : \mathbb{R} \to \mathbb{R}$ by

$$\eta(s) := \begin{cases} 0 & \text{if } s \le t \\ \frac{s-t}{r} & \text{if } t \le s \le t+r \\ 1 & \text{if } s \ge t+r \end{cases}$$

Then

$$\eta'(s) := \begin{cases} \frac{1}{r} & \text{if } t < s < t+r \\ 0 & \text{if } s < t \text{ or } s > t+r \end{cases}$$

Hence, for each $\varphi \in C_0^1(\Omega, \mathbb{R}^N)$, we have:

$$-\int_\Omega \eta(u(x)) \operatorname{div} \varphi \, dx = \int_\Omega \eta'(u(x)) \nabla u \cdot \varphi \, dx = \frac{1}{r} \int_{E_t(u) \setminus E_{t+r}(u)} \nabla u \cdot \varphi \, dx.$$

Therefore,

$$\frac{f(t+r) - f(t)}{r} = \frac{1}{r} \int_{E_t(u) \setminus E_{t+r}(u)} |\nabla u| \, dx$$

$$\geq \frac{1}{r} \int_{E_t(u) \setminus E_{t+r}(u)} \nabla u \cdot \varphi \, dx = -\int_\Omega \eta(u(x)) \operatorname{div} \varphi \, dx.$$

Letting $r \to 0$, we obtain

$$f'(t) \geq -\int_{E_t(u)} \operatorname{div} \varphi \, dx \quad \text{for almost every } t \in \mathbb{R}.$$

Then, taking the supremum over $\varphi \in C_0^1(\Omega, \mathbb{R}^N)$ with $\|\varphi\|_\infty \leq 1$, we conclude

$$f'(t) \geq \|D\chi_{E_t(u)}\|(\Omega).$$

from where, taking into account (6.119), we obtain that

$$\int_{-\infty}^{+\infty} \|D\chi_{E_t(u)}\|(\Omega) \, dt \leq \int_\Omega |\nabla u| \, dx = \|Du\|(\Omega).$$

Hence, taking into account (6.118), we complete the proof of (6.115) for $u \in BV(\Omega) \cap C^\infty(\Omega)$. Finally, let us see that (6.115) holds for $u \in BV(\Omega)$. By Theorem 6.35, given $u \in BV(\Omega)$, there exist $u_n \in C^\infty(\Omega) \cap BV(\Omega)$ such that

$$u_n \to u \quad \text{in} \quad L^1(\Omega) \tag{6.120}$$

$$\|Du\|(\Omega) = \lim_{n \to \infty} \int_\Omega |\nabla u_n| \, dx. \tag{6.121}$$

As seen above, for each $n \in \mathbb{N}$, we have

$$\int_\Omega |\nabla u_n| \, dx = \int_{-\infty}^{+\infty} \|D\chi_{E_t(u_n)}\|(\Omega) \, dt. \tag{6.122}$$

Now,

$$\int_\Omega |u_n(x) - u(x)| \, dx = \int_{-\infty}^{+\infty} \left(\int_\Omega \left| \chi_{E_t(u_n)}(x) - \chi_{E_t(u)}(x) \right| dx \right) dt.$$

Then, taking into account (6.120),

$$\chi_{E_t(u_n)} \to \chi_{E_t(u)} \quad \text{in} \quad L^1(\Omega) \quad \text{for almost every} \quad t \in \mathbb{R}.$$

Hence, by the lower semicontinuity of the variation,

$$\|D\chi_{E_t(u)}\|(\Omega) \le \liminf_{n\to\infty} \|D\chi_{E_t(u_n)}\|(\Omega).$$

Therefore, by Fatou's Lemma, (6.121) and (6.122), we obtain

$$\int_{-\infty}^{+\infty} \|D\chi_{E_t(u)}\|(\Omega)\, dt \le \liminf_{n\to\infty} \int_{-\infty}^{+\infty} \|D\chi_{E_t(u_n)}\|(\Omega)\, dt$$

$$= \liminf_{n\to\infty} \int_\Omega |\nabla u_n|\, dx = \|Du\|(\Omega),$$

which, taking into account (6.118), concludes the proof.

□

Theorem 6.39 (Isoperimetric Inequality) *Let E be a bounded Caccioppoli set in $\mathbb{R}^N$. Then,*

$$\mathcal{L}^N(E)^{(N-1)/N} \le I(N)\, Per(E), \tag{6.123}$$

where $I(N) = S(N)$.

Proof Since $\chi_E \in BV(\mathbb{R}^N)$, it is enough to apply the Sobolev inequality (6.107) to the function χ_E. □

Remark 6.15 We have seen that the isoperimetric inequality (6.123) is a consequence of the Sobolev inequality (6.107). Let us now show that the converse is also true. Let T_k be the truncation operator at levels k and $-k$, i.e.,

$$T_k(r) := \begin{cases} r & \text{if } |r| \le k \\ \\ k\,\text{sign}(r) & \text{if } |r| > k. \end{cases}$$

Consider the function

$$f(t) := \left(\int_{\mathbb{R}^N} |T_t(u)|^{N/(N-1)}\, dx \right)^{(N-1)/N}.$$

For each $h > 0$, we have

$$|T_{t+h}(u)| \le |T_t(u)| + h\chi_{E_t(u)},$$

so that

$$f(t+h) \le f(t) + h\left(\mathcal{L}^N(E_t(u))\right)^{(N-1)/N}.$$

It follows that

$$f'(t) \le \left(\mathcal{L}^N(E_t(u))\right)^{(N-1)/N}.$$

Then, taking into account the coarea formula, from (6.123) we deduce that

$$\left(\int_{\mathbb{R}^N} |u|^{N/(N-1)}\,dx\right)^{(N-1)/N} = f(\infty) - f(0) = \int_0^{+\infty} f'(t)\,dt$$

$$\le \int_0^{+\infty} \left(\mathcal{L}^N(E_t(u))\right)^{(N-1)/N}$$

$$\le I(N)\int_0^{+\infty} \mathrm{Per}(E_t(u))\,dt = I(N)\|Du\|(\mathbb{R}^N).$$

With a symmetrization argument, it can be shown that the best constant $I(N)$ for the isoperimetric inequality is the one obtained for the unit ball, i.e.,

$$S(N) = I(N) = \left(\mathcal{L}^N(B(0,1))\right)^{(N-1)/N} (Per(B(0,1))^{-1} = N^{-1}\sigma_N^{-1/N},$$

where σ_N is the volume of the unit ball in $\mathbb{R}^N$, $\sigma_N = \frac{\Gamma(\frac{1}{2})^N}{\Gamma(N+\frac{1}{2})}$.

We also have the following *Relative Isoperimetric Inequalities*.

Theorem 6.40 *Let E be a Caccioppoli set in $\mathbb{R}^N$. Then, there exists a constant C_N that depends only on the dimension N, such that, for every ball $B(x,r) \subset \mathbb{R}^N$, the following inequalities hold:*

$$\mathcal{L}^N(E \cap B(x,r))\,\mathcal{L}^N(B(x,r) \setminus E) \le C_N\, r^{N+1} P(E, B(x,r)). \tag{6.124}$$

$$\min\{\mathcal{L}^N(E \cap B(x,r)),\ \mathcal{L}^N(B(x,r) \setminus E)\} \le C_N\, r\, P(E, B(x,r)). \tag{6.125}$$

Proof Let $u := \chi_{E\cap B(x,r)}$. The average value of u over $B(x,r)$ is given by

$$(u)_{x,r} = \frac{\mathcal{L}^N(E \cap B(x,r))}{\mathcal{L}^N(B(x,r))},$$

and

$$\int_{B(x,r)} |u - (u)_{x,r}|\, dy = 2\frac{\mathcal{L}^N(E \cap B(x,r))\, \mathcal{L}^N(B(x,r) \setminus E)}{\mathcal{L}^N(B(x,r))}.$$

Applying the inequality (6.110), we obtain

$$2\frac{\mathcal{L}^N(E \cap B(x,r))\, \mathcal{L}^N(B(x,r) \setminus E)}{\mathcal{L}^N(B(x,r))} \leq C(N)\, r\, P(E, B(x,r)), \tag{6.126}$$

from which inequality (6.124) follows.

On the other hand, from (6.126) we deduce that

$$2\frac{\mathcal{L}^N(E \cap B(x,r))}{\mathcal{L}^N(B(x,r))}\left(1 - \frac{\mathcal{L}^N(E \cap B(x,r))}{\mathcal{L}^N(B(x,r))}\right)$$

$$\leq C(N)\, r\, P(E, B(x,r))\mathcal{L}^N(B(x,r)).$$

From this, since

$$\frac{\mathcal{L}^N(E \cap B(x,r))}{\mathcal{L}^N(B(x,r))} \in [0,1]$$

and $\min\{t, 1-t\} \leq 2t(1-t)$, we have

$$\min\left\{\frac{\mathcal{L}^N(E \cap B(x,r))}{\mathcal{L}^N(B(x,r))}, 1 - \frac{\mathcal{L}^N(E \cap B(x,r))}{\mathcal{L}^N(B(x,r))}\right\}$$

$$\leq C(N)\, r\, P(E, B(x,r))\mathcal{L}^N(B(x,r)),$$

which proves (6.125). □

The concept of trace for functions in Sobolev spaces, as we saw in Sect. 6.9, can be extended to the more general setting of functions of bounded variation. The following result holds, whose proof can be found, for example, in [17].

Theorem 6.41 *Let $\Omega \subset \mathbb{R}^N$ be a bounded open set with Lipschitz continuous boundary. There exists a bounded linear operator $T : BV(\Omega) \to L^1(\partial\Omega, \mathcal{H}^{N-1})$ such that*

$$\int_\Omega u \operatorname{div} \varphi\, dx = -\int_\Omega \varphi \cdot Du + \int_{\partial\Omega} (\varphi \cdot \nu) T(u)\, d\mathcal{H}^{N-1} \tag{6.127}$$

for every $u \in BV(\Omega)$ and $\varphi \in C^1(\mathbb{R}^N, \mathbb{R}^N)$, where ν denotes the outer unit normal to $\partial\Omega$. Moreover, for $\mathcal{H}^{N-1}$-a.e. $x \in \partial\Omega$,

$$T(u)(x) = \lim_{r\to 0} \fint_{B(x,r)\cap\Omega} u(y)\, dy.$$

The function $T(u)$ is called the *trace* of u on $\partial\Omega$. In the case where $u \in W^{1,1}(\Omega)$, we have that $T(u) = \gamma(u)$, where γ is the trace operator defined in Sect. 6.9.

Theorem 6.42 (Extension) *Let $\Omega \subset \mathbb{R}^N$ be a bounded open set of class C^1. Let $u_1 \in BV(\Omega)$, $u_2 \in BV(\mathbb{R}^N \setminus \overline{\Omega})$. We define*

$$\tilde{u}(x) := \begin{cases} u_1(x) & x \in \Omega \\ \\ u_2(x) & x \in \mathbb{R}^N \setminus \overline{\Omega} \end{cases}$$

Then, $\tilde{u} \in BV(\mathbb{R}^N)$ and

$$\|D\tilde{u}\|(\mathbb{R}^N) = \|Du_1\|(\Omega) + \|Du_2\|(\mathbb{R}^N \setminus \overline{\Omega}) + \int_{\partial\Omega} |T(u_1) - T(u_2)|\, d\mathcal{H}^{N-1}.$$

Proof Let $\varphi \in C_0^1(\mathbb{R}^N, \mathbb{R}^N)$ with $\|\varphi\|_\infty \le 1$. Then, taking into account Theorem 6.41, we have

$$\int_{\mathbb{R}^N} \tilde{u} \operatorname{div}\varphi\, dx = \int_\Omega u_1 \operatorname{div}\varphi\, dx + \int_{\mathbb{R}^N\setminus\overline{\Omega}} u_2 \operatorname{div}\varphi\, dx$$

$$= -\int_\Omega \varphi \cdot Du_1 - \int_{\mathbb{R}^N\setminus\overline{\Omega}} \varphi \cdot Du_2 + \int_{\partial\Omega} (T(u_1) - T(u_2))\varphi\cdot\nu\, d\mathcal{H}^{N-1}$$

$$\le \|Du_1\|(\Omega) + \|Du_2\|(\mathbb{R}^N \setminus \overline{\Omega}) + \int_{\partial\Omega} |T(u_1) - T(u_2)|\, d\mathcal{H}^{N-1}.$$

Therefore, $\tilde{u} \in BV(\mathbb{R}^N)$ and also

$$\|D\tilde{u}\|(\mathbb{R}^N) \le \|Du_1\|(\Omega) + \|Du_2\|(\mathbb{R}^N \setminus \overline{\Omega}) + \int_{\partial\Omega} |T(u_1) - T(u_2)|\, d\mathcal{H}^{N-1}.$$

To prove the equality, first observe that for every $\varphi \in C_0^1(\mathbb{R}^N, \mathbb{R}^N)$ it holds that

$$\begin{aligned} -\int_{\mathbb{R}^N} \varphi \cdot D\tilde{u} &= -\int_\Omega \varphi \cdot Du_1 - \int_{\mathbb{R}^N\setminus\overline{\Omega}} \varphi \cdot Du_2 \\ &+ \int_{\partial\Omega} (T(u_1) - T(u_2))\, \varphi\cdot\nu\, d\mathcal{H}^{N-1}. \end{aligned} \tag{6.128}$$

Therefore,

$$D\tilde{u} = \begin{cases} Du_1 \ \text{ on } \ \Omega \\ \\ Du_2 \ \text{ on } \ \mathbb{R}^N \setminus \overline{\Omega} \end{cases}$$

Then, using (6.128), we obtain

$$-\int_{\partial\Omega} \varphi \cdot D\tilde{u} = \int_{\partial\Omega} (T(u_1) - T(u_2))\, \varphi \cdot \nu \, d\mathcal{H}^{N-1},$$

from which it follows that

$$\|D\tilde{u}\|(\partial\Omega) = \int_{\partial\Omega} |T(u_1) - T(u_2)|\, d\mathcal{H}^{N-1},$$

and thus the equality is established. □

Corollary 6.11 *Let $\Omega \subset \mathbb{R}^N$ be a bounded open set of class C^1, and let $u \in BV(\Omega)$. We define*

$$E(u)(x) := \begin{cases} u(x) & x \in \Omega \\ \\ 0 & x \in \mathbb{R}^N \setminus \overline{\Omega}. \end{cases}$$

Then, $E(u) \in BV(\mathbb{R}^N)$ and

$$\|DE(u)\|(\mathbb{R}^N) = \|Du\|(\Omega) + \int_{\partial\Omega} |T(u)|\, d\mathcal{H}^{N-1}.$$

In particular, Ω has finite perimeter and

$$Per(\Omega) = \mathcal{H}^{N-1}(\partial\Omega).$$

Proof The first part is an immediate consequence of the previous result. To prove the last part, it suffices to apply the first one to $u = \chi_\Omega$. □

Theorem 6.43 (Embedding Theorem) *Assume that $N \geq 2$. Let $\Omega \subset \mathbb{R}^N$ be a bounded open set of class C^1. Then the embedding $BV(\Omega) \hookrightarrow L^{N/N-1}(\Omega)$ is continuous, and compact into $L^p(\Omega)$ for every $1 \leq p < \frac{N}{N-1}$.*

Proof Taking into account the previous corollary, if we apply (6.107) to $E(u)$, we obtain

$$\|u\|_{L^{N/N-1}(\Omega)} = \|E(u)\|_{L^{N/N-1}(\mathbb{R}^N)} \leq C\|DE(u)\|(\mathbb{R}^N)$$

$$= C\left(\|Du\|(\Omega) + \int_{\partial\Omega} |T(u)|\, d\mathcal{H}^{N-1} \right).$$

From this, taking into account the continuity of the trace operator, we obtain the continuous embedding $BV(\Omega) \hookrightarrow L^{N/N-1}(\Omega)$. Now, by Corollary 6.10, we have that the embedding $BV(\Omega) \hookrightarrow L^1(\Omega)$ is compact. Therefore, applying Hölder's inequality, we conclude that the embedding $BV(\Omega) \hookrightarrow L^p(\Omega)$ is compact for every $1 \le p < \frac{N}{N-1}$. □

Problems

6.1 Let $\Omega = B(0,1)$ be the open unit ball in $\mathbb{R}^N$, and define

$$f(x) := \|x\|^{-\alpha} \quad \text{for } x \in \Omega,\ x \neq 0.$$

For which values of $\alpha > 0$ is $f \in W^{1,p}(\Omega)$?

6.2 Let $\Omega :=]-1,1[\times]-1,1[$, and define

$$f(x,y) := \begin{cases} y & \text{if } y > 0 \\ \\ 0 & \text{if } y \le 0. \end{cases}$$

Prove that $f \in H^1(\Omega) \setminus H^2(\Omega)$.

6.3 Let Ω be the ball in $\mathbb{R}^2$ centered at the origin with radius $\frac{1}{2}$. For which values of $\alpha > 0$ is the function $f(x) := \left[\log\left(\frac{1}{\|x\|}\right)\right]^\alpha$ an element of $H^1(\Omega)$?

6.4 Let $\Omega := \{x \in \mathbb{R}^2 : |x_1| < 1,\ |x_2| < 1\}$. Define

$$f(x) := \begin{cases} 1 - x_1 & \text{if } x_1 > 0,\ \ |x_2| < x_1 \\ 1 + x_1 & \text{if } x_1 < 0,\ \ |x_2| < -x_1 \\ 1 - x_2 & \text{if } x_2 > 0,\ \ |x_1| < x_2 \\ 1 + x_2 & \text{if } x_2 < 0,\ \ |x_1| < -x_2. \end{cases}$$

For which values of $1 \le p \le \infty$ is f an element of $W^{1,p}(\Omega)$?

6.5 Let $f \in W^{1,p}(\Omega)$. Prove that

$$\nabla f = 0 \text{ a.e. on } \{f = 0\}.$$

6.6 Let $\Omega \subset \mathbb{R}^N$ be a connected open set, and let $f \in W^{1,p}(\Omega)$. Prove that if $\nabla f = 0$ a.e. in Ω, then f is constant a.e. in Ω.

6.7 Let $f \in L^p(\Omega)$, with $1 < p \leq \infty$. Prove that the following statements are equivalent:

(i) $f \in W^{1,p}(\Omega)$

(ii) There exists a constant $C > 0$ such that

$$\left| \int_\Omega f \frac{\partial \phi}{\partial x_i} \right| \leq C \|\phi\|_{p'} \quad \forall\, \phi \in \mathcal{D}(\Omega), \quad \forall\, i = 1, \ldots, N.$$

(iii) There exists a constant $C > 0$ such that for every open set $U \subset\subset \Omega$ and every $h \in \mathbb{R}^N$, with $\|h\| < \operatorname{dist}(U, \mathbb{R}^N \setminus \Omega)$, we have

$$\|\tau_h f - f\|_{L^p(U)} \leq C\|h\|.$$

6.8 Let $\Omega \subset \mathbb{R}^N$ be a bounded open set and $\psi \in C^1(\overline{\Omega})$. Prove that the mapping $f \mapsto \psi f$ is a linear and continuous operator from $H^1(\Omega)$ into $H^1(\Omega)$. Moreover, if $\psi > 0$ on $\overline{\Omega}$, then this operator is a surjective isomorphism.

6.9 Let Ω, Ω' be two open subsets of $\mathbb{R}^N$, and let $\Phi : \Omega' \to \Omega$ be a bijective mapping such that

$$\Phi \in C^1(\Omega'), \quad \Phi^{-1} \in C^1(\Omega), J_\Phi \in L^\infty(\Omega'), \quad J_{\Phi^{-1}} \in L^\infty(\Omega).$$

Prove that if $f \in W^{1,p}(\Omega)$, then $f \circ \Phi \in W^{1,p}(\Omega')$ and

$$\frac{\partial (f \circ \Phi)}{\partial y_i}(y) = \sum_{i=1}^N \frac{\partial f}{\partial x_i}(\Phi(y)) \frac{\partial \phi_i}{\partial y_i}(y), \qquad i = 1, \ldots, N.$$

6.10 Prove that $\delta \in H^s(\mathbb{R}^N)$ whenever $s < -\frac{N}{2}$.

6.11 Using the Fourier transform, prove that if $s > \frac{N}{2}$, then there exists a constant C that depends only on s and N, such that

$$\|f\|_{L^\infty(\mathbb{R}^N)} \leq C \|f\|_{H^s(\mathbb{R}^N)} \quad \forall\, f \in H^s(\mathbb{R}^N).$$

6.12 Prove that every distribution with compact support in $\mathbb{R}^N$ belongs to some $H^s(\mathbb{R}^N)$ for a certain $s \in \mathbb{R}$.

6.13 Prove that $W^{m,p}(\mathbb{R}^N) = W_0^{m,p}(\mathbb{R}^N)$.

6.14 Prove that if Ω is a bounded open set, then

$$W^{1,p}(\Omega) \neq W_0^{1,p}(\Omega).$$

6.15 Let $\Omega \subset \mathbb{R}^N$ be a bounded open set. Prove that the map $f \mapsto \|\Delta f\|_2$ defines a norm on $H_0^2(\Omega)$ equivalent to its usual norm.

6.16 Prove that Theorem 6.3 is not valid, in general, for $p = \infty$.

6.17 Suppose that $\Omega \subset \mathbb{R}^n$ is an open set with $\partial\Omega$ of class C^1, and let $1 < p < \infty$. Let $f \in L^p(\Omega)$. Prove that the following are equivalent:

(i) $f \in W_0^{1,p}(\Omega)$
(ii) There exists a constant $C > 0$ such that

$$\left| \int_\Omega f \frac{\partial \phi}{\partial x_i} \right| \leq C \|\phi\|_{p'} \quad \forall\, \phi \in \mathcal{D}(\mathbb{R}^N), \quad \forall\, i = 1, \ldots, N.$$

(iii) $f^0 \in W^{1,p}(\mathbb{R}^N)$.

6.18 Prove that the Hölder spaces $C^{k,\gamma}(\overline{\Omega})$, with respect to the norm $\| \ \|_{C^{k,\gamma}(\overline{\Omega})}$, are Banach spaces.

6.19 Let $f \in W^{1,p}(\mathbb{R}^N)$, with $N < p < \infty$. Prove that

$$\lim_{\|x\| \to \infty} f(x) = 0.$$

6.20 Prove that if $p > N$, then $W^{1,p}(\mathbb{R}^N)$ is a Banach algebra, i.e., if $f, g \in W^{1,p}(\mathbb{R}^N)$, then $fg \in W^{1,p}(\mathbb{R}^N)$ and

$$\|fg\|_{1,p} \leq C \|f\|_{1,p} \, \|g\|_{1,p}.$$

6.21 Let Ω be an open subset of $\mathbb{R}^N$. Prove that if $f \in W^{1,p}_{loc}(\Omega)$ for some $N < p \leq \infty$, then f is differentiable at almost every point of Ω. Deduce from this the classical *Rademacher's Theorem*, which asserts that every locally Lipschitz continuous function is differentiable almost everywhere.

6.22 Let $\Omega = B(0, \frac{1}{2}) \subset \mathbb{R}^2$ and

$$f(x) := \log\left(\log\left(\frac{2}{\|x\|}\right)\right), \quad x \in \Omega.$$

Prove that $f \in W^{1,2}(\Omega) \setminus L^\infty(\Omega)$. More generally, prove that for $N > 1$, if $\Omega := B(0, 1)$, then the function f, defined by

$$f(x) := \log\left(\log\left(1 + \frac{1}{\|x\|}\right)\right), \quad x \in \Omega \setminus \{0\},$$

belongs to $W^{1,N}(\Omega) \setminus L^\infty(\Omega)$.

6.23 Let $h \in L^1(\mathbb{R}^N)$ and let $\mathcal{B}$ be a bounded subset of $L^p(\mathbb{R}^N)$, with $1 \leq p < \infty$. Define $\mathcal{K} := \{h \star g \ : \ g \in \mathcal{B}\}$. Prove that for every bounded open set $\Omega \subset \mathbb{R}^N$, the set $\mathcal{K}|_\Omega := \{f|_\Omega \ : \ f \in \mathcal{K}\}$ is relatively compact in $L^p(\Omega)$.

6.24 Prove Theorem 6.22.

6.25 Let $\Omega \subset \mathbb{R}^N$ be a bounded convex open set such that $0 \in \Omega$ and

$$\lambda\Omega = \{\lambda x \,:\, x \in \Omega\} \subset \Omega \quad \text{for} \quad 0 < \lambda < 1.$$

Let $f \in W_0^{1,p}(\Omega)$ such that $\|f\|_{p^*} = 1$. For $k \in \mathbb{N}$, define

$$f_k(x) := \begin{cases} k^{\frac{(N-p)}{p}} f(kx), & x \in (\frac{1}{k})\Omega \\ 0, & x \in \Omega \setminus (\frac{1}{k})\Omega. \end{cases}$$

Prove that $\|f_k(x)\|_{1,p} = \|f\|_{1,p}$ and that $\|f_k\|_{p^*} = \|f\|_{p^*} = 1$ for all $k \in \mathbb{N}$. Deduce that the embedding $W^{1,p}(\Omega) \hookrightarrow L^{p^*}(\Omega)$ is not compact.

6.26 Prove that the trace operator γ satisfies the following properties:

(i) $|\gamma(f)| = \gamma(|f|)$ for all $f \in H^1(\Omega)$.
(ii) If $f, g \in H^1(\Omega)$ and $f \le g$ $\mathcal{L}^N$-a.e. in Ω, then $\gamma(f) \le \gamma(g)$ H^{N-1}-a.e. on $\partial\Omega$.
(iii) If $f \in H^1(\Omega) \cap L^\infty(\Omega)$, then $\gamma(f) \in L^\infty(\partial\Omega)$ and $\|\gamma(f)\|_\infty \le \|f\|_\infty$.

6.27 Prove that if $f \in H^1(\Omega)$ and $g \in W^{1,\infty}(\Omega)$, then $fg \in H^1(\Omega)$ and $\gamma(fg) = \gamma(f)\gamma(g)$.

6.28 Let $\mathcal{D}(\partial\Omega)$ be the space consisting of the restrictions to $\partial\Omega$ of functions in $\mathcal{D}(\mathbb{R}^N)$. Prove that:

(i) $\mathcal{D}(\partial\Omega)$ is dense in $H^{1/2}(\partial\Omega)$.
(ii) $H^{1/2}(\partial\Omega)$ is dense in $L^2(\partial\Omega)$.
(iii) $L^2(\partial\Omega)$ is dense in $H^{-1/2}(\partial\Omega)$.

6.29 Prove that the Poincaré–Wirtinger inequality also holds for $p = 1$ and $p = \infty$.

6.30 For $f \in L^1(\mathbb{R}^N)$, we denote its average over the ball $B(x, r)$ by $(f)_{x,r}$, i.e.,

$$(f)_{x,r} := \fint_{B(x,r)} f(y)\, dy.$$

Define

$$[f]_{BMO(\mathbb{R}^N)} := \sup \left\{ \fint_{B(x,r)} |f(y) - (f)_{x,r}|\, dy \,:\, x \in \mathbb{R}^N,\ r > 0 \right\}.$$

We say that the function f has bounded mean oscillation if

$$[f]_{BMO(\mathbb{R}^N)} < \infty.$$

Prove that every function $f \in W^{1,N}(\mathbb{R}^N) \cap L^1(\mathbb{R}^N)$ has bounded mean oscillation. More precisely, prove that there exists a constant $C > 0$ such that

$$[f]_{BMO(\mathbb{R}^N)} \leq C\|\nabla f\|_N, \quad \forall\, f \in W^{1,N}(\mathbb{R}^N) \cap L^1(\mathbb{R}^N).$$

6.31 Let $\Omega \subset \mathbb{R}^N$ be a bounded domain with C^1 boundary and $1 < p < \infty$. Let $\omega \subset \Omega$ with $\mathcal{L}^N(\omega) > 0$. Prove that there exists a constant C such that

$$\|f\|_{L^p(\Omega)} \leq C\left(\|\nabla f\|_{L^p(\omega)} + \|f\|_{L^p(\Omega)}\right) \quad \forall\, f \in W^{1,p}(\Omega).$$

6.32 Prove that the space $BV(\Omega)$ is not separable with respect to the norm $\|\cdot\|_{BV(\Omega)}$.

6.33 Suppose that Ω is a connected open set and $u \in BV(\Omega)$. Prove that if $Du = 0$, then u is constant in Ω.

6.34 Let $u \in BV(\Omega)$ and $\phi \in \mathcal{D}(\Omega)$. Prove that $\phi u \in BV(\Omega)$ and that $D(\phi u) = \phi Du + u\nabla\phi$ in the sense of distributions.

6.35 Prove that $u \in BV(\Omega)$ if and only if there exists a sequence $\{u_n\}$ of smooth functions in Ω such that

$$u_n \rightharpoonup u \quad \text{in } L^1(\Omega) \quad \text{with} \quad \sup_{n\in\mathbb{N}} \int_\Omega |\nabla u_n|\, dx < \infty.$$

6.36 Prove that for every $0 < \alpha \leq 1$ there exists a constant $C_\alpha > 0$ such that

$$\|u\|_{L^{N/(N-1)}(B(x,r))} \leq C_\alpha \|Du\|(B(x,r))$$

for every $x \in \mathbb{R}^N$, $r > 0$, and $u \in BV_{loc}(\mathbb{R}^N)$ satisfying

$$\frac{\mathcal{L}^N(B(x,r) \cap \{u = 0\})}{\mathcal{L}^N(B(x,r))} \geq \alpha.$$

6.37 Prove that $BV(\Omega) \cap L^\infty(\Omega)$ is dense in $BV(\Omega)$ with respect to the norm $\|\cdot\|_{BV(\Omega)}$.

Chapter 7
Elliptic Boundary Value Problems



7.1 Introduction

Although variational problems have been present throughout the history of science since antiquity, the oldest problem in the calculus of variations, known as *Dido's problem*, dates back to the time of the founding of the city of Carthage. It consists of finding, among all closed curves of the same length, the one that encloses the greatest area. Only after the development of calculus by Newton and Leibniz did the brothers Johann and Jakob Bernoulli, along with Johann's student, Leonhard Euler, lay the foundations of this new discipline. The first text on the Calculus of Variations was published by Euler in 1744 under the title *"A method for finding curves that have certain maximum or minimum properties"*.

In this work, Euler expressed his belief that *"every effect in nature follows a maximum or minimum rule"*. This belief in the universality of the Calculus of Variations as a tool was very widespread at the time. Note that in that same year, 1744, Maupertuis formulated his famous *Principle of Least Action*, which states:

> *Nature change occurs in such a way that the least amount of energy or action necessary for that change is expended.*

The pioneer of this line of thought was Fermat, who in 1662 deduced the law of refraction in geometric optics through his famous principle according to which *"nature always acts in the shortest way"*. Although some authors consider Fermat's study of light refraction as the first variational problem treated analytically, it is the *Brachistochrone Problem*, solved by Johann Bernoulli in 1697, that is considered the true beginning of the Calculus of Variations as a new theory. This problem was originally posed by Galileo in 1638 and consists of finding the curve along which a particle of mass m moves from a point A to a point B without friction under the action of gravity. The correct answer was given by Johann Bernoulli, who named

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J. M. Mazón, *Elements of Functional Analysis*, Universitext,
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this curve the Brachistochrone. To solve it, he formulated it as a variational problem in the following way:

By the law of conservation of energy, for every time t, the sum of kinetic and potential energy is constant, so we have:

$$\frac{1}{2}mv^2(t) + mGh(t) = \frac{1}{2}mv^2(0) + mGy_a = C,$$

where G is the gravitational constant, $v(t)$ the velocity of the particle at time t, and $h(t)$ the Y-coordinate at time t (we are assuming that the X-axis represents the surface of the Earth). The path followed by the particle can be represented as the graph of a function $y : [a, b] \to \mathbb{R}$. If $S(t)$ is the distance traveled at time t, then we have:

$$v(t) = \frac{dS(t)}{dt} = \sqrt{\frac{2C}{m} - 2Gh(t)} = \sqrt{2G}\sqrt{y_a - h(t)}.$$

Therefore, the time required for the particle to travel from A to B following the graph of the function y is given by:

$$T(y) = \int_0^{t_B} dt = \int_0^{S(t_B)} \frac{dS(t)}{v(t)}\, dt = \int_a^b \frac{\sqrt{1 + \left(y'(x)\right)^2}}{\sqrt{2G}\sqrt{y_a - h(t)}}\, dx.$$

Thus, the Brachistochrone problem consists of finding the function y, with boundary values $y(a) = y_a$, $y(b) = y_b$, that minimizes $T(y)$.

Another problem, also proposed by Galileo in 1638 and solved by the Bernoulli brothers, is to find the shape of a thin and heavy inextensible cable suspended by its ends. Assuming, as in the previous example, that the endpoints $A = (x_1, y_1)$ and $B = (x_2, y_2)$ lie in a vertical plane, and that the cable is the graph of a function $y : [x_1, x_2] \to \mathbb{R}$, then the potential energy is proportional to

$$\int_{x_1}^{x_2} y(x)\sqrt{1 + |y'(x)|^2}\, dx.$$

Moreover, the inextensibility of the cable tells us that

$$\int_{x_1}^{x_2} \sqrt{1 + |y'(x)|^2}\, dx = L,$$

where L is its length. Therefore, this problem consists in minimizing the functional $\mathcal{F} \colon C([x_1, x_2]) \to \mathbb{R}$,

$$\mathcal{F}(y) := \int_{x_1}^{x_2} y(x)\sqrt{1 + |y'(x)|^2}\, dx$$

over the set

$$\mathcal{A} := \{y \in C^1([x_1, x_2]) \ : \ y(x_1) = y_1, \ y(x_2) = y_2,$$

$$\int_{x_1}^{x_2} \sqrt{1 + |y'(x)|^2}\, dx = L\}.$$

We thus see that these classical problems of the Calculus of Variations can be formulated as follows: Let $L : [a, b] \times \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be a function, called the *Lagrangian* of the problem. Associated with the Lagrangian L, we consider the *energy functional*

$$I(y) = \int_a^b L(x, y(x), y'(x))\, dx.$$

The problem is to compute the minimum

$$\min_{y \in \mathcal{A}} I(y), \qquad with \quad \mathcal{A} \subset C([a, b]).$$

Not all problems considered in the Calculus of Variations were one-dimensional like the previous ones.

In the nineteenth century, variational methods were employed by Gauss and Dirichlet to determine the distribution of an electric charge on the surface of a conductor by requiring that the energy of the associated electric field be minimal, or by Riemann for the construction of analytic functions. These problems gave rise to what Riemann called the *Dirichlet Principle*, which states that:

There exists a unique function that minimizes the Dirichlet integral

$$\frac{1}{2} \int_\Omega |\nabla u|^2$$

among all functions $u \in C^1(\Omega) \cap C(\overline{\Omega})$ *that take a given value on the boundary* $\partial\Omega$. *Moreover, such function is harmonic in* Ω.

As we saw in the Introduction of Chap. 6, the Dirichlet Principle fell into disrepute and underwent a crisis. This crisis began to be overcome at the beginning of the twentieth century by the efforts of several mathematicians who did not wish to abandon the great mathematical tool that the Dirichlet Principle was. Among these mathematicians, D. Hilbert stands out, who gave a direct proof of the existence of a minimizer for the Dirichlet integral in two dimensions in the class of functions that have smooth values on the boundary; and H. Lebesgue with the introduction in the Calculus of Variations of the concept of semicontinuity which had been defined by R. Baire.

Problems in the Calculus of Variations can be formulated in the following general form: let $\Omega \subset \mathbb{R}^N$ be an open set and $L : \overline{\Omega} \times \mathbb{R} \times \mathbb{R}^N \to \mathbb{R}$ the Lagrangian of

the problem, $L(x, z, p)$. Let $X(\Omega)$ be a space of real-valued functions on Ω, and $\mathcal{A} \subset X(\Omega)$. Consider the energy functional of the problem $I : X(\Omega) \to \mathbb{R}$,

$$I(u) = \int_\Omega L(x, u(x), \nabla u(x))\, dx.$$

The problem to be solved is to compute the

$$\text{Min}\{I(u) \ : \ u \in \mathcal{A}\}. \tag{7.1}$$

During the first two hundred years of its history, the method used to solve problem (7.1) was the so-called *indirect method*. This method, based on the optimistic idea that every minimization problem had a solution, consists first in determining conditions that this minimum must satisfy. As in the case of real-valued functions of a real variable, the first step is to look for the critical points of the functional I, i.e., the functions $u \in \mathcal{A}$ that satisfy

$$I'(u) = 0, \tag{7.2}$$

where I' represents a certain "derivative" of I. This equation, called the *Euler–Lagrange equation* of the problem, is the following quasi-linear partial differential equation in divergence form

$$-\sum_{i=1}^{N} \big(L_{p_i}(x, u(x), \nabla u(x))\big)_{x_i} + L_z(x, u(x), \nabla u(x)) = 0 \quad \text{in} \quad \Omega. \tag{7.3}$$

Since the Lagrangian associated with Dirichlet's Principle is

$$L(x, z, p) = \frac{1}{2}\|p\|^2,$$

its Euler–Lagrange equation is the Laplace equation:

$$-\Delta u = 0 \qquad \text{in } \Omega.$$

If we instead take the Lagrangian

$$L(x, z, p) = \sqrt{1 + \|p\|},$$

then the corresponding energy functional

$$I(u) = \int_\Omega \sqrt{1 + \|\nabla u(x)\|^2}\, dx$$

represents the area of the surface defined by the function u. Its Euler–Lagrange equation is

$$-\operatorname{div}\left(\frac{\nabla u}{\sqrt{1+\|\nabla u\|^2}}\right)=0 \quad \text{in } \Omega,$$

an equation known as the minimal surface equation.

As we have already mentioned, after the crisis caused by Weierstrass's example, the Calculus of Variations revived, particularly through the contributions of Hilbert and Lebesgue, which led to Tonelli's introduction of the so-called *direct method* in the Calculus of Variations. This method consists in directly proving that the problem has a minimum, using compactness and the concept of semicontinuity introduced by Baire as fundamental tools. When applying the direct method, in order to ensure the necessary compactness, it is essential to enlarge the function space and to admit functions that are not differentiable in the classical sense.

For example, when dealing with the Dirichlet integral, the appropriate function space is the Sobolev space $W^{1,2}(\Omega)$, and in the case of the minimal surface equation, the suitable space is $BV(\Omega)$, the space of functions of bounded variation. This fact was first pointed out by Hilbert in his famous lecture at the International Congress of Mathematicians in Paris in 1900, where he proposed his famous 23 problems for the future of mathematics. Problem 20 states:

> *Does every regular variational problem have a solution, provided certain regularity conditions on the boundary are assumed, and provided, if necessary, that the concept of solution is appropriately extended?*

Since we must admit as solutions functions that are not differentiable in the classical sense, we are then led to the so-called *regularity problem*. This is Hilbert's nineteenth problem, which asks:

> *Are the solutions of regular problems in the Calculus of Variations necessarily always regular?*

Many partial differential equations arising in Physics and Engineering come from a *variational principle*. Typically, there is a class of admissible solutions and an energy functional associated with them; the goal is to minimize the energy in order to find the solution to the problem. The corresponding *Euler–Lagrange equation* for the minimization problem yields the partial differential equation. In this chapter, we study some examples of this general situation. With the help of Sobolev spaces and abstract variational formulations (Stampacchia's and Lax–Milgram's theorems), we will find the "weak solutions" of certain boundary value problems for elliptic equations. For the Dirichlet problem for the Poisson equation, we will establish the *Dirichlet–Hilbert principle*.

We will also study, using Sobolev embedding theorems, the *regularity problem*, i.e., the degree of smoothness of the weak solutions and under what conditions they are classical solutions. Finally, we present the weak and strong maximum principles and conclude the chapter with a spectral study of elliptic operators.

7.2 Abstract Variational Problems

In this section, we study some abstract variational problems that will serve as the foundation for our subsequent study of elliptic boundary value problems. In this section, $(H, (\,|\,))$ will denote a real Hilbert space.

One of the most classical abstract variational results in Functional Analysis is the *Theorem of Best Approximation* (Theorem 3.2), which addresses the problem of minimizing the distance function in a Hilbert space.

Let $B : H \times H \to \mathbb{R}$ be a bilinear map. With a similar argument to that used to characterize continuous linear maps, one can easily prove that B is continuous if and only if B is bounded, i.e., if there exists a constant $M > 0$ such that

$$|B(x, y)| \leq M\|x\|\,\|y\| \qquad \forall\, x, y \in H.$$

The smallest constant M satisfying the above inequality is denoted by $\|B\|$.

Definition 7.1 A bilinear form $B : H \times H \to \mathbb{R}$ is said to be *H-elliptic* (or *H-coercive*) if there exists a constant $\alpha > 0$ such that

$$B(x, x) \geq \alpha\|x\|^2 \qquad \forall\, x \in H.$$

Note that if B is a continuous and H-elliptic bilinear form, then

$$|x|_B := \sqrt{B(x, x)}$$

defines a norm on H equivalent to its original norm.

Theorem 7.1 (Stampacchia's Theorem) *Let B be a continuous and H-elliptic bilinear form, and let $K \subset H$ be a nonempty closed convex set. Then, for any $F \in H'$, there exists a unique $x \in K$ such that*

$$B(x, y - x) \geq \langle F, y - x\rangle \qquad \forall\, y \in K. \tag{7.4}$$

If, in addition, B is symmetric (i.e., $B(x, y) = B(y, x)$ for every $x, y \in H$), then x is the solution to the variational problem

$$\begin{cases} x \in K \\ \\ J(x) = \min\limits_{y \in K} J(y), \end{cases} \tag{7.5}$$

where

$$J(y) := \frac{1}{2}B(y, y) - \langle F, y\rangle.$$

Proof By the Riesz–Fréchet Theorem (Theorem 3.4), there exists a unique $f \in H$ such that

$$\langle F, y\rangle = (f|y) \qquad \forall\, y \in H.$$

On the other hand, for each $x \in H$, the mapping $y \mapsto B(x, y)$ is a continuous linear form, so applying the Riesz–Fréchet Theorem again, there exists a unique $Ax \in H$ such that

$$B(x, y) = (Ax|y) \qquad \forall\, y \in H.$$

Clearly, A is a linear operator on H and, furthermore,

$$\|Ax\| = \sup\{|(Ax|y)| \ : \ \|y\| \le 1\} \le \|B\|\,\|x\|. \tag{7.6}$$

Therefore, $A \in \mathcal{L}(H, H)$. Moreover, due to the H-ellipticity of B, we have

$$(Ax|x) = B(x, x) \ge \alpha\|x\|^2 \qquad \forall\, x \in H. \tag{7.7}$$

With this new notation, the problem consists of finding $x \in K$ such that

$$(Ax|y - x) \ge (f|y - x) \qquad \forall\, y \in K. \tag{7.8}$$

Let $\rho > 0$ be a constant to be fixed later. Then (7.8) is equivalent to

$$(\rho f - \rho Ax + x - x|y - x) \le 0 \qquad \forall\, y \in K. \tag{7.9}$$

Hence, our problem consists of finding $x \in K$ such that

$$x = P_K(\rho f - \rho Ax + x). \tag{7.10}$$

Therefore, if we define $S : K \to K$ by

$$Sy := P_K(\rho f - \rho Ay + y),$$

we are looking for a fixed point of S. Now, by the Banach Fixed Point Theorem, it will be enough to find a $\rho > 0$ such that S is a strict contraction. We have

$$\|Sy_1 - Sy_2\| \le \|\rho f - \rho Ay_1 + y_1 - (\rho f - \rho Ay_2 + y_2)\|$$

$$= \|(y_1 - y_2) - \rho(Ay_1 - Ay_2)\|.$$

Thus, taking into account (7.6) and (7.7), we obtain

$$\|Sy_1 - Sy_2\|^2 \le \|y_1 - y_2\|^2 - 2\rho(Ay_1 - Ay_2|y_1 - y_2) + \rho^2\|Ay_1 - Ay_2\|^2$$

$$\leq \|y_1 - y_2\|^2 - 2\rho\alpha\|y_1 - y_2\|^2 + \rho^2\|B\|^2\|y_1 - y_2\|^2.$$

Therefore, if we take $0 < \rho$ such that $0 < 1 - 2\rho\alpha + \rho^2\|B\|^2 < 1$, we conclude that S has a fixed point.

Now suppose that B is symmetric. Then $B(x, y)$ defines a new inner product on H whose associated norm

$$|x|_B := \sqrt{B(x, x)}$$

is equivalent to the original one. Applying the Riesz–Fréchet Theorem in this new Hilbert space, there exists $g \in H$ such that

$$(f|y) = \langle F, y\rangle = B(g, y) \quad \forall\, y \in H. \tag{7.11}$$

From which it follows that

$$\frac{1}{2}|y - g|_B^2 = \frac{1}{2}B(y - g, y - g) = \frac{1}{2}B(y, y) - \langle F, y\rangle + \frac{1}{2}|g|_B^2 = J(y) + \frac{1}{2}|g|_B^2.$$

Then, since $\frac{1}{2}|g|_B^2$ is constant, minimizing $J(y)$ over K is equivalent to minimizing $|y - g|_B^2$ over K. Now, by the optimal approximation theorem, the minimum of $|y - g|_B^2$ (and consequently of $J(y)$) over K is characterized as the unique element $x \in K$ such that

$$B(g - x, y - x) \leq 0 \qquad \forall\, y \in K.$$

Or equivalently, by (7.11), x is the unique element of K satisfying

$$\langle F, y - x\rangle \leq B(x, y - x) \qquad \forall\, y \in K.$$

Consequently, x is the unique solution to the variational problem (7.5). □

Sometimes, it is very useful to use the following equivalent formulation of (7.4).

Theorem 7.2 (Minty's Theorem) *Suppose we are under the hypotheses of Stampacchia's Theorem. Then, the following problems are equivalent:*

$$\begin{cases} x \in K \\ \\ B(x, y - x) \geq \langle F, y - x\rangle \qquad \forall\, y \in K. \end{cases} \tag{7.12}$$

$$\begin{cases} x \in K \\ \\ B(y, y - x) \geq \langle F, y - x\rangle \qquad \forall\, y \in K. \end{cases} \tag{7.13}$$

Proof Suppose that $x \in K$ is a solution of (7.12). Then, for every $y \in K$, we have

$$B(y, y-x) = B(y-x, y-x) + B(x, y-x) \geq B(x, y-x) \geq \langle F, y-x \rangle,$$

which implies that x is a solution of (7.13).

Conversely, suppose that $x \in K$ is a solution of (7.13). Given any $y \in K$, since for $t \in (0, 1)$ we have $x + t(y-x) \in K$, it follows that

$$B(x + t(y-x), t(y-x)) \geq t\langle F, y-x \rangle \quad \forall\, y \in K.$$

Dividing by t and letting $t \to 0$, we obtain

$$B(x, y-x) \geq \langle F, y-x \rangle \qquad \forall\, y \in K,$$

which proves the result. □

Remark 7.1 In the previous theorem, we have seen that the solution of the variational problem (7.5) is the unique element of K that satisfies the inequality (7.4). For this reason, inequalities of the type (7.5) are called *variational inequalities*.

The simplest variational inequality is given by the following example. Let $f : [a, b] \to \mathbb{R}$ be a smooth function, and let $x_0 \in [a, b]$ be such that

$$f(x_0) = \min_{x \in [a,b]} f(x).$$

Then we have:

(i) $f'(x_0) = 0$ if $x_0 \in]a, b[$; (ii) $f'(x_0) \geq 0$ if $x_0 = a$; (iii) $f'(x_0) \leq 0$ if $x_0 = b$. Consequently, x_0 satisfies the variational inequality

$$f'(x_0)(x - x_0) \geq 0 \qquad \forall\, x \in [a, b].$$

Now suppose that $K \subset \mathbb{R}^N$ is a closed convex set and $f : K \to \mathbb{R}$ is a smooth function. Again, the point $x_0 \in K$ that satisfies

$$f(x_0) = \min_{x \in K} f(x)$$

is characterized by the fact that it satisfies the variational inequality

$$\nabla f(x_0) \cdot (x - x_0) \geq 0 \qquad \forall\, x \in K.$$

Example 7.1 Let $\Omega \subset \mathbb{R}^N$ be a measurable set and, for $\phi \in L^2(\Omega)$, consider the set

$$K := \{v \in L^2(\Omega) \ : \ v \geq \phi \quad \text{a.e. in } \Omega\}.$$

Clearly, K is a closed convex subset of $L^2(\Omega)$. Define

$$B(u, v) := \int_\Omega u(x)v(x)\, dx.$$

Since B is a bilinear, symmetric, and coercive form on $L^2(\Omega)$, for each $f \in L^2(\Omega)$, by Stampacchia's Theorem there exists a unique $u \in K$ that satisfies the variational inequality

$$\int_\Omega u(x)(v(x) - u(x))\, dx \geq \int_\Omega f(x)(v(x) - u(x))\, dx \quad \forall\, v \in K. \tag{7.14}$$

Let us show that $u = \max\{\phi, f\}$. Indeed, since $v - \phi \geq 0$ for every $v \in K$, if we define $\tilde{u} := \max\{\phi, f\}$, then we have

$$\begin{aligned}
&\int_\Omega \tilde{u}(x)(v(x) - \tilde{u}(x))\, dx \\
&= \int_{f<\phi} \tilde{u}(x)(v(x) - \tilde{u}(x))\, dx + \int_{f\geq\phi} \tilde{u}(x)(v(x) - u(x))\, dx \\
&= \int_{f<\phi} \phi(x)(v(x) - \phi(x))\, dx + \int_{f\geq\phi} f(x)(v(x) - f(x))\, dx \\
&\geq \int_{f<\phi} f(x)(v(x) - \phi(x))\, dx + \int_{f\geq\phi} f(x)(v(x) - f(x))\, dx \\
&\geq \int_\Omega f(x)(v(x) - \tilde{u}(x))\, dx.
\end{aligned}$$

Taking $K = H$ in Stampacchia's Theorem, we obtain the following result.

Theorem 7.3 (Lax–Milgram's Theorem) *Let B be a continuous and H-elliptic bilinear form. Then, given $F \in H'$, there exists a unique $x \in H$ such that*

$$B(x, y) = \langle F, y\rangle \qquad \forall\, y \in H. \tag{7.15}$$

Moreover, if B is symmetric, then x is the solution of the variational problem

$$\begin{cases} x \in H \\ J(x) = \min\limits_{y\in H} J(y), \end{cases} \tag{7.16}$$

where

$$J(y) := \frac{1}{2}B(y, y) - \langle F, y\rangle.$$

Proof It is enough to apply Stampacchia's Theorem and observe that, if $y \in H$, then $z = y + x \in H$, so that

$$B(x, y) = B(x, z - x) \geq \langle F, z - x\rangle = \langle F, y\rangle,$$

and taking $z = x - y \in H$,

$$B(x, -y) = B(x, z - x) \geq \langle F, z - x\rangle = \langle F, -y\rangle,$$

so that $\langle F, y\rangle \leq B(x, y)$, and consequently, $\langle F, y\rangle = B(x, y)$. □

Remark 7.2 Suppose we are under the conditions of the Lax–Milgram Theorem with B symmetric. Let $\Phi(y) := B(y, y)$. Then,

$$\frac{\Phi(y + z) - \Phi(y)}{\|z\|} = \frac{B(y + z, y + z) - B(y, y)}{\|z\|} = \frac{2B(y, z) + B(z, z)}{\|z\|}.$$

Hence,

$$\Phi'(y)(z) = 2B(y, z).$$

Therefore,

$$J'(y)(z) = B(y, z) - \langle F, z\rangle,$$

so that

$$J'(x) = 0 \iff B(x, z) = \langle F, z\rangle \quad \forall\, z \in H.$$

We thus have that, in the terminology of the Calculus of Variations, the equation

$$B(x, z) = \langle F, z\rangle \quad \forall\, z \in H$$

from the Lax–Milgram Theorem is the Euler–Lagrange Equation of the minimization problem

$$J(x) = \min_{z \in H} J(z).$$

7.3 Variational Formulation of Elliptic Boundary Value Problems

In this section, we will apply the abstract results from the previous section to obtain existence and uniqueness results for certain elliptic boundary value problems. We begin by studying the Dirichlet problem for the Poisson equation.

Let Ω be a bounded open subset of $\mathbb{R}^N$. Consider the homogeneous *Dirichlet problem*:

$$\begin{cases} -\Delta u = f & \text{in } \Omega \\ \\ u = 0 & \text{on } \partial\Omega, \end{cases} \tag{7.17}$$

where $f : \Omega \to \mathbb{R}$ is a given function. A *classical solution* of (7.17) is a function $u \in C^2(\Omega)$ that satisfies (7.17) pointwise.

Suppose that u is a classical solution of (7.17). Given $\phi \in \mathcal{D}(\Omega)$, multiplying the equation $-\Delta u = f$ by ϕ, integrating, and applying Green's formula (Theorem A.4), we obtain

$$\int_\Omega f\phi = -\int_\Omega \Delta u\, \phi = \int_\Omega \nabla u \cdot \nabla\phi - \int_{\partial\Omega} \phi \frac{\partial u}{\partial \eta},$$

and since $\phi = 0$ on $\partial\Omega$, we get

$$\int_\Omega \nabla u \cdot \nabla\phi = \int_\Omega f\phi \qquad \forall\, \phi \in \mathcal{D}(\Omega).$$

Now, since $u \in C^2(\Omega)$ and $u = 0$ on $\partial\Omega$, it follows that $u \in H_0^1(\Omega)$. Moreover, by the density of $\mathcal{D}(\Omega)$ in $H_0^1(\Omega)$, we have

$$\int_\Omega \nabla u \cdot \nabla\phi = \int_\Omega f\phi \qquad \forall\, \phi \in H_0^1(\Omega). \tag{7.18}$$

A function $u \in H_0^1(\Omega)$ that satisfies (7.18) is called a *weak solution* of problem (7.17). Therefore, every classical solution is a weak solution. Furthermore, if u is a weak solution of (7.17) and we know a priori that $u \in C^2(\overline{\Omega})$, then u is a classical solution.

Indeed, since $u = 0$ on $\partial\Omega$, by Green's formula (Theorem A.4) we have $-\Delta u = f$ in the sense of distributions. Now, since $\mathcal{D}(\Omega)$ is dense in $L^2(\Omega)$, it follows that $-\Delta u = f$ as functions in $L^2(\Omega)$, and then, since $-\Delta u \in C(\overline{\Omega})$, the equality $-\Delta u = f$ holds point-wise, and consequently u is a classical solution.

As a consequence of the abstract results from the previous section, we have the following existence and uniqueness theorem for weak solutions.

Theorem 7.4 *Let $\Omega \subset \mathbb{R}^N$ be a bounded open set, and let $f \in L^2(\Omega)$. Then there exists a unique weak solution $u \in H_0^1(\Omega)$ to the Dirichlet problem (7.17). Moreover, u is a solution to the variational problem*

$$J(u) = \min_{v \in H_0^1(\Omega)} J(v),$$

where

$$J(v) = \frac{1}{2} \int_\Omega |\nabla u|^2 - \int_\Omega f v.$$

Proof Consider the symmetric bilinear form on $H_0^1(\Omega)$ given by

$$B(u, v) := \int_\Omega \nabla u \cdot \nabla v.$$

We have

$$|B(u, v)| = \left| \int_\Omega \nabla u \cdot \nabla v \right| \leq \sum_{i=1}^N \left| \int_\Omega \frac{\partial u}{\partial x_i} \frac{\partial v}{\partial x_i} \right|$$

$$\leq \sum_{i=1}^N \left\| \frac{\partial u}{\partial x_i} \right\|_2 \left\| \frac{\partial v}{\partial x_i} \right\|_2 \leq \|u\|_{H_0^1(\Omega)} \|v\|_{H_0^1(\Omega)},$$

so B is continuous. Moreover, by Poincaré's inequality,

$$B(v, v) = \int_\Omega \nabla v \cdot \nabla v = \|\nabla v\|_2^2 \geq C \|v\|_{H_0^1(\Omega)},$$

which shows that B is $H_0^1(\Omega)$-elliptic. Now, the mapping

$$F : H_0^1(\Omega) \to \mathbb{R},$$

defined by

$$F(v) = \int_\Omega f v,$$

is linear and continuous. Therefore, by the Lax–Milgram Theorem, there exists a unique $u \in H_0^1(\Omega)$ such that

$$F(u) = B(u, v) \qquad \forall\, v \in H_0^1(\Omega),$$

i.e., u is the weak solution of problem (7.17). Moreover, since B is symmetric, u is also the solution of the variational problem

$$J(u) = \min_{v \in H_0^1(\Omega)} J(v),$$

where

$$J(v) = \frac{1}{2} B(v, v) - F(v) = \frac{1}{2} \int_\Omega |\nabla v|^2 - \int_\Omega f v.$$

□

Remark 7.3 Note that the above result also holds if $f \in H^{-1}(\Omega)$. One simply needs to replace $\int_\Omega f v$ by $\langle f, v \rangle$, where $\langle \cdot, \cdot \rangle$ denotes the duality pairing between $H^{-1}(\Omega)$ and $H_0^1(\Omega)$.

Now consider the non-homogeneous Dirichlet problem

$$\begin{cases} -\Delta u = f & \text{in } \Omega \\ \\ u = g & \text{on } \partial\Omega, \end{cases} \tag{7.19}$$

As in the previous case, we will formulate this problem in weak form. We assume that $f \in L^2(\Omega)$ (or more generally that $f \in H^{-1}(\Omega)$). Since we need to apply the Trace Theorem (Theorem 6.26), we must assume that the open set Ω is of class C^1. Then, if we assume that $g \in H^{1/2}(\partial\Omega)$, there exists a function $\tilde{g} \in H^1(\Omega)$ such that $\gamma(\tilde{g}) = g$. We define

$$K_g := \{v \in H^1(\Omega) \ : \ v - \tilde{g} \in H_0^1(\Omega)\}.$$

It is easy to see that K_g is a convex and closed subset of $H^1(\Omega)$.

Definition 7.2 We say that $u \in H^1(\Omega)$ is a *weak solution* of problem (7.19) if $u \in K_g$ and the following condition holds:

$$\int_\Omega \nabla u \cdot \nabla v = \int_\Omega f v \qquad \forall\, v \in H_0^1(\Omega). \tag{7.20}$$

As in the homogeneous case, by applying Green's formula, it is easy to see that every classical solution of (7.19) is a weak solution, and that if $u \in C^2(\Omega)$ is a weak solution, then u is a classical solution.

To prove that problem (7.19) has a weak solution, one might consider the bilinear form

$$B(u, v) := \int_\Omega \nabla u \cdot \nabla v$$

and apply Stampacchia's Theorem. However, this form is not $H^1(\Omega)$-elliptic. We proceed as follows.

Suppose that u is a weak solution of (7.19). Then $w = u - \tilde{g} \in H_0^1(\Omega)$, and we have

$$\int_\Omega \nabla w \cdot \nabla v = \int_\Omega f v - \int_\Omega \nabla \tilde{g} \cdot \nabla v \quad \forall\, v \in H_0^1(\Omega). \tag{7.21}$$

Now, if $F : H^1(\Omega) \to \mathbb{R}$ is defined by

$$F(v) := \int_\Omega f v - \int_\Omega \nabla \tilde{g} \cdot \nabla v,$$

we have

$$|F(v)| \leq \big(\|f\|_2 + \|\nabla \tilde{g}\|_2\big)\, \|v\|_{H_0^1(\Omega)},$$

which implies that $F \in (H_0^1(\Omega))'$.

Moreover, as a consequence of Poincaré's inequality, B is $H_0^1(\Omega)$-elliptic. Then, by applying the Lax–Milgram Theorem, there exists a unique solution $w \in H_0^1(\Omega)$ of (7.21). Hence, $u := w + \tilde{g}$ is a weak solution of (7.19).

Furthermore, the solution is unique: if u_1 and u_2 are two solutions of (7.19), we have

$$\int_\Omega \nabla(u_1 - u_2) \cdot \nabla v = 0 \quad \forall\, v \in H_0^1(\Omega).$$

Taking $v = u_1 - u_2 \in H_0^1(\Omega)$, we obtain

$$\int_\Omega \big(\nabla(u_1 - u_2)\big)^2 = 0.$$

Then, as a consequence of Poincaré's inequality, we get $u_1 = u_2$, so the solution is unique.

Moreover, the solution u depends continuously on the data f and g, i.e., there exists a constant $C > 0$ such that

$$\|u\|_{H^1(\Omega)} \leq C\big(\|f\|_2 + \|g\|_{H^{1/2}(\partial\Omega)}\big). \tag{7.22}$$

Indeed, by the continuity of the trace operator, there exists a constant $C_1 > 0$ such that

$$\|\tilde{g}\|_{H^1(\Omega)} \leq C_1 \|g\|_{H^{1/2}(\partial\Omega)}.$$

Taking $v = w$ in (7.21) and applying Poincaré's inequality, we obtain

$$\int_\Omega |\nabla w|^2 \le \|f\|_2 \, \|w\|_2 + \left(\int_\Omega |\nabla \tilde{g}|^2\right)^{\frac{1}{2}} \left(\int_\Omega |\nabla w|^2\right)^{\frac{1}{2}} \le$$

$$\le C_2 \|f\|_2 \left(\int_\Omega |\nabla w|^2\right)^{\frac{1}{2}} + \|\tilde{g}\|_{H^1(\Omega)} \left(\int_\Omega |\nabla w|^2\right)^{\frac{1}{2}}.$$

Therefore,

$$\left(\int_\Omega |\nabla w|^2\right)^{\frac{1}{2}} \le C_3\big(\|f\|_2 + \|g\|_{H^{1/2}(\partial\Omega)}\big).$$

From here, applying Poincaré's inequality again, we have

$$\|w\|_{H^1(\Omega)} \le C_4\big(\|f\|_2 + \|g\|_{H^{1/2}(\partial\Omega)}\big).$$

And since $u = w + \tilde{g}$, from the above we deduce (7.22). We have thus established the following result.

Theorem 7.5 *Let $\Omega \subset \mathbb{R}^N$ be a bounded open set of class C^1. Given $f \in L^2(\Omega)$ and $g \in H^{1/2}(\partial\Omega)$, there exists a unique weak solution u of problem (7.19). Moreover, there exists a constant $C > 0$ such that*

$$\|u\|_{H^1(\Omega)} \le C\big(\|f\|_2 + \|g\|_{H^{1/2}(\partial\Omega)}\big).$$

We now study the Dirichlet problem for a general second-order elliptic operator. Let $\Omega \subset \mathbb{R}^N$ be a bounded open set, and let

$$P(x, D) = \sum_{|\alpha| \le 2} a_\alpha(x) D^\alpha$$

be a second-order linear differential operator. If the coefficients $a_\alpha \in C^1(\Omega)$ for $|\alpha| = 2$, then the operator $P(x, D)$ can be written in the form:

$$P(x, D) = -\sum_{i,j=1}^{N} D_i\big(a_{ij}(x) D_j\big) + \sum_{i=1}^{N} a_i(x) D_i + a_0(x). \tag{7.23}$$

An operator written as in (7.23) is said to be in *divergence form*. Operators in divergence form are the most natural for applying the *energy method* based on integration by parts. When an operator is written in the form (7.23), we will always assume that

$$a_{ij}, a_i, a_0 \in L^\infty(\Omega), \quad \text{for } i, j = 1, 2, \ldots, N.$$

The operator $P(x, D)$ given by (7.23) is said to be *uniformly elliptic* in Ω if there exists a constant $\theta > 0$ such that

$$\sum_{i,j=1}^{N} a_{ij}(x)\xi_i\xi_j \geq \theta\|\xi\|^2 \quad \forall\, \xi \in \mathbb{R}^N, \ \text{and a.e. in } x \in \Omega. \tag{7.24}$$

If we assume the symmetry condition $a_{ij} = a_{ji}$, then the ellipticity condition (7.24) means that for almost every point $x \in \Omega$, the symmetric matrix $(a_{ij}(x))$ is positive definite, and its smallest eigenvalue is greater than or equal to θ.

Let us consider the homogeneous Dirichlet problem associated with the operator $P(x, D)$, i.e., the problem:

$$\begin{cases} -\displaystyle\sum_{i,j=1}^{N} D_i\big(a_{ij}(x)D_j u\big) + \sum_{i=1}^{N} a_i(x)D_i u + a_0(x)u = f & \text{in } \Omega \\ \\ u = 0 & \text{on } \partial\Omega. \end{cases} \tag{7.25}$$

Suppose that $f \in L^2(\Omega)$ and that u is a classical solution of problem (7.25). As in the case of problem (7.17), i.e., by multiplying by test functions and applying integration by parts, we find that u satisfies

$$\int_\Omega \left(\sum_{i,j=1}^{N} a_{ij} D_i u D_j v + \sum_{i=1}^{N} a_i D_i u v + a_0 u v \right) dx = \int_\Omega f v \, dx \tag{7.26}$$

for every $v \in H_0^1(\Omega)$. A function $u \in H_0^1(\Omega)$ that satisfies (7.26) is called a *weak solution* of problem (7.25). Therefore, every classical solution of problem (7.25) is a weak solution. The bilinear form B on $H_0^1(\Omega)$, defined by

$$B(u, v) := \int_\Omega \left(\sum_{i,j=1}^{N} a_{ij} D_i u D_j v + \sum_{i=1}^{N} a_i D_i u v + a_0 u v \right) dx \tag{7.27}$$

is called the bilinear form associated with the problem (7.25). Note that with this terminology, a function $u \in H_0^1(\Omega)$ is a weak solution of the problem (7.25) if it satisfies

$$B(u, v) = (f|v) \qquad \text{for every } \ v \in H_0^1(\Omega), \tag{7.28}$$

where $(\,|\,)$ denotes the inner product in $L^2(\Omega)$.

To obtain existence and uniqueness of the problem (7.25) via the Lax–Milgram Theorem, it is necessary that the bilinear form B be continuous and $H_0^1(\Omega)$-elliptic. For this purpose, we need the following *energy estimates*.

Lemma 7.1 *Suppose the operator $P(x, D)$ given by (7.23) is uniformly elliptic and that B is the bilinear form associated with the problem (7.25). Then there exist constants $\alpha, \beta > 0$ and $\gamma \geq 0$ such that*

$$|B(u, v)| \leq \alpha \|u\|_{H_0^1(\Omega)} \|v\|_{H_0^1(\Omega)} \tag{7.29}$$

and

$$\beta \|u\|^2_{H_0^1(\Omega)} \leq B(u, u) + \gamma \|u\|^2_{L^2(\Omega)} \tag{7.30}$$

for all $u, v \in H_0^1(\Omega)$.

Proof To prove (7.29), it is enough to observe that if $u, v \in H_0^1(\Omega)$, then

$$|B(u, v)| \leq \sum_{i,j}^{N} \|a_{ij}\|_\infty \int_\Omega |\nabla u| \, |\nabla v| \, dx + \sum_{i=1}^{N} \|a_i\|_\infty \int_\Omega |\nabla u| \, |v| \, dx$$

$$+ \|a_0\|_\infty \int_\Omega |u| \, |v| \, dx \leq \alpha \|u\|_{H_0^1(\Omega)} \|v\|_{H_0^1(\Omega)}.$$

To prove (7.30), if $u \in H_0^1(\Omega)$, and taking into account the ellipticity condition (7.24), we have that

$$\theta \int_\Omega |\nabla u|^2 \, dx \leq \int_\Omega \sum_{i,j}^{N} a_{ij} \, D_i u D_j u \, dx$$

$$= B(u, u) - \int_\Omega \left(\sum_{i=1} a_i D_i u u + a_0 u^2 \right) dx$$

$$\leq B(u, u) + \sum_{i=1}^{N} \|a_i\|_\infty \int_\Omega |\nabla u| \, |u| \, dx + \|a_0\|_\infty \int_\Omega u^2 \, dx.$$

Now, given $\epsilon > 0$, applying Young's inequality (Lemma 2.3), we have

$$\int_\Omega |\nabla u| \, |u| \, dx \leq \epsilon \int_\Omega |\nabla u|^2 \, dx + \frac{1}{2\epsilon} \int_\Omega |u|^2 \, dx.$$

Then, choosing $\epsilon > 0$ such that

$$\epsilon \sum_{i=1}^{N} \|a_i\|_\infty < \frac{\theta}{2},$$

we obtain

$$\frac{\theta}{2}\int_\Omega |\nabla u|^2\,dx \le B(u,u) + C\int_\Omega u^2\,dx,$$

where C is a suitable constant. From here, taking into account Poincaré's inequality

$$\|u\|_{L^2(\Omega)} \le R\||\nabla u|\|_{L^2(\Omega)},$$

we easily obtain (7.30). □

Theorem 7.6 *Let $P(x,D)$ be a uniformly elliptic operator in Ω of the form (7.23). Then, there exists a number $\gamma \ge 0$ such that for each $\lambda \ge \gamma$ and each $f \in L^2(\Omega)$, the problem*

$$\begin{cases} P(x,D)u + \lambda u = f & \text{in } \Omega \\ \\ u = 0 & \text{on } \partial\Omega \end{cases} \tag{7.31}$$

has a unique solution $u \in H_0^1(\Omega)$.

Proof Let γ be the constant that appears in the energy inequality (7.30). For $\lambda \ge \gamma$, consider the bilinear form on $H_0^1(\Omega)$ defined by

$$B_\lambda(u,v) := B(u,v) + \lambda(u|v),$$

where B is the bilinear form associated with the operator $P(x,D)$, and $(\ |\ )$ denotes the inner product in $L^2(\Omega)$. Clearly, B_λ is the bilinear form associated with the operator $P_\lambda(x,D)u := P(x,D)u + \lambda u$. Now, as a consequence of the energy estimates (7.29) and (7.30), B_λ satisfies the hypotheses of the Lax–Milgram Theorem, which concludes the proof. □

As a consequence of the previous result, we have that if the differential operator has no first-order derivatives, i.e., if $a_i = 0$, then the corresponding Dirichlet problem has a unique weak solution. More precisely, we have the following result.

Corollary 7.1 *Suppose that the functions $a_{ij} \in L^\infty(\Omega)$ satisfy the ellipticity condition (7.24) and that $0 \le a_0 \in L^\infty(\Omega)$. Then, for each $f \in L^2(\Omega)$, the problem*

$$\begin{cases} -\displaystyle\sum_{i,j=1}^{N} D_i\big(a_{ij}(x)D_j u\big) + a_0(x)u = f & \text{in } \Omega \\ \\ u = 0 & \text{on } \partial\Omega \end{cases} \tag{7.32}$$

has a unique solution $u \in H_0^1(\Omega)$.

Proof It is enough to observe that in this case one can take the constant γ from the previous result to be zero. □

Remark 7.4 Note that the previous result is also valid if $f \in H^{-1}(\Omega)$. One simply needs to replace $\int_\Omega f v$ with $\langle f, v\rangle$, where $\langle \cdot, \cdot \rangle$ denotes the duality pairing between $H^{-1}(\Omega)$ and $H_0^1(\Omega)$.

As in the case of the Laplacian (Theorem 7.4), if we assume that the matrix $(a_{ij}(x))$ is symmetric, i.e., $a_{ij}(x) = a_{ji}(x)$ for each $i, j = 1, \ldots, N$ and for almost every $x \in \Omega$, then the unique weak solution u of problem (7.32) is the unique minimizer of the variational problem:

$$J(u) = \min_{v \in H_0^1(\Omega)} J(v),$$

where

$$J(v) = \frac{1}{2} \int_\Omega \left(a_{ij}(x) D_i v D_j v + a_0(x) v \right) dx - \int_\Omega f v.$$

Let us now consider the general problem (7.25). In some cases, the bilinear form B given by (7.27) is not $H_0^1(\Omega)$-elliptic, so the Lax–Milgram Theorem cannot be applied to solve problem (7.25). However, with the help of the Fredholm Alternative Theorem—applicable thanks to the compact embedding provided by the Rellich-Kondrakov Theorem, we have the following alternative.

Theorem 7.7 *Let $P(x, D)$ be a uniformly elliptic operator in Ω of the form (7.23). Then, one of the following alternatives holds:*

(i) For every $f \in L^2(\Omega)$, there exists a unique weak solution $u \in H_0^1(\Omega)$ of the problem

$$\begin{cases} P(x, D)u = f & \text{in } \Omega \\ \\ u = 0 & \text{on } \partial\Omega \end{cases} \tag{7.33}$$

(ii) There exists a nontrivial weak solution $u \neq 0$ of the homogeneous problem

$$\begin{cases} P(x, D)u = 0 & \text{in } \Omega \\ \\ u = 0 & \text{on } \partial\Omega. \end{cases} \tag{7.34}$$

Moreover, if alternative (ii) holds and we denote by N the subspace of $H_0^1(\Omega)$ consisting of weak solutions of (7.34), then this space is finite-dimensional, and

$$N^* := \{ v \in H_0^1(\Omega) \ : B(u, v) = 0 \quad \forall\, u \in H_0^1(\Omega) \},$$

N and N^* have the same dimension.

Finally, problem (7.33) has a weak solution if and only if

$$(f|v) = 0 \quad \textit{for all } v \in N^*.$$

Proof By Theorem 7.6, we know that there exists a $\gamma \geq 0$ such that, for every $g \in L^2(\Omega)$, there exists a unique function $u \in H_0^1(\Omega)$ solving the problem

$$B_\gamma(u, v) := B(u, v) + \gamma(u|v) = (g|v) \quad \forall\, v \in H_0^1(\Omega). \tag{7.35}$$

When (7.35) holds, we write

$$u = R(g).$$

Observe that $u \in H_0^1(\Omega)$ is a weak solution of (7.33) if and only if

$$B_\gamma(u, v) = (\gamma u + f|v) \quad \forall\, v \in H_0^1(\Omega), \tag{7.36}$$

which is clearly equivalent to

$$u = R(\gamma u + f). \tag{7.37}$$

Now, letting $K := \gamma R$, we can rewrite (7.37) as

$$u - K(u) = h, \quad \text{with } h = R(f). \tag{7.38}$$

We now show that the linear operator $K : L^2(\Omega) \to L^2(\Omega)$ is compact. Indeed, from the energy estimate (7.30), we have that if u satisfies (7.35), then

$$\beta \|u\|^2_{H_0^1(\Omega)} \leq B_\gamma(u, u) = (g|u) \leq \|g\|_2 \, \|u\|_2 \leq \|g\|_2 \, \|u\|_{H_0^1(\Omega)},$$

from which it follows that there exists a constant $C > 0$ such that

$$\|K(u)\|_{H_0^1(\Omega)} \leq C \|g\|_2 \quad \forall\, g \in L^2(\Omega).$$

Hence, using the Rellich–Kondrakov Theorem (Theorem 6.25), which ensures that the embedding of $H_0^1(\Omega)$ into $L^2(\Omega)$ is compact, we deduce that K is a compact operator. Then, applying the Fredholm Alternative Theorem (Theorem 4.14) to the operator K, we obtain the following alternatives:

(a) For every $h \in L^2(\Omega)$, the equation $u - K(u) = h$ has a unique solution $u \in H_0^1(\Omega)$,
(b) The equation $u - K(u) = 0$ has a nontrivial solution.

Now, from the above reasoning, (a) is equivalent to (i), and (b) is equivalent to (ii). Moreover, if (b) holds, the Fredholm Alternative Theorem also guarantees that

$\mathrm{Ker}(I-K)$ is finite-dimensional, which means that under alternative (ii), N is finite-dimensional. Furthermore, $\mathrm{Ker}(I-K)$ and $\mathrm{Ker}(I-K^*)$ have the same dimension. On the other hand, it is easy to see that

$$\mathrm{Ker}(I-K^*) = N^*.$$

Finally, the Fredholm Alternative Theorem also tells us that $\mathrm{Ran}(I-K) = [\mathrm{Ker}(I-K^*)]^\perp$, which implies that problem (7.33) has a solution if and only if $h = R(f) \in [\mathrm{Ker}(I-K^*)]^\perp$. Now, if $v \in \mathrm{Ker}(I-K^*)$, then

$$(h|v) = \frac{1}{\gamma}(K(f)|v) = \frac{1}{\gamma}(f|K^*(v)) = \frac{1}{\gamma}(f|v).$$

Consequently, problem (7.33) has a solution if and only if $f \in (N^*)^\perp$. □

Remark 7.5 As a consequence of the previous theorem, we have that if we can prove that problem (7.34) has the zero solution as its only weak solution, then for every $f \in L^2(\Omega)$, problem (7.34) has a unique weak solution. In the case where $a_0 \geq 0$ a.e. in Ω, we will see, as a consequence of the maximum principle, that this situation occurs.

The study carried out here for second-order operators can be extended to strongly elliptic operators of order $2m$. See, for example, the textbook by A. Friedman [19] or Dautray-Lions [12]. This study is strongly based on a result by Gårding, known as the *Gårding inequality*, which states that the bilinear form B associated with a strongly elliptic operator of order $2m$ is $H_0^m(\Omega)$-elliptic relative to $L^2(\Omega)$, i.e.,

$$B(u,v) + \alpha\|u\|_2^2 \geq \|u\|^2_{H_0^m(\Omega)} \quad \forall\, u \in H_0^m(\Omega).$$

The consideration of strongly elliptic operators of even order is due to the fact that it is known (see for example [2]) that strongly elliptic operators with real coefficients are of even order.

We now study the Neumann problem. Let $\Omega \subset \mathbb{R}^N$ be a bounded open set with smooth boundary $\partial\Omega$, and let η denote the unit outward normal to $\partial\Omega$. For $f \in L^2(\Omega)$, consider the *Neumann problem*:

$$\begin{cases} -\Delta u + u = f & \text{in } \Omega \\ \dfrac{\partial u}{\partial \eta} = 0 & \text{on } \partial\Omega \end{cases} \tag{7.39}$$

Let $u \in C^2(\overline{\Omega})$ be a classical solution of (7.39). Then, if $v \in C^1(\overline{\Omega})$, multiplying the equation in (7.39) by v and integrating by parts yields:

$$\int_\Omega \nabla u \cdot \nabla v + \int_\Omega uv = \int_\Omega fv.$$

Now, by density, the above equality holds for every $v \in H^1(\Omega)$. This leads to the following definition of weak solution.

A function $u \in H^1(\Omega)$ is called a *weak solution* of the Neumann problem (7.39) if it satisfies:

$$\int_\Omega \nabla u \cdot \nabla v + \int_\Omega uv = \int_\Omega fv \quad \forall\, v \in H^1(\Omega). \tag{7.40}$$

From the above, we see that every classical solution of problem (7.39) is also a weak solution. Conversely, suppose that $u \in H^2(\Omega)$ is a weak solution of problem (7.39). Then, applying the integration by parts formula (6.88) we get:

$$-\int_\Omega v\,\Delta u + \left\langle \frac{\partial u}{\partial \eta}, \gamma(v) \right\rangle_{\partial\Omega} + \int_\Omega uv = \int_\Omega fv \quad \forall\, v \in H^1(\Omega). \tag{7.41}$$

Taking $v \in \mathcal{D}(\Omega)$ in (7.41), it follows that:

$$-\Delta u + u = f \quad \text{in } \mathcal{D}'(\Omega).$$

Then, since $f \in L^2(\Omega)$, we have:

$$-\Delta u + u = f \quad \text{a.e. in } \Omega.$$

Therefore,

$$\left\langle \frac{\partial u}{\partial \eta}, \gamma(v) \right\rangle_{\partial\Omega} = 0 \quad \forall\, v \in H^1(\Omega).$$

From this, since $\gamma(v) \in H^{1/2}(\partial\Omega)$ and $H^{1/2}(\partial\Omega)$ is dense in $L^2(\partial\Omega)$, we conclude that:

$$\frac{\partial u}{\partial \eta} = 0 \quad \text{in } L^2(\partial\Omega).$$

Consequently, if we assume that the weak solution u of problem (7.39) is in $C^2(\overline{\Omega})$, then u is a classical solution of problem (7.39).

Let π be the bilinear form on $H^1(\Omega)$ associated with problem (7.39), i.e.,

$$\pi(u, v) := \int_\Omega \nabla u \cdot \nabla v + \int_\Omega uv, \quad u, v \in H^1(\Omega).$$

Then we have:

$$\pi(u, u) = \|u\|^2_{H^1(\Omega)} \quad \forall\, u \in H^1(\Omega).$$

Therefore, π is $H^1(\Omega)$-elliptic, and applying the Lax-Milgram Theorem, problem (7.39) has a unique weak solution. Moreover, this solution is the element $u \in H^1(\Omega)$ that minimizes the energy functional:

$$J(v) := \frac{1}{2}\left(\int_\Omega |\nabla v|^2 + \int_\Omega v^2\right) - \int_\Omega fv.$$

We thus obtain the following result.

Theorem 7.8 *Let $\Omega \subset \mathbb{R}^N$ be a bounded open set of class C^1 and let $f \in L^2(\Omega)$. Then, there exists a unique weak solution $u \in H^1(\Omega)$ to problem (7.39). Moreover, u is given by*

$$J(u) = \min_{v \in H^1(\Omega)} J(v),$$

where

$$J(v) := \frac{1}{2}\left(\int_\Omega |\nabla v|^2 + \int_\Omega v^2\right) - \int_\Omega fv.$$

Nonhomogeneous Neumann problems of the form

$$\begin{cases} -\Delta u + u = f & \text{in } \Omega \\ \\ \dfrac{\partial u}{\partial \eta} = g & \text{on } \partial\Omega. \end{cases} \tag{7.42}$$

can also be studied using variational methods. In this case, assuming that $f \in L^2(\Omega)$ and $g \in H^{-1/2}(\partial\Omega)$, a function $u \in H^1(\Omega)$ is said to be a *weak solution* of problem (7.42) if it satisfies

$$\int_\Omega \nabla u \cdot \nabla v + \int_\Omega uv = \int_\Omega fv - \langle g, v\rangle_{\partial\Omega} \qquad \forall\, v \in H^1(\Omega),$$

where $\langle \cdot, \cdot \rangle_{\partial\Omega}$ denotes the duality pairing between $H^{-1/2}(\partial\Omega)$ and $H^{1/2}(\partial\Omega)$. Using the techniques above, one can prove that problem (7.42) has a unique solution (see Problem 7.4).

Elliptic problems with mixed boundary conditions (see Problem 7.6) or with *Robin boundary condition* (see Problem 7.5), i.e., the condition $\frac{\partial u}{\partial \eta} + \alpha u = 0$ on $\partial\Omega$ with $\alpha > 0$, can also be studied via variational methods.

As we did for the Dirichlet problem, one can study the Neumann problem for general uniformly elliptic operators in dimension 2, i.e., for operators of the form (7.23) that satisfy condition (7.24). In this case, the boundary condition takes the form

$$\sum_{i,j=1}^{N} a_{ij} \frac{\partial u}{\partial x_j} \eta_i = 0,$$

where $\eta = (\eta_1, \ldots, \eta_N)$ is the unit outer normal to $\partial\Omega$.

So far, we have studied elliptic boundary value problems in bounded open sets. In the case where the domain is unbounded, a similar analysis can be carried out, but the boundary condition must be replaced by a growth condition at infinity. For example, the condition $u \to 0$ as $\|x\| \to +\infty$ corresponds in the weak formulation to the requirement that $u \in H^1(\mathbb{R}^N)$. Consider, for example, the problem

$$\begin{cases} -\Delta u + u = f & \text{in } \mathbb{R}^N \\ u \to 0 & \text{as } \|x\| \to \infty. \end{cases} \tag{7.43}$$

Suppose that u is a classical solution of problem (7.43). Then $u \in C^2(\mathbb{R}^N)$ and $\lim_{\|x\|\to\infty} u(x) = 0$. Therefore, if we multiply the equation by a function $\varphi \in \mathcal{D}(\mathbb{R}^N)$ and integrate by parts, we obtain

$$\int_{\mathbb{R}^N} f\varphi = \int_{\mathbb{R}^N} \nabla u \cdot \nabla \varphi + \int_{\mathbb{R}^N} u\varphi.$$

From this, taking into account the density of $\mathcal{D}(\mathbb{R}^N)$ in $H^1(\mathbb{R}^N)$, we obtain that u satisfies:

$$\int_{\mathbb{R}^N} fv = \int_{\mathbb{R}^N} \nabla u \cdot \nabla v + \int_{\mathbb{R}^N} uv \qquad \forall\, v \in H^1(\mathbb{R}^N). \tag{7.44}$$

A function $u \in H^1(\mathbb{R}^N)$ that satisfies (7.44) is called a *weak solution* of problem (7.43). Thus, every classical solution of problem (7.43) is a weak solution. Moreover, since the bilinear form

$$B(u, v) := \int_{\mathbb{R}^N} \nabla u \cdot \nabla v + \int_{\mathbb{R}^N} uv, \qquad u, v \in H^1(\mathbb{R}^N)$$

is clearly $H^1(\mathbb{R}^N)$-elliptic, applying the Lax–Milgram Theorem, we conclude that for every $f \in L^2(\mathbb{R}^N)$, problem (7.43) has a unique classical solution.

To conclude this section, we are going to study an obstacle problem using variational inequalities. Let us consider an elastic membrane that at rest occupies an open set $\Omega \subset \mathbb{R}^2$ and is fixed on the boundary $\partial\Omega$. If a certain obstacle pushes

it from below, the equilibrium position u minimizes the elastic energy given by the Dirichlet integral

$$J(v) = \frac{1}{2}\int_\Omega |\nabla v|^2\, dx.$$

Now, if ϕ is the function in Ω describing the obstacle—function which obviously must satisfy $\phi \le 0$ on $\partial\Omega$—the only admissible deformations are those satisfying $v \ge \phi$ in Ω. Therefore, if we denote

$$K = \{v : \Omega \to \mathbb{R} \ : \ v = 0 \ \text{ on } \partial\Omega, \ v \ge \phi \ \text{ in } \Omega\},$$

the equilibrium position of the membrane under the tension exerted by the obstacle ϕ is the function u that solves the variational problem

$$J(u) = \min_{v\in K} J(v).$$

To rigorously treat this problem, we shall formulate it in the context of Sobolev spaces, and we shall do so in the N-dimensional case. Thus, we assume that Ω is a bounded open subset of $\mathbb{R}^N$. Recall that if $\phi \in H^1(\Omega)$, then the functions

$$\phi^+ = \max\{\phi, 0\}, \quad \text{and} \quad \phi^- = \max\{-\phi, 0\}$$

also belong to $H^1(\Omega)$. When we say that a function in $H^1(\Omega)$ is less than or equal to zero on the boundary of Ω, we mean the following. Let $\phi \in H^1(\Omega)$; we say that

$$\phi \le 0 \quad \text{ on } \partial\Omega$$

if and only if $\phi^+ \in H_0^1(\Omega)$.

For $\phi \in H^1(\Omega)$ with $\phi \le 0$ on $\partial\Omega$, consider the set

$$K_\phi := \left\{ v \in H_0^1(\Omega) \ : \ v \ge \phi \text{ a.e. in } \Omega \right\}.$$

Theorem 7.9 *Let $\Omega \subset \mathbb{R}^N$ be a bounded open set and let $\phi \in H^1(\Omega)$, with $\phi \le 0$ on $\partial\Omega$. Then there exists a unique solution u of the variational inequality*

$$\begin{cases} u \in K_\phi \\ \displaystyle\int_\Omega \nabla u \cdot \nabla(v-u)\, dx \ge 0 \quad \forall\, v \in K_\phi. \end{cases} \tag{7.45}$$

Moreover, u is the unique solution of the variational problem

$$J(u) = \min_{v\in K_\phi} J(v), \tag{7.46}$$

with

$$J(v) = \frac{1}{2}\int_\Omega |\nabla v|^2\, dx.$$

Proof It is clear that it suffices to verify that the bilinear form

$$B(u, v) = \int_\Omega \nabla u \cdot \nabla v\, dx$$

and the set K_ϕ satisfy the hypotheses of Stampacchia's Theorem (Theorem 7.1) in the Hilbert space $H_0^1(\Omega)$. Now, according to the proof of Theorem 7.4, it is enough to show that the set K_ϕ is a nonempty closed convex set. Clearly, K_ϕ is convex, and since $\phi \leq 0$ on $\partial\Omega$, we have $\phi^+ \in K_\phi$. Let us now show that K_ϕ is closed in $H_0^1(\Omega)$. Let $v_n \in K_\phi$ such that

$$v_n \to v \quad \text{in} \quad H_0^1(\Omega). \tag{7.47}$$

We must show that $v \in K_\phi$. From (7.47) it follows that

$$v_n \to v \quad \text{in} \quad L^2(\Omega),$$

so, extracting a subsequence if necessary, we may assume there exists a set $A \subset \Omega$ of Lebesgue measure zero in $\mathbb{R}^N$ such that

$$v_n(x) \to v(x) \quad \forall\, x \in \Omega \setminus A. \tag{7.48}$$

On the other hand, since each $v_n \in K_\phi$, for every $n \in \mathbb{N}$ there exists a null set $A_n \subset \Omega$ such that

$$v_n(x) \geq \phi(x) \quad \forall\, x \in \Omega \setminus A_n. \tag{7.49}$$

Then, if

$$B := \left(\bigcup_{n\in\mathbb{N}} A_n\right) \cup A,$$

we have that B is a null subset of Ω, and from (7.48) and (7.49) it follows that

$$v(x) \geq \phi(x) \quad \forall\, x \in \Omega \setminus B,$$

which implies that $v \in K_\phi$, and the proof is complete. □

Remark 7.6 If u is the solution of the obstacle problem (7.45), the set

$$C_\phi := \{x \in \Omega \ : u(x) = \phi(x)\}$$

is called the *coincidence set*, and its boundary, which like u is one of the unknowns in the obstacle problem, is called the *free boundary*. Obstacle problems are among the classical free boundary problems, which appear in many applied mathematics contexts (see, for example, [10, 34, 44]).

Let us now determine the Euler–Lagrange equation associated with the variational problem (7.46).

Theorem 7.10 *Suppose we are under the hypotheses of Theorem 7.9. Then, if u is the solution of problem (7.45), we have*

$$-\Delta u \geq 0 \ \text{ in } \ \Omega, \ \ i.e. \ \int_\Omega \nabla u \cdot \nabla \varphi \, dx \geq 0 \ \ \forall\, 0 \leq \varphi \in \mathcal{D}(\Omega). \tag{7.50}$$

Moreover, if $u - \phi$ is continuous in Ω, then the set

$$\Omega_\phi := \{x \in \Omega \ : u(x) - \phi(x) > 0\}$$

is open, and therefore

$$-\Delta u = 0 \quad \text{in } \ \mathcal{D}'(\Omega_\phi). \tag{7.51}$$

Proof If $0 \leq \varphi \in \mathcal{D}(\Omega)$, then $v = u + \varphi \in K_\phi$, and by taking this test function in (7.45), we obtain

$$\int_\Omega \nabla u \cdot \nabla \varphi \, dx \geq 0,$$

which proves (7.50). To prove (7.51), given a function $\varphi \in \mathcal{D}(\Omega_\phi)$, we take $\epsilon > 0$ sufficiently small so that

$$u \pm \epsilon\varphi - \phi > 0.$$

Then $u \pm \epsilon\varphi \in K_\phi$, so taking these functions as test functions in (7.45) we obtain

$$\int_{\Omega_\phi} \nabla u \cdot \nabla \varphi = 0,$$

which proves (7.51). □

Remark 7.7 The hypothesis that $u - \phi$ is continuous in Ω holds in the case where ϕ is continuous, since then, by regularity results given in [34], it follows that $u \in W^{2,\infty}(\Omega)$.

7.4 Regularity of Weak Solutions

In the previous section, we saw that weak solutions of elliptic boundary value problems belong to Sobolev spaces of order lower than the order of the differential operator. We also saw that when weak solutions are sufficiently smooth, they are classical solutions. In this section, we will study the regularity problem, i.e., how smooth the weak solutions are. The problem consists in examining whether weak solutions belong to Sobolev spaces of higher order, since in that case, the continuous embedding theorems studied in the chapter on Sobolev spaces provide the smoothness of such solutions. The smoothness of weak solutions depends on two factors: the smoothness of the initial data and the smoothness of the domain boundary.

We will restrict ourselves to studying the regularity of solutions to the Dirichlet problem for the equation $-\Delta u + u = f$. Similar results hold for uniformly elliptic second-order operators (see [15]).

We begin with the case $\Omega = \mathbb{R}^N$.

Theorem 7.11 *Let $u \in H^1(\mathbb{R}^N)$ such that*

$$-\Delta u + u = f \in L^2(\mathbb{R}^N)$$

in the distributional sense. Then $u \in H^2(\mathbb{R}^N)$ and there exists a constant $C > 0$, independent of f, such that

$$\|u\|_{H^2(\mathbb{R}^N)} \leq C \|f\|_2. \tag{7.52}$$

Moreover, if $f \in H^m(\mathbb{R}^N)$ with $m \in \mathbb{N}$, then $u \in H^{m+2}(\mathbb{R}^N)$ and

$$\|u\|_{H^{m+2}(\mathbb{R}^N)} \leq C \|f\|_{H^m(\mathbb{R}^N)}. \tag{7.53}$$

Proof Since $u,\ \Delta u \in L^2(\mathbb{R}^N)$, applying Plancherel's Theorem (Theorem 5.20), we have that $\hat{u} \in L^2(\mathbb{R}^N)$ and

$$\widehat{\Delta u} = \hat{u} - \hat{f} \in L^2(\mathbb{R}^N).$$

Now, $\widehat{\Delta u} = -\|\xi\|^2 \hat{u}$. Therefore, $(1 + \|\xi\|^2)\hat{u} \in L^2(\mathbb{R}^N)$, and as a consequence of Theorem 6.5, it follows that $u \in H^2(\mathbb{R}^N)$. Moreover, using Parseval's identity, we have

$$\|u\|^2_{H^2(\mathbb{R}^N)} = \int_{\mathbb{R}^N} (1 + \|\xi\|^2)^2 |\hat{u}(\xi)|^2 \, d\xi = \int_{\mathbb{R}^N} |\hat{u}(\xi) - \widehat{\Delta u}(\xi)|^2 \, d\xi$$

$$= \int_{\mathbb{R}^N} |\hat{f}(\xi)|^2 \, d\xi = (2\pi)^N \int_{\mathbb{R}^N} |f(\xi)|^2 \, d\xi,$$

from which we deduce (7.52).

Now suppose that $f \in H^1(\mathbb{R}^N)$. Then, for each $i = 1, \dots, N$

$$-\Delta\left(\frac{\partial u}{\partial x_i}\right) + \frac{\partial u}{\partial x_i} = \frac{\partial f}{\partial x_i} \in L^2(\mathbb{R}^N)$$

in the distributional sense. Therefore, applying the previous result, we obtain $\frac{\partial u}{\partial x_i} \in H^2(\mathbb{R}^N)$ for each $i = 1, \dots, N$, so $u \in H^3(\mathbb{R}^N)$. The general case follows by induction. □

Let us now consider the case $\Omega = \mathbb{R}^N_+ = \{(x_1, \dots, x_N) \in \mathbb{R}^N \ : \ x_N > 0\}$.

Theorem 7.12 *Let $u \in H^1_0(\mathbb{R}^N_+)$ and $f \in L^2(\mathbb{R}^N_+)$ be such that*

$$\int_{\mathbb{R}^N_+} \nabla u \cdot \nabla v \, dx + \int_{\mathbb{R}^N_+} uv \, dx = \int_{\mathbb{R}^N_+} f v \, dx \quad \forall\, v \in H^1_0(\mathbb{R}^N_+).$$

Then $u \in H^2(\mathbb{R}^N_+)$ and there exists a constant $C > 0$, independent of f, such that

$$\|u\|_{H^2(\mathbb{R}^N_+)} \le C \, \|f\|_2. \tag{7.54}$$

Moreover, if $f \in H^m(\mathbb{R}^N_+)$ with $m \in \mathbb{N}$, then $u \in H^{m+2}(\mathbb{R}^N_+)$ and

$$\|u\|_{H^{m+2}(\mathbb{R}^N_+)} \le C \, \|f\|_{H^m(\mathbb{R}^N_+)}. \tag{7.55}$$

Proof Given $v \in H^1_0(\mathbb{R}^N_+)$, we define by reflection

$$\tilde{v}(x', x_N) := \begin{cases} v(x', x_N), & \text{if } x_N > 0 \\ \\ -v(x', -x_N), & \text{if } x_N < 0. \end{cases}$$

Since $u \in H^1_0(\mathbb{R}^N_+)$, we know that $\tilde{u} \in H^1(\mathbb{R}^N)$. Let us show that

$$-\Delta\tilde{u} + \tilde{u} = \tilde{f} \quad \text{in } \mathcal{D}'(\mathbb{R}^N). \tag{7.56}$$

Given $\phi \in \mathcal{D}(\mathbb{R}^N)$, we have

$$-\int_{\mathbb{R}^N} \tilde{u} \, \Delta\phi + \int_{\mathbb{R}^N} \tilde{u}\phi$$

$$= -\int_{\mathbb{R}^N_+} u(x', x_N)\Delta\phi(x', x_N) + \int_{\mathbb{R}^N_-} u(x', -x_N)\Delta\phi(x', x_N)$$

$$+\int_{\mathbb{R}^N_+} u(x', x_N)\phi(x', x_N) - \int_{\mathbb{R}^N_-} u(x', -x_N)\phi(x', x_N)$$

$$= -\int_{\mathbb{R}^N_+} u\Delta\varphi + \int_{\mathbb{R}^N_+} u\varphi,$$

where

$$\varphi(x', x_N) = \phi(x', x_N) - \phi(x', -x_N), \quad x_N > 0.$$

Now, since $\varphi(x', 0) = 0$, we have that $\varphi \in H_0^1(\mathbb{R}^N_+)$. Therefore,

$$\int_{\mathbb{R}^N_+} u\Delta\varphi + \int_{\mathbb{R}^N_+} u\varphi = \int_{\mathbb{R}^N_+} \nabla u \cdot \nabla\varphi + \int_{\mathbb{R}^N_+} u\varphi = \int_{\mathbb{R}^N_+} f\varphi = \int_{\mathbb{R}^N} \tilde{f}\phi,$$

and consequently

$$-\int_{\mathbb{R}^N} \tilde{u}\,\Delta\phi + \int_{\mathbb{R}^N} \tilde{u}\phi = \int_{\mathbb{R}^N} \tilde{f}\phi,$$

which proves (7.56). Then, applying the previous theorem we obtain that $\tilde{u} \in H^2(\mathbb{R}^N)$ and

$$\|u\|_{H^2(\mathbb{R}^N_+)} \le \|\tilde{u}\|_{H^2(\mathbb{R}^N)} \le C\|\tilde{f}\|_{L^2(\mathbb{R}^N)} \le 2C\|f\|_{L^2(\mathbb{R}^N_+)},$$

which proves (7.54).

Just as in the proof of the previous theorem, if $f \in H^1(\mathbb{R}^N_+)$, then for each $i = 1, \ldots, N-1$, we have

$$-\Delta\left(\frac{\partial u}{\partial x_i}\right) + \frac{\partial u}{\partial x_i} = \frac{\partial f}{\partial x_i} \in L^2(\mathbb{R}^N_+)$$

and

$$\frac{\partial u}{\partial x_i} \in H_0^1(\mathbb{R}^N_+).$$

Therefore,

$$\frac{\partial u}{\partial x_i} \in H^2(\mathbb{R}^N_+), \quad \text{for } i = 1, \ldots, N-1.$$

In particular,

$$\frac{\partial^2 u}{\partial x_i \partial x_j} \in H^1(\mathbb{R}^N_+), \quad \text{for } i = 1, \ldots, N-1,\ j = 1, \ldots, N.$$

Now,

$$\frac{\partial^2 u}{\partial x_N^2} = u - f - \sum_{i=1}^{N-1} \frac{\partial^2 u}{\partial x_i^2} \in H^1(\mathbb{R}^N_+).$$

Thus, all second partial derivatives belong to $H^1(\mathbb{R}^N_+)$, and so $u \in H^3(\mathbb{R}^N_+)$. The general case follows by induction. □

For a bounded open set, we have the following regularity theorem.

Theorem 7.13 *Let $\Omega \subset \mathbb{R}^N$ be a bounded open set with boundary of class C^2, and let $f \in L^2(\Omega)$. If $u \in H_0^1(\Omega)$ is a weak solution to the Dirichlet problem*

$$\begin{cases} -\Delta u + u = f \text{ in } \Omega \\ u = 0 \qquad\quad \text{on } \partial\Omega, \end{cases}$$

then $u \in H^2(\Omega)$ and there exists a constant $C > 0$, depending only on Ω, such that

$$\|u\|_{H^2(\Omega)} \leq C \|f\|_{L^2(\Omega)}.$$

Moreover, if Ω is of class C^{m+2} and $f \in H^m(\Omega)$, then $u \in H^{m+2}(\Omega)$ and

$$\|u\|_{H^{m+2}(\Omega)} \leq C \|f\|_{H^m(\Omega)}.$$

In particular, if $m > \frac{N}{2}$, then $u \in C^2(\overline{\Omega})$, so u is a classical solution.

Sketch of the Proof We will only outline the main ideas of the proof of this result. For a detailed proof, see [7]. The proof is quite technical and is divided into two parts: regularity in the interior and at the boundary.

We take a family of local charts $\{V_j, \Phi_j\}_{j=1}^k$ associated with the boundary of Ω, and, associated with the covering $\{V_j\}_{j=0}^k$, with $V_0 = \mathbb{R}^N \setminus \partial\Omega$, we consider a partition of unity $\{\psi_j\}_{j=0}^k$. Then we write

$$u = \sum_{j=0}^{k} \psi_j u = \sum_{j=0}^{k} u_j.$$

We first prove the regularity in the interior, i.e., that $u_0 \in H^2(\Omega)$. Since $\text{supp}(\psi_0) \subset \mathbb{R}^N \setminus \partial\Omega$, define

$$\overline{u}_0 := \begin{cases} \psi_0(x)u(x), & \text{if } x \in \Omega, \\ 0, & \text{if } x \notin \Omega, \end{cases}$$

then $\overline{u}_0 \in H^1(\mathbb{R}^N)$. We will show that

$$-\Delta\overline{u}_0+\overline{u}_0 = \psi_0 f-2\nabla\psi_0\cdot\nabla u-(\Delta\psi_0)u =: g_0 \in L^2(\mathbb{R}^N) \quad \text{in } \mathcal{D}'(\mathbb{R}^N). \tag{7.57}$$

Indeed, given $\phi \in \mathcal{D}(\mathbb{R}^N)$, since $\phi\psi_0 \in \mathcal{D}(\Omega)$, applying Green's formula we obtain:

$$\begin{aligned}
\int_{\mathbb{R}^N} g_0\phi &= \int_\Omega f(\psi_0\phi) - 2\int_\Omega (\nabla\psi_0\cdot\nabla u)\phi - \int_\Omega (\Delta\psi_0)u\phi \\
&= \int_\Omega \nabla u\cdot\nabla(\psi_0\phi) + \int_\Omega u\psi_0\phi - \int_\Omega (\nabla\psi_0\cdot\nabla u)\phi + \int_\Omega (\nabla\psi_0\cdot\nabla\phi)u \\
&= \int_\Omega u\psi_0\phi + \int_\Omega (\nabla u\cdot\nabla\phi)\psi_0 + \int_\Omega (\nabla\psi_0\cdot\nabla\phi)u \\
&= \int_{\mathbb{R}^N} \overline{u}_0\phi + \int_{\mathbb{R}^N} \nabla\overline{u}_0\cdot\nabla\phi = \int_{\mathbb{R}^N} (-\Delta\overline{u}_0+\overline{u}_0)\phi,
\end{aligned}$$

which proves (7.57). By applying Theorem 7.11, it follows that $\overline{u}_0 \in H^2(\mathbb{R}^N)$, and thus $u_0 \in H^2(\Omega)$. Furthermore,

$$\|u_0\|_{H^2(\Omega)} \le \|\overline{u}_0\|_{H^2(\mathbb{R}^N)} \le \|g_0\|_{L^2(\mathbb{R}^N)} \le C\left(\|u\|_{H^1(\Omega)} + \|f\|_{L^2(\Omega)}\right).$$

Now, since

$$\int_\Omega \nabla u\cdot\nabla v + \int_\Omega uv = \int_\Omega fv \quad \forall v \in H_0^1(\Omega),$$

taking $v = u$ we get

$$\int_\Omega |\nabla u|^2 + \int_\Omega u^2 = \int_\Omega fu,$$

which implies

$$\|u\|_{H^1(\Omega)} \le \|f\|_{L^2(\Omega)}.$$

Thus,

$$\|u_0\|_{H^2(\Omega)} \le C\|f\|_{L^2(\Omega)}.$$

This proves the interior regularity. It remains to prove the boundary regularity, i.e., that $u_i \in H^2(\Omega)$ for $i = 1,\dots,k$. This part is more technical. The functions $u_i \in H_0^1(\Omega\cap V_i)$ satisfy

$$-\Delta u_i = \psi_i f - \psi_i u - 2\nabla\psi_i\cdot\nabla u - (\Delta\psi_i)u =: g_i \in L^2(\Omega\cap V_i) \quad \text{in } \mathcal{D}'(\mathbb{R}^N).$$

Using the maps Φ_i^{-1}, this equation is transported to the domain

$$Q_+ = \left\{(x', x_N) : |x'| < 1,\ 0 < x_N < 1\right\}.$$

Unfortunately, under this change, the operator $-\Delta$ becomes a second-order elliptic operator with variable coefficients, so we cannot directly apply Theorem 7.12, but rather an adapted version for such operators. After applying this result, we return to the domain $\Omega \cap V_i$ using Φ_i. □

7.5 The Maximum Principle

One of the most characteristic properties of second-order elliptic operators is the so-called *maximum principle*.

Theorem 7.14 (Strong Maximum Principle) *Let $\Omega \subset \mathbb{R}^N$ be a bounded domain. If $u \in C^2(\Omega) \cap C(\overline{\Omega})$ satisfies $\Delta u \leq 0$ in Ω (that is, u is subharmonic in Ω then is constant or*

$$u(x) > \inf_{\partial\Omega} u \qquad \forall\, x \in \Omega. \tag{7.58}$$

Proof Let $m = \inf_{\partial\Omega} u$. We write $\Omega = \Omega_1 \cup \Omega_2$, where

$$\Omega_1 := \{x \in \Omega \ :\ u(x) = m\} \quad \text{and} \quad \Omega_2 := \{x \in \Omega \ :\ u(x) > m\}.$$

Since u is continuous, we have that Ω_2 is open. We will show that Ω_1 is also open. In that case, since Ω is connected, we must have either $\Omega = \Omega_1$, in which case u is constant, or $\Omega = \Omega_2$, in which case the inequality (7.58) holds and the result is proven.

Let $x \in \Omega_1$, and let $r > 0$ be such that $B(x, r) \subset \Omega$. Then, since $\Delta u \leq 0$ in Ω, applying Problem 5.45, we have

$$m = u(x) \geq \frac{1}{\sigma_N r^{N-1}} \int_{\|x-y\|=r} u(y)\, dS(y),$$

which implies

$$\int_{\|x-y\|=r} (u(y) - m)\, dS(y) = 0.$$

Hence, $u(y) = m$ for all $\|x - y\| = r$. Since this argument can be repeated for every $0 \leq \rho < r$, it follows that $u(y) = m$ for all $y \in B(x, r)$, and consequently Ω_1 is open. □

The strong maximum principle also holds for uniformly elliptic second-order differential operators. This result is due to Hopf, and a proof can be found, for example, in [15] or [21].

We now state a maximum principle for weak solutions of strongly elliptic second-order operators in divergence form.

Theorem 7.15 (Weak Maximum Principle) *Let $\Omega \subset \mathbb{R}^N$ be an open set of class C^1. Suppose that the functions $a_{ij} \in L^\infty(\Omega)$ satisfy the ellipticity condition (7.24), and that $0 \le a_i \in L^\infty(\Omega)$, for $i = 0, 1, \ldots, N$. Let $f \in L^2(\Omega)$ and $u \in H^1(\Omega) \cap C(\overline{\Omega})$ such that*

$$\sum_{i,j=1}^{N} \int_\Omega a_{ij}(x) D_j u \, D_i v + \sum_{i=1}^{N} \int_\Omega a_i(x)\, u D_i v + \int_\Omega a_0 uv = \int_\Omega fv \tag{7.59}$$

for every $v \in H_0^1(\Omega)$. Then the following holds:

(i) If $f \ge 0$ in Ω and $u \ge 0$ on $\partial\Omega$, then $u \ge 0$ in Ω.
(ii) Assuming $a_0 = 0$, we have

$$f \ge 0 \ \text{ in } \ \Omega \ \Rightarrow \ u \ge \inf_{\partial\Omega} u \ \text{ in } \ \Omega,$$

$$f = 0 \ \text{ in } \ \Omega \ \Rightarrow \ \inf_{\partial\Omega} u \le u \le \sup_{\partial\Omega} u \ \text{ in } \ \Omega,$$

Proof We will prove the result for the case in which $a_i = 0$ for $i = 1, \ldots, N$. The general case is technically more involved; for a proof, see [21].
(i) Since $u \ge 0$ on $\partial\Omega$, we have $u^- = 0$ on $\partial\Omega$, hence $u^- \in H_0^1(\Omega)$. Therefore,

$$\sum_{i,j=1}^{N} \int_\Omega a_{ij}(x) D_j u \, D_i u^- + \int_\Omega a_0 u u^- = \int_\Omega f u^-.$$

Now, since $u = u^+ - u^-$, $\operatorname{sop} u^+ \cap \operatorname{sop} u^- = \{x \in \Omega \,:\, u(x) = 0\}$, and on this set $D_i u^+$ and $D_i u^-$ are zero almost everywhere (see Theorem 6.2), from the above we obtain

$$-\sum_{i,j=1}^{N} \int_\Omega a_{ij}(x) D_j u^- \, D_i u^- = \int_\Omega a_0 (u^-)^2 + \int_\Omega f u^- \ge 0.$$

On the other hand, by the ellipticity condition (7.24) and the Poincaré inequality, we have

$$\sum_{i,j=1}^{N} \int_{\Omega} a_{ij}(x) D_j u^- \, D_i u^- \geq \alpha \int_{\Omega} \sum_{i=1}^{N} (D_i u^-)^2 \geq M \|u^-\|_{H_0^1(\Omega)}.$$

Thus, $\|u^-\|_{H_0^1(\Omega)} = 0$, so $u^- = 0$, and consequently $u = u^+ \geq 0$ in Ω.
(ii) Assume $a_0 = 0$ and $f \geq 0$ in Ω. Let

$$m = \inf_{\partial\Omega} u.$$

Since $a_0 = 0$, the function $u - m$ also satisfies (7.59). Then, as $u - m \geq 0$ on $\partial\Omega$, applying (i) yields that $u - m \geq 0$ in Ω, i.e.,

$$u \geq \inf_{\partial\Omega} u \ \text{ in } \ \Omega.$$

Finally, if $f = 0$, then $-u$ also satisfies (7.59), and applying the previous result to $-u$ gives

$$u \leq \sup_{\partial\Omega} u \ \text{ in } \ \Omega.$$

which completes the proof of the theorem. □

Remark 7.8

(i) As a consequence of the regularity theorems and the continuous Sobolev Immersion Theorems, hypothesis $u \in C(\overline{\Omega})$ is satisfied by weak solutions in many cases. For example, if Ω is an open set of classes C^2, we know that if $f \in L^2(\Omega)$, then $u \in H^2(\Omega)$. Hence, if $N \leq 3$, we have $u \in C(\overline{\Omega})$ as a consequence of Theorem 6.20.

(ii) If $a_0 \geq 0$, as a consequence of the previous theorem we have that if $f = 0$ and $u = 0$ on $\partial\Omega$, then $u = 0$ in Ω. Therefore, in this case, applying Theorem 7.7, we have uniqueness of weak solutions to the problem

$$\begin{cases} -\displaystyle\sum_{i,j=1}^{N} D_i(a_{ij}(x) D_j u) + \sum_{i=1}^{N} a_i(x) D_i u + a_0(x) u = f & \text{in } \Omega \\ \\ u = 0 \quad \text{on } \partial\Omega. \end{cases}$$

To finish this section, we will see that a weak maximum principle can be given without assuming the continuity of the solution. For this, we need to introduce the concept of positivity in the sense of $H^1(\Omega)$.

Definition 7.3 Suppose that $\Omega \subset \mathbb{R}^N$ is a bounded domain. Let $u \in H^1(\Omega)$ and $A \subset \overline{\Omega}$. We say that the function u is *nonnegative on A in the sense of $H^1(\Omega)$*, and we write $u \geq 0$ on A in $H^1(\Omega)$, if there exists a sequence $u_n \in W^{1,\infty}(\Omega)$ such that $u_n \geq 0$ on A, and $u_n \to u$ in $H^1(\Omega)$.

We write:

$$u \leq 0 \quad \text{on } A \text{ in } H^1(\Omega) \quad \Longleftrightarrow \quad -u \geq 0 \quad \text{on } A \text{ in } H^1(\Omega).$$

$$u \leq v \quad \text{on } A \text{ in } H^1(\Omega) \quad \Longleftrightarrow \quad v - u \geq 0 \quad \text{on } A \text{ in } H^1(\Omega).$$

Since v can be constant, we can give the following definition:

$$\sup_A u := \inf\{k \in \mathbb{R} : u \leq k \text{ on } A \text{ in } H^1(\Omega)\}.$$

Since for a fixed $A \subset \overline{\Omega}$, the set

$$\{u \in H^1(\Omega) : u \geq 0 \text{ on } A \text{ in } H^1(\Omega)\}$$

is a closed convex set, in the definition of $u \geq 0$ on A in $H^1(\Omega)$ it is sufficient to choose the sequence $u_n \rightharpoonup u$ weakly.

The following relation holds between $\geq$ in $(H^1(\Omega))$ and $\geq$ almost everywhere in Ω.

Proposition 7.1 *Let $\Omega \subset \mathbb{R}^N$ be a bounded domain, $u \in H^1(\Omega)$, and $A \subset \overline{\Omega}$. Then:*

(i) If $u \geq 0$ on A in $H^1(\Omega)$, then $u \geq 0$ almost everywhere in A.
(ii) If $u \geq 0$ almost everywhere in Ω, then $u \geq 0$ on Ω in $H^1(\Omega)$.
(iii) If $u \in H_0^1(\Omega)$ and $u \geq 0$ almost everywhere in Ω, then there exist functions $u_n \in W^{1,\infty}(\Omega)$ with $u_n \geq 0$ in Ω such that $u_n \to u$ in $H_0^1(\Omega)$.
(iv) If A is an open subset of Ω and $u \geq 0$ almost everywhere in A, then $u \geq 0$ on every compact subset $K \subset A$ in $H^1(\Omega)$.

Proof

(i) By hypothesis, there exist $u_n \in W^{1,\infty}(\Omega)$ with $u_n \geq 0$ on A, such that $u_n \to u$ in $H^1(\Omega)$. Then we can extract a subsequence $u_{n_j} \to u$ almost everywhere in Ω, and thus $u \geq 0$ almost everywhere in A.

(ii) Let $v_n \in W^{1,\infty}(\Omega)$ be such that $v_n \to u$ in $H^1(\Omega)$ and almost everywhere in Ω. Set $u_n := v_n^+$. Then, since $u = u^+$ almost everywhere in Ω, we have:

$$\|u_n - u\|_2 = \|v_n^+ - u^+\|_2 \leq \|v_n - u\|_2,$$

so $u_n \to u$ in $L^2(\Omega)$. Moreover,

$$\int_\Omega |\nabla u_n|^2 \leq \int_\Omega |\nabla v_n|^2 \leq C \quad \forall n \in \mathbb{N},$$

which implies that the sequence $\{u_n\}$ is bounded in $H^1(\Omega)$. Hence, it has a weakly convergent subsequence. Consequently, there exists a subsequence $u_{n_k} \rightharpoonup u$ weakly, and since $u_{n_k} \geq 0$ almost everywhere in Ω, we conclude that $u \geq 0$ on Ω in $H^1(\Omega)$.

(iii) The proof is similar to the previous one, starting with $v_n \in W^{1,\infty}(\Omega)$.

(iv) Let $K \subset A$ be a compact set. Let φ_ϵ be a regularizing family with $\text{sop}(\varphi_\epsilon) \subset B(0,\epsilon)$. Take $\epsilon_n \downarrow 0$, with $\epsilon_0 < \frac{1}{2}\text{d}ist(\mathbb{R}^N \setminus A)$. Then, if $w_n := u \star \varphi_{\epsilon_n}$, we have that $w_n \geq 0$ almost everywhere in K. Since A is open, we can find a function $\phi \in \mathcal{D}(\mathbb{R}^N)$ with $0 \leq \phi \leq 1$, such that

$$\phi = 1 \quad \text{on } \{x \in \mathbb{R}^N : \text{d}ist(x, K) \leq \frac{1}{2}\text{d}ist(\mathbb{R}^N \setminus A, K)\},$$

$$\phi = 0 \quad \text{on } \mathbb{R}^N \setminus A.$$

Let $v_n \in W^{1,\infty}(\Omega)$ with $v_n \to u$ in $H^1(\Omega)$. Then,

$$u_n := \phi w_n + (1-\phi)v_n \in W^{1,\infty}(\Omega),$$

and we have that $u_n \geq 0$ in K and $u_n \to u$ in $H^1(\Omega)$, so $u \geq 0$ on K in $H^1(\Omega)$.

□

Proposition 7.2 *Let $u \in H^1(\Omega)$ and suppose that*

$$\sup_{\partial\Omega} u = \inf\{m \in \mathbb{R} : u \leq m \text{ on } \partial\Omega \text{ in } H^1(\Omega)\} < \infty.$$

Then, for each $a \geq \sup_{\partial\Omega} u$, we have

$$(u-a)^+ \in H_0^1(\Omega) \text{ and } (u-a)^+ \geq 0 \text{ on } \Omega \text{ in } H^1(\Omega).$$

Proof To prove that $(u-a)^+ \in H_0^1(\Omega)$, it will be enough to show that there exist $v_n \in W^{1,\infty}(\Omega)$ with compact support such that $v_n \rightharpoonup (u-a)^+$ weakly in $H^1(\Omega)$. Let's see that this is true. By hypothesis, for each $m \in \mathbb{N}$, we can find a function $u_n^m \in W^{1,\infty}(\Omega)$ such that

$$a - u_n^m + \frac{1}{m} > 0 \text{ on } \partial\Omega$$

and

$$a - u_n^m + \frac{1}{m} \to a - u + \frac{1}{m} \quad \text{in } H^1(\Omega).$$

Then, setting $u_n := u_n^n$, we have

$$a - u_n + \frac{1}{n} > 0 \text{ on } \partial\Omega$$

and

$$u_n \to u \quad \text{in } H^1(\Omega).$$

Therefore, if we define $v_n := (u_n - \frac{1}{n} - a)^+$, then $v_n \in W^{1,\infty}(\Omega)$, v_n has compact support (note that the functions v_n are continuous), and $v_n \rightharpoonup (u-a)^+$ weakly in $H^1(\Omega)$. Hence, $(u-a)^+ \in H_0^1(\Omega)$. Finally, since $(u-a)^+ \geq 0$ almost everywhere in Ω, by the previous proposition we conclude that $(u-a)^+ \geq 0$ on Ω in $H^1(\Omega)$. □

Theorem 7.16 *(**Weak Maximum Principle**) Let $\Omega \subset \mathbb{R}^N$ be a bounded domain and let $\varphi \in H^1(\Omega)$. If $u \in H^1(\Omega)$ is the weak solution to the Dirichlet problem*

$$\begin{cases} -\Delta u = 0 & \text{in } \Omega \\ u = \varphi & \text{on } \partial\Omega, \end{cases} \tag{7.60}$$

then

$$\|u\|_{L^\infty(\Omega)} \leq \sup_{\partial\Omega} |\varphi|.$$

Proof We may assume that $\sup_{\partial\Omega} |\varphi| < \infty$. We will prove that

$$\sup_{\partial\Omega} \varphi - u \geq 0 \text{ a.e. in } \Omega, \tag{7.61}$$

since then, as $-u$ is the weak solution of the problem with boundary value $-\varphi$, it follows that

$$\sup_{\partial\Omega}(-\varphi) + u \geq 0 \text{ a.e. in } \Omega,$$

and consequently,

$$\sup_{\partial\Omega} |\varphi| - \|u\|_\infty \geq 0 \text{ a.e. in } \Omega,$$

which completes the proof.

Let $a = \sup_{\partial\Omega} \varphi$. Since u is the weak solution of (7.60), we have $u - \varphi \in H_0^1(\Omega)$, hence $a = \sup_{\partial\Omega} u$ (see Problem 7.11). Then, by the previous proposition we have $v := (u-a)^+ \in H_0^1(\Omega)$. Therefore, using (iii) of Theorem 6.2, we obtain

$$0 = \int_\Omega \nabla u \cdot \nabla v = \int_\Omega |\nabla v|^2.$$

From this, applying Poincarés inequality, we conclude that $v = 0$, and thus (7.61) holds. □

Remark 7.9 With a proof similar to that of the previous theorem, one can prove an analogous result by replacing the operator $-\Delta$ with a uniformly elliptic operator $P(x, D)$ of the form

$$P(x, D) = -\sum_{i,j=1}^{N} D_i\big(a_{ij}(x) D_j\big).$$

7.6 Spectral Theory of Elliptic Operators

The aim of this section is to study the spectrum of uniformly elliptic operators. Although we will restrict ourselves to the Dirichlet problem, many of the results have analogues for the Neumann problem.

Theorem 7.17 *Let $P(x, D)$ be a uniformly elliptic operator in Ω of the form (7.23). Then, there exists a finite or countable set $\Gamma \subset \mathbb{R}$ such that the problem*

$$\begin{cases} P(x, D)u = \lambda u + f & \text{in } \Omega \\ \\ u = 0 & \text{on } \partial\Omega \end{cases} \tag{7.62}$$

has a unique weak solution for each $f \in L^2(\Omega)$ if and only if $\lambda \notin \Gamma$. Moreover, if Γ is infinite, then $\Gamma = \{\lambda_n\}_{n=1}^{\infty}$, with $\lambda_1 \le \lambda_2 \le \ldots$, and

$$\lim_{n\to\infty} \lambda_n = +\infty.$$

Proof Let γ be the constant from Theorem 7.6. Without loss of generality, assume $\gamma > 0$. Let $\lambda > -\gamma$. As a consequence of Theorem 7.7, the problem (7.62) has a unique weak solution for each $f \in L^2(\Omega)$ if and only if $u = 0$ is the unique weak solution of the problem

$$\begin{cases} P(x, D)u = \lambda u & \text{in } \Omega \\ \\ u = 0 & \text{on } \partial\Omega, \end{cases} \tag{7.63}$$

or equivalently, if $u = 0$ is the unique weak solution of the problem

$$\begin{cases} P(x, D)u + \gamma u = (\gamma + \lambda)u & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega. \end{cases} \tag{7.64}$$

Taking into account the proof of Theorem 7.7, we have that (7.64) holds when

$$u = \frac{\gamma + \lambda}{\gamma} K u, \tag{7.65}$$

where $K : L^2(\Omega) \to L^2(\Omega)$ is a compact linear operator.

Now, that $u = 0$ is the only solution to (7.65) is equivalent to

$$\frac{\gamma + \lambda}{\gamma} \notin \sigma_p(K). \tag{7.66}$$

Therefore, the problem (7.62) has a unique weak solution for each $f \in L^2(\Omega)$ if and only if (7.66) holds. Consequently, since the operator K is compact, it suffices to apply the Riesz–Schauder Theorem (Theorem 4.15) to complete the proof. □

Remark 7.10 The set Γ from the previous result is denoted by $\sigma_D(P(x, D))$ and is called the *spectrum of the operator* $P(x, D)$ with Dirichlet boundary conditions.

In particular, the Dirichlet problem

$$\begin{cases} P(x, D)u = \lambda u & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega, \end{cases}$$

has a nontrivial solution $u \neq 0$ if and only if $\lambda \in \sigma_D(P(x, D))$. In this case, λ is called an *eigenvalue*, and the corresponding function u an *eigenfunction*.

For simplicity, from now on we will restrict ourselves to the case $P(x, D) = -\Delta$, although similar results (see, for example, [15]) can be obtained with analogous proofs for uniformly elliptic symmetric operators of the form $P(x, D) = -\sum_{i,j=1}^{N} D_j(a_{ij} D_i)$.

Theorem 7.18 *Let $\Omega \subset \mathbb{R}^N$ be a bounded open set. Then*

$$\sigma_D(-\Delta) = \{\lambda_n\}_{n=1}^{\infty},$$

with

$$0 < \lambda_1 \leq \lambda_2 \leq \dots \quad \text{and} \quad \lim_{n\to\infty} \lambda_n = +\infty.$$

Moreover, there exists an orthonormal basis $\{w_n\}_{n=1}^{\infty}$ *of* $L^2(\Omega)$ *consisting of eigenfunctions, i.e., each* $w_n \in H_0^1(\Omega)$ *is a weak solution of the problem*

$$\begin{cases} -\Delta w_n = \lambda_n w_n & \text{in } \Omega \\ \\ w_n = 0 & \text{on } \partial\Omega. \end{cases}$$

In the case where $\partial\Omega$ *is of class* C^∞*, we have that* $w_n \in C^\infty(\overline{\Omega})$.

Proof Given $f \in L^2(\Omega)$, let $Kf \in H_0^1(\Omega)$ be the unique weak solution of the problem

$$\begin{cases} -\Delta u = f & \text{in } \Omega \\ \\ u = 0 & \text{on } \partial\Omega, \end{cases}$$

i.e.,

$$\int_\Omega \nabla(Kf) \cdot \nabla v = \int_\Omega f v \qquad \forall\, v \in H_0^1(\Omega). \tag{7.67}$$

By the Rellich-Kondrakov Theorem, the operator $K : L^2(\Omega) \to L^2(\Omega)$ is compact. Furthermore, for $f, g \in L^2(\Omega)$, we have

$$\int_\Omega (Kf)g = \int_\Omega \nabla(Kf) \cdot \nabla(Kg) = \int_\Omega f(Kg),$$

and by Poincaré's inequality,

$$\int_\Omega (Kf)f = \int_\Omega \nabla(Kf) \cdot \nabla(Kf) \geq C\|Kf\|_{H_0^1(\Omega)} > 0 \quad \text{if} \quad f \neq 0.$$

Thus, K is a compact, self-adjoint, and positive operator on the Hilbert space $L^2(\Omega)$. Therefore, applying the Spectral Theorem (Theorem 4.21), there exists an orthonormal basis $\{w_n\}_{n=1}^{\infty}$ of $L^2(\Omega)$ and a sequence $\{\mu_n\}_{n=1}^{\infty}$ with $\mu_1 \geq \mu_2 \geq \ldots \geq \mu_n \to 0$, such that

$$K w_n = \mu_n w_n \qquad \forall\, n \in \mathbb{N}.$$

If we define

$$\lambda_n := \frac{1}{\mu_n},$$

then $w_n = K(\lambda_n w_n) \in H_0^1(\Omega)$, and from (7.67) it follows that

$$\int_\Omega \nabla w_n \cdot \nabla v = \lambda_n \int_\Omega w_n v \qquad \forall\, v \in H_0^1(\Omega).$$

Therefore, w_n is the unique weak solution of the problem

$$\begin{cases} -\Delta w_n = \lambda_n w_n & \text{in } \Omega \\ \\ w_n = 0 & \text{on } \partial\Omega. \end{cases}$$

Finally, let us show that if $\partial\Omega$ is of class C^∞, then $w_n \in C^\infty(\overline{\Omega})$. Since $w_n \in H_0^1(\Omega)$, by applying Theorem 7.11, we have $w_n \in H^3(\Omega)$. Then, applying Theorem 7.11 successively, we obtain $w_n \in H^m(\Omega)$ for all $m \in \mathbb{N}$, and hence, by the Sobolev embedding theorem, we have $w_n \in C^\infty(\overline{\Omega})$. □

Remark 7.11

(i) To obtain that the eigenfunctions $w_n \in C^\infty(\Omega)$, it is not necessary to impose any regularity on the boundary of the open set Ω (see Problem 7.15).

(ii) If we consider the inner product in the space $H_0^1(\Omega)$

$$(u|v) := \int_\Omega \nabla u \cdot \nabla v,$$

then

$$\left\{ \frac{w_n}{\sqrt{\lambda_n}} \right\}_{n=1}^\infty$$

is an orthonormal basis of $H_0^1(\Omega)$. Indeed:

$$\left(\frac{w_n}{\sqrt{\lambda_n}} \middle| \frac{w_m}{\sqrt{\lambda_m}} \right) = \frac{1}{\sqrt{\lambda_n}} \frac{1}{\sqrt{\lambda_m}} \int_\Omega \nabla w_n \cdot \nabla w_m = \lambda_n \int_\Omega w_n w_m = \delta_{n,m}.$$

Therefore,

$$\left\{ \frac{w_n}{\sqrt{\lambda_n}} \right\}_{n=1}^\infty$$

is an orthonormal system in $H_0^1(\Omega)$. Moreover, if $u \in H_0^1(\Omega)$ and $(u|w_n) = 0$ for every $n \in \mathbb{N}$, then

$$0 = \int_\Omega \nabla u \cdot \nabla w_n = \lambda_n \int_\Omega u w_n.$$

From this, since $\{w_n\}$ is a complete system in $L^2(\Omega)$, it follows that $u = 0$, and hence the system is a basis of $H_0^1(\Omega)$.

To provide a variational characterization of the eigenvalues λ_n and the eigenfunctions w_n, it is very useful to consider the so-called *Rayleigh quotient*, defined by

$$R(v) := \frac{\displaystyle\int_\Omega \nabla v \cdot \nabla v}{\displaystyle\int_\Omega v^2} = \frac{\||\nabla v|\|_2}{\|v\|_2}, \qquad \text{for } \; v \in H_0^1(\Omega).$$

Clearly, we have

$$\lambda_n = R(w_n) \qquad \forall\, n \in \mathbb{N}.$$

We have the following variational characterization.

Theorem 7.19 *Let $\{\lambda_n\}$ be the eigenvalues of $-\Delta$ in $H_0^1(\Omega)$, and let $\{w_n\}$ be the corresponding eigenfunctions. Then, the following formulas hold:*

$$\lambda_1 = \min\{R(v) \; : \; 0 \neq v \in H_0^1(\Omega)\} \tag{7.68}$$

$$\lambda_m = \max\{R(v) \; : \; 0 \neq v \in \mathrm{L}IN\{w_1, \ldots, w_m\}\} \tag{7.69}$$

$$\lambda_m = \min\{R(v) \; : \; v \perp \{w_1, \ldots, w_{m-1}\}\} \tag{7.70}$$

$$\lambda_m = \min\left\{ \max_{v \in W} R(v) \;\; : \;\; W \subset \; H_0^1(\Omega), \; \mathrm{d}im(W) = m \right\} \tag{7.71}$$

for $m \geq 2$.

Proof Let $E_m := \mathrm{L}IN\{w_1, \ldots, w_m\}$. If $v \perp E_{m-1}$, then the Fourier series of v in $L^2(\Omega)$ with respect to the basis $\{w_n\}$ is given by

$$v = \sum_{k=m}^{\infty} \alpha_k w_k, \quad \text{with} \quad \alpha_k = \int_\Omega v w_k.$$

If we take

$$v_l := \sum_{k=m}^{l} \alpha_k w_k,$$

we have that

$$v_l \to v \quad \text{in} \quad L^2(\Omega). \tag{7.72}$$

On the other hand, the Fourier series of v in $H_0^1(\Omega)$ with respect to the basis $\left\{\frac{w_n}{\sqrt{\lambda_n}}\right\}_{n=1}^{\infty}$ is given by

$$v = \sum_{k=m}^{\infty} \frac{\beta_k}{\sqrt{\lambda_k}} w_k,$$

with

$$\beta_k = \frac{1}{\sqrt{\lambda_k}} \int_\Omega \nabla v \cdot \nabla w_k = \frac{1}{\sqrt{\lambda_k}} \int_\Omega v w_k = \frac{1}{\sqrt{\lambda_k}} \alpha_k.$$

Hence,

$$v_l \to v \quad \text{in} \quad H_0^1(\Omega). \tag{7.73}$$

As a consequence of (7.72) and (7.73), we have that $R(v_l) \to R(v)$. Now,

$$R(v_l) = \frac{\displaystyle\sum_{k=m}^{l} \alpha_k^2 \int_\Omega \nabla w_k \cdot \nabla w_k}{\displaystyle\sum_{k=m}^{l} \alpha_k^2} = \frac{\displaystyle\sum_{k=m}^{l} \alpha_k^2 \lambda_k}{\displaystyle\sum_{k=m}^{l} \alpha_k^2} \geq \lambda_m.$$

Therefore, $R(v) \geq \lambda_m$, from which we deduce (7.70), since

$$w_m \perp \{w_1, \ldots, w_{m-1}\} \quad \text{and} \quad R(w_m) = \lambda_m.$$

Taking $m = 1$ in the previous proof, we obtain (7.68).

If $v \in E_m$, then

$$v = \sum_{k=1}^{m} \alpha_k w_k,$$

so that

$$R(v) = \frac{\displaystyle\sum_{k=1}^{m} \alpha_k^2 \lambda_k}{\displaystyle\sum_{k=1}^{m} \alpha_k^2} \leq \lambda_m.$$

Therefore,

$$\max\{R(v) \ : \ v \in E_m\} \leq \lambda_m,$$

but since $R(w_m) = \lambda_m$, we have proved (7.69).

Finally, let W be a vector subspace of $H_0^1(\Omega)$ of dimension m. Then, there exists $w \in W$ such that $w \perp \{w_1, \ldots, w_{m-1}\}$. Consequently, by (7.70), we have

$$\max\{R(v) \ : \ v \in W\} \geq \lambda_m.$$

Now, since this is true for every vector subspace W of $H_0^1(\Omega)$ of dimension m, we obtain

$$\min\left\{ \max_{v \in W} R(v) \ : \ W \subset H_0^1(\Omega), \ dim(W) = m \right\} \geq \lambda_m.$$

From this, taking into account (7.69), we deduce (7.71). □

Note that the characterization (7.71) is intrinsic in the sense that it is independent of the sequence of eigenfunctions.

Proposition 7.3 *If $w \in H_0^1(\Omega)$ is a nonzero element that satisfies the relation $R(w) = \lambda_1$, then w is an eigenfunction corresponding to the eigenvalue λ_1.*

Proof We can assume without loss of generality that $\|w\|_2 = 1$. Let $v \in H_0^1(\Omega)$ and $t \neq 0$. Then, since $w + tv \in H_0^1(\Omega)$, applying (7.68), we have that $R(w+tv) \geq \lambda_1 = R(w)$, i.e.,

$$\frac{\displaystyle\int_\Omega \nabla(w + tv) \cdot \nabla(w + tv)}{\displaystyle\int_\Omega (w + tv)^2} \geq \int_\Omega \nabla w \cdot \nabla w = \lambda_1.$$

Then,

$$t^2 \int_\Omega \nabla v \cdot \nabla v + 2t \int_\Omega \nabla w \cdot \nabla w \geq \lambda_1 \left(2t \int_\Omega wv + t^2 \int_\Omega v^2 \right).$$

Dividing by $2t$ and letting $t \to 0^+$ and $t \to 0^-$ respectively, we obtain

$$\lambda_1 \int_\Omega wv = \int_\Omega \nabla w \cdot \nabla v,$$

and since this holds for every $v \in H_0^1(\Omega)$, the result follows. □

The first eigenvalue λ_1 is usually called the *principal eigenvalue*. In the next result, we will see that this eigenvalue is simple, i.e., its associated eigenspace is one-

dimensional and contains a strictly positive function. Thus, this result resembles the classical Perron–Frobenius Theorem for matrices, which ensures that every positive and irreducible matrix has a strictly positive spectral radius which is a simple eigenvalue and has an associated strictly positive eigenvector (see [20]). There is an infinite-dimensional version of the Perron–Frobenius Theorem in Banach lattices, the Krein–Rutman Theorem (see [49]), which, when applied to the operator K in the proof of Theorem 7.18, yields a proof of the following result. Here, however, we follow a different approach from [33].

Theorem 7.20 *Let $\Omega \subset \mathbb{R}^N$ be a bounded domain with C^∞ boundary. Then, the principal eigenvalue λ_1 is simple and the corresponding eigenfunctions do not change sign in Ω. Therefore, we can choose the eigenfunction $w_1 > 0$.*

Proof Let $w \in H_0^1(\Omega)$ be an eigenfunction corresponding to the eigenvalue λ_1. Since $\partial\Omega$ is of class C^∞, we have $w \in C^\infty(\overline{\Omega})$. We have

$$\int_\Omega \nabla w \cdot \nabla v = \lambda_1 \int_\Omega wv \quad \forall\, v \in H_0^1(\Omega).$$

Now, since $w^+, w^- \in H_0^1(\Omega)$, as in the proof of Theorem 7.15, we obtain

$$\int_\Omega \nabla w^+ \cdot \nabla w^+ = \lambda_1 \int_\Omega (w^+)^2 \tag{7.74}$$

and

$$\int_\Omega \nabla w^- \cdot \nabla w^- = \lambda_1 \int_\Omega (w^-)^2. \tag{7.75}$$

Suppose, by contradiction, that w does not have constant sign in Ω. Then $w^+ \neq 0$ and $w^- \neq 0$. From (7.74) and (7.75), it follows that

$$R(w^+) = R(w^-) = \lambda_1.$$

Therefore, by the previous proposition, w^+ and w^- are eigenfunctions associated to the eigenvalue λ_1. Then, by applying the strong maximum principle, we would have that w^+ and w^- are strictly positive, which leads to a contradiction. Thus, w has constant sign, and we can choose $w_1 > 0$.

Finally, suppose that λ_1 is not simple. Then there exists an eigenfunction w associated to the eigenvalue λ_1, orthonormal to w_1. Now, since w has constant sign, we have

$$\int_\Omega w_1 w \neq 0.$$

This leads to a contradiction, and therefore, λ_1 is simple. □

Remark 7.12 Note that the hypothesis about the smoothness of the boundary of the open set Ω has only been used to obtain that $u \in C^\infty(\overline{\Omega})$. Now, to apply the strong maximum principle, we only need that $u \in C^2(\Omega) \cap C(\overline{\Omega})$, and this can be achieved with less regularity on the boundary; for example, since $w_1 \in H^2(\Omega) \cap C^\infty(\Omega)$, if $N \leq 3$, by continuous embedding we have that $u \in C(\overline{\Omega})$.

Problems

7.1 Prove that the *Sturm–Liouville problem*

$$\begin{cases} -(pu')' + qu = f & \text{in }]0,1[\\ u(0) = u(1) = 0, \end{cases}$$

where $p \in C^1([0,1])$, $q \in C([0,1])$, and $f \in L^2(]0,1[)$ are given data satisfying

$$p(x) \geq \alpha > 0 \quad \forall\, x \in]0,1[,$$

has a unique solution.

7.2 Prove that for each $f \in L^2(]0,1[)$ and $a, b \in \mathbb{R}$, there exists a unique $u \in H^2(]0,1[)$ solution to the problem

$$\begin{cases} -u'' + u = f & \text{in }]0,1[\\ u'(0) = a, \quad u'(1) = b. \end{cases}$$

7.3 Suppose that $\Omega \subset \mathbb{R}^N$ is a bounded domain with smooth boundary and $f \in L^2(\Omega)$. Prove that the Neumann problem

$$\begin{cases} -\Delta u = f & \text{in } \Omega \\ \dfrac{\partial u}{\partial \eta} = 0 & \text{on } \partial\Omega \end{cases}$$

has a weak solution if and only if

$$\int_\Omega f(x)\, dx = 0.$$

7.4 Prove that for each $f \in L^2(\Omega)$ and $g \in H^{-1/2}(\partial\Omega)$, the problem (7.42) has a unique weak solution.

7.5 Let $\Omega \subset \mathbb{R}^N$ be a bounded open set with smooth boundary, $\alpha > 0$, and $f \in L^2(\Omega)$. Provide a variational formulation of the Robin problem

$$\begin{cases} -\Delta u = f & \text{in } \Omega \\ \dfrac{\partial u}{\partial \eta} + \alpha u = 0 & \text{on } \partial\Omega \end{cases} \tag{7.76}$$

and prove that this problem has a unique weak solution.

7.6 Let $\Omega \subset \mathbb{R}^N$ be a bounded open set with smooth boundary and $f \in L^2(\Omega)$. Suppose that $\partial\Omega = \Gamma_1 \cup \Gamma_2$, with $\Gamma_1 \cap \Gamma_2 = \emptyset$, and that Γ_1 has strictly positive $(N-1)$-dimensional measure. Provide a variational formulation of the mixed problem

$$\begin{cases} -\Delta u = f & \text{in } \Omega \\ u = 0 & \text{on } \Gamma_1 \\ \dfrac{\partial u}{\partial \eta} = 0 & \text{on } \Gamma_2 \end{cases} \tag{7.77}$$

and prove that this problem has a unique weak solution.

7.7 Let $\Omega = (1, +\infty)$. Find $f \in L^2(\Omega)$ such that the problem

$$\begin{cases} -u'' = f & \text{in } \Omega \\ u \in H_0^1(\Omega) \end{cases}$$

has no solution.

7.8 A function $u \in H^2(\Omega)$ is said to be a weak solution of the *biharmonic equation*

$$\begin{cases} \Delta^2 u = f & \text{in } \Omega \\ u = \dfrac{\partial u}{\partial \eta} = 0 & \text{on } \partial\Omega \end{cases} \tag{7.78}$$

if

$$\int_\Omega \Delta u \, \Delta v = \int_\Omega f v \qquad \forall\, v \in H_0^2(\Omega).$$

Prove that for every $f \in L^2(\Omega)$ there exists a unique weak solution of the biharmonic equation (7.78).

7.9 Let $\Omega \subset \mathbb{R}^N$ be an open set and let $u \in H^1(\Omega)$ such that

$$\int_\Omega \nabla u \cdot \nabla \varphi = \int_\Omega f\varphi \qquad \forall\, \varphi \in \mathcal{D}(\Omega).$$

Prove that if $f \in H^m(\Omega)$, then $u \in H^{m+2}_{loc}(\Omega)$. In particular, if $f \in C^\infty(\Omega)$, then $u \in C^\infty(\Omega)$.

7.10 Let $\phi(x) = \frac{3}{4} - x^2$. Compute the solution to the obstacle problem

$$\begin{cases} u \in K_\phi = \{v \in H^1_0(]-1,1[) \ : v(x) \geq \phi(x) \ \ a.e. \ \ x \in]-1,1[\} \\ \\ \displaystyle\int_{-1}^{1} u'(x)(v'(x) - u'(x))\,dx \geq 0 \quad \forall\, v \in K_\phi. \end{cases}$$

7.11 Let $\Omega \subset \mathbb{R}^N$ be a bounded domain. Prove that if $u, v \in H^1(\Omega)$ with $u - v \in H^1_0(\Omega)$, then

$$\sup_{\partial\Omega} u = \sup_{\partial\Omega} v.$$

7.12 Let $\Omega \subset \mathbb{R}^N$ be a bounded domain. A function $u \in H^1(\Omega)$ is called a supersolution of the equation

$$-\Delta u = 0 \quad \text{in } \ \Omega$$

if it satisfies

$$\int_\Omega \nabla u \cdot \nabla v \geq 0 \quad \text{ for every } \ 0 \geq v \in H^1(\Omega).$$

Prove that if u is a supersolution, then

$$u(x) \geq \inf_{\partial\Omega} u \quad \text{ for almost every } \ x \in \Omega.$$

7.13 Let $\Omega \subset \mathbb{R}^N$ be a bounded open set with smooth boundary and let $f \in L^2(\Omega)$. Prove that if $u \in H^1(\Omega)$ is a weak solution of the Neumann problem

$$\begin{cases} -\Delta u + u = f & \text{in } \ \Omega \\ \\ \dfrac{\partial u}{\partial \eta} = 0 & \text{on } \ \partial\Omega, \end{cases}$$

then

$$\inf_{\Omega} f \leq u(x) \leq \sup_{\Omega} f \qquad \text{a.}e.\ x \in \Omega.$$

7.14 Let $\Omega \subset \mathbb{R}^N$ be a bounded open set and $P(x, D)$ a uniformly elliptic operator in Ω. Prove that if $\lambda \notin \sigma_D(P(x, D))$, then

$$\|u\|_2 \leq C\|f\|_2 \qquad \forall\, f \in L^2(\Omega),$$

where $u \in H_0^1(\Omega)$ is the unique weak solution of the problem

$$\begin{cases} P(x, D)u = \lambda u + f & \text{in } \Omega \\ \\ u = 0 & \text{on } \partial\Omega. \end{cases}$$

The constant C depends only on λ, Ω, and the coefficients of the operator $P(x, D)$.

7.15 Prove that if $\Omega \subset \mathbb{R}^N$ is a bounded open set and $u \in H_0^1(\Omega)$ is a solution of the eigenvalue problem

$$\begin{cases} -\Delta u = \lambda u & \text{in } \Omega \\ \\ u = 0 & \text{on } \partial\Omega, \end{cases}$$

with $\lambda \neq 0$, then $u \in C^\infty(\Omega)$.

Appendix A
Some Calculus Results

In this appendix, we provide a quick outline of some calculus results.

Theorem A.1 (Mean Value Theorem) *Let $\Omega \subset \mathbb{R}^N$ be an open set and $f : \Omega \subset \mathbb{R}^N \to \mathbb{R}^M$ a differentiable function on Ω. Let $x, y \in \Omega$ such that $[x, y] \subset \Omega$. Then the following inequality holds:*

$$\|f(y) - f(x)\| \leq \sup_{z \in [x,y]} \|Df(z)\| \, \|y - x\|.$$

Definition A.1 Let $\Omega \subset \mathbb{R}^N$ be an open set, and let $k \in \mathbb{N}$. We say that $\partial\Omega$ is of *class* C^k if for every point $x^0 \in \partial\Omega$ there exists an $r > 0$ and a C^k function $\gamma : \mathbb{R}^{N-1} \to \mathbb{R}$ such that

$$\Omega \cap B(x^0, r) = \{x \in B(x^0, r) \, : \, x_N > \gamma(x_1, \ldots, x_{N-1})\}.$$

When $\partial\Omega$ is of class C^k for all $k = 1, 2, \ldots$, we say that $\partial\Omega$ is of class C^∞. If the function γ is only Lipschitz continuous, we say that $\partial\Omega$ is Lipschitz.

Frequently, we need to perform a change of coordinates near a point $x^0 \in \partial\Omega$ so that, with respect to the new coordinates, the boundary becomes flat. To be more precise, if $x^0 \in \partial\Omega$ is fixed and γ is the function such that

$$\Omega \cap B(x^0, r) = \{x \in B(x^0, r) \, : \, x_N > \gamma(x_1, \ldots, x_{N-1})\},$$

we define

$$\phi^i(x) := x_i, \qquad \text{for } i = 1, 2, \ldots, N-1$$

$$\phi^N(x) := x_N - \gamma(x_1, \ldots, x_{N-1}),$$

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and we write $y = \phi(x) = (\phi^1(x), \ldots, \phi^N(x))$. Similarly, we take

$$\psi^i(y) := y_i, \qquad \text{for } i = 1, 2, \ldots, N-1$$

$$\psi^N(y) := y_N + \gamma(y_1, \ldots, y_{N-1}),$$

and we write $x = \psi(y) = (\psi^1(y), \ldots, \psi^N(y))$. Then, $\phi = \psi^{-1}$, and the map $x \mapsto \phi(x) = y$ flattens $\partial\Omega$ near x^0.

Definition A.2 If $\partial\Omega$ is C^1. then along $\partial\Omega$ is defined the *outward normal vector field*

$$\eta = (\eta_1, \ldots, \eta_N).$$

Let $u \in C^1(\overline{\Omega})$. We cal

$$\frac{\partial u}{\partial \eta} := \eta \cdot Du$$

the outward normal derivative of u.

Theorem A.2 (Green–Gauss Theorem) *Let $\Omega \subset \mathbb{R}^N$ be a bounded open set with boundary $\partial\Omega$ of class C^1 and $u \in C^1(\overline{\Omega})$. Then*

$$\int_\Omega \frac{\partial u}{\partial x_i}\, dx = \int_{\partial\Omega} u\eta_i dS \quad i = 1, \ldots, N.$$

We have the following integration-by-part formula.

Theorem A.3 (Integration-by-Part Formula) *Let $\Omega \subset \mathbb{R}^N$ be a bounded open set with boundary $\partial\Omega$ of class C^1 and $u, v \in C^1(\overline{\Omega})$. Then*

$$\int_\Omega \frac{\partial u}{\partial x_i} v\, dx = \int_\Omega \frac{\partial v}{\partial x_i} u\, dx + \int_{\partial\Omega} uv\eta_i dS \quad i = 1, \ldots, N.$$

Theorem A.4 (Green Formulas) *Let $\Omega \subset \mathbb{R}^N$ be a bounded open set with boundary $\partial\Omega$ of class C^1 and $u, v \in C^2(\overline{\Omega})$. Then*

(i)

$$\int_\Omega \Delta u\, dx = \int_{\partial\Omega} \frac{\partial u}{\partial \eta}, dS$$

(ii)

$$\int_\Omega \nabla u \cdot \nabla v\, dx = -\int_\Omega u \Delta v\, dx + \int_\partial \frac{\partial v}{\partial \eta} u\, dS,$$

Appendix B
Measure Theory

In this appendix, we provide a quick outline of some fundamentals of measure theory.

Let $(\Omega, \mathcal{B})$ be a measurable space, i.e., Ω is a set and $\mathcal{B}$ is a σ-*algebra* in Ω, i.e., $\mathcal{B}$ is a collection of subsets of Ω satisfying:

(i) $\emptyset \in \mathcal{B}$,
(ii) $A \in \mathcal{B} \Rightarrow \Omega \setminus A \in \mathcal{B}$,
(iii) $\bigcup_{n=1}^{\infty} A_n \in \mathcal{B}$ whenever $A_n \in \mathcal{B}$ for all $n \in \mathbb{N}$.

A *measure* μ on $(\Omega, \mathcal{B})$ is a map $\mu : \mathcal{B} \to [0, \infty]$ that satisfies

(i) $\mu(\emptyset) = 0$,
(ii) whenever (A_n) is a countable family of disjoint members of $\mathcal{B}$, then

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n).$$

On occasion when it is desirable to call attention on the particular σ-algebra and measure under consideration, we shall write $(\Omega, \mathcal{B}, \mu)$.

The sets $A \in \mathcal{B}$ with the property that $\mu(A) = 0$ are called the *null sets*. We say that a property holds a.e. (or for almost all $x \in \mathcal{B}$) if it holds everywhere on $\mathcal{B}$ except on a null set.

We say that the measurable space $(\Omega, \mathcal{B}, \mu)$ is σ-finite if there exists a countable family (Ω_n) in $\mathcal{B}$ such that $\Omega = \bigcup \Omega_n$ and $\mu(\Omega_n) < \infty$ for all $n \in \mathbb{N}$.

A function $f : \Omega \to \mathbb{R}$ is called measurable if for every open set A of $\mathbb{R}$, $f^{-1}(A) \in \mathcal{B}$.

We assume that the reader is familiar with the notion of an integrable function $f : \Omega \to \mathbb{R}$, and the concept of an integral; see, e.g. [6, 24, 46, 47, 57]. In the Euclidean space $\mathbb{R}^N$, the most important measure is the Lebesgue measure $\mathcal{L}^N$ integral, whose

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integral we will denote as

$$\int_{\mathbb{R}^N} f(x)dx.$$

If $f : \Omega \to \mathbb{C}$, we denote by $\Re f$, $\Im f$ the real and imaginary parts of f, respectively, i.e.

$$f(x) = \Re f(x) + i\,\Im f(x), \qquad \text{for each } x \in \Omega.$$

Hence, $\Re f$ and $\Im f$ are real-valued functions. A function f is said to be *measurable* if its real and imaginary parts are measurable, and f is said to be *integrable* in Ω if both parts are integrable. In this case, we write $f \in L^1(\Omega, \mathbb{C})$, i.e., $f \in L^1(\Omega, \mathbb{C})$ if and only if $\Re f, \Im f \in L^1(\Omega)$. Moreover, we define:

$$\int_\Omega f := \int_\Omega \Re f + i \int_\Omega \Im f.$$

If $f : \Omega \to \mathbb{C}$, its modulus is the real-valued function $|f|$ defined by:

$$|f|(x) := \sqrt{(\Re f(x))^2 + (\Im f(x))^2}.$$

We have that $L^1(\Omega, \mathbb{K})$ is a Banach space respect the norm

$$\|f\|_1 := \int_\Omega |f|\, d\mu.$$

As usual, we identify two functions that coincide a.e. The following are the most important results on the convergence of integrable functions.

Theorem B.1 (Dominated Convergence Theorem) *Let f_n be measurable functions defined almost everywhere on Ω such that $f_n \to f$ almost everywhere in Ω. Suppose that there exists $g \in L^1(\Omega, \mathbb{K})$ such that $|f_n| \le g$ almost everywhere in Ω for every $n \in \mathbb{N}$. Then $f \in L^1(\Omega, \mathbb{K})$ and moreover:*

$$\lim_{n\to\infty} \int_\Omega f_n\, d\mu = \int_\Omega f\, d\mu.$$

Theorem B.2 (Monotone Convergence Theorem) *Let $\{f_n\}$ be a monotone sequence of functions in $L^1(\Omega, \mathbb{R})$ whose integrals form a bounded sequence. Then, $\{f_n\}$ converges almost everywhere to a function $f \in L^1(\Omega, \mathbb{R})$, and moreover*

$$\int_\Omega f\, d\mu = \lim_{n\to\infty} \int_\Omega f_n\, d\mu.$$

Theorem B.3 (Fatou's Lemma) *Let $\{f_n\}$ be a sequence of positive functions in $L^1(\Omega)$ such that $f_n \to f$ a.e. in Ω. If there exists a $K \in \mathbb{R}$ such that*

$$\int_\Omega f_n \, d\mu \leq K \quad \forall n \in \mathbb{N}.$$

For almost all $x \in \Omega$ we set $f(x) = \liminf_{n\to\infty} f_n(x) \leq +\infty$. Then $f \in L^1(\Omega)$ and

$$\int_\Omega f \, d\mu \leq \liminf_{n\to\infty} \int_\Omega f_n \, d\mu.$$

Let $(\Omega_1, \mathcal{B}_1, \mu_2)$ and $(\Omega_2, \mathcal{B}_2, \mu_2)$ be two measures spaces that are σ-finite. The structure of the measurement space $(\Omega, \mathcal{B}, \mu)$ can be defined in a standard way on the Cartesian product $\Omega = \Omega_1 \times \Omega_2$.

The two following results are related the integral in Ω with the reiterated integrals.

Theorem B.4 (Fubini's Theorem) *If $f \in L^1(\Omega_1 \times \Omega_2)$, then*

$$\int_{\Omega_1\times\Omega_2} f(x, y) \, d\mu(x, y)$$

$$= \int_{\Omega_1} \left(\int_{\Omega_2} f(x, y) \, d\mu_2(y) \right) d\mu_1(x) = \int_{\Omega_2} \left(\int_{\Omega_1} f(x, y) \, d\mu_1(x) \right) d\mu_2(y).$$

Theorem B.5 (Tonelli–Hobson Theorem) *Suppose that $f : \Omega_1 \times \Omega_2 \to \mathbb{R}$ measurable. If one of the iterated integrals*

$$\int_{\Omega_1} \left(\int_{\Omega_2} |f(x, y)| \, d\mu_2(y) \right) d\mu_1(x), \qquad \int_{\Omega_2} \left(\int_{\Omega_1} |f(x, y)| \, d\mu_1(x) \right) d\mu_2(y)$$

exists, then $f \in L^1(\Omega_1 \times \Omega_2)$.

Definition B.1 Let $(\Omega, \mathcal{B})$ be a measurable space. A function $\nu : \Omega \to \overline{\mathbb{R}}$ is called a *signed measure* if it satisfies:

(i) $\nu(\emptyset) = 0$,
(ii) ν takes at most one of the values $+\infty$ and $-\infty$,
(iii) if $E_n \in \Omega$ is a sequence of pairwise disjoint sets, then

$$\nu\left(\bigcup_{n=1}^{\infty} E_n \right) = \sum_{n=1}^{\infty} \nu(E_n),$$

where the equality means that the series $\sum_{n=1}^{\infty} \nu(E_n)$ converges absolutely if $\nu\left(\bigcup_{n=1}^{\infty} E_n\right) \in \mathbb{R}$ and diverges otherwise.

Definition B.2 Given two measures μ_1, μ_2 on the measurable space (X, Σ), they are said to be *mutually singular*, and we write $\mu_1 \perp \mu_2$, if there exist disjoint sets $A, B \in \Sigma$, with $X = A \cup B$, such that $\mu_1(A) = \mu_2(B) = 0$.

Theorem B.6 (Jordan Decomposition Theorem) *Let ν be a signed measure on the measurable space $(\Omega, \mathcal{B})$. There exists a unique pair of mutually singular measures ν^+ and ν^- such that $\nu = \nu^+ - \nu^-$. Moreover, if ν_1 and ν_2 are measures such that $\nu = \nu_1 - \nu_2$, then $\nu^+(E) \leq \nu_1(E)$ and $\nu^-(E) \leq \nu_2(E)$ for every $E \in \Sigma$.*

The measures ν^+ and ν^- of the previous theorem are called the *positive variation* and *negative variation* of ν, respectively. The measure $|\nu|$ defined by

$$|\nu|(E) := \nu^+(E) + \nu^-(E) \quad \text{for each } E \in \Sigma$$

is called the *total variation*. The representation of ν as $\nu^+ - \nu^-$ is called the *Jordan decomposition* of ν.

Definition B.3 Let $(\Omega, \mathcal{B}, \mu)$ be a measurable space an ν a signed measure on the measurable space $(\Omega, \mathcal{B})$. It is say that ν is *absolutely continuous* respect to μ, and it write $\nu \ll \mu$, if $\nu(E) = 0$ for each $E \in \mathcal{B}$ with $\mu(E) = 0$.

Theorem B.7 (Radon–Nikodym Theorem) *Let $((\Omega, \mathcal{B}, \mu)$ be a σ-finite measure space and let ν be a signed measure on the measurable space $((\Omega, \mathcal{B})$. Then, if ν is absolutely continuous with respect to μ, there exists a μ-measurable function f_0 such that*

$$\nu(E) = \int_E f_0 \, d\mu \quad \textit{for each } \ E \in \Sigma.$$

Moreover, the function f_0 is unique in the sense that if g_0 is another μ-measurable function with the same property, then $f_0 = g_0$ μ-almost everywhere.

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